

Master Beekeeping Complete Course Content

Parts I–IV • Clean Reading Edition

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Part I: Honey Bee Evolution, Biology, and Behavior

The Evolution of the Honey Bee as a Social

Insect Honey bees are one of the most fascinating animals on Earth. While an individual honey bee cannot survive on its own, a colony of honey bees, working together as a cooperative unit, can thrive. In this module, you will explore how honey bees evolved from solitary wasps into one of the most advanced examples of social living on the planet.

WATCH

The Honey Bee Colony as a Superorganism

■ A honey bee colony is a Key Points captivating thing. A mature colony in A superorganism is a group of the peak of summer can contain related, social individuals working 30,000 to 50,000 individuals, all together to effectively cooperate working together as a cooperative unit as one large organism. that functions as an integrated whole. The individual members cannot survive on their own. The queen, who is the mother of all the bees in a colony, depends on her worker daughters for building and defending the nest, keeping it stocked with food, rearing the brood, fighting parasites and pathogens, and everything else. Meanwhile, the workers depend on their queen mother to lay all the eggs the colony needs to produce its workers, drones, and young queens. As a superorganism, the bees are able to benefit from social living. They forage very efficiently by communicating the location of rich food sources to their nest mates, they regulate the temperature of their home, and they reproduce through swarming. Swarming is a remarkable example of a colony's social integration, because it requires finding a new home site that all of the bees agree upon. For successful beekeeping, it is important to understand these and other traits that have evolved with the social life of these bees. Watch the video to learn about some more of the social behaviors honey bees have evolved that make them such successful organisms. Video Transcript So thousands of individual organisms are working together and acting like one large organism, or a superorganism, and they're cooperatively collecting food, they're mounting defenses against intruders and they're coordinating behavioral responses to a changing environment. So in order to do all of this socially, complex communication is required. And honey bees use three forms of communication: acoustic — touch and vibration, chemical and visual. And the most famous form of communication with honey bees — the waggle dance, actually combines some of these three forms. Another really neat thing about this social behavior in honey bees is social immunity. So diseases can be easily transmitted when we have lots of individuals in close proximity to each other, like honey bees in a hive. And so, social immunity is using social behaviors to try to reduce the disease spread or reduce parasites that are in the colony. So honey bees some of them may groom each other to remove mites from each other's backs, or they can detect brood that's infected with diseases and actually remove them from the colony, a form of hygienic behavior. And for some infections like chalkbrood infections, they can actually develop a social fever. And this is where they physically raise

the temperature of the brood to the point where it gets too hot for the chalkbrood infection to live. So these are just some of the really cool things that these very complex socially behaving honey bees are doing.

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Other Superorganisms

Honey bees are not alone in this style of advanced social living in which offspring "stay at home" for life to live with one or both of their parents. Termites, ants, some wasps, naked mole rats, and a few other organisms also live this way. These social colonies are often referred to as superorganisms because they are many individual organisms that function together as one large organism. Sometimes the behavior of an individual member of the colony may seem confusing or counterintuitive, but makes sense when it is understood as part of the functioning of a superorganism. Photo: Giant nest built by termites (an example of a superorganism). Photo credit: Graham Thompson

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What Does It Mean to Be a Honey Bee?

■ All living organisms can be classified scientifically into different categories to Western honey bees are referred help us understand which to as *Apis mellifera* because of characteristics they share and which how they are classified are different. We give organisms As a Master Beekeeper, you need scientific names based on their place to be able to distinguish between in this classification system. This honey bees and the other insects system is useful in understanding why they are frequently confused with Western honey bees are called *Apis mellifera*, identifying how they are unique compared to other bees and insects, and describing the immense diversity of living organisms we see around us. This classification system is broken down into the following groups: kingdom, phylum, class, order, family, genus, and species. As we move down the groups from kingdom to species, we observe more similarities among organisms' appearances, behaviors, and genetics. How does the honey bee fit in with the rest of life on Earth? Kingdom At its most basic level, life is classified into five different kingdoms: animals, plants, fungi, bacteria, and archaea (which are other small, single-celled organisms). Honey bees, along with all other insects, are in the animal kingdom. The five kingdoms of life: Animals Plants Fungi Bacteria Archaea Phylum The next division is at the phylum level. There are many phyla (plural of "phylum") within the animal kingdom. Honey bees are members of the phylum called Arthropoda, so we call them arthropods. All arthropods are invertebrates, as they do not develop a backbone and instead have a tough outer body shell called an exoskeleton. Arthropods have segmented bodies and pairs of jointed appendages. Some members of the phylum Arthropoda: Class The next division is at the class level. Within Arthropoda, there are several classes, including Insecta, Arachnida (e.g., spiders and Varroa mites), and Crustacea (e.g., crustaceans), to name a few. Honey bees belong to the class Insecta (Insects), which is the most diverse class of animals on Earth. Insects are defined by having six legs across three body parts (head, thorax, and abdomen), compound eyes, and a pair of antennae. Some members of the class Insecta: Order Within the class Insecta, there are many different orders of insects. Bees are in the order Hymenoptera, which includes ants, bees, and wasps. Insects in the order Hymenoptera usually have two pairs of wings joined together by small hooks. Females have a hardened ovipositor (egg-laying organ) that may be used for piercing, stinging,

or reproduction. They are also the biggest-brained insects, hence they are considered the "intelligentsia" of the insect world. Aside from similarities in body structure, some Hymenopterans share other traits with honey bees: they have similar social living strategies with a queen and cooperative helpers, they build nests and store food, and they sting to defend their colonies. Some characteristics can cause people to confuse other bees and wasps with honey bees. Some members of the order Hymenoptera: Family Within the order Hymenoptera, there are nine families of bees that consist of about 20,000 documented bee species throughout the world! Honey bees are within the family Apidae. All bees within this family carry pollen on their legs. It is the largest family of bees and includes bumble bees, carpenter bees, and many others in addition to honey bees. In addition to honey bees, large carpenter bees (left) and bumble bees (right) also belong to the family Apidae. Genus The last two categories comprise the species name. Every organism has a two-part species name. The first part is the genus name and the second part is the species name. Bumble bees, for instance, are all part of the genus 'Bombus', while honey bees are all part of the 'Apis' genus. Organisms within the same genus often look similar to each other. Species The Western honey bee has *Apis mellifera* for its species name. What does "mellifera" mean? Melli means "honey" and fera means "bearer", so it was named as the "bearer" or maker of honey. There are eight species of honey bee (members of the genus *Apis*) found throughout the world, and some that you may be familiar with are the Eastern honey bee (*Apis cerana*), the dwarf honey bee (*Apis florea*), and the giant honey bee (*Apis dorsata*). These different species have different body sizes, behaviors, and nest designs, but they all have a 'honey bee' look to them with similar colors and banding patterns, and they all make honey. Most importantly, organisms need to be the same species in order to reproduce. Western honey bees can only mate and create offspring with other Western honey bees. However, closely related species can share parasites and diseases. *Apis cerana* is the original host of *Varroa mites* and *Nosema ceranae*, but these parasites are able to thrive in colonies of the closely related Western honey bee. Different species of honey bees (from left to right): *Apis dorsata*, *Apis florea*, and *Apis mellifera*.

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Distinguishing Honey Bees From Other Insects

Some insects are often confused with honey bees, especially wasps, other bees, and flies. As a Master Beekeeper, you must be able to distinguish honey bees from other organisms. You may receive calls from the public about swarms and unwanted nests, and it is important to identify whether the insects in question are actually honey bees in order to determine how you can be of service. If you receive one of these calls, ask the person to describe the nest and the appearance of the insect, and to send a photo if they have one. If the insect in question is a wasp and a nuisance, you can refer them to a pest control company. However, if the location of the colony is not a threat to anyone, you can suggest they let the colony be, as wasps are important natural enemies of many insect pests.

Organism	Nest	Appearance	Food
Yellow Jackets (Wasps)	Some species nest underground; black banding; nectar; larvae are others build round hairless, slender carnivores nests covered in waist; often have papier-mâché-like yellow legs; social material in high colonies of many locations; no honey individuals; body is produced or about the same size stored in the nest as honey bee	Paper Wasps (Wasps)	Open 'papery' Yellow and black or
	Adults feed on hexagonal cells; no brown and black nectar; larvae are honey is produced banding;		

hairless, carnivores or stored in the nest slender waist; social colonies of many individuals; body is often larger than honey bees Do not create nests; Yellow and black or Adults feed on do not live socially brown and black pollen and nectar; Syrphid Flies banding; single pair larvae are of wings that don't carnivores fold when resting; large eyes that almost touch; some hair on head and thorax; very short antennae; body can be smaller than or similar in size to honey bees Paper wasp photo credit: Sarah Blucher

READ

The Evolution of Honey Bees

■ How did bees evolve? Before bees Key Points existed, there were several species of Honey bees evolved from wasps. These wasps, just like wasps carnivorous, solitary wasps today, hunted prey to feed their young. Bees first evolved about 130 million Evidence suggests that millions of years ago, shortly after flowering years ago, one species of wasp started plants first evolved feeding their young pollen from flowers. These wasps were the ancestors to modern bees, who feed on nectar and pollen. In a way, you can say that bees are vegetarian wasps. Over time, they evolved hairs on their bodies and special pollen- transporting apparatuses that allow them to collect pollen efficiently. All species of bees within the family Apidae carry pollen on their legs. Other species carry pollen inside their bodies or on the undersides of their abdomens. (Photo credit: Jennifer Tsuruda) Fossil records of bees preserved in amber indicate that bees have existed for at least 100 million years, but they are estimated to have first evolved around 130 million years ago, shortly after flowering plants first appeared on Earth. Plants evolved colors, odors, shapes, and food rewards that attract bees, while bees evolved mechanisms to collect pollen, which help plants reproduce through pollination. This period is known as the Cretaceous period, and was a time when dinosaurs roamed the earth. Humans did not evolve for another many million years. While bees have been around for roughly 130 million years, the oldest fossils of honey bees (the bees in the genus *Apis*) date back to around 60 million years ago. They occupied the so-called Old World: what is now Asia, Africa, and Europe. There is still debate over where precisely honey bees arose within the Old World. Recent genetic analyses suggest that all honey bees (genus *Apis*) actually originated in Asia, and *Apis mellifera* later diversified into Europe and Africa. They were introduced to the New World for the first time in the 1600s by European settlers. Now, they have been introduced to all continents except Antarctica. The native habitat for *Apis mellifera* included rainforest, savannah, desert and other relatively warm climates in Africa, Asia, and southern Europe. These broad geographical ranges and differing climates allowed for several subspecies (races) of honey bees to evolve adaptations specific to their region. Honey bees have adapted to a variety of climates, including the colder temperate regions of North America. Some subspecies are better able to withstand cooler climates and store food for their year-round needs; others are better able to avoid or combat pests and diseases that are more prevalent in warmer climates. (Photo credit: Ben Rulison). Eusociality: How Honey Bees Are Truly Social ■ You may look at your hive and wonder Key Points how this bustling honey bee society of Honey bees are eusocial, meaning tens of thousands of individuals that individuals depend on societal working together evolved from a organization to survive and cannot solitary wasp-like organism. Honey bee live on their own colonies are examples of the most advanced form of social living on the planet. This level of organization is called eusociality. "Eu-" means true or genuine. Individuals fully depend on a societal

organization to survive and cannot live on their own. There are three defining characteristics of a eusocial organism: Overlapping generations: More than one generation is living together. For honey bees, there are, for the most part, two generations present in one colony: a mother (the queen) and her offspring (workers and drones). Cooperative brood care: Individuals share the duty of raising young that is not theirs. Cooperative brood care is rare in the animal kingdom. Only a few species care for their young, and the work is usually done by the mother. In the case of honey bees, the queen lays the eggs, but the larvae, pupae, and young adults are cared for by their older sisters. Reproductive division of labor: In honey bees, reproduction is divided among the members of a colony. The queen is the sole egg layer under normal conditions, and workers do not usually reproduce. The workers are not fully sterile, for they can still lay unfertilized eggs that can develop into males, but they normally do so only if they lose their queen. The three members of a bee colony (queens, workers, and drones) are divided by their reproductive ability and sex. Honey bee colonies exhibit all three characteristics of eusociality.

WATCH

How Did Eusocial Societies With Distinct Castes Evolve?

Having multiple distinct forms of the same sex — like the queen and worker castes in honey bee females — is very rare in the animal kingdom. Evolutionary biologists have spent a lot of time studying other close relatives of honey bees to find out how such a caste system evolved in the first place. This video explores the prevailing theory. Video Transcript So how did these honey bee colonies get to have such distinct castes between queens and workers? The main theory is that the ancestor to honey bees, the solitary wasp-like organism, actually did all of the jobs that she needed to do in life by herself. She collected her own food, she built her own nest, she mated and laid her eggs and then provisioned all of those eggs with food, similar to how solitary bees live today. The resources that were available to her determined how much time and energy she was going to allocate to these different tasks. So it's not like she could be collecting food and laying eggs at the same time. There was only a short period of time where she could do each of these tasks. Importantly, once her offspring actually emerged from those eggs they left the nest. They didn't stay and hang around for any period of time to help the mother. So at some point in the honey bee's evolution, tens of millions of years ago, there was a mutation or a genetic change that occurred. And likely, it was probably more than one. And this genetic change allowed the offspring to remain at the nest with her mother after emerging instead of flying away and starting her own nest. And this was the first time that we start to see overlapping generations present in the nest, the mother generation and the daughter generation. And we also start to see cooperative brood care, as the daughters would cooperate in rearing the brood, or their younger sisters. This change and other genetic changes were favored by natural selection and persistent in the population, allowing these family groups to start to develop, with a mother bee and helper daughter bees evolving. So over time the mother was sort of allowed to start specializing in reproduction and laying eggs, while the daughters, this kind of frees up time for them to start specializing in other helping strategies. Over time natural selection sort of continued to favor this and then these rigid castes were finally in place. Their evolved, there's reproductive division of labor between the queen and reproduction and the workers in other tasks. In the case of honey bees, these castes have become so rigid that female bees must function either as a queen or as a worker. It's not like a queen bee can revert back to being a worker and vice versa. A worker can't psychologically change to become a

queen. Some other species of bees like bumble bees aren't quite as distinct and rigid. They still have queens and workers, but if a queen suddenly dies, a large worker can actually start to dominate her sisters and it is physiologically possible for her to mate and lay eggs. Whereas it is not physiologically possible for workers to mate.

WATCH

Kin Selection

The evolution of a non-reproducing worker caste is one of the most extreme examples of altruism in nature, and has fascinated biologists for centuries. The term 'altruism' refers to the fact that the majority of female bees in a colony suppress their own reproduction and instead invest energy into helping another individual (their mother, the queen) reproduce. On the surface, this appears to go against nature. Reproducing to pass its genes on to the next generation is, biologically speaking, the most important thing an organism can do – it is usually the measure of success for living things. Charles Darwin, who first proposed the theory of evolution, wondered how individuals who refrain from reproducing could evolve. If an organism does not have any offspring, how do its genes persist in the world? And why would some bees refrain from reproducing in order to raise another bee's young instead? In *The Origin of Species* (1859), he describes the non-reproducing caste of honey bees as "one special difficulty, which at first appeared to me insuperable, and actually fatal to my theory". It was not until the 1960s that a clear explanation arose: kin selection. Watch the video below to learn about kin selection: Video Transcript Workers inhibit their own reproduction in order to help the queen reproduce. And it's really costly for them to invest so much energy into rearing the queen's offspring. It's costly for them to go out and collect pollen from flowers, and risky because they might come in contact with predators. It's costly for them to produce royal jelly in their glands and to feed that to the offspring and to keep the brood warm. It's also costly for them to keep on checking on the brood and making sure that they're developing okay and that they're not getting any diseases. And most of all, one of the biggest costs is that they sacrifice their own lives to sting intruders to protect the colony. So all these things are really energetically expensive. So why are workers behaving so selfless? The answer is kin selection. Even in humans, the want to help is more likely with kin or your relatives than non-kin. Think of how protective you feel if someone wants to fight one of your siblings. Or a time when you might put yourself in danger or in a risky situation to try to save one of your relatives. Another good example of family helping is when there's a new baby in the family and aunts and nieces and other relatives will come together and help raise the baby. Kin selection is a strategy where individuals reduce their own reproductive success or survival in order to increase the reproductive success of someone else. And in the honey bee colony it's the worker who's reducing their reproduction to help their mom, the queen, reproduce. On the surface this appears really selfless, but at the genetic level you can start to see why it's advantageous and why this behavior likely evolved. Being related means you share genes with someone and the amount of genes that you share influences your helping behavior. And humans and bees are different. They differ in how they inherit their genes and how related they are with their kin. So let's start with humans. Individuals inherit genes from their parents, so you have genes in common with all your relatives: your parents, your grandparents, your siblings and your own children. And the closer you are related, the more genes you're actually sharing. So you share more genes with your mom than you would with your great, great grandmother, for example.

Humans are diploid organisms. This means that they have two sets of each chromosome in all of their cells. So their skin cells, their bone cells, their muscle cells, they all have two sets of chromosomes. One set is inherited from your mom, and one set is inherited from your dad. The exception is when you're looking at your gametes. Your egg cells or sperm cells only have one set of chromosomes. So because you're inheriting one set from your dad, and one set from your mom, this means that 50% of your genes are shared with your mom, and 50% of your genes are shared with your dad. And also as a result, 50% of your genes are shared with your sibling. Looking at a family tree, we can see the average degree of genetic relatedness. And this just goes to show again how many genes you're sharing. So, mothers and daughters share 50%, fathers and daughters share 50%. And same with your brother, they share 50% with their mom and 50% with their dad. Honey bees have a slightly different genetic system and it's called haplodiploidy. The females are similar to humans where they come from a fertilized egg so a sperm and an egg come together and they have two set of chromosomes. So they're also diploid. But, the males are different. Instead, males come from unfertilized eggs. They only inherit one set of chromosomes. That from their mother. So they are called haploid. They have half the number of chromosomes than the females in the colony do. So drones are haploids and all the females, workers and queens are diploid. A haplodiploid system actually makes bees more related than 50%. So this is where we see the difference from humans. In fact, if a queen only mates with one male, workers will share 75% of their genes. So how does that even work? Well, both drones and queens are passing on one copy of their DNA to create diploid daughters. And since the drones only have one set of chromosomes to begin with, all the copies that he's passing on to each daughter are going to be identical. So, sisters are sharing 100% of genes that they receive from their father. But with the queen she's diploid. She has two sets of each copy of each gene in her DNA. And she gives one copy to each of her daughters. So one sister might inherit the same set of chromosomes or some of the same genes as the other sister. But sometimes they receive different copies. Across the thousands of genes that they're inheriting the amount of times they end up sharing the same copies averages out to about 50%. It's similar to tossing a coin. There's two options. You can have heads or you can have tails. If you and your sister are each flipping the coin over time, over thousands of tries, the average amount of time that you are going to get the same copies is about 50%. So sisters share about 50% of their genes that they receive from their mother. So a daughter of a singly mated queen who has the same dad will share on average 75% of their genes; 100% from the dad side and 50% from the mom side averages out to 75%. Okay, so what if a worker were to lay her own eggs? If a worker were to lay her own egg she would pass on 50% of her DNA. Workers can't mate or store any sperm so they're producing unfertilized haploid drones. They're always drones. So if we're thinking about it, what is the better avenue to get the most genes to persist in the population? Is it to have your own offspring, where you're passing on 50% of your genes? Or is it to help rear your sisters, who are going to be sharing 75% of your genes? It's to help raise your sisters. And if we are looking at a honey bee colony, we see that the queen is able to lay many many eggs a day, up to a thousand eggs a day, so it is a really good way to pass on your genetic information even though you yourself aren't reproducing or the workers themselves are not reproducing.

READ

Polyandry

■ When honey bees first evolved, queens Key Points were monogamous, meaning they Queen honey bees usually mated with just one male. This meant with between 12 and 50 males, that all of her daughter bees which helps to increase genetic were highly related to one another and diversity among her offspring, they had 75% of their genes in common. colony's workers. This high level of relatedness helped Genetic diversity improves colony drive the evolution of eusociality. health and productivity. Once eusociality was established, Relatedness among sisters however, other colony traits also decreases with polyandry, but it is evolved. One such trait is polyandry. still more advantageous for Polyandry is the practice of females workers to rear the queen's mating with multiple (poly) males offspring (their brothers and (andry). Queen honey bees are sisters) rather than their fellow promiscuous, typically mating with sisters' offspring (their nephews). between 12 and 50 males. By mating with multiple males, a queen greatly increases the genetic diversity among her daughters, the workers in the colony, thus improving the colony's health and productivity. Queens mate with multiple drones. The Impact of Polyandry on Relatedness When queens mate with multiple males, the relatedness between sisters decreases. While they share 75% of their genes when they have the same father, daughter bees only share 25% of their genes when they have unrelated fathers. Under these circumstances, one might expect workers would produce their own offspring instead of helping rear their sisters. Yet workers still refrain from reproducing, despite the fact that they would now be more related to their offspring than their half-sisters. How come? One possible reason has to do with the high reproductive output of the queen. The queen can lay 1,000 to 1,500 eggs per day, while a laying worker lays only a few eggs per day. Even though workers are less related to the queen's offspring than their own, they can still effectively pass on more of their genes by helping the queen. Another possible reason has to do with the relatedness between a worker and her sisters' offspring. Although it would be advantageous for a worker to rear her own offspring, it would not be advantageous for her to rear her sisters' offspring. If several workers began laying eggs in the colony, one worker would invest energy in rearing many of her nephews as well. In this situation, it is more advantageous for sisters to invest energy and resources into rearing their mother's offspring (with whom they share 25% of their genes) than their sisters' offspring — their nephews — with whom they only share 12.5% of their genes. A specific social behavior has evolved to discourage workers from reproducing. This behavior is called worker policing, and it refers to workers consuming the eggs that are laid by their sisters. In the rare instances where workers do lay eggs, their sisters will eat them so that they never develop. We've talked about the genetic relatedness among several family members in a honey bee colony, and it is possible to calculate these values for all their relatives. It is, however, beyond the scope of this course to have you calculate genetic relatedness among family members. The take-home message is this: Understanding genetic relatedness among family members in a bee colony can help us learn how and why honey bees evolved such a social living strategy.

WATCH

The Importance of Wild Bees

■ There is incredible diversity in bees and Key Points striking differences in the way they There are 20,000 species of wild look, behave, build their nests, and live bees in the world alone or together. Although honey bees are the species most commonly used for pollination in agriculture, many species of bees also have important roles in pollinating food crops and wildflowers. Across the world, there are

over 20,000 documented species of bees! Watch this video to hear Bryan Danforth discuss the many other bees (besides honey bees) that are found in the world. Video Transcript So honey bees are not the only bees in the world. In fact, there are over 20,000 described species of bees in the world. In North America, we estimate about 4,000 described bees. California hosts over 1,000 species of bees and French-speaking Europe, hosts nearly a 1,000 species of bees as well. In New York state, we have recently tallied up the numbers and come up with a total 420 species. So, while these are extraordinarily diverse, and if you compare the species richness of wild bees to other familiar large vertebrate groups, bees are comparable to fishes in the vertebrates with about 20,000 species. There are about five times as many bees as there are mammals and there are about three times as many bees as there are birds. So bees are an enormously diverse group. They're a group that arose over 120 million years ago. That's well before the extinction of the dinosaurs and it's a period when the earliest flowering plants were just evolving on earth. I'll use two terms throughout this presentation. One is native bee to describe bees that are native to a region or of the world. And wild bee to describe all the non-managed bees that we'll be covering in today's presentation. Bees occur on all continents except Antarctica and they're particularly diverse in desert habitats such as the circa Mediterranean region, western Australia, Southern Africa, and western North America. By comparison, honey bees encompass only eight species in the genus *Apis*. And one of those, the managed honey bee, is *Apis mellifera*. Now it turns out wild bees are actually very important agricultural pollinators and we've learned this fairly recently, from studies in crops like apples, peaches, cherries, blueberries, cranberries, strawberries, squashes and pumpkins. Wild bees are a large component of the fauna in these agricultural habitats. And in New York, for example, we have 120 species of wild bees visiting apple trees, just alone. Apple growers, for example, in New York and other agricultural producers are well aware of the role of these wild bees. And they do a very good job of trying to maintain a diverse wild bee fauna in and around their cultural habitats. Native bees and the wild bees we're talking about today have coevolved with many different host plants. And one of the interesting things about native wild bees is that they are often narrow host plant specialists, unlike the honey bee, which is a very broad generalist visiting many different host plant families. So what do we know about these other wild bee species? First of all, they are extraordinarily diverse in anatomy and behavior. There are a tiny species the size of a fruit fly in arid regions of western North America and there are enormous carpenter bees the size of a bird that one can see flying in, more or less, tropical habitats. Bees come in a range of colors and in different amounts of hairiness. But they're quite diverse anatomically and behaviorally. The vast majority of bees are solitary. So our estimate is that about 77% of the bee species on earth are solitary. A single female constructs her own nest. She guards that nest against parasites and predators. She gathers pollen and nectar for her offspring, and she lays eggs. She's usually active for a very short period of time. Somewhere between two to three weeks. In contrast to a honey bee colony which is active for months throughout the summer. And again, many of these solitary wild bees are narrow host plants specialists, visiting just a few host plant genera or even species. Now these wild bees build nests in a variety of substrates, including soil, hollow stems, wood, snail shells, some of them build above ground nests, and some of them build nests in solid wood, those are the carpenter bees. So let's start with the soil nesters which are the largest proportion of wild bees on earth. These are bees that burrow into the soil. They can build nests from just a few inches deep to up to nine feet deep. There are records of these bees building nests that go down nine feet into the soil. At

the bottom of their nest, they will construct a series of underground brood chambers which they provision with pollen and they'll waterproof these brood chambers with glandular secretions or floral oils. And then they'll provision that brood chamber by making many trips, usually five to ten foraging trips. They'll lay an egg on that collected pollen and then they'll close that brood chamber and start the whole cycle again. A typical solitary bee produces about one offspring per day, but some very rapid foragers can do five to even six offspring per day. And an average wild bee doesn't live for very long. She might produce 10, 12, 15 offspring over the course of her entire life. Another group of solitary bees are the cavity and stem nesters. These include the mason bees, the leaf cutter bees, and the wool carder bees. And we call these renters because they occupy cavities that were created by another organism or just cavities available in the environment. And what they do is they construct their brood cells within these cavities by modifying the cavity slightly with mud, leaves, plant fibers, even flower petals and once they've constructed their brood chamber with those materials they then provision the brood chamber with pollen and nectar just like the ground nesting bee would do. They lay an egg and then they close the brood chamber and carry on with another brood cell the next day. Another group of wild bees are the wooden pith-nesters. And these include the large carpenter bees as well as the small carpenter bees. These are bees that construct a burrow in solid wood or in pithy stems. And carpenter bees have an enormous head with stout mandibles for burrowing into solid wood and constructing a series of burrows. and then from within those burrows they construct their brood cells. They provision their brood cells again with pollen and nectar, they lay an egg and they close the brood cell. And they will construct multiple brood cells throughout their lives. And finally, the last group of wild bees are the resin bees and these are bees that actually construct above-ground nests from resins combined with pieces of stone or chewed up leaves. Again, they use a variety of floral products, but they actually construct their nest above-ground, and they attach it to a plant or a stone, almost like a little flower pot. And then they forage out from that nest collecting pollen and nectar, and they bring those floral resources back to their nest, they provision their brood cells, they lay an egg, and they close that brood cell off. So those are the resin bees in the family megachilidae. And again these are all solitary, like most of the bees I've told you about so far. Now one of the really fascinating groups of bees are the kleptoparasites. These are bees that very few people know about. They comprise 13% of wild bee species, and almost 25% of the family Apidae which is the group that includes the honey bee. These are bees that don't construct a nest. They don't gather pollen and nectar. They sneak into the nest of other bees. And they lay an egg discretely in the brood cell of that host and then they leave the nest as quickly as possible so as not to alert the host to their presence. Their egg hatches and the first instar larva usually seeks out the host egg or early instar larva and they kill it with greatly enlarged mandibles. And, then the klepto parasite larva consumes the pollen provisions gathered by the host female. So these are really similar to the cuckoo birds that some of you might be familiar with that are known for birds where the cuckoo lays eggs in another birds nest and then flies away and hopes that that bird will take care of its offspring. So these are called cuckoo bees or kleptoparasites or brood parasites. And you can often see them if you look around bee nesting sites, if you look near the ground, you'll see what look like wasps almost flying near the ground and looking for nests of their hosts and then entering those nests to lay their eggs. So, we don't know whether these are good pollinators but they're certainly fascinating bees that have adopted a very unusual lifestyle. And then the last category of bees are the Social Bees, which of course would include the honey bee. In this group only includes 9% of the

bees in the world. Social bees other than honey bee would include bumble bees, sweat bees and stingless bees in the tropical parts of the world. Stingless bees are comparable to honey bees in terms of social complexity. They build enormous colonies with thousands of workers. They are aggressive defenders of their nest. They communicate about the location of resources. They've actually been managed in various parts of the world for honey production, just like we manage honey bees today in North America. It's a little more difficult to manage a stingless bee colony for honey production, so they're not as widely managed as the honey bee. So while honey bees are fascinating and extremely effective pollinators in all kinds of crop systems, we shouldn't forget that wild bees are important pollinators, but also fascinating from a biological perspective. And what we're discovering with our work on wild bees of apples is that conserving wild bees is an important approach for agricultural producers who want to maximize the pollination of their crops. It's also important for environmental conservation and for carrying on sustainable agricultural practices.

READ

Conserving Wild Bees

■ There is a lot of interest today in bee Key Points conservation. Honey bees are not. Some wild bee species are alone in facing impacts from parasites, experiencing population declines, diseases, nutrition, and pesticides; and several have been added to some wild bee populations are also the Endangered Species List facing challenges from these factors. Their conservation is equally important for maintaining natural ecosystems and sustainable agriculture. Wild bees contribute to overall ecosystem health and the biodiversity of native flowers, and as beekeepers, we are regarded as stewards of all pollinators and advocates for healthy environments. Although honey bees comprise only 9% of bee species (and aren't native to North America), they have become the face of pollinator declines in the media. Master Beekeepers can use this opportunity to help educate the wider public about all kinds of bees and pollinators. The beekeeper's voice is an important one. In the Northeast U.S., there are 53 species of wild bees whose populations are currently in decline. In 2016, several species of bees were added to the Endangered Species list for the first time in history. These bees were seven species of yellow-faced bees that nest in small cavities and stems in Hawaii. In early 2017, the rusty patched bumble bee, native to eastern North America, was also added to the list. With most research focused on the factors affecting the health of honey bees, we are only recently identifying how wild bees are impacted by some of the same factors. Wild bees, for instance, are more susceptible to habitat loss than are honey bees. While wildflowers generally only provide food for honey bees in the U.S., they also provide nesting sites for some native bees. Habitat loss in the form of agricultural intensification and urbanization has been identified as one of the main driving factors for reducing some wild bee populations. The impact of various pesticides on wild bees is only beginning to be investigated. We are learning that pesticides affect different bee species differently — some are much more sensitive to certain chemicals than honey bees. We are also starting to better understand the role of certain parasites and pathogens in wild bee health. At least two are shared between honey bees and bumble bees; *Nosema ceranae* and Deformed Wing Virus can infect both genera ('genera' is the plural of genus). Research on how pesticides and pathogens are impacting native bee populations is currently underway at Cornell and other institutions.

MODULE WRAP-UP

The Evolution of the Honey Bee Colony as a

Superorganism In this module, you explored the remarkable evolution of honey bees from solitary wasps to one of nature's most sophisticated social organisms. Understanding how individual bees function as part of a larger, interconnected colony helps beekeepers appreciate the complex adaptations that make honey bees such successful and fascinating creatures.

Honey Bee Reproduction

Reproduction in honey bee colonies occurs at two levels: The entire colony can reproduce through swarming. Individuals within a colony can reproduce by mating and having offspring. This module focuses on individual reproduction and covers everything you need to know about the biology and behaviors involved.

READ

Passing on Your Genes

■ Through reproduction, bees pass on their genetic material to their offspring. This course uses a variety of terms when talking about genetic material. An organism's DNA is coiled up into material: genome, chromosome, gene, chromosomes and DNA. These words are all related, but mean slightly different things. Genes are sections of DNA that code for particular proteins. Genes get familiar with these terms so that are passed down to offspring we can better understand honey bee through reproduction and genetic characteristics. Beekeepers can selectively breed for the characteristics they desire. Basic Genetics A genome is an organism's complete set of genetic material. Many honey bee behaviors are set of genetic material. governed by multiple genes. Deoxyribonucleic acid (DNA) is the name of the molecule that carries this genetic material. The honey bee genome is different from the human genome and the genomes of all other organisms because each species has differences in their DNA that make them unique. The genome includes information for a variety of elements, the most important being genes. The honey bee genome was sequenced in 2006 and is one-tenth of the size of the human genome. An organism's DNA strands are incredibly long. For instance, if a human's DNA was stretched out it would be three meters in length! In order to fit all the DNA into each cell of an organism's body, it is tightly coiled up into molecules called chromosomes. Honey bee DNA is distributed across 16 different chromosomes: Queens and workers. Diploid females carry two sets of chromosomes, so each cell (except their egg cells) contains 32 chromosomes total. Drones. Haploid males carry only one set of chromosomes, so each cell (including their sperm cells) contains 16 chromosomes total. An organism's DNA serves as the blueprint for its body. Different regions of the DNA strands are the written instructions for different genes. The honey bee genome contains around 10,000 different genes, so each chromosome contains the instructions for many hundreds of different genes. Small particles within cells read the gene instructions and build their corresponding proteins. For example, a gene with instructions to build an eye cell would get translated into an eye cell protein. Like most other animals, honey bees reproduce by creating egg cells and sperm cells. These cells contain the parents' chromosomes, so all offspring that are produced inherit their parents' genes. How would a beekeeper breed for the trait they desire?

Many of the traits we observe in honey bees are due at least in part to genes, so beekeepers can selectively breed for characteristics they desire. For example, there are many genes involved in nectar collecting, in temperament, and in hygienic behavior. Bee breeders can try to improve their operation's honey production, gentleness, or colony health by breeding queens from colonies that have these traits. Honey bee behaviors are complex because they tend to be governed by multiple genes and the environment also influences their behaviors, but nonetheless beekeepers have been breeding for traits like these for decades - long before we knew about DNA. All honey bees are recognizably honey bees, but no two are identical. While they all contain 10,000 or so genes in their chromosomes, the versions of those genes can be different. To illustrate this point, let's think of an example in humans: eye color. Everyone has a gene that codes for eye color located on the same position on one of their chromosomes, but people can have different colors of eyes: some have brown, while others have blue or green. This is due to different versions of the eye color gene. Individuals inherit their specific gene versions from their parents. Scientists refer to different gene versions as "alleles". Now let's look at an example with honey bees: temperament. Let's imagine that we have a bee yard with many hives in it, and the colonies vary greatly in their temperament. Some colonies are very gentle, while others are much more prone to sting. In the honey bee genome, there is one gene that codes for temperament (in reality temperament is governed by many different genes, but we will keep this example simple). Some colonies carry a version of this gene that makes them more aggressive, while other colonies carry a version of this gene that makes them gentle. Over the years, we can improve the temperament of our entire bee yard by selecting which gene versions are going to be passed down to subsequent generations. We can allow the queens from gentle colonies to reproduce, and can prevent the queens from aggressive colonies from reproducing (by replacing them with queens raised from our gentle colonies). Over a few bee generations, we can have a yard of entirely gentle colonies, as that gene version becomes predominant over the other version in the population. By understanding honey bee reproduction in this way, beekeepers can start to breed for characteristics they desire - whether their goal is to increase honey production or to help improve honey bee health in an effort to be a better steward to the environment.

READ

The Molecules of Bee Life

DNA and chromosomes are important molecules of bee life, but there are others to consider as well. We don't often think of tiny microscopic molecules when we think of honey bee reproduction, development, and growth, but in order to understand how these processes occur we need to take a step deeper than what we see with the naked eye. It's interesting to know that honey bees reproduce and develop in unique ways as animals, but it is even more valuable to understand how they do it. Chemical compounds in the form of molecules are key players in development and reproduction of all living organisms, for they enable communication among the cells inside a body and among the individuals within a honey bee colony. The molecules important to honey bee reproduction, development, and sociality that we will cover in this course are proteins, neurotransmitters, hormones, and pheromones. Proteins form the building blocks of life (human, honey bee, and every other living thing) because they are involved in so many functions. For example, many organs and tissues (e.g. hearts and muscles) are made primarily of different types of proteins. Even the immune system

relies on proteins to function. Proteins also play a big role in development and reproduction. The instructions for building proteins are written in DNA (deoxyribonucleic acid), found in most cells within an organism. Neurotransmitters are chemicals released by neurons in the brain to send very fast signals to other neurons. They are the important messenger molecules of the nervous system and are involved in many functions ranging from muscle movements to learning and memory in the bee. Many neurotransmitters in honey bees are also present in humans, such as dopamine, octopamine, and serotonin. ■ Neurotransmitters ■ Hormones Hormones are chemicals produced in special glands that are secreted into the blood (called the hemolymph, in bees) to be carried to specific organs and tissues to influence how these structures function. Hormones travel within the body and do not exit it. In the honey bee, as in humans, hormones are involved in reproduction, development, and the general functioning of the body (e.g., stress hormones). Pheromones, like hormones, are also chemicals created by specialized glands, but they are secreted outside of the body to elicit a behavioral or physiological response by another individual of the same species. Insects, vertebrates (including humans), plants, and even yeast commonly use pheromones as a means of communication. Some pheromones function as airborne molecules (odors) that are detected through the sense of smell, while others function as surface-borne molecules that are secreted onto the outside of the body where they can be detected through taste or close-range smell. Some honey bee pheromones are even transmitted throughout the colony by worker bees engaging in trophallaxis (transferring or exchanging fluids or food mouth-to-mouth). ■ Pheromones ■

WATCH

Mating

In order to pass their genes on to their offspring, honey bees mate. Queens and drones are the only colony members that mate. Mating activity in honey bee populations typically follows a seasonal cycle that coincides with swarm season. When colonies swarm, they leave behind new virgin queens. Drones are produced amply in spring and summer to mate with these queens. In northeast North America, this mating activity peaks in the spring and early summer but continues to occur to a lesser degree throughout the summer as queens are superseded and late swarms occur. Watch this video for details of where and how mating happens. Video Transcript So when a virgin queen is sexually mature, she's ready to take a mating flight and she'll take between one and three mating flights where she leaves the hive and flies to the specific geographical location called the drone congregation area. And this is an area high in the sky. It's between 50 to 250 feet up and it's just essentially a cloud with hundreds of drones that are waiting for a queen to arrive. So queens will fly up to nine miles to get to these drone congregation areas. Even if there's one near her own hive, she wants to fly far. And this is so that she reduces the chances of mating with her brothers and other relatives. And it's actually quite effective. Only a very small percentage of queens are mating with drones from the same apiary. So drones actually get a little bit of a bad rep. People often say that drones are lazy and they don't do too much, and the word drone is literally synonymous with the word lazy. But when it comes to reproduction, drones are anything but lazy. What they do is they leave their colony many times a day to fly to this drone congregation area, where they hang around for about half an hour waiting for a queen to show up. After about a half an hour, they're very hungry, so they have to return to the hive refuel, and they set out again. So they do this several times, just hopelessly waiting for a queen and seeing if

they're going to get lucky. So a really cool thing about drone congregation areas is that queens and drones from colonies in a given region actually return to the same drone congregation area year after year. In fact, right here next to Dyce, we have a drone congregation area that people have observed for decades. But we really have no idea what attracts drones and queens to these congregation areas to begin with, or how they become established, or how they know to continue going to these spots year after year. We do know that there are some traits that make drone congregation areas more attractive. So for instance, open areas are more attractive than areas that are near dense forests or urban areas and congregation areas that are above slopes that are facing southward are more attractive than slopes facing any other direction. But beyond that, really a lot more research needs to be done and drone congregation areas still have a lot of mystery to us. Okay, so let's talk about the act of mating. We have hundreds of drones in these drone congregation areas and suddenly a queen arrives, this is the moment they've been waiting for. And the drones can detect the queen with two ways. First is sight, they have very large eyes. They have to be able to distinguish one queen in a cloud of hundreds of drones. And the second is smell. So queens are continuously producing a combination of pheromones that are collectively called queen substance or queen pheromone. And it acts as a sex pheromone to males. It's very attractive, and it makes them want to mate with her. So once they see her or smell her, the race is on. And they are trying to get to the queen. And out of hundreds of drones in a drone congregation area she only mates with between 10 and 20 other males. So the competition is fierce. The actual endeavor of mating is very quick, it lasts less than five seconds. Normally the drone's penis is tucked underneath its body, but it gets everted when it's ready to mate, and inserted in the queen's vagina. Less than five seconds later, he ejaculates, and this actually ruptures his penis. His body falls away and he dies, and the tip of the penis remains in the queen. The next drone to meet with the queen will remove the tip with this specialized structure on his own penis, and then he will mate with her. So consecutively she'll mate with 10 to 20 drones and then when she's finally finished mating with all the drones she's going to, she returns to the hive and the last tip remains. And some beekeepers actually witness this and they can see this, and it's called the mating sign. And it says to them that they have a mated queen. So it's actually a good thing if you happen to see this. The workers in the colony will remove the last tip. So after the queen returns to the hive from her one to three mating flights she'll never mate again. All the sperm that's stored inside of her is the sperm she's going to use to fertilize all of her eggs. And it's going to be stored inside of her for several years. You can tell that she's mated because her abdomen is larger, it's more full of sperm. She's actually moving slower in the colony because she's heavier, and she won't fly again unless she's going to swarm. After she's returned from a mating flight, two to four days later she'll begin laying eggs, and this is what she continues to do for the rest of her life. Most mating flights take place in the afternoon, usually between 2:00 and 5:00 p.m. While queens can fly quite far to drone congregation areas (DCAs), on average they will typically fly 1.2 to 3 miles (2 to 5 km). Drones, on the other hand, will visit much closer DCAs that average 0.3 to 1.2 miles (0.5 to 2 km). If drones do not get to mate with a virgin queen on the first flight they take, they will continue to take up to six 30-minute flights each day as the weather permits. In between flights, they return to the hive to refuel on carbohydrates.

READ

Reproductive Systems

■ The physiology of reproduction varies Key Points by sex and by caste. Review the Queens have a very high information on each tab below to reproductive output and signal understand the reproductive their fertility to the colony through physiology of the drone, the queen, queen pheromone and the worker bee, as well as the role Worker honey bees have similar each plays in the process of reproductive systems to queens, reproduction. but they have fewer ovarioles and are not able to mate Drone The Reproductive System of a Drone Drones become sexually mature and ready to mate after nine to 12 days of adult life. The drone's penis (called an endophallus) only everts when he latches onto a queen to mate; it is normally tucked inside the drone's abdomen and is not visible. Because the bees are mating while flying, the drone's endophallus also contains a pair of claspers for gripping onto the queen. In a successful mating, 6 to 12 million sperm travel through the endophallus and into the queen's vagina. Many fewer sperm than this make it to a queen's sperm storage sac (the spermatheca); the rest exit her body. Queen The Reproductive System of the Queen The queen bee is an egg-laying machine for nearly her entire lifetime: In the right conditions, she lays about one egg every minute, which translates to 1,000 to 1,500 eggs a day and about a half-million eggs in her two- to three-year lifespan, although this can vary enormously. Some sources have stated that some queens can lay over 2,000 eggs a day! When the quality or quantity of her egg laying declines and signs of her fertility weaken, the workers supersede her with a new queen. Queen eggs laid in cells. The ovaries in queen bees are two large, tube-like organs that are comprised of 150 to 180 ovarioles each. All of the queen's eggs are produced in these ovarioles. When an egg is released, it leaves an ovariole, slides down an oviduct, passes by the spermatheca, moves through the vagina, and exits the queen's body from her vagina. The spermatheca is a large sac that stores sperm from all the drones a queen has mated with; about 5 to 6 million sperm total. To fertilize an egg, a small duct at the entrance of the spermatheca opens as the egg is passing by to allow sperm to exit and fertilize the egg. In this way, queens are able to control whether they lay fertilized eggs that will develop into workers or unfertilized eggs that will develop into drones. There is generally enough sperm in a spermatheca to last for a few years of egg laying; only a few sperm are released to fertilize each egg. Note that a queen who has just mated may, for a time, lay multiple eggs into a cell, but within a couple of weeks, she should settle into a productive and sustainable pattern of egg laying. The queen (marked) signals her fertility by producing a blend of pheromones, collectively called "queen pheromone." Workers pass along this information about her fertility to other bees in the colony by transmitting the pheromone to them. Workers near the queen brush this pheromone blend off the queen's body with their antennae and lick it off with their tongues, then walk through the hive spreading it to one another by making antennal contacts and via trophallaxis. In this way, they can constantly assess and communicate the presence and fertility of the queen. As the queen ages and her fertility wanes, she produces less queen pheromone. Once worker bees can no longer detect queen pheromone at a minimum threshold, they start to construct supersedure cells to replace her. If the queen dies or is removed from the hive by a beekeeper, the workers come to realize she is gone because of the lack of queen pheromone being passed among the bees. In response, they begin constructing emergency queen cells to rear a new queen. (Photo credit: Bee Informed Partnership) Worker The Reproductive System of the Worker While queens and workers have the same basic biological equipment for reproduction, they differ vastly in their ability to lay eggs. Under normal circumstances, when the hive is queenright, worker bees are effectively sterile. They have two ovaries just as queens do, but these ovaries are small, with only two

to 12 ovarioles each, in striking contrast to the 150 to 180 ovarioles in each ovary of a queen. Similarly, a worker's spermatheca is tiny and cannot store sperm. Because of this, workers cannot mate and do not tend to lay eggs. In circumstances when they do lay eggs, their eggs are unfertilized and develop into haploid drones. There can be a few laying workers even in a queenright hive, but rarely do they make up more than 1% of the worker population, and the eggs that they lay are usually eaten by other workers. If the queen dies, workers can detect the absence of queen mandibular pheromone in as little as 20 minutes. If the colony is not successful in rearing a new queen from emergency queen cells, some of the workers' ovaries may activate, and they can start laying eggs, becoming what are called "laying workers." The young workers (nurse bees) in particular are affected by shifts in this pheromone, as they are much more likely than the foragers to be in the hive and in contact with the queen. It is these young house bees that can become reproductive. In the absence of queen pheromone, up to 30% of young workers in the colony may start to produce and lay eggs! The reproductive output of laying workers is very different from queens. Laying workers only lay a few eggs here and there — not nearly the 1,000 or so per day that a queen is able to lay. Because workers have short abdomens that cannot reach the bottom of a cell, their eggs tend to be on the sides of cells rather than at the bottom. Also, there is often more than one egg per cell. Beekeepers often look for the eggs to confirm the presence of the queen — but Master Beekeepers must be able to inspect the egg pattern and identify whether the eggs were laid by workers or by a queen. Workers will often lay eggs in worker cells, but because they cannot mate, all the eggs they lay will develop into drones. The drones' larger bodies will cause the cell cappings to extend outward and become bullet-shaped. These bullet-shaped cells, in combination with multiple eggs on the side of cells, is a good indicator that there are laying workers present in the colony. Worker eggs in cells look like several small white eggs crowded together. Worker eggs laid in worker cells develop into drones. Because drones are larger, they form "bullet"-shaped cells when they are capped. Finally, an important part of reproductive anatomy that is different in honey bees compared to most other insects is the ovipositor, or the egg-laying organ. In most other insects, the ovipositor is used to lay eggs; in honey bees, however, it has adapted to become a stinger. This is why only female bees can sting: Their stinger is a modified egg-laying organ! Instead of laying eggs from the ovipositor, honey bees lay eggs from the vagina.

MODULE WRAP-UP

Honey Bee Reproduction

In this module, you examined the dual nature of honey bee reproduction, exploring in detail how individual bees mate and produce offspring within the colony. Understanding these fundamental biological processes and behaviors is essential for successful colony management and provides crucial insight into the complex reproductive dynamics of honey bee societies.

Honey Bee Development

All bees begin life as an egg. From there, some become workers, others queens, and others drones. As adults, bees continue to change in response to the social and physical environment of the colony and the natural world. In this module, you will learn all about how bees develop and how their physiology and behavior change throughout their lives.

WATCH

The Stages of Development

Honey bees go through an ordered pathway of development as they transition from egg, to larva, to pupa, to adult. Although there is slight variation in development time from bee to bee, one is able to generally track how many days it takes queens, workers, and drones to mature through the various phases. Being able to identify the different stages of development, knowing how long each stage lasts, and predicting the emergence of the different castes will come in handy for you as a beekeeper. Such knowledge is directly applicable, for instance, to queen rearing and to management practices that reduce Varroa mite loads. Watch this video for more about how honey bees develop from egg to adult.

Video Transcript So by now, as a beekeeper, you're probably familiar with the basic development of honey bees. Honey bees begin life as an egg that hatches into a larva that becomes a pupa, and then undergoes metamorphosis to form an adult. But it's really important, as a master beekeeper, to know how the developmental timing differs among the castes. So for all three castes, they all take three days as an egg and six days as larva. The amount of time as a pupa is what is different. Drones take 15 days to pupate, workers takes 12 and queens only take seven days despite the quality and quantity of resources that goes into creating them. So why would this be? There's actually a few reasons why natural selection would favor a quick development time in queens. So think about a time when a colony is producing lots of queen cells, like when they're swarming. The first queen to emerge is going to be the queen for that colony, and as soon as she emerges she's going to kill all the other rival queens. So it's actually advantageous to develop as quickly as possible. Also developing quick as a queen gives you a little bit of a headstart to start growing your colony, collecting food and gets you better prepared to survive winter. And even though this might be a difference of a day or two, this can make an impact if it's perfectly timed with a nectar flow. And lastly, if a colony finds itself suddenly queenless, if the queen dies or something else happens to her, then they need to produce a new queen quickly. And if you have a quick development time for queens, this ensures that the colony isn't queenless for very long. Okay, so what is exactly happening during development? Three days after the egg is laid it hatches and a small larva crawls out. And a one-day-old larva is actually really tiny. It's about the same size as an egg, but you can detect it with your eye. The main function of a larva is to eat. In fact, its stomach is almost the entire length of its body. And it's fed many meals a day by nurse bees. Workers are first fed royal jelly and then they're fed bee bread, which is pollen mixed with honey and some saliva and fermented by microorganisms. After it's eating, it's growing, and it needs to shed its skin to accommodate its bigger body. So after six days of eating, growing, molting, eating, growing and molting, it's finally ready to pupate. And during this time, workers are going to cap the cell with wax, the larva spins a cocoon around its body, and it doesn't eat anymore until it becomes an adult. While it's a pupa, it starts to develop features that make it look more like a honey bee and less like a grub. Its three body segments — the head, thorax, and abdomen, are becoming distinct. And you can start to see its mouth parts, its eyes, its legs and antennae. And the eyes will start to take on color. Instead of being white they start to turn purple, which is referred to as the purple-eyed stage of development, and then they'll eventually become black. And after seven to 15 days, depending on if it's going to be a queen, a worker or a drone, it has its final molt and it chews its way out of its cell. It actually leaves its cocoon behind in the cell, and it's ready to begin life as an adult bee. And you can actually identify these

one-day-old bees that have freshly emerged in the colony, because they have really fuzzy heads and thoraces, and they're actually not able to sting at this stage. So they're pretty cute. To witness and understand the full process of bee development (including under the pupal cappings!), watch this video by Anand Varma. It shows the 21 days of a worker's development from egg to emergence, condensed into one minute.

TOOL

The Stages of Development Timeline

Take a look at the developmental  Download the Tool timeline tool to get a better sense of Use this chart as a developmental the stages of development for all timeline reference for honey bees. honey bees. Understanding developmental timing can give you information about the queen's fertility, which is more valuable than actually seeing her in the hive. The presence of queen-laid eggs is a sign that the colony's queen has been there in the past one to three days, so she is most likely in the hive. If there are no eggs or young larvae during a time when you would expect the queen to be laying, but you see older larvae, you can infer that it has been six to nine days since she has laid any eggs. You should then do a thorough check to see whether she is in the colony and anything else you can learn. Similarly, understanding the timing for queen emergence is critical if you are selling queen cells or introducing them into queenless colonies. You want to make sure that emergence is timed for a day or two after they have been introduced to a colony, rather than before! You'll notice in this course that we simplify the amount of time it takes the different colony members to pupate. Scientific observations find queens take an average of five to six days, workers take six days, and drones take six to seven days. In this tool, we report all colony members take approximately six days to pupate, but just know there can be some variation.

READ

How Sex Is Determined

■ What determines whether a bee Key Points develops into a male or a female? The ratio of females (workers) to Honey bees, as well as the other males (drones) in the colony is insects in the order Hymenoptera, dynamic, depending on colony have a different sex determination size, comb architecture, and queen system than humans. fertility. In humans, both males and females Inbreeding can result in high inherit two sets of each of their numbers of diploid males because chromosomes in their cells, one set of the sex determination system in from each parent. One particular pair honey bees. of chromosomes is the sex chromosomes, and these are what determine a person's sex. A person inheriting an X chromosome from its mother and a Y chromosome from its father develops into a boy, while a person inheriting an X chromosome from both its mother and its father develops into a girl. In honey bees, chromosomes also play a role, but there are interesting differences. Before we describe the role chromosomes play, let's talk about the ratio of male to female brood produced in a colony. The queen has control over which eggs she fertilizes by opening her spermathecal duct to release sperm or keeping it closed so no sperm is released. So what makes her fertilize an egg or not? It is actually the worker bees that build comb that affect the ratio of worker offspring to drone offspring in the colony. They build large cells (approximately 6.4 mm wide) to accommodate drones and small cells (approximately 5.2 mm wide) to accommodate workers. Drone cells (bottom) are slightly larger in

diameter than worker cells (top) The ratio of workers to drones depends largely on the population of the colony. Once the colony has at least 4,000 or so workers, it can begin investing in drone production, and comb builders will begin constructing larger cells for drones. The queen walks along the combs and assesses which type of egg she should lay in each cell based on its size. She sticks her head and thorax into empty cells and measures their size using her front legs. Small cells stimulate her to lay a fertilized egg, whereas large cells stimulate her to lay an unfertilized egg. Honey bee sex determination is usually explained simply in this way: Fertilized eggs yield offspring that inherit two sets of chromosomes (so are diploid organisms) and develop into females, while unfertilized eggs yield offspring that inherit only one set of chromosomes (so are haploid organisms) and develop into males. However, it actually goes one step beyond this. Let's explore the Complementary Sex Determination gene next.

READ

The Complementary Sex Determination Gene

It is not the number of sets of chromosomes a bee inherits that determines their sex; rather, it has to do with the gene version (also called an allele) they possess at a specific gene position on a certain chromosome. The special gene is the Complementary Sex Determination (CSD) gene, and this is what ultimately determines sex in honey bees. If a bee receives just one copy of this gene (from the mother's egg), then a male is created. Yet if a bee receives two copies of this gene (one copy from their mother's egg and one copy from their father's sperm), then it generally means a female is created. Here is the key, however: A female's two copies must be different versions. If a bee receives two different versions of the gene, then she develops normally as a female, but if the two versions happen to be identical, the bee will develop as a male — even though it has two sets of chromosomes. We call these "diploid males," but you rarely see diploid male adults walking around the colony because they usually get detected and cannibalized by their sisters while they are still larvae. Sex determination in honey bees depends on the CSD gene (horizontal yellow, green, and gray rectangles) located on a specific chromosome. A colony formed from a queen that mated with two drones yields six potential offspring. The number and version of the gene inherited determines each offspring's sex. There are several implications this has for the colony. If many larvae receive the same two versions of this gene and get removed, the brood pattern will end up looking spotty. Populations with largely unrelated honey bees usually have higher genetic diversity. In these instances, there are many versions of the CSD gene in the population, and the chances of a bee inheriting two identical versions are low, so diploid males are rare. Yet in populations that are heavily inbred, with low genetic diversity, the likelihood of inheriting the same version of the CSD gene is much higher. These highly inbred bees will likely produce a spotty brood pattern because the workers are removing so many diploid males. For this reason, it is important that honey bee breeding programs do not result in inbreeding. Spotty brood pattern can be the result of inbreeding. Photo credit: Bee Informed Partnership

READ

Diploid Male Occurrences

If spotty brood can indicate health issues in your hive, how many larvae can we normally expect will receive two copies of the same gene in a normal, healthy population? The number of different versions

of the CSD gene varies in normal populations, but studies have found it tends to range between six and 17. Let's imagine one population has 10 different versions of the CSD gene. Each of the colored circles in the above illustration represents different CSD gene versions. Our queen has two of these 10 possible versions, one on a specific chromosome she inherited from her mother and the other from her father. So let's say our queen mates with 10 males, and each male has a unique version of this gene. We can draw all the potential fertilized offspring this queen can have. In this scenario, there are 20 different combinations that are possible, and of all these combinations, there are two cases where the offspring inherits identical gene versions. These two offspring — representing 10% of all the offspring the queen lays on average — would be the ones that are removed and cannibalized. This goes to show that in normal populations, we can always expect to see some spotty brood due to the sex determination system in honey bees. It would be rare to find an absolute perfect brood pattern on a frame.

WATCH

How Caste Is Determined

For generations, people have wondered what determines whether a bee will become a queen or a worker. How is the decision made? What factors are involved? There are still some mysteries surrounding the exact molecular changes that occur during the process of caste determination, but scientists are getting close to piecing together the whole picture. This video addresses our current understanding of honey bee caste determination. Video Transcript Interestingly, caste determination in honey bees is due to environmental factors instead of genetic factors. This means that a young, female larva can become either a worker or a queen. Think about that for a moment. They differ vastly in their appearances and in their behavioral repertoires and an order of magnitude of a difference between their lifespan. Workers only live for a few weeks and queens live for a few years. Yet they arise from an identical DNA blueprint. The key is that there is a short time period during which a larva determines to either become a queen or a worker. And this time period is between 12 and 36 hours, from when it hatches as an egg. So anything beyond 36 hours the larva has already been destined to develop the way that it was meant to whether it's a worker or a queen, there is no switching. And this has implications for a colony if it suddenly finds itself queenless. If the queen dies, workers need to try and rear emergency queens out of larvae that is already existing in the colony. And they could only do this from larvae that are less than 36 hours old. So if there aren't any young larvae in the colony it's considered hopelessly queenless unless a beekeeper intervenes and introduces a new one or a frame with young brood on it. So what determines this developmental pathway of a larva? What is that environmental factor? It's actually royal jelly. And royal jelly is a food that's secreted from the hypopharyngeal glands and the mandibular glands of nurse bees. It's milky in its consistency and it contains water, sugars, salts, fatty acids, other lipids, vitamins, free amino acids, and proteins. So basically it's highly nutritious. And queens and workers actually receive different amounts of royal jelly, and this contributes to why they are different castes. So queens receive royal jelly for their entire period as a larva, all six days. But workers are fed royal jelly only for the first two to three days as a larva, and then they transition to eating bee bread. Also, queens are fed approximately ten times more often than worker larvae are fed. About 1,500 times, versus 150 times. So, essentially, some components of royal jelly are starting a cascade of reactions in the bee's cells that begin to drive this differential

development. And it turns out that there are hundreds of genes that are involved. Hundreds of genes are being turned off and turned on, and worker larvae, and queen larvae as they're developing. And although queens and workers have the same DNA blueprint to begin with, the take-home message is that genes of this blueprint are turned on and off differently. And these can be genes that are involved in life span or body size or other factors as well, like reproduction, and there's just many that are involved and it's a very cool and complex process.

READ

Molecules and Food Involved in Caste Determination

There are slight differences in the jelly fed to workers and queens. The popular name for all milky brood food, whether it's fed to queen or worker larvae, is "royal jelly." It's worth noting, however, that scientists have determined that, while the substances are largely the same, there are slight differences between the jelly fed to workers and the jelly fed to queens. You may sometimes hear the terms "worker jelly" and "royal jelly" used to differentiate between the two. Royal jelly has a lower moisture content and a higher sugar content than worker jelly. It also contains higher amounts of a fatty acid called 0-hydroxy-2-decenoic acid. The jelly fed to different-aged brood changes slightly over time as well; e.g., the moisture content, sugar content, and mineral content in the jelly are different depending on whether it's fed to a larva that is one day old, two days old, three days old, and so forth. The consumption of royal jelly triggers responses in hundreds of genes and proteins; many hormones are also involved. Caste determination is complex and dynamic, with scientific research still unfolding. For the purposes of this course, we will cover only two components that are well studied for their role in caste determination. Juvenile hormone is a hormone found in all insects, but it plays an important role in honey bee caste determination. The hormone is high in emerging queens and low in emerging workers. Eating lots of royal jelly as a larva stimulates a high production of juvenile hormone, which in turn contributes to the development of a queen with large ovaries comprised of many ovarioles, as well as a long lifespan. Vitellogenin is a protein that is produced by the fat body in the honey bee and found in the ovaries, among other tissues. In the ovaries, it functions to help make eggs. Larvae with high vitellogenin levels in their hemolymph develop into queens. Consuming royal jelly as a larva increases the level of vitellogenin in the developing queen, ensuring that she can create many eggs throughout her lifetime. Vitellogenin levels in queens remain high as adults, which likely plays a role in contributing to their long lifespans. On the other hand, larvae with low vitellogenin levels in their hemolymph develop into workers. As adults, this protein remains at low levels in their ovaries, contributing to their reduced fertility. The quantity of royal jelly a larva consumes initiates cellular reactions that lead to the development of either a queen or a worker bee.

READ

Intercastrates

A queen is successfully reared from a female larva under three conditions: When the larva is exclusively fed royal jelly When it is fed copious amounts of royal jelly When the larva is selected at a young age to start receiving this special queen diet If even one of these three conditions isn't met, the resulting queen that develops is an "intercaste." Intercastrates can develop when they are fed small amounts of royal jelly as larvae, when their diet is altered to differ in the ratio of sugars or other

nutrients that are naturally occurring in royal jelly, or when the larva that starts receiving a queen diet is older. Inter castes can develop from larvae that are as old as four days! Inter castes have characteristics of both queens and workers. The length of their abdomen, the size of their spermatheca, and the number of ovarioles for inter castes are between those of a worker and a queen. Some inter castes develop corbicula on their legs for pollen storage, while others develop with a shriveled spermatheca and are therefore incapable of storing sperm. If an inter caste develops with more queen-like features than worker-like features, they can act as a queen for at least a short period of time in the hive — mating, laying eggs, and attracting a retinue of workers — but they tend to be superseded by workers in the same season. Naturally occurring inter castes are rare in a colony and only result if workers are forced to rear an emergency queen from older larvae. They can occur, however, if beekeepers are grafting older larvae when they are queen rearing.

WATCH

How the Worker's Role Transitions Throughout Its Life

The tasks that workers specialize in allow the colony to perform many functions at once. The nest can be constructed and cleaned, food can be collected and stored, brood can be fed, and the colony can be guarded - all at the same time! This is made possible because workers specialize in different tasks at different times. As a worker ages, she learns and performs different roles. Watch this video to learn how tasks change over a worker's life. Video Transcript The queen and worker castes represent a division of labor and reproduction. But this isn't the only division of labor we see in a colony. Labor is also divided up among workers who transition through many tasks or duties as they age. And this transitioning and specializing in different tasks at a time is called polyethism. And the tasks that they perform depend on the age of the bee and also the needs of the colony. So roughly speaking, workers begin their first half of their life as house bees, performing tasks inside the nest and then they spend their second half of their lives as field bees, performing tasks outside the nest. The very first thing that a bee does when she emerges from her cell is she cleans her cell and cleans the other cells around her. Then she transitions into capping brood and tending to the brood where she's called a nurse bee or attending to the queen where she's called the retinue, or makes up the retinue. Then she transitions into receiving nectar that's coming into the colony, cleaning debris in the colony, packing pollen and comb building. So just general hive maintenance. When she's finally ready to start transitioning outside of the hive, they will perform tasks that are right in the intermediate, right at the hive entrance. So they can participate in ventilation at the entrance or guarding the entrance against intruders. And then when they're finally ready they take their first foraging flight. And as foragers they can collect a variety of different things. They can forage for nectar, pollen, water, resin or even new nest sites if a colony swarms. And they usually remain as foragers until they die of natural causes. A really neat thing is that foragers specialize in what they're collecting. So they may be specialize as water collectors or as propolis collectors, although they can be flexible if the need arises. It's really interesting to note that there is a variation as to when a worker is beginning a new task. So, a worker that's finally becoming a forager might do that after 14 days, whereas another worker might do that after 19 days. It's not so rigid or set in stone. It's flexible. And the really interesting thing to know as a master beekeeper, and where it's relevant, is that workers with different duties are found in different parts of the nest. So nurse bees are usually found on the brood comb, because they're tending to the brood. And that's usually toward

the bottom or center of the hive. Guards are usually found at the entrance of the hive. And comb builders are found wherever there's new comb being built, which could basically be anywhere within the colony. And when home, foragers can also be found throughout the colony. but they're most often clustered below the brood area, so that they can easily enter and exit the hive. And this is important to know because it might impact your decisions when you're inspecting colonies, when you're monitoring for certain parasites or diseases and also for swarm control.

TOOL

Worker Bee Ages and Tasks

As you just learned, the worker bee's ■ Download the Tool role transitions throughout its life. This Download this tool for more tool will help you understand the information on polyethism. average ages that workers perform each task.

READ

Molecules Involved in Worker Polyethism

The structures and chemicals in honey bee brains change as they age and as they adopt new behavioral repertoires. The shifts in worker duties are regulated by hormones, proteins, and neurotransmitters within the bee's hemolymph and neurons. Some of these molecules may be familiar to you, as they are also involved in caste determination. It is not uncommon in biology for molecules to be involved in more than one function. Changes in worker duties are caused by subtle changes in the levels of these and other hormones and proteins, which are in turn regulated by the social environment of the colony. The presence and fertility of the queen, the population level, the amount of brood present, and the amount of food coming in, can all influence what task a worker is going to perform. Here we will touch upon four molecules involved in worker polyethism: Juvenile Hormone JH (Juvenile hormone) influence the transition from in-hive tasks (like nursing) to outside tasks (like foraging). Newly emerged worker bees have low levels of this hormone in their bodies, but the level increases steadily throughout the first 15 days or so of a worker's life. At this point, juvenile hormone assists in the shift from tasks inside the hive to tasks outside, thus promoting the shift to foraging. You may be wondering why this hormone is called 'juvenile' hormone if it is actually highest in bees. It turns out, the progression of juvenile hormone levels are actually opposite in honey bees compared to other insects. In other insects, this hormone is highest in juveniles and decreases as the insect ages into adulthood. Vitellogenin In addition to facilitating egg production in the ovaries, vitellogenin is present in the hypopharyngeal glands, the fat body, and the hemolymph. In the hypopharyngeal glands of nurses, vitellogenin helps create royal jelly to feed to brood. Nurse bees have high levels of vitellogenin in their hemolymph and their hypopharyngeal glands, and the levels decrease as bees age and become foragers. This makes sense because it is nurses that feed larvae, while foragers do not tend to the brood. It is this rapid decrease in vitellogenin, combined with the increase in juvenile hormone, that is associated with the behavioral switch from nurse to forager. Dopamine Dopamine is a neurotransmitter that is present in many organisms, including humans. Neurotransmitters are chemicals released by neurons in the brain to send fast signals to other neurons. In honey bees, low levels of dopamine stimulate nurses to care for brood. So which bees in the hive likely have low levels of dopamine? You guessed it – younger nurse bees. Their levels are much lower than those found in foragers. It is the

presence of the queen (and her pheromones) that regulates dopamine production in worker bees' brains. Nurse bees (rather than foragers) come in contact with queen pheromone, which sets off a molecular pathway in the nurse bee brain to reduce the production of dopamine and thereby stimulate nursing behavior. Pheromones Several pheromones are known to play a role in the regulation of workers' role transitions throughout their life. For example, the foraging pheromone and especially one of its component, ethyl oleate, is emitted by returning foragers and it suppresses a premature switch to foraging roles in young bees. Here is another example: the brood pheromone emitted by the developing larvae signals the need for nursing. This pheromone can delay the onset of foraging behavior in worker bees, ensuring that the brood is well-cared for by young bees. These, and other pheromones, assist workers in their assessment of the needs of the colonies (more foragers? more nurses? etc.) and they contribute to the remarkable flexibility exhibited by honey bees when it comes to role allocation in the colony.

READ

Social Environment Influence on Worker Tasks

How does the social environment influence worker tasks? Although the molecules involved in worker polyethism change as a bee ages, and this influences the transition to different tasks, age is not the only factor that determines when workers will be house bees and when they will be field bees. If that were the case, colonies wouldn't be able to respond to changes in their environment. So, in addition to age, there are also strong effects of the social environment. Honey bees are able to detect what is going on inside the nest and alter their behavior accordingly. For example, although house bees transition to become field bees at around 15 days of age, the exact age at which this happens is adjustable. If a high percentage of the foragers are removed from a colony (e.g., through a pesticide poisoning event), their sisters will recognize that the colony no longer has enough bees to bring in pollen and nectar. Some nurse bees will transition to becoming foragers earlier than they would otherwise. In the reverse, if the nurse bee population suddenly declines, some foragers will revert to being nurses. More broadly, worker bee behavior is flexible enough to take on tasks that are needed to help the colony run smoothly. Worker bees spend a considerable amount of time performing tasks, but they actually spend most of their time being inactive and walking around the nest. It is likely that they are assessing their colony's state at this time, sensing social and environmental cues of the colony condition. This ability to read the colony's situation and then respond accordingly makes these honey bee societies flexible and resilient. For example, during a nectar flow, foragers will recruit more bees to help receive the food through a tremble dance, and brood rearing increases. When there is not enough storage space for holding the incoming nectar, more bees will engage in comb building. A remarkable example of this ability to detect and respond to a colony's needs is the work of heater bees. Heater bees are workers found in the brood nest that generate heat through shivering as a way to incubate pupae at the proper temperature. While brood needs to develop at a certain temperature (91°F to 97°F) to become healthy adults, they can tolerate slight variations. Interestingly, these small variations can impact the behaviors of these bees after they emerge. For instance, bees that develop at the highest end of the range (closer to 97°F) transition to become foragers at an earlier age, are more likely to perform undertaking tasks, are better learners, and perform more dances than bees that develop at the lowest end of the range (closer to 91°F). Beyond this range, however, pupae fail to develop properly or

die.

READ

Bee Lifespan

While healthy queen bees live on average two to three years (a very long time for an insect!), worker lifespans are a lot shorter. Their lifespan depends on a number of factors, including their genetics, the activities performed throughout their lives, and the season. During spring and fall, workers typically live for one to two months. In the summer, they live much shorter lives, typically two to five weeks. Bees that live throughout the winter are the longest living bees. These bees are called winter bees, and can live for four to five months! They are needed to survive over winter so they can regulate the temperature in the brood nest and help rear the first brood at the end of winter. Winter bees are generated within the colony in autumn from the end of August through November and they differ in their physiology compared to summer bees. Winter bees have lower levels of juvenile hormone (analogous to nurse bees who spend their time performing in-hive tasks) than summer bees, and they have much larger hypopharyngeal glands and larger and more numerous fat body cells than summer bees. The fat body is a collection of tissues comprised of many fat body cells that store fat and glycogen (stored sugar) to release for energy production in times of food shortage. It is located mostly in the abdomen, but also spans the thorax and head. Drones that do not mate with queens will live an average of three to four weeks. Their lifespans may be shortened as winter approaches because workers usually evict drones from the hive prior to winter. The colony does not expend precious food stores on feeding drones that do not aid in survival over the winter. Drones usually get evicted by their sisters in the fall.

MODULE WRAP-UP

Honey Bee Development

In this module, you examined the remarkable journey of honey bee development, from egg to adult. You learned how environmental and social factors determine whether eggs develop into workers, queens, or drones, and how adult bees continue to adapt and change their roles throughout their lives. This knowledge of bee development and physiology reveals the dynamic nature of colony life and the impressive adaptability of honey bee behavior.

How Honey Bees Communicate

If you think about it, you'll realize that all living things need to communicate. Birds sing, wolves howl, and even fish use body language. We typically think communication is all about our own uniquely complicated use of the voice, but most forms of life use other sensory modalities to communicate. It is really no surprise that eusocial insects like honey bees are elaborate, multi-sensory communicators. Even though we only understand a fraction of their language, that tiny part is fascinating! Bees live together in the dark. It is reasonable to assume that vision is not a sense they rely heavily on for communication. Instead, bees mostly use the senses of "hearing" (vibration detection), taste, smell (pheromones), and touch to "speak" with each other. The famous biologist Karl von Frisch first described a type of bee communication that he called a "dance" in the 1940s, work for which he won a

Nobel Prize. Subsequent studies of different forms of physical bee communication — all called “dances” — were conducted by a number of his students and others, such as Martin Lindauer, Jürgen Tautz, and Tom Seeley. In 1960, the communication signal “Queen Mandibular Pheromone” was discovered, opening the doorway to the captivating world of bee pheromones. In the years since this discovery, over 28 uniquely different honey bee pheromones have been described. In this module, you will learn some of the ways honey bees communicate important information through pheromones and dances, and you will learn how to recognize different communication behaviors in your own colonies.

READ

Pheromone Communication

■ Honey bees have evolved one of the most complex systems of pheromone communication. The honey bee colony is a communication known in nature. chemically busy environment, with This communication regulates pheromones secreted by the functions ranging from reproduction queen, brood, and workers. to foraging, enables communication Pheromones elicit behavioral among the colony members within a changes in members of the same hive, and occasionally permits species. communication between colonies. The pheromones emitted by an individual bee can change over the course of its lifetime or as a function of its reproductive status. Of the many chemical signals that bees are known to emit, only a small fraction of them are well understood. Queen-Produced Pheromones Let us start with the most important pheromones, the ones produced by the queen. Why are those the most important? First, it is known that when the queen is old or sick (low pheromonal signal) or it dies (no pheromonal signal), workers are driven to rear new queens from young brood within 12–24 hours. Second, the removal of the queen in absence of young brood soon leads to the decline of the colony: the workers stop performing their activities and start to lay unfertilized eggs that develop into male adults (drones); the colony becomes disorganized, unfit, dirty, susceptible to diseases and prey of predators; it rapidly depopulates and goes toward a certain death. The queen signal includes 40-100 compounds produced from multiple glands: the mandibular gland, tergal glands, Dufour’s gland, Koshevnikov gland and tarsal glands. Those chemicals attract drones during mating flights. They stimulate foraging and brood rearing, while inhibiting queen rearing and ■ suppressing ovary development in workers. They stabilize swarms (along with the Nasonov pheromone produced by workers), and much more. How can ONE individual affect the physiology and behavior of thousand of individuals dispersed in the hive? The answer is that the queen signal attracts workers to surround and tend to the queen. This retinue of workers that are constantly touching and licking her pass along the pheromone to the rest of the colony through body contact, antennating (touching antennae), and trophallaxis. The transfer of these pheromones enables the bees in the hive to know that the queen is alive and fertile, thus providing social cohesion to the colony. Five of the dozen components of the queen signal are secreted by the mandibular gland; these are collectively known as queen mandibular pheromone (QMP). This is the best-understood, behaviorally and chemically, pheromone in honey bees. A strong signal indicates that the queen is healthy and productive, which inhibits the development of workers’ ovaries. It also attracts Workers detect queen pheromone by licking workers to surround her in the hive and antennating the queen. Photo credit: Bee or in a swarm and stimulates them to Informed Partnership. tend to her. Queen mandibular pheromone also helps regulate the transition for nurse bees becoming forager bees by delaying the onset of foraging. What about the

other compounds emitted from other glands? They are less well known, behaviorally and chemically. We know about the queen's Dufour's gland which is located near the egg laying device of the queen, aka the ovipositor. As the queen lays an egg, she secretes a unique waxy, non-volatile, mix of ~ 20 compounds that ends up covering her eggs. This unique chemical signature allows nurse worker bees who inspect the eggs later using antennal contact/mouthpart contact to identify queen-laid eggs. Why is that important? Because this allows nurse workers to identify worker-laid eggs, since those do not have the same chemical signature. As you know, workers are not supposed to lay eggs, and there are many pheromones present in the hive to deter them from having functional ovaries, yet every now and then, and more under some situations, workers lay a few eggs. Those eggs do not bear the queen's chemical signature thus workers destroy them. This is called worker policing. Another portion of the queen signal that has been characterized is what oozes from the queen's tarsal glands in the last segment of her legs. The footprint queen's secretion is oily, colorless and composed of ~ 12 non-volatile substances that are extruded through openings on the tarsi when the queen is walking onto the comb, as she explores the hive for empty cells she can use to lay her eggs. Those secretions deposited as the queen walks throughout the hive signal to the colony that there is a functional queen present and that she is active. This in turn maintains colony stability, contributes to the inhibition of queen cell construction by workers and also to the inhibition of worker reproduction.

Brood-Produced Pheromones While larvae do not seem to do much other than eat and grow, they actually influence the behavior of nurses and help ensure their own care through brood pheromone. This pheromone is released by the larvae and detected by the nurse bees. It gives the nurses information about a specific larva's sex, caste, and age, which in turn influences the nurses to provision them different. Larvae release pheromones to provide nurses amounts and types of food. It also with information. inhibits worker ovary development and can affect the onset of foraging in nurse bees. Young nurse bees that detect brood pheromone delay their transition time to becoming foragers, while older nurse bees that detect the pheromone are stimulated to "speed up" their transition to foraging.

■ **Worker-Produced Pheromones** The Nasonov pheromone is the primary means of aggregation and orientation for honey bees. The pheromone is released from the Nasonov gland, located near the end of the abdomen in workers. There are three primary behavioral contexts where the Nasonov pheromone is used: marking the nest entry to assist returning bees, aggregation during swarming, and marking a water source. The ability of adult workers to release this pheromone increases with age; peak production occurs when the bee is about 28 days old. This pheromone is unique to workers, as queens and drones do not contain a Nasonov gland. Synthetic Nasonov pheromone has been manufactured into a lure that can be used by beekeepers to attract swarming bees to unoccupied hives. Some of the same compounds in Nasonov pheromone are also found in lemongrass oil, so beekeepers can also use lemongrass oil to increase the chances of catching a swarm. The forager pheromone is derived from fermented nectar, which foragers regularly consume when they are out collecting resources. They process a chemical in the nectar and exude it through their exoskeleton. Bees inside the hive smell this chemical when the foragers return. This pheromone delays the onset of foraging in nurse bees: When there are numerous foragers and therefore high levels of the pheromone, the transition is delayed. By contrast, when there are few foragers and the pheromone is low, some nurse bees are stimulated to transition into foragers to help maintain the field force.

■ **Alarm Pheromone** When a worker bee is disturbed by a predator or threat to the colony, she emits alarm pheromone to mark the

location of the potential threat and to rally others to aid in defense. She releases alarm pheromone from her mandibular gland and a gland beside the stinger. When stinging, the gland near the stinger produces the pheromone and it coats the sting shaft. When the bee stings the enemy, the alarm pheromone "marks" the sting position so other bees can quickly locate it. Promptly Alarm pheromone coating the sting shaft removing the stinger and smoking helps mark intruders for easy identification by other bees. the area helps to mask the smell of this pheromone to prevent future attacks. Note that dark colors, erratic movements, and certain odors, such as leather, can elicit an attack response prior to alarm pheromone being released. Alarm pheromone is one of the honey bee pheromones that is easily smelled by humans. If you've smelled it before, you may have thought it smells a bit like bananas. Interestingly, one of the main components in alarm pheromone, isoamyl acetate, is the same compound that gives bananas their sweet smell. ■

WATCH

Honey Bee Dances — The Waggle Dance

The various forms of physical (touch) communication among honey bees are called "dances." Karl von Frisch described what has become known as the "waggle dance," a behavior used by a successful forager to share information about the location and scent of a rich food source with other foragers. In this video, Dr. Tom Seeley, the Cornell professor who has dedicated much of his career to studying honey bee communication and behavior, describes this fascinating behavior. Video Transcript Okay, hello. What I want to share with you now is the waggle dance of the honey bee. This is a very special communication signal. And what makes it special is that it's a signal that honey bees produce that shows other individuals the location of a rich food source and there's only one other species that has that ability with the honey bee. That's ourselves, human beings. We do that when we give somebody directions to find a good grocery store, for example. And the honey bees are doing that to show their hivemates the locations of rich food sources, flower patches. Now, one thing I'd find helpful in explaining the waggle dance is to understand that you can think of this as the bee performing a miniaturized reenactment of her flight out to the flower patch. And when I say a miniaturized reenactment of the flight, that's actually realistic because as the bee goes through the movement of the waggle dance which consists of her doing the waggle motion in the center of the formation of the dance and then she circles back and does another waggle run. She's also buzzing her wings. And she is pointing you in the direction that indicates the direction to the food source and the length of that waggle run, the duration of it, how long each waggle run lasts, whether it's a half a second or several seconds, indicates the distance. So it's a lot like the bee making a miniature flight out to the food source. Let's look at this waggle dance. Okay. You can see that as this bee walks forward she's wagging her body back and forth. In this case she's pointing. Well I should explain that this is a bee doing the dance on the side of a swarm. So it's a vertical surface. And the bee is dancing. As she walks forward doing the waggle motion, she's pointed 90 degrees to left of straight up. And again, this is on a vertical surface, the side of a swarm of bees. So she's telling other bees the direction and as we'll see in a second, 90 degrees to the left of straight up means the direction of the flowers is 90 degrees to the left of the direction of the sun. We'll come back to that in another diagram. The other thing I want you to notice now is that she's consistent in the direction she faces, as she does the wagging, and she's consistent in how long each waggle run lasts. In this example, each waggle run lasts about eight tenths of a

second. And so that indicates a food source about 800 yards away. Okay, let's take a look at how this information is encoded. As I mentioned, the direction is indicated by the angle of the waggle run. And here we've got an example where the bee is, as she does her waggle runs, you can see that she would be walking upward on the comb but 40 degrees to the right of straight up. And what she's telling the other bees following her dance is that when they go outside of the hive, the flowers are 40 degrees to the right of the direction of the sun. So inside the hive, up is the reference direction. Outside the hive, the sun's direction is the reference direction. So it's a very simple translation from what the bee experiences in the hive to what direction she should fly outside the hive. And then the distance is indicated by the duration of the waggle run. You can see that in the chart. So our little bee that was dancing eight-tenths of a second you can see that corresponds to a distance of about, well I see, 500 meters really. And it's just like doing a long flight. If the food source is a long ways away, each waggle run lasts for longer and longer amounts of time. So, that's how the waggle dance works. And you might sometimes hear about two types of dances being done in the hive, a round dance and a waggle dance. That's what people used to think but we now understand that these round dances are just waggle dances where the waggle portion of the dance is very very short. It's the same signal, it just looked different before people had means of video recording the dances and studying them closely. A waggle dancer uses scents, vibrations, and repetition to enhance her communication. A successful forager performing a waggle dance will carry on her body the scent of the flowers she has found; this helps other bees find the exact flowers to which she is guiding them. The number of waggle circuits communicates the quality of a food source. A dancing bee produces only a dozen or so circuits if the food source that she has found is good but not great, but she produces hundreds of circuits if it is a great food source or if the colony is desperate for what she has found (nectar, pollen, water, or even propolis). Her excitement about the food source is also communicated by how rapidly she conducts her dance.

WATCH

Honey Bee Dances — The Shaking Signal

Like most animals, bees need sleep. Foragers that aren't busy foraging can be found fast asleep on the margins of the cluster or up in the honey supers. You can tell a bee is sleeping because she is very still on the comb. If you poke at her with your finger, she doesn't react. Researchers have found that the brain activity of sleeping bees is reduced, just as it is for people. During a honey flow, most foragers are at work bringing in nectar and pollen all day, and they only sleep at night. When nectar sources are scarce, however, or when the weather is bad, foragers have little to do. It is then that you find sleeping foragers while doing a daytime hive inspection. During this time there are also many foragers inside the hive that are awake but aren't participating in any tasks. When a bee finds nectar-laden flowers early in the morning, after a dearth, or following a period of bad weather, she wants to recruit sister foragers, many of whom may be asleep or are not already occupied with tasks inside the hive. She must therefore motivate some of them to get a waggle dance audience, which she does by grabbing inactive bees with her forelegs and shaking them. This shaking motion is accomplished by the bee vibrating her abdomen. Dr. Tom Seeley calls this behavior the "shake and wake." The foraging bee will first unload her nectar, shake a few inactive foragers, and then fly back out to collect another load of nectar. She repeats this routine a number of times before she has enough of an audience to perform her waggle

dance. After the waggle dances begin, it is only a very short time — about an hour — until many foragers may be found at the food source. Most often, the shaking signal is performed on the waggle dance floor. The signal “activates foraging preparations,” so the earliest foragers motivate others to receive information from dancers. The shaking signal doesn’t only occur during foraging; for instance, workers shake the queen during swarm preparation, inducing her to stop egg laying and to become more active on the comb. It is thought that the shake signal is a general one used to alter what bees are doing. In this way, the shake signal means “Switch tasks!” In the context of foraging, the shaking signal tells sleeping or inactive bees to get ready to “watch” a dancer and then go forage as well. In this video, a bee performs the shaking signal.

WATCH

Honey Bee Dances — The Tremble Dance and “Stop”

Signal As increasing numbers of foragers bring their bounty back to the hive during the start of a honey flow (abundant nectar supply), they deliver their harvest to receiver bees. If a forager cannot find a receiver bee, she must recruit one using the “tremble dance.” Another dance performed by foragers, specifically nectar foragers, the tremble dance is used to recruit additional receiver bees to unload nectar from incoming foragers at the start of a honey flow. If a forager that has found an excellent nectar source has a hard time finding a receiver bee to unload her when she returns to the hive, then she will perform a tremble dance instead of a waggle dance. She trembles her body and walks throughout the hive so that middle-aged bees (ones older than nurse bees but younger than foragers) can pick up on this communication signal and learn of the need for more bees to function as nectar receivers. Once more bees are engaged in nectar receiving, the nectar foragers will get unloaded promptly when they get home, enabling them to more efficiently exploit the richly flowing nectar sources that they have found. The average duration of a tremble dance is around 30 minutes! Over time, there will be a rise in the number of nectar receivers, so the colony gradually adjusts to handle the rich nectar sources that the foragers have discovered. Witness a forager performing a tremble dance in this video. A worker bee (located slightly right of center in the clip) performs a tremble dance. The “stop” signal Eventually, as a food site is failing or too crowded with bees, returning foragers convey this information to other foragers in the hive. They will press themselves up against others and give a short, high-toned beeping noise. This beeping sound has the effect of stopping waggle dances if the dancer has been “beeped.” In this way, waggle dances, shaking signals, tremble dances, and beeping help to stabilize and coordinate the flow of nectar and its processing in the colony, and form the basis of the ways in which bees “talk” to one another to regulate food gathering and storage.

READ

How Are Dances Perceived in the Dark?

Dance followers stand still during the dance and “listen” with their antennae stretched out stiffly at an angle of between 90 to 120 degrees to each other. The dancer’s rhythmically moving abdomen displaces each follower’s antennae. Because followers use their own gravity detectors to determine where they are on the comb, they can calculate the position of the dancer and thus the direction to the resource that the dancer is advertising. Bees choose particular places on the comb to conduct their waggle dances. Most dances occur on areas of empty cells about 10 cm square — quite a small area

considering a hive typically has an overall comb area of five square meters! Motivated foragers congregate at the dance floor to learn from dancers where the best forage might be. It has been discovered that the wax on their “dance floor” has unique vibrational properties that help to transmit dance vibrations to the maximum number of “listeners”! One experiment showed that bees dancing over empty cells recruited three to four times as many foragers as those dancing on sealed cells. Normally, a dancing bee creates strong vibrations using her uncoupled flight muscles, which are synchronized with the left and right movements of her abdomen during the dance. However, when the dancer first “calls” for attention but does not yet have a following of interested foragers, she does not use her flight muscles. During this early phase of dancing, she is signaling to distant recruits only through vibrations she introduces to the comb with her legs. Every waggle generates a pull on the edges of the cell she is standing on, and that pull sets up a vibration through the wax. Bees looking for dancers can feel comb vibrations at some distance away and, in response, come looking for the dancer. The comb dance floor can be “tuned” by changes in temperature and by the addition of propolis to the rims of the cells. Vibrational properties of the comb are slightly dampened by the beekeeper’s four-sided frame. On combs where bees dance, the bees often improve the vibrational property of the comb by constructing large gaps between the edge of the comb and the sides or bottoms of the frames. The wax on the dance floor is also chemically marked by the bees, as has been shown by experiments in which the dance floor is cut out and put back into a different place in the hive. The bees search for the relocated piece of comb before starting to dance again! They do not search for the “old dance floor” for more than a day, suggesting that its chemical signature dissipates. Several versions of the waggle dance are variously referred to as the round dance, sickle dance, or transition dance. All are variations of the same dance, modified to indicate distance from the hive to the food source. It has been shown that the different races of bees (Carniolan, Italian, German, and Caucasian) have different “dialects” or “accents” of waggle dances that vary in the number of waggles per run. There is also some evidence that certain odors are released by dancers’ abdomens during the waggle dance to help motivate followers. When these substances were experimentally released into a hive, more foragers left to collect food.

MODULE WRAP-UP

How Honey Bees Communicate

In this module, you explored honey bees’ sophisticated communication systems, from their complex use of pheromones to their intricate dance languages. Learning to recognize these communication behaviors in your own colonies enhances your beekeeping skills and deepens your appreciation for these remarkable social insects. [Back to Table of Contents](#)

Honey Bee Biology

Have you ever wondered how bees function as living organisms and navigate the world around them? In this module, you will explore the biological systems of honey bees that are needed for them to survive and succeed in their environments.

READ

The Nervous System

■ The honey bee nervous system is an intricate network of many nerve cells. The central nervous system of the honey bee is called the brain and is formed of a "body," where most of the ganglia are found, and "processes," thin filaments. Honey bees have a capacity for learning and memory that is quite long (in some mammals, they surpasses all other insects. can be several feet long!). These processes are called "dendrites" or "axons" depending on whether they receive input or convey output, respectively. The brain of the bee is where most of her neurons are located, and their processes, both axons and dendrites, extend along a "nerve cord" (analogous to our spinal cords) through the bee's thorax and abdomen, passing through a number of neuron cell body clusters called "ganglia". The brain and the ganglia act as "control centers", connected by nerve fibers to every part of the honey bee's body, including wings, muscles, legs, and even cuticular hairs. Although the bee brain coordinates all the body's functions including the activity of the ganglia, it is the ganglia that exert local control. This is why honey bees can continue to walk, flap their wings, and even sting for a short while after they are beheaded. The brain of a worker bee is tiny, and yet it contains nearly one million neurons, so it is not surprising that a worker honey bee's capacity for learning surpasses that of all other insects! They are able to communicate with their nest mates, learn directions, and remember a vast array of smells, sights, and tastes. Honey bees also have impressive mathematical abilities. They are able to count up to four and can understand the concept of zero. After a long foraging trip visiting many flowers, they can determine how to "beeline" home. The bee brain is organized into clusters of neurons that are called lobes. Different lobes of the brain receive inputs from different sensory stimuli, and in turn, each lobe controls a different function or activity in the body. For example, the antennal lobe receives inputs from olfactory stimuli (smells) and the optical lobe receives inputs from visual stimuli. A very important lobe in the brain is the mushroom body (indicated in orange in the illustration), named so because it resembles the shape of a mushroom. The mushroom body is the center for learning and memory. Sensory information that is processed by different lobes in the brain (like smells and tastes) are interpreted in the mushroom body where it is learned and remembered. This lobe takes up 20% of the bee brain in volume, while only 2% in a fruit fly. This gives an idea of the vast capabilities of honey bee learning and memory as an insect. It is useful to realize that the bee's nervous system as a whole operates according to a very simple principle. Information (input) is perceived by sensory nerve cells (neurons) that convey the information to the brain, eliciting a chemical or motor reaction by other neurons (output). This very basic organization "input—(reaction)—output" is found in all animal nervous systems, including that of humans. The sensory input of the bee's nervous system includes the tactile senses (temperature, pressure, gravity), the olfactory system (smell), the gustatory system (taste), the auditory system (hearing), and the visual system. The neurons concerned with delivering output are either "motor" or "endocrine" and they govern muscle contractions and the secretion of chemicals, like hormones. On the following pages we will take a brief look at some of the inputs and outputs of the honey bee's nervous system.

READ

Sensory Systems: Mechanoreception

"Mechanoreception" is the ability of an animal to detect and respond to certain kinds of stimuli, such as touch, pressure, temperature, or vibration. Mechanoreception permits animals to navigate their world,

react to mechanical stimuli, and even understand their place in space by perceiving gravity. Animals have evolved a clever way to discriminate among different sensory inputs by developing biological structures that are specifically attuned to certain physical stimuli. The most complex of these structures is the eye, whereas the simplest sensory input structure is merely a naked neuron that reacts to being touched or bent. For bees, sensory information is perceived by many small sensory organs or “mechanoreceptors” that are located on the periphery of their body, just below the cuticle, and on the tarsi (feet), mouthparts, wings, and antennae. These sensory organs are called sensilla (singular “sensillum”). They are small, hair-like structures that contain at least one neuron. Sensilla that perceive tactile information enable honey bees to recognize and learn the feeling of different objects, navigate different surfaces, land on flowers, and physically interact with their nestmates. Bees can perceive gravity. Bees have groups of sensory hairs called “hairplates” that are essentially a hair attached to a neuron. Hairplates are found on antennal joints, in the neck joint between head and thorax, the waist joint between the thorax and the abdomen, and in joints of their legs. Deflection of these hairs is how sensory neurons attached to them detect gravity. Humans detect gravity in the same way. Hairs located in our inner ears bend when tiny crystals lying over them are pulled by gravity, triggering sensory neurons attached to the hairs. Bees use their sense of gravity to orient their waggle dance inside the nest so they can communicate resources to other bees. Bees also use gravity to build their combs straight down. This has implications for how level a hive body should be: if a hive is slanted to one side, bees may build comb across the frames, which can become a nuisance for you as a beekeeper trying to move frames. It is important that a beekeeper ensures hives are not slanted ■ downward to the left or right in the bee yard. Tipping your hive forward to enhance drainage out the entrance is fine, because the angle is parallel to the direction of the combs. The perception of gravity helps bees build their combs straight. Bees can hear. When you hear a sound, your ear is detecting vibrations in the air. These vibrations are actually minute changes in pressure, like ripples on a pond, and just like the ripples they occur with different frequencies and amplitudes. Sometimes you may notice that you feel the floor vibrate when someone turns up the bass on their subwoofers. The power of the air’s vibrations can be transferred to other substrates. This fact is a key to understanding how hearing works in both bees and humans. Bees can hear both substrate vibrations (i.e. comb vibrations) and airborne vibrations (e.g. the sounds of dancing bees). They detect surface vibrations with pressure detectors in their legs (more specifically their tibia) called “subgenual organs”, and they discern airborne sounds using another type of sensory hair structure found on their antennae called a “Johnston’s organ”. The quality of the bee’s hearing differs from our human sense, but the basic neural mechanism is very similar. ■

WATCH

Sensory Systems: Olfaction (Smell)

The sense of smell is arguably the most important sense for honey bees. They rely on smell to help them find and learn important food sources, to communicate through pheromones to their other nest mates, and even to discriminate against non-nest mates that might be trying to enter their nest. Each honey bee colony develops a unique odor profile that is created by individuals secreting hydrocarbon molecules from their exoskeletons. By smelling these hydrocarbons, bees can determine who is a nest mate and who isn’t. Watch this video clip to learn more about olfaction in honey bees. Video Transcript

The sense of smell is called olfaction and honey bees have an incredible sense of smell. They have hundreds of olfactory receptors on their antennae that all bind to different odor molecules. An antenna can even determine the direction of an odor. So whether it's the right or left antenna that's sensing the odor more strongly, that tells them exactly which direction to fly to zone in on a flower. Olfactory receptors are found beneath these hairlike structures called sensilla located on the antennae. And odors bind to these receptors and stimulate a nerve signal to the brain, which can interpret exactly what they've smelled. When we sequenced the honey bee genome, we found that they have an expansive olfactory system with more olfactory receptor proteins than the other insects than we've sequenced so far. So the fruit fly and the mosquito. So for instance, the fruit fly has 62 olfactory receptor proteins, the mosquito has 79, and the honey bee has an impressive 170 olfactory receptor proteins. And this isn't surprising, given the importance of smell to honey bees. They need to be able to distinguish between their nest mates, and non-nest mates, to smell and interpret all of the different pheromones that are found in the colony, and even to recognize many different types of flowers that they are encountering in their everyday lives. And honey bees actually have a very good memory when it comes to smells. When the brain is processing the olfactory information from the antennae, they can send the nerve signal to other lobes that help them learn and remember this smell. And one of the major lobes that does this is the mushroom body. It's the center for learning and remembering in the bee's brain. So this is important because they want to be able to remember which flowers are really good nectar sources. So, for instance, maybe in the fall they can smell that distinct smell of goldenrod and know that this is associated with a really good food reward. And actually, humans have sort of been taking advantage of this, in training honey bees using classical conditioning, similar to the Pavlov dog. And, what they do is that they expose honey bees to a certain smell, even if it is a smell they would never come in contact with in nature. And then after they expose them to the smell, they'll give them a reward, usually sugar syrup. And honey bees learn really quickly, often after just one or two presentations of this odor food reward system, to associate the odor with this reward. And researchers will actually use this technique to measure learning and memory in bees. And some people have even trained bees to sniff out landmines and illegal drugs using this technique. Though neither of these applications has really proven to be practical yet. Okay. So researchers can even look at this technique to look at how certain substances might impair learning and memory in bees. So for example, if honey bees are exposed to or consuming a particular pesticide, does this have an impact on their learning and memory? By doing this training, we can see whether or not bees are unable to learn an odor associated with a food reward, or if they learn it more slowly than normal, or if they forget it quickly than they normally would. And even though these sort of presentations for testing, learning, and memory and response to different factors are done in a lab, it actually has practical real world applications because learning and memory is so important for honey bees to navigate in their world, to collect food, and to cooperate socially. So it's just cool to know that by understanding the olfaction system in the honey bee, we're actually able to apply this technique to have broader implications in the bee's ability to function normally in the real world.

READ

Sensory Systems: Gustation (Taste)

The sense of taste is related to olfaction. In bees, as in humans, the sense of taste is less well-developed than the sense of smell. Bees have about 10 gustatory receptor sensilla types on their mouthparts, tips of their antennae, and tarsi that are specialized to taste a few chemicals, most of them sugars. Humans have around 10,000 taste buds that detect about six tastes. But the most impressive gustatory system that we know of is found in the catfish, which has an amazing 175,000 or more taste buds scattered all over its body!

WATCH

Sensory Systems: Vision

Honey bee vision is well adapted to enable these insects to perform all the functions necessary to thrive. They are able to detect the movement of predators, fly safely, find flowers and other resources, navigate based on polarized light, and use visual cues to find their way back to the hive. Drones also use their eyes to find queens on their mating flights. However, inside the dark hive, bees rely more on their senses of smell, touch, and vibration detection than on their visual sense. Video Transcript Honey bee vision takes place in two types of eyes. The first type are the ocelli. These are three eyes that are formed in a triangular pattern near the top of the bee's head and here you can see the middle ocelli. The ocelli detect light intensity, they can't form an image. But detecting light intensity is needed for orientation during foraging, and also to regulate behavioral and hormonal changes that occur between daytime and nighttime. Each ocellus has a lens with retinal cells beneath it. The next type of eyes are the ones that we commonly refer to as bee eyes on the side of the head. And these are compound eyes and they're much more complex. They're actually comprised of thousands of individual, independent facets with lenses. And this doesn't mean that honey bees see thousands of repeated pictures. But the lenses each form a larger picture, kind of like pixels in a computer screen. The structure of these eyes means that honey bees actually have a low visual resolution. They can't really see great at distances the way that humans can. But that's okay, because they get really close to the things they need to see, like flowers, the nest entrance, intruders. So the effective resolution is quite high. But anything beyond about two yards is actually blurry for a honey bee. Drones have a bit better vision. They have twice the number of facets and lenses as worker bees. Why do you think that's the case? It's to find the queen in the drone congregation areas. So vision is really important for honey bees because it allows them to detect color, to detect light polarization and also to detect movement. Perception of color occurs with the compound eyes. This is what's detecting the color of light that's entering it. And the colors that bees see are a little bit different from the colors that humans see. Human eyes can detect combinations of red, green and blue. They can essentially see these colors on this wavelength spectrum. But because bees have different pigments, they can see combinations of green, blue and ultraviolet. Ultraviolet being beyond violet or beyond purple on the color spectrum. So they can actually see these colors on the spectrum. And so you can tell that bees actually can't see red, and they interpret that as black. And their sensitivity to ultraviolet light helps them when they're orienting the sun as a compass, even when the sun is sometimes behind a cloud. Another neat thing is that because they're sensitive to ultraviolet light, they can also see flowers differently than humans can. So some flower petals actually reflect the ultraviolet light and they appear brighter to bees and they kind of have these really cool patterns on the petals that humans actually can't see. So when they're detecting polarized light, what happens is they're able to detect these bands of light polarization that

are in the sky. And basically, air molecules actually scatter particles of light so that you can see this band that's at a 90 degree angle to the sun. And as the sun moves throughout the day so does this band. Always at a 90 degree angle. So as long as bees can see this band of polarized light they can tell where the sun is. And it's a special arrangement of the light sensitive molecules in the compound eyes that enables bees to see where this skylight overhead is polarized and where it isn't. And this is really important, obviously, because it helps bees detect the position of the sun even when it's hidden behind a cloud. But bees must still see some patches of blue sky in order to determine the sun's direction. So if it's completely cloudy, you won't find any foragers about, they'll all be inside the hive. And the sun serves as a really important visual reference for navigating during flight. They need to be able to tell exactly where the flowers are that they're going to, how to get back to their hive, and especially, how to communicate this important flower patch in relation to the sun to their other worker bees through the waggle dance. Another thing that they're able to perceive with vision is movement. And even though honey bee eyes lack detail, these can actually discriminate fast movements better than humans can. They can discriminate them out of 0.03 of a second, incredibly fast. And this is really important for detecting predators. They can sense any mammals that are moving in front of their colonies so they can have a really quick attack. They could also sense when bee keepers are moving sort of agitated or quickly in front of their colonies too. A cool thing about movement is that this helps them determine how fast they're flying. So, the speed that objects are moving past their eyes or the optic flow helps honey bees sense how fast they're traveling and the distance that they're actually traveling from the hive, which is also important not only so they can find their way back home, but also to communicate it through the dance language to their sisters — the other worker bees. Vision is really important for learning and memory in bees. On the first journey from a hive, all bees whether they're drones, workers or queens, they perform an orientation flight. And they're trying to learn what their house looks like from the outside and how they can find their way back home. And they leave their hives and they sort of perform very short flights that are kind of hovering around the hive, and they're taking in what their hive looks like and other landmarks around them so that they won't get lost. And they have to be able to remember this. Honey bees will use the patterns of polarized UV light and the position of the sun to learn the patterns of the sun's movement during the day. And this is a really cool other thing about memory and it's something that you don't often think about. But for instance let's say that you are a worker bee and there's this really good patch of flowers directly east from your hive. So in the morning when you go and forage on those flowers, you know that you're going to fly directly toward the sun which is in the east. But let's say you come back to the hive and you're just hanging around until the afternoon when you're ready to go back out to that patch of flowers. This time the sun is in the west. So instead of flying directly toward the sun they have to know to fly directly away from the sun to get to those flowers. So bees are actually able to compensate for the change in the sun's position relative to the earth and remember how the sun is moving during the day. They're really good at using it as a directional reference even though it's changing which I find really fascinating.

WATCH

How Understanding Honey Bee Vision Can Make You a

Better Beekeeper Studying an organism's biology is a great way to learn how we can maximize its health and success in the real world. By applying a few simple principles we learned about honey bee

vision, we can reduce drift and confusion among our colonies in our own bee yards. These tips are applicable in a variety of beekeeping circumstances from colony arrangement in your yards to moving your colonies specific distances. Watch this video to learn some tips on what you can do. Video Transcript Another navigational technique that honey bees use is landmarks. This is really important to know as a master beekeeper because it impacts what you're doing with your colonies. So it's important to have landmarks that are sort of large, clear, and distinctive near the hives. This helps bees find out what their hive looks like and how they can get back home. It could be something as simple as putting the hive near a tree or painting your hive bodies different colors. Or at Dyce, we have sort of these unique symbols that are near the hives to help the bees. So the normal foraging radius or the average foraging radius of a honey bee is about two miles. So they're pretty good at learning a lot of the landmarks within that area. So if you're going to move your colonies you need to keep this in mind. If you're moving your colonies beyond two miles that's not a problem. A bee will leave her hive, she'll look around, not recognize anything and know that she has to relearn new landmarks and the landscape as she flies about because she's in an unfamiliar place. However, if you're moving your hive within two miles this is when confusion can arise. Now the hive position has changed, relative to familiar landmarks. A worker bee will leave the colony, she'll look around and she'll kind of remember some of the things that she's seeing and then she'll fly back to where the hive used to be. But now it's changed, so they're still recognizing their landmarks, but now they're not able to orient the same way that they did before. So we recommend to reduce confusion, move your hive small increments of a few feet at a time or move them over winter if you can when the bees are inside for many months and individual bees have to learn their surroundings again come spring. Another way where landmarks are good is in reducing drift. So honey bees drift from their home hive to another hive, especially if these hives are close together because there's not enough landmarks or visual cues to distinguish among the hives. And this isn't necessarily good, especially with disease transmission or transmitting Varroa mites. So you'd like to reduce drift as much as possible. And you might think, well isn't that what guard bees are for? Are they going to keep out all of the non-nest mates? And they do keep out a lot of the non-nest mates. But it turns out that if a forager is coming into a neighboring hive full of pollen or nectar, they're more willing to be accepted into that colony. So it's almost like the guard is saying okay, you're bringing food to the party, you can come on in. And so it turns out that there's actually a fair bit of drift that's occurring between colonies. And especially if you have hives in a row, the hives at the end of those rows tend to be comprised of a lot of drifting bees. And when we think about how hives are actually or how nests are actually spread out in nature in tree kind cavities in these wild bee colonies, they're really spread out. And it's nothing like how humans actually keep honey bees really compact together in an apiary, sometimes with 10 or 20 hives really close together, or even on pallets. So just having extra landmarks so that they can distinguish which hive is theirs is going to help them out a lot.

READ

The Muscular System: Flight

The bee's body has many dozens of muscles that help it move, but by far the largest muscles are those associated with flight, located in the thorax. Honey bees flap their wings incredibly fast (a whopping 240 times per second!) allowing them to reach speeds of 20 miles per hour! Although they appear to only have two wings while they are flying, they actually have four that hook together in pairs (with hamuli) so

they can beat in synchrony and reduce drag and turbulence. Having wings hooked together by hamuli is a characteristic of Hymenopteran insects, as are membranous wings (hence the name of the order Hymenoptera). Veins in the wings add strength to the thin membranes. Honey bee hamuli The way bees and other insects effect flight is one of the most amazing marvels of bioengineering. Bees use both their cuticle and muscles to beat their wings. The cuticle has a tendency to retain its shape, yet it is also somewhat flexible. This allows it to act like a springboard when it is deformed - it has stored energy that is released when it regains its shape. Two massive sets of muscles in the thorax of the bee are attached to the inside of the cuticle as shown in the diagram below. These sets of muscles squeeze the thorax to deform it alternately, either along the body axis, or side-to-side. This causes the wings, which are attached to hooklike wing-processes of the thoracic cuticle, to flap up and down. Thus the power stroke of the wings is aided by the natural recoil of the cuticle.

READ

The Respiratory System

Instead of relying on lungs like humans, honey bees breathe through a series of tubes called tracheae (singular: trachea) throughout their bodies. Tracheae function much like lungs, carrying oxygen to muscles and organs and carrying carbon dioxide away from them. You can watch a bee breathe. She expands and contracts her abdomen like an accordion. This rhythmical movement brings the air in through the thorax and eventually, out through the abdomen. Air enters the tracheal system mainly through holes in the bee's thorax called "spiracles". From there, the air is drawn throughout the body via smaller and smaller tracheae, much like capillaries. These tiny tracheae come within just a few cells' width of every living cell in the body of the bee, ensuring that adequate gas exchange can occur. After gas exchange occurs, the tracheae consolidate and enlarge again, then exit the body of the bee through several pairs of smaller abdominal spiracles, one pair per segment, for a total of 10 pairs (three thoracic pairs and seven abdominal pairs). Tracheal mites, a pest of honey bees, enter through the large thoracic spiracle and reside and reproduce in the tracheae, impairing honey bees' ability to breathe. Bees suffering from tracheal mite infections are unable to fly, because the heavy congestion of mites in their tracheae means they cannot get enough air to supply their muscles with the oxygen needed to respire and provide the energy for flight. Dissected honey bee tracheal tube full of various life stages of the tracheal mite. Photo by Christina Wahl

READ

The Circulatory System

The circulatory system of the honey bee is called "open" because it has no capillaries. In contrast, the circulatory system of humans, containing millions of capillaries, is called "closed". This is because the blood is contained within the blood vessels of humans whereas in honey bees it leaves the arteries and circulates freely through the bees' bodies. Humans have fluid moving freely around the body's cells also, called "lymph". Thus the blood in the bee is aptly named "hemolymph" since it functions both as blood and as lymph. Hemolymph is kept moving a few ways: 1) by the bee's heart, located in its abdomen, 2) by the large blood vessels with their muscular walls, and 3) by general body muscle contractions as the bee moves. When bees are inactive for long periods of time – such as during the winter – they can stimulate circulation by contracting the muscles in their abdomens. Hemolymph that

is in the abdomen of the bee enters the heart through the valves, where it gets pumped through the aorta and empties out of the aorta into the body cavity near the brain. As the bee moves, the hemolymph will make its way back to the abdomen. The cycle repeats. In this way, the hemolymph is able to move throughout the bee's body. The circulatory system has five main functions: Transporting energy from digested food to the rest of the body cells to sustain function and life Bringing waste from cells to the excretory organs to be removed Lubricating body movements Conferring immunity through cells that attack pathogens Carrying hormones from the glands where they are produced to the organs and tissues they are intended to target

READ

The Digestive System

The digestive system of animals is like a “tube inside a tube”: Food goes in one end, fecal material comes out the other, and the gut tube’s walls along the way not only function to absorb nutrition and eliminate waste but also to protect the body’s inner tissues from assault by foreign pathogens. Just like an animal's skin or cuticle (the outer “tube” in our “tube inside a tube” analogy), the gut tube functions as a barricade. It is divided into three parts: the foregut, midgut, and hindgut. Foregut The foregut is located in all three body segments — the head, thorax, and abdomen. Food (nectar, honey, water, and pollen) enters at the mouth, located in the head. Bees use their proboscis to bring liquids into their mouth. The proboscis is formed of several mouthparts that the bee brings together to form an airtight tube for drinking. The hairy middle part is what we think of as the bee’s “tongue.” The salivary glands of the bee, which are also part of the foregut, open on its upper surface. Salivary glands are distributed in both the head and thorax. The salivary glands in the thorax are Wahl also called thoracic glands. When the proboscis is not in use, it is folded back under the bee’s “chin.” When she wants to feed, she swings the proboscis forward and brings its several parts together into a tube. The bee then rapidly moves her tongue back and forth, bringing the liquid up through the proboscis and back into her pharynx (throat). If you have a dissection scope at home, you can watch a bee drink. It helps to visualize the process if you give her a colored sugar syrup to drink. During drinking, the outer parts of the proboscis remain stationary, acting simply as the walls of the “straw” through which liquid is drawn in. To chew bee bread, a bee uses its mandibles to break it down into small pieces that it then places in its mouth. ■ Swallowed nectar, honey, and bee bread pass through the esophagus and enter the last part of the foregut called the crop (also called the honey stomach), located in the abdomen. The crop is a storage area for nectar, honey, or water. Bees can partially fill their crops with honey before leaving the hive on foraging rounds to use as an energy resource, or they can fill it with nectar or water while foraging to bring back to the hive. A valve called the proventriculus is located at the base of the crop; when shut, it prevents the nectar that a bee has collected from moving into the midgut and being digested. The crop can swell enormously from the collection of nectar, displacing other abdominal organs. A bee that has brought red syrup home in her crop. Photo credit: Christina Wahl Midgut While a bee collecting nectar shuts her proventricular valve to keep nectar in her crop, a bee that is eating and digesting honey or bee bread will open her valve so food can pass from the crop to the ventriculus, an organ somewhat analogous to both the human stomach and small intestine. Here, food is digested and nutrients are absorbed. The ventricular walls are only one cell thick. They are protected from the abrasive qualities of ingested pollen by an onion-like, noncellular membrane that wraps around and around the inner

ventricular walls, called the peritrophic membrane. It is here in the ventriculus that pathogens such as *Nosema* and various viruses break through the peritrophic membranes and infect the ventricular cells.

■ **Hindgut** ■ At the rear end of the ventriculus is another valve, called the ventricular valve, which separates it from the hindgut. The hindgut begins with a short intestine, a narrow tube from which the 100 or more Malpighian tubules emerge. Malpighian tubules are bee versions of kidneys. They secrete nitrogenous waste from the hemolymph into the hindgut tube for elimination. Little is known of the function of the bee's intestine. It is much shorter and anatomically more simplistic than the small intestines of animals that rely on the small intestine for nutrient absorption. The hindgut ends with the rectum, where solid waste materials are stored for eventual excretion. A wintering bee, unwilling to defecate inside the hive, will store fecal material in the rectum for many weeks. When she is eventually able to relieve herself, the rectum can be enormously engorged, displacing most of the other abdominal organs. Bee feces contain undigested pollen fat and husks; dead cells from the midgut; undigested products from food supplements; some pesticide residues; and, in the case of a disease or parasitic infection, materials that might infect other bees. Bees, therefore, generally do not void waste inside the nest and instead take cleansing flights outside the nest to keep the colony clean and to reduce the potential for disease transmission. The bees' ability to avoid defecation within the nest is especially important in winter, for then they can be confined inside for weeks or even months. Finding fecal deposits inside the hive or in abnormally high amounts on the hive exterior (dysentery) signals a problem, as dysentery is a symptom of nutritional issues and some diseases. Orange bee feces on snow from a winter cleansing flight. It is healthy and normal for bee feces to be evident outside the hive in winter, generally after a window of warmer weather when they were able to take cleansing flights en masse.

MODULE WRAP-UP

Honey Bee Biology

In this module, you discovered the essential biological systems that enable honey bees to survive and thrive in their environments. This journey through their fundamental life processes showed you how honey bees function as living organisms and interact with the world around them. [Back to Table of Contents](#)

How Honey Bees Construct and Maintain

Their Nests To care for honey bees more effectively, we need to understand their uniquely constructed nests. In particular, the nests of wild honey bee colonies reveal a lot about the bees' needs and preferences, despite regional and racial variations. Much of the information included in this module was derived from studies by Cornell University of wild colonies and swarms being installed in hives. In this module, you will investigate how bees construct and maintain their homes, what characteristics they prefer, and how they keep them livable and conducive to healthy development.

READ

The Building Blocks of a Nest

Honey bees are one of the few animals on Earth that are able to make by bees' bodies and plant resins architectural constructions from that are processed into propolis, materials produced by their bodies. The properties of wax enable it to be an ideal material for building comb, storing honey, and communicating. Wax Production Wax is composed of over 300 individual compounds and is considered energetically "expensive" for the bees to create. The book "The Hive and the Honey Bee" (2015) estimates that a colony must consume anywhere from five to 20 pounds of honey in order to produce one pound of wax. Wax production is stimulated by two things: excess nectar and a colony's need for more storage space. Bees only build comb under these two conditions. Because comb building is so expensive for the colony, bees typically only produce wax when there is an excess amount of carbohydrates coming into the hive, like during a honey flow or when feeding on 1:1 syrup. Furthermore, the bees will only build comb when they have a need for more storage space; most of the current comb in the hive must be in use before the colony invests in constructing more. Some examples of conditions where bees need more space are during a honey flow, when they need more room to store the incoming nectar, or when the queen is laying plenty of eggs and the colony needs more room to accommodate the growing brood nest. Worker bees between 12 and 18 days old produce most of a colony's wax (making comb building one of the final roles of house bees before they transition to foraging). Their wax glands (four pairs) are located on the undersides of their abdomens, growing when wax is being produced and shrinking when it is not. If comb building is urgently needed after a bee has transitioned into another job, the bee can reactivate her glands to once again be a wax producer. Liquid wax is excreted from these glands and deposited onto plates called wax mirrors, where it hardens into wax scales or flakes as it dries. Every 12 hours, each bee's gland can form one new scale. The bees then scrape off the hardened wax scales, chew them, add saliva, and manipulate them. Worker's wax glands secreting wax. Photo with their legs. Once the wax is the credit: Jennifer Tsuruda right consistency, the bees transfer it to the comb, where it is worked over by other bees to construct comb and cappings. Comb Comb-building workers determine how much comb will be built to accommodate worker brood and how much will be built for drone brood. They walk along the brood comb already constructed to evaluate the amount of drone comb that is present. If there is already sufficient drone comb available for the queen to lay in, they will produce worker comb; if not, they will build drone comb. Honey bees will rebuild or repair damaged comb as needed. One behavior that is associated with wax construction and repair is festooning. This is where inactive bees hang between combs, linking themselves together by holding legs with one another. Comb is often repaired beneath the bees that are festooning. In a hive, wax will darken over time, particularly in the brood nest, where it is littered with pollen, pupal cocoons, larval feces, and other debris. Wax also absorbs lipophilic (fat-loving) substances, which include pesticides, antibiotics, and other toxins. Because this litter and these contaminants accumulate in beeswax combs, it is a good idea to replace them regularly. Our recommendation is to replace the two oldest frames (or the two with the darkest combs) in each hive body with fresh foundation every year. A good recordkeeping tip used by many beekeepers is to write the year on the top bars of each frame to easily identify which are the oldest to rotate out. You can work these combs toward the edges of the brood nest so they are not full of brood when you would like to remove them. Honeycomb is commonly referred to as an architectural marvel: Its shape uses a minimum amount of material for maximum strength and storage space. The combs themselves are generally parallel and upright in accordance with gravity, with the cells tipped upward slightly to help

keep in their contents. Comb serves not only as a place for food storage and brood production but also as a medium for communication within the hive by transmitting vibrations. The nest interior is dark, so sound and vibrations are important for communication. Propolis Propolis is used to caulk and seal holes around the perimeter of the hive, protect the hive from moisture, and even cover/mummify intruders that cannot be removed (e.g., mice). Bees also add a thin layer of propolis to the ■ surface of newly constructed combs to provide structural support. Notice this layer of orange propolis on the photo of the comb to the right. In the nests of wild colonies, a layer of propolis often covers all the inner walls of the nest cavity. Propolis even provides the colony with antimicrobial protection, and research has determined that creating a propolis envelope around the nest cavity can actually help protect the colony from some Propolis used to seal the hive inner cover to pathogens. This is an important the top hive body. Photo credit: Bee Informed advance in understanding the Partnership function of what has long been considered just a sticky and unsightly annoyance! Propolis is collected by specialized foragers who carry home resin from a range of trees (poplar, aspen, cottonwood, pine, and others) in their pollen baskets. This resin is mixed with wax to create propolis. It can be sticky or stiff, depending on its origin and temperature.

READ

How a Nest of a Honey Bee Colony Is Organized — Location

and Foraging Area Honey bee colonies can and do thrive in many locations, including forests, cities, and agricultural areas, in part because honey bees fly so far from their homes to collect food. Estimates vary, but an individual forager may fly up to six miles from the nest, although they usually fly shorter distances when resources are available (one to two miles). Even with foraging trips extending out only two miles, the foraging area of a honey bee colony is more than 12 square miles, or over 8,000 acres. This flight range is important for management: It affects how beekeepers respond to pesticide applications, how many colonies they can place in one apiary, where and how far apart they locate multiple apiaries, whether they must feed closely packed colonies to supplement their forage, and so forth. Natural areas have long been recognized as prime locations to keep honey bees due to the variety of floral resources. Agricultural areas can also have excellent resources and yield bumper honey crops, but some can put the bees at risk for pesticide exposure. More recently, cities have proven to also be a good habitat for honey bees (and for many other wild bee species), although, of course, this can vary enormously. Cities tend to have high floral diversity from the presence of gardens, parks, green roofs, and community gardens; this and other factors have contributed to the rise in popularity of urban beekeeping. Beekeeping was legalized in New York City in 2010. Beekeepers in urban areas, however, need to be especially cognizant that their bees don't bother neighbors and passersby.

WATCH

How a Nest of a Honey Bee Colony Is Organized — Bee

Space We'll start out small to understand the organization of a honey bee colony's nest. Combs in wild honey bee nests and managed honey bee hives alike are built about $\frac{3}{8}$ " apart from one another, a dimension known as bee space. This leaves just enough room for bees to pass one another on opposite combs. If the open space between two combs is greater than $\frac{3}{8}$ ", then the bees will tend to

build comb (commonly called burr comb) to fill this extra space. If there is a space less than 3/8", bees will likely fill it in with propolis and won't use this space at all. Bee space has implications for beekeeping equipment. It prevents the combs from getting stuck to each other and to the hive walls, and therefore allows the beekeeper to inspect a colony without needlessly ripping apart the combs. Indeed, the invention of a movable-frame hive by Reverend Lorenzo L. Langstroth depended on understanding bee space. Before designing hives that adhere to the dimensions of bee space, it was nearly impossible to remove the combs to evaluate the health and productivity of the colony without destroying the comb. Being able to pull out each comb one by one without greatly disturbing the colony caused a fundamental change in the way people kept bees. It just goes to show how much beekeepers can learn by studying the biology of honey bees. The patent for the Langstroth hive was filed in 1852, and the hive design spread throughout North America and Europe, ushering in the modern era of beekeeping. Watch this video to observe how nests and hives of honey bees are typically organized.

Video Transcript Typically brood is raised in a spherical arrangement in the bottom and center of the nest surrounded by a thin layer of pollen and bee bread. Honey is stored around and above the brood nest, hence the name of a honey super. Super means over or above. When you pull all frames from a broods nest in a time when there's ample nutrition and ample brood production, you can expect to see ideal frames that have brood in the center with arcs of pollen surrounding the brood, and an arc of honey surrounding the pollen. Seeing frames like this always gives the beekeeper a good feeling that there's lots of resources coming in. Sometimes however, a colony can become honey bound. This is where a nectar flow is occurring, but there's not enough space for the colony to store it above the brood nest. In these cases, the bees who receive the nectar will deposit it in empty cells in the brood nest, thereby restricting the area where the queen can lay. The cells that are being filled with nectar are usually cells that the queen would lay eggs in. And this can reduce the amount of overall brood that's being produced and if there's not enough available space to continue rearing brood, the colony may swarm. Similarly, colonies can become pollen bound, if there's a very large pollen flow in your region. We'd recommend regularly checking in the amount of pollen or honey in your hives, and where they're being deposited, to make sure that bees have enough space. If your colony is pollen or honey bound, the beekeeper can make more space in the brood nest by switching out honey filled frames, with frames of drawn comb or foundation. The honey filled frames can then be moved to a higher box if those boxes hold the same size frames or they can be placed in a freezer and then thawed, and put back in the same colonies later in the season when food reserves are low. Reinserting them back into the same colony that they came from will help reduce the transmission of any disease spores that may be present. It's important to note that over time the organization of the hive changes. Bees move up vertically within their nest, so in summer and autumn the brood nest will be located in the bottom deep of the hive with honey stores above. But throughout the winter, the bee cluster moves upward to consume the honey stores. By springtime the clusters occupying the upper deep and there's virtually no stores or bees on the bottom deep. The colony will become congested if they've got no more room to grow vertically or to store honey. And this has implications for management. Bees tend not to store any honey below the brood nest so beekeepers will often switch the positions of the two deep boxes in spring to bring the brood nest back down to the bottom deep and giving them more storage space above. And this practice of rotating your deeps to ensure the brood nest is brought to the bottom is referred to as rotating hive bodies or reversing hive bodies. And it allows the bees to use their space in

a continuous, seasonal cycle. It's really important that you first check all the frames in the colony to make sure the brood nest isn't split between the two deeps. If that's the case, you can accidentally disrupt the brood nest when you're rotating hive bodies. And this will make it hard for the bees to keep all the brood warm and can result in chilling of the brood. The colony's vertical movement over time is also a consideration in winter. Honey stores should be located above the cluster going into winter and the beekeeper should remove the queen excluder so the queen isn't left behind as the cluster is moving upwards. The position of queen cups and queen larvae are also created in specific areas of the nest. So within the brood nest queen cups tend to be toward the bottom of the cones or frames and facing downward. Queen cups are present at most times during the year, but are most abundant in the spring swarm season. These special cups are called queen cells once eggs have been laid in them. Queen cells that are produced in preparation for swarming, called swarm cells, are often located at the bottom of the comb. The exceptions to this general rule are with supersedure cells and emergency queen cells. Supersedure cells are created while the original queen is still alive, but the workers would like to supersede her with a new queen. These cells are the most often found in the center of the comb although there are also exceptions to this as well. Emergency queen cells are built when workers abruptly find themselves without a queen and they must convey larvae that were originally intended to be workers into queens instead. And these cells can be located wherever there are worker larvae. The timing of the production of drone comb is just as important as its location. It's built only after the hive has become well established. The Cornell bee researcher Michael Smith calls drone comb construction a sign that the colony is undergoing puberty. That is, it's starting to invest in reproduction by beginning to produce drones. And these drones are helping the colony reproduce because they're passing on the colony's genes by mating with queens in the region.

READ

How a Nest of a Honey Bee Colony Is Organized — Nest

Characteristics Understanding a honey bee nest characteristics is very helpful for beekeepers. Let's explore the precise architectural elements that make up their homes. Size/Volume Wild honey bee colonies living in hollow trees occupy an average nest cavity size of about 10 gallons and range from five gallons to about 15 gallons. This is much smaller than a typical Langstroth hive, in which two deep brood chambers and one medium honey super amount to about 30 gallons. Bees in temperate climates tend to prefer the entrances to their homes to be small (an area only two to three square inches wide), south-facing, high (more than 15 feet off the ground), and positioned at the bottom of the nest cavity. These characteristics help the colony to defend its nest against predators and intruders, to regulate its nest Honey bee nest entrance high in a tree. Photo temperature, and to store and ripen credit: Thomas Seeley enough honey to survive winter in the cold climate of the northeastern U.S. Beekeepers can manipulate hive entrance size with simple tools like an entrance reducer. One colony will often continuously occupy a cavity and can do so for decades. Swarms will preferentially occupy a previously used cavity, indicated in part by ■ Entrance ■ Previous Use ■ the presence of comb. Bait hives with comb are more successful in luring swarms than those without.

READ

Hive Styles

The Langstroth hive (photo on left) is the style that we will be discussing throughout the course. There are many variants of the Langstroth hive, which are also popular in Europe, and you can order or build Langstroth hives that fit eight frames or ten. Some other hive styles that you may be familiar with include the top-bar hive (seen in the middle photo and sometimes called the Kenyan top-bar hive) and the Warré hive (in the photo at the right). The best hive style for your bees depends on factors such as climate, goals, and budget, but all hive styles and management regimes require regular inspections.

READ

Temperature Control in the Nest

Unlike most insects, honey bees are capable of generating heat by exercising their flight muscles, both during flight and when they are stationary inside their nests. Given the ability of individuals to generate heat, it is not surprising that a whole colony of honey bees is able to keep the center of its nest (the brood area) at a constant and elevated temperature, approximately 92°F to 97°F (33°C to 36°C). This ability to regulate the temperature of their environment is called thermoregulation. Thermoregulation helps the colony's brood to develop properly; temperatures outside of this range lead to developmental and behavioral abnormalities or failure to emerge as adults at all. This capacity for regulating the temperature of the nest has enabled *Apis mellifera* to thrive in a wide range of climates around the world, including the cold temperate regions of Europe and North America. Bees can regulate the temperature of their nests in two ways: passive regulation and active regulation. Passive thermoregulation refers to temperature control effected through the structure of the nest itself, while active thermoregulation is temperature control effected through behaviors performed by the bees. Bees expend energy performing active thermoregulation.

READ

Temperature Control in the Nest — Heating

Honey bee colonies use several techniques to keep the brood nest portions of their nests warm. This allows them to be active inside the nest and ready to fly in cool weather without first requiring the sun to heat the nest from the outside. This also allows them to survive and even rear brood when the outside temperature drops to frigid lows during winter. Passive strategies to maintain a warm brood nest include nesting in cavities whose walls act as insulation and shelter them from cold winds. The entrance is usually near the bottom of the nest cavity to keep in warm air (as hot air rises), and cracks are sealed with propolis to keep out cold drafts. Wax and honey that surround the brood nest also provide additional insulation. Active thermoregulation largely pertains to the incubating of brood. Heater bees cover the brood nest, and some may press their warm bodies against the capped brood, increasing the temperature much in the same way that a chicken incubates an egg. Others crawl into empty cells headfirst to heat adjacent brood from the side. In these ways, the brood is kept at approximately 92°F to 97°F. Beekeepers should take care not to open up the brood nest in cold weather, exposing and chilling the developing bees. We generally recommend not to expose brood to temperatures below 59°F, and to not expose brood to rain, which can cause them to become chilled and fail to develop. The most notable active thermoregulatory behavior that bees use to keep warm is the formation of a winter cluster. When ambient temperatures drop below 57°F, the colony clusters together as a group to stay warm. The brood nest is warmed, but the rest of the hive or tree cavity

interior is not. While the winter cluster is not technically in hibernation, many other colony functions, such as foraging and water collection, cease when bees are clustering. Photo: The winter cluster. Bees on the inside of the cluster generate heat by shivering. They contract and relax their flight muscles without flapping their wings. At the center of the cluster, bees can move around, tend to brood (if there is any), and eat honey. Bees around the edges of the cluster act as insulation to prevent the heat from dissipating. Bees take different roles in the cluster at different times, moving from the center to the edge and back over hours or days. If there is a respite in the cold, individual bees may leave the hive for brief cleansing flights, often their only opportunity to defecate for several months. The cluster as a whole will also move upward to the honey stores when the temperature is warm enough to give them access to food. If the temperature stays too cold for the bees to break the cluster and move, the bees can starve despite there being plentiful honey stores in the nest. Finally, an interesting note is that the heating behavior of bees can be lethal to predators. The Asian giant hornet is a predator of honey bees in Asia, and bee stings do not penetrate the hornet's cuticle. Instead, bees will perform balling behavior as defense instead. They crowd around individual hornets and heat them to death! Honey bees will overheat the Asian Giant Hornet to kill it.

WATCH

Temperature Control in the Nest — Cooling

Honey bees have also adapted several techniques to help cool their nests on hot days. Watch the video to learn some of these strategies. Video Transcript On hot days, bees also need to be able to cool the hives so that it doesn't exceed a rough ceiling of 97 degrees. Above that point, the heat can kill the brood and adults and can soften beeswax combs. Most cooling of the colony is active, the main passive mechanism is the ventilation that's achieved through the entrance of the nest and any upper entrances provided by the beekeeper. Bees use a variety of active methods to reduce heat. First, just as clustering helps the bees keep themselves warm, dispersing helps them cool the colony. So on warm days bees will actually move away from the brood to other areas of the hive, to help dissipate the heat. Bees can also fan their wings to increase ventilation within the hive. They gather along the inner walls of the hive and along the entrance and fan their wings to expel warm air from inside. The motion of the wings is actually different from when they're flying. They are moving air horizontally, as opposed to vertically. Fanning bees face the hive entrance and point the tips of their abdomens downward, and then fan their wings. This is in contrast to when they're fanning the nasonov pheromone to orient other bees to the entrance. In this case, the abdomens are pointed upward. So fanning actually helps create air currents that bring in cool air to circulate through the hive. If fanning alone doesn't cool the nest adequately, forager bees collect water for evaporative cooling. They pass the water off to water receiver bees who place it on the cappings of brood cells or in open cells. Fanning with their wings then evaporates the water and cools the nest. If the temperature continues rising in the nest, more bees will turn to receiving, which signals to water foragers that they need to increase their water collecting and recruit even more bees to fetch water. Bees can only collect water during the daytime, so during especially hot nights it might be difficult for them to control the temperature inside the nest. Finally, the worker bees may beard on the outside of the hive, where they literally hang out on the front in a beard shape. Leaving the nest helps increase ventilation and also reduces the amount of heat that's generated by workers inside the hive. Bees may also beard after miticides are applied or if a fume

board is inserted as the bees are making an effort to escape the chemicals. But it's most commonly seen when the colony is overheating. Honey bees are resourceful organisms that have adapted a variety of behaviors to help them survive in temperatures that exceed the thermal range necessary for brood development and adult survival. Additionally, bees in the wild carefully select nest sites that will help with thermoregulation. These passive and active strategies have allowed them to be successfully managed nearly everywhere in the world.

MODULE WRAP-UP

How Honey Bees Construct and Maintain Their

Nests In this module, you learned about the intricate ways honey bees build and maintain their homes. Your investigation of their preferred nesting characteristics and colony maintenance behaviors revealed how these insects create environments that support healthy development and colony success. Back to Table of Contents Thank You and Farewell Congratulations on completing "Master Beekeeping Part I: Honey Bee Evolution, Biology, and Behavior." I hope that you now have a deeper understanding of how honey bees evolved and why they behave the way they do. The material covered in this course serves as a foundation to understand and Emma Walters apply the practical, hands-on aspects of beekeeping. I hope it has Honey Bee Extension Associate Dyce Lab of Honey Bee Studies met your expectations as you support your colonies' survival and Cornell University success. From all of us at Cornell University and eCornell, thank you for participating in this course. Sincerely, Emma Walters

Downloadable Resources You may wish to download and save these files for future reference. The Stages of Development Worker Polyethism Kin Selection Bee Anatomy Recommended Books There are more books written about honey bees than any other animal on the planet, besides humans. With so many to choose from, we've compiled a list to help guide you in your readings. There are no required readings for this course; all examination questions will be based on the online material only. If you are interested in delving deeper into topics we've discussed, here are some great resources: The Evolution of the Honey Bee Colony as a Superorganism "The Buzz About Bees: Biology of a Superorganism" (2008) by Jürgen Tautz "The World History of Beekeeping and Honey Hunting" (1999) by Eva Crane "Honey Bee Ecology" (1985) by Thomas Seeley "The Social Behavior of the Bees" (1974) by Charles Michener Wild Bees "Bees: An Up-Close Look at Pollinators Around the World" (2015) by Sam Droege and Lawrence Packer "A Sting in The Tale: My Adventures with Bumble Bees" (2013) by Dave Goulson "The Bees of the World" (2000) by Charles Michener How Honey Bees Communicate "Honey Bee Democracy" (2010) by Thomas Seeley "Pheromones of Social Bees" (2000) by J.B. Free "The Dancing Bees" (1953) by Karl von Frisch Honey Bee Biology "The Hive and the Honey Bee" (2015) by L.L. Langstroth, et al. "The Lives of Bees" (2019) by Thomas Seeley "Honey Bee Biology and Beekeeping" (2013) by Dewey Caron and Lawrence Connor "Bees and Beekeeping: Science, Practice, and World Resources" (1990) by Eva Crane "The Biology of the Honey Bee" (1987) by Mark Winston

Part II: The Science and Art of Beekeeping

Safety and Biosecurity in the Bee Yard

Keeping yourself and your visitors safe in the bee yard is a top priority. Honey bee stings can cause a wide range of reactions, some of which can be life-threatening, and learning how to prevent or respond to them will help you enjoy beekeeping for many years to come. The safety of your bees is also a priority. Following biosecurity practices minimizes the spread of infections, parasites, and predators, helping keep your colonies healthy. In this module, we will learn how to keep both you and your bees safe in the bee yard.

READ

Preventing Stings

■ Beekeepers are liable to get stung at least Key Points occasionally. By now, you probably have a few Accept that some stings are pictures of your swollen hands or face! If it inevitable. weren't for their stingers, most bee colonies Protective equipment is a personal wouldn't get to keep so much of their honey choice requiring a balance of risk from predators and intruders, and they could and ease of movement. starve to death. Communicate with potential As best you can, let those stings tell you visitors that there are honey bees something. Colonies that sting a lot relative to in the area and provide visitors other colonies may be experiencing extra pest with appropriate protective pressure, or need to be moved, requeened, or equipment. even culled outright. If all your colonies sting a lot, you may need to retool your inspection style: how you do it, when you do it, or even what you wear. Stinging is something to be avoided, but not beyond reasonable measures. If you don't let yourself accept that some stings are likely inevitable, then you probably won't get comfortable enough to really know and understand your bees. Protective equipment to prevent stings in your own bee yard is a matter of personal choice. It's a balancing act. On the one hand, you should not endanger yourself or others just to prove that you're capable of handling thousands of stinging insects without protection. On the other hand, you shouldn't insulate yourself so much that you are clumsy or immune to any ill effects that you are having on your bees. Different veil styles offer different levels of convenience and protection. It's important to have a veil handy even if you don't use it, for the rare occasion that a hive suddenly becomes too defensive to handle easily. Gloves are a matter of personal choice- some, like nitrile gloves, are thin enough to permit more tactile sensation; others, like kidskin or rubber gloves, will protect you better, but at the cost of your dexterity. Cost and waste are also factors. Other, low-tech protective clothing measures abound. Some beekeepers tuck their pants into their socks or stretch rubber bands around their pant legs to prevent bees from crawling up into their clothing. As a Master Beekeeper, you may be bringing new beekeepers or members of the public to your bee yard. Many people unfamiliar with honey bees are afraid, and some may have unexpected reactions to stings if they have not been stung before. Even if you do not generally use it, you must always have protective

clothing available for visitors — a veil, gloves, a suit, or clothing with long sleeves and pant legs — and use your best judgment about your visitors' tolerance and comfort around bees. You should also exercise safety precautions — again, more than you might generally — to reduce the chance of stings. Avoid inspections in poor weather, when a hive is shaded, or during a nectar dearth. Use adequate smoke and instruct your visitors not to block the hive entrance. Pay attention to warning signs from your bees, such as pinging your veil or louder buzzing, and close your colony for another day if you see any of these signs. Lastly, post a sign at your bee yard(s) informing readers that live honey bees are in the area; this should include a contact phone number for the public to be able to contact you. Many beekeepers keep their bee yards out of sight, especially from roadways, to limit their view from the public.

READ

Bee Sting Reactions

■ There are two general types of reactions to bee Key Points stings. Be careful about restricting blood flow to the site of the sting. Local Reactions Local reactions at the sting site will usually hurt Systemic reactions occur within for a few minutes, with milder pain for about 10 minutes to an hour of being one day. The area may be swollen, red, and itchy stung and affect the body well for two to three days, with the swelling beyond the site of the sting. increasing after the first day before it decreases. Age is an important factor for bee Taking oral antihistamines, such as Benadryl, sting allergies. and applying ice to the sting site will help reduce swelling. In rare cases, numbness at the sting site can occur, even for an extended period. Be careful about restricting blood flow to the site of the sting; if you are stung on the hand, you may want to remove any rings on your fingers immediately afterward. Systemic Reactions Systemic reactions occur within 10 minutes to an hour after being stung and affect the body well beyond the sting site. A variety of reactions can occur, either on their own or in combination: a large rash, hives, swelling at sites other than the sting site, dizziness, fainting, wheezing, vomiting, and abdominal pain. Anaphylactic shock, or anaphylaxis, is the most severe reaction, where the tongue swells, and the throat begins to close, making breathing difficult and possibly leading to a loss of consciousness and respiratory failure. Call 911 for immediate medical attention if you see or experience any of these reactions. Emergency responders can provide epinephrine (adrenaline) to the patient. At the hospital, other treatments may include steroids to reduce inflammation and inhalers to open the airways. Anaphylaxis is caused by a chemical overreaction in the body's immune system in response to an allergy. A person typically does not have an allergic reaction the first time they are stung; the first sting stimulates the immune system to produce specific antibodies for an allergen in the venom. After the second sting, these antibodies recognize the venom allergen and bind to it. This, in turn, stimulates white blood cells to release histamine and other chemicals, which then cause anaphylaxis and associated symptoms, such as swelling and an inability to breathe. Bee sting allergies can be unpredictable. Many people with a history of systemic bee sting reactions do not experience severe reactions the next time they are stung; however, they may again in the future. One important factor is age: bee sting allergies are more common in adults than in children, 1% of children and 3% of adults. Adults over 40 years old are more likely to experience life-threatening reactions and die than children. Bee clubs or universities with apiculture programs generally do not carry epinephrine auto-injectors, such as EpiPens®, or generic alternatives, in case visitors are stung and have a

systemic reaction. This is because EpiPens® require individual prescriptions. The wrong dose — e.g., an adult dose for a child — or an unnecessary injection can cause serious harm. As a beekeeper, it's a good idea to talk to your doctor about getting your own EpiPen®. Even if you don't have a history of systemic bee sting reactions, an anaphylactic allergy can develop at any time. It's a good safety measure to keep an EpiPen® on hand when you are beekeeping, for the rare instance you may need it.

READ

Hygienic Management Practices

As a Master Beekeeper, you should follow hygienic management practices that reduce the risk of disease transmission and pests in your operation. The social nature of bees usually helps them thrive, but some characteristics of social living may also make them more vulnerable. Bees may drift from one colony to another, or rob honey from a weak colony, putting them at risk of catching whatever illnesses are present in that colony of bees. Robbing or drifting can spread parasites and pathogens, including Varroa mites, viruses, Nosema, and American foulbrood. Spacing apart your colonies and placing them near landmarks so bees can easily identify them will help reduce drifting. If you keep your hives on pallets, having each face a separate direction can help as well. Wash your hands and sterilize your hive tool between bee yards to prevent pathogen transmission. If a colony is visibly infected with disease, clean your hands and hive tool before moving on to the next one. To clean your tool, scrape off propolis and wax, then immerse it in a disinfectant — alcohol, bleach, or a still- burning smoker (be careful not to burn yourself when you take it out!). Colonies infected with American foulbrood require immediate attention, and your first line of defense is preventing the spread of disease. Any hive tools used in infected hives need to be scraped clean and disinfected thoroughly with alcohol or bleach — though these solutions do not kill American foulbrood spores, they can dilute them. These hive tools should then be inserted into a still-hot burning smoker or fired with a blow torch. Keep your bee yards clean. Leaving burr comb, exposed frames, or equipment in the yard can attract other pests and stimulate honey bees to rob. Clean your protective clothing regularly as well by throwing your suits and jackets into the washing machine. Some veils need to be removed from the suit and hand-washed separately. If you are visiting someone else's bee yard, do not bring your hive tool or other equipment; use theirs instead. The same logic goes for when you have visitors in your yard. If you have staff or helpers in your operation, train them on safety, hygiene, and emergency procedures.

WATCH

Robbing

Robbing is the name for honey bees collecting — or stealing — honey or nectar from another colony instead of from flowers. It most commonly occurs when flowers are not producing much nectar. It is especially pronounced in late summer or early fall when colony populations have grown, and there are lots of foragers out looking for food. The smell of honey, particularly from an open hive, attracts foragers. Weak colonies are a relatively easy target to get some sweet food. Robbers will sneak in the entrance, or fight the guard bees to get in, quickly rip the cappings off the stored honey, slurp up honey, and fly back to their colonies. They will communicate to their sisters the location of the great food source they found, and their sisters will come to check it out themselves. In a short period of time, you

can get a lot of robbers from nearby hives targeting the weak colony. Robbing can happen gradually over a few days, or it can escalate within a few minutes or hours to kill an entire weak colony. Watch this video to learn how to identify robbing behavior. Video Transcript When your colonies are getting robbed, this is something that must be dealt with immediately. Because there's negative outcomes for both colonies; the weak colony that's getting robbed, and the stronger colony that's doing the robbing. So the weak colony that's getting robbed is going to lose their food stores, and part if not all of their adult bee population. An extensive robbing can actually wipe out an entire colony altogether. The stronger colony that's doing the robbing is also going to be at risk for developing certain infections. And this is because robbers have to walk across the brood nest in order to get to the honey stores. And when they're walking across a brood nest, they might come in contact with spores, bacteria, mites and other infectious agents that they're bringing back to their own hives. And the reason why this is likely a problem with robbing is because weak colonies might be weak because they have certain parasites and pathogens. So how do you know if a colony is getting robbed? What are the signs and symptoms? Robbing bees actually have a very characteristic behavior to them. There is usually frenzied activity at the front of the entrance, and also the robbers have this kind of exploratory behavior where they're hovering around the entrance, swaying to and fro, looking for a place to enter and sneak past the guards. And they won't just be hovering around the front entrance, but they'll also be hovering around any cracks or any outer cover, anywhere where they're attracted to the smell of honey that's being exuded from the colony and looking for additional entrances to access it. So once a robber lands at the entrance of a hive, ideally, they're going to be detected and attacked by guard bees. So the guard bees are the ones that are patrolling the front entrance looking to see if the intruders are coming in. And they can tell when a honey bee is not a nest mate because they have unfamiliar odor and usually strong colonies can do. The more populated the colony the more guard bees are present and the better protected they are against robbers. But, weak colonies that don't have a very big population also don't have many guards to patrol the entrance. So, when a guard bee detects a robber, they release an alarm pheromone, which stimulates the older forager bees to exit the hive and sting the intruders. And this fighting at the entrance is noticeable. This is a major symptom of hives that are being robbed. You'll see bees grappling and stinging each other. Also within an hour or maybe a few hours, you'll start to see the buildup of wax cappings on the bottom board of the hive and also on the ground in front of the entrance. You'll also start to see dead bees that are accumulating on the ground, the bees that are dying in fight essentially, and also in front of the entrance. Another thing that you might notice on your colonies are dirty light brown footprints. And these are the footprints of the robbers that are landing on the entrance of the hive and also on the front entrance as they're tracking out honey. And these footprints are noticeably different than bee fecal matter, which is more of an orangey or dark brown color, and more in a drip shape than an actual footprint shape. But these footprints are really noticeable. They're more the color of honey because they're literally honey that's being trailed out of the colony. So the reason why robbers are leaving cappings and dirty foot prints behind is because they're in a hurry to escape with their honey. If you go into a colony that has been robbed, you might be able to tell, because you'll notice that the cappings are ripped off of the honeycomb. And you may also notice a reduction in population size from the last time you went and inspected the colony. But it's important to note that not all robbing isn't this frantic, escalating war that ends in colony death. Sometime robbers are just gradually taking honey from another hive and the can go undetected by the bees in that hive

and also by the bee keeper. And the way that this works is that the robber bee will sneak past the guards the first couple of times that they go in the hive, they manage to get past them and the more they're entering the colony, the more they're starting to acquire the colony odors, and the less the guards are detecting them. And over time, they eventually come to smell as nest mates, they're not attacked by the guards at all. And they become these chronic robbers, stealing a little bit of honey here and there. So let's take a look right now at this film clip, where we can see one of my poor colonies that got robbed at the end of summer and start to identify what these symptoms are. Okay, so while we're looking at this hive we see that there's a lot of frenzied activity. And if you're just walking past you might think there's just a strong honey flow going on, there's nothing too serious happening. But, when we look closer we'll start to notice that we are seeing fighting at the entrance. Look closely and you'll see a few groups of bees that are actually grappling. They're biting each other. They're stinging each other. And here we get a closeup where we can really tell. There's a couple bees totally fighting it out right now. And you can even see this yellow jacket who is a little opportunistic in thinking, there seems to be a lot of distractions going on and this colony is weak. Maybe I can sneak in and steal a little honey for myself. On the front of the hive entrance, here, you'll notice some of the fecal matter that I talked about. You can see the characteristic drip, but I want you to differentiate that from the dirty footprints that you're seeing on the entrance here. This is after a few hours of robbing and actually this colony didn't end up making it. And after a few hours you'll notice there's a pile of dead bees on the ground and these are the bees that essentially died in combat while trying to fight off the robbers. So those are the characteristic symptoms of a colony that's really being robbed extensively.

READ

How to Prevent Robbing

Robbing behavior can increase a colony's susceptibility to parasites and diseases. Robbing is a relatively common occurrence during a dearth, but beekeepers should try to minimize their colonies' susceptibility to robbing. Colonies that get robbed are usually weak to start with, with fewer guard bees to defend the entrance than strong colonies. Apiaries with many colonies or where colonies are kept close together are more susceptible to robbing behavior. Robbers can quickly smell honey and access weak colonies more readily than they would in their nests that are spread out in the wild. Keeping the colony population strong and healthy is best achieved by ensuring there is a fertile queen present, providing bees with adequate nutrition, and maintaining low parasite levels. When providing sugar syrup, be mindful of how you are doing so. Having open feeding areas close to hives or Boardman feeders can attract other bees to the colonies being fed and promote robbing behavior. We generally recommend an inside feeder, especially when feeding bees during a dearth. If you prefer open barrel feeding, place your feeding barrels a couple of hundred yards away from your bee yard to divert robbers away from your colonies and to the feeding station instead. If you need to inspect colonies during a dearth to do necessary management, keep your inspections brief. While in the hive, try not to open capped honey with your hive tool or as you pull out a frame. In addition to the risks to the colony, bees that have experienced robbing can be more aggressive in the following days or weeks. Reducing the potential for robbing will help keep both your bees and you safer.

WATCH

How to Stop Robbing

What if, despite your efforts, some of your colonies end up getting robbed? It is important to act fast. If the robbed colony is quite weak or if you don't intervene within a few hours, it can die in as little as one day because of the loss of life to fighting. Colonies that have survived being robbed are more likely to be defensive in the following weeks. Use caution when opening up these hives. Watch this video to learn what measures you can take to stop robbing once it has started. Video Transcript So how do we go about stopping robbing? What do we do if we are witnessing these symptoms in our own colonies? There's a few things that we can do. So to deal with robbing the main thing that we want to do right away is to reduce the entrance to the colony to the size of one or two bee bodies. So this is about half an inch in width. And the idea is that this opening is big enough for the colony's foragers to be able to enter and exit. But small enough so that the guards can easily defend it. And using a wooden entrance reducer is fine but ideally it would best if you could use a mesh screen. So if you could staple a mesh screen across the front entrance leaving just this tiny opening in the middle for the guards to defend, this is still going to allow ventilation to happen through the hive. And, also, it's not directing robbers directly to that small entrance where they're able to smell the honey odors. Instead, they can smell the honey odors throughout the entire entrance and they are more distracted. And this gives the opportunity for the foragers to be able to enter and exit out of that small hole that they can easily learn. Alternatively you can purchase a robbing screen from your local bee supply company to use, and these screens can be left in place for the entire period of a nectar dearth to prevent robbing and they have a similar idea. They distract the robbers from finding the true entrance to the colony. It might also be beneficial to place a wet sheet over the colony, something like a bed sheet, and to do this for one to two days. And you want to completely cover the entrance and drape the sheet all the way down to the ground. So here, the resident bees are still able to find their way underneath the sheet back into the colony, but the robbers are directed away from the colony entrance, and they're directed away from any cracks that may be exuding honey odors in hopes that in a day or two they'll move onto another food source. The main consideration with this is that having a bed sheet over the hive can put the colony at risk for overheating. So we recommend that you use a light colored sheet instead of a dark sheet and that you re-wet the sheet if it dries out so the colony doesn't get too hot. And don't leave the sheet on for more than a day or two. And finally, if these two measures using an entrance reducer and using a sheet don't stop the robbing behavior, then you might just want to move the hive entirely to a new location, away from the apiary at least two miles away if you can, and if you have the land to do so. And this is something that you might want to consider from the get go if you already know that the colony is weak and will most likely die from being robbed. Sometimes, having the entrance reducer and the wet sheet just still proves futile and moving the colony is the only way to save it.

READ

Minimizing Competition Between Your Hives and Native Bee

Species The Western honey bee (*Apis mellifera*) was introduced to North America in the 1600s, and it is now considered livestock, like cows and sheep. In contrast, native bees are wildlife. There are ~4,000 described bee species in the USA, and ~450 bee species in New York State alone. When you place managed honey bees in a landscape, you're adding a large, social, efficient, high-density foraging organism into the environment that includes thousands of foragers per colony. What are the

impacts on the native bee fauna? Several studies in the last 10 years, conducted in the USA, have looked at the possible competition between managed and native bees for flower resources (nectar/pollen). While some studies found evidence of such competition, others did not. Another impact explored focused on disease spillover, the sharing of honey bee pathogens/parasites with native bees. While many native bees were shown to acquire honey bee pathogens when visiting flowers previously visited by honey bees, there is no conclusive evidence that those pathogens/parasites are detrimental to those native bees. Even though no clear detrimental impact has been reported from having hives next to native bee populations, most experts agree that these effects are difficult to study and document; thus, the possibility of negative interactions between managed honey bees and native bees remains a concern. Here are practical actions to reduce competition for flower resources and disease spillover. 1. Place your apiaries in agricultural or human-dominated landscapes such as orchards, mixed farmland, suburban gardens, and roadside plantings. 2. Provide or enhance forage by planting high-nectar, high-pollen patches near the apiary to absorb some foraging pressure. You could also include native wildflowers, so native bees also benefit. 3. Reduce the likelihood of disease spillover by maintaining strong colony health because healthy bees shed fewer pathogens.

MODULE WRAP-UP

Safety and Biosecurity in the Bee Yard

In this module, you gained essential knowledge about safety in the bee yard, both for humans and honey bees. Your study of sting prevention, reaction management, and biosecurity practices equipped you with vital skills to protect yourself, your visitors, and your colonies. These fundamental safety protocols form the foundation for successful, sustainable beekeeping.

Inspecting the Colony

Colony inspections are key to managing healthy colonies. Regularly monitoring colonies gives you information about their population dynamics, the condition of the queen, whether and when to treat for pests and diseases, and when to add supers for honey production. In this module, you will learn how to perform a colony inspection with all this information and more.

READ

Preparing to Inspect Your Hives

Your motivations for keeping bees will directly impact how often you inspect your colony. If you are interested in learning about and observing bees, you may want to inspect more often, but remember that frequent disturbance to the colony may unnecessarily disturb the bees or set back their productivity. We recommend inspecting your colonies and recording your observations and management activities every 10 days to two weeks in the spring and summer, less in the fall and winter. This cycle of inspection can allow you to spot and remove swarm cells and identify any disease issues in an early state. This inspection frequency allows you to monitor population dynamics, including swarming; identify disease or pest issues in an early state; evaluate your colonies' nutrition; and more. This schedule is more frequent than in the past because honey bees face more intense parasites and

other pressures today. If you have too many colonies to inspect that often (over 50 or so), aim to inspect a representation of the colonies in each yard every time to get a snapshot of your operation. For instance, if you run 1,000 colonies, aim to inspect 100 colonies (10%), randomly selected. In this case, make sure to at least inspect the entrances of all the hives in the yard to see if anything is obviously wrong that requires further investigation. Have an objective For each inspection, you should have a clear objective before you open your hives: Are you monitoring for diseases? Checking whether you should add supers? Evaluating the colony's nutrition? Having a specific purpose will help you plan which equipment to bring, prepare you for how long you will need to keep the hive open, and help you stay on top of your management tasks. Keep a stocked toolkit so that you are prepared anytime you see something unfamiliar, unusual, or worth documenting. Some items worth keeping in your toolkit — beyond your smoker, fuel, matches/lighter, hive tool, and protective gear — are the following: Notebook or record-keeping binder Pen or pencil Queen cage and marking pen ■ Stock your toolkit ■ Disease ID handbook, such as Penn State's Field Guide to Honey Bees and their Maladies Disease sampling equipment: quart-sized ziplock bags, Sharpie® marker, toothpicks, a bottle of rubbing alcohol, at least 70% Vita® American foulbrood or European foulbrood test kits Camera or a camera phone to record interesting things There are many other useful tools you can implement in your inspections that you may decide to use throughout your beekeeping endeavors. Beekeeping catalogs are full of products that promise to make inspections easier: by easing the weight of heavy supers, holding frames, etc. Some are very useful; some aren't. Ask around before purchasing anything expensive or that you haven't used yet. Check the weather Before going into the apiary, check the weather to make sure it is conducive to working bees. Consider checking several days in advance so you don't miss a crucial weather window to get something done in the hive. By now, you should be familiar with the best weather conditions to inspect a colony: sunny skies, little wind, and temperatures above 59°F (15°C). Opening the colony in cooler temperatures can chill the brood. In cloudy weather, more foragers stay inside the hive since they are unable to use the sun to orient themselves while flying. Foraging bees are more likely to sting than their nurse bee sisters. You'll likely notice their increased defensiveness when inspecting your colonies while more of them are home. ■

READ

What to Look For When Inspecting

A colony inspection should be a quick procedure, taking about five to ten minutes. Click on the tabs to get an overview of the factors to consider in assessing the health and productivity of the colony.

Population Management

1. Population: How many adult bees are in the colony? Is the number appropriate for the season?
2. Queen status: Is the queen laying worker eggs? Is there evidence that the bees are intending to supersede her or swarm? Is there evidence of laying workers?
3. Brood quality and quantity: Is the brood solid or spotty? Is it sufficiently abundant for the time of year? Is brood of all ages present — eggs, larvae, and capped brood?
4. Population dynamics: If you compare the apparent brood rearing activity with the natural attrition rates of foragers at this time of year, does the population seem to be increasing or decreasing?
5. Drone population: Is drone production seasonally appropriate? Are there a disproportionate number of drones?

Nutrition

1. Floral sources: What flowers are blooming in your area at the moment, and what pollen sources are your bees bringing in? Are there environmental conditions that may be impacting the bees' nutrition, e.g., prolonged bad weather or a

drought? 2. Foraging behavior: Are there sufficient numbers of bees foraging? Seeing 10 to 20 bees entering with full pollen baskets per minute is an indication that lots of resources are coming into the colony. Are bees bringing in pollen or propolis in their pollen baskets or nectar/water in their honey stomachs? 3. Food stores in the colony: How much pollen/bee bread and nectar/honey is stored? Is this amount appropriate for the season? 4. Larval nutrition: Are the young larvae floating in abundant pools of royal jelly? Pests, Parasites, and Diseases 1. Do you see any symptoms of brood disease in the colony? Not all problems will be visible, but brood that is discolored, twisted, melted-looking, partially cannibalized, or has punctured cappings may indicate a problem. 2. Do you see any adult bees with deformities or behavioral abnormalities? Do you see any Varroa mites or pests such as hive beetles, wax moths, or the larvae of beetles and moths? It is important to note that for Varroa, without sampling procedures that follow specific protocols, such observations alone do not necessarily provide a full picture of the mite pressure within a colony. 3. Do you see other pests — or evidence of such pests — present in the honey supers or other areas of the hive? 4. Are the bees acting more defensively than usual?

WATCH

Walk Through a Hive Inspection

To illustrate a colony inspection, the well-known California beekeeper and bee researcher Randy Oliver from ScientificBeekeeping.com will take us through the steps. He will start by inspecting the outside of the hive and then pull out a few frames from each box to get an idea of the status of the general categories we outlined. We recommend pulling two to three frames from each honey super and at least four frames from each brood box to gauge the health and productivity of the hive. For a quick inspection, start with the highest box and work your way down. Starting at the top, quickly checking some of the frames, then working your way downward, helps reduce aggression because you are not opening and moving all the boxes from the start. Working this way can help minimize overall disturbance to the colony and the number of bees flying about. While you are working through the hive, you will inevitably squish some bees. Try to squish as few as possible, and always try to be mindful of the queen's presence to avoid accidentally hurting her. Move slowly and quietly to reduce defensiveness, as bees can react to quick, jerky movements. Watch this video to see Randy Oliver conduct a colony inspection. Video Transcript Okay so I'm coming up to this hive right now. First thing I do is I look at the entrance. I would look for bees on the ground here. First thing you look for bees with deformed wings which may be walking away from the hive. That would tell you you have a high viral overload in your hive and that deforming virus is going epidemic. I don't see any dead bees here, that's a good thing here. They were flying a minute ago. This pollen load's coming into the hive, which indicates most likely we have a queen inside this hive. And I can't see any right now but there are, when you look at there was a pollen load coming in, looking at the foragers coming back you can see that the abdomens of the foragers are distended. That the plates of their abdomens are pulled apart. That means that they're getting nectar out there. Okay so there's a pollen forager right there. That pollen forager got very little nectar on, because you can see that its abdomen is still shrunk up right there. I've been seeing other foragers coming back either with or without pollen with their abdomens more distended indicating that there's nectar inside their crops. The reason I'm looking at this, it tells you before you start what you expect to see in the hive and how the temperament of the colonies are

going to be. If you have a lot of the nectar foragers coming back to the hive, the colonies going to probably be gentle as pussycats. If they're all coming back instead with their abdomens shrunk up and you see bees acting as guards on the entrance, then you need to put on a veil, put on more gear, make sure your smoker is going very well. I don't see any guards at all on the entrance right now. If I did, you'd see bees standing with their two front legs up and their wings cocked back out at 45 degrees, rather than over their back. And their heads would follow all of my movements, so if I move my hand here you'd see bees follow that movement. If you see guards like that, that means there's a nectar dearth on or there's robbing going on or there's wasps or something bothering the hive and that colony is going to be far more defensive and the beekeeper should be aware. They're going to have to be ready for a defensive colony. But there's no indication right here that this colony. Okay, see that bee right there, that's a guard. I don't know if you saw his behavior right there. Okay, when I moved my finger his head followed my finger movement, his wings were out, and he flew out and investigated my finger, or she did. No other, a forager bee would not be interested in that at all. Okay so what we're going to do now, we're going to smoke the colony a little bit. We smoke in order to change the behavior of the guard bees. Now, it's not necessary to drown a colony in smoke, smoke is toxic to bees, it's very disruptive to the colony. After you smoke a colony, they stop making honey for a day. The only place where there's guard bees is near the entrance and on the periphery of the clusters, so you don't have to blow smoke into the center of the cluster. Now when you smoke a colony make sure your smoker's going well first. White cool and dense smoke and then you only need to put a small amount of smoke on the entrance just enough to disrupt the guards. So usually just one little puff like this across works. But notice if I smoke right here the smoke does not go in. The bees are exhausting air on this side. If I smoke right there, the smoke went into the hive. So be aware of, make sure the smoke, actually, a small amount goes in the hive. But it just takes a little bit right there. Now, we'll go up to the top of the hive. And with no vibrations, I'm going to very gently lift this lid up and put a puff of smoke under the lid. And I pick the lid up carefully without making any vibration to the hive, put the lid upside down on the ground. This is a place where you can put your boxes without having to pick up grass or rocks on them. Now use your hive tool. You want to be careful as you break the propolis that you don't want to make a lot of vibration to the hive, a lot of snaps. And again, a little smoke underneath, using the lid or the inner cover to direct the smoke down into the hive. What we have on here is a honey super on the top. Now the last beekeeper, who put this box in here, did not put the frames close together. You see the spacing right here, on the end bars? These are called Hoffman self-spacing frames - this spacer right here, and they're for a purpose. If you put your frames close together now you have bee space right here. If you leave them like this you have more than bee space. It is not so critical in the honey supers but in the brood nest I like to maintain the bee space. This hive right here, the way these frames are arranged, the next beekeeper to come in and after these bees have filled these spaces full of propolis, it is almost impossible to get a frame out. So the beekeepers have set themselves up for a problem. When we close a hive up I always shove all the frames over to one side and then back, like this, making sure they're very tight. And I leave the bee space in each one of these four corners right here to make sure that when I come back in the hive, I can add these 3 eighths of an inch space up to get a full 3 quarters of inch of clearance to pull my first frame out. I watch to see if the bees are aware of me, if they are looking at me. If they're looking at me, they're aware of me. If you want to be able to do inspections without wearing a lot of gear, you have to make sure the bees are never aware of you. So the key thing

is, if they're looking at you, they're aware of you. If you're seeing faces looking at you, put a little smoke on them and turn those faces around. In a second it'll just be tails. You don't have to smoke down into the cluster, just on the edges because the only place the guard bees are going to be is on the periphery right here. Here we'll take a look at another frame here now. Okay, here's a frame of larvae here. That right here what you see, see the difference in the larval sacks, can you see that there's large larvae and small larvae there? That tells me that the queen laid some eggs and the eggs were likely removed by the bees or the larvae died. You should see an even progression of age so that's an indicator to look a little closer to figure out why you would have larvae of different ages side by side. This is a center frame in the brood box. There's no pollen band at the top, but that's typical in a center frame. Okay, this sign right here, this arc of even egg laying and the solid brood and then filled in with fresh young larvae, this indicates a very nice laying queen. So just at this point, this look right here tells me everything I need to know about the queen. I don't know anything else about the queen. This tells me right now, that's a great laying queen in this colony. Now, I can look over, if I want. And at this point, I'd be done with my inspection, this would be all done. I've already confirmed that it's a good queen. I don't see any major signs of disease. I see good colony nutrition. Close the hive up and be done with it. Turn these guard bees away. Or those aren't guard bees, they're bees at the entrance. This is a much stronger colony than the last one we were in. Get a little smoke each time you crack a box, immediately right at that point. You want to prevent the bees from going into defensive behavior before they do. So that super is about half full of honey. You can just tell by weight as you pick it up. I pry everything nice and slowly, to avoid making, this super is full of honey, plenty of weight to it. Okay, so I see a few faces looking at me here. I'm going to go ahead and turn those faces around. You can see how immediately the bees are turned around. Now I see tails instead of faces. I'm not seeing any white wax being produced right now. So there's not a nectar flow on, a strong one at this time. So the honey that I felt in those supers was honey put on previously. Okay beautiful laying pattern right here. I see solid brood right here, I see a step down pattern of the larvae. Tons of jelly around those larvae. Look at them, they're just swimming in jelly. Very high colony nutrition indicating plenty of pollen coming in. Nice band right here, falling apart a little bit but not a big deal here. I'm looking for any kind of anomaly here. Again, always look for the anomaly. I don't see any cells open. I don't see any pupae or larvae dying right here. So, I'll put this tray to the side. Now I'll go to the next frame, and I can already see the brood panel looks lovely on this frame here. Okay, that's about as good as it gets right here. Lovely brood pattern right here. Seeing a little bit of Varroa sensitive hygiene right here. Notice the anomaly, my eye just immediately goes to the one cell on this frame that has a problem which is this one right here, and this is not a problem it's a good thing - the Varroa sensitive hygiene. That the bees that's one of their mechanisms to deal with Varroa and viruses. But beautiful brood frames here. A little bit of back filling with bee bread here and one more Varroa sensitive hygiene right here. But notice you just ignore all the rest of the bees on the comb and the rest of the cells you look for the anomaly, the ones that aren't doing what you expect. Okay so, if you are in bees everyday you realize you never have to, or rarely ever have to actually see the queen. You look at the queen's work right here. This tells me everything I need to know about the Queen right here. Much more than how the Queen looks. She has a beautiful pattern. The brood is in a nice step at an age. See this hole right here? There's a round hole as this pro-pupus being sealed. So that's totally normal. Nothing wrong with that. Another beautiful brood pattern here. I'm seeing fresh eggs off in the corner, so we have a laying queen. We don't need to know anything else about the

queen whatsoever. This colony looks like it's in perfect condition. Doing very, very well. The only thing it's going to need to do is to replace this brood in the center frames with honey, and move the brood nest down lower in the hive, in preparation for winter. You do that on the goldenrod flows around here? Yes, the goldenrod flows should do that. So, beautiful colony, beautiful inspection, let's see if we see anything else on the next frame. Notice how little smoke you need to use on these bees. When I look right here, there's no bees looking at me, no smoke needed. As soon as I see heads looking at me and eyes follow me, then I will use smoke. Another nice frame of brood right here. Larvae looking quite good. Good survival of the larvae. No sign of any disease in either the adult bees or in the brood. These little queen cups right here are quite normal. Some races of bees like the Russians build these all the time. They're just preparation. It doesn't mean anything at all. Not unless they've put jelly into them that means they're actually talking about supersedure, or thinking about supersedure. Now these bees are balled up right here so instead of crushing them, I'm just going to shake a few of them off. Put this comb right back in. And squeeze all those combs together tight. I don't really care much for these plastic frames for this reason. They're a little difficult to squeeze. Okay, so everything's squeezed. See, I still see a gap right here, so I'm going to really put some pressure on that. Squeeze that gap out, there we go. There we go, now this hive is ready to put back together.

READ

Outside the Hive

Before even opening the hive, you can learn a lot from observing the exterior of the hive and the activity of bees at the entrance. **Predator Evidence** — Skunks, Bears, Raccoons Skunks have been scratching the ground in front of this hive. Check the outside of the hive for damage or signs of predators. Skunks often scratch the ground in front of a hive or at the entrance to lure out curious bees. Bees that come to check out the sound get eaten. In this way, skunks can hasten the depopulation of weak colonies. If you see scratch marks on the ground or your hives, elevate your hive on a stand so the skunks can no longer reach the entrance. You can also put small mammal fencing around your apiary to keep them out. Bear damage is much more obvious: knocked-over hives and ripped-up supers. In their search for brood and honey, they can kill colonies entirely. If there are bears in your area, you will probably need to install an electric fence around your apiary. ■ Another often overlooked predator is the raccoon. Raccoons are very intelligent animals that, like skunks, eat bees at the entrance of the hive. However, they go one step beyond skunks and remove the covers from hives and pull out frames. They can eat everything on the frames, right down to the foundation, and leave the frames strewn about in the bee yard. Unlike bears, however, they do not shatter boxes or frames. Raccoons will keep coming back night after night unless you install an electric fence that is low enough to the ground that they can't crawl underneath it. **FecalMatter** Look at the exterior of the hive for bee fecal matter. Bees defecate outside the colony to maintain a clean home and reduce the spread of infection to their nestmates. A little bit of fecal matter on the outside of the hive is normal; in fact, some fecal matter visible on snow in winter is a good sign, since it means that the bees found a mild day to ■ defecate outside after a long period of confinement. But excessive drips of orange or brown stains on the front of a hive are a signal that something is wrong. **Guard Bees** The last thing to look for before you open the colony is guard bees. They stand at the entrance, often holding their front legs up. They check out bees entering the hive — and may even be checking you out. Wave your hand slowly in front of the entrance. If a guard

bee follows your hand, she is watching you. You'll need to use smoke to break her attention. ■

WATCH

What Can We Learn From Bees at the Entrance?

Observing bees at the hive entrance can give us several indicators of the colony's health. There's so much information you can get from just looking at the outside of the colony! Now, when you open the colony, you are better prepared for what you might find. Video Transcript So we're still standing at the outside of the hive, but after we've checked the exterior of the hive to look for any evidence of predator damage or fecal matter, now let's turn our attention to the bees. So let's focus on the bees that are flying around the hive. Are they gentle, are they unusually aggressive, what is their temperament like? If they are more aggressive than normal, then it can be a sign that did experience some predator attack or something like extreme heat in the summer might be bothering them. And if the hive is unusually aggressive, it's up to the bee keeper to figure out what the cause is and determine the stressor that's causing this behavior. So keep that in mind as you're inspecting your colony. Now let's look at any indication of adult bees that are found on the ground, or at the entrance of a hive. Are you noticing dead bees? Are you noticing just a few dead bees or is there a big pile of dead bees? Do you see any bees that have deformed wings, or any other deformities? Are there bees that are slowly crawling or twitching? A few dead bees are a normal sight to see but piles of them can be a sign of either intense robbing or a pesticide poisoning event. It's important as a master bee keeper to learn that there are normal seasonal fluctuations for dead bees. Bees are naturally dying over winter and early spring and their sisters are cleaning out their dead bodies on warmer days whenever they get the occasion to. So it's not worrisome to find a lot of dead bees on the bottom board or the front of the hive in winter or early spring. You may even expect to see a few dead bees scattered in the snow, which is a sign that they were performing their cleansing flights, but it may have been too cold for them to make it back to the hive. And these bees are usually accompanied by bee feces sprinkled across the snow too. However, if you're seeing large numbers of dead bees in late spring, summer or fall, then this signals a problem. Bees that are twitching or extending their proboscises, clumped in a big pile, or scattered across the ground near the hive, in the times of year that I just described, are indicative of a pesticide poisoning event. If you're witnessing this, take photos or videos and report it as soon as you can. It's even helpful to take a sample of these bees and put it in a ziplock bag, or a jar and freeze it. Contact your state or provincial apiarist or your local extension associate for information on the correct agency to contact. You may also notice that instead of seeing large piles of bees or lot's of dead bees, you're only seeing a few bees on the ground, or at the entrance, that are twitching or crawling. And this can be a sign of viruses. There are some viruses like paralysis viruses that cause twitching. And others, like deformed wing virus, that can cause a lot of crawling behavior. And it's not often that bee keepers witness these symptoms in virus ridden honey bees. But, in some cases, you can actually see it. And the reason why it's not so common, is because infected bees will actually remove themselves from the colony so that they're not infecting their other sisters or any of the brood. Okay, so turning our attention away from dead adults, let's think about any dead brood we're seeing at the ground or at the entrance. Take note of any discolored larvae or pupae that have been dragged out of colony as the bee is performing hygienic behavior. So this is when bees are detecting that there's an infection or a parasite infestation or a virus in the brood and they're actually removing those dead bodies. If you're seeing

excessive numbers of dead brood at the entrance, then this can indicate that you have a high mite load and the bees are doing a lot of this hygienic behavior to clean it out. So what you want to do is probably take another sample of these dead bees. You can put them again in a zip-lock bag or a jar but this time add 70% alcohol to it. Something like rubbing alcohol and you can send it to the USDA Beltsville lab in Maryland for free analysis. And then follow up by doing a mite check in the colony to see if you do have mite loads and if this is probably what caused this issue. You might also see brood that looks like little pieces of chalk or little white mummies which is indicative of chalk brood.

Video Transcript You can learn a lot by watching the bees entering and exiting the hives. And as a beekeeper, it's really enjoyable actually just to sit there for a few moments before you open your colony and see what the bees are doing. See what they're bringing in, see if there's any issues that might need management, and just to watch them go about their day-to-day lives. If the weather is sunny and warm and there's flowers in your area producing nectar and pollen, the bees should be entering and exiting the hive regularly. If you see lots of activity notice what they're doing. Are they bringing in nectar? A way that you can tell this is by looking at their abdomens. Their abdomens may look a little distended or slightly stretched out and hanging downward as they fly into the entrance because they're weighed down the nectar. And this takes a few times to be able to pick up. So what you're looking at are the bands on the abdomens and seeing if they're stretched out more than normal, and comparing it to workers that are just walking around inside the colony. But it's something that you can detect if you're looking closely. Are they bringing in pollen? This one's much easier to tell because you can just see the pollen baskets on their legs. Take note of the color in the pollen baskets and you can begin to associate what different flowers they're collecting the pollen from and what's actually blooming in your area at different times of year, and what is a main food source for the bees. Reduced activity can be a sign of a problem, however. The colony could have a queen issue, it could have a disease or be weak from some other cause, so you want to look for clues when you open it up. One really neat indicator about pollen coming into your hive is that it can actually indicate that the colony is queenright. So you can know this before even opening the colony and looking for the queen or signs of eggs. And the reason for this, is because bees collect pollen to use as a protein source, to feed her young larvae. So if you're not seeing a lot of pollen coming in, when you know there's pollen around, then that could be the indication of a queen issue. Okay, so looking at this activity of bees at the entrance, if you're noticing that there's more activity than normal, take note of whether it's orderly or if it's frantic. If it's really busy but it's orderly, meaning that the bees are still landing on the entrance board and they're walking right in, then this could just be a sign that there's a strong honey flow on. And this is normal and it's actually a good sign. But if you're seeing more frantic behavior, bees flying all around the entrance, or fighting occurring at the entrance, then this could be a sign your colony is actually getting robbed. Sometimes, you may see bees that are flying around the entrance, but they appear to be floating almost, and there's not any fighting that's occurring, and you're seeing lots of bees just floating and hovering around the entrance and around the hive. And this is actually indicative of the orientation flights that the new forager bees are taking to learn where their hive is located and other landmarks surrounding them. So take a few moments just to watch these bees and to see exactly what is going on with them, whether or not it's an issue or if it's a good thing. There are a few other things that you might sometimes see at the entrance, but you certainly won't come across them every time you're doing an inspection. One of these things you might see are bees fanning at the entrance, and this is usually a sign that the colony is hot. So,

they could be pointing their abdomens downward, flapping their wings to circulate air and increase ventilation in the hive, that's what fanning looks like. And on hot summer days you can actually help the bees by providing them with a good source of water and improving their ventilation. So you can increase ventilation by giving them a screen bottom board or drilling in an upper entrance in the top hive body or even by propping open the outer cover above the inner cover with a stick or a shim just to try and release some of that hot air and moisture out of the hive, much like the way that hot air is released out of the chimney in a house. Another behavior that you might witness and that I actually got a lot of questions about is wash boarding behavior, and this is where bees are taking one step forward and one step backward repetitively and they're licking the surface beneath them. And they do this behaviour for hours without stopping. And bees will washboard at any time during the day, they'll do it past dark, they'll do it when it's hot out, they'll do it all the time. And it's actually a common behaviour, and it doesn't signal that anything is wrong. But we don't fully understand why bees perform washboard behaviour. We think that it may help smooth the entrance of the hive. Or it could just be a sign they're performing general maintenance activities, because bees who are washboarding also tend to engage in other nest maintenance activities such as building comb. And the last thing that you might see with activity on the outside of the hive is bearding. And bees may beard on the outside of the hive for two different reasons; either they're too hot or the hive is too crowded. So they can beard, just like wash boarding, at any time of the day or even after dark. And on really hot summer evenings, you'll see bearding after dark because all the foragers are home, but it's really hot and it's not sunny, so they can't go out and collect water to cool off. And bearding isn't a behavior to be too concerned about if it is hot outside, it's a natural thing that they do. But the bees will be gentlest if they're not hot and bothered, so you might want to manage your colonies to help the bees regulate the hive temperature, or provide them with adequate space if they need it. So for temperature regulation you can do the same things to improve ventilation I mentioned before, giving them a screen bottom board, an upper entrance or improving heat release from the top of the hive by propping up the outer cover and by giving them some water. Now colonies that beard may also do so because they're really crowded. So either they're out of space to grow, or just out of space to distribute themselves to help keep the brood cool on hot days. So to increase the space for a bustling population, you can either add honey supers or you can expand your brood nest to two boxes if it's currently one. It's important first to do a colony inspection and look for swarm cells that may have started getting constructed before you add a super. If they are confined and there's no more space for them to grow, then they could have started thinking about swarming and building these swarm cells. So you don't want to get to the point where you're giving them more space just to find a few days later that a lot of the bees leave in a swarm. So make sure that you check and rule out the fact that they're swarming before you give them additional space.

READ

Inspecting During Less-Than-Perfect Conditions

■ You might not always be able to inspect, Key Points manage, or treat your colonies on beautifully if you need to open the hives in sunny days; however, if something needs to be less-than-perfect conditions, the done to keep a colony alive (for example, usual precautions apply, but even feeding during cold weather), then you should more so. do it — just be quick. If you need to open the Different amounts of smoke hives when the bees are stressed or when their are required depending on your

behavior has higher stakes — e.g., when you are goals, the bees' disposition, and demonstrating a hive inspection to visitors — the amount of time you plan to the usual precautions apply, but even more so. spend with the hive open. Extra smoke will help calm the bees, especially if you smoke for a few minutes before opening During a nectar dearth, only the hive. You might apply a few puffs of smoke inspect your colonies when you to the entrance and under the outer cover absolutely must, and do it as quickly, wait a minute, and then open the hive. quickly and efficiently as possible. In the same way, you might puff smoke between hive bodies before lifting off the topmost one. Depending on your goals, the bees' disposition, and the amount of time you plan to spend with the hive open, different amounts of smoke are necessary. Inspecting During a Nectar Dearth If your area is experiencing a mid-summer nectar dearth, robbing may start during an inspection. In severe instances, robber bees try to enter your colony as soon as you stick your hive tool in to separate the boxes! Only inspect your colonies when you absolutely must, and do it as quickly and efficiently as possible. Do not smoke the entrance before opening the colony, as you want the guard bees present to fend off intruders. Stack the boxes squarely as you remove them so there aren't any open cracks for robbers to enter, and keep an extra outer cover or sheet on top of them until you are finished. The goal is to limit the areas robbers can access a hive. If severe robbing starts, close the hive and try again another day.

READ

Inspecting Honey Supers

■ The reason to inspect honey supers is to make Key Points sure that bees are filling them and that you are Inspect honey supers to ensure adding them at the right pace throughout the bees are filling them and that you season (not so slowly that the bees run out of are adding them at the right pace room to store honey, and not so quickly that the throughout the season. bees cannot defend the far corners of their Assess honey supers for signs of hive). If honey production is your goal, add pests and moisture issues. supers as needed during a nectar flow to maximize the nectar your bees take in and Become familiar with the nectar allow you to separate different honey varietals. flows in your region. Lift off the outer and inner cover and peer into the honey supers. If you see bees poking their heads between frames, watching you, give them some smoke to divert and disorient them so they won't be defensive. Adding Honey Supers Pull out a few frames from your honey supers to see how they are being filled. If most of the comb has been drawn, bees are working on the comb, and the frames are being filled, there is probably a nectar flow on, and you'll need to add another super soon. Seeing fresh light- colored wax on the top bars of the frames is also indicative of a nectar flow. Some beekeepers remove and extract supers throughout the season; others continue adding supers and extract only once or twice a year. This choice depends on how much time and hive equipment you have. If supering during a nectar flow, we typically recommend placing a new super underneath the top super to draw the bees into it, a practice called bottom supering. If you add a super on top of one that is already full, the bees may not use it as readily. If you don't use a queen excluder, it's a good idea to leave one super full of capped honey between the brood nest and the next super you are adding. The super of capped honey can act as a barrier to dissuade the queen from moving up into the supers. Instead of adding a super of drawn comb or foundation below a honey-filled super, some beekeepers choose to checkboard their supers instead. With this practice, honey frames from the full super and foundation or drawn comb from the empty super are rearranged so that they create an alternating pattern in both

supers. The bottom super contains frames in this pattern: honey, foundation, honey, foundation, etc., while the top super contains frames in this pattern: foundation, honey, foundation, honey, etc. Checkerboarding supers is done to encourage bees to draw out frames of foundation. It also helps prolong or prevent swarming if done before the brood nest begins to be backfilled with honey. It does so by interrupting the full super's solid band of honey above the brood nest, freeing up more storage space for honey above and for brood below. Fresh white wax on the top bars of the frames indicates nectar flow. Supering usually begins here in the Northeast around mid-May. Once a super is half full of nectar, you can add one to the colony. Don't wait until all the honey in the super is fully capped before adding another one. If a super is three-quarters full or nearly full with nectar, some beekeepers may add an additional two or even three supers to their colonies during a strong nectar flow to minimize the number of trips they need to take to the bee yard. We would suggest adding multiple supers only if you have a large bee population that can defend that much space. If you add a honey super that only contains frames of foundation and no drawn comb, don't immediately separate it from the rest of the hive with a queen excluder. You can add an excluder later, once the bees have drawn out comb on some of the frames. The reason for doing this is that bees can sometimes be reluctant to build comb in a new super with undrawn foundation if the excluder is put on at the same time. But if the excluder is left off initially, the bees readily draw out the foundation when there is a honey flow, and since they have begun working on it, they will continue building and using that super once you put the excluder on. Otherwise, the beekeeper who adds both the excluder and undrawn foundation super on at the same time may come back to find the super empty, the brood boxes completely full of honey, and the colony ready to swarm or has already swarmed. If the colony has a vigorous young queen, the super may end up with a few eggs, but usually it doesn't. Just make sure to check that your queen isn't in the super before adding the excluder!

Looking For Signs of Pests

You may see pests or their signs in the honey supers since they congregate where there are fewer bees. Pests in the honey supers are not usually an issue during the active season, particularly if the bee populations are strong enough to defend the space, and if colonies are not weak from other issues. In great numbers, however, small hive beetle larvae can cause the honey to ferment and look slimy in appearance. Wax moth on a honey frame

Assessing Moisture

Look at the inside of the inner and outer covers, the hive walls, and the tops of the frames. Do you see any moisture problems? Moisture promotes the growth of mold, which can appear as black, blue, or gray streaks or spots in the woodenware or on the honey cappings. If any of the wood has condensation or looks moldy, improve ventilation at the top (either by creating an upper entrance or propping up the outer cover with a shim), tip the hive slightly downward to allow water to flow out the entrance of the hive, and make sure your covers are free of holes or cracks. Replace any moldy frames. If the moisture problem persists, consider relocating your hives to a new location with dry ground and ample sunlight. Hives sitting in standing water or soggy ground are more likely to experience moisture problems, as are hives with tall grass or weeds at the entrance. Another moisture issue specific to honey is fermentation. Normally, after bees collect nectar, they evaporate the water until the moisture content is below 18%, before capping it. But in very humid climates or during heavy nectar flows, bees may cap honey before the moisture level is low enough. In this case, the natural yeast population of the honey rises and ferments the honey (converting the sugar into alcohol and carbon dioxide). Fermented honey looks bubbly or foamy, due to the carbon dioxide, and often smells yeasty. You can remove the affected frames and rinse out the fermented honey with water. The combs

are fine to reuse after they are dry. **Understanding Nectar Flows** It's important to familiarize yourself with the nectar flows in your region. Depending on the nectar flows in your area, you will see variations in the color of the honey, from water white to chocolatey brown. There will also be variations in smell, taste, and texture. If there is a strong nectar flow from one source in your area (which changes from year to year), most of the honey in a given super will likely come from that source, and you may be able to market it as monofloral honey, such as basswood, clover, Japanese knotweed, or goldenrod honey. Keeping track of the starting and ending points of major honey flows in your area over the years will help you plan for supering and extraction. You may also notice variations in honey cappings. Sometimes bees will leave air between the honey and the wax capping, giving the wax a convex, white appearance. Other times, the cappings lay flat against the honey and appear darker. No one knows the reason behind the difference, although there appear to be genetic predispositions; it does not affect the quality or taste of the honey.

READ

Starting Your Inspection of the Brood Nest

During your inspections, you should carefully inspect at least four frames from each brood box. It is important to base your understanding of the quality and health of the queen on multiple frames, as early stages of a disease might only affect a few cells that can be easily missed by only inspecting one frame — but this aim for thoroughness should always be balanced by the imperative not to disturb your colony for too long, for their sake and yours! Throughout this course, we will refer to the following numerical positions of specific frames in a hive. **Long description** ■ There are ten frames in the hive. The frames in the hive are numbered one through ten, with one being the first frame on the left, and ten being the last frame on the right. Start your inspection by removing one frame first and setting it aside to make room to manipulate the others. We usually try to remove the #2 frame first so that we remember its position later when we are ready to close up the colony, but you can choose to remove whatever frame is most convenient for you. We recommend against removing the #1 frame first, as this can roll any bees, including the queen, between the hive wall and the first frame, possibly damaging them. Carefully set aside the first frame you've removed to make space and reduce the risk of crushing bees as you inspect a few other frames. Aim to only keep two frames out of the hive at all times — the first one you set aside and the one you are looking at. This helps you keep all frames in the right orientation and reduces the risk of squishing bees. Also, try to inspect your frames while holding them above the brood nest. If the queen falls off a frame, she will likely fall back into the hive instead of on the ground.

READ

What to Look For When Inspecting the Brood Nest

Click the accordions to explore what to look for as you are examining the brood frames. **Adult Bees** First, look at the adult bees on the frames. Is the population strong? If there are sufficient numbers of bees to nearly cover both sides of each frame in the brood nest, the colony has a strong population. In contrast, if there are not enough bees to cover the brood, it may signal a problem and cause other issues. For instance, the brood is at risk of being chilled in a colony with a low population. Next, look for indications of diseases on the adult bees, such as hairlessness, greasiness, deformed wings, or

twitching movements. Estimate the proportion of drones in the adult population: too many may indicate you have a drone-laying queen or laying workers. During peak drone production in late spring and early summer, about 5% of the adult population in a hive will be drones. As you inspect the frames, scan for the queen to see if she has a retinue of bees responding to her pheromones. Work to make sure she doesn't get harmed during the inspection. If you plan on pulling more frames or doing anything that might stress or harm her (e.g., sampling bees for mites), you can ensure her safety by gently caging her and placing her in a shady spot while you work, and reintroducing her to the top brood chamber at the end of the inspection. If you cage her, be careful not to forget about her! ■ **Brood Health** If you have a strong population with many adults covering the frames, it may be difficult to assess the brood pattern beneath the bees. In these cases, you can gently blow on them or place your hand over them so that they move out of the way, or shake the bees off into the hive body you are working in. When inspecting brood frames, hold the frame with the sun at your back and angle it so the sun shines directly into the cells. A healthy brood frame has a solid pattern of brood (with few open cells), with arcs of pollen and then honey surrounding the brood. The frame will resemble a rainbow, with brood in the center, pollen arcing around it, and honey as the topmost layer. Some empty cells are normal: 6% to 17% of larvae are generally removed in a healthy colony (those that inherit identical same sex determination gene versions), but a higher percentage than that can indicate that the queen is poorly mated, inbred, or failing from age, or that the brood may be diseased or afflicted by parasites. Diseased brood can take on a variety of symptoms depending on the parasite or pathogen in question. Instead of pearly white larvae, diseased larvae may be coffee-colored, brown, gray, black, or appear chalk-like. Their position may also change from being curled up nicely in their cells to being twisted, pointy, melted, or mummified. Infected pupae show other symptoms: perforated or concave pupal cappings are indicative of diseases, as are pupae that have been uncapped and chewed down by nurse bees. Lastly, bees that die upon emergence with their proboscises extended are indicative of some diseases and pesticide poisonings. Always be on the lookout for any signs of brood disease, even if symptoms are seen in only one cell. Take photographs and notes of any issues. If the disease is unfamiliar to you or resembles American Foulbrood, send samples for free analysis to the Beltsville Bee Lab in Maryland, and label the affected colony so you can identify it later. Lastly, check for deposits in empty cells in the brood frame. Take note of any black scale that does not scrape off with a toothpick (indicative of American Foulbrood) or small patches of white crystals on the ceilings of brood cells called guanine; these crystals are indicative of high Varroa mite levels. ■ **Brood Pattern and Sex** The queen's pattern of laying is important. A healthy hive has brood of all ages: eggs, larvae, and capped pupae. A good fertile queen will usually begin laying eggs near the center of the frame, and work her way outward in concentric circles. The result is brood aging in a "step down" pattern: larvae the same age sit next to each other, and moving outward from the center, the larvae are one day younger, then two days younger, and so on. As the fully developed bees emerge, the queen should fill those newly open cells with eggs. If you cannot find any eggs after a thorough inspection, you should take measures to replace the queen or introduce a frame with eggs on it so the bees can rear their own queen. The sex of the brood is important. The colony starts investing in drone production once its population reaches around 4,000 individuals. For colonies of this size or bigger, drones are most plentiful during swarm season, when males are needed to mate with the newly emerging queens. Here in the Northeast, swarm season peaks from about mid-May to mid-July. Drone production ceases toward the end of

summer as the colony begins to invest most of its energy into preparing for winter by building up its honey stores. On average, about 17% to 20% of comb in a colony is drone comb (about two full frames' worth in a 10-frame hive body). This is the "natural" average in wild colonies and in foundationless hives, but beekeepers will likely find less drone comb in their hives if they are using frames of worker cell foundation. Drone comb will usually be built along the bottom or sides of the frames, not usually in the center of the brood nest. Drone comb can also be found in the burr comb on the tops of the frames or in other areas. If you see a high number of capped drone cells late in the summer (representing well over 5% of the drone population) or in colonies with weak ■ populations, those colonies may have a drone-laying queen. A drone-laying queen will still lay only one egg per cell, while laying workers will lay multiple eggs in the cells. This brood pattern is not great. There are many empty cells, and larvae of different ages are next to each other. Queen Cups and Queen Cells

Examine the frames for queen cups and cells. Queen cups are, by definition, small wax cups that are empty. It is normal to find a few of these throughout the year, as the colony keeps them on hand in case it needs to raise a new queen or swarm. a couple queen cups do not signal any impending changes in the colony. This is true even if you find them in different places in the nest, as they may be torn down and rebuilt several times throughout a season. However, colonies will ramp up the production of queen cups in preparation of swarming. In this case, you can see many queen cups, usually close to a dozen, usually located along the lower edge of the brood nest. If you see an egg or larva in the cup, it is now a queen cell, indicating that the bees are seeking to raise a new queen. The placement of the cell can indicate why. Generally, many queen cells along the bottom of a frame signal that the colony is preparing to swarm, and are known as swarm cells. In contrast, a few queen cells near the center of the frame are usually either queen supersedure cells or emergency queen cells. These can be used to replace an existing queen that may be failing, or to make up for the loss of a queen. The presence of queen supersedure cells is usually a sign that there is an issue with the queen and the workers are naturally replacing her. ■ A swarm cell placed near the bottom of the brood nest.

Brood Nutrition Take note of the brood nutrition. One to two frames' worth of pollen/bee bread in each brood box is ideal for brood rearing. However, some colonies may not always have that much on hand because they are using it up relatively quickly after they collect it. Observe the diversity of pollen in the colony. Does it appear that bees are foraging on pollen from a single source, or are there multiple sources of pollen (evident by various colors) present in the hive? Document any observations of entombed pollen, where bees cover a cell containing pollen with propolis and wax. Entombed pollen has been found to contain high levels of certain pesticides. Besides looking at bee bread, you can get a good idea of a colony's nutritional levels based on the amount of royal jelly surrounding young larvae. Abundant white pools of royal jelly signal that there is a lot of protein available for nurses to synthesize into royal jelly, whereas colonies with little or no jelly visible around the larvae may benefit from receiving a protein supplement. ■ A young larva with a good-sized pool of royal jelly. Pollen is present in the cell beside it.

Condition of the Frames Lastly, examine the condition of the comb. Is the comb old and/or dark? An old comb can harbor old pupal cocoons, disease spores, and pesticides, with the latter two posing a risk to colony health. Each year, aim to replace your two oldest frames in each hive body to keep your hives in optimum health. However, do not remove more than four frames from a hive at once, as the Bee Informed Partnership found this can have negative effects on the colony's population. Removing more than four frames from a hive at once will divert the bees' resources to building comb rather than raising brood or producing

honey. There is a tradeoff between reducing the risks to the hive by discarding old, possibly contaminated, comb and requiring the bees to expend more of their energy. ■ This brood frame is five years old and should soon be replaced with a newer frame. To make all the cells visible so you can identify potential diseases or issues, or to make later inspections easier, you may want to remove any burr comb (comb anywhere other than on the faces of the frames) from the bottom of frames or the sides of the hive as you go, placing any scrapings into a bucket to take with you. Leaving comb in the bee yard attracts other bees and pests. Keep the frames in each hive body tightly squeezed together or spaced in accordance with bee space to prevent the bees from building more burr comb. Once the Colony Inspection Is Complete As you finish inspecting frames, place them back in the boxes in the same orientation and order that they were in to ensure that the brood nest is intact — again, this is easily accomplished if you only take one to two frames out at a time. The inspection is complete when you have accomplished everything you set out to do. After the inspection, consider what sort of management the colony needs: do the bees need more room to store honey, or do you need to take steps to prevent or manage swarming? Take note of when you need to return to your colonies for the next routine inspection or to follow up on anything you saw or did today.

WATCH

Inspection Tips

To help you conduct your colony inspections, we've included some tips that can improve your skills in the bee yard. Video Transcript And, one of the tricks that I use here, is the easiest way to throw your back out, is by trying to lift this frame right here, because the propolis holds it down, and all the lifting is done by the muscles in the small of your back. Very, very easy way to pull your back out. So, grab on to the frame here, put your knuckles down here, and lift with your fingers. So my middle finger pushes down, when my index finger and thumb lift this up. At this point, immediately when you clear it, put your middle finger underneath the end, that way if you pinched a bee, you can swap your fingers without dropping the frame. This super is full of honey, plenty of weight to it. And again, I'm putting all the weight onto my knee, none on my back at all, so I don't strain my back. You do this 100 times a day, 200 times a day, you can blow your back out very easily. Video Transcript Another thing to show you is, if you have trouble looking, you can just shake the bees off the frame. That gives you a chance to look at it. The bees don't get bothered at all by the shaking. You see them just flying around but that does not initiate any kind of defensive behavior whatsoever. And I'll show you the shake one more time. So I'm going to rest the frame on my middle finger here and put my thumb above it. If your thumb touches the frame will swing wildly and you can throw the queen off. You're going to balance the frame up and down between your middle finger and your thumb. Ready? Okay, see that? That was three taps and it reverses the direction of the bees twice. So they just shear right off. Now watch what happens if I put my thumb down and hold it. When it does it flips the frame. You have to do this straight up and down, snap it. So it goes straight and the bee shear's straight down. And notice it doesn't cause any kind of stinging at all so feel free anytime you want to do that. Notice the bees are just completely ignoring me right here. Bees don't mind at all being disturbed on the combs. You can push them right out of the way, as long as you don't crush them. There's another way. If you're looking for eggs a lot of times, we'll just blow on those bees and they can move out of the way. So, get used to it being around your bees. Treat them gently, treat them with respect. None of these things that I'm doing elicit defensive

behavior whatsoever in the bees. You hear some noise but they're not interested in stinging in any way. They're just going to, these are young bees that just don't know where to go, they're going to mill around for second. They'll smell their way in and they'll find their way back to the hive. One of the things you can do is when you put a box down, put it on kitty corner, so now you only have four little tiny points of contact. And then as you twist it any bees that are on top of the edge of the box will be pushed out of the way rather than crushed. Video Transcript And I get my smoker going here. I see it. Right now the smoker has turned grey, I wouldn't even start in the hive. You've got to come in here. One of the things to do is take your hive tool and open up the bottom. And then lifting the embers off the bottom, see how that immediately allows it to start smoking? Works best is slow, long strokes like this and that will get the smoke going. Okay, now we have a smoker ready for the hive. Don't make a mistake of just charging ahead if your smoker is not putting out white smoke. Stop, go back, do what I just did until white smoke is coming. Turn these guard bees away. Oh those aren't guard bees, those are bees at the entrance.

TOOL

Beekeeping Calendar

Timing is critical for good hive management. By  Download the Tool now, you should have some familiarity with the Use this Sample Beekeeping seasonal cycles of a honey bee colony. You have Calendar to help track hive seen the exciting activity in spring and summer, conditions throughout the year. as well as the quiet periods of overwintering. Beyond that, knowing bee biology, predicting cycles of pests and diseases, monitoring the succession of flowering forage, and staying attuned to other seasonal changes helps beekeepers to keep their hives alive and healthy. These biological and phenological observations tell beekeepers when they need to supplement a dearth of good forage, help the bees take advantage of a good honey flow, or ensure that a hive is prepared for winter. Developing this environmental awareness is also one of the great pleasures of keeping bees. Here we present a sample beekeeping calendar that describes the goings-on in a hive and environmental conditions more broadly during different months of the year, and the associated management practices that support colony health and productivity throughout the year.

WATCH

A Year in the Bee Yard

Timing, location, and local conditions make all the difference between a successful honey harvest and a disappointing season. Watch this video to learn more about the beekeeping calendar and why it matters. Video Transcript Our calendar has a focus on the northeast, in part because that's where we are here in Dyce Lab, Ithaca New York, And you have to pick somewhere. But every area is a little different. It's well- known saying that all bee keeping is local, that's what keeps it interesting, keeps it fun from year to year. The calendar is also geared toward small scale honey production. That's not what all beekeeping is anymore and especially not for a master beekeeper. And so we have included information about producing nucs, rearing queens, providing pollination services. But those are more specialized and you'll want to create your own calendars according to your own schedule and your own business goals for that. Here in the northeast the active season is short but it tends to be more predictable than other areas. For example, the orange blossom honey flow in Florida can vary by as

much as two months but here in New York State the blooming of the American basswood trees, the linden, doesn't usually vary by more than a week or two around July 4th. But the northeast is a big area with lots of variation. They include the US hardiness zones, the regional hardiness zones from all the way from the 3 to 7. And things continue to change. Climate change means that there are new crops here and new flowering dates. The apple bloom here in New York is about two weeks earlier than it was a hundred years ago. And even the nutritional value of some plants is changing because of climate change. One study found that goldenrod has less protein when there's a higher concentration of carbon dioxide in the air. And so, the flowering plants that we have listed, are the ones that are most abundant here in the Northeast, not necessarily the most nutritious. For example, dandelion is an incomplete protein source that's missing some amino acids that are key for your bees. But, because of when it blooms, and relative to other plants, it's pretty likely to be foraged by your bees at that time of year. So as you embark on your beekeeping, pay attention to what's in bloom. That includes more unusual stuff, like a late flowering broccoli in your garden. I've got hives at a friend's place with broccoli that goes all the way until October or November. And witch hazel, on the other end of the calendar, as early as February or March. You'll want to keep records of blooms and other seasonal changes. And because that season, our season is short, the weather conditions actually play an outsized role for honey production. Because if the weather, say is bad and keeps the bees in the hive for all around July 4th, they're going to miss that basswood flow even if it was abundant. So your record will be a great source of information for your calendar in years to come.

WATCH

Colony Assessments

At different points during the season, you may want to conduct a colony assessment, which is more detailed than an inspection. An assessment helps you to quantify the growth and productivity of the colony by pulling out every frame to estimate and log the following: 1. Number of frames of adult bees. 2. Number of frames of brood and proportion of eggs, larvae, and capped brood. 3. Number of frames of honey. 4. Number of frames of pollen. 5. Number of frames of drawn comb and foundation. Watch this video to learn more about the process of conducting colony assessments and the circumstances where it could be beneficial to you and your bees. Video Transcript Assessments have several uses for experienced beekeepers. And I find that as I'm getting to know my bees more and get more confident in my colonies I'm starting to ask more questions, and want to do more testing with my bees to find the answers to those questions that I have. So with assessments you can actually quantify how productive your colonies are. You're quantifying the number of frames of bees, the amount of pollen you have in your hive, the amount of brood that you have, or the amount of honey that you have. And so, you can sort of engage in a small research project with your colony to figure out some of these questions. So, for instance, you might want to know if a new pollen substitute on the market is worth trying, and if it's actually going to help build your brood production or build your adult population. And for this you can feed some of your colonies as pollen substitute and leave some of your others colonies without it which will be your control group and after a certain amount of time, you can assess the amount of pollen and the amount of, sorry, the amount of bee bread that they would be converting it into. The amount of brood and also the amount of adults in your colony to see whether or not it's worth spending the money to invest in this pollen substitute. You might also wonder whether or not the crops that your bees are

pollinating with pollination services. if it's negatively affecting brood production. And you can get to learn which crops are more healthy for your bees and which ones might not be and may need some additional management to make sure that they remain healthy when they're in those landscapes. So for instance you can be measuring the amount of brood that's produced before they go into this crop and afterward to get an idea of how that's being quantified. And lastly if you're interested in breeding some queens assessments will let you know which colonies have queens that are laying early, which ones are building up quickly and which are heading colonies that produce a lot of honey. So in your beginning days of bee keeping you might not start asking these questions. But certainly after a few years you start to think about these things and assessments can help you find those answers. We generally recommend to conduct two full colony assessments during the year anyways, regardless of whether or not your testing a question you might have. And we recommend that you do an assessment in the spring, right after the bees have come out of winter, to see what tape of shape there in, whether or not they're strong enough to split or whether or not they have a good viable queen. And then also to conduct an assessment in late summer, before the weather turns cooler, to make sure that your queen is still viable and has a good laying pattern. And if not, then this is the time that you can re-queen her with enough time before fall to ensure that she gets accepted and that the colony survives winter. So these types of assessments and quantifying everything that's in your hive is time consuming. It's much more time consuming than conducting an inspection. But they do provide very valuable information to the master beekeeper. Okay, so when you are accessing a colony, you are basically pulling out all of the frames and quantifying these different factors on it: bees, brood, pollen, honey. You can just do the ones you are interested in or you can do all of them. And the way that it helps me to do it when am conducting an assessment, is to picture the frame being broken down into quarters or sixths, so that you can actually estimate the fraction. And then we quickly record the fractions for each frame. We just write it down on our record sheet and then we can add it up in the margin at the end and figure out the quantity of all the brood and all the pollen and all the bees in one hive. So to assess frames of bees, sometimes it's tricky to actually be looking at a frame if your bees aren't very still on the frame. And if they are still then it's fine. You can break them up into fractions and continue doing it that way. But if they are moving around or more flighty, than an easy way to assess the frames of bees is just to tip up each hive box and lay it on its side to quickly estimate how many frames are covered by bees from the bottom. And it's a simple as it sounds. One frame of bees will be completely covered with bees on both sides. So, estimating these frames of bees and other materials is also subjective. And even when I'm in the field sometimes what I interpret as being three quarters of a frame of pollen someone else may interpret as a slightly different fraction. So for the sake of consistency, it's best that the same person conduct each assessment for the colony. And since assessments take more time than inspections, it usually takes us on average about 15 minutes to do an assessment versus just a few minutes to quickly go through and inspect your colonies. The bees might become more defensive and they're a bit more disoriented. So we recommend wearing more protective clothing than you normally would. And you need to be watching to make sure that you're colonies aren't getting robbed. When you're having the boxes totally separated, it can be easy for robbers to get in. So it's also helpful to stack your boxes squarely and put an outer cover on the outside, just so that there aren't any options or room for robbers to sneak in. And it's really handy to have your record keeping materials ready with you, when you go out and do your assessments. Having a check off list. Being able to quickly write down fractions will

help you in your efficiency much more than if you're writing everything word for word. Okay so let's go back to assessing the number of frames of bees. Because this is actually the most difficult part of an assessment I find. So when you're assessing your frames of bees we usually work the colonies a little bit differently before inspecting our colony. When we conduct an inspection, we are usually starting at the top and working our way down. But when we're assessing our frames of bees, we usually start from the bottom and work our way up. And the reason why we start from the bottom is because if we started from the top, as we're smoking the bees down, we're driving them down into that bottom box. And by the time you get down to the bottom box, it's literally overflowing with bees. So you can't really get an accurate response or accurate estimate. However, some people still prefer to work from the top down just to reduce aggression and disorientation and it is possible to still conduct assessments working from the top down. In these cases, the key is in the amount of smoke you're using. So smoke the hive very lightly so that you are not driving the bees down to the bottom. Again, wear extra protective gear if need to. But you just want to make sure that you're able to get the accurate quantification of frames of bees.

WATCH

Keeping Good Records

Whether you are a hobbyist or a large-scale beekeeper, record keeping is a crucial component of beekeeping. Monitoring your colonies and documenting the dates of your inspections and manipulations helps you predict and meet your colonies' needs, but it can also be a source of learning and enjoyment in its own right. Record keeping has a range of applications and is particularly useful for managing honey production, pest and disease outbreaks and progression, queen health/productivity, and swarming and population dynamics. With written or photographed information, you can track a colony's population, health, behavior, and production over the course of a season. From year to year as well, records can help you compare the same colony's performance across years and predict what to expect at different times of the year. In short, good record-keeping allows you to manage your colonies more precisely for your goals. Watch these videos to learn how record-keeping can benefit your operation. Video Transcript Record keeping is important no matter your goals and motivations, whether you have 3 hives or you do migratory pollination contracts every year. There's a ton you can track and a ton you can learn from just one metric. For example a lot of people track their hives weights over the years or within a given year and that helps you see honey flows, maybe a swarm event and making sure they're putting on enough weight for winter. But even simple records can help you see your problem, for example, monitoring how defensive or gentle a hive is and whether that changes suddenly. In the short-term, records could help you plan for your next inspection. Do you need to return sooner rather than later to check on a new queen or to resample after a mite treatment? Do you need to bring equipment that you don't normally bring and you don't have available in the bee yard? I think most importantly for us though it's pest and diseases, is what record keeping kind of helps you keep a handle on. Which of my treatments are working? When is an apparent disease, when is the outbreak cyclical or when does it just follow bad weather or a period of confinement? If you lose a colony your records will tell you if anything was off in the previous weeks or months even, or whether you're not seeing something that you really ought to and you need to go ask somebody else. And finally, of course, records help you compare colonies in the same year. Colonies naturally vary, that's why we tell

novice beekeepers to have at least two when they start. It depends, each one varies on how fast they build up, how much honey and pollen they're storing, their temperament, and much more. And records will help you determine when a hive seems to have a problem or whether it's always lagging behind one of its neighbors. Finally, if you do want to do breeding, record keeping will help you learn which colonies have the traits you're most interested in, whether there's one that's more productive or more gentle or even more hygienic. As a lot of people are trying to breed for hygienic bees these days. But without records, you'd never be able to breed successfully. With your records, you might be able to do small experiments or tests in your area. You probably can't do thoroughly scientific experiments. But you can certainly test out different types of equipment for your own use. On a bigger scale, you might coordinate with people in your club or even a local citizen science project and record the data that they need for their own data sets. So we've got a few tips. Some of them are pretty obvious but we still think they bear mentioning. First of all, stay consistent with your abbreviations in the type of data you're taking. Of course your own system is going to evolve over time, but you can't compare say 15 frames of bees, that looks crowded, to a lot of activity, to 2 deeps. Those are all different ways of measuring more or less the same thing - population dynamics. And so using a data sheet will prompt you to record the same things every time and to count them in the same way. Plus they also make records more legible to others so you can share them with your club or maybe your local extension associate. We have got a few sample data sheets we are sharing with you. But there are lots of others out there and you can of course create your own. As you record data you want to stick to the facts. If you can see if something is wrong but you can't tell why, don't jump to conclusions. Just write down everything you can about the symptoms clearly so you can ask somebody else. If you have a guess mark it as such but it's not helpful to write down that you have AFB and then go back later and realize of course you didn't. Finally, there's a lot of other helpful tools and supplements you can use. You can take photos, make drawings, take samples, even do a voice recording that you transcribe later. As always, you'll want to include the date, the hive you found it in, and any other relevant information. And you can send that to your honey bee extension associate or just a local club member. And of course you have to make sure you don't mind getting your camera or camera phone a little bit sticky. Video Transcript Be aware that there's a learning curve to beekeeping and the learning curve never ends. You will be learning until the day you die. The question is whether you want to actually learn something. There's a lot of first year beekeepers 20 years in a row. First year beekeepers who never really know what they're doing. One of the things about learning is keeping records. What we use is just a simple Excel spreadsheet, a yard log, where we can fill in a list of all of our yards. And we can put the date down and what we did on that date. What the mite counts were on that date, whether we fed them, did a mite treatment, and observations on bloom. Your first several years, just a little spiral bound logbook for each yard to talk about what was blooming and put down the date and what the bees were doing. That could be extremely valuable in later years then because then you can look back for patterns and see what to expect in those yards. You just can't trust your memory at all. Our memory just does not work for that. And we often, like last year we had a problem with Varroa, late this season and my sons, one my sons, the accusatory son said, "well dad it's because you did this and this and because we didn't get the mite treatment on." So I just calmly say, get the yard log, let's take a look. And he grabs that, he looks at it. I could tell by how long that he looked at it that he realized he was wrong about that. Anyway, we have a documented record of when we did mite treatments and what we did and what the

mite counts were. And then I can say, "ah, now that we got that straight let's have a tailgate meeting, a learning experience, and see what we can learn and what we're going to do differently next year. So I very much encourage everybody to keep some kind of simple record keeping. Make it simple enough that you will actually do it. If you make it too complicated that you're not going to do it, then don't do it. Make it a simple record keeping.

TOOL

Record Sheets

You might be wondering — what kind of ■ Download the Tool records should you keep? The answer to this Record keeping sheets: question depends on your operation, as the • Colony Inspection sheet importance of certain types of records will vary • Varroa Mite sheet with different beekeepers' goals. But at a • Apiary Information sheet minimum, you should keep basic information • Hive Information sheet on your individual hives and apiary • Contact Information sheet organization, records of ongoing inspections, and logs of monitoring for Varroa mites and management regimes. You might also choose to track nectar and pollen flows, particularly if you are producing varietal honey. For rearing queens, you can keep colony breeding record cards. Businesses of course need to keep financial records, inventory, and many other logs. We have developed several record keeping guides that we use here at the Dyce Lab, available for download. We have tried to find a middle ground between small- and large-scale beekeepers, which is to say by leaving room for enough detail to learn from the colonies year after year, but not so much detail that such record keeping is impossibly time consuming. On the inspection sheet, in particular, we have left room for different sorts of information. On your record sheets, you can also note the temperature and weather, as this will help with evaluating the bees' temperament during inspections. Your record sheets should not be limited to those provided here. Make your own sheets as necessary. As the season ends, consider preparing an archive with all the data sheets combined, or entering them into a computer database such as Excel for easy analysis.

Colony Inspection Sheet If you do no other record keeping, the colony inspection form, or a variation of it, should be a standard part of your beekeeping routine. You will update it continually throughout the active season, taking note of the population, nutrition, queen status, and indications of disease. There is room to briefly describe the quality of the brood pattern (whether it is spotty, large/small — and relatedly, whether the brood nest needs to be expanded — whether there are signs of a laying worker, etc.). From there, you should make comments on the state of the queen and her laying, note if you see her or her eggs, whether she is marked or not (only relevant if she was marked and does not now appear to be, which ■ means that the old queen was likely superseded), and whether there are queen cups or cells. You might also note if they are supersedure cells or emergency cells. You should also estimate the stores available to the bees in the form of honey and pollen, and whether there is extra room available/unused in the hive. These will, of course, fluctuate over the season, but knowing their storage levels (past) and needs (present and future) will help you keep them nourished and healthy throughout the year. Finally, there is room to take notes on a range of indicators. Population management refers to any actions you have taken to split or merge a colony, requeen it, move frames to another hive, etc. Pests and diseases refer to any signs you may see in the hive and notation of any treatments or equipment, e.g., a screened bottom board. There is admittedly some overlap with the Varroa monitoring sheet, but in part, because Varroa needs to be monitored consistently, and in part

because we recommend that Varroa be an apiary-level consideration (rather than just at the level of the colony). The nutrition and hive products category refers to whether you need to or have already added supers, fed sugar syrup, extracted honey, installed a pollen trap or a queen excluder, or other relevant actions. The general notes category captures anything else of interest, e.g., who performed the inspection (if multiple people are working in the same apiary), the bees' temperament, activity in and around the hive (e.g., bees bringing in lots of pollen), whether you added any equipment, or took any actions like cleaning the bottom board or reversing the brood chambers. You can include notes if samples from the colony were taken. Indicate what type of samples you took and where you sent them for analyses (e.g., brood comb sample sent to USDA Beltsville Bee Lab). Varroa Mite Monitoring and Treatment Sheet Varroa mites are behind much of the recent concern over bee health, directly and indirectly. Controlling Varroa is key to healthy and productive colonies and requires ongoing monitoring of mite levels. Use the Varroa mite record-keeping sheet to aid your monitoring. Our monitoring sheet includes a column for mite counts. Common monitoring techniques involve sampling a ½ cup of bees, which is about 300 bees. But many treatment thresholds are specified at the number of mites per 100 bees, or a percentage of mites. To avoid confusion, we have included both ratios here. There are a variety of different monitoring methods (powdered sugar rolls, alcohol washes, etc.), which is why you should indicate which method you are using. You can compare your methods and mite counts to neighboring beekeepers. ■ The sheet then includes room to describe your treatment method(s). Note that this can include chemical treatments as well as cultural management practices for those practicing treatment-free beekeeping or Integrated Pest Management. Timing is critical for both chemical and cultural treatments (anyone who has ever left drone comb in a frame too long knows this well!). Notes might include information about the conditions (e.g., temperature, since temperature affects the efficacy of some chemical methods, or the presence of a lot of brood since many treatments only work on mites present outside of brood cells). Finally, any good monitoring and treatment regime also needs follow-up to see whether a given treatment was effective, which can be filled out in the next line below.

Apiary Information Sheet The apiary information record-keeping sheet only occasionally needs to be filled out and can be left at the front of a binder. It includes information about multiple apiaries (name or code for hive IDs, location, contact information for landowners, the number of hives in that location, and the dates they are moved in and out, and whether the yard requires any ongoing maintenance. There is room for other notes about nearby bee yards, forage and honey production, and any other measures you might want to track. Like the apiary information sheet, the hive information record-keeping sheet should be updated once a year or as necessary. It tracks individual colonies, with information about their origins (split, nuc, package, swarm, other), the colony's longevity, the origins of the queen(s) (supersedure, swarm, purchased — from where), and the queens' stock and/or subspecies. It is handy to keep a contact information sheet in the bee yard in case you need to quickly make a call to diagnose a problem or ask a landowner about changes in the site. We have included the information for the USDA Beltsville Bee Lab in Maryland; most other contacts will vary from state to state and beekeeper to beekeeper. ■ Hive Information Sheet ■ Contact Information Sheet ■

READ

Record Keeping Tips

To help you record your colony inspections, we've included some tips that can make your records meaningful for future reference. **Make Record Keeping Convenient** A key to accurate record keeping is to label all your colonies and yards with an identification system that clearly distinguishes them from one another. If you anticipate moving hives between apiaries, be sure to use a system that can account for that. You may want to label individual hive bodies (at least the brood chambers). You can paint it on, write it with a permanent marker, burn it on, or scratch/etch it in. If you do not anticipate moving any hive bodies between hives, you can put a waterproof label on the hive lid or make a map of your apiary with the hive IDs clearly labeled. Some beekeepers label frames within a colony with the year they were made using different colors, markers, notches, etc. This can be useful in allowing you to discard the oldest frames in each hive body every year, to reduce contaminants in your hives. Don't wait until after you've returned from the bee yard to take notes because you could forget important information. **To make record keeping a consistent and immediate habit, keep your record sheets in the bee yard or with your smoker and** **Make Record Keeping Consistent** **other equipment.** They can be stored in a weatherproof container in a nearby shed, or even beneath your hive(s) in a zip-lock bag. This not only ensures that you have them, but also that you are using the same notebook every time. Keep writing implements handy — try tying one onto a clipboard or binder with a string. Finally, having extras of everything — writing implements, paper, clips, etc. — will undoubtedly come in handy. **Use Technology** There are also an increasing number of digitized forms and apps such as 'Hive Tracks' or 'Beetight'. If you want to input information directly to a smartphone or other device, you need to consider the conditions in your bee yard. You will need connectivity or a system that doesn't need to be online, or can be uploaded later. You may want to avoid a lot of typing, particularly if you have sticky fingers! ■

READ

Tech-Enhanced Beekeeping

Tech-enhanced beekeeping, also known as precision apiculture or smart beekeeping, is one of the fastest-growing areas of apiculture research and commercial innovation. It uses sensors, AI robotics, and data analytics to monitor hive health (temperature, humidity, sounds, weight) in real time and provide early warning for pests or disease. These technologies aim to shift management from reactive to proactive to improve yields and bee survival. Those expensive tools can be useful, yet it is important to note that they detect indirect signals that are claimed by the companies that sell them to be correlated with various colony conditions. Some of those correlations are clear, e.g., an increase in hive weight and nectar flow/abundance of honey in the hive. Others are less clear; e.g., some sensors claim to reliably measure Varroa infestation levels. Current tech-enhanced beekeeping should be best seen as an early-warning and triage system, not as a diagnostic or decision-maker. Those tools might reduce unnecessary hive inspections, but they should be complemented with direct observation of brood, disease, or mites. Below is an overview of some useful technologies and how they are used by some beekeepers right now. **Hive Scales and Weight Sensors** These measure nectar flow and honey harvest readiness (gain), absconding or colony loss or swarming (sudden weight drop). This helps detect nectar flows without opening the hive, alerts beekeepers to feed during dearth, and helps commercial operations know when to move hives. These are placed inside the brood nest or upper super and can reveal the brood presence (brood nest stays ~34 to 35°C), queen failure (temperature fluctuations), colony stress or starvation (temperature decreases). Honey bees maintain humidity and

temperature remarkably tightly, so changes recorded by those sensors indicate biological problems days to weeks earlier than visual inspection. ■ Temperature and Humidity Sensors ■

Acoustic/Vibration Monitoring (AI-Powered) Microphones or accelerometers can be used to listen to colony sounds. They can detect queenlessness (distinct acoustic signature), swarming preparation (piping, quacking, wing-beating patterns), and colony collapse signals (reduction of sounds altogether). Those devices must be coupled with machine-learning models that were trained to evaluate the "acoustic fingerprint" of hive states. Individual bees can be equipped with Radio-frequency identification (RFID) tags. Those tags are tiny, passive chips that can be attached to individual bees (usually on the thorax) and record their movements and behaviors through automatic tracking. Useful information includes the forager lifespan, the homing success rates, the sublethal pesticide effects, and the orientation drift. ■ RFID Tagging for Individual Bee Tracking ■

READ

AI for Beekeepers

Besides assisting tech-enhanced beekeeping, AI offers all beekeepers equipped with a computer and Internet a wide range of capabilities for complex, high-powered situations and personal use. Beekeepers can tap into AI tools to help understand science and to bring that knowledge into the public sphere. You likely have access to free AI tools. As of 2025, the capabilities and rapid evolution of AI threaten to make any recommended tools outdated by 2026, so ask your classmates or facilitator for suggestions on which tool to try first. Below are some ways AI can be useful to you as a beekeeper. Understand Science Reading scientific papers is hard, unless you do it all the time (and sometimes even then). If you want to understand the context of some academic research, upload the document to an AI chat (or give it a link to a publicly available paper) and ask it questions about it. If you can't seem to get started on a document, citizen science idea, or presentation, you can use an AI chatbot as a sounding board or to set your thoughts off in a new direction. The AI tool will present ideas for you to explore further or reject outright. Either way, you will have external input to push you to think in different directions. Not everyone can put together sentences with ease, avoid grammatical errors, or argue about the Oxford comma. Use AI as your copyeditor. Note that the suggestion here says "copyeditor," not "writer." Use your own voice and your own thoughts first, and then the AI as the polish. ■ Jumpstart Your Thinking ■ Polish Your Writing ■ Make Global Connections ■ Query an AI tool for non-English sources, and then ask it to translate it for you. You can even use AI translation services to conduct a conversation with someone in another language. You can also make your own documents and presentations available in other languages at different reading levels.

MODULE WRAP-UP

Inspecting the Colony

In this module, you mastered the essential elements of colony inspection. These inspection techniques will serve as your foundation for maintaining thriving colonies.

Honey Bee Nutrition

Bees use the nectar and pollen from many kinds of flowers to fulfill their nutritional needs. Foragers from just one colony can visit over 200 different plant species in one year! A Master Beekeeper learns the important floral resources in his or her region, understands the nutritional needs of the colony, and knows when and how to supplement this natural forage through feeding when necessary. In this module, we will help you make sure you are meeting the nutritional needs of your colonies.

READ

Honey Bee Nutrition Overview

Nutrition is intricately tied to proper brood development and honey bee health. Colonies with adequate nutrition can rear healthy brood, have well-functioning immune systems, and store food for times of shortage. Unlike solitary organisms, honey bee foragers are tasked with the duty to fulfill all the nutritional requirements for themselves and for their nestmates. They must ensure they gather food that contains sufficient carbohydrates, proteins, vitamins, minerals, and lipids so the colony can thrive. Furthermore, nutritional requirements vary depending on the age of the bees and the season. Brood and young bees require a protein-rich diet, while older bees feed mainly on carbohydrates. Honey bee colony nutrition is a dynamic process, and one arena in which beekeepers can sometimes intervene to improve health and productivity. What happens if the colony has inadequate nutrition? Honey bees' nutritional needs may not always be provided for with ever-increasing reductions in natural habitat, growing agricultural monocultures, and changes in climate. Management practices also affect nutrition, e.g., when bees are used for pollination services of specific crops. Inadequate nutrition can have several physiological and behavioral impacts on bee health. Bees that receive inadequate pollen during development live shorter lives as adults, transition to become foragers much earlier in life, and forage less often than bees that receive adequate pollen. These bees also perform fewer waggle dances to communicate good food sources to their nestmates, and when they dance, the information they convey is not always accurate. Insufficient protein during winter bee development results in reduced fat bodies, which shorten their lifespan. Colony survival can be in jeopardy if too few bees survive the winter. A colony will react to food shortages by rearing fewer brood and sometimes cannibalizing brood to acquire their nutrients. The multitude of stressors affecting bees today can also interact with nutrition. Poor nutrition can reduce bees' resistance to stress: they are less resistant to *Nosema apis* infections, Varroa mite infestations, and pesticides. In the same way, honey bees infected with *Nosema ceranae* live longer if they have more pollen available.

WATCH

Nutritional Requirements — The How and Why of Bee Food

Although natural forage (pollen and nectar) is almost always better than substitutes, beekeepers can help reduce nutritional stress by supplementing bees in times of need. At the Master Beekeeper level, we are delving deeper than how to feed your colonies to focus on the why. Why is it important to feed during certain periods? Why should you sometimes use 1:1 sugar syrup and sometimes 2:1? Is honey always better than sugar syrup? Why do your colonies need pollen from a variety of floral sources? Understanding the principles behind nutrition and feeding will help you anticipate your colony's needs. Video Transcript Honey bees consume carbohydrates in the form of honey and sugar syrup. And they need carbohydrates all year long to maintain the adult population and to fuel their day-to-day activities.

And even before a bee is about to take her energetically expensive foraging trip, she'll consume a little bit of honey to get some carbohydrates to fuel her. And, she'll store this honey in her crop, and digest it during the duration of her trip. And you can even tell which bees are about to take a foraging trip by looking at the hive entrance. And those that are taking foraging trips actually don't fly straight out of the entrance, but instead they sort of dip down before flying upward because they are weighed down by their food. So carbohydrates fuels everything for the way that adults are living, but it can also affect the behavior of adult bees too. So if you're having steady or increasing amounts of nectar, so a nectar flow, this can actually stimulate the colony to build comb and to start rearing brood. And the idea is that they're perceiving that there's a lot of nutrition in their environment, so this is a good time to start having larvae and boosting their population size. So, if you want to encourage comb building, or brood rearing, when there's limited nectar in the environment, then you can supplement your colonies with light sugar syrup to stimulate them the same way. So how much honey does a colony need? Well here in the Northeast, to fuel all their activities, colonies here, on average, consume about 95 pounds of honey during the summer. And during the winter, in order to stay warm and to begin rearing brood and melt in winter, colonies will consume on average another 50 to 90 pounds of honey. And there's a lot of variation on how much they're eating in the winter. This depends on the strength of the colony, the environmental temperatures and the amount of brood rearing that they're actually doing. So I often get asked how many pounds of honey do I need to make sure that my colonies have over winter? And I usually recommend to aim to provide them with 100 pounds of honey. So you wanna go on the safe side of things. So, while 50 pounds might be okay for some winters, it's often not going to cut it. So, aim to have 100 pounds of honey on your hive. Anything that's left over can be given to splits in the spring or can be extracted. Video Transcript Protein is a really important nutrient for reproduction in the colony. Protein is needed for brood production because brood is eating protein in the form of royal jelly and in the form of bee bread. And it's also needed for the development in young adult bees. So these are usually the nurse bees. And they need to consume bee bread in order to complete developing their hypopharyngeal glands, which are producing royal jelly to feed to the brood. So the building blocks of proteins are amino acids, and there's 20 amino acids that are needed to build all the proteins necessary for bee life. 10 of these amino acids are manufactured in the bees' bodies. But the remaining ten must be acquired through protein either by pollen ideally or through pollen substitute, which bees are going to then convert into bee bread and consume. So how much protein does a larvae need anyways to develop properly? It turns out that one larva needs about one cell of bee bread to develop. And a typical colony will consume about 50 pounds of pollen in total, in the form of bee bread, each year. So if you're the type of person that likes to calculate exactly how much pollen you need to be feeding your bees based on the number of cells that brood then go around ahead. But if you just want a general rule of thumb, then you can ensure that every deep hive body that has brood in it, has at least one to two frames worth of pollen or bee bread. So this includes full frames of pollen, but also pollen that's making up that rainbow arch around the brood. But sometimes, it's not always easy to base your colony's nutrition on the amount of pollen that you're seeing in the colony, alone. And a really good way to tell if brood has enough protein is by checking the amount of royal jelly in each brood cell that has larva. So, when you're looking in the cells that have young larva, you want to make sure that you're seeing a small pool of glistening white jelly, with a larva floating on top. And when pollen is especially plentiful, when there's a pollen flow on, you may see larvae with tons of royal jelly surrounding it. And

this is what we call wet brood, and these larvae have ample amounts of nutrition to develop into really good, healthy bees. But if pollen stores are low, you'll see very little jelly or the cell will be completely dry. And in this case we recommend supplementing your colonies with a pollen substitute just to make sure that they do have those nutrients to develop okay. Video Transcript The requirements and rules of vitamins and minerals for honey bees is not well understood. We do know, however, that several vitamins and minerals are necessary for insect life. And vitamins appear to be especially important for brood production and for the development of nurse bees hypopharyngeal glands. So, where do bees get these vitamins and minerals from? They mostly get them from pollen. And there's only trace amounts that they're getting from nectar. And there's starting to be this interest in supplementing your colonies with vitamins or minerals or even probiotics. And research is beginning to investigate whether there's positive effects of doing so. But right now the nutritional science is not yet developed enough for us to make clear-cut recommendations for using any of these supplements to promote colony health. Video Transcript Honey bees acquire their fats through the pollen, they don't actually get any fats from nectar in their diets. And fat content of pollen depends largely on the flower species just as sugar content in nectar depends on the flower species. And fats are stored in the bee's fat body which is a collection of tissues comprised of many fat body cells that are mostly found in the abdomen of the bee, surrounding the reproductive and digestive organs. But they're also distributed across the abdomen or sorry, they're also distributed across the thorax and the head. And here these tissues are storing fat and glycogen, which is stored sugar, to release for energy during periods of food shortage. And the biggest time here in the northeast that this is critical for is over winter. So fat nutrition is very important for wintering bees. And you want to make sure that bees who are reared in late summer and early fall have adequate fats in their diet. Because these bees are going to be the hive's winter bees. So you want to make sure that their fat bodies are large enough to provide energy throughout the entire winter. If they don't have large enough fat bodies to sustain them, then these winter bees can die before the winter is even over. So, it's critical that bees have access to adequate pollen from August to November in order to help them build their fat bodies. And they're building these fat bodies starting as larvae and continuing until they're young adults. So, if there isn't ample pollen at this time of year, then we would advise you to provide them with a pollen substitute in late summer or early fall. And sometimes, in your area, at least here in the northeast, we get a really big pollen flow of goldenrod. And this supplies a lot of pollen to bees right in that late summer period of when winter bees are starting to grow or develop. But this isn't predictable every single year. Sometimes, if you're not having that guaranteed pollen flow, then these are the times that you want to intervene as a beekeeper.

READ

Natural Forage

The flowering plants in nature provide bees with the nutrients they require in the form of nectar and pollen. Nectar Nectar is mostly sugar and water, but not all nectars are created equal. Plants vary in the amount of nectar they produce, the time of day it is produced, and the concentration of sugar in that nectar. Environmental conditions (such as drought) can also affect the quality and quantity of nectar available in a plant or across a landscape. Indeed, the concentration of sugar in the nectars of different species of flowering plants varies from 5% to 92%. Honey bees largely use the sugar concentration of nectar to guide their foraging decisions - nectars with high sugar concentrations are perceived as high

quality to bees. The main sugars found in nectar are sucrose, glucose, and fructose. Nectar also has small amounts of minerals, vitamins, fats, and enzymes. Because of this, honey is more nutritious for bees than syrup from cane or beet sugar, which only has trace amounts of minerals. Darker honeys, like buckwheat, contain more minerals than lighter honeys. However, these mineral-rich (darker) honeys tend to make bees produce more fecal solids, which can be a problem in winter or other periods of confinement when they cannot take cleansing flights. That is one reason some beekeepers choose to harvest all honey from the hive and heavily feed their colonies sugar syrup in the fall. Some nectar can also be poisonous to bees — and humans — due to very small amounts of naturally-occurring toxins. The most famous are rhododendron and azalea nectar, but despite their widespread presence in North America, poisoning incidents are rare. This field of red dead nettle makes great forage for honey bees. Photo credit: Jon Ryan Pollen Like nectar, not all pollen is created equal. Protein amounts in the pollen of different species can vary from 5% to 40%. Pollen must contain at least 20% protein to fulfill the nutritional requirements for bees. Pollens containing at least 25% are considered excellent protein sources. Only a few forage plants have been analyzed to determine their protein contents, as the costs for these analyses are quite high. Examples of poor protein pollens are cypress (2.5%), blueberry (13.9%), and sunflower (12.9%), in contrast to the good and excellent protein pollens of vetch (24%), white clover (25.9%), viper's bugloss (34.9%), and the Californian flower Padre's Shooting Star (61.7%). Pollens also vary in their amino acid content. Some, like canola, have sufficient amounts of all of the 10 essential amino acids that bees are not able to produce. In contrast, others may have low amounts or not have certain amino acids altogether. For example, pine pollen is deficient in four essential amino acids, dandelion pollen is deficient in two, and weeping willow and corn pollen are deficient in one. Because of this, honey bees cannot rear larvae on any of these pollens alone, no matter how much they consume. So, a colony collects pollen from a variety of flowers, and the amino acids from the different sources complement each other. Table 1. Amino acid requirements for honey bee nutrition (based on DeGroot (1953)).

Proportion Required	Amino Acid in Diet (%)
3	Arginine
1.5	Histidine
3	Lysine
1	Tryptophan
1.5	Phenylalanine
1.5	Methionine
3	Threonine
4.5	Leucine
4	Isoleucine
4	Valine

TOOL

Valuable Pollen Plants

Timing is also an important part of pollen's value to bees. Some flowering plants, like dandelions, do not have the most nutritious plants categorized by their pollen value, but because they flower early in spring when little else is available, they are very valuable to colonies. The early pollen acquired from dandelions helps colonies build up in early spring, and have a strong forager population ready for the first big nectar flow. Therefore, it is important not to mow the dandelions in your yard if you can tolerate their appearance. You might also encourage your neighbors to keep their dandelions or at least avoid spraying them, to keep unnecessary pesticides out of your colonies. Please review honey bee-visited plants categorized by their pollen value tool. In the USA, these plants are all important for honey production, whether because of the nectar's abundance, timing, or sucrose concentration. The table describes the value of the pollen of these plants, based on whether bees often collect pollen from them, if they are abundant, and whether the pollen is available during a season when not much other pollen is available, e.g.,

dandelion flowers in early spring. Note that this table does not evaluate the nutritional value (protein or amino acid content) of that pollen. The information from this table is from Eva Crane's book, "Bees and Beekeeping," and DC Somerville's report, "Nutritional value of bee-collected pollens."

READ

Learn the Blooms in Your Area

■ As beekeepers grow to better understand bees, Key Points they also become more connected to the Through simple observations, environment around them. Through simple beekeepers can learn the observations, beekeepers can learn the important honey plants and pollen important honey plants and pollen plants that plants that their bees are visiting. their bees are visiting. Each year, the foraging patterns of colonies vary, based on weather and how it affects the availability of different floral resources over the course of the year. Some years, plants bloom successively, and bees can be observed bringing in pollen from many different species. In other years, periods of poor weather can delay the bloom of early flowering plants until their bloom times correspond with later flowering plants. In these times, bees may visit the early flowering plants less than normal. The observations of when and what is blooming year to year are integral to being a successful beekeeper. We encourage you to identify the flowering plants that contribute to your honey crops. The colors and flavors give you clues about which plants they come from. There is a great variety in honey colors here in the Northeast and in other parts of the world. For instance, basswood honey and fireweed honey are water white, alfalfa honey and orange blossom honey are white to light amber, blueberry honey is light amber to amber, tulip poplar honey is dark amber, and buckwheat honey is dark brown. Clockwise from top: basswood, Japanese knotwood, and clover honey Flavors range from caramel and marshmallow to mint and even bitterness. Honey flavor is determined by a synergistic interplay between the chemistry of the nectar source, enzymatic and ripening processes by bees, environmental conditions, and post-harvest handling by humans, all of which collectively create honey's extraordinary diversity of tastes and aromas. It is important to recognize that, despite of the many factors involved in determining the flavor/aroma of honey, the primary determinant of honey flavor is the chemical composition of the nectar from which bees collect sugars, namely the ratio of different sugars present in the nectar (ex: sucrose, fructose, glucose) as well as the presence of allelochemicals (ex: terpenes, phenolic compounds). Plants vary widely in that respect, both at the species level, but also within plant species, since the particular profile of these sugars/allelochemicals will change based on the environment these plants experience as they grow, for example, soil type, temperature, and rainfall. Master Beekeepers should also become familiar with the plants from which their bees are collecting pollen. Different plants produce a variety of different colored pollens, and there is even variation in the colors of pollen from the same species of plant. This can depend on how long ago the pollen was released by the plant, the amount of dust or other particles incorporated in it, and the amount and color of the nectar/honey the bee uses to moisten the pollen to pack it in her pollen baskets. We encourage you to observe bees collecting pollen and take note of the color of their pollen loads and the type of flower they are on. There are many field guides for identifying wildflowers. In the same way, as you inspect your colonies, notice the colors of pollen stored in the comb. Here is a pollen color and bloom chart to help get you started. Information from this chart is from the book "The Pollen Loads of the Honey Bee" by Dorothy Hodges. Long description ■ Left Column — Spring Seasons Early Spring: This section features plants

like snowdrop, alder, hazel, yellow crocus, blue crocus, elm, poplar, willow, Siberian squill, and almond. Their pollen colors range from light yellow and orange to darker greens and blues, with hexagons displaying the spectrum of each plant's pollen hues. Early to Late Spring: Includes plants such as red deadnettle, plum, wild cherry, pear, crab apple, dandelion, Norway maple, and apple. The associated pollen colors expand to include reds, light to dark yellows, greens, oranges, and browns. Late Spring to Early Summer: Lists plants like common beech, garden bluebell, birch, sycamore, white horse chestnut, red horse chestnut, hawthorn, buttercup, oak, lilac, and mountain ash. The pollen colors show a broad palette that includes shades of yellow, orange, green, blue, and brown hues. Right Column — Summer Seasons Early Summer to Summer: Plants listed are lupine, sainfoin, raspberry, oriental poppy, white clover, blackberry, red clover, asparagus, birdsfoot trefoil, and vipers bugloss. Their pollen colors are widely varied, ranging from light yellows and oranges to darker hues like deep blues and greens. Summer: Features plants like wild honeysuckle, cornflower, knapweed, hogweed, borage, and leek. The corresponding pollen colors continue the diversity, including yellow, green, blue, orange, and brown tones. Summer to Late Summer: Lists plants such as onion, buckwheat, mullein, field thistle, evening primrose, purple loosestrife, dahlia, Virginia creeper, tobacco, goldenrod, and aster. Their pollen colors maintain consistency with earlier sections, showcasing a wide array of yellows, greens, oranges, blues, and browns. The chart provides a comprehensive overview of the plants that bloom across each period and their respective pollen appearances, aiding in identifying when they produce pollen and its characteristics. The seasonal groupings are visually distinguished by colored boundaries around each section.

READ

When to Feed Carbohydrates

Because natural forage is not available or sufficient all year round, you may find that you need to feed your bees, particularly in the following situations: In winter, if stores are low. In early spring, before much is in bloom, to provide food if their winter stores are low and to grow the population by stimulating brood rearing. In spring or summer, after installing a nuc, package, or swarm into a hive. In spring or summer, if there is a local dearth of flowering plants and little nectar/honey in the hive. In late summer and fall, to ensure sufficient winter stores, particularly if you have taken off honey. When rearing queens, for cell builders and mating nucs (if you do not have honey frames to spare). During and after requeening, to encourage acceptance of the new queen (if there is no nectar flow). When you want the bees to draw out foundation, and there is no nectar flow. When administering certain treatments to colonies. For example, Terramycin to help control European foulbrood is mixed with powdered sugar, Fumagillin for Nosema control is dissolved in sugar syrup, and oxalic acid for Varroa control is dissolved in syrup when applied using the dribble method.

READ

How to Feed Carbohydrates

There are several ways to feed your bees carbohydrates. Explore the options below to learn more. Frames of Honey The easiest way to feed bees is with frames of honey that you have saved from earlier in the season (that you froze) or took from another hive after making sure that they are pest- and disease-free. It is important to only feed bees honey from your own disease-free colonies; do not feed

them honey purchased from a grocery store or from another beekeeper, as there could be disease spores present (American Foulbrood or Nosema). In the early 1990s, Dr. Roger Morse and K. Steinkraus screened 82 random samples of Canadian and American honey from honey producers and found 8.5% of those samples contained spores. It is hard to know how comparable those levels are to those of honey produced today, but it certainly demonstrates that there is some risk to this practice. If you do not have extra honey frames on hand, sugar syrup helps meet bees' carbohydrate needs. Additionally, bees eating sugar syrup produce less fecal matter than when they eat honey, especially the dark, mineral-rich honeys. This is most important in winter when the bees must go long stretches without a chance to defecate outside the hive. Therefore, feeding your colonies a heavy sugar syrup in autumn (usually until the end of October in the Northeast) is a fine practice. Only feed your bees refined cane or beet sugar (both are sucrose), liquid sucrose, or high fructose corn syrup (fructose). Never feed bees mannose, raffinose, galactose, or lactose sugars; these sugars are toxic to them. Other sugars, such as brown sugar, molasses, powdered sugar, maple syrup, or raw sugar, contain impurities that are indigestible to bees. Powdered sugar contains cornstarch to prevent clumping, which can pose problems with bee digestion, so it should not be used as the main method of bee feed (although in small quantities, like when using it to monitor Varroa mites, it does not appear to have detrimental effects on the colony). Refined sugar, because it is refined of all other components, is the best. ■ Types of Sugar and Syrup ■ Even refined sugar can be problematic if it is overheated. Overheated sugar syrup may caramelize if the water is boiling when you mix in the sugar, or if the mixture cooks too high and for too long. The bees cannot digest the caramel and may kill them.

Syrup Concentration You probably know that you should feed your bees a syrup of one part sugar to one part water in the spring, and two parts sugar to one part water in the fall. But why? Light sugar syrup — one part granulated sugar (cane or beet) to one part water — supplements bees' natural forage in the spring and stimulates them to draw comb. Its sugar level is similar to that of rich nectar, so colonies respond as if there is a nectar flow coming in: workers draw comb and the queen lays eggs. When feeding light syrup at the end of winter or early spring, feed continuously until the bees begin bringing in nectar. This ensures the colony is adequately fed and is well prepared for when the nectar flow starts without any interruption: honeycomb is being constructed, and the queen continues to lay eggs. Stop feeding as soon as the nectar flow starts and the bees begin bringing in more than they are consuming. Otherwise, the bees may store the syrup in supers that you later extract. Such honey is considered adulterated by law. If bees have a surplus of sugar syrup in their brood nest, they may move some of it upward to be stored in supers if they need to create more room for the queen to lay. In late winter/early spring, you can also consider putting a solid feed (dry sugar, fondant, or a candy board) in the hive. They will consume this solid sugar and not store it; however, it does not stimulate brood rearing in the way that syrup does. In the fall, the beekeeper should instead use heavy syrup (two parts sugar to one part water). Syrup with a lower ratio of water to sugar is easier for the honey bees to evaporate the water from and cap for their winter stores; they may not have time to evaporate a lighter syrup before the cold weather hits, and their efforts may raise moisture levels inside the hive. Excess moisture can condense on the inner cover above the cluster and drip down onto the bees, potentially killing your colony mid-winter, so the less moisture in the hive when temperatures are cold, the better. The extra water in unripe honey can also bulk up the honey bee hindgut quicker, so bees consuming this honey may need to take a cleansing flight sooner. If the weather isn't conducive, this can lead to dysentery in

the hive. ■ **High Fructose Corn Syrup** ■ Some beekeepers feed their bees high fructose corn syrup instead of sugar syrup. On a large scale, this option is cheaper and more convenient than sugar syrup, as it is already in liquid form. Nor, despite concerns, does it have detectable quantities of neonicotinoids, an insecticide applied to virtually all corn in the United States. But bees fed high fructose corn syrup produce less wax and have a smaller body mass than those fed sugar syrup, and their colonies have smaller populations compared to bees fed sugar syrup. Additionally, if high fructose corn syrup is stored improperly (above 105°F), the sugars inside can dehydrate, producing hydroxymethylfurfural (HMF), which is toxic to bees even at relatively low doses. Adding Stimulants

Some beekeepers add essential oils to their sugar syrup. For instance, lemongrass oil is very attractive to honey bees and can stimulate them to consume sugar syrup when it is mixed in, as it contains some of the same components as Nasonov pheromone: citral and geraniol. Honey B Healthy is the most common feeding stimulant on the market. It contains lemongrass and spearmint oil. Adding this stimulant to the syrup also helps delay the growth of mold. But there is not enough scientific evidence to suggest that adding other essential oils is beneficial to colony health. There is a variety of equipment that you can use as feeders for sugar syrup. What type is right for you depends on your colonies' needs, your management goals, the size of your operation, the equipment you have available, the costs, and logistical concerns such as how often you can refill the feeder. Feeders can be placed inside or outside the hive. Top feeders and bucket feeders placed upside down on top of colonies (above the inner cover and below the telescoping or outer cover) are the easiest way to feed. They can hold several gallons at a time and can be refilled without opening the hive beyond lifting off the outer cover. A Boardman feeder is also convenient, but its location outside the hive can encourage robbing and make it susceptible to freezing in cold temperatures and to moldy growth in the summer heat. It is also small. Inside the hive, frame feeders (which are the size and shape of frames) are popular. They can hold one or two gallons of syrup at a time and may need to be refilled often, which requires opening the colony. With some practice, you can learn to refill these feeders without fully opening the hive. Frame feeders also require a sort of ladder (a screen or textured surface) for the bees so that they do not fall in and drown. ■ **Feeders** ■ While top feeders require no extra equipment and can simply be placed on top of the hive, a bucket feeder usually requires the beekeeper to have an additional empty deep hive body on top of the colony to set the bucket feeder into. Others may cut a hole in the top cover to set the bucket over, and when the bucket is not in use, insert a plug into the hole. Methods vary depending on the operation size. One caution with bucket feeders: the feeder should be level and sealed properly; otherwise, the liquid feed may immediately drain out of the feeder and could drown your bees. Feeding in open barrels is common for commercial beekeepers in North America. In this feeding practice, a barrel is filled with sugar syrup, straw is placed inside the barrel to act as a flotation device for the bees to stand on without drowning, and it is left in a yard for all the colonies to feed on. The open barrels or buckets are usually placed out of the line of sight of any colonies a few hundred yards away so that the bees have to actively forage to find the site initially like any other nectar source. This method is less time-consuming for the beekeeper. In addition, if barrels are located outside the bee yard, it can help direct robbers and wasps away from weak colonies that are susceptible to robbing. Some provinces in Canada have banned open barrel feeding, with the concern that it could facilitate disease spread, encourage robbing behavior if the barrels are placed close to hives, and concern for the public should they observe the frenzied bee activity that barrel feeding incites. There is also concern that populated

colonies collect more syrup than the weaker colonies that need it most, leading to the strong getting stronger while the weak get weaker. To offset this, weak colonies can be fed directly with frame feeders at the same time that open barrels are available. Since the strong colonies are feeding at the barrel, robbing should not pose an issue. To date, there is no evidence to support these concerns, as, to our knowledge, no scientific studies have compared barrel feeding with other feeding regimes. Finally, the barrels may also attract mammal pests or be blown over. You may need to experiment with a few different feeders to find what works best for you. Regardless of which model you choose, you need to monitor the system carefully, especially to detect and prevent robbing at individual hives and at the feeders, to check whether they need to be refilled or dumped, and to remove dead bees and pests. Sugar syrup can become moldy or ferment after a few days in warm weather, so if the bees are not actively taking up the syrup, you should remove it and clean the feeder with hot soapy water and rinse thoroughly. If the feeder is not properly washed out, it can contaminate any syrup you add to that feeder again. Different feeder styles are shown below. From left to right: bucket feeder, frame feeder, hive top feeder, Boardman feeder.

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Dry Feeding Carbohydrates in the Winter

When feeding bees during winter (when they ingest it directly rather than storing it), do not use sugar syrup; solid sugar is better. Bees can hold in their feces for a long time — but only up to a certain point. The water in the syrup is the problem: as the solids already in the digestive tract absorb it, they expand and may push the bees past their limit. Bees also need to consume a greater volume of diluted syrup to receive the same amount of carbohydrates as they would if they consumed a small volume of pure dry sugar. Dry sugar also does not stimulate brood rearing, as it does not resemble nectar. This is good for winter feeding since you do not want to stimulate brood rearing until mid- to late winter. In the Northeast, beekeepers usually resume syrup feeding in March, as the occasional warm days will offer bees opportunities to take cleansing flights. Several options for dry feeding include using dry granulated sugar, candy boards, fondant, or winter patties with a small amount of protein added.

Granulated Sugar The first method of using dry sugar, usually referred to as the “Mountain Camp Method,” is simply to pour a few pounds of dry granulated sugar onto a sheet of newspaper or wax paper on the top bars of the frames beneath the inner cover. A rim or spacer is added below the inner cover to accommodate the large lump of sugar. The dry sugar can also be placed on top of the inner cover, but the bees may not be able to break their cluster to retrieve it there, whereas if it were on the top bars, the bees would have easy access to it. The dry sugar has the additional benefit of absorbing moisture moving up through the hive, and as it does so, the sugar hardens on the newspaper. This helps reduce moisture problems in the hive. Dry sugar is not used during the warmer months of the year when bees are actively flying outside the hive, as they tend to carry the sugar granules out of the hive. You will then notice piles of sugar on the landing board or in front of the hive, which may attract other pests and predators.

Candy Boards A candy board looks like an inner cover with a deep (1" to 2") rim into which you pour hot bee candy. Let it cool and harden before placing it on top of the hive, in place of the inner cover. Recipes for candy boards vary, but the basics (sugar + water) are the same. You can order pre-filled candy boards from most bee supply companies, or you can make them at home.

Fondant Alternatively, you can feed your bees fondant, which can be found at some bakeries or

purchased from bee supply companies. Be sure to buy fondant that only contains water, sugar, and corn syrup. Do not purchase any with food coloring or other additives. Place thick patties of fondant on wax paper on the top bars for bees to reach directly; they will chew through the paper and carry it out of the hive. Without the wax paper, the fondant could drip down onto the cluster, so it should always be placed on wax paper. Winter Patties Lastly, winter patties (such as Pro Winter Patties®) are similar to fondant. They are comprised of water, sugar, corn syrup, a very small amount of pollen substitute, and sometimes essential oils used as a feeding stimulant. These are usually purchased in one- pound patties in waxed paper and should be placed right on the top bars of the frames in late winter. The sugar provides energy to sustain them throughout the winter. In contrast, the small amount of pollen substitute provides them with some protein, so they can rear some brood in late winter without building up as fast as they would with a regular pollen patty. Any time you feed your bees in winter, be sure to do it on a warm day or as quickly as possible to avoid chilling the bees. There usually is no perfect time of day to open a hive in the winter, but try to feed on days when the sun is shining, temperatures are mild, and there is little to no wind.

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Feeding Protein

In the Northeast, pollen is available from early March, sometimes late February, through the beginning of October. Depending on the weather variations year to year, and the species of flowering plants in your region, there can be periods during the bee season when there is a pollen dearth — a period of very low pollen availability. If your bees do not have enough pollen, you will see lots of dry larvae with little jelly and little pollen stored in the hive. In this case, you may decide to feed them bee bread, pollen, pollen supplements, or pollen substitutes. Timing is important: Supplementing brood rearing with pollen substitute in late winter and early spring can help a colony build up its population before spring nectar flows, allowing more bees to forage for nectar and produce more honey. Many beekeepers feed pollen substitute in late winter, usually starting in mid-February and continuing until mid-March if their goal is to produce strong colonies for making splits or spring pollination. In autumn, however, bees use the reduction in natural pollen as a cue to increase their production of winter bees. If fed pollen substitute to prolong the protein coming into the hive, a colony continues rearing summer bees until that incoming protein slows down. For this reason, it is not necessary to feed pollen substitute in September or October, as it continues to promote the production of short-lived summer bees; however, it is not harmful to the colony either. It is best if colonies have three to five frames of bee bread stored in the hive at the beginning of November, entering winter. If honey production is your goal, you want to time the growth of the forager population with the nectar flows in your region. Keeping in mind that it takes a worker bee 21 days to develop from an egg, and an additional 20 days or so until she has transitioned to become a forager, a mid-May nectar flow requires ample brood to develop by the beginning of April at the latest. It is advisable to start even earlier. Many beekeepers in the Northeast begin feeding pollen supplements or substitutes as early as February, when the colony starts building its population. Similarly, if you manage your bees for pollination services, you also may plan to supplement your colonies' protein to stimulate brood rearing and guarantee the growers a minimum number of frames of bees. This should begin a minimum of 45 days before the crop in question is blooming if there is not a lot of brood already present. Again, it is advisable to start earlier if the crop

your bees are pollinating blooms at a time when colonies may not be at their peak strength, such as apples in May. On the other hand, feeding may not be necessary if your bees are going to pollinate crops that bloom when colonies normally have strong forager populations, such as pumpkins in mid-July. Lastly, if you are keeping honey bees for minimal pollination in your garden, modest honey production, and just the joy of working with bees, then you may not wish to feed protein in the spring to build strong colonies. Colonies naturally adjust their brood production in accordance with what pollen is available in the environment. We do, however, recommend feeding sugar syrup in spring, when the colony needs it, to keep it alive until regular nectar is coming in, and feeding protein if there is a strong summer pollen dearth in your region. In summary, feeding protein is recommended in the following situations, depending on your goals: In early spring, to kick-start brood production, if you want to build the strength of your colonies for summer. Beekeepers typically begin feeding pollen substitute five to six weeks before natural pollen is available. In spring or summer, if there is a local dearth of flowering plants, young larvae do not have adequate jelly in their cells. When rearing queens, for cell builders and mating nucs, if you do not have pollen frames to spare. Before, during, and after providing pollination services to ensure adequate nutrition, especially if the hives will be placed in an extensive monoculture field without much floral diversity. Preparing pollen supplement patties to feed colonies in early spring.

READ

Protein Feeding Options

There are several different options for supplementing your bees with protein. Stored Pollen As in feeding carbohydrates, the most straightforward way to feed bees protein is with frames of bee bread that you have saved from earlier in the season or taken from another colony (after making sure that they are pest- and disease-free). If you do not have extra frames on hand, pollen or pollen substitutes can help meet their protein, fat, vitamin, and mineral needs. Fresh pollen is the most nutritious of your options, and pollen substitutes the least. You can collect fresh pollen using a pollen trap during a pollen flow and then use it later during a dearth. There are a variety of models on the market, such as the Sundance™ pollen trap and the Superior Pollen Trap™. Most models have two settings, one to collect pollen “activated,” and one that allows foragers to bypass the trap and bring pollen into the hive, “inactivated.” In general, we recommend activating a trap only when you see many foragers returning with pollen baskets — 10 to 20 entering the hive per minute — and only for a few days at a time, with several days between uses. If you want to collect pollen for longer, you will need to provide the colony with a supplement or substitute to fulfill their protein requirements. Depriving the colony of too much pollen may slow their brood production or lead to malnourishment. Monitor your colonies regularly to ensure there is ample royal jelly around the larvae and that the colony is not under stress from the trap — perhaps overheating if the trap is blocking a screened bottom board or preventing adequate ventilation at the entrance. To store pollen to use later to feed your bees, dry and freeze it. Pollen can be lightly dried in a low-humidity room, spread out on a cookie sheet for a few days, placed in an open container in a frost-free freezer for a few weeks — the freezer draws out moisture — or dried in a food dehydrator. Pollen dried for too long in a food dehydrator can cause the pollen to become too dry and grainy, so care should be taken as to how long pollen is dried. Pollen is dry enough when it can be gently rubbed between your thumb and index finger without disintegrating into dust. Once the pollen is

dry, store it in sealed Ziploc bags in the freezer. Pollen stored at room temperature loses 76% of its nutritional value after one year, but dried and frozen, it can retain its nutritional properties for up to 11 years. We generally recommend keeping pollen trapped from each colony separate from each other, as pollen can contain disease spores and other contaminants. The easiest way to do this is to put all the pollen from one colony in a large Ziploc bag and label it. If you are harvesting pollen from many colonies and this practice is unrealistic, at the very least, make sure not to harvest pollen from colonies that are diseased or have experienced a pesticide poisoning event. To feed pollen to a colony, you can make a pollen patty by mixing the trapped or purchased pollen with sugar and water, two to three parts pollen to one part heavy sugar syrup, or enough to give it the consistency of dough), or you can pour pollen pellets into drawn comb and spray sugar water on the comb to make them stick in the cells. The bees will process it into bee bread. If you are purchasing pollen for feed, it is best to purchase irradiated pollen. This pollen has undergone a process to remove disease spores. Pollen Supplements Pure pollen is the most nutritious avenue to feed bees, but the second-best method is to provide them with a pollen supplement. Pollen supplements are a mixture of a pollen substitute — a product that contains no pollen — and natural pollen. The added pollen makes the supplement more attractive and nutritious to the bees, and they are more likely to consume it. To prepare a pollen supplement patty, mix pollen substitute purchased from a bee supply company with at least 10% pollen — preferably more — that you have trapped or purchased. A pollen supplement patty placed on top of the brood box. Pollen Substitute Lastly, if you do not have pollen, you can feed bees the pollen substitute alone. There are a variety of protein substitutes available from bee equipment companies. Some examples include Bee Pro®, FeedBee®, MegaBee®, UltraBee®, and Pro Winter Patties® — this last one contains only a small amount of protein and is fed in late winter; see the previous section. Beekeepers have long made their own pollen substitutes — the most common ingredients being soy flour, skim milk powder, and dried brewer's yeast — but this must be done with care, and we generally recommend purchasing substitutes from bee supply companies. Some ingredients must be quite specific to be safe for honey bees: for example, skim milk powder must be lactose-free since lactose is toxic to honey bees. Others must be processed in a specific way: for example, soy flour must be processed using the expeller method and heated to remove certain enzyme inhibitors that can be unhealthy to bees. Manufactured pollen substitutes are analyzed and tested to ensure that they contain the necessary amino acids and vitamins, minerals, and fats needed for optimal honey bee nutrition. These products can be fed to the bees in many ways, but they are most commonly fed in a dry pollen feeder outside the hive (that the foragers find) or placed directly into the colony as a patty sandwiched between pieces of wax paper to retain its moisture. Honey bees collecting dry pollen substitute. Photo credit: Jennifer Tsuruda Protein Feeding Considerations When you are feeding bees protein in the colony, location is important. Place pollen patties directly above the brood nest so that they are accessible by nurse bees. If you are inserting pollen frames, the type of brood you place them beside matters. Bees are more likely to use protein that is adjacent to open larvae than protein next to capped brood. This stands to reason, as foragers usually deposit their pollen loads in cells adjacent to open larvae. Finally, feeding sugar syrup at the same time can also stimulate bees to eat the pollen. When feeding protein in late winter and early spring to promote early spring brood rearing, supply the bees continuously until they are regularly bringing in natural pollen to allow for the maintenance of brood that is already developing.

Honey Bee Nutrition

In this module, you explored the diverse nutritional needs of honey bee colonies and the vast array of floral resources they utilize. Your study of regional plants, colony nutrition, and supplemental feeding strategies will help you make informed decisions about supporting your colonies' dietary requirements throughout the year.

Techniques in Managing Colony Population

A colony's population changes throughout the season. Maximizing a colony's health and achieving your beekeeping goals requires understanding the needs of its changing population. In this module, we will provide an overview of managing colony population dynamics.

READ

Annual Population Cycle of a Honey Bee Colony

Successful beekeeping is fundamentally based on understanding the annual cycle of a honey bee colony. The population ebbs and flows with the seasons, and the division of young and old, male and female, summer and winter bees, changes over time. The following illustration demonstrates the population of a healthy honey bee colony in the Northeast that does not swarm. Long description ■

The graphic shows the rise and decline of the adult honey bee population throughout the winter. In January, the population is around 5,000, which steadily grows until it peaks in mid- June at 40,000 to 50,000 bees. It then slowly declines until mid-August. After that, it takes a sharp decline, and once again reaches around 5,000 bees in December. In winter, the colony has a population of around 10,000 workers, which is steadily dropping. They are clustered in a ball about the size of a basketball, keeping the small brood nest warm. In mid to late January in the Northeast, the queen begins laying the first eggs of the new season. Around the first of March, the colony reaches its lowest population as the winter-bee cohort that emerged last fall dies off. But the brood increases steadily, tended to by these overwintered bees. The colony can consume up to one frame of pollen per week to sustain this brood rearing — all this before anything outside the hive is blooming. With the first early blooms of the season and incoming nectar comes a rapid increase in brood rearing. The newly-emerged adults boost first the nurse, and then the forager population, and the colony grows very quickly, quadrupling or more by the end of June. This degree of growth requires favorable conditions. In some years, the availability of forage is delayed by poor weather, and colonies exit winter with a weak population. If resources are not available to support early colony growth, old winter bees can die off faster than new bees can replace them, and the colony population drops. This is referred to as “spring dwindling”, and it can imperil the survival of the colony. In this case, the beekeeper must intervene by feeding the colony until good forage is available and the bees are numerous enough to take advantage of it. As healthy colonies with plentiful resources grow rapidly in the spring, congestion in the nest can initiate swarming behavior. Thus, the period of most swarms here in the Northeast is from mid- May to mid-July. As July winds down, the population begins to decline. It is not uncommon for there to be a dearth of nectar and pollen at this point in the summer. Brood rearing slows, and foragers born during the earlier “baby boom” die off. In some areas, there is a final fall flow with incoming nectar and pollen in August and

September. This nectar can be critical for colonies to store for winter. The pollen helps provide a small increase in brood rearing, and the population experiences one final, small boost. At this point in time, from mid-August to mid-September, there is a second swarm season in this area. When incoming pollen starts to lessen, it is a signal for the colony to begin rearing winter bees. The summer bees die off, and brood rearing ceases around November. Winter weather typically arrives — and stays — in the middle of this month, with the colony again forming a cluster, and slowly decreasing in size until next January, when the cycle repeats itself.

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Reasons for Splitting Colonies

In times of population increase or peaks, you may wish to split your colonies. Taking a strong colony and dividing it into two smaller colonies is a very simple practice, though there are steps to ensure that both resulting colonies (the splits) are healthy and productive. Once you are confident in making splits in your operation, you can become a much more self-sufficient beekeeper. There are a few reasons why you may want to split colonies:

- **Replace Your Winter Losses** Some degree of colony loss, especially over winter, is expected. In the U.S., beekeepers report that winter losses of around 17% are acceptable. These losses can be made up by purchasing more packages or nucs, but the most affordable option is splitting strong survival colonies. Beekeepers who replace their losses through splitting and manage these splits to become strong colonies that earn them acceptable profits are practicing sustainable beekeeping. Growing your operation through splitting colonies is a slower process than simply ordering many packages; however, it is arguably the best way to expand. Increasing slowly allows you to learn more about beekeeping, become comfortable adjusting to managing more colonies, and evaluate whether and when you need to hire help. Colonies can typically be split once a year if they are strong enough, but more than once a year may result in colonies that are too small or have too few resources to survive over winter. Splitting a colony is essentially an artificial swarm — taking around half of the bees and placing them in a new home. Splitting a colony can reduce the instinct for colonies to swarm, although it cannot always prevent it. If your colony has already produced many swarm cells, or if they are very genetically predisposed to swarm, they may swarm regardless. Splitting a colony to prevent swarming only helps if it has not yet begun swarm preparation, that is, before the queen has laid eggs in the swarm cells.
- **Grow Your Operation**
- **Prevent Swarming**
- **Reduce Varroa Mite Levels** Performing splits to reduce Varroa mite levels has become increasingly popular in recent years. It is a treatment-free method that can be used on its own or in combination with other cultural or chemical treatments. Performing a split can be effective at reducing mite levels because it creates a break in the brood cycle in the part of the split that does not receive the queen. When the colony is broodless, Varroa mites are unable to reproduce, and their population decreases. The most effective way to use splitting to reduce Varroa levels is by removing the queen altogether and leaving both resulting colonies to rear their own queens. In this case, both colonies reap the benefits of an interruption in the brood cycle.

READ

How To Make a Proper Split

Making a split is simple. Splits should only be performed on strong colonies: crowded with bees and at least six frames of brood of any age and four frames of mixed food stores. Remove four to five frames and all the bees they contain from the original hive and insert them into a new hive body. Aim to move over two to three frames of brood (all stages) and two frames of food (honey and bee bread). Check each frame for the queen so you can make sure she remains in the original hive, or so you can remove her altogether if you want both colonies to rear their own queens. If you do not have at least one frame of pollen, cumulative from the food frames and the brood frames, for each of the two resulting colonies, you can supplement your colonies with pollen, pollen supplement, or pollen substitute. Insert a caged queen or a queen cell into the new colony, or let the colony rear a new queen. For the latter, there must be nurse bees, pollen, and eggs, to guarantee that there are larvae sufficiently young to still develop into queens; less than 36 hours. The easiest way to ensure there are young enough larvae is to have eggs present. Replace the now missing frames in your original colony with drawn comb or foundation. Put these in the end positions of the hive bodies regardless of where you pulled the other frames out from. It is best not to divide the remaining brood nest or to separate the brood nest from other frames of food stores. Similarly, add drawn comb frames or foundation to fill up the rest of the split. Drawn comb gives them a head start to fill them with stores and brood, but if you are uncertain about your comb sanitation, or do not have any on hand, you can use foundation. Feed your split, and add an entrance reducer for at least the first few weeks, just as you would for a newly established colony. Finally, shake three frames of nurse bees from the original colony into the split. It is important to have enough bees to keep that brood warm. Nurses congregate in the brood nest to accomplish this. The original colony was strong, so it should be able to tolerate the loss of these bees. If you are keeping this colony in the same bee yard, shake an additional frame of bees into your split. Any foragers in the split are going to fly back to the original colony, and you need to make sure enough bees will remain. The colony with fewer foragers will adjust as well as it can. Some nurse bees become early-onset foragers and start collecting pollen and nectar. The colony may build up its honey stores more slowly and needs to be fed before winter. If your goal is honey production and you have the land, we recommend moving splits into a separate bee yard at least two miles from the original colony. Any foragers in the split will learn their new surroundings and not travel back to their original colony. These foragers can start collecting nectar immediately. Should you transport your colonies that rear their own queens, make sure you do it before the colony has begun rearing queen cells, as queens are very fragile during their development and may be injured during transportation. Return to the split two weeks later to see if the colony is getting established and whether the queen is laying. A strong colony (top) can be split into two smaller colonies (bottom). Long description ■ The graphic shows how a strong colony can be split into two smaller colonies. The top of the strong colony has frames one through three filled with honey, frame four filled with pollen, frames five and six filled with open or capped brood, frames seven and eight are filled with pollen, and frames nine and ten are filled with honey. The bottom of the strong colony has frame one filled with honey, frame two filled with honey, frame three filled with pollen, frames four through seven filled with open and capped brood, frame eight filled with pollen, and frames nine and ten filled with honey. The two split hives have one stack of frames. Each has frame one as a syrup feeder frame, frame two is drawn comb/foundation, frame three is pollen, frames four through six are filled with open/capped brood, frame seven is pollen, frames eight and nine are drawn comb/foundation, and frame ten is honey.

WATCH

Splitting Colonies

Watch this video to see a split being performed. Video Transcript So we're going to go through the steps of performing a split. We're starting with a strong colony. Ours is comprised of two deep hive bodies but you can certainly split a strong colony in one deep hive body as well. The first thing we do is separate the hive bodies and look for the queen. Once we find her, we're going to put her in a cage for safekeeping. And we can't forget to put her back into the original colony at the end. Our goal here is to move over the following frames from our parent colony. Two to three frames of brood, one frame of honey, two frames of pollen and then fill the remaining spaces with drawn comb. So here is a good brood frame. We got capped brood on top, larvae on the bottom, some eggs. In here is a really good honey frame. We got capped honey on the top with some open nectar on the bottom. And here is an example of our pollen frame. So we can see some pollen in here. Okay so let's assemble our split. First we're going to put a feeder frame in the number one position and we're going to fill this with light sugar syrup. After that, we're going to put in a frame of drawn comb, followed by a pollen frame. After this one, we're going to put in three frames of capped brood. And even though you can't see it in this video I continue to fill up this box to make sure that we have another pollen frame, two more frames of drawn comb and then a final frame of honey in the number ten position. And we're going to shake about four frames of additional bees in it to help keep all that brood warm that we just transferred over. You can introduce a clean cell or clean cage between the two frames of brood but we're going to let this colony rear its own queen. It's got eggs and pollen and lots of young bees to help with the process. Because it's a small colony, we're going to add an entrance reducer and we're going to move it into another yard over two miles away. So, Eleanor is going to take this box after it has been stapled to the bottom board and move it. Regardless of whether the colony is rearing its own queen or if you're introducing one we recommend leaving the colony for a week without opening it. So the bees have some time to adjust and not be disturbed. After one week, it's good practice to come back and check on it to make sure that all is well in your split.

READ

Alternate Ways to Split a Colony

There are many alternate ways to split a colony, but the method that we describe in this course is the easiest, requires the least equipment, and is the least time-consuming. Other methods may not require you to find the queen in advance, but require multiple trips to the bee yard. We encourage you to try different variations and see what works best for you in your operation. All splits are based on the same few principles: The new “mini” colony must have all the necessary resources: brood, honey, bee bread, nurses, and foragers. The original colony needs to retain these resources, too — you do not want to deplete it as you want to end up with two functioning colonies. The colony with fewer resources and foragers may need to be fed. Foragers that are moved into the split will likely return to the original colony if the split is not moved at least two miles away. Nurse bees are required to tend to the brood in the split. The split needs to be provided with a new queen or rear its own. To rear one, eggs need to be present in the new colony. The best time to perform splits is early in the beekeeping season, around the same time as swarm season. In the Northeast, this is April through June. Splitting early allows bees

to have time to build their population and collect food for several months, which increases their chances of overwintering successfully. If you wish to split later in the season, monitor the new colony closely and feed them as needed so they can build sufficient stores and a large population for winter.

READ

Nucleus Colonies

Nucleus colonies (nucs) are small colonies contained in small hive bodies with only two to five frames. They are created by pulling frames and bees from larger colonies. Nucleus colonies are good to have on hand for extra resources — brood, eggs, nurse bees, honey, bee bread, or queens — and we encourage beekeepers to keep a few "resource nucs" for this purpose throughout the beekeeping season. Nucs can also be used for mating queens that you have reared or sold for profit. Lastly, they can be very useful for queen introductions. If you have a failing queen in one of your colonies and plan to replace her, it is good practice to place her in a nuc rather than killing her. This way, if the colony rejects your new queen, you can temporarily place the failing queen back into the colony until you are ready to order a new one or try again. While there are many benefits to making nucs, they do require maintenance. If bees and resources are not taken from a nucleus colony and its population continues to grow, it may swarm. Swarming preparations usually begin once the nuc has reached a strength of four to five frames of bees. To delay swarming, a frame of brood can be swapped with drawn comb from another hive. Some beekeepers use their nucs to establish new full-size colonies. This has a double benefit of preventing swarming and of growing their operation. Nucs are ready to be inserted into an 8 to 10 frame hive body once they have reached four to five frames of bees. After the colony has grown for three to four weeks in a full-sized hive under favorable conditions in spring and early summer, you can pull another nuc out of it. In this way, you can continually grow your operation and have nucleus colonies on hand.

WATCH

How to Make a Nuc

Watch this video to learn the proper procedure and considerations for making a nucleus colony. Video Transcript Let's go through the process of making a nucleus colony. We're going to start out with a very strong colony that's at least in one deep hive body. And, in this example, ours is actually very strong, it's across three deep hive bodies. We start out with an empty nucleus colony, and we're going to separate the hive bodies of our strong colony, so that we can see what resources we have to work with. And here we have Eleanor in the bee yard helping us assemble this nuc. So we first put in one frame of honey right at the edge of nuc box. And we're going to follow this up with a frame of drawn comb. The drawn comb is going to give the queen more space to lay, or for the colony to fill it with food stores. It'll take time to fill up this frame, so the colony won't swarm in the very near future. After this the next frame we are going to add is a frame of brood. And this frame contains mixed brood, there is capped brood, open larvae and eggs. The frame after that is going to be another brood frame. In this case this one has much more capped brood to contribute to the colonies population once it emerges. The last frame we're going to add is a frame of pollen. In this case, this frame is only partially filled with pollen, and the rest is simply drawn comb. Again, giving the bees a bit more space to work with. And some beekeepers will forego the frame of pollen altogether, in favor of a full frame of drawn comb or

foundation. Having more drawn comb gives the colony more room to grow and can prolong the time it takes for a bee keeper to perform more management. And we here are providing a frame of pollen because we want the nuc to rear its own queen and it needs protein to do so. With a nuc, we don't shake additional bees into the box, we want them to grow their own mini colony slowly. So we put on the lid and we'll be moving it to another bee yard to minimize the amount of bees that will fly back to the parent colony.

READ

The Composition of Constructed Nucs

Your constructed nuc should have the following frames in these positions: Long description ■ The graphic shows the orientation you should have your frames in for a constructed nuc. There are five frames. Frame one is honey, frame two is drawn comb, frames three and four are mixed brood, and frame five is pollen. While we give this formula for making nucs, the formula can vary depending on the resources you have available and the amount of time you can afford to spend managing your nucs. For instance, some beekeepers do not provide a frame of pollen but will instead provide a frame of foundation to slow the growth of the colony before they need to conduct swarm prevention measures. Bees are quite resilient animals and can tolerate variations in nuc arrangements, so follow this procedure as a guide, but don't feel obligated to always follow it to a tee. Double Nucleus Colony A double nucleus colony is the size of a standard deep Langstroth hive body but can house two four-frame nucs separated by a solid divider board. The double nucleus box has two entrances, on opposite sides from each other, so that each colony has an entrance. Having entrances on opposite sides reduces the chance of bees drifting from one nuc colony to the other. Double nucleus colonies made up in spring or early summer can be supered once they have nearly filled their frames of drawn comb and have drawn out and partially filled any frames of foundation. Before adding a super, add a queen excluder. Workers from both nucleus colonies contribute to honey production in the shared super above. Supers are usually removed in early fall, and the two nucleus colonies overwintered in the double nuc box.

READ

Wintering Nucs

Many beekeepers also make up nucs in mid to late summer, overwinter them, and use the surviving ones the following spring to replace any full-sized colonies that perished over the winter. This is a good way to be more self-sufficient and potentially save money. Creating nucs in mid to late summer has the added advantage of breaking the brood cycle and reducing Varroa mite levels. In general, nucleus colonies do not winter quite as well as full-sized colonies because they do not form as big a cluster. Nevertheless, it is possible to overwinter many nucs and have them available for next spring. The most success in overwintering nucs is to make sure they have a strong population of four or five frames of bees entering winter. The cluster should be about the size of a football, compared to a basketball-sized cluster for full-sized colonies. Nucs also require two full frames of honey to consume during winter. In mid-September, feed a heavy syrup to build up these carbohydrate stores if they are low. The colony's final weight should be close to 50 pounds. There are three common ways to overwinter nucs in North America. We encourage you to try one or more methods and find the style that works best for you. 1. A

double nucleus colony can be placed on top of a strong full-sized colony in early November. In the southern U.S., where winters are milder, nucs are often separated from the full-sized colony by a double screen board or a screened inner cover to prevent bees from mingling between the two colonies. Heat generated by the bottom colony can pass through the screen and help warm the cluster in the nucleus colonies. However, moisture from the bottom colony may also rise upward into the nuc, so this practice is less common in the Northeast and Canada, where lower temperatures generate more moisture. In the Northeast, double nucleus colonies are often placed on top of a strong full-sized colony, separated by a regular bottom board. Some heat can still transfer to the top colony. It is important that the bottom colony has an upper entrance to provide ventilation and to allow bees to perform their cleansing flights in the event the bottom entrance gets plugged with bees. It is also important that upper ventilation be provided to the nucleus colonies, usually in the form of a small hole drilled into the top half of the box and covered with mesh wire. The nucleus colonies are usually provided with insulation board placed on top of their covers and along the two longest sides, and are wrapped in tar paper. 2. A nucleus colony can overwinter on its own or can be placed adjacent to or stacked on top of other nucleus colonies to share some heat. It is usually provided with insulation and wrapped in the same manner as a full-sized colony, and is left outside, sheltered from the wind. 3. Nucleus colonies can be overwintered indoors, where they are less susceptible to temperature fluctuations and cold and do not require wrapping or colony insulation. Indoor wintering facilities are more common in the Canadian prairies, but some U.S. beekeepers will use them as well. Indoor wintering facilities should be maintained at 39°F to 41°F, dark, and ventilated to remove excess water vapor and carbon dioxide produced by the colonies.

READ

Merging Colonies Overview

If you find that one of your colonies has a small population in late summer or early fall, fewer than eight frames of bees per brood chamber hive body, you may wish to merge this colony with another to give those bees a better chance of surviving. Before merging colonies, you need to ask why the colony is weak in the first place. Rule out the possibility of disease as best you can with a thorough inspection. The last thing you want to do is introduce a disease from your weak colony into your strong, healthy one. Look for brood diseases and any other indication that your bees may be sick. Know your levels of Nosema and Varroa, which cannot be interpreted by a visual inspection alone. If you are unsure how to measure disease levels yourself, you can send a sample to the USDA Beltsville Bee Lab in Maryland for a free analysis. The colony may have been weak due to a failing queen, poor nutrition, predation, etc.; just be sure it was nothing contagious. Now you can get to the main event. You can merge two weak colonies, a weak colony with a strong colony, a nuc with another full-sized hive, or a newly caught swarm with another colony. Indeed, if you catch a swarm anytime from mid-July to September in the Northeast, we recommend merging it with another colony, as it likely will not have time to prepare for winter on its own. Merging is most often done in late summer or early fall when beekeepers are starting to prepare their colonies for winter. By this time, you should know which colonies are strong and which ones should be merged to boost their chances of survival. Merging can also be done in spring if you have one or more colonies that are weak coming out of winter, often from spring dwindling, when the population of winter bees is dying at a higher rate than new bees are emerging.

READ

Merging Colonies — The Newspaper Method

The easiest way to merge colonies is the newspaper method. If two colonies are merged and the bees from those colonies interact with each other immediately, they are likely to attack each other and the queen(s). The colonies need to meet gradually and become accustomed to each other's odors to accept each other as nest mates. To prolong their introduction, separate the colonies with newspaper. Long description ■ The graphic shows a stacked hive with two boxes containing a strong colony at the bottom and one box with a weak colony at the top. They are separated by a red line, which indicates a newspaper. Frame ten in the weak colony is a honey or a syrup feeder. It will take the bees a few days to chew through. By the time they do, their odors will have mingled, and they will be more likely to accept each other. Remove the queen from one colony. Consider making a nuc colony with her, to keep a nuc on hand, or in case the other queen is killed in the merge. If you are uniting two weak colonies, you can choose which queen you want to keep or let them fight it out. Remove the outer and inner covers from the other hive, the one you will add bees to. This colony usually has the queen you want to keep in it if it is the stronger colony. Place one or two layers of newspaper on top of the open colony. Do not let newspaper hang over the edge; in the event of rain, it can bring moisture into the colony. If the week's forecast calls for hot weather, cut a slit in the newspaper with a razor or scissors to improve ventilation. Do not use a hive tool, as the slit it creates is too large and allows bees from the two colonies to mingle too quickly. Rearrange the frames in the queenless hive so that they are consolidated to one deep — or two mediums if you use only mediums in your operation — with brood, honey, and bee bread. Shake all the bees into this single box. Place this hive on top of the newspaper. If you do not have at least one frame of honey in this hive, add one from another healthy colony or supplement it with sugar syrup. This colony will not be able to fly freely for a few days, so you need to supply it with food. Make sure there are no other entrances for the bees to use. If there are, seal them up. Leave the combined colony alone for one full week. The bees will begin to chew through the newspaper and begin mingling in one to two days. At the end of the week, check on the colony to make sure the bees have chewed through the newspaper and are mingling in harmony. Confirm that the queen is alive and laying. Foragers from the merged colony will continue to gather at the location the hive used to be. If they are gentle, they will find their way into other colonies on their own. If they are defensive, you can collect them by placing an empty super or hive body in the old location, returning at night when the bees are inside, and killing them with soapy water. Continue to catch them this way every day until they cease to be a problem.

WATCH

Merging Colonies

Watch this video to see a merge being performed between two colonies. Video Transcript Alright. So we're going to merge a weak colony with a strong one. The weak colony was originally in two hive bodies. But we actually consolidated it into one hive body by rearranging some frames and shaking all the bees inside. Because it's much easier to merge a colony that's in one hive body compared to two. But before we even started thinking about merging these colonies, we gave the weak colony a very thorough inspection to make sure it didn't have any disease symptoms. And we determined that the

colony was weak because the queen was failing. So here the weak colony is on the right and the strong colony is on the left. We want the queen from the strong colony to remain, so we're going to remove the queen from the weak colony, before we get started. Eleanor found it here in the frame and we're going to put it in a cage. And I like to transfer here temporarily into a nuc in case the queen from the strong colony gets injured or killed during the merge, I have a backup queen. Next, we're going to lay a single sheet of newspaper on top of the strong colony and cut a small slit in it to give the bees a head start at chewing through the paper and to allow the colony's odors to begin mixing. Then we place the weak colony on top of the newspaper. We have ensured that this hive body has at least one full frame of honey in it. Because the bees won't be able to exit the hive for a couple of days, they need to have food available to tide them over until they have successfully merged. The last step is to trim the edges of the newspapers so that there isn't any paper sticking out between the colonies. We do this because, if it happens to rain, the newspaper will actually absorb the water and bring it into the colony. And we don't want excess moisture getting into these hives. After one week, we're going to open up the weak colony and peer down to verify that the newspaper has been chewed through, and that the two colonies are successfully mingling as one.

READ

Merging a Nuc With an 8 to 10 Frame Colony

This can be accomplished in two ways: 1. Take all the frames out of a nuc and place them in a deep. Fill up the rest of the frames with drawn comb, making sure that they have at least one frame of honey for food to tide them over until they have merged. Proceed with the newspaper method. If you are performing this merge in spring, simply allow the colony to grow into those additional combs. If you are performing the merge later in the year, such as late summer, feed heavily to encourage the bees to fill the additional combs with food in preparation for winter. 2. Take the frames out of the nuc and place them in a full-sized deep without adding any additional frames. Proceed with the newspaper method. After the bees chew through the newspaper, you will need to consolidate the best frames from the nuc and the uniting colony so that they fit in a standard-sized hive. Four or five frames will be left over. They can be stored or used for other colony management.

MODULE WRAP-UP

Techniques in Managing Colony Population

In this module, you examined how colony populations shift throughout the seasons and how these changes influence colony health. Understanding these population dynamics will help you to better manage your colonies and achieve your beekeeping objectives.

Swarming

Swarming is the subject of books, debates, folklore, and more. Most beekeepers today try to prevent swarms and keep their bees where they want them, with strong populations making lots of honey. Swarming can cost beekeepers a honey crop, inadvertently spread pests or disease, cause fear and consternation among the public, and risk the survival of the parent colony if the new queen does not

develop and mate properly. Furthermore, it can be difficult to prevent if early signs or triggers are not recognized. On the other hand, with careful management and a little bit of luck, swarming can also be a way to expand your colony count, provide a brood break for managing Varroa mites or other maladies, requeen with a minimum of fuss, and/or spread the genes of relatively fit colonies around in the local environment. Swarms may also present an opportunity to you as a Master Beekeeper to educate the public. In this module, we will take a comprehensive look at swarming.

WATCH

Colony-Level Reproduction — Swarming

Swarming is colony-level reproduction, when one- to two-thirds of a colony's population — up to tens of thousands of bees — leave to found a new colony. The swarm consists of worker bees of all ages and a small percentage of drones, and it includes the colony's queen. It leaves behind developing queen cells for the original colony to requeen itself. The original colony experiences a long break in the brood cycle as a new queen emerges, reaches sexual maturity, takes her mating flight(s), and begins laying eggs. It takes about six weeks for new bees to emerge in the original colony. In this video, Tom Seeley discusses the process of swarming. Video Transcript What I want to share with you now is some information about swarming. And one thing I want to emphasize right at the get go, is that swarming is completely natural and it's good for a honey bee colony. It actually helps them with their disease control but even more importantly it's a natural part of a colony's reproductive process, colonies reproduce by swarming. So even though in your beekeeping classes you may have been taught that you want to prevent swarming - you want to take on this measure or that measure to stop swarming, you don't have to do that and you can let colonies swarm and your colonies will actually be a little better for them. They may not make as much honey but they'll be healthier. How does swarming work? Well the starting point is when a colony becomes strong enough to divide and reproduce itself. When it does, and that's usually in May or June. The nurse bees in the colony start rearing additional queens and they also interestingly put the mother queen on a diet. She will need to lose 30% of her body weight so that she can fly when the swarm leaves the hive. At the same time as the nurse bees are doing their thing, some of the foragers, some of the oldest foragers, start going out of the hive and no longer going to flowers but looking in the woods for potential nest sites and these are the nest-site scouts. And then a week or so later after those things have started on a day of good weather, the nest-site scouts who are outside, sensing that the weather conditions are good, trigger the bees that will be in the swarm to leave the nest and they fly out en mass and form the swarm cluster. From here, the nest-site scouts continue their process of choosing the home site. And once they finish choosing that home site, that can take a day or two, the nest-site scouts then organize the movement of the swarm to launch into flight and fly to its new home. And, that's the end of that swarming process. Meanwhile, back in the original nest of course, the young queens are emerging and if the colony is still strong because a lot of brood has emerged, the colony can emit an after swarm. But if the colony is actually weak at that point, the workers just let the new queens fight it out among themselves to see who will inherit the old nest. One thing I want to stress about the activity of the scout bees is that as they choose the swarms new home to set up the new colony, this is a life or death decision. Because they have to find a home site that gives their new colony good protection. Now a dream home for honey bees has a couple of key features. One is that the cavity that they move into is sufficiently spacious. It has to be 1 to 1.5 cubic

feet of volume to give the bees enough room for their combs, for having a good brood nest, enough space to store up the 40 or 50 pounds of honey the colony needs to get through the winter. So, cavity volume is critical. What they are seeking is about the size of one deep hive body. The other thing that also matters is the entrance. And the entrance has to be ideally for the bees. It's high off the ground. At least ten or 15 feet up off the ground. And this means that it's well above the snow level in the winter. It may make it less vulnerable to predators like bears. They also like the entrance to be small, so it's not a drafty home site, and it can be defended easily against things like yellow jackets, wasps. And they like the entrance to face south. Probably, I'm not sure, but probably because that helps the bees do their cleansing flights in the winter, if the entrance is sunny and southerly. I'll say a little bit about how the scouts actually make their decision, how they make the choice of their new home. It's a competition among scout bees that have found different sites. And it's a competition that's done by the bees adjusting the strength of the dances that they produce to advertise the site that they're visiting. I want to stress this is a friendly competition. All of these scout bees, have the shared interest of finding the best site for the colony's new home, for the swarm's new home. But it is a competition. And, the way it works is the bees adjust their dancing. That is, to say, how long a bee will perform a dance to advertise the site is adjusted to the goodness of the site. The better the site, the longer the dance. And, this means that, the discussion or the debate might start out with two bees coming back from two sites. One better, one poorer. The bee coming back, the scout coming back from the better site will do a dance that lasts three minutes. And the bee coming back from the poor site, a dance that lasts only one minute. Well if you came back a few hours later, you would find that there are many more bees dancing for the better site than the poor site, because there were many more bees that were able to attend the dance of the Bee from the better site. And the process then snowballs in favor of the better site because of the positive feedback of the buildup of the bees, the scouts for the better site. And all of those bees are still dancing three times as stronger, more strongly from the poor site. And finally, the process ends when one of the sites attracts enough, has reached what we call the quorum. One of the sites has accumulated enough scout bees visiting it, that the scout at that site can sense, they look around, they see there's lots of scout bees here, that means this site has had a lot of interest, sufficient interest to tell them that that site is going to should be chosen for their new home. Once the quorum has been reached, the swarm flies as a group to the new nest site.

READ

Swarming Timing and Initiation

The timing of swarms follows a seasonal pattern. Mid-May is the beginning of the major swarm season for the Northeast and Ithaca, NY, more specifically. Nearly 80% of swarms issue between mid-May and mid-July. Warm, clear, and calm days tend to be especially “swarmy”; even more so if they follow a few days of bad weather. A modest number of swarms issue during a smaller, second swarm season from mid-August to mid-September. These later swarms, if captured, require much feeding and care to store enough food and grow in population to overwinter successfully. Because so many factors contribute to population buildup and floral resources, some years see more swarms than others. Swarming is stimulated by crowding of adult bees in the entire hive or crowding by brood in the brood nest. In other words, swarming is initiated when there are high numbers of bees relative to the size of the hive. Bees can perceive this crowding in several ways, including a lack of ventilation or raised temperatures within

the hive. Their perception of crowding is also related to a decline in the concentration of queen mandibular pheromone (QMP), the pheromone that signals that the colony is queenright. This decline may be due to its dilution across a large population or because the brood nest is plugged with bee bread and nectar and the queen is restricted from laying (laying affects the amount of pheromone she produces). Abundant food resources coming in, whether through a nectar flow or through feeding, may also prompt swarming. As in many of the relationships among population, communication, and resources within a hive, this one is redundant: If resources are so abundant as to fill up the brood nest, thereby restricting laying, causing the amount of queen pheromone to decrease, and stimulating swarming, they are also abundant enough to support a colony that will lose a significant proportion of its workers and the honey they load up on before they leave the hive. Crowding and an abundance of resources are common conditions, particularly as colony populations build in spring. Because they are not managed to prevent it, feral colonies are estimated to swarm nine years out of ten. This may also be the case because of the smaller cavity size of most feral colonies. Finally, some subspecies and stocks swarm more than others. Africanized honey bees swarm more often than their European counterparts, although often with smaller numbers of bees. This is partly because their populations tend to grow faster. They also migrate by way of swarming, often for long distances, which partly explains how quickly they moved north after their initial release in Brazil. Russian bees are known for swarming as well. German and Buckfast bees generally swarm the least.

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The Preparation and Process of Swarming

Since ancient times, beekeepers have recognized that the process of swarming is the way honey bee colonies propagate themselves. Before the invention of moveable-frame hives that allow beekeepers to do splits and utilize other techniques for increasing their operation size, apiaries were increased chiefly by collecting and/or managing swarms. There are benefits to keeping bees in this way. When it swarms, a colony accomplishes more than just reproduction: Swarming requeens the mother colony (since the old queen is the one that leaves with the swarm), and requeening is an act known to reduce symptoms associated with several bee diseases. It promotes a brood break in both the mother colony and the swarm (daughter colony) that can help reduce mite loads in both. The swarm leaves behind toxins and pathogens that may have accumulated in the mother hive. It provides an opportunity for the swarm to settle in a more suitable location. Every swarm after the first one — and there may be three or four afterswarms, depending on the weather and strength of the colony — is headed by a young queen, ready to work hard to grow her new colony. In northeastern North America, some swarms start to issue in mid-May, but the swarm season really begins in earnest during June. To learn which colonies will be likely to swarm, inspections should begin during April or May, as soon as the weather becomes favorable and dandelions are in bloom. Signals that stimulate the swarm impulse Colonies begin reproducing when environmental conditions are favorable to support the new “daughter” colony’s rapid growth and ample food storage before winter comes. This period typically begins in spring. The strength of a colony’s swarm impulse, as we currently understand it, might be described as the sum of several trigger signals. Assuming that the weather has become favorable, meaning that nectar and pollen are abundantly entering the hive, the chief signals appear to be: The rapid growth in the number of workers and amount of comb The resulting abundance of young adult bees Congestion in the brood

nest The reduced transmission of queen pheromone When these signals collectively reach some as-yet-undefined threshold, the colony begins its swarm preparations. We don't yet understand how the different signals combine in different ways to trigger these swarm preparations. For example, a colony ■ under a young queen might grow vigorously in plenty of space and then swarm, while a neighboring colony that appears to be very similar in these same respects does not swarm. Why? Less-visible signals caused one colony to prepare to swarm but not the other. These differences could include faster transition from nurse to worker bee in the non-swarving colony (resulting in less abundant nurses; i.e., "young adult bees") or a better laying pattern conducive to less brood nest congestion. These less-obvious changes have caused many experienced beekeepers to fail to recognize when certain colonies are about to begin swarm preparations. Drone brood is produced A strong colony with many workers and worker brood in May will be building abundant drone brood, while a weak one will have little, if any, drone brood. Queens will not be raised and swarming will not occur in the absence of mature drones. Also, as soon as virgin queens reach sexual maturity, they need drones to mate in their environment so that they can begin laying worker brood as soon as possible. Although virgins do not mate with drones from their own colony, bee colonies are all depending on drones to mate with their virgins that other colonies provide. Therefore, the timing for drone production must precede that of queen production. An immature drone develops, pupates, and emerges in 24 days, but that newly emerged drone is still immature and unable to fly or mate. At about nine to 12 days of age, drones are sexually mature and are ready to mate. Thus, the time taken from when the first drone eggs are laid until those drones are able to mate with a virgin queen is, at a minimum, 33 to 36 days. However, it only takes 16 days for a queen to fully develop and emerge, and she sexually matures and is ready to leave the hive on her mating flight about one week later. So we expect to start seeing the colony produce drone brood at least 10 to 13 days before the earliest queen cells appear. ■ The hive becomes congested Congested hive conditions that initiate swarm cell production include: Worker volume (the space worker bees occupy) is greater than the free volume of the hive (there are more than 2.3 workers per mL of hive space). The volume of workers can be so high that bees will congregate outside the hive at night in these conditions. More than 90% of the brood comb is in use (unavailable for egg-laying). There is a large proportion of young (nurse) bees in the colony. Colonies preparing to swarm are often seen to have workers crowded into the brood area, even when there is room for them elsewhere in the nest. Not all congested colonies build swarm cells, and some colonies with plenty of space do build swarm cells! It turns out that all three of the above conditions must meet some as-yet-undefined threshold before swarm cells are built and other preparations for swarming commence. The next swarm preparation is the most obvious one for the beekeeper to recognize: the production of many queen cells! Workers construct plenty of queen cups in response to the perceived decline in QMP and to the other triggers we mentioned earlier. The queens also perceive the decline in QMP (as it returns to them after it has been distributed throughout the colony), and this stimulates them to lay eggs in queen cups, which then become queen cells. Empty queen cups are generally present whether or not a colony is preparing to swarm; their presence along the lower third of the frames, including between hive bodies, is part of the normal "architecture" of a hive. But bees ramp up the production of these cups during the preparation of the swarming process. However, the presence of cups on their own is not a signal of an imminent preparation to swarm; what matters is whether there is an egg or larva inside one or more. Queen cells for swarming can be very numerous (there are usually

at least 10), and are referred to as “swarm cells” due to the time of their appearance, their location along the lower edges of the brood nest, and their generally — though not always — large size. The stronger the colony, the more swarm cells are constructed. The colony often does not wait for a new virgin to emerge before it swarms; in fact, they usually swarm shortly after at least one of the new queen cells has been capped and before a virgin has emerged. If the larva in a queen cell is two days old, for example, the colony could ■ Queen cells are produced ■ swarm as early as one week later. If one or more of the queen cells are capped, the colony has probably swarmed already. This fact makes it very important for the beekeeper to locate a mature queen in the hive before cutting out all of the swarm cells! It is unclear why swarm cells are often produced in large numbers, when at best only three or four queens from that number may eventually head up a colony. It is often said among beekeepers that workers will “select the best queen from a number of queen cells,” and this may, in a sense, be true, as the first queen to emerge will kill all her sisters if she finds them. Queens about to emerge will pipe (make a chirping sound) in their cells, and these shrill calls may provoke any emerged queens to find and kill the singer. This murderous phenomenon favors the genetics of early maturation among queens: she who emerges first has the best chance of surviving. Selection over many generations thus favors queens that emerge in less and less time. This is a desirable feature if a queenless colony must rear an emergency replacement! But if that first queen kills her unemerged sisters, how can there be afterswarms? Afterswarms only occur in very large and strong colonies. Perhaps the sheer size of a large colony containing many swarm cells reduces the likelihood that the first virgin queen to emerge will be able to locate and destroy all her competition, thereby leaving enough cells for subsequent afterswarms. Another possibility is that the workers may control the behavior of their queens in ways we as yet have not observed and do not understand. Meanwhile, the drones continue to develop their flight ability until they are able to congregate at drone congregation areas, at which point they are competent to mate with virgin queens. Scout bees begin to look for nest sites In addition to raising queens and drones, scout bees begin to search for suitable nesting sites well in advance of the swarming event. Scout bees are almost all foragers, ■ but instead of looking for bright flowers that offer food, they start seeking out dark crevices that could be good nest sites. How scouts are recruited and which workers assume the job of scouting is unknown. The queen prepares to leave To produce all those additional workers needed for a colony to swarm, the old queen must be in full production mode. This means that she is devoting most of her body’s energy to egg production and egg-laying, and it means that her abdomen is heavy with eggs. To be ready to fly with the swarm, the queen must make physical changes. A few days before the swarm is cast, the queen reduces or sometimes even stops laying eggs. Her physiology alters: New eggs do not mature in her ovaries or descend the oviducts, and she becomes thinner and more restless on the combs, preparing herself for flight. The queen’s retinue of attendant workers assists in her weight-loss process. One way in which they do this is to decrease feeding the queen. Another way is to “vibrate” the queen by hanging onto her forelegs while rapidly moving their abdomens up and down — a “shake” that alerts her to change activity — over a period of about two weeks. Workers will also vibrate each other in this way prior to swarming. ■

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The Swarm Event

Bees prefer to swarm when the day is sunny and calm, and when temperatures are between 75 and 85° F (24 to 29° C). A beekeeper observing a strong colony with little activity at the entrance on the chosen day when nearby colonies are busily at work could reasonably expect that colony is about to swarm. As the swarming impulse reaches its peak, with ample pollen and nectar coming in, plenty of worker bees and brood, and both drones and queen cells in the hive, the bees will be inside on the combs, engorging themselves with honey in preparation for their journey. The old queen's restlessness is communicated to the whole colony by her rapid movements over the combs. The workers also cause some commotion, piping and performing "buzz-runs" where they briefly dash through a crowd of bees, vibrating their wings. Such a colony becomes noisy. The bees flying outside face the entrance, where some are seen flying in and out. Generally, between the hours of 11 AM and 3 PM, they depart in a mass exodus that is awesome to observe. As the swarm issues from the hive, you can stand beside the flurry of activity and the bees will simply ignore you. Forming a temporary cluster The emerging swarm has not yet decided where their new "home" will be. During the time needed to choose a permanent nesting site, the swarming bees cluster temporarily with their queen, usually on a tree branch not too far from where they emerged. Bees mark this location with Nasonov pheromone to help the members of the cluster orient to it. This temporary cluster is called a bivouac. In apiaries that have existed in the same location for many years, beekeepers report that emerging swarms use these same temporary cluster sites over and over again — a useful tip for the apiculturist who hopes to collect swarms! Because the temporary cluster site is not protected, the bees must hurry to find a suitable place to set up their new colony. If they are unable to find a good shelter, they will build their combs in the open. Such colonies do not survive Northeastern winters. In the wild (absence of beekeepers and buildings), the places bees choose are often ■ hollow trees, but they also consider other locations such as crevasses in rock outcroppings. With knowledge of the type of shelter bees are seeking, beekeepers can lure swarms into "bait boxes." Scout bees are looking for specific features in their new home. We will cover these details further along in this module. Evaluating potential nest sites There are about 200 to 300 scouts in each swarm. They fly from the temporary cluster to search in all directions. Although scouts begin searching for new nesting sites before the swarm emerges, the final decision is made after the swarm has formed its temporary cluster. The scout bees spend time examining each site very thoroughly. They fly around the interior of the cavity, inspecting every nook and cranny. When a scout locates a place she is enthusiastic about, she returns to the cluster and performs a waggle dance on its surface. Without the vibratory properties of a brood comb, this dance message doesn't penetrate the swarm very well. It is no small matter to communicate with up to 20,000 clustered bees! The ones in the middle of the cluster will have no idea that dances are happening at the surface and won't get any messages. Compounding this problem is that other scouts may also come back with information about other good sites. They also convey their information through dance on the swarm's surface. How does the colony learn which spot is the best one? How do they know to leave their bivouac and go to that place? To convince the colony that she has located a good home, a scout must recruit some of her sisters to return to the potential nest site with her. By flying to and fro, and by dancing to recruit more bees to visit, she can eventually persuade as many as 5% of her sisters to become enthused about that new place. They return to advertise the same site through dance. The more recruits that dance, the more bees are persuaded to check out that place the dance tells them about. A less desirable spot will produce lackluster recruitment and fewer dancers. Some of the new recruits remain at the location

and spend their time marking it with Nasonov pheromone. ■ Departing for their new home ■ Eventually, all the dancers stop dancing and force their way into the cluster of bees, piping or “quacking” as they go. This high-pitched sound is made by wing muscles, and it is transmitted to all the bees the piping scout comes into contact with. Each bee that is contacted in this way begins to raise her body temperature, as hanging in an exposed bivouac for a while can cause them to get a bit cool. Within about 10 minutes, the entire colony has heated up and is ready to fly. When the swarm reaches an internal temperature of about 95° F (35° C), the bees leave, suddenly, with a tremendous buzzing roar. They form a large, loose cloud that becomes organized by the scouts, who fly back and forth in a straight line through it. The axis of their flight path indicates the direction the swarm will take. As more and more bees recognize the flight path, the swarm shapes itself into a thick cigar, and off they go to their new address! A swarm may only take a few hours to find a suitable nesting site or it may be hanging onto a branch for several days before flying off to its new home. The scouts’ process of evaluating, describing, and deciding on a site is beautifully described in Tom Seeley’s book, “Honeybee Democracy.”¹ 1. Seeley, Thomas D. *Honeybee Democracy*. Princeton University Press, 2010.

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Swarming or Not?

Several behaviors might look like swarming to the uninitiated. The most common and the most easily confused are orientation flights and robbing. When the population is growing rapidly, and hundreds or even thousands of new bees are emerging every day during the active season, orientation flights may look like swarming, because of the sheer number of bees looping around in front of the hive. In contrast to swarming, however, they are simply returning to the hive entrance; the hive entrance is not flooded with bees trying to join the swarm. Robbing may also look like swarming, except bees are trying to enter the hive, not leave it (including through cracks between supers), and it is accompanied by fighting at the entrance. Other behaviors that may look like swarming include bearding, a common phenomenon in the summer months where the bees cluster on the outside of the hive to cool off or vent fumes; cleansing flights after a period of confinement (look closely and you can see the bees’ defecate); and absconding. Absconding is a mass exodus from a hive, when all or most of the population leaves — not to reproduce or to alleviate crowding but as a survival mechanism in the face of various threats and pressures (from pests and diseases or something threatening the bodily integrity of the hive). Look carefully at the bees’ behavior to identify whether, in fact, you are watching a swarm. Remember also that each of these easily confused behaviors is likely to take place more often during one time of the year: orientation any time of the year, but most impressively during the times of greatest population growth; robbing during a nectar dearth or the end of summer; and bearding on the hottest days of the summer, evenings, or immediately following a mite treatment where fumes are being released in the hive. In the same way, swarming is a specific seasonal phenomenon. While checking out heightened activity at a hive entrance, this beekeeper found himself become the gathering spot for a swarm!

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Swarm Prevention

Preventing swarms is difficult because the act of swarming is fundamental to the instinct for survival in the honey bee. It is, after all, the urge to reproduce. When conditions are right, the swarm impulse can become so strong that bees will overcome almost any obstacle a beekeeper can contrive to prevent it. Basically, a good swarm prevention technique is aimed at reducing the future (not imminent!) impact of one or more of the signals that the colony uses to begin its preparations for swarming. The method you choose will also depend on your beekeeping plan. Are you trying to produce as much honey as possible? Do you want to limit the number of colonies you are managing or grow your operation? Are you interested in producing nucs for yourself or to sell to others? The answers to these questions will help determine the best method for swarm prevention in your apiary. You know that swarm season is coming, so be prepared and take steps before queen cells appear. The classic way this is done in the Northeast begins at the first colony inspection, when the hive bodies in strong colonies are often reversed to reduce brood nest congestion and bottom boards are cleaned. After that, inspect colonies every week or 10 days. When you see drone brood appear, add honey supers as necessary and consider requeening or making splits, depending on your management goals. Let's cover a few more details for these practices.

Lower the Colony Population Shake some nurses into weaker colonies Shake a couple frames of nurse bees (not including the queen) into other colonies that would benefit from a population boost. This method equalizes the colonies in your yard without increasing the number of colonies. Before doing this, confirm to the extent possible that you are not inadvertently spreading disease or pests by moving around these adult bees. Because of this, it is advisable to confine this method to colonies in the same bee yard.

Make splits or nucs To reduce the population of bees in a colony, you might make a split or pull a nuc out of it. This method allows you to increase the number of colonies you manage, but reduction in the season's total honey yield may result, depending on the weather and how aggressively you split.

■ **Provide More Space** ■ If there is unused space beneath the brood nest, you can reverse the two bottommost hive bodies, which is routine for many beekeepers in spring. After that, timely supering allows bees to store most honey in supers boxes instead of plugging the brood nest. In some cases, just adding a super on top may not be enough, although it certainly helps, provided that the foundation on the frames is already drawn out. Sometimes colonies temporarily ignore frames of foundation, as they do not offer immediate comb for queen laying and food storage. If the brood nest is still getting congested with honey, move some frames of honey out of the brood nest and into a super. Replace the removed frames with drawn comb to make space more immediately available in the brood nest for the queen to lay. Lastly, you can try expanding the brood nest, where you place frames of drawn comb between the existing frames of the brood nest. Be careful about doing this and chilling the brood nest when there are cooler temperatures.

Increase the Bees' Perception of QMP Increase the amount of perceived QMP in the colony by requeening with a young queen with pheromones stronger than those of her predecessor. This method results in no increase in operation size; you start with one colony and end with one colony. Finally, the perception of crowdedness is due in part to sensory signals of rising temperatures, limited ventilation, and a decline in the concentration of QMP. You can add more room, an upper entrance, or a screened bottom board to increase ventilation and lower the temperature.

■ **Adjust the Bees' Perception of Crowdedness** ■

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Swarm Control

Once you see swarm cells, the bees are motivated and will attempt to swarm. If you are not able to prevent these conditions from arising, you may need to try to prevent the act of swarming itself. Swarm control is trying to stop the process once the colony has already constructed swarm cells. There are a few strategies you can employ to try to stop the colony from swarming. Whichever method you choose should be accompanied by practices that help to reduce the swarm preparation instinct (e.g., providing more space) so that the colony does not continue to try to swarm several times. What you can do at this point that might help reduce the chance your colony swarms: 1. Remove the old queen and cut out all but a couple of swarm cells. Removing the old queen may not stop a swarm, but it does increase the likelihood that your colony will not swarm. You have induced a period of “queenlessness” in the colony. The brood nest congestion is relieved as sealed brood hatches, and removing the old queen simulates conditions in a post-swarm colony. Be prepared to check the colony in about two weeks to determine that there is a newly mated queen there. You can relax when you see fresh eggs and young brood. 2. You can cut out all queen cells if you do not find any that have been capped. Even a single capped cell could indicate the colony has already swarmed. If any cells are capped, you must first confirm the queen is still in the colony before cutting the cells. Seeing eggs is not enough, as the queen can continue laying some eggs up until the day she leaves. You must physically see her. This is a laborious way to prevent increase and maintain a single colony, but it may be your best option if you missed the early appearance of queen cells and it is too late to requeen, or if you did requeen and they are trying to swarm anyway. Make certain that you have found and isolated the queen by putting the frame she is on in a nuc box or by caging her temporarily as you ransack the brood nest. Some beekeepers use the cells they cut out in their newly made nucs and splits. You must find every last queen cell. They can be in any nook or cranny among the combs, so be sure to smoke or brush away the bees as you look, or breathe on them to get them to clear the combs, because a layer of bees can easily conceal queen cells. Return to the colony every five days to continue cutting cells throughout the swarm season until the impulse has passed. Each time you must isolate the old queen to protect her. (Yes, this is very labor intensive!) This colony will swarm any day now, if it hasn't already. The two swarm cells in this photo are already capped.

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Ineffective Practices for Swarm Control

Clipping the queen's wing Some beekeepers clip the queen's wing to prevent her from leaving with a swarm. Yet this does not always prevent the colony from swarming and, as the swarm leaves the colony, the queen remains in the hive or gets lost in the grass. The swarm usually returns to the hive, only to leave again as soon as a virgin queen emerges from a swarm cell. Note: Clipping a queen's wing can help inform a beekeeper of her age. The left wing is usually clipped in years that end in odd numbers, and the right wing is usually clipped in years that end in even numbers. This serves as a backup identifier if her paint dot fades. **Placing an excluder below the brood nest** Some beekeepers place a queen excluder between the bottom board and a lowest hive body in an effort to prevent the swarm from leaving. The idea is that the queen won't be able to fit through the excluder to exit, so perhaps the bees will stay inside. Unfortunately, this method does not work. The swarm instead leaves with the first virgin that emerges and can fit through the excluder. An excluder in this location can also reduce ventilation and cause the colony to overheat. Neither of these practices works, and neither

mitigate the triggers that lead to a colony initiating the swarm process.

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My Colony Swarmed! Now What?

If you see no new eggs or young brood when inspecting your colony that has swarm cells, or if you find a queen cell that has been opened, your colony has probably swarmed. There will also be fewer bees in a post-swarm hive, but that is not always evident to the beekeeper. Of course, you will want to know if there is a queen in the post-swarm colony, but in such a situation, the best thing to do is close up the hive and wait. If a virgin is present, doing an inspection can increase the chance that you injure her, that she gets lost, or that the bees fail to accept her. In two to four weeks, return to check for new brood. If you are especially anxious to ensure that the colony becomes queenright quickly, you can transfer a frame with eggs and very young brood to the colony. If there is no virgin present, the bees will immediately build emergency queen cells using the brood in the frame you've transferred to them. If there is a queen present, the bees will simply rear the brood in the frame you've given them.

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Should I Let My Colonies Swarm?

A good beekeeper is a good neighbor to other beekeepers and the public and is motivated to help protect the health of native pollinators as much as possible. Our philosophy at the Dyce Lab is to prevent swarms from becoming loose in the wild to the best of our ability. In suburban or urban areas, it can be stressful to the public and your neighbors to have bees take up residence in buildings or in trees close to public walkways. More than that, feral colonies that are unmanaged can introduce viruses and other diseases to your neighboring beekeepers' colonies as well as to wild bees. Uncontrolled swarms (those that leave without being caught and hived) cannot be managed for diseases and mites.

Therefore, we do not recommend allowing colonies to swarm naturally and consider it an essential beekeeping skill to take steps to prevent swarming and to recognize when a colony is preparing to swarm then take appropriate counteractive measures. As Tom Seeley¹ says, swarming is indeed a healthy, natural behavior for honey bees. This act of reproduction can reduce Varroa mite loads, as 20% to 35% of adult mites leave with the swarm. Swarming also provides one to three weeks of a brood break (a period where there is no sealed brood), which can further reduce mites. The process results in a young queen in the colony, and queens produced during swarming are often high quality because they developed from good nutrition. Luckily, there are ways that beekeepers can simulate these phenomena without allowing swarms to get loose in the wild. Through creating intentional brood breaks, producing splits, and requeening, colonies can reap these same rewards without the risks. 1. Seeley, Thomas D. Honeybee Democracy. Princeton University Press, 2010.

WATCH

Catching and Baiting Swarms

Catching and baiting swarms is a fun and free way to obtain bees, although, of course, it's not foolproof. Catching swarms in a bait hive is the easiest method. Such hives should be set up in advance of the full-on swarming season since the scouts are already looking for new nest sites a week

or more before the swarm issues from the parent colony. During swarm season, you can check on your bait hives every week or so to see if you need to rehome any bees. As in all facets of beekeeping, there are many styles of bait hives that you can build or buy, but most derive from the same principles. Watch this video to learn what characteristics bees prefer in bait hives and how you can set some up. Video Transcript Now, what we've learned about how swarms choose a home is not only fascinating in terms of its basic biology but it's also very useful information for the beekeeper. Because it means that we now know how to put out bait hives that fulfill the bee's home site preferences so that these bait hives are very attractive to swarms, and so the beekeeper can catch swarms. And this is a way to get free bees, freebies. Let's review what matters to the bees. There's six things that they're seeking, at least six things that matter. They want the cavity to be sufficiently spacious. We mentioned already, about one cubic foot which is about 40 liters. And with respect to the entrance: they want the entrance small, they want it high off the ground, they want it near the bottom of the cavities so that it keeps the heat in the nest during the winter, and they want it to face south. And there's one more thing that matters to the bees. If they can find a cavity that comes pre- equipped with combs from another colony that already lived there, but died out, then that is very attractive to the bees as well, because that gives them a tremendous economy in their comb building. They can move right in, it's like moving into a furnished apartment. Okay, so with that knowledge, what do we do? How do we make a bait hive? Well, one way to do it is to use an old hive body and put in some old combs. But if you do that, you'll need to make sure it has only a small entrance. Only about two or three square inches is needed. You can do that by drilling a hole in the side with bait hive, or use a bottom board and just put block of wood to reduce the size of the entrance. The small entrance is very important to the bees. Another thing that's really important is you want to get that bait hive above the ground, 10 or 15 feet will make it much more attractive than leaving it at the ground level. You want to have it face south. And you may want to, make sure that it's visible. And what I mean by visible is, when I put up my bait hives, I put them on the edges of woods, rather than deep in the woods. And why is that? It's because, when you think about what is a scout bee doing when she's looking for a home site? She's going from one tree to the next, to the next, looking for knot-holes. And I think the scout bees find these dark knot-holes, or dark openings more easily if it's on the edge of the woods and not deep in the woods. I also put my bait high in places that are shaded. I don't have real strong evidence about this but I do think the bait hive is more attractive if it's not overheated by sitting in the full sun. You might wonder, should you be putting something inside? Well, I mentioned the combs, putting in frames of old combs to give it the aroma and the attraction of a site filled with combs, so that's one thing. Another thing that's good is to put in the pheromones, the bee lures. These are not essential but they're helpful and the reason they're helpful, I know from my experiments, is that they make the site, your bait hive, much more conspicuous to the scout bees if you've put in one of these swarm lures. Some bee keepers have asked me can I just put in frames as foundation? I would say yes you can, because that will still have the aroma of beeswax, especially if it's beeswax foundation. But also put in a comb or two of, a frame or two with old combs, to give it their full odor. And one thing I want to stress about this whole process of trapping swarms with bait hives is that, in my experiences, the swarms that you get are excellent bees. Why is that? It's because only good strong colonies with good queens are able to become strong enough to swarm. You might be worried you may have heard the wild swarms you'll catch are going to be too stingy. Well in my experience that's extremely rare, but if it happens big deal, you just requeen your colony. And I do

want to stress, I've had really beautiful bees with my bait hives, really, really beautiful bees. But I do want to caution you, the trapping success varies from year to year. Last summer, I put up ten bait hives and I caught ten swarms, so 100% success. The year before, I had 0% success. There just weren't many swarms produced. So it's profitable in the sense you get free bees. But it's also a lot of fun, it's kind of like magic, you put up this box and you can check it on your way home from work or something like that. And one day, sooner or later, you're going to find that a swarm of bees has moved into it, it has come to life, so you can have a lot of fun. There's one other way I'll mention too that you can have fun with the bees. You're all beekeepers, some of you will become bee trappers. If you want to become a thorough, complete bee enthusiast I encourage you to also consider becoming a bee hunter. Bee hunters go out and track down wild colonies of bees living in hollow trees. And that's a lot of fun — doesn't involve cutting down the tree, the thrill is in the chase, not in the kill. There's no killing of the bees or the colony, or collecting honey, taking honey out of these trees. You're just taking yourself on a natural treasure hunt. If you're intrigued by this possibility, I can steer you to a nice little book that's been written on this and published recently called *Following the Wild Bees*. Catching swarms that have settled on an object Every swarm is different, but all are probably a little further out of reach than they look! There is no way to provide specific directions; just remember basic safety guidelines (stings to you and onlookers, heights, tools), common sense, and patience. Some useful swarm-catching equipment includes the usual protective gear (veil and gloves), smoker, fuel, phone, and hive tool, plus pruners, extension-pole pruners or limb loppers, a long rope, a bucket or nuc box (ventilated to allow for transportation), bee brush, spray bottle with sugar water, ladder, drop cloth, extra frames and foundation, and duct tape. Some beekeepers use vacuum cleaners adapted to safely suck up bees. In many cases, you can shake or sharply rap the bees off a branch and into a box or bucket. If you manage to secure most of the bees in the swarm and particularly the queen, the rest of the bees should simply walk over in time. You can set them up in a new hive or combine them with an existing colony. A newly settled swarm builds comb very quickly, so you might use them to help draw out foundation in a hive that needs it. Be careful in handling a swarm, especially if you do not know its origins, whether it has been disturbed, or how long ago it left its parent colony. While swarms are typically gentle (with full stomachs and no resources to defend), a dry swarm that has been clustered for a few days and digested the honey it brought with it may be defensive. A swarm that has been sprayed with insecticide but not killed may also be defensive. Furthermore, since the window to catch swarms can be quite limited, you may need to work in weather conditions that are less than ideal. If you are using a swarm as an opportunity to educate curious onlookers, it will be a much more positive learning experience for everyone if you handle them safely. There are several ways to catch swarms, and every situation will require your analysis to decide how to proceed. Shown below, from left to right: Bees can be shaken off a branch into a hive body, they can be brushed gently off a surface into a hive body (made easier if the queen is first found and placed in the hive body), or, if the object can be moved, it can be placed in front of a hive body with a frame of comb. The odor of the comb will draw bees into the hive as their new home. Photo credit for middle and right photos: Jon Ryan

Get your name on a swarm list! To get your name on a swarm list in your area, talk to your local beekeeping club, nearby extension offices, emergency responders, and pest control companies. Note that layman's language is often imprecise around swarms. If someone calls you to pick up a swarm, try to confirm whether it is indeed a swarm of honey bees or if it is an established feral colony of honey bees, a wasp's nest, or something else

(asking for a photo may help). Your first opportunity to educate the public may be on the difference between honey bees and wasps. In New York State, the Dyce Lab maintains a swarm catcher directory. If you would like to be added to the directory, email dycebees@cornell.edu. Keeping a swarm in its new hive If you have captured a swarm from its bivouac and re-housed it, you'll be anxious to prevent that swarm from abandoning the home you have just put them in. Sometimes swarms do abscond from their new hives. You should give them several frames of drawn foundation and several frames of plain foundation (for room to draw comb, which they are primed to do once they have a new home). The queen will take a few days to begin laying eggs. Try not to disturb the colony while it is becoming settled. If you are really anxious to know all is well, some beekeepers recommend listening with an ear against the hive body. If you hear gnawing and knocking sounds, they are building wax, and all is well! Perhaps the easiest way to ensure that the bees don't abscond is to provide them with a frame of unsealed brood from another colony at the time you establish them in their new hive. Bees almost never abandon brood, so they should stay put.

MODULE WRAP-UP

Swarming

In this module, you studied swarming from multiple perspectives — both its challenges and opportunities. You learned how swarms can impact honey production, colony health, and public relations, while also discovering their potential benefits for colony expansion, pest management, and genetic diversity. This comprehensive look at swarming will help you manage this natural process and educate others about this fascinating aspect of honey bee biology.

Queen Management

The queen bee is the sole female reproducer and the mother of all the other bees in the colony. The health and productivity of a honey bee colony is inherently linked to the queen. A healthy queen lays over a thousand eggs a day in a solid brood pattern, giving rise to a strong population of 40,000 or more workers by early summer. A strong population also helps protect colonies from some pests and diseases that can take advantage of low honey bee populations. Beekeepers must always be aware of the queen's status and quality as they inspect their colonies, since failing or lost queens can put the colony at risk of dying. In this module, we will go over important concepts and management practices to make sure that your colony has a healthy, fertile queen.

READ

Queen Status

Since you first began to keep bees, you have known that the queen is the most important member of the colony. The queen is intricately tied to colony performance. Her health and fertility are responsible for the colony's population as well as the proportion of workers and drones in the colony. This workforce, in turn, influences nearly all aspects of colony performance: the amount of nectar and pollen coming into the colony, its likelihood of overwintering successfully as a strong population, its susceptibility to pests, and whether it will be able to reproduce through swarming. The importance of a

healthy, productive queen cannot be emphasized enough. Yet queens have been failing at high rates in recent years. Historically, beekeepers would replace queens every two or so years. Now, Bee Informed Partnership surveys show that as many as 50% of beekeepers are replacing failing queens every six months, and commercial beekeepers report queen failure as the most common cause of overwintering mortality. The reason behind the widespread queen failure in the U.S. is largely unknown but thought to be the result of multiple interacting stress factors, including pathogens, reduced sperm viability of the drones a queen mates with, and pesticides. Failing queens pose a big risk to the colony. Colonies that experience a queen event, where a queen fails or dies, are 3.1 times more likely to die within two months than a colony without such a problem. It is therefore very important — and even more so today — that a beekeeper is always aware of the queen status in a colony. If queen failure is detected early, intervention can help save the colony. At the height of the season, regular inspections every 10 days to two weeks should include observations of the brood quantity and quality, which gives you an indication of queen fertility. A poor or failing queen lays a spotty brood pattern or has brood of different ages adjacent to one another. Interpret what the brood pattern tells you about the queen with regard to the time of year. There may not be eggs in mid-autumn as the colony winds down in preparation for winter or in periods of a dearth when there is not enough nutrition to support brood development. With knowledge and practice, beekeepers will be able to identify, analyze, and respond to queen issues in their colony in a timely manner, allowing them to maintain healthy colonies year after year. How can I test the health of my queens? If you are interested in knowing how healthy your queens are, you can have them tested through the North Carolina State Queen and Disease Clinic. You can have queens screened for *Nosema*, seven different viruses, and insemination quality. The fees for each service, as well as more information on submitting samples, can be found at the NCSU Queen and Disease Clinic.

READ

Signs of Queenlessness

Queens can die for a variety of reasons: predators, old age, getting lost on a mating flight, pesticide exposure (including some miticides), pests or disease, or mishandling by a beekeeper. Recognizing when a colony is queenless and taking steps so that the queen is quickly replaced is an important skill to have as a Master Beekeeper. Roaring In less than an hour, bees recognize that their queen is gone because queen mandibular pheromone (QMP) is no longer being passed throughout the colony. They become very agitated and fan their wings. A beekeeper can detect this within an hour if the colony is small, such as a nuc, or within a few hours if it is a full-strength colony at peak population. This fanning is more pronounced than the usual fanning to cool the hive and is characterized as “roaring” because of how loud it is. At this initial time, the bees are more defensive and more likely to sting. The bees then begin to create emergency queen cells. Nurses enlarge the cells of several larvae that are less than 36 hours old to the shape and size of a queen cup and feed them copious amounts of royal jelly. However, if the queen died without leaving behind young worker larvae in the hive — perhaps in late fall after brood production had ceased for the year or another interruption — the colony will become hopelessly queenless unless the beekeeper intervenes by introducing a new queen or a frame of eggs/very young larvae. ■ Emergency Cells ■ Laying Workers After three to four weeks of queenlessness, some of the nurse bees may activate their ovaries and lay eggs, becoming known as laying workers. Again, this is because QMP is no longer detected. In queenright colonies, workers’ ovaries are inactivated. The

presence of this pheromone signals to them that there is a fertile queen that is laying eggs and passing on the colony's genes. Without this pheromone, up to 30% of workers may begin laying their own eggs. While healthy queens lay a single egg neatly at the bottom of the cell, workers often lay eggs on the side of the cell — their abdomens are not long enough to reach down to the bottom — and several different workers will lay eggs in the same cell, so you will see multiple eggs in one cell. Their eggs are scattered in cells haphazardly throughout the brood frames. All bees that develop from worker-laid eggs are drones because workers do not mate and therefore cannot fertilize their eggs. This cohort of drones is a final biological attempt by the colony to reproduce, as these drones can then pass on the colony's genes by mating with other queens in the area. Without intervention, such a colony eventually dies out. Since these are drone brood, further evidence that a colony has laying workers is the presence of scattered, capped drone cells throughout the worker brood area, that take the appearance of "bullets" because larger drones are developing in small worker-sized cells. Once they hatch, these drones tend to be smaller than their queen-mothered brothers for this same reason. Laying workers, however, are not necessarily a sign of queenlessness: Sometimes workers lay eggs even in the presence of the queen (about 0.01% of them), so one or two cells with multiple eggs are not a concern. Often, other nurse bees consume those eggs before they can hatch into larvae, but this "policing" behavior breaks down in advanced stages of queenlessness. ■ It is important to note that sometimes newly mated queens lay multiple eggs in a cell in the first few days of egg-laying as she gets into the swing of things. The pattern of these eggs differs from worker-laid eggs in that queen-laid eggs are at the bottom of the cell and often parallel to each other. After a couple of days, the queen should begin laying one egg per cell. A worker-laid egg. When To Expect You'll Observe These Symptoms During every colony inspection, you should be verifying the presence of a laying queen. If you are inspecting all of your colonies every 10 to 14 days, you can identify a queenless colony before laying workers become an issue. If you are inspecting only a proportion of your colonies during each inspection, check all colony entrances for signs of reduced pollen coming in or fewer bees foraging, and listen for roaring. Inspect colonies that exhibit any of these symptoms. If, upon inspecting the colony, you cannot find the queen or eggs, or if you see multiple eggs in a cell during a regular inspection, here are the clues that your colony may be queenless: ■ Signs of Queenlessness and Time Characteristic Time Observed In the following Roaring bees hours/days In the following Increased defensiveness days In the following Emergency queen cells days to three weeks After three to Multiple eggs per cell four weeks After about one Bullet-shaped drone cells month After about two A high number of drones months Drones that are smaller After about two than normal months Dwindling adult After about two population months

WATCH

Considerations for Requeening a Queenless Colony

As a beekeeper, you already know by now how to introduce eggs, young larvae, a queen cell, or a caged queen into a colony to address queenlessness. As a Master Beekeeper, it is important to understand timing. Sometimes not enough time has passed for a colony to rear its own queen from an existing egg or young larva. Sometimes too much time has passed and a population of laying workers has become established, making the introduction of a new queen difficult if not impossible. Watch this video to learn how to critically analyze a queenless colony so that you can make an informed decision on how best to manage it. Video Transcript So requeening a queenless colony isn't just putting in a

queen cell or a queen in a cage, closing up the colony, and walking away. As a master beekeeper, we want to look at the other considerations that we need to take when we're requeening a queenless colony. So for instance, knowing the developmental time of bees, knowing that they begin life as an egg for three days, and as a larvae for about six days, before they're capped brood. We can actually estimate the last time the queen was present. And knowing the last time the queen was present can let us weigh our options of how we are going to deal with requeening this colony. So now examine the age of larvae or brood the you are able to see. If you're seeing absolutely no capped brood then you know that it has been a few weeks the queen has been in there. If you are seeing larvae, for instance, then you know that it's been between nine days and maybe four days since the queen has been in there. And if you are seeing the eggs then you are okay the queen is in there and you don't need to continue with anything. So if you are seeing young larvae, larvae that look like they're less than 36 hours old, so about the size of an egg, and you're positive that there's no queen, then you know that you've caught this really early. The queen has been there as little as four days ago, and we actually have these young larvae that the bees can develop into a queen. So, if you want to have the colony requeen itself, you can close up the hive and walk away, return in one week to make sure that they have started building those emergency queen cells, and return two weeks after that to make sure that there is a new queen in your colony. The thing is, is that every time you're getting a colony to rear queens from young larva, you need to make sure that they have enough nutrition and especially protein. So make sure that the colony has ample pollen or provide them with a pollen substitute so that the nurses can rear some healthy queens with enough nutrition. Okay so in this case though, you might wonder why the queen died in the first place, why she failed, and you might be skeptical to have the colony rear a queen from her own young larva, in case there was a genetic issue. So, in this case, you might want to introduce your own queen of a specific genetic line or stock that you're interested in. And here you can insert a queen cage using the method that works for you and go back every few days to destroy any emergency queen cells that the colony is producing, and make sure that you go back and see that she's been released and accepted by the colony. Okay, so lastly if the queen hasn't been around for a few days or a few weeks even, but you're not seeing any laying workers, you can also introduce a frame of healthy young brood from a different colony. And with these young larvae that are less than 36 hours, again the queenless colony can rear a queen from those. So the frame that you're choosing should have eggs and young larva and lots of nurse bees on it. And the main thing is you want to make sure that this is healthy. You don't want to be introducing any sort of disease into the queenless colony. So here are the considerations for moving over a frame from one colony into your queenless colony. You want to make sure that you're moving the frame over as quickly as possible so that you're not leaving open brood out in the colony for more than a few minutes. If you need to go more than an hour or if you need to drive between apiaries with this frame, then put it into a new colony with bees to keep it warm and protected. And with bees developing around 95 degrees Fahrenheit, any deviation of a few degrees can cause developmental problems or death. So you want to make sure that they're not deviating in temperature for any given amount of time beyond a couple minutes. Now, also it's important that you're not just introducing a frame with eggs and brood on it, but that you're introducing nurse bees too. So your colony might have been queenless long enough to the point where any nurse bees that used to be in that colony have now developed into foragers. And now, there really isn't any bees that can care for and help rear a queen from this young larva that you're about to introduce. So,

the nurse bees that you're going to introduce are typically bees that are found on any brood frame. So, when you're moving over the frame of bees, keep the bees that are on it into that new queenless colony. You can also shake nurse bees off from other colonies, just to make sure you have enough in there. And again, supply them with pollen or pollen substitute if the reserves are low in the colony. Okay. So what if you're in a situation where the colony has been queenless for a long time? There isn't any capped brood in there and it's now at the point where you're seeing laying workers. And in this case, you can't simply introduce a new queen. The laying workers are likely to kill the new queen that you're going to introduce. So here you need to make sure that every laying worker is gone. And since laying workers are the young bees, usually nurses, since young bees can activate their ovaries but foragers can't, then you just need to make sure that all of the young bees are out of the hive. And this is fairly easy to do it because young bees don't usually fly and they haven't yet taken their orientation flight to locate their way back to the colony. So the procedure that you do is to remove every single frame from the hive and shake and brush all of the bees off, the nurses and the foragers, all of the bees off from the frames and do this about 20 feet away from your colonies. And you're going to also do this to the hive boxes for any bees that are standing on the side of the hive boxes. Then reassemble everything, put the frames back in the colony, and wait until the next day for all the forager bees to return and then introduce your queen. So the forager bees will fly back and all the young bees that have activated their ovaries, all those laying workers are actually going to die outside because they can't find their way back to the colony. And in this case, some of the foragers will revert back to being nurse bees to care for any of the eggs and larvae that the new queen is laying. But it might be helpful to shake some additional nurse bees into the colony just to make sure you do have some of those present. And then, keep an eye on the colony over the next few weeks to make sure that the new queen is establishing herself and that brood is being tended to. An alternative method to requeening a colony with laying workers requires a bit more equipment and time but is also quite effective: Introduce one frame of open brood with nurse bees into the laying worker colony every few days. The open brood releases brood pheromone, which helps suppress worker ovary development. It takes time to suppress the workers' ovaries and for them to be willing to accept a new queen; sometimes up to three weeks. As new open brood frames are added, the beekeeper can check for the construction of supersedure cells. This indicates the colony is ready for a queen. The beekeeper can now either 1) allow the supersedure cells to develop — the most popular option, 2) destroy them and introduce a queen of their choice, or 3) merge the colony with a queenright colony using the newspaper method.

READ

Tip for Introducing a New Queen

In recent years, nearly 30% of newly introduced queens die within six months in the United States. Either they are immediately rejected or they are superseded after a few weeks. Queen acceptance can be particularly challenging when installing packages because the queen in the cage is not related to the bees in the package. To improve queen acceptance, introduce a frame of young brood and eggs at the same time you are introducing a new caged queen. Research from North Carolina State University finds this practice increases queen acceptance by 200%. In a 2021 study, researchers tracked queen acceptance in colonies for 12 weeks after introducing them. They compared queen acceptance rates among colonies that received 1) a frame of young brood and eggs, 2) brood ester pheromone, and 3)

no brood or pheromone in the control group. They found 86.7% of queens that were introduced along with brood were accepted, compared to 53.3% of queens that were introduced with brood pheromone, and compared to only 33.3% of queens that were introduced in the conventional way without any brood or pheromone.

READ

Requeening to Improve Health and Productivity

It is common practice among some beekeepers, especially commercial beekeepers, to routinely requeen colonies to improve health and productivity. Requeening in response to a problem can occur at any time of the year, but routine requeening is generally done in early spring or at the end of summer/beginning of fall. Requeening in the spring can be easier than requeening in late summer or early fall, as colony populations are lower and it is easier to find the old queen. Also, because a new queen is most easily accepted when there is food coming into the hive, it can be easier to requeen during a spring nectar flow or to feed the colony at the same time. Spring is also the time when colonies are most inclined to swarm and produce new queens. Requeening in fall, however, is gaining in popularity as a way to interrupt the brood cycle and therefore reduce Varroa mite levels. The colony also enters winter with a young, vigorous queen. Here, we do not cover how to requeen a colony, but rather when and why a beekeeper may want to requeen. Reduce Impacts From Pests and Disease Chalkbrood, sacbrood, small hive beetles, wax moths, and robbing from other honey bees can be overcome or handled better by colonies with strong populations. In fact, wax moths and small hive beetles tend to be secondary symptoms of an underlying issue and are not normally problems on their own. For chalkbrood, requeening is a recommended course of action if the colony does not clear the fungal infection on its own in a few weeks or months. A new queen that is freshly mated should start laying eggs and building up the population to overcome these issues. Requeening in summer to reduce Varroa mites works because it creates a break in the brood cycle. As mites reproduce in honey bee brood, this break means that mites cannot breed for this time. This has the effect of driving down their numbers and making them more vulnerable to mite treatments that target them while they are in their phoretic stage (outside of the brood cells, on the bodies of the bees). Letting the bees requeen themselves guarantees the largest break in the brood cycle since it takes many days before a new queen develops, hatches, mates, and starts laying eggs. You can also requeen with a queen cell or a mated queen in late summer. This won't cause a break in the brood cycle but may provide other advantages, such as more control over the genetics; e.g. if you introduce a queen of hygienic stock. Requeening in ■ summer to reduce Varroa can be accompanied by a mite treatment that works best in broodless periods. Maintain High Levels of Brood Production Beekeepers can also requeen their colonies routinely to ensure sustained high levels of brood production. The queen stores the sperm from her mating flight(s) in her spermatheca for her entire life and never again mates. Her ability to lay fertilized eggs, therefore, declines as she ages, and after a couple of years, it will decrease to the point that she will naturally be superseded by the colony. Queens are the most productive egg layers in their first season, and for this reason, many beekeepers replace the queen annually to maintain high brood production levels. A downside to this method is that it removes selection pressures on queens to be good layers for beyond one year. If queens are replaced every year, it does not matter whether they continue to be any good in Year 2 or Year 3, so we do not seek out queens that can remain fertile for

many years. Instead, we may be selecting for queens that are good layers during the first year but dwindle quickly after that. Purchasing new queens is expensive, and rearing your own is time-consuming, so queens that perform well for multiple years are desirable. If you can afford to experiment, consider letting your queens age to see how many years they remain productive. You can then try breeding from your long-lived productive queens and, in time, help create a stock of hearty queens that have strong brood patterns year after year. Of course, this long-term fertility also depends on the quality and quantity of sperm that she stores upon mating; breeding for a desired trait is never that simple! You might consider replacing the queen if brood is spotty or there is not the expected amount of brood in the brood chamber at the right time of year (in the Northeast, high in early spring and summer, low during nectar and pollen dearths, and slightly higher again in early fall). This is especially critical if the number of adults being produced is not enough to replace the adults dying from old age or other causes. Be sure to first rule out the possibility of pests, disease, or pesticides in causing irregular brood or a reduced adult population. Keeping notes of brood, pollen, nectar, and population levels in your inspection records help you see if these issues have been occurring gradually and could be associated with poor genetic stock or an old queen, or if they have occurred suddenly and are likely due to some other environmental agent.

■ **Improve Weak Population or Spotty Brood Pattern**

■ Sometimes reduced egg-laying is a result of the colony being honey bound. Check to see what the empty cells in the brood nest are filled with: If they are filled with nectar instead of brood, this can be a sign that your colony is honey bound and the queen simply does not have room to lay. This can happen with infrequent supering or during a strong nectar flow if bees are bringing in nectar faster than the queen fills cells with eggs. If there are no empty cells for them to deposit their nectar in supers, they will do it in the brood nest. If the queen does not have room to expand the brood nest because honey is taking up too much space, rearrange some frames or introduce drawn comb to make sure there is enough room for the queen to lay eggs and check back in a couple of days to verify she is laying. The colony may have already begun preparing to swarm; check for swarm cells. Note: Colonies can become honey bound relatively quickly in the spring, especially if you are feeding them. Keep an eye on the brood chamber to make sure there is still sufficient room for the queen to lay. Reduce

Defensiveness Ongoing defensive behavior has a genetic basis, so requeening with a gentler stock can mitigate it. For example, Italians are known for being docile bees. Many beekeepers may tolerate defensive bees that may have other desirable traits; e.g., hygienic behavior. In this case, the benefits of the stock may outweigh the cons of being stung regularly. It is usually up to the beekeeper what kind of bees they prefer to work with; however, there are two exceptions where very defensive behavior is not tolerated. The first is if you find you have Africanized bees. If you suspect that you have Africanized bees, contact your state apiculturist or local extension associate immediately. Depending on the state, you may be required to requeen or destroy colonies that are confirmed to be Africanized. Africanized bees are rare in the Northeast. The second instance where defensiveness is not tolerated is in urban beekeeping. While many cities, like New York City, are embracing beekeeping, there are often rules about the temperament of your bees because these areas have high human populations and neighbors are near. Requeening with gentle stock can be mandatory upon public complaint, and keeping the colony specified distances from property lines and public areas is also often required or recommended. As a Master Beekeeper, you need to be sensitive to the fears of the public and the reputation of bees and beekeepers at large.

■ **Reduce Swarming** ■ Requeening with a young queen can help reduce the

swarming impulse in a colony if that is something that you wish to prevent. The urge to swarm increases when the ability to detect queen pheromone decreases and young queens produce more pheromones than older queens. To be effective, requeening must happen in early spring before the colony has begun preparing to swarm and should be complemented with other methods, such as providing more space for the bees. Certain stocks of bees are genetically more likely to swarm, such as Russians, so requeening with stock less prone to swarming behavior may also be helpful.

READ

Managing a Two-Queen Colony

Sometimes beekeepers may run a colony with two queens to maximize honey production. In theory, two queens are producing twice as many bees and, therefore, more honey and other hive products can be made. The major trade-off is that managing a two-queen colony can be time consuming to set up and manage properly, so it is more popular among sideliners and hobby beekeepers than commercial beekeepers. What happens when two queens are present in the same colony? In nature, there is sometimes more than one queen present in the colony during supersedure or swarming, but it is usually only temporary. Under normal circumstances, when a queen finds another queen in the hive, she grabs onto her with her mandibles, climbs on top of her, and stings her to death. Workers can also reject a second queen through balling behavior or withholding food from her. Because of this behavior, free-roaming double queen colonies have only yielded mixed levels of success. Physically removing the queens' mandibles so they cannot position themselves on another queen to sting her and removing older workers, which tend to ball and kill queens the most, make the colony more stable, but it is unlikely for beekeepers to undergo such measures. By far the easiest way for beekeepers to maintain a two-queen colony is by keeping them separated. In this way, a two-queen system is essentially created by performing a split, adding a queen to it, placing the split on top of the parent colony, and separating the two with a queen excluder. How productive are two-queen colonies at producing honey? Little research exists that compares honey production between two-queen systems and single-queen systems. The most extensive study is from a USDA report that compared annual honey yields of the two types of colonies. It found that two-queen colonies averaged 112 pounds more than single-queen colonies each year. However, it also found that two-queen systems require 50% more time to maintain than single-queen systems. They are difficult to inspect with brood boxes between several heavy honey supers, can become quite tall, and are prone to becoming honey bound. Additionally, using this system to maximize honey production requires extensive planning ahead to set up — around two months before a honey flow — to maximize the number of foragers available to collect nectar.

READ

How To Make a Two-Queen Colony

Here, we outline the steps for making a vertical two-queen system, the most popular kind. Other people have experimented with other designs of two-queen colonies. Start with a colony housed in two deep hive bodies that have at least eight frames of brood. Separate the hive bodies and place the queen in the bottom deep. Manipulate the frames in the colony so that there are approximately five frames of mixed brood flanked by approximately five frames of honey and pollen. Long description ■ Two deep hive bodies with eight frames of brood. The top body has the frames in the fourth, fifth, and sixth

position. The bottom body has them in the third through seventh position. The queen is in the bottom frame among the mixed brood. Place a new deep hive body on top of this bottom deep. To this new hive body, add approximately four frames of honey flanked by approximately six frames of drawn comb. Place a Snelgrove board, also known as a division screen or double screen board, above this newly added hive body. A Snelgrove board has two screens of mesh that bees cannot pass through and a notch in the top for bees to enter and exit. Place it on the colony such that bees can enter and exit the uppermost brood chamber, the third hive body that you are about to add. Add the top hive body from the first step on top of the Snelgrove screen — this is acting as your split. Place three frames of capped brood — with the nurse bees on it — in this hive body and flank it with two or three frames of honey and pollen. Fill the rest of the hive body with foundation or drawn comb. Replace any frames you removed from the bottom colony with foundation or drawn comb at the periphery. A Snelgrove board. Photo credit: Betterbee Introduce a mated queen to the top hive body. The screen on the Snelgrove board prevents the two queens from stinging each other and allows heat from the colony below to rise and help keep the brood warm until it emerges. It also allows the colonies' odors to mix so they become accustomed to each other's scents. Bees cannot pass through the screens to venture into the other colony. Long description ■ A three deep hive body. The top body is separated by a Snelgrove board. This hive body has drawn comb in the first, second, ninth and tenth frame positions, honey/pollen in the third, seventh, and eighth positions, and capped brood in the fourth through sixth position. One queen is among the capped brood in the top body. Another queen is in the body hive body. There are two hive entrances. One above the Snelgrove board, and one below. Timing is important for this to work if you want the honey flow to coincide with both colonies having a strong forager population. Setup should therefore happen around two months prior to the flow beginning; they are at their strongest production eight to nine weeks after being assembled. In the Northeast, setting it up in late April or early May is best if the weather permits. Remember that it can take around three weeks for emerged adults to become foragers. Having a surplus of foragers is the key to this system yielding lots of honey production. When the honey flow begins and both colonies are populated with bees of all ages, remove the Snelgrove board and replace it with a queen excluder. Bees should begin filling the middle hive body with honey and drawing out any foundation and filling drawn comb. Start supering both colonies for honey production — super above the queen excluder on top of the bottom hive and above the top hive. Long description ■ A hive with multiple honey supers. There is one hive body on the bottom with three honey supers on top, then a queen excluder, then another hive body, then two more honey supers. There is one queen in each of the individual hive bodies — two queens in total. You may need to manipulate frames or reverse hive bodies below the excluder again in the summer if you find the lowest brood nest has moved upward. Try to keep the queen as low in the first colony as possible so that she does not encounter the new queen through the excluder. This system is not maintained over winter. In the fall, after removing the supers for honey extraction, either combine the colonies into one strong colony (remove one queen) or separate them into two individual queenright colonies.

READ

Using Local Queens

There is much discussion today about the benefits of locally adapted bees. The term “locally adapted” refers to bees that are well-suited to their specific environment, although this is not a strict definition,

and often advertisements for “local stock” or “locally adapted stock” simply refer to bees that have thrived in a specific region for at least two years with minimal intervention. Likewise, queens from colonies that have survived at least two winters are sometimes termed “survivor stock,” although this does not specify what sort of mite management regime has been used. Because of the advantages of locally adapted bees, we recommend that you try rearing some queens from your best colonies or consider buying local queens over ones originating from long distances. However, it can be difficult to produce large numbers of queens in the Northeastern U.S. by this time of the year. Many commercial beekeepers here need their colonies to be strong by spring for pollination of early crops or to take advantage of the first nectar flows, so they often depend on queens bred in the South. Indeed, most large-scale queen and package producers are concentrated in the warmer regions of the U.S., like South Carolina, Hawaii, Georgia, California, and Florida. Because winters are mild or nonexistent in these areas, queens can be produced, mated, and available to ship in April or May, when many beekeepers are looking to requeen or install packages. Furthermore, as stocks of Varroa-tolerant and Varroa-resistant bees are being developed, queens with those genetics may not yet be available in your local area.

READ

What Are The Benefits of Using Local Bees?

There are many benefits to using local bees. They can thrive in your climate. Environmental adaptation is important. In general, many beekeepers like to purchase queens, packages, and nucs from the same region they wish to keep them in because the bees may be well suited to that climate, especially if the populations have been persisting without the introduction of outside genetics. Honey bee populations that have lived in cold, northern climates for many generations have adapted to prepare for winters that stretch from November to April by building up large food reserves and ceasing brood production in mid- to late autumn. The most locally adapted bees in a region are often the wild bee populations that have been residing in tree cavities for many generations without human intervention. In the United States, there has been a lot of genetic mixture within the bee population over time, so sourcing bees that are truly adapted to your local climate can prove difficult. Locally adapted colonies tend to be more productive than non-local colonies. While research is needed to investigate this relationship in the U.S., several large-scale studies in Europe have compared locally adapted bees to bees that are not locally adapted in the same region. Collectively, these studies have found that worker population levels, overwintering success, honey production, queen lifespan, and overall colony lifespan are found to be greater in the locally adapted colonies. Purchasing local bees can help reduce shipping stress. Queens can be exposed to fluctuating temperature spikes during long-distance shipments, and these spikes may reduce their reproductive output by killing some of the sperm stored in their spermatheca. ■ They can be more productive. ■ It can reduce transportation stress. ■ It may help reduce the spread of pathogens. ■ The prevalence of many pathogens and parasites impacting bees today is due to the global and national network of honey bee markets and transportation. Small hive beetles, *Nosema ceranae*, and Varroa mites all originated outside the U.S., entered at specific sites, and are now prevalent throughout the country as bees are transported for pollination in different states, and queens and other bees are shipped in the mail to new regions. Rearing your own certainly does not mean that they will be free of infection, as these pathogens are quite widespread, but it may help reduce the

introduction of further infections or strains. It supports beekeepers in your community. Purchasing bees from local beekeepers enriches your agricultural economy and supports your beekeeping community. In addition, local producers that are selecting for specific traits can help contribute good bee genetics to their environment. ■

MODULE WRAP-UP

Queen Management

In this module, you focused on the vital role of queen bees in colony success. Your study of queen health, productivity, and colony dynamics will help you develop crucial skills for evaluating queen quality and implementing proper management practices. These essential techniques for maintaining healthy, fertile queens will help ensure strong colony populations and long-term success in your apiary.

Variation Within *Apis Mellifera*

Throughout your experience beekeeping, you may have come across many different kinds of honey bees. Some are gentler than others, some swarm more than others, and some may be better equipped to survive year-round in your climate. Honey bees even exhibit variation in their coloration, from golden Italian bees to dark German bees. In this module, we will explore these variations and their causes, introducing you to the different kinds of honey bees, *Apis mellifera*, in the U.S. and around the world.

TOOL

Honey Bee Subspecies and Lineages

After honey bees first evolved in Asia, they ■ Download the Tool dispersed throughout Africa, Asia, and Europe. Check out this Honey Bee Seven genetically distinct populations, called Subspecies Table to learn more lineages, arose in the Old World: African (A), about subspecies and lineages. Western and Northern European (M), Eastern European (C), Eastern and Central Asian (O), Western Asian (Y), Madagascar (U), and Egypt (L). Lineages differ in their geographical range, their behaviors, and their morphology (size, shape, color, etc.). Each lineage contains one or more subspecies; indeed, all modern subspecies and stocks of bees have descended from these lineages. Subspecies are often referred to as races of honey bees; we will stick to the term “subspecies.” Genetic differences in morphology and behavior among different subspecies arose through natural selection after populations were geographically isolated and unable to mate freely with other populations for a period. Note that subspecies names are an extension of species names. Each subspecies belongs to the species *Apis mellifera*, but they have an additional name afterward. While different species cannot mate to produce viable offspring, different subspecies can. For instance, *Apis mellifera* and *Apis cerana* cannot produce offspring together, but *Apis mellifera mellifera* and *Apis mellifera ligustica* can. Their offspring would be a hybrid of the two subspecies and possess a mix of genes and characteristics from each of the parent subspecies. There are at least 27 recognized subspecies of honey bees in the world, shown in the Honey Bee Subspecies Table. Some scientific debate exists about the actual number of subspecies, and as genetic analytic tools improve over time, additional subspecies may surface. For this course, you only need to familiarize yourself with the subspecies highlighted in orange in the middle column of

this table.

READ

Apis Mellifera in North America

Apis mellifera mellifera (M) (“German bees” or “German black bees”) was the first honey bee introduced to North America, in 1622. It was the only honey bee kept here for almost 250 years, until *Apis mellifera ligustica* (C) (“Italians”) was introduced in 1859, followed by *Apis mellifera carnica* (C) (“Carniolans”), *Apis mellifera caucasica* (O) (“Caucasians”), and many other subspecies depending on the origin of immigrants. The map below shows when seven subspecies were introduced to the mainland United States between 1859 and 1922: *A. m. ligustica* (1859), *A. m. lamarckii* (1866), *A. m. carnica* (1877), *A. m. cypria* (1880), *A. m. syriaca* (1880), *A. m. caucasica* (1882), and *A. m. intermissa* (1891).

Long description ■ This image shows a historical map depicting the introduction of different *Apis mellifera* (honey bee) subspecies to North America. The map focuses on the central and eastern United States, with state boundaries visible and coastal areas shown in gray. The legend identifies three distinct lineages of honey bees. The map includes dates ranging from 1860 to 1910, showing when different subspecies were introduced to specific locations. Various introduction points are marked across the map, with dates such as: 1866, 1868, and 1865-1868 in the central region 1880-1882 in some areas 1902-1910 and 1905-1910 in other locations. The distribution pattern suggests a gradual introduction of different honey bee subspecies across multiple decades, with various entry points throughout the eastern and central United States. The dates and locations demonstrate the systematic introduction and spread of honey bee populations during the latter half of the 19th century and early 20th century. Although several subspecies of the Western honey bee were brought to the U.S. since the 1600s, a 2023 study showed that approximately 93% of all current U.S. honey bees are from lineages primarily found along the northern Mediterranean coast and mostly from TWO subspecies: *A. mellifera ligustica* (Italian honey bee) and *A. mellifera carnica* (Carnolian honey bee). This means that current U.S. honey bees are not very genetically diverse; this is potentially making them less able to fend off threats from new enemies and pesticides. Efforts to increase the genetic diversity of U.S. honey bees have been particularly thwarted by the ban on live honey bee imports from all countries in 1922 when tracheal mites were discovered in Europe. Honey bees are no longer allowed to be imported from Europe, with an exception of the importation of Russian queens for a few years in the late 1990s for a USDA Varroa-resistant breeding program. Until 2019, queens and packages could still be easily imported from Canada, yet imports from Canada (or any other country, in fact) are now allowed only under conditions defined by U.S.-based federal regulatory authorities (USDA/APHIS). For example, certain life stages (queen bees, package bees, drone semen) may be permitted, depending on their health (absence of notifiable diseases like AFB-American foulbrood), physical inspection, and documentation criteria (several export/import permits needed from both Canada and the USA). Even though humans have been hunting honey and keeping honey bees for thousands of years, bee breeding is a relatively recent practice. Honey bee reproduction was not well understood until 1845, and the practice of mating specific queens and drones did not begin to occur in a controlled fashion for another hundred years. Beekeepers bred the different subspecies available to them together, combining gene pools through artificial or natural insemination of queens with selected drones/sperm over multiple generations to produce bees with traits they desired. In North America, we call these bees

that were intentionally bred stocks. They represent a small amount of the large genetic variation found in the bee lineages. Many have been so thoroughly bred with each other that they differ greatly from the original subspecies brought over hundreds of years ago. While most bees in North America today are descended from Italians, they are not the same pure Italian bees first introduced in 1859. They have been bred with other subspecies and have genes associated with these other subspecies, although they are called Italians since they resemble *Apis mellifera ligustica* most closely, with regard to their coloration and other traits. Other stocks, like Buckfast bees, are a hybrid of *Apis mellifera ligustica* and *Apis mellifera mellifera* and contain a mixture of genetic traits from each. Anyone who breeds bees to maintain specific traits is essentially creating a stock of honey bees. There are many stocks available for purchase in the U.S. from bee breeders, with new ones constantly in development. Italian bees (Photo credit: Bee Informed Partnership) Carniolan bees (Photo credit: Bee Informed Partnership) Traits Associated With Subspecies and Stocks Different subspecies and stocks are known for different traits. The Honey Bee Subspecies and Stock Traits Table includes the most common past and present stocks in the U.S. It outlines the general traits of each stock. We have also included notes on their origin, rarity, other behavioral traits, and appearance. Keep in mind that there can be a lot of variation in traits within a stock, so this table serves as a general guide, not a guarantee.

WATCH

Africanized Honey Bees

The Africanized honey bee (AHB), colloquially called “the killer bee,” is a hybrid of *Apis mellifera scutellata* (A) and *Apis mellifera mellifera* (M). Note that “Africanized” refers to hybrid bees and “African” refers to pure *Apis mellifera scutellata* (or other African subspecies) in their home region. These bees are known for their highly defensive nature. In 1956, *A. m. scutellata* were introduced to an experimental apiary in Brazil for a program meant to breed bees adapted to a tropical climate. Within a year, however, 26 queens, along with several swarms of *A. m. mellifera*, were accidentally released from their enclosure. They mated with other *Apis mellifera* subspecies in the vicinity and moved both south and north, sometimes hundreds of miles a year, reaching California in 1985 and Texas in 1990. They occupy several Southern and Southwestern states of the continental U.S. (Arizona, New Mexico, California, Nevada, Utah, and Oklahoma) and, more recently, Southeastern states (Louisiana, Arkansas, Florida, and North Carolina). They are therefore not a major concern of Northeastern beekeepers, except for the occasional hot hive that is traced to a Southern queen or package. The rapid spread of Africanized bees is due to the relative success of their genes. More than half of the genome in most Africanized honey bees (AHB) is of *A. m. scutellata* descent. Traits from European bees have tended to be lost or less expressed over time due to several factors. AHB colonies tend to be smaller, but they grow faster and swarm more. They also produce more and faster drones, meaning that queens of all subspecies mate disproportionately with AHB drones. Queens fathered by AHB drones also emerge earlier and can kill their rivals within their queen cells. AHB swarms sometimes practice “nest usurpation,” meaning they invade other colonies and replace the queen with their own. Africanized bee swarm. The name “killer bee” comes from their tendency to sting with less provocation and in greater numbers than other subspecies. Their nest sizes are often smaller than European bees, but they defend a larger area around that nest. The nests are also more likely to be in ground cavities, less visible, and therefore less easily avoided. The bees can pursue intruders well beyond the vicinity

of their hives; up to a quarter mile. An estimated thousand humans and more animals have died from a large quantity of stings. These have generally been in situations where they could not run away (because of young or old age, or confinement; e.g., a dog being chained). The stings of individual AHB are no more painful or venomous than those of other honey bees, but they are generally greater in number. The same precautions and reactions apply: Leave the area, cover your face/head, do not swat at the bees. Because Africanized bees do pursue and persist, jumping in water to hide is not recommended; they may wait until you emerge for air and continue to sting. Identifying AHB can be difficult: In short, any colony that is excessively defensive needs to be kept away from humans and animals and may need to be destroyed. In this case, there is no need to identify whether the colony is Africanized or not; it is simply unsuitable in an area inhabited by humans or livestock. If the colony needs to be identified, however, some behavioral and morphological traits distinguish European bees from AHB: AHB workers tend to be smaller in body size than European bees — although hybrids may be just as big — and their wings have different vein patterns. Behaviorally, in addition to those traits already discussed, individuals tend to transition to foraging earlier in their lives and have a lower threshold of sugar concentration for foraging, meaning that they bring back more watered-down nectar/resources. Finally, they are more resistant to some pests; e.g., Varroa mites. This may be in part to their relatively high use of propolis, their tendency to swarm, or other factors. In general, however, this is not a surefire route to identification. Genetic testing is the most reliable method to determine whether your bees are indeed Africanized. Watch this video to learn more about Africanized honey bees and how to approach a colony you suspect may be Africanized. Video Transcript Okay if you live in a southern area Texas, Florida, southern California, those areas largely have been invaded by the Africanized bee. That's a feral bee and they're a more defensive bee. And you have to make a choice then of whether you're going to costly requeen your colonies with gentle European stock or whether you're going to work with the Africanized stock. And I have friends who do both. If you're going to work with Africanized bees, you can't be in a tight neighborhood. It's just too dangerous to your neighbors. Your neighbors have the right to have safety for them and their kids on their property, and Africanized bees are too unpredictable. If you can get a location that is away from close neighbors, then the Africanized bees are a very robust bee, don't need treatment for Varroa mites, tend to be very well adapted to the environment and can be kept. Other than that, you're going to want to run European stock and have your queens marked. You might be able to get by with the F1 daughters, the first generation of Europeans mated with Africans but by F2 they typically become a bit too touchy. If you are working with the Africanized bees it's a different kind of bee keeping. We generally will wear protective gear, but you can learn how to handle them. The same principles that apply to working with European bees to avoid passing their response threshold—the threshold where they initiate a defensive reaction, which is the stinging— works the same for Europeans and Africans. You learn to move slowly, use an adequate amount of smoke, don't disturb the colony any more than you need to, slow down when it calls for it, and be ready to close the colony up. Okay, if you're in an area where there may be Africanized bees, be aware that the temperament from colony to colony can vary tremendously. Some Africanized bees can be as gentle as pussy cats. Some are like red-hot fire crackers. And you need to be able to be prepared for them to react very, very quickly. And some of the things you do, you don't want to put any kind of disturbance in front of a hive of Africanized bees. You don't put a box down. Don't put your smoker down. They don't want to see any kind of disruption in the

front. And then, be very much aware after you smoke the entrance. Watch the entrance as you touch the lid. So you're going to pick the lid up, smoke it, and keep an eye on that entrance. If you suddenly see a firehose of bees coming out that entrance, that means you better stop what you're doing or you better be geared up because they are suddenly going to sting everybody in the immediate surroundings. So take the lid up, protect your face with the lid, then look right here. You'll see they'll start bumping off your hands. So bees typically are pretty polite. They'll often give you warning bumps, so you'll feel bees coming up and bouncing off your hands or bouncing off your face without stinging. The good beekeeper is aware of that. You don't just keep charging on, they're telling you something. They're saying back off sucker, use more smoke, put on more gear something. But we are going to defend this hive. And have that awareness. And this is the same with European bees. Many beginning bee keepers because they are all armored up in a bee suit feel that they can just charge ahead in a hive. Some alarm from them gets released in the air either from crushing bees or bee stinging, and then things exponentially progress, they snowball into an ugly situation. It's unfortunate because a bad sting incident for a beginning beekeeper sticks in your mind. It takes away the joy of beekeeping. Beekeeping should be pleasant like this all the time. So be aware all the time. As I'm reaching over a hive, move slowly, kind of as though you're doing Tai Chi. Nice slow movements. No sudden movements at all. If you move your hand quickly like that, that will often draw bees and that's what a good test to see if the bees are defensive. Just do a fast wave like this and see if they bounce off of your hand. That will tell you that they are a little bit more defensive. If you suspect that you do have Africanized bees, contact your state or provincial apiculturist immediately. Depending on the state, you may be required to requeen or destroy colonies that are confirmed to be Africanized. North Carolina State University offers genetic analysis for determining whether bees are Africanized. Visit their Queen Clinic website for details. The USDA recommends that beekeepers in areas with AHB periodically requeen their colonies with European-mated queens.

MODULE WRAP-UP

Variation Within *Apis Mellifera*

In this module, you discovered the diverse world of honey bee varieties and their distinct characteristics. Your exploration of different bee stocks revealed how these variations affect colony management and survival. Use this knowledge of honey bee diversity to help inform your breeding and stock selection decisions.

Queen Rearing With the Doolittle Method

The ability to rear queens is a key skill to begin breeding bees or simply to raise queens for your own colonies without needing to purchase them every year. Whether your goal is to rear queens on a large scale for profit or on a small scale to keep bees sustainably in your backyard, the Doolittle method is our recommended route. This module outlines the steps and considerations you should take with this method to be able to successfully rear queens. Rearing queens can benefit your beekeeping operation because it allows you to have queens on hand to replace failing or missing queens in your colonies. If you like to requeen your colonies annually to improve their productivity, you can have your own source of queens to do so. Being able to rear your own queens instead of relying on purchasing queens is a

great way to become a more self-sufficient beekeeper.

READ

The Doolittle Method

There are many ways to rear queens depending on the number of colonies you manage, the number of queens you need, and the amount of time, money, and effort you want to invest. If you are a hobbyist who only wants to have a couple of extra queens on hand, you can simply pull nucs from your strong hives and allow those nucs to rear their own queens — make sure to include frames with eggs in the nuc. This practice becomes even more convenient if your strong colonies have begun building swarm cells. Move some young swarm cells into a nuc and allow the bees in the nuc to finish raising those queens. If you are only looking to rear a few queens and can do it in this way, it's a great approach! A paper published in 2019 found when queens lay eggs directly in queen cups, the daughter queens have a larger body mass and more ovarioles than queens grafted from eggs laid in worker cells. In this course, however, we cover a more advanced method of queen rearing called the Doolittle method, based on grafting larvae that develop into queens. This method is used most often by beekeepers who require more than a few queens at one time. This method scales well; you can rear several queens or hundreds at once. It also allows you to select larvae from your best colonies and ensure that they have abundant resources to develop into strong and healthy queens. Rearing queens by grafting entails carefully moving young larvae from a regular worker brood frame into manmade queen cups. Grafting lets you be in control of the timing: You know exactly how old the cells are at each stage and when they are ready to be inserted into new colonies or sold as is. Yet it requires some specific tools, fine motor skills, and much practice. Injuring the young larvae in any way, drying them out, or placing them in the wrong position in the queen cup will result in failure. You shouldn't expect a 100% success rate, although with practice you might get close. Although this course focuses on grafting, there are many methods to rear queens that do not involve grafting. Your knowledge of these methods will not be tested here, but if you are interested, you can read up on them in "Queen Rearing and Bee Breeding," the comprehensive book by Harry Laidlaw and Robert Page (1997). These methods include but are not limited to the Miller method, the Alley method, and the Smith method. The Doolittle method of queen rearing. Overview of the Doolittle Method 1. Very young larvae are selected from a queen mother colony and are placed (grafted) into manmade queen cups. 2. These new queen cells are inserted into a cell starter for 24 hours, where nurse bees feed them with abundant royal jelly. 3. After 24 hours, the larvae have doubled in size. The larvae are then transferred to a cell finisher, where workers continue feeding them and eventually cap the cell. 4. After nine to ten days, the queen cells are nearly finished developing and can be harvested one to two days before they will emerge. 5. The capped, nearly mature queen cells can be placed into new hives as is, but most are placed in small hives called mating nucs. Mating nucs permit queens to emerge from their cells to sexually mature, take mating flights, and begin laying eggs. 6. After queens have mated, beekeepers locate the queen in the mating nuc; if she is mated and laying a good pattern, she is removed to be sold or placed in a full-size colony.

READ

Biological Considerations

Honey bee biology is key to queen rearing. You need to think about timing, resources, and bees' instincts to requeen themselves, and be aware of other instincts that might stymie your efforts! Timing A female larva develops into a worker or a queen based on the nutrition she receives in the first 12 to 36 hours after hatching from the egg. Queen rearing, therefore, requires specific timing. Whether you leave it up to the bees or graft larvae yourself, the process starts with eggs or larvae less than 36 hours old. Seasonal considerations are important, too. While it is possible to rear queens from late April to the end of September, swarm season is the easiest time to rear queens. It is during this period that there are abundant food resources in the colony, rising colony populations, and drones to mate with in the area. When you start to observe drones being produced in your colonies, you have the green light to get started! In the Northeast, swarm season generally starts in mid-May and ends in mid-July. Queen rearing is also easier during a honey flow. Before and after swarm season or during a dearth, you may need to supplement your queen rearing operation with more food and you may not rear as many queens. Colonies require resources to make new queens. There need to be sufficient nurse bees to care for and feed royal jelly to developing larvae. To create royal jelly, the nurse bees need to have access to adequate pollen and nectar, which, in most queen rearing operations, requires feeding. In short, developing healthy queens requires lots of nurse bees and lots of food. Honey bees will create queens under three conditions. First, if the current queen is aging or otherwise failing, they will create a new queen to supersede her. The old queen's pheromones signify her age since the amount she produces diminishes over time. It is also thought that workers can detect a reduced rate of egg-laying by their queen. This prompts them to build supersedure queen cups. Second, if the queen is accidentally killed or lost, the colony will undertake emergency queen replacement. ■ Resources ■ Instinct to Create Queens ■ When a queen is removed, workers can begin to detect that she is missing after as little as 15 minutes. If there are eggs or larvae less than 36 hours old in the colony, they will mold a worker cell with an egg or young larva in it into a larger queen cell. If there are no larvae young enough, the colony will not be able to rear a new queen and it will die as its bees age and are not replaced. Finally, when the colony population grows bigger than the space available to it in the hive, it prepares to swarm and creates several high- quality swarm cells. Colonies will usually produce many more swarm cells than supersedure cells. They can create a few swarm cells every day for a week or so leading up to swarming. To stimulate bees to produce queens, the beekeeper needs to create these conditions. The easiest to do are the latter two: causing a queen emergency and simulating swarming conditions. This colony was crowded and produced multiple swarm cells; the cell on the left is capped. The cell on the right will be capped within a few days. Instinct to Kill Queens When bees need a new queen, they generally produce many queen cells as insurance that at least one of those cells will successfully rear a queen. The first queen to emerge killsthe other queens still developing in their cells, and if multiple queens emerge, they fight to the death until one queen remains. So, if you want to rear multiple queens, the nearly mature cells must be separated to permit the queens to emerge from their cells without competition. ■

READ

Queen Grafting

Queen grafting entails choosing larvae of a selected age and placing them in manmade queen cups. It takes fine motor skills, practice — around 50 tries — as well as a few tools. Rearing queens through

grafting lets you be in control: You know exactly how old the cells are at each stage and when they are ready to be inserted into new colonies or sold as is. Once you have the practice down, you can expect a success rate of around 80% to 85%. Picking up the young larvae incorrectly, injuring them in any way, drying them out, or placing them in the wrong position in the queen cup will result in failure. Nurse bees check all the larvae once they are inserted into cell building starters, and nurses will avoid feeding the larvae that have any of these issues. Any problems will be visible within a day because they won't feed the larva or draw out the cell.

Selecting and Preparing Larvae To Graft Pull out a frame of brood that contains the right age larvae to graft; 12 to 36 hours. These larvae will be around the same size as an egg, will be laying in a slight crescent shape, and will have a small amount of jelly in their cells. Choose frames from a colony that contain eggs and search in the cells close to those eggs to find the very young larvae. Some beekeepers make the process easier by confining the queen to a certain spot on empty comb with a "push in" cage so she is forced to lay eggs in that spot. After three days, the eggs hatch, and the beekeeper can confidently graft from these larvae knowing they are the right age. When you are ready to select the brood frame you are going to graft from, place the queen on another frame and gently brush the remaining bees off. Be very careful not to let the larvae dry out as you bring the frame indoors; otherwise, they will not develop properly. Ideally, your breeder colony will be located close to your grafting room so that the larvae are exposed to wind and dry conditions for the shortest time possible. If there is more than a minute or so of a walk, carefully wrap the brood frame in a damp towel to keep it moist and protected. Bring the frame into your grafting room. Grafting should be done indoors as working with brood frames outdoors can attract pests and robbing if there is honey on the brood frame. The room conditions should be warm — ideally warmer than room temperature — and should be very bright to properly see the young larvae for grafting. Having a bright lamp to shine directly into ■ **The Grafting Room** ■ the brood cells is very handy. A damp cloth should be on hand to lay over the queen cups on the grafting frames as you insert larvae to prevent them from getting too dry. It is important to maintain a clean area for grafting. Surfaces should be disinfected ahead of time, and the grafting tool should be cleaned between colonies.

Tools Needed Grafting tool — there are a variety on the market; try several to see which you prefer Grafting frame — a frame with three bars and 45 cups Lamp Something to elevate your frame to be at the appropriate angle and comfortable height Glass vial with previously collected royal jelly and dropper (optional) Headset with a magnifying glass (optional) Damp towel to lay over the part of the brood frame you are not working on

The grafting tool used to pick up larvae. ■ **Queen cups attached to a bar.** If you look closely in the cups, you will see larvae of the appropriate age settled on top of a small pool of royal jelly.

Grafting Frame Assemble your cups on the bars. Modern equipment from bee suppliers allows you to slide plastic cups onto a bar, and you can choose how many cups to put on each bar and how many bars you will insert into your colonies. Grafting frames vary in the number of cups they can hold, but they generally have three frames that each hold 15 cups, allowing you to insert up to 45 grafted larvae into one cell building colony at one time. The maximum recommended cups to have on one frame is 45: 15 cups per bar, three bars per frame. If you want fewer queen cells than that, you can adjust it however you want. The nurse bees in the cell-builder colonies tend to accept the cups better if they are first placed in a colony for a day to be "conditioned" by the bees or if they are lightly painted with melted wax. ■ **Grafting With Royal Jelly** ■ Before placing the larvae in the queen cup, many beekeepers will add a small amount of royal jelly mixed with water in a 1:1 ratio to the bottom of the cup to prime the cell. This gives the larvae

a gentler landing in the cell by floating it off into royal jelly. Royal jelly can be collected ahead of time by the beekeeper from swarm cells or other queen cells and stored in a glass jar in the freezer for a few months before being used. Royal jelly is collected from two- to three-day-old queen cells by scooping it out with a royal jelly spoon and discarding the queen larva. Dilute the jelly in a 1:1 ratio with warm water in a small vial or cup and use a dropper to prime each cup with a small amount of this solution — less than a full droplet — immediately before grafting. After you have primed your cell, use your grafting tool to carefully scoop out larvae that are 12 to 36 hours old. At first, you may find it easier to break down some of the cell walls to access the larvae until you get good at the scooping method. It is critical that the larvae enter the cups in the same orientation that they were in the cells. Larvae lay on their sides and breathe through the spiracles on the side facing upward. They should not be jarred on the side of their original cells as you remove them, as this can injure them. The scooping needs to be very gentle. Do not flip the larva over, do not submerge it in the royal jelly, and do not shake it. Flipping them over or submerging them in royal jelly blocks these spiracles and causes suffocation. Very carefully, place the larva into the center of the queen cup. Some grafting tools scoop up the royal jelly and the larva, while others scoop up mainly the larva. Some grafting tools, like the Chinese tool, have a small plunger to help release the larva. You will figure out what method and tool work best for you as you practice. If you suspect you injured a larva, discard that one and try it again with a new larva. Most commercial queen rearers dry-graft; that is, they do not prime the cell cups with royal jelly first. They carefully scoop up the desired larvae and some of the jelly it is floating in, and they place it into a dry queen cup gently. With this method, the cup contains the larva as well as a small amount of the royal jelly it was resting in. After you have completed one bar of grafting, place it upside down (the larvae will remain in the cups) on a damp towel while you finish grafting the other bars. This will prevent the young larvae from drying out. After you have completed grafting your frames, you are ready to insert it into your cell starter box.

WATCH

Queen Grafting Example

In this video, Brian Evans performs grafting. Video Transcript Okay; so now we've prepared our finishers, we've prepared our starters; now it's time for us to put a graft on. And I have gone ahead and pre-selected a frame of the right age larvae from a hive of a bee queen, this one in particular was artificially inseminated, so we have some pretty good genetics here that we want to keep and expand upon. This whole top region is all fairly good, aged larvae. There's some little pockets of older stuff but mostly it's one to two day old larvae, which is optimal. So as I'm grafting, a frame like this is fairly easy to go through. But what I try to do initially is go through and find what I believe is the youngest larva possible first. Then once I have all that exhausted cannot find anything else that I think is very young then I will kinda step up the age a little bit and look for stuff that is a little bit larger than what I was looking for prior. And then I'll start collecting all of those from the frame. And hopefully by the time I've done that, I've fulfilled my graft. Because you don't want to keep getting bigger and bigger and bigger because eventually you're going to start running into the three day mark and that's where you're going to start running into problems. Now it's very important to point out that it is a lot easier to find eggs three days prior to when you're going to graft, that way you can age those eggs properly. If you are going to graft and you go out to your yard and you start looking through hives haphazardly, you may get lucky

and find that right frame of brood. Or you could spend hours out there chasing larva that may or may not be the right age. In addition, having a frame of mixed larva of varying ages is going to take you a lot more time to look through and age those larvae correctly. And if there are enough you may be able to fulfill your graft. However if you run into a situation where you've used up all your one and two day-old larva and there's maybe some three day-old larva left and you've kind of use that up and all of a sudden your graft is not finished but your frames are finished, now you have to go back out and hopefully try and find another frame. So recognizing how much is on here, not pushing yourself to use that 3 day-old larva once you hit that two day stage and you can't find anything else, do yourself a favor, take this frame back to where it came from and try and find another one. Here to remove the larva from this comb and to deposit it into these queen cups, I prefer to use what's known as the Chinese needle which you've probably already heard of. It's got a small, I believe this is a horn material here that's very flexible at its tip. And it also features a small plunger to release the larva from the tongue. Now, the other style of grafting tool is known as the German dental needle. I generally don't use unless I am forced into a situation where I need to graft from fresh drawn comb. Occasionally a queen will draw or the bees will draw fresh wax and the queen will need to use that for the first time, sooner or later she's going to need to lay that up with eggs. Now if you found those and said, "wow, look at this, it's perfect eggs. I know in three days I'm going to have some really nice larva." If it's brand new foundation and there have been no bees born from that comb, it's going to be very difficult for you to use this Chinese needle because you're going to encounter wax in the bottom of the cell. And this tongue will actually bend over backwards. And you will be very frustrated and you will, very quickly, give up beekeeping altogether. So, in order to avoid that, you need to either pretend that frame does not exist or if you are determined to use that larva, you may want to consider trying the German dental tool. I can tell you that I have tried the German tool and it is very tricky to use. But is something that if you pick it up, can be very valuable in those situations. But it's good to try at least both ways to see how well you do. Because, as I said, if you've got some fresh foundation that has fresh wax, the German tool is really going to get you ahead if you need to use that larva. So here, what I've done is, we have just a wood bar with a one-eighth inch slot in it. I've attached 15 JZBZ plastic cups to this bar. And then as an added measure what I have done is I have dipped these into molten wax almost up to the tips but not flowing over into the cups. This will serve two purposes, the first of which is, it is going to help the bees make your cells just a touch bigger. They're not going to make them beautiful, they're not going to make them the best cells in the world. But the cells, the outer parts of the cells themselves will look nice, they will have enough wax stored right here. Whereas, if they are Carniolan or a race of bees or even a hive of bees that's having trouble drawing wax, this will help them make nice wax. You will find that your cells consist of less dark wax they have to recycle and they'll look kind of bright and pretty. And my favorite feature of dipping these in wax is the cell cups don't fall out of these very easily. And sometimes you will run into a cell bar where the cups just fall right out, you can't keep them in, they're shooting out, you can push them down they shoot back out. Very frustrating. However the wax makes it a non issue because now they're pretty well soldered into our cell bar. So then we take our larva and with our tool we go down into the bottom of this and with my handy-dandy miner's light I can have a very strong light right here. Some people prefer to use the magnifying glass with a florescent light, that is also acceptable to use. I find this to be a little more convenient. So you just shine this light down, you may have to adjust your light a little bit. But as soon as you find where that good larvae is you just

reach down there with the tool and just as it touches a little bit at the bottom, should be good enough, you withdraw that. And then check your tool, make sure the larva is on it, come to your cell cup. And then you want to place it towards the bottom, where this in this, here. And as you're pushing this plunger, you want to draw that across so that it actually deposits the larva without pushing the larva too much. As you're taking your tool back across and pushing this plunger down, it deposits the larva very, very, very nicely, very gently. The advantage to this tool is that, in addition to being more gentle to the larva, you're actually removing a little bit of the royal jelly from the bottom of the cells, which will gain you a little bit of time as you're grafting, and they will not dry out as easily. As you're grafting, it's very, very critical to keep a nice wet towel nearby because when you're finished with your bar, you're going to have to set this down where that larva will not dry out. Because if it dries out the bees will remove it. Then I take my grafting frame here, which has three notches in it, and again you may only wish to raise 30 cells to which you only need to fill two of these slots with. But this I'm doing three. You want to start at the lowest slot, which you can see these cells are facing up right now, but we're actually looking at the bottom of the frame. So starting here at the top, we're going to push this cell bar in. Just like that, keeping it centered as closely to the center as possible. The reason that I like to start this way and I work my way down is if a little bit of wax falls down, if there's anything that's falling down is not going to fall into any cells. So as I keep adding my cell bars down and down, I'm just keeping anything from the top that's falling down from falling into the cups. You want to keep contamination to a minimum. Actually you should keep your contamination down to zero, because the bees will remove that larvae as soon as they find it. So, once you have these frames filled, now it's out into the starter.

READ

Considerations When Placing Your Graft in Cell Builders

There are several things to take into consideration when placing your graft into cell builders. Placing Your Grafting Frame in the Cell Starter Immediately after you are finished filling your queen cups, add the bars to your grafting frame and carry it out to the bee yard covered in a damp towel. Slowly and carefully insert the frame into the middle position of your cell starter. One by one, quickly inspect the other frames in your cell starter to make sure there are no other young larvae present that will compete with your queen cells. Once the grafting frame is inserted, it is critical that you do not disturb or shake the colony. Developing queens are very sensitive and can easily be damaged or killed from rough handling. Handle grafting frames as little as possible and with extraordinary care. For the highest quality of queens, we recommend only inserting one grafting frame in a cell starter. Each grafting frame can contain up to 45 queen cells; three rows each with 15 cells. A very strong finisher, three to four deep hive bodies tall, can provide for two grafting frames, but too many queen cells may mean there are not enough nurse bees or resources to adequately feed them. Fewer healthy, strong queens are preferable to many poor queens. ■ Placing Your Grafting Frame in the Cell Finisher ■ After 24 hours have passed, carefully lift the grafting frame out of the cell starter. The queen cells will have more wax added to them and the larvae will be larger and provisioned with lots of jelly. The queen cells will be covered with nurse bees. Gently brush the nurses off with a bee brush. You can use a little smoke as well, but do not drive smoke directly into the open cells. Place the grafting frame into the cell finisher and, again, be very gentle when doing so. Removing Your Grafting Frame From the Cell Finisher Nine days after you introduce the queen cells to the cell finisher, the queen cells are ready to harvest. They

will be ready to emerge from their cells in two days. Queen cells can either be inserted directly into mating nucs or can be placed into an incubator set to the proper temperature and humidity to keep them warm until you are ready to ship them to customers. If you are going to be away from the bee yard the day the queens are to be removed, place queen cell protectors over each cell on your grafting frame prior to inserting it into the cell finisher. Without a queen cell protector, the first queen to emerge will quickly destroy all the other queen cells in your cell finisher. Queen cell protectors can be bought from your local bee supplier. With them, the queens will emerge in their protectors and workers will feed them through the mesh/holes until you have time to return to the colonies and can harvest the queen cells. ■ Cells are removed from the cell finisher after nine days. They are then transported in a protective box or incubator. (Photo credit: Jon Ryan) They are inserted in mating nucs.

WATCH

Cell Builders — The Cell Starter

Healthy queens can be reared by using cell builders, which are colonies used to build queen cells. Grafting frames containing larvae are inserted into cell-building colonies. The workers will provide them with royal jelly and draw out and cap the cells. These cell builders can be either cell starters or cell finishers, and each is prepared one day in advance of when they will receive larvae. Strong cell builders are overflowing with bees. Photo credit: Ben Rulison The Cell Starter The purpose of a cell starter is to provision the developing queen cells with ample early nutrition to get them off to a good start. Queen larvae are only placed in the cell starter for one day. When nurses are primed to raise queens, they produce a lot of royal jelly. Yet as they age, their ability to sustain an intense level of feeding this many queens decreases. For this reason, after 24 hours, queen cells are often transferred to a cell finisher. The cell starter colony can be a swarm box — a nuc with a screened bottom to allow ventilation and no entrance — or it can be a deep hive body on a bottom board with an open entrance. The advantage of using a swarm box is that fewer bees are needed. Remove the queen from the cell starter 24 hours before you introduce the grafting frame to simulate an emergency queen-rearing situation. The bees will be eager to quickly start rearing queens from the larvae you introduce. The cell starter should also have a large population of bees — especially nurse bees — packed to the point that they beard at the entrance. Provide it with frames of capped brood that will soon emerge and continue contributing to the population. There are no larvae (open brood) in the cell starter. This ensures that all of the nurse bees' resources go toward the queen cells you introduce. It also prevents the colony from raising its own queens. Supply the cell starter with abundant bee bread. Frames of pollen/bee bread placed next to the queen rearing frames allow the nurse bees to use this protein to produce royal jelly to feed the queen larvae. Protein supplement/substitute should be supplied if little pollen is naturally available, and sugar syrup (1:1) should be supplied to feed the adult bees. Lastly, if using a swarm box with no entrance, place a small sponge soaked with water inside the colony, and place the swarm box in a cool, dark space so the crowded colony can remain cool. If using a deep hive body, you may need to provide extra ventilation if the weather is quite warm. Long description ■ A hive box with ten frames. Frame 1 is a syrup feeder frame. Frame 2 is honey. Frames 3 and 4 are capped brood. Frame 5 is pollen. Where Frame 6 usually is, there is an open space for a grafting frame. Frame 7 is pollen. Frames 8 and 9 are capped brood. Frame 10 is honey. Watch this video to see how we set up a cell starter at the Dyce Lab. Video Transcript It's the day before we're going to graft. So we're going to set

up our cell starter. We have started with a very strong colony, and two deep hive bodies. Our cell starter is going to be a ten frame hive body. Alternatively, you can also use a swarm box as a cell starter. The resources will be in the same proportions for a swarm box, but fewer. So here is our starter colony and two hive bodies. We're going to remove the top hive body and place it on the bottom board. After we do this, we're going to look for the queen to make sure that she's secured in this hive body. So we found her on this frame and we're going to put her back in. We don't want to to be on cell starter otherwise she will kill our queens cells. This hive body that I'm working in is not going to be our starter colony. We're actually going to use other hive body to the left of me and we're going to rearrange the frames in our cell starter so that we have the right resources and conditions for layering a queen I have Eleanor, who's going through some of these frames to find ones that we're looking for in particular. The first frame that we're going to add is a honey frame, on the far right side, in the number ten position. Beside that frame, we're going to add two frames of cap brood. And we check these frames in advance of inserting them to make sure there aren't any young larvae or eggs on them. And then after these two frames of cap brood, the next frame that we're going to put in will be a frame of pollen. And on this frame in particular, there's lots of pollen on the side that's closest to me and more brood on the side that's away from me. So we're going to position this frame so that the pollen is going to face the grafting frame and the will face away from it. The frame that's going to go beside this pollen frame is going to be our grafting frame, so we'll a space for that. And then we're going to mirror the frames on the left side of the colony. If you don't have enough honey, you can actually place the syrup feeder in the number one position. And then after we've got all our frames in, we're going to shake three frames worth of nurse bees into the colony to make the population very strong and to have a lot of nurses to feed the developing queen. We're going to smoke down the colony, to make sure that when we add our lid, we're not going to be squishing any bees. We'll put our inner cover one, and we'll put our outer cover on. And then our starter will be complete. Complete. After 24 hours we return to the colony. Today is the day we're inserting our grafting frame. We smoke down the bees, and we want to make sure that the population in our starter colony is still nice and strong. So we see lots of bees at the entrance, lots on the inner cover, I'm anticipating that not a lot of bees flew away since we shook them in yesterday. So yes, it looks nice and strong, ready to build some queen cells. So that open space in the middle is actually where we're going to be inserting our grafting frame. But before we put it in, we're going to do one final check of our brood frames to make sure that there's no young larvae that will compete with our queen cells that we might have missed finding yesterday. Once we verified that there weren't any, or once we remove them if there were, we can insert our graphing frame gently. So I'm not able to find any young larvae or eggs, so I think we're good to go with this one. Eleanor's going to put in our grafting frame and she's going to handle it very gently to make sure that we're not disturbing the larvae or shaking them or mishandling them. And once that frame is inside, we're ready to close up the colony and we'll come back in another 24 hours to move our grafting frame out of this cell starter and put it into a cell finisher.

WATCH

Cell Builders — Cell Finisher

After your queen cells have been placed in the cell starter for 24 hours, they can be transferred to a cell finisher. The cell finisher can be made up a day in advance at the latest, but many beekeepers ensure

the finisher is ready about a week in advance. Cell finishers are queenright and are very populous. Crowded conditions initiate swarm preparation; once queen cells are introduced, the bees quickly contribute to their development. Cell finishers should be strong colonies that are between two and four deep hive bodies tall. The queen is kept in the bottom one or two deeps of the colony, confined with a queen excluder. With her are frames of honey, capped brood, and drawn comb in which she can lay. Above, the queen excluder are frames of bee bread and honey as well as larvae of mixed ages. The 24-hour-old queen cells are placed among the frames of larvae in the top hive body. The presence of larvae draws nurse bees up into this box to provide for them and for the queen cells. In this video, Brian Evans, the head of queen rearing for Kutik's Honey Farm, sets up a cell finisher. Video Transcript Let's join queen rearer Brian Evans in assembling a cell finisher. In this example, we're starting out with a strong colony with a brood nest spanning two deep hive bodies. Some beekeepers instead use very large colonies with brood nests spread across three or even four deep hive bodies. And those huge colonies are typically prepared weeks or even a season prior in advance, so that they can get so large. Our double deep cell finisher needs to be assembled at least 24 hours in advance of when we're going to insert our queen cells. First, let's separate the two brood chambers. We'll take this top box off and place it on top of an empty hive body so as to not squish any bees. Our next step is going to be to look for the queen. We want to make sure that she ends up in the bottom hive body. If she is in the top body, she will kill all the queen cells that we insert. As we're pulling out frames to find her, we're also taking note of what's present on those frames to determine which box they're going to end up in. After checking a few frames, Brian's found her. She's paint marked with a white spot and is on the left hand side of the frame scurrying around the edge there. Now that we know where she is, we're going to place her back into the bottom box. Once we've done that, we're going to rearrange the frames in the colony, such that the top box, which Brian is filling out here, consists of food stores in the end positions. On the left side, in the number one frame position, he has a frame feeder that he fills with sugar syrup. On the number ten frame position on the far right side, he has a frame of mixed pollen and nectar. Moving his way toward the middle of the box, he inserts frames of eggs and open larvae. And typically, the age of larvae is oldest in the further positions and gets younger toward the middle of the box. After he's picked all the frames he's going to insert, he's going to move them around a little bit so that there's going to be an opening for our grafting frame that's going to go right in the middle. And the bottom box, although you don't see him assembling it in this video, it's also going to receive several frames of honey in the end positions and at least three frames of capped brood in the middle positions. Once we've rearranged all the frames, a queen excluder is added to keep the queen separated from her young larvae and from the queen cells we're going to insert. And there's an extra black frame on the ground because we've left one frame space to accommodate the grafting frame, which you can see here. We place the top box back on top and all the extra bees that are on that frame can be shaken right into the colony. And after that, our cell finisher is going to be complete. We're going to let the cell finisher sit for at least 24 hours before we insert our grafting frame. Because we want to allow the colony to settle into its new arrangement and we want to allow time for the nurse bees to move up to the top box so that they'll be available to feed our queen cells once we insert them. You can reuse finisher colonies to rear many rounds of queen cells throughout the season, as long as the colony remains strong. But because the cell finisher is crowded, contains ample food stores, and is queenright, it is at risk of swarming. With each round of queen cell finishing, you will likely need to

remove swarm cells and take other preventative measures so that it does not swarm. After larvae in the top box become capped, they should be moved below the queen excluder. Once they emerge, the queen will have space to lay again. After she lays in those frames, they should again be moved back up to draw nurse bees to the top box. In this manner, these frames are rotated every nine to ten days. The continuous emerging capped brood contributes a steady supply of nurse bees to care for each round of queen cells, and the rotation of frames ensures the queen always has room to lay and the colony does not swarm. Colonies used for queen rearing need to be managed more intensely than colonies used for honey production. Queen cells are left in the cell finisher for nine days to complete development. They are removed two days before they will emerge, which gives the beekeeper time to insert them into a mating nuc or a queenless colony, or to ship them to a customer. It is important that the cells are removed before the first queen emerges; otherwise, she will kill the rest of the queens still in their cells. Excellent queen cells, ready to be harvested. (Photo credit: Jon Ryan)

WATCH

Mating Yards

After the queen cells are one to two days away from emerging, they can be placed in mating nucs. These are queenless nucleus (or mini nucleus) colonies where the queen will emerge from her cell, take her mating flight(s), and begin laying eggs. Mating nucs are small colonies so that it is easy to find the queen after she has returned from her mating flight. These colonies also do not require a lot of resources, which makes them easier for beekeepers to create and manage. The queen emerges from her cell on Day 16 (Day 1 is the day she was laid as an egg). She takes approximately one week to become sexually mature then takes her mating flight(s) if weather conditions are good. Two days after she returns from her mating flight, the queen should begin laying eggs. Initially, she may lay multiple, parallel eggs in some cells until she gets into a regular laying pattern. Video Transcript In a perfect pocket, the rear most, next to the feeder, this comb would consist mostly of honey and pollen, and maybe just a little patch of brood. Whereas these two front frames, I like to see a lot more brood but maybe a little bit of honey and pollen around the periphery of where the brood is being reared. But generally like to try to keep three frames full of bees but not so much that they're overflowing. So also when you're building these mating nucs in the springtime initially, when your evening and night time temperatures are going to be going down into the 40s and possibly into the 30s. When you place these queen cells, you want to make absolutely sure, even if you've got to kill a couple of sealed brood, you want to keep these queen cells right on that brood because all the nurse bees are going to be gathered to keep that brood warm. And you also want them to keep the queen cell warm. However this is towards the end of the summer in New York State, so I don't really expect a lot of night time temperatures to dip down much past 55, even I think one night we had 50. So when I go to put the queen cells in these things, what I actually do is I open them up in a diamond shape just like this. And so all of these boxes will be opened as a diamond shape, which allows me to come in with the queen cell and put the queen cell in. So by this time of year it makes it fast to put the cells in, but I'll also just put the cell right off to the side here where I'm not hurting any of the brood and it's kind of on the honey. There's enough bees in here now going, where they're going to keep her warm regardless, even if it's a little bit cooler at night. But I don't expect the temperatures to really drop dramatically at night, so these queens should still be okay. Video Transcript Okay, so we've put the cells in here approximately 10 to

14 days ago. We've checked a couple of these boxes and determined that all these queens should be mingling. So now we are going to go through and inspect each of these four pockets here to hopefully find a mated queen. And we have an inner cover that's basically made out of burlap. The good part is you can kind of smoke through the burlap, pre-smoke the bees. And carefully peel this burlap back, hold it down. We have ourselves a couple of queen cages. And we'll begin to look for the queen. So, let's see. The way that I like to do this, is I do one frame at a time because I'm a little bit more clumsy than other people. And in my search for eggs and or a queen, now I have started to see eggs here. We've got some multiple egg laying. Here but I'm not going to assume a lame worker just yet because that's the first frame. Again the key to catching these queens is stealth and the less noise, the less you get these bees disturbed, the easier it's going to be to find the queen. I already saw the eggs and I can already see this queen right here running around. Which the other advantage of these types of hives is there's not as many bees so it's a lot easier to find the queen a lot quicker. So now to catch the queen what we need to do is obviously not use gloves because that could very well crush her. So with our bare hand we are going to find a good space that the queen is in. And she's going to keep playing her game here. But we basically, right up behind her, grasp her by the wings. Now, to mark the queen, what we need to do is grasp her by her thorax. Now, you can see her legs are dangling in the air because she's looking for something to grab onto, which is going to be my finger. So as soon as she grabs on to the thumb, kind of position her around. You don't want to catch her legs in your fingers, you want to make sure they are down by her sides. So now we will just take our marking pen, just put a little mark on her. And now that she is in my fingers, dropping her upside down, deposit her right into the cage. Video Transcript A beginner could probably, comfortably keep approximately, I would say, 50 boxes which would yield about 200 holes, which is not going to put a lot of pressure on raising queen cells. And it's also not going to put a lot of pressure on catching the queens because all of this takes time. So you want to make sure that you don't overshoot what you're actually, you know capable of. Because you could very easily get into trouble, because, again, if you don't catch these queens in a timely manner, they will get strong, and they will leave, and you'll have to start over basically. They'll leave the brood and eggs behind, but mostly, due to the size of these pockets, because there's not enough bees to actually divide normally, as a swarm normally does, the whole pocket will just leave. They'll just take all the honey they can, and just get out of there. So you need to absolutely catch these queens no more than 14 to 18 days after you put the cells in, depending on mating weather. But generally, the general rule is probably 14 to 18 days. That's probably the key point I would say.

International Queen Marking Color Code The colors that queens are marked typically coincide with the year, which makes it easy for a beekeeper to identify her age. Because queens don't typically live past five years (although it has happened in history!), the colors repeat themselves starting in Year 6. The international color code is outlined below.

Color	Year Ending
White	1 or 6
Yellow	2 or 7
Red	3 or 8
Green	4 or 9
Blue	5 or 0

Marking queens also makes it easy to tell when a queen has been superseded or if she has left with a swarm. If you find an unmarked queen upon inspecting the hive, you know that your original queen is no longer there.

TOOL

Queen Rearing Schedule

Now that we have covered the complete ■ Download the Tool process of queen rearing, please review the Use this Queen Rearing Schedule Queen Rearing Schedule. This schedule to remind yourself of the steps to outlines the steps of queen rearing, day by day. rear a queen. The left column outlines the days, with Day 1 being the day the egg is laid. The next column outlines the activities of the beekeeper, followed by the activities of the bees in the colony. The rightmost column is the stage of development of the queen.

MODULE WRAP-UP

Queen Rearing With the Doolittle Method

In this module, you explored the fundamentals of queen rearing using the Doolittle method. Learning to produce your own queens marks an important step toward becoming a more self-sufficient and skilled beekeeper.

Breeding for the Traits You Desire

If you're interested in maintaining a few colonies that are better able to survive the winters in your region or thinking about starting a large-scale breeding program to sell mite-resistant stock for profit, having a fundamental understanding of honey bee breeding is required. In this module, we begin to explore how to breed honey bees and what traits beekeepers can select for.

READ

Breeding Bees

Honey bee breeding is a powerful tool at the disposal of beekeepers. The goal of bee breeding is to increase desirable traits in a bee stock while removing undesirable traits. Just as horse breeders can breed the fastest horses and farmers can breed the juiciest tomatoes, beekeepers can breed bees for a variety of different traits. Let's say you have 10 colonies. Two of them produce a lot of surplus honey for you to harvest, three of them produce very little surplus honey, and the remaining five produce a moderate amount. Breeding bees (queens and drones) from those two productive colonies may help you produce more colonies that express the same trait — good honey production — in the following generation. Of course, breeders can only select for traits that arise, at least in part, from genes. These traits include gentleness, a good laying pattern, a reduced tendency to swarm, fast spring buildup, good wax production, some elements of hardiness in cold winters, high honey production, hygienic behavior, grooming behavior, an instinct to bite or otherwise incapacitate Varroa mites, and more. It is more complicated than that, though. Some traits are easier to select for than others, depending on particular qualities of the genes. A complex trait like honey production is governed by many genes, and it can be difficult to select multiple genes at the same time. Furthermore, the environment plays a big role in behavioral traits. The forage available within its flying distance, the amount of precipitation, and the weather conditions during nectar flows all influence how much honey a colony stores. Lastly, we often want many beneficial traits in our bees. Not only do we want them to be good honey producers, but we also want them to grow quickly in the spring, be robust in the face of cold winters, and be resistant to major pests and diseases. Selecting for all these traits, which are governed by many

different genes, while also considering how outside environmental factors are influencing the expression of those traits, is complicated and requires trade-offs. No one bee stock can excel in every single trait, so it is up to the breeder to decide what traits to prioritize. If you are new to bee breeding, try selecting one trait to begin with, then build up your experience over several years. There are many ways to approach breeding. Some methods include the inbred-hybrid breeding system, the circular mating system, and the mass selection system. These systems for extensive breeding efforts will not be covered here; what we will do instead is cover the primary principles behind breeding. From these principles, Master Beekeepers can develop their own approaches. Variation in honey production among colonies. Photo credit: Jon Ryan

WATCH

Genetic Diversity

While many breeders select for one or more traits in their honey bees, an emerging trend is to instead promote colonies that have high genetic diversity. This is important for organisms to adapt to environmental changes and other pressures within one generation or over time. Genetic diversity — having many different gene versions present in a colony — is a result of the queens mating with many different drones. Some breeders strive to promote genetic diversity in their colonies. Instead of removing undesirable gene versions in a population through selective breeding, they establish populations of bees with many different gene versions and, as a result, many different traits. Different workers carry these different gene versions, so the colony as a whole is mixed. Genetic diversity has been shown to improve a colony's ability to survive disease, stabilize the nest temperature in changing environments, store more honey, and build comb faster. Colonies with queens that have mated with many different males have lower rates of American Foulbrood infection than colonies with monandrous queens (queens that have mated with only one drone). Genetic diversity also leads to more productive colonies. In one study, colonies with polyandrous queens produced more new workers and drones, and therefore had more foraging workers returning to the hive every minute, stored about 40% more food, and were more likely to survive the winter than colonies with monandrous queens. Swarms from colonies with polyandrous queens also found new nest sites faster. Watch this video to learn more about the benefits of genetic diversity. Video Transcript Let's take a look now at why genetic diversity in honey bee colonies is so important to their health and well being. And by genetic diversity, what I mean is diversity among the genes that are carried by the worker bees. It's the workers that do virtually everything in a honey bee colony except lay the egg. And as we'll see, it's very important that those workers, that those bees in the workforce are not all identical, genetically, but that they have differences among themselves. Now, where does that genetic diversity among the worker bees come from? Well, it comes from, the shocking, really shocking promiscuity of queen honey bees. On average they go out and mate with between 15 and I guess it's between 10 and 20 males. And they do this in their mating flights and they can get lots and lots of sperm from different males. And they combine this, mix it all up and store it away in their spermatheca. So when a queen lays an egg and she doles out a few sperm, well, as she lays egg, after egg, after egg, it's a different father for the different eggs because she's got sperm from 10 to 20 different males, or even more. Now why is that important, what's the payoff to a colony of having genetically diverse workers? Well one of the big ones has to do with it enhancing a colony's ability to deal with diseases. Diseases are constantly evolving and the

bees also have to constantly evolve their defenses against these diseases. This is called the Red Queen phenomenon, it's from Alice in Wonderland where Alice is running along constantly and she's chasing the Red Queen and she wonders why am I running all this time and then she realizes because the Queen is always running. Well the Queen is like these diseases. Diseases are always evolving new ways to infect the bees and the bees have to constantly evolve new ways to defend themselves from these diseases. Now let me give you an example of how important this diversity plays out for one disease. And I'm going to talk about American foulbrood. This is sort of the worst disease or probably the most virulent disease, bacterial disease that honey bees encounter. And the way we looked at the role that diversity plays in giving colonies resistance to American foulbrood is we did an experiment where we set up groups of colonies. In some colonies the queens were instrumentally inseminated with the sperm of one male and the other the queens were instrumentally inseminated in sperm from ten different males. And we would then infect the colonies, all of the colonies we sprayed them with a solution of American foulbrood spores. So they were all exposed to this disease in the same way, and what we found was that if the queens were, in the colonies that were the queens were multiply inseminated, there were very few cells per frame of brood that were infected with American foulbrood - basically, zero. But if the colonies were headed by queens that were just singly inseminated with the sperm of just one male, there were a few colonies that had very low infection rates of American foulbrood, but there were other colonies that had very high levels of infection rate. Evidently, in some of those colonies, many of those colonies, the workers just didn't know how to deal with American foulbrood. And it also, these results also show that you don't have to have all the workers in a colony that have the ability to clean out American foulbrood, you just need a small fraction of them. So it's really, it's a very dramatic example of the ability of the colony to deal better with disease, by having the queen multiply inseminated. This high genetic diversity among the workers also has a payoff in terms of foraging. And as you can see here, what we've got plotted here is the weight gain of colonies, for two groups of colonies. When one group, the colonies are genetically diverse indicated by the orange squares and those are again produced by colonies that are instrumentally inseminated with the sperm from multiple males, versus colonies that instrumentally inseminated with the sperm of one male and if you follow these two groups of colony all summer long, the colonies that are genetically diverse workers gain a lot more weight, build much more comb, thrive, and survive for the winter vastly better than do the colonies without that genetic diversity. You might wonder why that is, well it's an interesting story, and there are many parts to it. But the part that I like the most is the fact that, if the queen is mating with multiple males, there's some fraction of her workers that just get up out of bed and go to work early in the morning. They're not, they don't work fast. They don't work long, and more wisely or anything like that but they just get going in the morning. And when they go out and they come back and start the dancing, they got the whole ball rolling. They get the colony humming sooner in the beginning of the day than the colonies that don't have such workers. So, that's another in which the genetic diversity can help our colony with its health and well being.

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Inbreeding

Inbreeding, or the production of offspring from closely related individuals, leads to a loss of genetic diversity. Sometimes breeding efforts can lead to inbreeding, when breeders remove so many gene

versions from a population that only a few remain. When inbreeding occurs at length, there are a limited number of gene versions in a given population. Select mutations may become more common. If colonies are faced with a new selection pressure (such as a drought or a bad infection), an inbred colony may have no workers with traits that can survive that event, and the colony could die out. Inbreeding also increases the likelihood of larvae inheriting two identical versions of the sex determination gene, which causes them to be cannibalized by their sisters. Less dramatically, if certain desirable traits are missing from the gene pool, a beekeeper using only an inbred population to select from for breeding purposes may find that none strongly display the trait of interest (for example, none may be especially gentle or especially good honey producers). No matter how carefully bees are bred, the desired traits cannot be produced unless the gene versions associated with those traits are present. You can avoid inbreeding by making sure that your queens come from genetically diverse lines and periodically requeening from different stock as well as by ensuring that virgins can access a large and diverse population of drones, whether through mating flights or artificial insemination. It is best to use unrelated drones from different colonies to mate with your queens to reduce possible inbreeding. To reduce the chances of inbreeding, it is ideal to use five to 10 colonies as your queen mother colonies and 10 to 15 colonies as your drone mother colonies.

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Overview of Breeding Bees

There are many methods to breeding bees, but they all boil down to mating the queens from your best colonies with drones from your other best colonies. You can allow them to mate naturally or you can use artificial insemination to insert semen from chosen drones into your chosen queen. Here are the main steps to bee breeding: Evaluate traits in your colonies to determine which colonies you will select as your queen mother colonies and drone mother colonies. To do this, set up an isolated apiary with the colonies you wish to evaluate. This apiary is called your testing apiary. The colonies that receive the highest evaluations will be your queen mother and drone mother colonies. Take careful notes on how your colonies perform with regard to your trait(s) of interest over time. Graft larvae from your queen mother colonies and place them into cell builders to mature. Place your drone mother colonies in yards one to three miles away from your mating nucs — this is the typical distance queens travel to drone congregation areas. If you can, have your drone mother apiaries spread out from one another in a circle, with your queen mother apiary located in the center of the circle, so no matter which way the virgins fly, they are likely to encounter a drone congregation area that contains your drones. If you do not live in a remote location, your queens will likely mate with other drones in the area — not just your drone mother colonies' drones. Make sure the timing aligns so that there will be sexually mature drones in your drone mother colonies when your virgin queens are ready to mate with them. The earliest you can time your first graft of the year is to coincide it with when much of the drone brood has reached the stage where they have developed purple eyes. Open a couple developing drones to check their eye color. Place mature queen cells into mating nucs one day before the queens are ready to emerge. Virgin queens will emerge and, within a week or so, they will mate with drones in the surrounding area. Once the queens are mated and laying, introduce them into full-sized hives or sell them. For any queens you keep, continue to evaluate their traits over time. Keep breeding in this way for several generations until the trait you are interested in is consistently expressed in your daughter queens. Once

a stock is established — say, one with good hygienic behavior that confers resistance to Varroa mites — it must be maintained through continued breeding. If it is not, this trait will eventually be weeded out as the queens carrying hygienic genes mate openly with drones carrying other genes. Some regions have cooperative breeding programs, where all beekeepers in the area maintain bees with certain traits, saturate the region with these drones, and perpetuate these traits in the population. Breeding bees with specific traits is a continuous effort. If you only want to breed bees for yourself, then you probably do not want to continuously maintain breeding lines. In this case, you can do small-scale breeding in your bee yard and, every few years, purchase a queen with the traits you desire from a reputable breeder. That way, if open mating has weeded out some of the traits you selected, you can reintroduce them.

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The Role of Drones

Female honey bees — both queens and workers — inherit half of their genes from their mother and half of their genes from their father. Therefore, selection of good quality drone mother colonies is equally as important as selection of the queen mother colonies. Add drone comb foundation to each of your drone mother colonies in your mating apiary well in advance — at least 35 to 40 days before drones are needed — to boost the number of drones in preparation for mating. Drones take 24 days to develop from egg to adult plus an additional nine to 12 days to sexually mature and be ready to mate. Remember that if you are inserting undrawn drone comb, the bees will need more time and resources to draw it out. Also, keep in mind colonies will only start investing in producing drones if they perceive they have the resources to begin investing in reproduction. These colonies should be healthy, strong, and have access to good nutrition. Under good conditions, you can expect up to 20% of comb to be built for drone rearing and about 5% of the bees to be drones. So in a strong colony in early June that may have around 20,000 bees, about 1,000 of them may be drones. Drone mother colonies may need to be fed honey/syrup and bee bread/pollen/pollen substitute in early spring to build the population. Simulating a nectar flow in a strong population encourages queens to lay drones, and good nutrition is critical for the first few days after emergence for drones to develop high-quality and abundant sperm.

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Controlled Mating vs. Open Mating

Controlled Mating Controlled mating is the surefire way to know the genealogy of your bees because you know exactly which drones the queen is mating with. Attempts at controlled mating began as early as the 1700s, when beekeepers confined queens and drones in glass dishes, hoping they would mate. As we know, queens and drones require specific conditions for mating and do not mate in confined spaces naturally. The past several hundred years have seen many advances in controlled mating, and now it can be accomplished in two ways: **Isolated mating yards:** Some beekeepers have isolated areas where they set up their breeding apiaries. These areas could be in mountain valleys or on islands, separated by at least nine miles — the maximum flight distance — from other wild and managed honey bee colonies. Only the colonies that they wish to breed are present in these areas; drone congregation areas only have drones from those colonies. **Artificial queen insemination, also known as instrumental insemination:** If you do not have access to an island or mountain valley, it will be difficult to ensure your

queen is only mating with the desired drones. In this case, you can accomplish controlled mating using instrumental insemination. The first instruments for queen insemination were developed at Cornell University in 1927; the technique has been successfully implemented since the 1940s and refined over time. Open Mating You may not have access to an isolated island or mountain valley; you may instead live in an area with many colonies in your bees' flight range. You may also not have the desire to learn artificial queen insemination and purchase the necessary equipment. In cases like this, your queens must mate openly. Open mating is the natural mating process, wherein queens mate with 10 to 20 adult drones in drone congregation areas. Most beekeepers rely on open mating, but there is no guarantee that their queens will mate with drones of their desired stock. To combat this, try to saturate your area with drones exhibiting the traits you desire by setting out many drone mother colonies and by putting drone comb frames in your drone mother colonies. If you contact surrounding beekeepers, you can collectively bring in colonies with desired traits and saturate the area with good drones. An openly mated queen returns with her mating sign. Photo credit: Jennifer Tsuruda

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Evaluating and Comparing Colonies' Traits

Your inspection records may reveal which colonies exhibit traits that you like. Colonies that you like should be moved to a testing apiary to be evaluated for the traits of interest and considered for drone mother and queen mother colonies. If the traits are heritable, then the offspring colony should resemble the parental colonies. Selection of traits through a breeding program like this — and repeated every year for the continual improvement of colonies in your apiary — will, over time, generate your own desired stock of honey bees. Indeed, the colonies from your selected queens, and drones should also eventually be evaluated to ensure they have the trait(s) you want. In the testing apiary, reduce drifting as much as possible. Drifting bees from neighboring colonies can make up as much as 30% of the bees in a colony! We do not want to assess the traits of these adopted neighbors. Make sure the colonies are several yards apart and each colony has a unique landmark to help orient the bees. Do not arrange the colonies in rows or in a regular way. Do not interchange brood or bees between colonies, either. These practices ensure that the bees performing the behavior in a particular colony are from the same genetic line as the queen in the colony. Assessments A system for evaluating your colonies helps you choose which ones to breed from. The system will also evaluate your success to determine whether the new colonies did, in fact, inherit the traits you wanted. Assess all the colonies in your testing bee yard and keep detailed records of the scores they receive. For some traits, it takes two or more years to thoroughly assess a colony's performance; e.g., queen longevity or winter survival. If possible, keep the colonies in the same apiary to reduce geographical factors, and treat them similarly: Feed them the same amount if supplementing in spring or fall and overwinter them using the same methods. Next, develop a scale for assessing traits to help you keep records of how the colonies perform. These scales can vary as you see fit, but they traditionally look something like this. Scale for assessing traits

Unacceptable	Poor	Satisfactory	Good	Excellent
0	1	2	3	4

For example, colonies with no frames of surplus honey might receive zero points, one to five frames receive 1 point, six to ten frames receive 2 points, 11 to 15 frames receive 3 points, and 16 or more frames receive 4 points. Assess your colonies yourself to reduce evaluation bias from other people and assess all of them during the same time period to reduce the influence of seasonal or other environmental factors. Since

there is no North American standard for evaluating traits, there is a lot of variation in evaluations and record keeping. Here, we have provided some examples of assessment methods on a small selection of traits.

Honey Production The most common way to assess honey production is by waiting until the end of the beekeeping season and weighing your honey supers after you have removed them. Record the weight of the honey supers for each colony, extract the honey, reweigh the empty supers, and subtract the second number from the first. This gives you the total weight of extracted honey. You can also count the number of frames of honey produced by their colonies. You can do this on a scale of 0 to 5, as shown above, or just use the total number of frames. Although these two ways are the most accurate, they require that you wait until the end of the season to compare colonies. Alternatively, you can keep track of hive weights during a honey flow. Honey production is directly correlated with short-term weight gain. Use a hive scale to weigh the colony gains at the beginning of an expected honey flow. Weigh all the colonies again after the honey flow is over and subtract the initial weight. Be sure to collect all colony weights on the same day and within a narrow time window. Evaluate each colony's performance using the scale you have devised — either 0 to 5 or another scale.

Gentleness Gentleness is usually assessed with leather. Honey bees have an aversion to the smell of leather and are prone to sting it. You can attach a piece of leather to a stick and use it to provoke the bees. Wave it around the entrance of the colony and remove the inner and outer covers and wave it around the open hive. Do this for an equal amount of time for each colony — usually two minutes — and count the number of stings in each piece of leather. Score the colonies using the scale you have devised.

Hygienic Behavior Hygienic behavior is the ability of adult bees to detect dead brood and their tendency to uncap the cells and remove/chew the dead brood quickly. Hygienic behavior lowers the likelihood of brood diseases like American foulbrood and chalkbrood because infection levels can be lowered if the affected brood is removed quickly. Hygienic behavior is also associated with Varroa resistance. Varroa mites reproduce in capped brood cells, so detecting and removing infested larvae can help reduce mite infestation levels. Testing for this hygienic behavior does not involve introducing any of these diseases into your colonies. Pull out a frame with capped brood from the colony you would like to assess. Kill a small section of the capped brood cells with liquid nitrogen or by pricking them with a pin. Replace the frame in the hive, and return to assess it after 48 hours. Bees with good hygienic behavior should remove all or almost all the brood that was killed. Bees with poor hygienic behavior leave the dead brood or are much slower to remove it. Score each colony's performance using the scale you have devised.

Swarming Historically, beekeepers tended to select against swarming tendency in colonies. Now that we know swarming is beneficial in reducing Varroa mite populations, more beekeepers are now more open to having their colonies swarm regularly. If you wish to select against swarming, it can easily be quantified by whether a colony swarmed and by the presence of swarm cells in the colony. During each inspection, look for queen cups or queen cells. Colonies that initiated the swarming process despite having adequate room in the hive are not selected as breeding colonies, whereas colonies that do not initiate swarming are in the running. Make sure that the colonies being assessed have the same amount of room — number of hive bodies — available on the same dates.

Brood Quantity The amount of brood a queen produces throughout the year can be assessed relatively easily. For example, a colony may have brood present on five frames. One frame is covered entirely with brood (100%), two frames contain 50% brood each, and two others contain only 25% brood each. The remaining five frames in the colony are honey, pollen/bee bread, drawn comb, or

empty. This colony would therefore contain $100\% + 50\% + 50\% + 25\% + 25\% = 250\%$ or 2.5 frames of brood. You can also change these scores to values of 0 to 5, depending on the evaluation system you use. Brood Quality Good queens produce solid brood patterns. High-quality brood is continuous on the frame (not spotty) and arranged in concentric circles. This means that the queen begins laying in the center of the frame and continues in a circular fashion, moving outwards. You can rate brood quality on a scale of 0 to 5, based on the degree of spottiness and neighboring brood of mismatched ages. You may want to take into account the age of the queen. Additional Traits Additional traits that you might assess include early spring buildup, overwintering success, propolis production, pollen collection, and Varroa loads.

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Evaluating Multiple Traits Across Multiple Yards

Evaluating one trait at a time is straightforward. If you are interested in more than one trait, fill out assessments for each trait of interest and add up the total score of the colonies. Colonies with the highest scores will be the queen mother and drone mother colonies. Here is an example of an assessment sheet for three colonies: A7, A9, and A14. The beekeeper wants to select for honey production, gentleness, hygienic behavior, and reduced Varroa loads. Assessing Multiple Colonies for Multiple Traits

Colony code	A7	A9	A14	Date assessed	7/15/16	7/15/16	7/15/16	7/15/16
Honey production	2	4	0					
Gentleness	1	4	3					
Hygienic behavior	4	0	3					
Varroa score	2	2	1					
Total	9	10	7					

Honey production: No frames of surplus honey (0 points), one to five frames (1 point), six to ten frames (2 points), 11 to 15 frames (3 points), 16 or more frames (4 points). Gentleness: No stings in the leather patch (4 points), one sting (3 points), two stings (2 points), three stings (1 point), four or more stings (0 points). Hygienic behavior: Removed 80% to 100% of dead brood in 48 hours (4 points), removed 60% to 79% (3 points), removed 40% to 59% (2 points), removed 20% to 39% (1 point), removed 19% or less (0 points). Varroa: Mite loads exceeded 12% by the end of the season (0 points), mite loads reached 9% to 11% (1 point), mite loads reached 6% to 8% (2 points), mite loads reached levels between 3% and 5% (3 points), mite loads stayed below 3% (four points). These scores reflect mite loads as a trait the beekeeper wants to discourage. If you value these traits equally, your best colony would be Colony A9, which received an overall score of 10. You can also select for one trait above all else. If you estimate that you favor, say, hygienic behavior twice as much as any other trait, double the scores from that row and leave all the other scores the same. Your new table looks like this. Assessing Multiple Colonies for Multiple Traits

Colony code	A7	A9	A14	Date assessed	7/15/16	7/15/16	7/15/16	7/15/16
Honey production	2	4	0					
Gentleness	1	4	3					
Hygienic behavior*	8	0	6					
Varroa score	2	2	1					
Total	13	10	10					

*Twice as important as other traits. Now the best colony is Colony A7, which received a score of 13. These are just examples of assessment and record keeping. You can alter these calculations to make your records work for you. As you begin to evaluate your colonies for multiple traits and assign different priorities to them, the math gets a bit more complicated. Evaluating Colonies From Multiple Testing Apiaries If your test colonies are set up in two apiaries, it might be difficult to evaluate and reliably compare traits. Take honey production as an example: If one apiary had much more forage surrounding it than the other apiary, the colonies in the first apiary would have a natural advantage in producing more honey. At the end of the season, you rate your colonies for honey production (using the scale described above), and these are your results: Evaluating Colonies From Multiple Testing

Apiaries Apiary A: Apiary B: Good honey flow Poor honey flow Colony code A1 A2 A3 A4 A5 B1 B2 B3 B4 Assessment 2 3 3 5 5 1 1 2 4 Apiary 3.6 2 average Standard 1.3 1.4 deviation* *The standard deviation calculates the variation among hives. In examinations, we will provide you with standard deviation values. Colonies in Apiary A produced more honey on average than colonies in Apiary B. Based only on the amount of surplus honey created, you might think that the best colonies are A4 and A5 (both creating 17 to 20 frames of honey and receiving a score of 5). However, they had an advantage because they were in an apiary that had particularly good forage and rainfall. If they were in a different apiary, would they have performed as well? We cannot know based on that score alone. To eliminate the apiary effect on honey production, you can calculate Z scores for each of your colonies. The colony with the highest Z score is the one you should choose as your breeder queen. Using Z scores is not the only way to compare colonies across several apiaries, but this method is fairly simple. Z scores synthesize the individual score of each colony, the apiary average score, and the standard deviation (which measures the variation among colonies). You do not need to know how to calculate standard deviation for this course, but you are required to plug in numbers to calculate a colony's Z score. The formula is simple: $Z = (\text{individual score} - \text{average}) / \text{standard deviation}$ Here are the Z scores for each colony. Evaluating Colonies From Multiple Testing Apiaries Apiary B: Apiary A: Poor honey Good honey flow flow Colony A1 A2 A3 A4 A5 B1 B2 B3 B4 code Assessment 2 3 3 5 5 1 1 2 4 Apiary 3.6 2 average Standard 1.3 1.4 deviation Z score -1.23 -0.46 -0.46 1.07 1.07 -0.71 -0.71 0 1.42 Colonies with negative scores performed worse than average. Colonies with positive scores performed better than average. Colonies with scores equal to zero performed average. Based on these scores, we can clearly see that our best colony was Colony B4, with a score of 1.42. They produced more surplus honey than average given their location in the apiary with the poor honey flow. If you were wondering what colony to breed from, this is it! The worst colony for honey production was Colony A1, with a score of -1.23.

MODULE WRAP-UP

Breeding for the Traits You Desire

In this module, you gained foundational knowledge in honey bee breeding principles. From selecting for winter survival to developing mite-resistant stock, your study of breeding fundamentals will prepare you for both small-scale colony improvement and potential commercial breeding programs. Thank You and Farewell Congratulations on completing "Master Beekeeping Part II: The Science and Art of Beekeeping" I hope that you will apply the systematic approach, practical techniques, and knowledge you learned to achieve your desired outcomes. The material covered in this course is intended to help you gain confidence in your ability to analyze different situations and make Emma Walters informed management decisions. I hope it has met your expectations as Honey Bee Extension Associate you support your colonies' survival and success. Dyce Lab of Honey Bee Studies Cornell University In addition, I hope that your beekeeping journey is allowing you to enjoy the unique pleasure that comes from connecting with bees and nature. View how Randy Oliver describes what inspires him about beekeeping: Video Transcript It's not for the money, obviously. It's a tough job to do. It's for the lifestyle. You've gotta love it. And as a lifelong naturalist and biologist, I love being in touch with nature. You're very much, just like any other form of agriculture, you're very much in touch with the seasons, with the rainfall, the temperature, with the plants that are blooming. So it's a lifestyle in our modern age where

you're living very, very close to the land, you're very much aware every single day what's happening in nature with the bees. One of the stories is my younger son, he was 16 and he was working with me out in the bee yard an April day, beautiful day, perfect conditions. And we're going through this bee yard and he says, "Dad I know this is going to sound kind of weird, but on a day like this when the bees are this happy, it's like I get high. It's just like, it's this weird feeling." And I put down my hive tool, walked over, I gave him a big hug and I said, Ian, "You just became a beekeeper. You've got the bee vibe!" You're feeling that vibe of the bees. And that's what beekeeping is, you get in touch with that vibe of the colony. It's almost like a sixth sense. And when you have that, and Ian's had that since then. Ian has just got the vibe of the bees. He can open a hive, take the lid off, he goes oh, there's something wrong with this hive, or this hive's doing great. You have that feeling of the hive. That's what a beekeeper can aspire to do, to really become one with a part of nature, the bees, which are so much involved with everything else out there. Enjoying working outdoors. Enjoying producing an honest, healthy food crop. Performing pollination services and leaving a very green footprint on the planet earth. Beekeeping leaves an incredibly green footprint on the planet. So it's a lifestyle that you can be very very much, very proud of. From all of us at Cornell University and eCornell, thank you for participating in this course. Sincerely, Emma Walters

Downloadable Resources You may wish to download and save these files for future reference. Beekeeping calendar for the northeast Varroa mite record keeping sheet Apiary record keeping sheet Hive information record keeping sheet Contacts record keeping sheet Amino acids necessary for honey bee nutrition Honey bee-visited plants categorized by their pollen value Pollen color and bloom chart Honey bee subspecies Honey bee stock traits Queen rearing schedule Recommended Books There are more books written about honey bees than any other animal on the planet, besides humans. With so many to choose from, we've compiled a list to help guide you in your readings. There are no required readings for this course. All examination questions will be based on the online material only. However, if you are interested in delving deeper into topics we've discussed, here are some great resources: Honey Bee Biology "The Hive and the Honey Bee" (2015) by L.L. Langstroth, et al. "Honey Bee Biology and Beekeeping" (2013) by Dewey Caron and Lawrence Connor "Bees and Beekeeping: Science, Practice, and World Resources" (1990) by Eva Crane "The Biology of the Honey Bee" (1987) by Mark Winston "The Lives of Bees" (2019) by Tom Seeley Inspections and Colony Management "Increase Essentials" (2014) by Lawrence John Connor "Honey Bee Producer Guide to the National Bee Farm-Level Biosecurity Standard" (2013) by Canadian Food Inspection Agency "The Beekeeper's Handbook" (2011) by Diana Sammataro, Alphonse Avitabile, and Dewey Caron "Better Beekeeping: the Ultimate Guide to Keeping Stronger Colonies and Healthier, More Productive Bees" (2011) by Kim Flottum "The New Complete Guide to Beekeeping" (1994) by Roger Morse "At the Hive Entrance" (1985) by H. Storch "A Year in the Bee Yard" (1983) by Roger Morse Swarming "Swarm Essentials" (2014) by Stephen J. Repasky "Honey Bee Democracy" (2010) by Thomas Seeley Bee Breeding and Queen Rearing "Queen Rearing Essentials" (2015) by Lawrence John Connor "Queen Bee: Biology, Rearing and Breeding" (2010) by David Woodward "Successful Queen Rearing Short Course" (2006) by Marla Spivak and GS Reuter "Queen Rearing and Bee Breeding" (1997) by Harry H. Laidlaw Jr. and Robert E. Page Jr. "Practical Queen Production in the North" (1985) by Carl Jurica Bee Hunting "Following the Wild Bees" (2016) by Thomas Seeley

Part III: Managing Pests and Diseases

Honey Bee Immunity

When we talk about honey bee health, we often focus on the numerous—and increasing—diseases and pests that make bees sick. But bees have developed a variety of ways to protect themselves against invasions and infections, including internal defenses, individual protection strategies, and, most importantly, social behaviors that protect the entire colony. In this module, we will explore how honey bees resist pest and disease exposure, how they deal with infections, and how beekeepers can provide help when needed.

READ

Individual-Level Defenses

■ Bee colonies have persisted for thousands of years despite attacks from Stingers, grooming behavior, and from such diverse actors as bacterial and viral pathogens, insect pests, mites, individual defenses that protect bears, humans, and, more recently, bees from outside invaders. chemical exposure. Many of their strategies are based in group defenses and cooperation, but let us first explore the simplest ways that each bee keeps itself safe and its body healthy. Preventing Invaders You have surely experienced a bee's most familiar form of defense: its sting! It might be strange to think of a bee's sting as immunity, but a bee uses its sting to keep out invaders. When the colony allows fewer external objects and organisms to enter the hive, it reduces the risk of damage and disease. To keep even smaller attackers out, the bee performs self-grooming (called autogrooming), regularly removing foreign objects and pathogens from its body. Basic hygiene, like showering for humans, removes disease agents from the outside of the bee's body. This is important both for cleaning and for spreading cuticular hydrocarbon secretions (skin oils). Cuticular hydrocarbons are distinct for bees from the same colony and, as such, are used as cues for bees to distinguish their nestmates from unrelated bees trying to rob the hive. Grooming is so important that honey bees have evolved specialized grooming structures. One special groove on the inner forelegs has a dense row of comb-like spines that are perfectly fitted to run along an antenna. Bee "skin" itself is a successful blockade against the outside world. Called the cuticle, it separates the inside and outside of the bee, preventing the weather and physical abrasions from damaging the bee. The cuticular hydrocarbons form a thin wax coating on the bee's cuticle that protects it from water and keeps internal moisture from escaping. The cuticle is made of chitin and proteins; it does not break down in water nor is it digestible by mammals. This makes it a reliable protective layer. The honey bee's cuticle provides a protective layer. Photo credit: Jennifer Tsuruda. Disease-causing viruses and bacteria (pathogens) can enter a bee's body through the food it eats or from contact with flowers, hive parts, nestmates, and the fecal matter of other bees. Once inside the body, the bee's gut plays a big role in preventing bacterial or viral infection. The physical barrier offered by the gut lining controls what can move between the gut and the insect's organs. Just as the gut lining can allow nutrients to enter the body, it can also prevent pathogens from entering. Further, having healthy, symbiotic bacterial communities in the gut can offer

protection for the bees, and the acidic pH of some parts of the honey bee gut can help make it an inhospitable place for bacterial and fungal pathogens.

READ

Attacking Pathogens Inside the Body: the Honey Bee

Immune System ■ Honey bees possess an efficient but Key Points simplified immune system compared Unlike humans and other to vertebrates. They lack an adaptive invertebrates, honey bees do not immune system (no antibodies or produce antibodies. memory B/T cells) but have a robust Honey bees have fewer immunity- innate immune system that is related genes than do other activated within minutes or hours. For insects. example, some cells in honey bees' hemolymph have the role to destroy pathogens by physically surrounding (nodulation) and consuming them (phagocytosis). Also, bees facing bacterial pathogens can quickly produce antimicrobial compounds that kill or inhibit the growth of bacteria in the body (examples include abaecin, apidaecin, hymenoptaecin, and defensin). Honey bees can even defend themselves against viral intruders by activating RNA interference (RNAi), a widespread defense mechanism in animals that results in silencing of viral genes. Note that such responses are weakened when honey bees are faced with multiple stresses at once. For instance, innate immune responses are weaker when honey bees are under moderate levels of Varroa mite stress or when they are stressed with the bacterial pathogen that causes American foulbrood. Part of the honey bee's immune system includes specific cells (purple) that consume and destroy bacteria (black). Although honey bees, unlike mammals, do not produce antibodies or have immunological “memory,” it was recently shown that queens can still transmit a form of immune “memory” to their offspring. This phenomenon is called transgenerational immune priming (TGIP), and although its mechanisms are not fully understood at the moment, TGIP is currently leveraged in the first commercialized “vaccine” for honey bees (and insects!) that was approved by the USDA in 2023 for protection against American foulbrood. Killed bacteria are fed to worker bees, who then feed it to the queen. As a result, immune signals or fragments get passed into her eggs. The offspring exhibit a faster or stronger immune response as a result of having the pathogen previously encountered by their mother. Honey bees do not only rely on individual immunity; in fact, they have only one-third as many immunity-related genes as other insects. How do honey bees stay protected from attacks, with fewer genes dedicated to the job? The secret lies in social immunity: Honey bees have an advanced social life. Instead of investing in their individual defenses, like many solitary insects do, honey bees have evolved colony-level defenses that provide immune defenses to the group as a whole.

READ

Colony-Level Defenses: Social Immunity

■ When groups of organisms live Key Points together instead of alone, diseases Honey bees have evolved social spread more easily among individuals; behaviors that reduce the just think about how quickly infections incidence and spread of pests, can spread in schools. It is not predators, and pathogens. surprising, then, that many strategies for protection have developed at the colony level instead of at the individual level. Some social immunity practices are preventative; others are responsive to problems once they arise. Let's explore some of the behaviors that constitute social immunity. Grooming In addition to autogrooming, bees groom one another (“allogrooming”) to remove

foreign objects and pathogens from their adult nestmates. Bees can remove items from hard-to-reach places on another bee's body. Grooming behavior has a genetic basis to it, so some colonies may have more grooming bees than other colonies. Some bees clean up their hives to keep disease from spreading. “Undertaker” bees are a subset of middle-aged bees (two to three weeks old) tasked with removing corpses and debris to maintain low levels of dead, diseased, and unwanted materials in the hive. Also, because bees are fast learners and improve at repeated tasks, designated “undertakers” make hive management more efficient. Honey bees are unique from many other species in that they can control their genetic stock on relatively short time scales by replacing their queen. Remember, the queen is the primary egg-layer in the colony. If she is not ■ Undertaking ■ Requeening ■ laying sufficient brood, then there are not enough worker bees to keep up brood rearing, nursing, and cleaning. Furthermore, colonies that are under stress from parasites and pathogens may construct supersedure cells to replace the queen. The reasons are unknown, but it could be that the queen herself is sick or that the colony is attempting a “fresh start.” Absconding Absconding is the act of an entire honey bee colony leaving their nest or hive completely. If mite levels or other disease or pest levels are too high, bees may leave their current location and search out another one. This can also be triggered by hive damage, repeated bear attacks, or other environmental problems. The location of the nest site is a crucial decision made by bees; they become “immune” from bears and other attackers — including humans — when they select a nest site high off the ground, with only a small entrance hole to defend. This strategic decision in nest location makes them harder to detect, access, and destroy. ■ Siting the Nest ■

WATCH

Propolizing the Nest

■ You are already quite familiar with Key Points propolis. Bees make propolis by Propolis contains antimicrobial gathering plant resins and mixing properties. them with wax. Propolis reduces drafts and water damage. And just as it slows down your access to the hive, it also blocks or slows many small insects, pests, bacteria, fungus, and any number of other agents of disease and destruction. Bees can use propolis to “entomb” these pests and predators, which keeps the hive clean and minimizes the spread of disease — they'll even fully propolize mice or snakes that die in the hive. The presence of propolis has also been shown to reduce the expression of honey bee immune-related genes. This is not a problem; it means that the colony is healthier in general and the immune systems of honey bees are less stressed when propolis is present. Propolis also has antimicrobial properties, although its exact degree of antimicrobial activity varies based on the plant from which the resin came. There is some evidence that honey bees increase their resin collection following pathogen pressure. So even though as a beekeeper you may be tempted to think of propolis as a nuisance, it's actually a big help to your bees. Dr. Marla Spivak, Professor at the University of Minnesota, studies the beneficial properties of propolis. Watch this video to understand more about how propolis keeps colonies healthy. Video Transcript So, propolis is, or propolis, however you like to pronounce it, is composed of resins that bees collect from trees and other plants in their environment. And they collect the resins on their back legs and they bring them back to the nest, where they deposit them inside the nest. They sometimes mix the resin with wax. And once it's inside the nest, we call it propolis, or propolis. And the resins are produced by certain trees as a defensive compound. So, they protect the leaf buds, the new leaf buds,

from infection, from UV light, and from herbivory. So, the bees are very good at cuing in on where these resins are, what trees produce the resins. And if they're available to them, then the bees go to them, they scrape the resins off the tree, or off the leaf buds, mostly, stuff it on their hind legs, and return to the nest. In a feral, natural nest cavity, like in a tree log, they would put this propolis all around the inside of the tree, surrounding their brood nest and their honey. So, they would basically paint the walls, or varnish the walls, with propolis. In our Langstroth boxes, the inside of the box is very smooth. It's plain wood, and so they don't stick the propolis there as they would normally in a tree cavity. Instead, they put it between the boxes or under the frame rests, where it causes a lot of angst for beekeepers. Makes it very difficult to manage a bee colony when they stick everything together. It turns out these resins and propolis are highly medicinal. So, we've known of the medicinal qualities of propolis for humans since antiquity, but no one really, or very few researchers, looked at the properties of propolis for bee health. So, we began by looking at how the propolis envelope, if we could put one inside of modern bee boxes, Langstroth boxes, if we could give the bee colony a propolis envelope, how might it affect them. And the first thing we looked at is their immune function. So, we can measure antimicrobial peptide, gene expression for these peptides, and quantify what happens when a colony has a propolis envelope and when it does not. So, we started with healthy colonies and we give them propolis envelope. And the way we did it, to begin with, is we made a propolis extract. You take propolis scrapings, you dissolve them in ethanol, 70 percent ethanol, then you filter it and you have this propolis extract. Then we can make sure we know the concentration of the extract. We painted it on the inside of small, side frame nuc boxes. Put bees in there, and after seven days' exposure to the envelope, the propolis envelope, we removed those bees, now at seven days, and looked at their immune gene transcripts for these antimicrobial peptides. And what we found is that the propolis envelopes seemed to lower the microbe load in that nest cavity, which then helped the bees reduce the need to express their immune genes. So, the immune system is very costly for an organism to express. If you are constantly fighting allergies, for example, it's at a cost to your body. And so, for bees, it's the same way. So, having the immune system run quietly, or not having to be ramped up all the time, is a really good thing. So, we think that the propolis envelope for a healthy colony is just killing off microbes in the nest and helping the bees keep an efficient and quiet immune system. But if you make a colony sick, with chalkbrood or American foulbrood, the colonies with this propolis envelope are then able to really mount a strong immune response. And, somehow, the properties of the propolis reduce the infection load in the colony. So, they don't eliminate the disease, they just reduce the amount of chalkbrood and American foulbrood that you see inside the colony. So, we don't know if the mode of action is by contact with the propolis, or if it's through the volatiles, or both. We do know that bees are not eating the propolis, so it's not per os, it's not through ingestion. So, there's lots of other questions we're asking. We know that if you challenge a colony with chalkbrood, in other words, infect a colony with chalkbrood, and you watch and count the number of resin foragers, after chalkbrood infection, the colony increases the number of resin foragers. But if you challenge them with American foulbrood, they do not increase the number of resin foragers. So, chalkbrood's fungal pathogen and American foulbrood's a bacterial pathogen. So, is there a difference? We really don't know. Most of the resins they're collecting are from cottonwood trees, or in the genus *Populus*. So hybrid poplar, balsam poplar. And then there's some other trees. The antimicrobial properties change depending on the species of resin, but it can also change within, for example, cottonwood, some trees that, the properties are higher

than others. But, in general, it seems that the bees are collecting what's probably good for them, or at least what's accessible for them. One kind of mysterious thing is we know that white spruce, or some of the conifers, have really high antimicrobial value, but the bees are not collecting them, at least in our area. So, that opens a question, why not? Is it, like I said, just accessibility, or maybe, is it toxic? We don't, we really don't know. The take home message for beekeepers is that bees really need a propolis envelope. We don't know, our measurements of when they just stick it between the cracks in the boxes and the frame rests. Those were our control colonies, and we know that they still got chalkbrood and AFB infection at higher levels. So, it seems to be something about the whole propolis envelope within the nest, that has the best properties and is the most helpful. So, what we do is take propolis traps, the kind, they're black plastic traps, grids basically, that you put, normally, you put on top of the colony to collect propolis. We cut them up and staple them into the insides of the boxes. Or you could use unfinished lumber, but it has to have a lot of good cracks and crevices in it. You could take a router and make grooves in the boxes. Roughing it up with sandpaper is not enough. So, think about the interior of a tree log, and how rough that surface is. And that's what the bees like to attach the propolis to. The bees, the colony needs to have this whole lining around, especially, the brood nest.

READ

The Microbiome

■ Often when we think of bacteria, we think of the ones that cause problems. The honey bee microbiome is the community of beneficial bacteria that improve health, microorganisms found in the gut. such as probiotics that help people. Newly emerged honey bees digest food. Probiotics are the acquire their gut bacteria from beneficial microorganisms that reside their hive environments and in our digestive systems, and nestmates. consuming probiotics contributes healthy bacteria to these normal Healthy gut bacteria can microbial communities. Scientists are outcompete harmful bacteria in only beginning to understand all of the the bee's body. different ways that these complicated, Honey bee probiotics require more tiny communities help keep humans rigorous testing of their safety and and other organisms healthy. Honey efficacy. bees also have a distinctive community of beneficial microbes living in their gut that support their health, a small community referred to as their “microbiome.” Though larvae and pupae have some beneficial gut bacteria, they do not yet have the diverse community that adult bees do. However, once the pupa has developed into an adult and chewed its way through its cell capping, it quickly comes into contact with all sorts of bacteria. During the first four to six days of its life, it receives food from other nurse bees, interacts with other bees, and touches hive material and bits of feces. Interestingly, if an adult bee doesn't touch any fecal matter, it doesn't acquire its entire gut community. This may come as a surprise, since normally honey bees are thought to only defecate outside the hive. But there are some feces in the hive, at very low levels, and bees' contact with it is helpful in developing healthy gut bacteria. Through trophylaxis and other means, the honey bee acquires bacteria to cultivate its microbiome. The Internal Team of Healthy Microbes There are several distinct groups of bacteria that are always present in the symbiotic honey bee community. We call them “symbiotic” because both the bacteria and the honey bees benefit from living together. Scientists think that because they are always present and are quite different from other bacteria, these species have been coevolving with honey bees for many years. Different bacteria live in different sections along the bee's gut, where they also

perform different jobs. For instance, the healthy lactic acid bacteria in the bee's digestive tract can kill American foulbrood bacteria and other harmful bacteria sometimes present in nectar. When a healthy symbiotic community is thriving, there is not much space for “bad” bacteria to settle in. This is both because the healthy microbes physically use up space and resources, and because they fight against other bacteria that try to invade. Some species of gut bacteria can also help with detoxification in the gut and digesting food. For instance, specific microorganisms are needed to help them digest bee bread. This understanding of beneficial honey bee gut bacteria is new. Despite the need for more research, commercial companies have already begun to create and sell probiotics to feed bees. Most of these products contain a mixture of lactic acid bacteria and yeasts, but they have not yet been rigorously tested in the field. Further scientific experiments are needed to test the long-term safety of these products and to ensure that they don't interact negatively with other products that beekeepers are putting in their colonies.

READ

Beyond Immunity—Beekeeper Intervention

■ Despite the impressive individual and Key Points group strategies that honey bees use The most effective approach to against attackers, they still need help managing pests and parasites is from beekeepers sometimes. When by using Integrated Pest the pressures and challenges get too Management. high, we can intervene to help. As a beekeeper, you are a caretaker of colony health. How do we know when the challenges are too high? That question is fundamental to the approach that you should be taking to help your bees. This approach, and the approach we will be taking for this entire course, is Integrated Pest Management (IPM). Integrated Pest Management uses a variety of different techniques, applied at different times depending on what the situation calls for. It can be broken down into three actions: Step 1: Support Honey Bee Immunity First, we want to help bees help themselves. For example, we provide them all of the conditions required for health, like access to flowers, supplemental nutrition if necessary, a clean and dry environment, protection from heat and cold, etc. These pre-conditions also include maintaining strong genetic stock (e.g. stock that is adapted to your local environment or stock that is more resistant to certain parasites or infections). Other strategies include cultural controls that are based on an understanding of pest/parasite and honey bee biology. Providing bees with this good head start is not enough. We have to be aware of the pest and parasite populations in the colony. IPM relies on careful monitoring to evaluate 1) whether pests or parasites are affecting the colony, 2) how those levels are changing over time, 3) whether or not the level is high enough to harm the colony and require intervention, and 4) whether or not a given intervention was effective. Once pests or parasites reach a certain level or threshold predicted to cause harm to the colony, the beekeeper can intervene with a chemical treatment and/or additional cultural controls. It is important to note that chemicals and antibiotics are the last, not the first, resort of IPM. The method you use for any pest or parasite situation should be chosen to minimize the risk of harm to honey bee, human, and environmental health. Targeted intervention can be highly effective for managing certain parasites, such as the Varroa mite. Treating only when necessary and before the parasite population gets too high ensures that you are not treating before you need to, but that you aren't treating too late, either. This minimizes any side effects caused by treatments and minimizes natural and synthetic chemicals in the hive and environment. It ■ Step 2: Track the Pest or Parasite

Population ■ Step 3: Targeted Intervention by the Beekeeper ■ also prevents the beekeeper from spending money and time on treatments that don't help. We will apply an IPM approach in treating different pests and parasites throughout this course, outlining specific monitoring and intervention strategies. Practicing IPM requires more careful planning, but in the long run it is better for both the bees and the broader environment.

MODULE WRAP-UP

Honey Bee Immunity

As we've seen, when it comes to honey bee health, the challenges from diseases and pests are numerous and growing. In this module, we've explored the various ways honey bees resist and combat pests and diseases. We've examined their natural defense mechanisms against exposure to threats, their methods for dealing with active infections, and the crucial ways beekeepers can intervene to support colony health. Understanding these aspects of honey bee health management is essential for successful beekeeping and maintaining strong, resilient colonies.

Insect Pests

Honey bee hives can attract a variety of pests that feed on honey, detritus, and other hive materials. Their impact on colony health can range from being a small nuisance to total destruction of colonies and damage to hive equipment. In this module, we will explore four common insect pests: small hive beetles, wax moths, yellowjackets, and hornets.

READ

Wax Moth Biology

Wax moths have plagued hives for centuries and are a familiar pest to beekeepers around the world, especially in warmer regions. They are largely considered a secondary pest because they mostly affect stored equipment or unused/unprotected areas of a hive (including hives in an apiary that die over winter). In a strong colony, they will build up slowly (if at all) and keep to corners or small crevices that the bees cannot reach. In weakened colonies, particularly queenless ones, they can create a problematic infestation and destroy brood combs within weeks or months, although they are not usually the direct cause of eventual death. While other moths might make a home in your hives, especially if you are trapping pollen, wax moths are by far the most common and most destructive. Wax moth webbing damage to a frame. Wax moths aren't all bad; the larvae (waxworms) are raised commercially for fish bait or feed for reptiles, birds, and small mammals because they are nutritious, easy to raise, and can survive at length without food. Recently, the greater wax moth has been discovered to eat plastic bags, decomposing polyethylene plastic films into ethylene glycol, which is biodegradable. Life Cycle There are two species of wax moths: greater (*Galleria mellonella*) and lesser (*Achroia grisella*). The greater wax moth is more common and more destructive, although the lesser one can still damage stored equipment. Here we will discuss both species together. Greater wax moth adult. Photo credit: Bee Informed Partnership Wax moths spend nearly all their lives inside hives. Eggs are laid in cracks and crevices or unprotected comb (e.g., in mostly empty honey supers or brood combs in weak

colonies) within the hive. They usually hatch in three to eight days in warmer temperatures, leading to shorter development times, although they may require an entire month at colder temperatures. The larvae (also called caterpillars or waxworms) then feed on the nearby nectar, honey, pollen, bee bread, and wax, as well as cocoons/skins from molting larvae/pupae in brood comb. Indeed, bee bread, shed larval skins, and pupal cocoons in the brood comb are a good source of protein for the developing moths. During this feeding phase, which can last for up to five months (depending on temperature and other conditions), the larvae also spin silky webbing into tunnels across the comb. The bees have trouble removing them from the tunnels. The larvae burrow into the comb and chew up woodenware, eventually making little round depressions where they spin silk cocoons and pupate. After a week or even two months, they emerge as adult moths and leave the hive for up to a few days to harden their wings and mate. The females then return to the hive to lay eggs, living for an average of 12 days, and the cycle begins again. Wax moth eggs deposited in a cell.

WATCH

Wax Moth Identification in the Colony

In hives or other unprotected equipment overwhelmed by wax moths, drawn frames are covered in grayish white webbing from the larvae, the drawn comb is tunneled through (under cappings), and woodenware is riddled with little depressions. Their feces (frass), small dark granules, may also litter the frames, caught in the webbing. Watch this video to learn how to identify when wax moths are present in your hives. Video Transcript Let's get familiar with diagnosing wax moth pests in our hives. Now, wax moths can be seen on the outside of our colony as we're approaching it to do a hive inspection, or they can also be found on the outer cover or inner covers of our hives. Greater wax moth adults are between half an inch and three quarters of an inch in size and are gray/brown in color. They have long, thin antennae. The larvae are three quarters of an inch long. Often, if you see larvae in your colony it'll either be a wax moth larvae or a small hive beetle larvae. Wax moth larvae are twice as big as small hive beetle larvae, and they can be differentiated from them by their prolegs. Looking at this wax moth larva here, we can see that it has three sets of legs on the front of its body, but it also has four prolegs along its abdomen. Prolegs are these small, stubby legs that help the larva grip onto surfaces. While small hive beetle larvae also have three sets of front legs, they do not have any prolegs. And lastly, they feel differently when you handle them. If you squeeze a small hive beetle larva, it'll feel rubbery and it won't be squishy. A wax moth larva, on the other hand, will squish very easily between your fingers, popping to release its internal contents. Wax moth larvae webbing is found most often on the surfaces of combs. As larvae walk along the combs, they can carry the webs along with them. And their fecal matter can be seen in their webbing. Cocoons can also be found on the surface of the frames or along the bottom of the frames where the wood is, where they can create marks and depressions in the woodenware, as seen here. This photo is a great example of a wax moth infestation. Here we can see four larvae webbing their fecal matter and a couple of cocoons. Inside of a honey bee colony, wax moth larvae will actually tunnel beneath the cell cappings. And sometimes we can see a line of darkened, concave cell cappings that tells us where these moths are tunneling. We may also see uncapped pupa, also known as bald brood, in a line as a result from their tunneling. Bees can detect that there is a wax moth larva beneath those cells, so they uncap them as a form of hygienic behavior. Bald brood is a symptom of a variety of diseases, but when several occur in a path like this,

it's quite indicative of wax moths. Try pulling out some of those pupae with your hive tool to see if there are wax moth larvae inside the cells. So, if you see any of these signs — the tunnels, or bald brood in a line, larvae, webbing, cocoons, or adult moths — this can tell you that there are wax moths in your colony. An opened wax moth cocoon to reveal a prepupa. Credit: Mary Kate Wheeler Wax moth depressions in the bottom bar of a frame. Credit: Mary Kate Wheeler Eggs: Light-colored eggs are laid in masses of 50 to 150 eggs but are generally too small to be visible. Larvae: The larvae are usually 3/4 inches (2 cm) long, although their size depends in part on food availability and temperature. They are white with a dark head, but they darken as they age. They have three pairs of legs and multiple body segments with small “prolegs.” Pupae: When wax moths pupate, their tough off-white cocoons are 1/2 inches (1.27 cm) and 3/4 inches (2 cm) long and usually stuck into oval depressions in the woodenware, although they can sometimes be found in comb or in debris on the bottom board. The depressions are chewed out of the woodenware and therefore remain after the cocoon is hatched. Adults: Greater wax moths are between 1/2 inches (1.27 cm) and 3/4 inches (2 cm) long, with a wingspan of 1 1/4 inches to 1 1/2 inches; lesser wax moths are a bit smaller. Both are mottled gray and fly mostly at night then hide in the dark during the day.

READ

Wax Moth Prevention

■ A strong honey bee colony will control Key Points the wax moth population within its Keeping colonies strong and hive by removing larvae and housed in the right-sized space preventing adults from re-entering the prevent wax moth infestations. hive. However, the bees are unable to Before storing equipment, freeze prevent an infestation when they are or heat it to the appropriate weakened and spending their limited temperature. Infested equipment energy on collecting resources and can be treated with PDB crystals. raising brood. The best protection is, therefore, keeping colonies strong and the hives right-sized (not bigger than the bees inhabiting it can protect), and only supering when the bees are bringing in nectar. To prevent an infestation in stored equipment, particularly brood frames, store it carefully; to the extent possible, avoid keeping equipment in dark and warm areas with little ventilation. Expose the boxes to light, tipped up so that light can shine between the frames. Freeze equipment before storing it to kill any wax moth adults, larvae, or eggs that may be present. The length of time varies with temperature; at lower temperatures (e.g., 5°F) you only need to freeze the equipment for two hours, whereas at higher temperatures (e.g., 20°F) you will need to freeze the equipment for four to five hours. After freezing your equipment, store it in hive body stacks flanked with telescoping covers, or keep them in airtight containers or bags to prevent reinfestation. Freezing equipment before storage. You can also heat the frames, although temperatures above 120°F will weaken wax and above 140°F will melt it. At 115°F, you should heat for an hour and a half. Even though this temperature is below the wax-melting threshold, be sure that the heat is evenly distributed so that no smaller area is raised above the melting point of wax.

READ

Wax Moth Treatment

There are no chemical treatments approved to kill wax moths and waxworms in an occupied hive. To address an infestation in unoccupied equipment, you can freeze/heat it as previously described. This is

preferable for a number of reasons, not least that it doesn't affect the honey. There are two chemical treatments available, if necessary. Honey supers can be fumigated with carbon dioxide (CO₂), which does not affect the honey. Hive bodies (but not honey) can also be fumigated with paradichlorobenzene (PDB) crystals if the above options are not feasible. PDB is not effective against eggs, however, and needs to be used continuously. Additionally, it lingers in beeswax and may be carcinogenic. For these reasons, using PDB crystals should be a last resort for the beekeeper, and we don't personally recommend it. Any frames stored with PDB crystals should be aired out for at least 48 hours before being reused in a hive since the comb can absorb the odor. Even after equipment is sufficiently aired out, some beekeepers still report being able to smell PDB. PDB crystals work best above 70°F, off-gassing continuously. The gas is heavier than air, so you place it at the top of a stack of equipment. To use the crystals, stack the boxes tightly, sealing cracks with tape and closing the bottom entrances if necessary. Use six ounces of crystals for five deep boxes or ten medium/shallow honey supers, placing a small pile on newspaper or another support on top of the topmost frames. Check on the equipment every four to six weeks to see if more crystals need to be added. It does not kill eggs and will therefore need to be continuously replenished if ambient temperatures are high enough to support moth development. Stacked hive bodies with PDB crystals. Photo credit: Betterbee

Biological control of wax moths is still in development; a parasitic wasp, a bacterium, and at least one virus all show some promise. These control methods may be available in the future.

READ

Small Hive Beetle Biology

Aethina tumida, known as the small hive beetle (SHB), was first confirmed in Florida in 1998, although it may have arrived as early as 1996. It is widespread across the U.S., particularly in the Southeast, and has more recently appeared in some Canadian apiaries. In recent years, several honey bee pathogens were found in SHBs, including replicative forms of deformed wing virus (DWV), suggesting pathogen spillover between managed honey bees and SHBs. The epidemiological role that SHBs may play in pathogen transmission to honey bees and other insects, as a possible dead-end host or vector, is unclear. Life Cycle Understanding the life cycle of the SHB is important for effectively using cultural and chemical strategies to limit an infestation. The SHB spends only a portion of its life inside the hive. Eggs are laid in the hive (a single female may lay between 1,000 and 2,000 eggs total) and usually hatch within three to five days if they do not dry out and die. Once hatched, the young larvae feed on the bees' resources (pollen, bee bread, pollen supplements and substitutes, honey, nectar, and brood) for 10 to 14 days, or longer if the food sources are poor. After this feeding period — when they can cause trouble for your bees — the larvae leave the hive to find soil in which to pupate. They tend to leave at night to avoid predators. They generally travel just a few feet from the hive and only 4 to 8 inches down into the soil, although there are records of greater distances. Sandy soil is more hospitable. In the soil, SHBs pupate in three to four weeks then emerge as adults, crawl out of the soil, and fly back into the hive, usually around dusk. Small hive beetles can locate bee hives by detecting the pheromones of adult bees and the smell of honey and pollen. They also introduce a yeast into the hive that mimics the smell of the honey bee alarm pheromone, attracting further SHB adults (low doses of alarm pheromone are common in the hive). In the hive, after about a week, SHBs are mature enough to mate. A female adult will then deposit eggs on pollen and brood cells or in masses in corners

of the hive — her ovipositor is long and flexible, which allows her to leave tiny eggs in tiny cracks. In total, the entire SHB life cycle is only four to six weeks, with multiple generations in a year. Adults can live up to six months. In the climate of the Northeastern U.S., the SHB does not reproduce during the winter months, although the adults can survive in the warm honey bee cluster. They may begin to reproduce quickly in the early spring when the honey bee colony is at its lowest population.

READ

Small Hive Beetle Effects

■ In the northeastern U.S., SHB are Key Points mostly tolerated by healthy colonies. Small hive beetles are mainly a problem when colonies are weak like wax moths, they are therefore often considered a secondary pest, as too much space to adequately defend colonies that are already weak or intense small hive beetle queenless. Small colonies, including infestations ferment honey, splits and nucs, or those with too much causing it to take on a slimy, space to defend can sometimes discolored, and frothy appearance. become overwhelmed by them. However, when an SHB infestation reaches high levels and larvae and adults are consuming honey and defecating in it, the honey ferments, becoming slimy and discolored and smelling of rotting oranges. Such honey may drip out of the cells and even out the bottom of the hive. It can no longer be eaten by humans and will be rejected by bees, although the combs and frames themselves are not affected. In extreme cases, larval feeding can also damage brood. Rarely is this observed in live colonies in the northeastern U.S., but it can be observed in colonies in year-round warm climates such as the southern U.S. African honey bees may abscond if SHB levels rise too high, but European honey bees will tolerate a much higher level of infestation before they reach the threshold for absconding. Small hive beetle larvae in a frame feeder. Small hive beetle larvae on a honey frame. Effects on honey The biggest issue with SHB in the Northeastern U.S. is when they infest honey supers. While in a live colony, honey bees are usually capable of deterring beetles to the point where larvae rarely slime up honey frames. But when beekeepers remove honey supers and temporarily store them before extracting, there are no longer bees to deter the beetles. If SHB are present in supers and the honey is not promptly extracted, beekeepers may see their honey become infested with larvae and lose part of their honey crop.

WATCH

Small Hive Beetle Appearance and Identification

There are several ways to identify if SHB are in your hive. Eggs SHB eggs are similar in shape and color to honey bee eggs but are smaller, at about 0.05 inches in length. They are usually laid in clusters, but sometimes a small puncture in the cap of a brood cell is visible from SHB oviposition. As larvae, SHB are whitish with brown heads and are much larger than the adult beetles. They closely resemble wax moth larvae, although the average wax moth larva is larger. Small hive beetle larvae grow to be 0.37 inches long by 0.06 inches wide. However, they can be distinguished by their number and behavior: A large, irregularly clustered heap in a corner is likely to be SHB. Even a single or small group of larvae can be distinguished by their legs. Both SHB and wax moth larvae have three pairs of legs next to their heads, but wax moth larvae have additional small prolegs. SHB larvae also have small spines and tougher outer bodies than the fragile wax moth larvae. If you squeeze one between

your fingers it feels firm, whereas wax moth larvae are tender and squishy. In contrast to adult beetles, older beetle larvae are attracted to light. ■ Larvae ■ Small hive beetle larva. Note the spines along its back and the absence of prolegs. Photo credit: Jamie Ellis Pupae The pupae are underground and therefore not often apparent to beekeepers; they are pearly white and darken over the three to four weeks that they pupate. SHB adults are about 0.2 inches long (females slightly longer than males) and 0.13 inches wide. They are dark brown, turning nearly black as they mature. They have six legs and small club-like antennae, with hard outer wings that protect them from bee stings, although their legs and antennae may not be visible if they have their heads tucked down. Adults prefer the dark and may stay near the tops of hives with screened bottom boards. Indeed, you may see them scatter across the inner cover or tops of the frames when you open up a hive. Watch this quick video clip to see small hive beetles scurry from the inner cover of a hive. ■ Adults ■

READ

Small Hive Beetle Prevention

■ SHB adults can fly up to eight miles in Key Points search of a hive, may be able to join Beekeepers should keep their honey bee swarms, and are further apiaries and honey houses free of spread in the moving of hives for combs, cappings, and wet frames pollination services or returning north that can attract small hive beetles. after wintering in the southern United Promptly replace any uneaten States. Additionally, adults can survive pollen patties and extract honey for up to ten days without food. from harvested supers. Keeping hives free of arriving adults is thus near impossible. Monitoring for SHB is generally based on a quick visual assessment when opening the hive. SHB will scatter to avoid the light, so looking immediately is key. Larvae are more likely to be found on unprotected frames and on the bottom board, or in pollen supplements or substitutes. There are no protocols for accurately measuring the level of an infestation, nor is there a threshold for treatment, since their ability to overwhelm a colony depends on many factors. Uneaten pollen patties can attract small hive beetles. Maintaining general hive health and implementing other cultural controls are the most important tools for SHB management. As for many bee pests and diseases, the first line of defense is keeping strong colonies without extra hive bodies that the bees do not use and cannot patrol. The bees will also defend themselves, although European stocks of bees are poorly adapted to SHB, given its African origins. Honey bees in a healthy colony may confine and guard bees in “beetle jails,” or the small corners and crevices of the hive where they are already hiding. When a beekeeper opens a hive, they may be unwittingly disturbing the guards and releasing the SHB. Beekeepers should keep their yards and honey houses free of deadouts, combs, cappings, wet frames, and other attractive debris to discourage beetles as well as other pests. Locating hives on clay or rocky soil will help lower the number of successfully pupating beetles. Within the hive, removing pollen supplements and substitutes if they are infested or within two weeks if they remain uneaten is also advised. Hive bodies and supers should not be added unless there are sufficient numbers of bees to fill and patrol them. Equipment should be kept in good condition to reduce the number of cracks and crevices in which the beetles can hide, and debris should be removed from the bottom board if it accumulates enough to hide adult beetles or larvae — and especially if it is so deep that it allows for pupation. Honey supers harvested from a hive with even a mild SHB presence must be extracted immediately (within one or two days). Keeping the honey house humidity below 50% can delay eggs from hatching or kill them from

desiccation outright. Infested equipment should be frozen for 48 hours at 30°F (-1°C), or a colder temperature (10°F or -12°C) for 24 hours. Combs with ruined honey can also be washed out and then frozen. Honey stored in hives over winter is not at risk of infestation; similarly, equipment can be safely stored in unheated spaces over the winter.

READ

Small Hive Beetle Treatment

■ Beetle traps can help keep SHB levels Key Points lower and should be the first mode of Small hive beetles should first be action to manage an infestation. Traps controlled with beetle traps. If available from bee supply companies these prove ineffective, pesticides use apple cider vinegar and/or oil to or ground drenches can be used. bait and kill adult beetles. DIY traps work on the same principle, using sugar, bananas, or other attractants to trap and kill SHB. The downside to these traps is it is easy to spill the oil when conducting hive inspections and tipping hive bodies. Beekeepers have also had success placing dry dustmop sheets (such as Swiffer pads) or Brawny Dine-A-Max towels on the top bars beneath the inner cover. As beetles run into the mop sheets/towels to escape bees, they get trapped. The beekeeper can replace the pads during the next inspection. Some species of nematodes that attack SHB while they pupate underground are commercially available, but research is still underway to determine where and when these are most effective. Beetle trap placed in a colony.

Synthetic Chemicals Two synthetic chemical treatments are available: one that is applied in the hive (CheckMite+™) and another that is applied to the ground in front of the hive (GardStar™). Note: Using chemical treatments to control small hive beetles should be a last resort if the infestation is still out of control and you are still observing significant SHB larvae damage after using beetle traps. We do not personally recommend using CheckMite+™ at any time. The active ingredient in CheckMite+ (coumaphos) remains in comb for many years, contaminates hive products, and can reduce survivorship and fertility in queens. Checkmite+ should not be used in queen-rearing colonies. The active ingredient in GuardStar (permethrin) is toxic to honey bees. CheckMite+™ CheckMite+ can only be used when honey supers are not on the hive and will not be for another two weeks after the treatment has ended. Cut corrugated cardboard into 4"x4" squares and then peel off one side (taping the other side to smooth it and discourage bees from chewing it up). Cut a CheckMite+ strip in half and staple the two pieces to the cardboard squares. This creates small crevices between the cardboard and strips where the beetles can be trapped. In hives with solid bottom boards, place the squares near the bottom of the hive with the strip face down; in hives with screened bottom boards, place the squares above the inner cover. Beetles will try to hide under it. These can be left in the hive for 42 to 45 days; the treatment should not be repeated more than twice in one year. Take precautions when handling CheckMite+: Follow the label carefully, wear chemical-resistant gloves (not standard beekeeping gloves), and dispose of the strips according to the label. GardStar is a ground-drenching method that kills beetles pupating in the soil near the hive. Note that the active ingredient in GardStar, permethrin, is toxic to honey bees and other animals and cannot be applied near water sources or ■ GardStar™ ■ other sites where wildlife gather food. First, cut/mow any grass to ensure that GardStar reaches the soil, and then use a watering can to apply water mixed with GardStar to the soil. Drench the area for at least two feet around the hive (and under it if the hive has screened bottom boards). It may need to be reapplied after one or more months. Because GardStar is toxic to bees, the soil drench

should be applied in the evening and care should be taken not to spray/drip any onto any part of the hive. Take necessary precautions when handling GardStar: Follow the label instructions carefully and wear eye protection.

READ

Yellowjackets

■ Yellowjackets can be a nuisance to Key Points colonies in late summer and early fall. Yellowjacket attacks can be They feed on honey and scavenge on prevented by taking measures to dead adults and brood on the bottom reduce colony robbing and by board. Normally, strong colonies can setting up wasp traps. defend themselves against yellowjacket intrusions, but weak colonies can be wiped out entirely. In some areas of the U.S., like the Pacific Northwest, yellowjackets can become an intense problem, even for strong colonies. From mid-August until mid-October, if you begin to see yellowjackets attempting to rob your colonies, manage your colonies as you would to eliminate robbing behavior: Place entrance reducers on the hive to reduce the size of the area to be defended, or install robber screens. Ensure your hive bodies do not have holes or cracks in them. Even if they are too small to be an entrance, wasps can be attracted to the odor of the honey. Refrain from feeding using open feeders, opening capped honey while working your hives, dripping uncured nectar, and keeping your colonies open for extended periods of time during inspections, as these odors can attract yellowjackets. For further protection, eliminate any yellowjacket nests that are in the vicinity of the apiary and hang yellowjacket traps at the edge of your apiaries. Environmentally friendly traps exist that rely on a pheromone lure instead of an insecticide. The wasps are attracted to the lure and are unable to escape. These traps can be quite specific to yellowjackets and avoid accidentally catching honey bees. Lastly, manage populous colonies that are better able to defend against intruders and threats — yellowjackets or otherwise.

READ

Hornets

Two invasive hornets with the potential to impact honey bees in North America have been detected in recent years. Their lifestyle consists of capturing and killing adult bees, often decapitating or dismembering them. These hornets can invade hives to consume brood or steal larvae/pupae as protein sources for their own brood. They use pheromones to recruit conspecifics to an identified hive target, magnifying damage. This increased predation risk may reduce honey bee foraging flights, reduce their efficiency, or force bees to avoid certain flight paths, which can be detrimental to pollination services. The loss of population, loss of brood, stress on defense, and resource drain can all increase the hive's vulnerability to other pests/diseases, thus weakening the bee colony. Vigilance, rapid response, trapping, and nest removal remain critical to prevent the establishment or re-establishment of highly damaging hornet species. All beekeepers should therefore be able to recognize these hornets and be ready to alert the USDA Animal and Plant Health Inspection Service. The Northern giant hornet (*Vespa mandarinia*), aka "Asian giant hornet" or "murder hornet," has attracted a lot of attention because of its capacity for mass attack on honey bee hives. In its native range (temperate and tropical regions of Asia), it is known to decimate honey bee colonies. A few hornets can kill many bees per minute due to their large mandibles and strong predatory capacity. V.

mandarinia was first confirmed in North America around 2019, with detections in British Columbia (Canada) and Washington state (USA). In recent official reports, authorities have claimed that the species has been eradicated from the U.S. since there were no confirmed sightings since around 2021. Despite the claimed eradication, vigilance continues (monitoring, traps) because reintroductions remain a risk. The yellow-legged hornet (*Vespa velutina*) is originally from Southeast Asia and became invasive in Europe in 2004. Workers prey on adult bees flying to and from hives or catch them near entrances (bee hawking). This hornet's voracious appetite can lead to severe losses in apiaries and diminish the effectiveness of pollination services, with considerable costs for beekeepers and agricultural yield. Native insects, including other wasps and flies, also suffer from predation and competition. The immediate impact is less extreme than *V. mandarinia* in terms of “slaughter phase” attacks on hive interiors, but cumulative predation pressure is still concerning, especially for weaker colonies, and Europe has been struggling with this invasive hornet ever since it was introduced. *V. velutina* was first confirmed in the USA in 2023, when a live specimen was found in Georgia and later in South Carolina. The hornet's colony was destroyed in both instances. Vigilance remains essential because new introductions are possible. *Vespa mandarinia* attacking a bee *Vespa velutina* attacking a bee

MODULE WRAP-UP

Insect Pests

We've now examined four significant insect pests that can threaten honey bee colonies: small hive beetles, wax moths, yellowjackets, and hornets. We've learned how each of these pests can affect colony health, from minor disturbances to potentially devastating colony losses and equipment damage. Understanding how to identify, prevent, and manage these common insect pests is crucial for maintaining healthy colonies and protecting valuable hive equipment.

Fungal Enemies — Nosema

Nosema disease affects the digestive system of adult bees. It is a very common condition throughout the world, though in the U.S. it is not a major cause of honey bee colony losses. In this module, we will describe this disease in detail and discuss what beekeepers can do to keep their colonies healthy.

Nosema Biology ■ Nosema disease is one of the most Key Points prevalent infections in honey bees. Nosema is a fungal gut parasite worldwide. It is caused by two species caused by two different species. of fungal parasites in the honey bee, *Nosema apis* and *Nosema ceranae*. Colonies can be infected with either species of Nosema or with both simultaneously, although the symptoms are different. However, both parasites can kill colonies that are unable to clear the infection. The New Nosema *Nosema apis* was first discovered in the early 1900s and was, for a long time, the only Nosema species infecting *Apis mellifera*. Yet in 1996, *Nosema ceranae* was discovered to be infecting the Asian honey bee, *Apis cerana*. Ten years later, the infection was observed in European honey bees, *Apis mellifera*, in Spain. Older museum samples confirmed *Nosema ceranae* switched hosts in the late 1970s. Since its first discovery in *Apis mellifera*, *Nosema ceranae* has increased in prevalence worldwide. It is now documented in North America, South America, Europe, Asia, Australia, and Africa. In the U.S. and Canada, *Nosema ceranae* has outcompeted *Nosema apis* almost entirely, perhaps

because it produces more spores in the bees' guts than *Nosema apis*. In the United States, nearly 100% of cases of *Nosema* infection are now from *Nosema ceranae*. The most recent survey of species prevalence in New York state showed that only 1.3% of infections were *Nosema apis* in 2011 and only 0.7% in 2012. The vast majority of infections are *Nosema ceranae*, though in some circumstances, colonies can be infected with both species at the same time. While *Nosema apis* is specific to honey bees, *Nosema ceranae* can infect other species of bees, most notably bumble bees. In addition, a variety of other species of *Nosema* can infect many different insects. It should be noted that the names "*Nosema apis*" and "*Nosema ceranae*" have officially changed to "*Vairimorpha apis*" and "*Vairimorpha ceranae*," respectively. This name change occurred in early 2020 because a genetic analysis revealed these species are actually members of the genus *Vairimorpha* instead of *Nosema*. For now, we will continue to use the classic names in this course because they are well recognized by beekeepers, but we will begin introducing the new names more often in our Cornell educational materials.

How Does a Bee Become Infected? For bees to become infected, *Nosema* spores must enter the midgut epithelial cells and produce environmentally resistant spores. These spores are then shed in the bee's feces, contaminating comb surfaces, hive walls, and water sources. Accordingly, *Nosema* can be spread through the fecal-oral route when bees ingest fecal matter that contains spores. Bees can also consume spores during trophallaxis (food exchange), through nestmate grooming, and by cleaning contaminated cells — all examples of within-hive transmission of the pathogen. The *Nosema* infection can also be transmitted from one hive to another. First, the pathogen can be sexually transmitted from spores present in drone sperm then transferred to a queen during mating. Second, bees moving between colonies (robbing/drifting) can introduce spores to other hives. Interestingly, the brood does not contract this infection. Once spores are consumed, they travel to the intestine where they become active and infectious. The active fungus feeds on the bee to acquire all the nutrients and energy it needs. It reproduces by producing spores, which then exit the bee's body in defecation. Infected bees can contain as many as 30 to 50 million spores in their midguts. The presence of spores does not cause symptoms in bees, but the active fungus can.

Nosema Symptoms ■ As the parasite uses up energy from the host bee, the bee has less energy. Symptomatic *Nosema* infections available to perform its daily tasks; the present as colonies that are weak. A parasite essentially starves her. Adult population, slow to build up, workers in this starved state consume and slow to produce brood and more food, have reduced fat bodies, honey, and have smaller hypopharyngeal glands. Dysentery is not a symptom of glands. Smaller hypopharyngeal *Nosema ceranae* glands make them less able or unable to produce royal jelly and tend to the brood. Infected worker bees usually begin foraging earlier in life, though the reason for this is not fully known. It could be that without the ability to tend to the brood, the bees transition to foraging. It could also be that infected bees take on the "riskier" role of foraging, leaving their healthy siblings to perform tasks inside the colony. It could also be that the infection alters hormones and neurotransmitters in the bee that are associated with worker polyethism. Inducing foraging behavior would help the parasite spread to other colonies and reproduce, as spores can be deposited on flowers to be picked up by other bees, and infected bees can drift into other colonies. Indeed, infected bees are less likely to return home compared to healthy bees, whether from disorientation or death. Although *Nosema*-infected bees begin foraging earlier in life, they are less efficient foragers, spending more time foraging but collecting less nectar and pollen compared to healthy adults. Infected bees usually die sooner than their non-infected nestmates — an average of

nine days sooner — although this varies greatly with the number of spores present in the gut. *Nosema* disease is not restricted to workers; all castes can be infected. Both queens and drones have a shortened lifespan, and their reproduction is impacted. Infected queens may cease egg laying and die after a few weeks and/or be superseded. Infected drones' sperm lose viability faster than normal. Adults infected with *Nosema apis* may exhibit crawling behavior and their abdomens may appear swollen. These symptoms are not usually seen in bees infected with *Nosema ceranae*. Worker bee crawling in the grass in front of the hive. Symptoms at the Colony Level *Nosema* infection is associated with reduced honey production, decreased adult population, and sometimes — though less commonly in the U.S. — colony death. As infected foragers die and nurse bees begin foraging rather than tending to the brood, there may be reduced brood production and the population may dwindle. To the beekeeper, this manifests as colonies that are slow to build up in spring and don't tend to produce much brood or honey, even if there is a queen present and a nectar flow. In some cases, the colony will create supersedure cells to replace the queen. These symptoms are most obvious to beekeepers when there are other colonies in the same apiary that are strong and producing honey, so they recognize when one or more colonies are doing poorly. Curiously, however, many colonies that have high *Nosema* spore counts are completely asymptomatic and not negatively impacted by the disease. In colonies that do die from the disease, the majority of adults die far from the hive. Inside the hive, there may be a few dead bees on the bottom board and only some young bees and the queen remaining. If the dead bees have not decayed, it is worthwhile to test these bees to determine their *Nosema* spore loads. Interestingly, there are some countries and regions outside North America where *Nosema* infection is responsible for high colony losses. It could be that colonies respond differently depending on the species and subspecies of *Nosema*, the presence of pesticides or nutritional stress, co-infection with viruses or other parasites, and the climate/environment surrounding the colony.

Dysentery
Beekeepers are often taught early on that the symptom of dysentery means a colony is infected with *Nosema*. While *Nosema apis* has for a long time been associated with dysentery, it's important to note that this symptom is not seen in *Nosema ceranae*. It is believed *Nosema apis* inhibits digestion and, as a result, bees become dysenteric — they defecate inside the hive or on the hive exterior instead of in flight. This dysentery has an advantage to the parasite, as it helps to spread spores to other nestmates in the colony. Yet this long-held belief that dysentery is a symptom of *Nosema apis* has recently been disputed, and further research is needed to illuminate this association. In North America, where *Nosema apis* is very rare, dysentery is often caused by nutrition issues. Honey bees can experience dysentery from consuming either fermented honey, dark honeys that have a high ash content, or unripened honey/sugar syrup with a high moisture content when they are confined in their hive for an extended period. These foods can fill the guts quickly and thus reduce the amount of time bees are able to "hold it" before needing to defecate. These periods are most often during winter or sometimes when bees are shipped in packages. Ensuring hives are dry inside and free of uncapped honey or sugar syrup during long periods of confinement is the best way to prevent dysentery from these causes. Dysentery in the hive.

WATCH

How Can I Tell If My Colony Is Infected?

The symptoms of Nosema are not unique to this disease. Reduced honey production, dwindling population, queen supersedure, and winter colony loss can also be symptoms of other colony diseases or problems. It is therefore difficult to tell from a simple inspection when a colony is infected with Nosema. The only ways to diagnose the disease are through microscopic examination or molecular analysis. Beekeepers can send samples to the USDA Beltsville Bee Lab in Maryland for a free diagnosis. Samples should include at least 100 bees that were collected alive. The lab provides the Nosema spore count per bee. The lab does not differentiate between Nosema species, but given that Nosema ceranae is so much more prevalent, distinguishing between the two species is largely unnecessary. Alternatively, beekeepers can count Nosema spores themselves by using a microscope. The most accurate method is to use a hemocytometer, a device usually used to count blood cells. Many bee clubs invest in purchasing a microscope for all their members to use. Many universities with apiculture extension programs also have microscopes for public use. The Dyce Lab for Honey Bee Studies, in Ithaca, New York, has microscopes available for beekeepers to diagnose Nosema for free. Contact the lab at dycebees@cornell.edu to set up a time. Diagnosing Nosema with a microscope just requires a couple of steps and is a quick and easy way to learn more about the health of bees. Watch this video to learn the procedure. Video Transcript Let's go through the steps of diagnosing a Nosema infection in a colony. First, we're going to need a few materials, one of them being a compound microscope that's capable of 400 times magnification. The next thing we're going to need are bees that we collected from a colony that we suspect has Nosema. And you want to collect a minimum of 25 bees. But the more, the better. And, in fact, 50 or 60 is an ideal number to collect. So, in this tube, I've collected 50 adult bees. We also need some tools to help us separate the abdomens of the bees from the rest of their bodies. So, we're not going to be including the thorax, the head, or wings. We're just going to be crushing and looking at abdomens. So, for this, you can use a pair of scissors. Smaller scissors, like the manicure scissors are a little bit better than these scissors, but this is what I have on hand today. Or you can try using tweezers and see how that works out. In addition, I have two paper bowls that I'm going to be keeping the abdomens in so that I can count and make sure that I have my 50 bees, which is what we're dissecting today. And then we need some materials to actually crush our abdomens to release the internal contents because we want to see the Nosema spores, which are found in the gut of the bee. So for that we have a jar that we're going to be using to crush the abdomens. And we also have a sandwich bag that we're going to put the bees in so that we're not making too big of a mess here. In addition to those materials, we also need water. And the amount of water that you're going to be adding to make a sort of gut spore solution depends on how many bees you're including in it. So, if you are using 25 bee abdomens, you're going to be adding 25 milliliters of water. If you're going to be doing 50 bee abdomens, like we are today, then you're going to use 50 milliliters of water. So, the ratio is one milliliter of water per one bee abdomen. Then you're going to need an eye dropper to pick up some of the solution that we're going to create so that we can drop it onto our hemocytometer, which also has a cover slip. And a hemocytometer is this nifty tool that's actually used to count blood cells in human patients. But in this case, we're going to be using it to count Nosema spores in bees. And so, if you look at the hemocytometer close up, you can see a small grid. And that's going to be very evident when we're looking at it under a microscope at 400 times magnification. And the small grid is going to allow us to easily quantify the spores. And from there, we can multiply it by a number to determine the infection rate in spores per bee. So let's go ahead and

pour our 50 bees into the small bowl so that we can begin dissecting their abdomens. And to do that, it's as simple as it sounds. We're just going to be separating the abdomens from the rest of their body. So, we're not including legs, wings, thorax, or the head. So, you may be wondering which bees you actually want to collect to gauge the infection level for an entire colony. And the bees that you want should be representative of the whole colony. You don't want to pick only foragers, and you don't want to pick only nurse bees. And the reason for that is, if we think about the biology of Nosema, we understand that once the infection starts in the gut of one bee, it will proliferate and produce more and more spores over time. So, the older the bee is, the higher the infection level will be. So, foragers should have much higher infection levels than nurse bees. So, that way, if we chose only foragers, then we would see a really high infection level, which might be way higher than what the overall colony is experiencing. And the opposite occurs with nurse bees. Bees that we would choose just off of the brood frame, for instance. So, to try to get kind of average age bees and not focusing only on foragers or only on nurses, a good place to take them from would be underneath the inner cover or the top bars of the frames. So, there's a few ways you can collect them. You can simply grab them by their wings, if you are confident in picking up bees, and put them into a tube or a bag, or if you have a little bee vacuum, you can suck them up pretty quickly with a vacuum. Or if you have a little tube, you can buy this online, you can simply scrape them along the top bar of the frame and collect some bees that way. Alright. So, we've got our 50 bee abdomens in this bowl. And the next thing that we're going to do is we're going to grind them all up to release all of the internal gut contents, including the spores. And if you have a pestle and mortar at home, that works really well to grind it up into a fine paste. But if you don't, using a sandwich bag and a jar will suffice. So, I pour all of the abdomens into this bag and I'm going to make sure that there's no air bubbles in there. And then I'm going to seal it up. And I'm just going to roll over it really well with this mason jar. And you can see that everything is getting squashed. Alright. So now, we're going to add our 50 milliliters of water to this bag. Then we're going to mix this all together. So, this is going to help suspend all of the spores. Now, we're going to use our eyedropper to pick up a small amount of the solution, trying to avoid any of the cuticle that could be floating around in this bag. And we're going to add a drop to each of our hemocytometer chambers. So, each hemocytometer has two chambers, or two grids. And we're going to be counting the spores in each of the grids and dividing it by two to get our average spore count. So, on my hemocytometer, I have my cover slip. And I'm just going to add a little drop right underneath, so that it will quickly cover the entire grid. And the cover slip is going to press on it to make everything one clean layer that's easy to see. So now, we're ready to look at this under the microscope and begin counting our spores. So, make sure that you read all the instructions that come with your particular microscope to learn how to use this properly. Basically, the way that I go about it is I start on the lowest magnification and I try to get as clear of an image as possible, and then I move up slowly to higher and higher magnifications until I'm at 400 times magnification. And you want to be able to clearly see the grid lines that are on the hemocytometer and then we're going to start looking for the spores. Okay, so this is what it looks like at a hundred times magnification. So we're not quite as far as we are planning on getting with 400 times magnification. This is kind of the intermediate step. And here we can clearly see the grid on the hemocytometer and we can also start to see Nosema spores. So, this colony is definitely infected. I haven't counted the spores yet, but I just wanted to describe to you what they look like. So, we'll see it much more clearly when we're at 400 times magnification. But at 100 times, you can see that they are

these very small ovals with a dark outline and a very bright, almost glowing, interior. At this level of magnification we could also see other items that are found in the honey bee gut. So, there's not just *Nosema* spores, but there's other things that the bees have been digesting. For instance, some pollen grains, you can start to see at this level of magnification. And the more you get familiar with looking through the microscope and if you turn to any resources that talk about honey bee microscopy and other things that you can find in their gut, you can learn to differentiate what are those *Nosema* spores from other items that you don't want to be counting. Alright, now that we've got our grid centered and we're getting comfortable with what we're seeing under the scope, let's go up to 400 times magnification. So, when you do this, you need to be a little careful because it's getting very close to the cover slip and you actually don't want it to puncture the glass. That can damage your microscope. So now that we're on it, we'll just take a second to focus. And you can adjust the background light to make it a little bit darker so that the spores appear to glow and that they're much more noticeable. And once you've got it tuned to exactly where you like it, you can see the spores very clearly. And just taking a look at them here, again, we see that oval shape with a sort of dark outline, and a very bright inside color. Looks like it's glowing. Alright. So now that we've got it focused just how we want it, we have our light adjusted so that we can easily see the spores that are appearing to glow, almost. And we can also see the grid that's showing up in the background. And when we're at this stage, we're finally ready to start counting the spores. And when you're using a hemocytometer, there's a very specific method that you use. You don't need to count every single spore on the grid or chamber, as that would take a very long time. Instead, what we're going to count just a subset of the squares that we can see. So, each hemocytometer is made up of two grids. Each grid, or chamber, consists of 25 large squares, and each of those 25 large squares consists of 16 smaller squares. So, what you're going to do is, you're just going to count the spores that are found in these squares. These five here. You're going to include all the spores within those squares and you're also going to include spores that are touching the top lines and the left lines. But you're excluding spores that touch the bottom lines and the right lines. And you're going to write down all the spores that you're counting in those five squares and add them all together. So, you do that for one chamber of a hemocytometer, and then you do that for your second chamber. So, you should have two spore counts. You're then going to average those spore counts together. In order to convert this to spores per bee, you're going to multiply that number by 50,000. So let's go through an example right now. Let's start with the first chamber. In my five squares, I saw 13, 12, 6, 9, and 11 spores, for a total of 51 spores. In our second chamber, the squares had 10, 9, 8, 11, and 12 spores, for a total of 50 spores. So, we're going to average these two chambers together. So, 51 plus 50, divided by two equals 50.5. And then we're going to multiply that by 50,000. And so, our answer is 2.5 million spores per bee. If you're going to be counting spores from another colony, it's really important to disinfect the hemocytometer, and cover slip, and even your tweezers, so that you're not introducing *Nosema* spores from one sample into the other. And there you have it. That's how you look at *Nosema* spores under a microscope to quantify how infected a colony is. Regardless of the method, it is good practice to check *Nosema* levels in colonies you suspect are infected. As mentioned earlier, most often these are colonies that have slow population buildup in spring and/or do not bring in much nectar during a honey flow. Monitor any infected colonies again six months later to determine if they have cleared the infection on their own. Typically, monitoring occurs once in early spring before colonies are supered, and again in autumn after supers are removed. These are the only times of year

that a treatment can be applied if the levels make such measures necessary.

READ

Prevalence and Seasonality

■ Nosema is a common disease in the Key Points U.S., though most colonies do not. In the U.S., almost all infections exhibit infection rates high enough to be caused by Nosema, which negatively impact colony health. Since ceranae and Nosema apis is now 2016, the NYS Beekeeper Tech Team very rare. facilitated by Cornell University has Nosema is most common in documented that approximately 86% spring and less common in of colonies in spring and 55% of autumn. colonies in autumn contain Nosema spores. The average infection rate in spring is 0.87 million spores per bee, while in autumn it is much lower at 0.23 million spores per bee. Although infections are less common and severe in autumn, colonies that have high spore loads at this time can sometimes die over winter. Nosema is highly seasonal in the Northeastern U.S. The graph below shows Nosema levels from hundreds of colonies sampled each spring and autumn between 2016 and 2019. Most infections occur in spring.

READ

The Interaction Between Nosema and Pesticides

■ A combination of multiple stress Key Points factors is widely believed to be a driver Nosema infections and honey for the increased rate of colony losses bees' vulnerability to them can be in recent years, and science is only exacerbated by exposure to beginning to uncover how different pesticides. stress factors interact. As bees are often exposed to Nosema and pesticides together in the real world, investigations into their combined effects are valuable, and more are needed. Exposure to imidacloprid, a neonicotinoid, increases honey bees' susceptibility to Nosema infection. Furthermore, bees experiencing combined stress from imidacloprid exposure and Nosema infection die at higher rates than bees exposed to either one alone, suggesting these two stressors interact synergistically. It has also been found that bees reared from brood comb with a variety of pesticide residues (including insecticides, fungicides, and miticides) are significantly more likely to develop Nosema ceranae infections as adults than bees reared in clean comb, which suggests that the environment in which bees develop influences their susceptibility to infections as adults. Rotating out old brood comb and taking measures to reduce pesticide exposure (which may be difficult depending on the apiary locations) can help reduce pesticide buildup in combs and bees' susceptibility to Nosema infections. Rotate out old combs to reduce the buildup of pesticide residues.

READ

Nosema Prevention

■ Nosema infections can be prevented Key Points in part by reducing the spread of Preventing the spread of Nosema spores and by keeping colonies requires properly caring for otherwise healthy. Choose an apiary susceptible colonies and site with good drainage and sun disinfecting contaminated hive during winter. Low moisture and equipment. adequate ventilation can help discourage fungi, including this fungal gut parasite, from proliferating. In spring, supplement colonies with syrup and pollen substitute if honey stores or larval jelly levels are low. If colonies do become infected, Nosema

spores can remain in comb for several years and reinfect new bees that are installed onto that equipment. Regularly replacing old comb with frames of foundation can help prevent *Nosema* spore accumulation in your equipment over time. Combs from colonies that died from *Nosema* infections should be disinfected before being reused. There are several ways to accomplish this: Expose combs and hive bodies to freezing temperatures for at least four consecutive days. This is the easiest method for disinfecting combs. While freezing equipment does not kill *Nosema apis* spores, it does drastically reduce the viability of *Nosema ceranae* spores. Fumigate combs and hive bodies with acetic acid. Apply two cups of 80% acetic acid to an absorbent pad and lay one pad along the top bars of a hive body. Hive bodies should then be stacked and sealed, left alone for one week, and then aired for one week before being used. Expose combs and/or hive bodies to heat. Hive bodies should be heated to at least 120°F for 24 hours or to 140°F for at least 15 minutes (though this temperature melts wax). Expose combs and hive bodies to gamma radiation.

READ

Nosema Management

■ Symptomatic colonies with infection Key Points rates higher than 1 million spores per Nearly all colonies that have high bee in spring should be moved to an spore counts in spring are able to isolated apiary to prevent *Nosema* naturally reduce or resolve their spread. Once the weather improves, infections by autumn. At this time, colonies start bringing in ample there are no treatments registered nutrition, and foragers are able to take for *Nosema ceranae* that are regular cleansing flights, colonies tend reliably effective. to improve and treatments are not usually necessary. Long-term monitoring of 465 colonies in 2017 and 2018 by the NYS Beekeeper Tech Team found that almost all colonies were able to reduce or resolve *Nosema* infections on their own without any beekeeper intervention. Fully 90.6% of colonies with high *Nosema* infections in spring reduced their infections to levels below 1 million spores per bee by autumn. Of the colonies that had any detectable levels of *Nosema* in spring, 36% were able to completely eliminate the infection by autumn. To support symptomatic colonies (weak populations and low food stores), we recommend supplementing them with syrup and pollen substitute. Beekeepers can even introduce frames of healthy brood and nurse bees from other colonies in an effort to boost the colony population. Good nutrition, especially adequate protein, is necessary for a healthy immune system. While good nutrition cannot prevent or reverse *Nosema* infections in bees, it can help bees tolerate the infection. Infected bees fed high-protein diets live longer and have higher survivorship compared to infected bees on low-protein diets. Measure the spore levels in these affected colonies again in autumn. More often than not, the colonies will have resolved the infection and can be returned to their original apiary. If the colony continues to have a high spore count and is still symptomatic, then consider applying a treatment. If colonies that had high autumn spore counts die from this infection, disinfect equipment before installing new bees. Good protein diets are important for a healthy immune system. Photo credit: Jennifer Tsuruda Naturally derived products are sometimes used to help resolve *Nosema* infections. One such product is called Nosevit and is made from oak bark extract. The mode of action is unknown, but Nosevit is commonly used in Europe. At this point in time, however, there is not enough scientific evidence to suggest Nosevit or essential oils are effective in controlling *Nosema* infections. It is not registered for *Nosema* treatment in the U.S., but U.S. beekeepers can purchase it online and are permitted to use it in their colonies. Another natural

supplement that is gaining popularity is HiveAlive™. This product contains seaweed extracts, thymol, and lemongrass oil, and is also fed to colonies by mixing it with sugar syrup. One study tested its efficacy against *Nosema ceranae* infection. This study supplied infected colonies with HiveAlive™ in autumn (in syrup), winter (in candy), and spring (in syrup) for two years and found *Nosema* spores were significantly lower (about 50%) in spring of both years, compared to a control group. While these results are very promising, this product requires additional testing, particularly for long-term efficacy. Like Nosevit, it is not registered for *Nosema* treatment in the U.S., but beekeepers can purchase and use it.

READ

Fumagillin Is Not a Reliably Effective Treatment

■ Only one commercially available Key Points medication has been formally tested in The treatment threshold for the lab and in the field. This treating *Nosema ceranae* is medication is fumagillin (product difficult to establish, but many name Fumagilin-B®, formerly Fumidil- institutions use 1 million spores per B®). Fumagillin is an antibiotic derived bee or greater. from the fungus *Aspergillus Long* used to treat *Nosema apis fumigatus*. Fumagillin was originally infections, Fumagillin may, in fact, isolated to control *Nosema apis* in exacerbate *Nosema ceranae*. honey bees. In the U.S., fumagillin can be purchased through bee equipment Fumagillin may have unwanted companies without the need for a side effects on both honey bees veterinary prescription. and humans and must be used according to directions. Is Fumagillin Effective? The short answer to this question is that it depends on the species of *Nosema* you are treating. Fumagillin has been used since the 1950s as an effective treatment for *Nosema apis* infections. It works by inhibiting an enzyme in the *Nosema* parasite, MetAP2, that is important for tissue repair and protein production. However, it only targets the parasite in its active state and does not kill spores, so even if the active parasites are killed, remaining spores in the colony can reinfect bees. Treating colonies infected with *Nosema apis* in autumn has been found to significantly increase colony survival the following spring. Colonies infected with *Nosema apis* and treated with fumagillin have also been found to increase brood production, honey production, and adult bee lifespan following treatment. When *Nosema ceranae* appeared in *Apis mellifera*, beekeepers began using fumagillin to control it. They assumed it would work similarly, but they experienced variable efficacy. In some cases, fumagillin appeared to clear the infection, while in other cases it did not. Beekeepers began reporting that after a colony was treated with fumagillin, the *Nosema ceranae* infection returned within a few months. A 2013 study investigated the efficacy of fumagillin on *Nosema ceranae*. The results of this study supported the observations from beekeepers in the field. After fumagillin was applied to colonies, the concentration of the antibiotic decreased as it naturally broke down over time. At high concentrations, the antibiotic was effective, but as the concentration of fumagillin decreased, the infection increased. In this study, the spore production in bees that had received fumagillin was 100% higher 20 days after they received the treatment compared to infected bees that did not receive any medication! This suggests fumagillin may actually exacerbate *Nosema ceranae* infections rather than reduce them. In the study, *Nosema apis* infection also increased as fumagillin broke down, but not nearly to the extent that *Nosema ceranae* does. Models estimate that following fumagillin treatment, *Nosema ceranae* spore counts may peak again two to five months later. Furthermore, a small-scale 2018 study by the Bee Informed Partnership found fumagillin was no more

effective at reducing *Nosema* infections than giving colonies nothing at all. New *Nosema* treatment options are needed. What Is the Threshold for Treating Colonies? Despite there being no reliably effective registered treatment for *Nosema*, some beekeepers do wish to try some form of treatment in autumn if their colonies are still sick, whether it be Fumagillin, Nosevit, or Hive Alive. In these circumstances, we recommend taking an integrated pest management approach and apply a treatment only when the parasite reaches a treatment threshold. Many institutions in Canada and the U.S. (including the Bee Informed Partnership) recognize the treatment threshold for *Nosema* to be 1 million spores per bee. This treatment threshold suggests if the average spore count in a colony sample is 1 million spores per bee or greater, the beekeeper can apply a treatment. In reality, there is no well-established treatment threshold for *Nosema*. It is difficult to establish a threshold for a variety of reasons: Colonies can have many infected individuals without exhibiting any colony-level impacts (e.g., no reduced honey production, no decrease in adult population, no colony death). Infection levels vary during the season and some colonies with high infections in early summer may resolve the disease on their own, without any treatment necessary. The seasonality of *Nosema* differs based on the geographic region. While the Northeast sees high levels in spring and low levels in autumn, other countries see different trends. Independent treatment thresholds would need to be established for different locations. Research suggests the number of infected bees in a colony might be a more important indicator of disease than the number of spores in a sample of bees. For the time being, beekeepers who wish to treat their colonies in autumn after seeing no improvement throughout summer can use the current threshold until further research can determine a more reliable figure. In other words, fumagillin can be applied in autumn if spore counts exceed 1 million spores per bee. Levels should be measured again following the treatment to evaluate efficacy.

Fumagillin May Pose Risks to Honey Bees With the prevalence of *Nosema* and the unreliable treatment threshold, many beekeepers used to routinely give fumagillin to their colonies, regardless of the spore levels, as an insurance method to prevent *Nosema* infections. While this was a common practice in the past, we are now learning that prophylactic treatments can impact honey bee digestive health. This is because the MetAP2 enzyme targeted by fumagillin is not specific to *Nosema* but is actually present in all living organisms, including honey bees and humans. Low levels of fumagillin (lower than the amount needed to control *Nosema*) alter important proteins in the honey bee midgut. Because of these potential consequences to bee health, we do not recommend using fumagillin prophylactically.

Residues in Honey and Human Health Impacts Fumagillin is toxic to humans and residues present in honey can pose a risk to human health. Fumagillin acts on the MetAP2 enzyme in humans as well and may induce abdominal cramping, diarrhea, and weight loss. For this reason, fumagillin cannot be applied while honey supers are present, although there currently is no maximum residue limit for fumagillin in honey. When fumagillin is present in honey, it usually degrades within a few days to a few weeks. The likelihood of finding fumagillin in honey is low when beekeepers apply it in accordance with the label. Fumagilin-B, the trademarked product, contains dicyclohexylamine (DHC) in addition to fumagillin. DHC is five times more toxic than fumagillin to humans and can cause damage to chromosomes in humans. It is currently unknown how long DHC remains in honey.

MODULE WRAP-UP

Fungal Enemies — *Nosema*

We've explored Nosema disease in detail, understanding how this common digestive condition affects adult honey bees worldwide. We've learned about its impacts and, importantly, the various management strategies beekeepers can employ to maintain colony health when faced with this disease. Knowing about Nosema will be invaluable to you when monitoring colony health and implementing appropriate preventive measures.

Varroa Mites

Varroa is the most serious and destructive parasite for honey bee colonies. It began infesting *Apis mellifera* colonies in the 1960s in Asia and has since spread worldwide to be a leading cause of colony losses. In this module, we explore the origins and lifecycle of Varroa, the symptoms associated with a colony impacted by this parasite, and a range of treatment methods. Understanding Varroa biology can help beekeepers make informed decisions on how best to control mites in their own colonies.

READ

The Rise of Varroa

■ Varroa mites are ectoparasites of the Key Points honey bee — “ecto” meaning “outside” Varroa mites were originally — that latch their mouthparts onto the parasites of the Asian honey bee. surface of a larva, pupa, or adult bees Varroa mites have since switched and feed primarily on the bees' fat hosts to the Western honey bee bodies. Varroa mites are visible to the and are now the most destructive naked eye. They are about the size and parasite for this species. shape of a sesame seed and reddish brown. They have eight legs (two of which are visible, protruding from one side of the body). These parasites depend entirely on honey bees to live; they acquire all their nutrients through feeding on the bee and reproduce exclusively inside honey bee brood. These mites can only parasitize honey bees and do not feed on other species of bees or other insects. There are two species of Varroa mites: *Varroa jacobsoni* and *Varroa destructor*. The two species differ slightly in appearance and can be differentiated by the naked eye: *Varroa destructor* is larger and more ovular (less circular) in shape than *Varroa jacobsoni*. Both of these species parasitize the Asian honey bee, *Apis cerana*, and have likely done so for hundreds — if not thousands — of years. During this time, the mites have co-evolved with the Asian honey bee and therefore do not pose a significant threat to colony health. In the 1960s, however, one of the two mite species, *Varroa destructor*, switched hosts to parasitize *Apis mellifera*, when colonies of those Western honey bees were moved into an area in Asia where Asian honey bees resided. (*Varroa jacobsoni* did not switch and are unable to reproduce in Western honey bee colonies.) *Varroa destructor* has since spread across the globe through natural migration (swarming) and commercial transportation. From an evolutionary perspective, this mite is still a relatively new parasite to the Western honey bee. Until the development of widespread genetic resistance and adoption of resistant stock by beekeepers, *Varroa destructor* poses a severe threat to honey bee colonies. In North America alone, hundreds of thousands of colonies die each year from Varroa pressures and related infections. Microscopic image of *Varroa destructor*. Photo credit: Bee Informed Partnership.

READ

Varroa's Entry Into the U.S.

■ Varroa was first introduced into the Key Points U.S. in Florida in 1987, when hives Varroa mites are ubiquitous in transported to Wisconsin from Florida nearly every country where honey were found overrun with mites. That bees are kept. fall, colonies across twelve different states began dying. Varroa had rapidly spread across much of the country before beekeepers had taken notice, and beekeepers and researchers were unable to prevent their further distribution. By the mid-1990s, every contiguous state in the U.S. had Varroa mites, and by 2007, they were introduced to Hawaii. Migratory beekeeping played a large role in the spread of Varroa mites from Florida and beyond. Not only did Varroa mites take a toll on the honey bee industry, but it also decimated feral colonies (honey bees found in the wild). Within ten years, it was reported that feral colonies had largely disappeared across the country because of Varroa infestations. Varroa mites are now ubiquitous worldwide, with only a few islands remaining free of the mite (examples: Isle of Man, and some islands in the Azores archipelago). Australia was one of the few major beekeeping nations free of the mite until 2022. Unfortunately, V. destructor was detected in New South Wales in June 2022. Initially an emergency eradication response was launched, involving hive destruction, surveillance, and control efforts. However, by September 2023, authorities concluded that eradication was no longer feasible, and shifted to a “transition to management” strategy. This means that Australia’s Varroa-free status is now compromised in many areas; it may still have some zones under exclusion or surveillance, but not the entire country.

READ

Life Cycle of Varroa

■ There are two stages in the life cycle of Key Points the Varroa mite: the reproductive Mites reproduce in honey bee stage and the phoretic stage (though brood cells. They prefer to the phoretic stage may be renamed in reproduce in drone cells as the near future). compared to worker cells. Reproductive stage: During the Varroa mites feed primarily on the reproductive stage, the mite is inside fat bodies of honey bee pupae and the honey bee brood cell, reproducing adults. They are found most and feeding on honey bee pupae. commonly on nurse bees. Phoretic stage: During the phoretic stage, the mite is found on the outside of the bee's body, feeding on the bee. Mites are most often found on the underside of the bees' abdomens, tucked between body segments. Reproductive mites found on a developing prepupa. Credit: Bee Informed Partnership Phoretic mite on an adult bee. Credit: Jennifer Tsuruda Mites enter honey bee colonies by hitching a ride on a forager bee that drifts into or robs an uninfested colony. They can also be introduced to colonies from beekeepers exchanging frames with bees or brood between hives, or from nucs and packages. Once inside the colony, a mite generally gets off the robber/drifter and crawls into an open brood cell with older larvae that will soon be capped. Mites enter worker brood cells around 20 hours before they are capped, and they enter drone brood cells around 40 hours before they are capped. Mites are found in drone brood cells about nine times more often than they are found in worker brood cells; this is likely because drones take three days longer to pupate, allowing more mites to reach maturity before the bee emerges. The mite hides inside the brood food until the cell is capped, and once the cell is sealed, it begins feeding on the pupa and transmitting viruses contained in its saliva through the open wounds from its bite. Three days after the cell is capped, the adult female mite (foundress) lays the first egg. This first egg is unfertilized and develops into a male (like bees, mites are haplodiploid). After the first (male) egg is laid, the foundress

then lays subsequent eggs every 30 hours. Each of these subsequent eggs (usually one to six in total) are fertilized eggs that will develop into females. After the mites hatch, they all feed on the pupa. The male is white to light pink in color and smaller than a female; the females are nearly white after hatching but darken as they age. Photo of two immature mites (white) on a pupa. Photo credit: Mary Kate Wheeler It takes mites six days to reach sexual maturity, at which point the females and males mate. When the bee emerges, the mated females that survive exit the cell on the bee's body. The male mites and immature females quickly die inside the cells due to desiccation. Because most bees emerge 12 to 15 days after being capped (depending on whether they are workers or drones), most of the mites do not reach maturity before the bee emerges. In fact, only an average of 1.3 new mites are released from each worker brood cell with a foundress mite, and an average of 2.6 new mites are released from each drone brood cell with a foundress mite. Some brood cells can contain multiple foundress mites, so more sexually mature daughter mites can emerge from these cells. During the phoretic phase, mites remain on adult bees feeding for up to 13 days (seven on average), moving from bee to bee, before they enter cells to reproduce. When the colony is broodless, they can remain on adult bees for much longer, and during the winter, they can survive five to six months in the bee cluster. They feed on nurse bees when given the choice between nurses and foragers. Nurse bees are not only in close contact with larvae (making it easy for mites to crawl into brood cells), but they also provide better nutrition to mites, allowing them to produce more offspring than if they were feeding on foragers. The life cycle of *Varroa* can differ depending on the species and race that is parasitized. *Varroa* mites are able to reproduce in both drone and worker brood in *Apis mellifera* colonies but reproduce exclusively in drone brood in *Apis cerana* colonies. In fact, any mites that enter worker brood cells in *Apis cerana* colonies remain infertile (they do not lay any eggs). This inability to reproduce in worker brood can really limit the population growth of *Varroa* in colonies and helps prevent them from reaching high infestation levels. In contrast, only 10 to 20% of mites entering worker brood cells are infertile in *Apis mellifera* colonies. The African honey bee, *Apis mellifera scutellata*, is in between, with 40% of mites entering worker brood cells remaining infertile.

READ

Varroa's Impact on Honey Bee Health — Feeding

Parasites feeding on their host can, of course, be taxing for the host. Scientists have recently discovered that *Varroa* mites feed predominantly on the bees' fat bodies. As a result, both pupae and adults parasitized by mites have reduced lipid levels. Lipids stored in the fat body are important for the development of winter bees and a colony's survival over winter. Indeed, bees parasitized by mites in autumn may not develop as winter bees and tend to live a shorter life. If infestation rates in a colony are high in late summer and early autumn, the colony can collapse in the following weeks or months as the population dwindles before or during winter. Unfortunately, *Varroa* population levels tend to peak at the same time that the colony is producing winter bees (from August to early October in the U.S.). Also, the fat body is much more than "fat storage"; it is a multifunctional tissue analogous to a combination of the liver, adipose tissue, and immune system in vertebrates. As a result, *Varroa* mites not only disrupt energy storage but also vital functions involved in immunity and detoxification. Infested bees then become immunocompromised and less able to detoxify the pesticides they encounter on a daily basis as they forage for nectar and pollen on flowers. Bees parasitized during their development also have

smaller hypopharyngeal glands and lower protein levels in their hemolymph as adults. This may impact their ability to effectively feed larvae when they are nurse bees.

READ

Varroa's Impact on Honey Bee Health — Transmitting

Viruses ■ Varroa is a vector for at least eight Key Points different bee viruses. In other words, a Varroa mites can transmit viruses mite can carry and transmit viruses to honey bees. The combination of from one bee to another. When a virus viruses and mites is more deadly enters the body, it can remain inactive to bees than either pressure is or it can become activated and alone. reproduce by making many copies of itself. Once it reaches a certain number of copies, the individual host can start to show symptoms. For instance, with one particular virus called deformed wing virus, the bee will develop the symptom of deformed wings once it has 10 billion copies of the virus in its body. Many of these viruses were present in honey bee colonies before Varroa mites were introduced, but they rarely caused symptoms. Varroa mites have been shown to activate some viruses in bees, meaning the viruses were present but not reproducing, but then begin to reproduce. In other words, where the viruses used to be harmless and asymptomatic, they begin to impair individual bee health and put stress on the colony as a whole. Now that Varroa mites are enhancing virus transmission, the combination of the mites and viruses has become a major issue for honey bee colonies. Indeed, a colony suffering from both is at a greater risk of death than if it were infected with a virus or mites alone. The next module explores these viruses in more detail and their effect on colony health.

WATCH

Varroa's Impact on Honey Bee Health — Parasitic Mite

Syndrome ■ Parasitic mite syndrome (PMS) is the Key Points most advanced stage of a Varroa virus Advanced stages of Varroa mite infestation. It has varied symptoms, but infestations cause Parasitic Mite is most often characterized by high Syndrome. mite levels, diseased brood infected In the Northeast, Varroa with one or more viruses, and adults populations peak in late summer with deformed wings (a symptom of and autumn. It is therefore critical deformed wing virus, transmitted by to monitor mites in August Varroa). Adult bees may also have through October. mites visible on their bodies. There may be signs of hygienic behavior: spotty brood, perforated cappings (cappings that have been completely chewed off, revealing “bald brood”) and chewed down brood. Parasitic mite syndrome usually first becomes evident in late summer (August in the Northeast). Once the colony has developed Parasitic Mite Syndrome, it can quickly collapse within 1-4 weeks. Watch this video to learn how to recognize PMS in your colonies. Video Transcript Parasitic mite syndrome is the advanced stage of a Varroa mite infestation. At this point in time, colonies have very high mite levels, well above the treatment threshold. And they're also accompanied by high levels of viruses that are transmitted by the mite. Here in the Northeast, parasitic mite syndrome typically occurs around August or September. So, it's critical that you conduct a colony inspection at this point in time. However, PMS can also begin earlier in the year if Varroa mite levels were not well managed sooner. It's really strange because the colonies that tend to experience parasitic mite syndrome usually appear the healthiest and the strongest from the outside. They're often colonies that have lots of honey supers, have collected lots of honey, and also have a lot of activity at the entrance. And it's not until you start investigating the colony

and inspecting the brood frames that you start to see symptoms. Right off the bat, you might start to notice frames that have a spotty brood pattern. And it's really important that you look at any cells that have abnormalities. With parasitic mite syndrome, we tend to see lots of hygienic behavior, which is the effort for bees to clear up the virus infections, and also the mite infestations that are affecting those pupae. So, you can see various degrees of hygienic behavior. You can start to see cell perforations where they've begun chewing off the cappings, you can see bald brood, which are those white headed pupae that haven't developed any coloration or hair yet, and you can also see chew down brood, which is where the bees are consuming, partially, some of those pupae that are in there. And then lastly, just those empty cells where bees have been pulling out the pupae, and fully getting rid of them from a colony. Before you shake any bees off the frame, take a look at the adults. With parasitic mite syndrome, we tend to see a low nurse bee to brood ratio, or not enough nurse bees to fully cover the brood that's found on that frame. In healthy colonies, there's enough nurses to cover that brood so that they can keep it warm, and they can tend to it, and feed any larvae. But in sick colonies, a lot of this brood is dying. So, they're not replacing those adult bees, or there's not enough colony recruitment from brood to adults. Also on the adults, you might notice deformed wings from high levels of deformed wing virus, or you might actually see some mites that are visible on the adults. And remember, by the time that you're actually noticing mites on adults, you likely have high levels of mites, because usually, you don't come across many mites when you're looking; most of them are found in the brood. Like other diseases, supersedure cells might also be present. And with parasitic mite syndrome, larvae can die at any age. So, we might see larvae that are just slumped down, or laying flat at the bottom of the cells, or we might see larvae that are kind of twisted or contorted. We might see dead larvae or pupae that are white, or that are beginning to turn brown as they decay and decompose, and they can start to take on a foul odor at this point in time. Once a colony develops parasitic mite syndrome, it can actually collapse within one to four weeks. So, it's really important that the beekeeper acts fast and puts on an effective mite treatment. And even if they do this, it might still be too late to save the colony. So, let's take a look at some photos of parasitic mite syndrome a little bit closer so that you can familiarize yourself with some of these symptoms in your own colonies, if you happened to see it, and then, let's join Randy Oliver in the bee yard, where he inspects a colony that has parasitic mite syndrome. Alright. So, in this first photo, the symptoms that we can see right away are a spotty brood pattern. And we can also see various degrees of hygienic behavior. So, we have one cell here that has a perforated capping, and then we have several cells with bald brood, which is the next level of hygienic behavior. And then, we even have a cell that shows chewed down brood, where this pupa has been chewed down partially, and only part of its body remains. Then we can also see some larvae that have died, and they're taking on sort of different shapes inside the cell. So, this larva here is slumped down in the cell. It's actually found at the top of the cell here. So, it's slumped and it's starting to change colors. In this next image, we can see a few adult bees that have Varroa mites on them. In this next image here, we can see a spotty brood pattern. Here we actually have the queen in this photo. But to the right of the queen, we can see an adult that has deformed wings. So, deformed wing virus. And the worker below the queen also has deformed wings. So, we have two adults here that show deformed wing virus. Again, we're seeing perforations in the cappings, we're seeing some chewed down brood, we're seeing some bald brood, and we're also seeing some larvae that are beginning to change colors in their cell and turning a little bit brown. This next photo is similar. We see a spotty brood pattern again, and some

larvae that are dead at various ages, and some chewed down brood. Here you can also tell that there are not enough bees to cover the amount of brood that's shown on the frame. In this next photo, we can see a larvae that slumped down in its cell, and you can actually see a Varroa mite that's feeding on it. So, with colonies that have parasitic mite syndrome, it's not uncommon to see mites inside open brood cells, and also see mite scurrying across capped brood. Another thing, if you look really closely, is you can see Varroa mite feces, which we call guanine crystals, on the top of this cell here. And as the mites are developing inside the cell, they're defecating. And so you can see them sometimes build up in these frames. And they're just small, white crystals. This next photo is classic parasitic mite syndrome. We see larvae that are dying at different ages and different orientations, and slumped in their cells. We can also see several mites that are in these open cells and a spotty brood pattern. This next photo is similar. We have bald brood, chewed down brood, a dead, slumped larvae, some perforated capping is here on the far right. And we can also see little bits of chewed brood, where not too much is remaining of that pupa that's been consumed. In this last photo, there is a bit of repetition, but it's really important that you're able to diagnose this if you come across that in your colonies. Here, we're seeing lots of the symptoms. We are seeing chewed down brood here. This is a chewed pupa. We're seeing larvae that are slumped in their cells. We are seeing mites that are feeding on the larva. And we're starting to see the larva changing colors. Let's join Randy Oliver in the bee yard to inspect a colony that has parasitic mite syndrome. And here is a classic example. We have a colony that originally appears very strong and healthy, and even the first few brood frames that we look at look pretty good. A solid brood pattern. It's not until we look a little bit closer that we start to see that this colony is close to collapsing. So, again, look right here. We have a supersedure cell going right here. I look up inside it. Okay. You can see the jelly in there. And the queen larvae in there. So, this colony is getting ready to supersede their queen. To me, I figured the bees know a whole lot more about what they're doing than I do, so I will not disrupt their supersedure process in any way. It's kind of interesting, if you look at this next frame right here, that makes you wonder why in the world they are superseding this queen because this looks like a beautiful frame of brood. You see a bunch of the queen cups right here. They're thinking at one time about swarming, but there's nothing in those cups. Now, the anomaly I see here, right here, I'm looking at it right here. Just look at these little holes right here. Why are there holes right there? And now I see right here. Okay. I think this is a white pupa. White, because there's no melonization. Which means this pupa right here was killed by the bees while it was still alive. Not a sick pupa. It put out a signal telling the bees that it had a virus, and the bees are removing it. They're called brood sensitive hygiene. Now, opposed to these pupae right here, which are melonized and discolored. They actually had an advanced virus, probably deforming virus, and the bees are removing them. This is very typical with colony with a lot of Varroa. You see these two cappings. There's one capping right there. One right there. One right there with a little tiny hole in here. These bees are uncapping these pupae because these pupae are infected with virus. Even though it looks healthy, it's likely going to be sick. Here's another one that is sick right here. And I can't tell, that looks maybe a little bit like chalkbrood, it's hard to tell. And this here is a dying one, a dying one, a dying one. So, this colony, despite the fact that it has a beautiful initial brood pattern, it likely has a very high Varroa load. All the indications are that this colony is nearing Varroa collapse. And we used to see a lot of bees with deformed wings at this period of time. And you can see right here, again. Okay. There's a dying one, dying one, dying one, dying one there, dying one here. So, this is a colony that is not being managed

adequately for Varroa mite. So, you can still turn this colony around at this time of year. This is August right now. You could clean this colony up with a good mite treatment, and this colony would have time to recover. This colony is greatly restricting the brood nest. They got a band of pollen, of bee bread here, but you can see they've restricted it. There's a dead, dying pupa right there. So, this is a typical colony in late summer. Varroa deformed wing virus epidemic building in it. Okay. I don't see any young larvae here at all. You can see the parasitic mite syndrome just collapse all these dying pupae here. So, this colony is the lack of young larvae. A little bit up in the upper box with very little young larvae, which means this colony is not recruiting at an adequate rate to keep the adult population up. If we did a shake into a tub of bees, you would see that most of the bees would fly out because they are older bees. We don't have very many young bees in this colony. Also, we don't have the type of bee-to-brood ratio we would expect. We'd expect to see more bees, adult bees, over this brood, which means that the recruitment rate of this colony is dropping off, again. You can see the proportion of them that are dying from deformed wing virus or from a virus which I suspect is deformed wing virus. So, our colony population is dropping rapidly. We won't have enough bees to keep the brood. If you have a cold night, you'll get chilling in the brood, and then you can go into a spiral, and do a pretty, pretty sudden collapse of the colony. If parasitic mite syndrome goes unnoticed, the beekeeper might simply wrap up the colonies for winter, thinking that everything is okay, and not realize until mid-winter or spring that the colony has died. So, how do you know if your colony died from Varroa mites? Well, there's a few telltale signs that you can identify. Colonies that die from high Varroa mites are usually the ones that have the strongest populations at the end of summer or in autumn. So, beekeepers are shocked in the spring to find that the remaining dead cluster is tiny. It's usually about the size of a grapefruit, or even a baseball. Often, just a queen with a handful of bees surrounding her. They're also surprised to see that there's no other dead bees in the hive besides this tiny cluster. You don't see bees on the bottom board, or any bees at the entrance on the ground, like you tend to see if bees died from moisture or starvation. And what happened is that the colony really died before the harsh onset of winter. And when weather was still warm enough, bees that die from high levels of viruses were able to leave the colony to die or any bees that were performing cleansing flights or foraging on any flowers that might still be blooming at that time were too weak to make it back. And lastly, it could be the case that undertaker bees were able to clean out a lot of the dead bodies for any bees that happened to die in the colony before the beekeeper is able to see any evidence of that. When we're looking at the brood frames, there's hardly any brood remaining. So, the queen was not alive in January or February to begin laying eggs and start getting ready for spring. There are small patches of spotty, capped brood cells, and they often have perforations, or signs of hygienic behavior from the colony that had a parasitic mite syndrome. You'll also notice that there's usually a lot of honey remaining on the hive. And again, the bees died before the harsh onset of winter. So, they weren't eating much of their stores. They definitely did not die from starvation in this case. And lastly, you might actually notice the Varroa feces, or the guanine crystals, in the cells. They're much easier to see now that there aren't any adult bees on the frame or much brood remaining. So, you'll see those white crystals that are sprinkled in those cells. So, these are some of the symptoms of colonies that have parasitic mite syndrome when they were still alive, and colonies that died from parasitic mite syndrome. So, you'll be able to identify if this has happened in your colonies. Ironically, strong colonies with lots of brood are often those that harbor the most mites. Before the arrival of Varroa mites, beekeepers often only inspected colonies when there was obviously a

problem. If they saw a strong colony in August with plenty of honey supers and activity at the entrance, they would not have thought to inspect them. Now, however, this time of year is critical to inspect colonies, particularly in colder climates.

WATCH

Varroa's Impact on Honey Bee Health — Colony Death

■ If left untreated, the Varroa mite Key Points population in a colony continues to The strongest colonies often climb. Eventually, it will likely harbor the most mites. overwhelm the colony, causing death between six months and two years. There is a lot of variation in the lifespan of untreated colonies because many variables can influence the Varroa population in a colony: climate, amount of worker brood, amount of drone brood, and genetic stock are just some factors that play a role. Nonetheless, it is important to understand how mites can kill colonies and to recognize the role that honey bee and Varroa mite population dynamics play. Watch this video to hear Randy Oliver describe how you can identify the Varroa mite population changes in a collapsing colony. Video Transcript When you look in your colony, understand a couple of things. Understand bee population dynamics and understand Varroa population dynamics. Honey bee population dynamics are based upon how many eggs the queen can lay a day, and how well the bees care for those larvae. And you wind up with a ratio of open brood, which is the larvae, to sealed brood, which should be about 50/50. And then you should have enough adult bees to cover that brood. Now, on a warm day, the bees won't be over the sealed brood because it gets too warm in there. But you get an idea of what you should see as far as the amount of bees and the amount of brood and the ratio of larvae to sealed brood. When a colony is starting to go downhill, those ratios get off. That's the first sign that there's a problem with that colony. So, a colony in growth mode should be full of young bees, lots of nurse bees, lots of young larvae being well fed. In the late summer, you typically have a decline. The queen starts to cut back and you're going to see a different ratio. As far as Varroa, understand Varroa starts off at its lowest level coming out of winter when the bees are first starting to rear brood because there's been Varroa mortality during the winter. And if you're in an area where there's no brood rearing during the winter, Varroa population drops about a half percent per day. Then, as the bees are rearing lots of brood in the early spring, and also drone brood, Varroa increases at its most rapid rate, increases at the most rapid rate when there's the largest amount of eight-day-old larvae in the hive, relative to the amount of bees. Plus the drone brood helps Varroa slightly. So, in early spring, Varroa increase at very rapid rate. And it grows exponentially. It roughly doubles its population once a month. And it depends upon what area you are in. So, knowing that it will roughly double once a month, you can project what the population is going to be. And then in the late summer, if you have a lot of feral colonies or other beekeepers who are not taking care of their hive, not managing them for Varroa, there will be collapses, a lot of this, what we call "colony mortality" now. If that's happening around you within flight range of your bees, a number of Varroa mites will get carried back to your hives, either through drift or through robbing, and can suddenly increase your mite population. It's very easy for a colony to pick up a hundred to 1000 mites in a matter of a couple of weeks to collapsing colonies in the area. So, be aware that mite infestation rate can accelerate in the late fall. To help illustrate how Varroa population dynamics leads to colony death, the figure below describes a hypothetical example of how Varroa levels rise and fall over the course of a year in an untreated colony in the Northeast US. In this example, the purple line represents the adult worker bee

population, while the orange line represents the adult mite population. At the beginning of June, the number of mites in the colony is not overwhelmingly high. In this example, there are around 1,000 mites and 39,000 bees, or 2.6 mites per 100 bees. At this point, the colony is growing and producing lots of honey. But as brood production increases, so too does mite production. The mite population lags behind the bee population, but increases quickly. Colony population peaks at the end of June. Once the July dearth hits, the queen begins to lay fewer eggs, less brood is produced, and the population begins to steadily decrease. A small population blip in August and September coincides with the fall flow and is when winter bees are largely produced. In this example of an untreated colony, the adult mite population reaches its peak from August to October. With less brood in the colony, many of the mites are now phoretic and fewer are reproducing. By September, there are around 5,000 mites in the colony and 35,000 bees, or 14 mites per 100 bees. This is at a level that threatens colony collapse. At this point, PMS is likely evident. The colony population is dwindling as adults die and a large fraction of the brood is unhealthy and does not reach adulthood. As the bee population dwindles, the colony is also susceptible to robbing, especially during late summer or fall dearths. Robbers entering this collapsing colony can help spread Varroa to other hives in the area. This colony is likely to collapse, reaching critically low population levels in November, and dying before the coldest days of winter arrives. Symptoms at the Colony Level Historically, many beekeepers only inspected colonies if they suspected something was wrong (limited activity at the entrance, a foul odor, dead bees outside the hive, etc.) During a peak Varroa infestation, there are few colony-level symptoms that would indicate the colony may soon collapse. Some spotty brood, chewed cappings, and chewed down brood may be evident. Bee populations remain strong until the colonies are extremely weakened by mite pressure, and Varroa infestations usually do not affect honey production until the point of extreme weakening. As the colony is growing in spring and summer, the mite population lags behind the honey bee population. During this time, the bees are likely still producing a normal honey crop. But this productivity can be deceiving! It is therefore critical that beekeepers' management practices include frequent inspections and mite monitoring. Remember, it is the strongest colonies that may harbor the most mites. Seeing a good honey crop might make the beekeeper think their mite levels are under control and the colony is healthy, but the colony must still be inspected and monitored for mites. Moreover, if nearby colonies have Varroa mites, robbing and drifting can significantly contribute to late summer and autumn mite levels and accelerate a colony's collapse.

READ

A Leading Cause of Colony Losses

Every winter, Varroa mites and their associated viruses kill hundreds of thousands of colonies around the world. Several studies conducted in North America and Europe have found that high mite levels in summer and/or autumn predict colony death over winter. One large-scale study in Ontario, Canada, determined how often colonies died from Varroa relative to other causes. Researchers monitored over 400 colonies and found that Varroa was the main culprit behind overwintering losses — the infestations accounted for over 85% of colony losses. This outweighed the contributions of other winter stressors, such as low population, lack of adequate food stores, or Nosema infections. Similarly, the NYS Beekeeper Tech Team, facilitated by Cornell University, has found that Varroa mites are the most significant predictor of winter colony loss for New York beekeepers. Our research in 2016 found that

mites are extremely common and difficult to manage in autumn. Nearly every colony of the 300 we sampled in September had Varroa mites (90% of 309), and 78% of the 60 beekeepers we worked with had colonies with Varroa mite levels above the treatment threshold. There is a stigma surrounding Varroa mites. Some people feel that beekeepers whose colonies die from mite infestations must not be good beekeepers because they can't keep their mite levels under control. Because of this misconception, many beekeepers often don't want to admit that their colonies have high levels of or have died from Varroa mites. It is therefore important that, as Master Beekeepers, we foster an open and nonjudgmental environment when speaking with others about Varroa. We can help beekeepers and the general public understand how common — and increasingly harmful — Varroa mites are in modern beekeeping and teach beekeepers how best to manage them. Only by working collectively can the scientific and beekeeping communities combat Varroa mites.

READ

IPM for Varroa

It is nearly impossible today to prevent Varroa mites from entering honey bee colonies. There are very few regions of the world that remain free of Varroa, and colony infestation is nearly unavoidable in areas where they are found. Instead of trying to eradicate Varroa completely, beekeepers are now focused on keeping their mite populations low all year long. The most effective strategy to combat Varroa mites is by taking an Integrative Pest Management approach. This is a proactive approach that allows beekeepers to manage their mites before they reach damaging levels. Central to this approach is being aware of colonies' mite levels throughout the year and using a variety of management options to keep levels low. This delays the need for treatments and reduces the amount of miticides introduced into the hive. It is helpful to develop a scheduling plan to incorporate regular monitoring and to have intervention tools and treatments on hand to apply as soon as they are needed. Over the next four pages, we discuss the steps of IPM for Varroa control in detail.

READ

Monitoring Overview

Monitoring should take place once a month from March to October in the Northeastern U.S. Accurately monitoring mites each month can help accomplish three things: 1. It informs you of the degree of infestation in your colony and allows you to track its progression, giving you information about the health of your colony. 2. It enables you to track the population growth of mites in your hives, allowing you to time your treatments when populations are at manageable levels, thus ensuring your treatments are most effective. It also helps you to time your intervention around other apiary tasks, such as removing honey supers and preparing your colonies for winter. 3. It allows you to observe mite levels before and after a treatment, which informs you as to whether or not that treatment was effective. Without regular monitoring, a beekeeper may assume a given treatment worked. If a treatment doesn't reduce the level of mites below the threshold, another treatment should be used promptly. How Many Colonies Should I Monitor? Monitoring mite levels in all of your colonies is ideal. But if your operation is large and this isn't realistic, randomly sample 10% of the colonies in each bee yard (for instance, if you have 30 colonies in a bee yard, sample at least three colonies). Maintain written records of your monitoring to track mite populations.

WATCH

How to Monitor

■ Learning your mite levels can be Key Points accomplished with the powdered The most effective monitoring sugar shake or the alcohol wash. Both strategies are to use either the methods are easy and can be powdered sugar shake or the incorporated into your regular alcohol wash. inspection schedule. These are the only techniques that have been shown to reliably estimate the number of mites in a colony. Watch the videos to learn how to conduct these monitoring techniques. Powdered Sugar Shake A powdered sugar shake helps beekeepers estimate the number of varroa mites in a honey bee colony by coating a sample of bees in powdered sugar, which causes the mites to fall off. Video Transcript Let's go through the steps of doing a powdered sugar shake to determine our mite levels in our colony. So, first we need a few things. We need a mason jar that we're going to switch out the original lid with mesh wire. And you want to use mesh wire that's smaller than an eighth of an inch. You want it to be big enough that Varroa mites can fall through it, but small enough that bees can't fit through it. And we find that if you're using an eighth of an inch hard wire, the bees' heads will actually get stuck in there. So, you want to use one that's a little bit smaller than that. You also need a measuring device. We're going to be measuring 300 bees, which is about half a cup of bees. And you can just use a baking cup that you buy from a baking supply store that's for half a cup, or what we like to use at the Dyce lab is one of these urine containers, I'll emphasize it's a clean urine container, that we purchased online at a lab company. And I like this because it holds 300 bees in it and it has a screw top. You also need a hive tool, a bag of powdered sugar, a container that you can shake your bees into, ideally with rounded edges so it's easier to scoop the bees, instead of using square edges. And then, you'll need a small dish. It can be a light colored bowl or a paper plate, just something that you're going to be able to see the mites on. And a little bit of water, which is going to help us visualize the mites as well. When we're determining our infestation level in a colony, we want to sample nurse bees. So, they'll be found in the brood nest. And nurse bees are more likely to be parasitized because they're constantly in contact with the brood, where mites are reproducing. And mites prefer to feed on nurse bees. So, we're going to pull out a frame from the brood nest and we want to make sure that there is brood on it. So, it can be both capped or open brood. And we want to make sure that there's not too much nectar. If we shake bees off of a frame with nectar, then the bees are going to get really wet and that'll help the mites stick to them. So, give it a really good check to make sure the queen isn't on there. We don't want to be sampling the queen. And then when you're ready, shake the frame of bees pretty vigorously into a bucket. It should just take one or two swift shakes. Right off the bat, all the forager bees are going to fly away, which is okay, because the nurse bees are left behind. They're not likely to fly, and so they're very easy to scoop up. And they're also pretty docile, so they won't sting you. I have them in my cup, here, for 300 bees. And I just like to keep the lid on while I get the rest of my materials organized. So, now we're transferring it into a mason jar, and I do this by tapping the containers so that the bees can't fly out very easily or walk out very easily. And then, I screw the lid on nice and tight. From there, we're going to add our powdered sugar. And we just use our hive tool to kind of guess, two and a half tablespoons, or about two heaping tablespoons of sugar with your hive tool. And some beekeepers will add the sugar to the mason jar first, but I like to have it falling on top of the bees so that they're easily coated with the sugar. If you're just looking and you notice that the bees aren't really well

coated, throw in a little bit more powdered sugar. From there, we're going to gently roll the bees to make sure that they're nice and coated, and then we're going to set them in the shade for a minute. The reason why we set them in the shade is because this will allow the bees to help get some of the mites off of their bodies. When the bees are heating up, the mites are more likely to come off of them. And also as they're shaking their wings, this can help dislodge some of those mites. We want to do this in the shade instead of in the sun because the sun will melt the sugar, allowing the mites to stick to the bees. So, leave them in the shade for one minute. And you can time this to make sure that everything is standardized every time you're sampling different colonies. After they've sat in the shade, you're ready to start shaking your bees into a bucket. And so, we do this for two minutes. We've got our timer, again, to make sure that it's the right amount of time. And we're going to shake them very vigorously so that you're hearing the bees thumping inside of the jar. All the mites during this time are going to fall through the mesh lid into the bucket. After two minutes, we'll pour a little bit of water into the bucket to more easily visualize the mites. In this monitor, we can see that there's five mites in the bucket. So, that's five mites per 300 bees, which translates to about 1.7 mites for 100 bees. We can put the dazed and confused bees back in the colony, and most of them will live, and their sisters will clean the sugar off of them. Until you get used to it, it can be kind of hard to tell what are mites compared to other debris that might come out of the mason jar. And for the first couple of times, you can bring a magnifying glass with you and you can identify mites based on how they look. So, they're going to be small, ovals, a cinnamon color. And you can even see some of their legs in the front that are sticking out. So, that's how you do powdered sugar shake.

Alcohol Wash The alcohol wash method is most reliable when the beekeeper rinses and shakes the bees three times to remove all the mites. One product on the market, the Varroa Easy Check®, makes it simple to perform alcohol washes quickly and with minimal equipment. If you don't have alcohol, you can substitute windshield-washer fluid or soapy water.

Video Transcript Let's go through the steps of how to do an alcohol wash to monitor for Varroa mites. The ingredients that you need to do this are the exact same as what we use for the powdered sugar shake, but this time, instead of powdered sugar, we're going to be using alcohol. And it's not just any alcohol from your cupboard. You want to make sure that you're using at least 70% ethanol, or isopropanol, or one of those alcohols that you can purchase from the first aid section of the drugstore. So, in order to do this, first, like we did with the powdered sugar shaker, we're going to be using nurse bees that are found in the brood nest. When we're looking for our frame of bees, we want to make sure that the queen is not on the frame. So, give it a careful check first. Then, we're going to shake the bees off the frame into a bucket. And it can help if you don't actually tap the bottom of the bucket because that can get some of the bees to be defensive with that loud sound. So, just shake them in the air and then the forager bees will fly out. The nurses will remain and we'll scoop them up with our little half cup scoop. From here, we're going to transfer these into our mason jar. And in this case, I like to have a mason jar with a regular lid instead of a mesh lid. We're going to take our ethanol, and I'm going to tap the bees down in the mason jar just to prevent them from escaping, and I'm going to pour ethanol on them. And the bees die pretty quickly. You want to make sure that you're adding enough ethanol that you're totally coating the bees, plus giving them a little extra layer of liquid on top. So, you can see how much alcohol is in this mason jar. Then, I'm going to put the lid back on and I'm going to vigorously shake this jar for one minute. And I have a timer going to make sure it's the same amount of time with each hive that I do this with. And shaking them is going to allow the mites to

dislodge from the bees' bodies. Then, I switch out this regular lid in the mason jar and I put on this mesh lid, the same one that we use for the powdered sugar shake. I'm going to shake this over a light colored container and all the mites are going to fall through the mesh into the container. It will be easy to visualize them. If you need to add a little bit more alcohol to give them a good rinse, then you can do that as well. In this example, we have five mites inside this bucket. So, that's five mites per 300 bees or 1.7 mites per 100 bees. And those are the steps of doing an alcohol wash. These two methods are accurate ways to monitor Varroa levels when these steps are followed. Shaking the container for less time, or missing any steps, can result in unreliable results. Which monitoring method you ultimately choose to use is up to you; try out both methods to see which one works best for your time availability, budget, and comfort level. Please note that according to a recent study, the powdered sugar shake was shown to be somewhat harmful to bees. According to that study, honey bee workers subjected to a powdered sugar shake were less likely to be recaptured five days after the procedure, compared to an untreated control. This suggests that the sampled honey bee workers may not die immediately, but die prematurely after a powdered sugar shake.

READ

Unreliable Techniques for Monitoring Mites

There are several other techniques that beekeepers do to gauge whether or not their colonies are infested with mites. These methods include visually inspecting adult bees, uncapping drone brood, performing an ether roll, and using a sticky board to observe natural mite drops. While these methods can tell you whether or not mites are present in your colonies, they cannot reliably provide a mite number that you can use to gauge the infestation level. This means that it is difficult for beekeepers to make treatment decisions based on their observations with these methods. Click the accordions to read more about the unreliable methods of interpreting mite levels in a colony.

Visual Observation Every now and then you may notice a mite on the body of an adult bee, usually on the thorax. Most often, however, they are not noticed on adult bees because they tuck themselves between the abdominal plates on the undersides of the bees to feed. In addition, about 80% of mites in a colony are inside the brood, while only 20% are phoretic. For these reasons, colonies can reach very high infestation levels before the beekeeper can spot mites on adult bees. Visual observations are therefore not a reliable indicator of the severity of a Varroa infestation. Indeed, the likelihood of observing mites increases as the infestation increases. In other words, by the time you see mites on bees, the infestation is likely severe. It is important to manage mites before the population is out of control. ■

Uncapping Drone Brood Uncapping drone brood using either your hive tool or an uncapping fork can reveal mites. With 80% of the mites in a colony reproducing in the brood — and a disproportionate number of those in the drone brood — the likelihood of finding mites in brood is higher than visually observing them on adults. Uncapping several drone brood cells during your inspection can help determine if mites are present in your colony. It does not, however, give an indication of whether or not your colonies are above the treatment threshold. Bees tend to construct drone brood along the bottom edge of a brood frame. This location makes it very easy to slice off a section of drone brood with your hive tool and use your fingers to break apart the brood cells to look for mites inside. An uncapping fork is also used, as the fork can pierce through drones to remove several at once, allowing you to look for mites. Wash your hands and tools before you work in other colonies to reduce the spread of viruses. ■

Sticky Board In this method, a sticky board purchased from a bee equipment supply company (or made at home using stiff paper and petroleum jelly) is used in conjunction with a screened bottom board to observe the natural mite drop of both dead and live mites in the colony. The sticky board is placed under the colony and examined 24 hours later, counting the number of mites stuck to it. The sticky glue or petroleum jelly prevents live mites from crawling back up to the colony after they have fallen down onto the board. Commercial sticky boards contain a grid to allow for easier counting of the mites. Many backyard beekeepers enjoy using the sticky board method because it does not kill any bees or require opening the colony. However, it is difficult to use the sticky board as an indicator of mite levels. For starters, scavengers such as ants can remove mites from the sticky board, interfering with your results. More importantly, the rate at which mites drop (without treatments) changes dramatically throughout the year. Many more mites drop when there is no brood compared to when there is brood present. Accurately estimating mite infestation levels requires you to know how much brood is in the colony and to make fairly advanced calculations to determine your mite levels. ■ On top of the difficulty in interpreting mite drop values, the sticky board method is the most time-consuming one. It requires two trips to the bee yard (once to insert the sticky board, and again to remove it after 24 hours). It is also time consuming to count the mites, as they are mixed in with the other debris that collects on a sticky board.

Ether Roll The ether roll method consists of collecting approximately 300 nurse bees in a mason jar and spraying them with two or three quick shots of ether (car- starter fluid). The bees quickly die. This amount of ether makes the bees and the inside of the jar sticky. The jar is shaken, removing mites from the bees' bodies that then get stuck to the sides of the jar. The number of mites stuck to the inside of the glass jar are then counted. This method is less reliable because it does not effectively remove all the mites from the bees, so typically it only allows the beekeeper to detect approximately 50-60% of the mites. This method is also riskier than others; ether makes bees defensive, and the highly flammable nature of ether makes it dangerous to use in close proximity to a smoker. ■

READ

Treatment Threshold

■ A key part of integrated pest Key Points management is determining a Beekeepers should take action treatment threshold. This threshold when mite levels meet or exceed indicates the level at which the two mites/100 bees April to July beekeeper makes decisions to control and three mites/100 bees August mites. If monitoring reveals that a to October. colony's mite levels are at or above the threshold, it is likely to die within six months to two years unless the beekeeper intervenes with a treatment. It's important to note that treatment thresholds are not constant. They can change over years, within the year, across regions, and in accordance with a beekeeper's management goals. When Varroa mites were first introduced to the U.S. in the 1980s, colonies could tolerate much higher infestations than they can today before collapsing — upwards of 11,000 mites. The treatment threshold during this time was high, between 10 and 20%, and mites could be effectively controlled with one or two treatments each year. Since then, however, colonies have become less able to tolerate such high infestations. Today, colonies are considered to have a severe Varroa problem once populations reach over 3,000 mites per colony. As a result, the treatment threshold is much lower (2 to 3%, or two or three mites per 100 bees) and colonies often require several treatments each year. Scientists don't fully understand why bees' tolerance to mites has been reduced over the years. One possibility is that the viruses

transmitted by mites have developed more virulent strains. Untangling the relationship between Varroa and viruses is an active area of research. The treatment thresholds in this course for the Northeastern U.S. combine recommendations from the Honey Bee Health Coalition and the Ontario Ministry of Food and Agriculture with evidence-based research from the NYS Beekeeper Tech Team. Varroa mite levels that exceed these thresholds require an immediate treatment; without one, they have been shown to predict colony death within six months to two years. From April to July, intervention is recommended when mites reach or exceed a 2% infestation (two mites per 100 bees). From August to October, intervention is recommended when mites reach or exceed a 3% infestation (three mites per 100 bees). 19 mites from monitoring a half-cup of bees. This example shows a 6.3% infestation. Autumn thresholds are set higher in the Northeast because Varroa mite populations are higher during this time than in spring. Also, more mites are phoretic in late summer and autumn because colony brood production is decreasing, which causes more mites to be observed when you monitor. Treatment thresholds can also vary regionally. Warmer states and countries may support brood rearing for longer periods during the year and so may have a lower threshold for mites. Similarly, a beekeeper's goals may influence their decision of when to treat. If a beekeeper has a very low tolerance for colony loss, they may apply a treatment at lower infestation rates than a beekeeper with a higher tolerance for colony loss. For instance, Randy Oliver is a proponent for treating colonies at a 1% infestation rate all year round. This takes into consideration his individual management goals as well as his yearlong warmer climate. Remember, however, that inadequately controlling your mite levels can not only cause colony losses in your operations but can also put your neighbors' colonies and wild honey bee colonies at risk. If your hives are located in an area outside of northeastern North America, consult your state/provincial apiculturist or extension personnel to discuss the treatment thresholds for your region.

READ

The Importance of Hygienic Behavior

■ Hygienic behavior is a trait where Key Points workers are able to detect and quickly remove infected or parasitized pupae. Using Varroa-resistant stock is the best way to give colonies a head start in controlling mite levels. Worker bees are able to detect the start of parasitized or viral-infected pupae inside capped cells based on odor and, as bees chew cappings and remove a result, chew open the cappings. First, pupae from infested cells. In doing a small hole is chewed in the capping, so, mites inside the cells dry out or the capping can be entirely chewed and die off. That bee or another one then drags the pupae out of the colony or consumes it within the cell. Because Varroa can only reproduce in capped cells, hygienic behavior can disrupt their reproduction. Once cells are open, the immature mites die from desiccation. While the term "hygienic behavior" refers to the quick and complete removal of pupae, beekeepers can often observe the intermediate stages of hygienic bees' attempts to control disease and mite infestations. For instance, a beekeeper may observe (1) perforations (small chew holes) in the cell cappings; (2) "bald brood" (with the intact, white, hairless pupa visible inside); and (3) chewed-down brood (in which the bees have partially consumed the pupa in the cell). It is currently not fully understood how honey bees detect infested cells, but studies have found they can detect altered brood pheromone, other odors that are released by sick or dead brood, and odors released by reproducing mites. In many cases, the act of chewing holes in cappings alone, then resealing these holes later, is enough to significantly reduce Varroa mite

reproduction within the cell. Several feral honey bee populations that persist with Varroa in the wild display an enhanced ability to uncap and recap parasitized cells. Photo of perforated cell cappings alongside hygienic behavior (empty cells). Photo credit: Bee Informed Partnership. Photo of bald brood. Photo credit: Bee Informed Partnership. Photo of chewed-down brood (blue arrow). Photo credit: Bee Informed Partnership. Photo of chewed-down brood. Photo credit: Bee Informed Partnership. While conducting your regular inspections, look for evidence of hygienic behavior, uncapping, or chewed-down brood. If you see cells with perforations and you're not confident that the bees are merely capping a larva's cell, use your hive tool to open them up and see what's inside. Remove some cells with bald brood or chewed-down brood using a toothpick or small twig. You can sometimes find mites (both mature and immature) inside the cells during this process. Hygienic behavior allows bees to detect not only Varroa-infested brood but also other brood diseases, such as chalkbrood, sacbrood, viruses, and American foulbrood. For this reason, it's a good management practice to open cells with perforations. While hygienic behavior can be alarming to see in your colony, it is actually a good behavior. The bees are demonstrating that they are able to detect an issue and attempting to control mites by cleaning out parasitized brood. Seeing a couple cells of hygienic behavior on a frame signals to the beekeeper that they should monitor for mites if they aren't already doing so on a monthly basis. Seeing many cells with evidence of hygienic behavior, chewed-down brood, bald brood, or perforations often indicates there is a high Varroa mite population in the colony and should serve as a warning sign to beekeepers that the colony may have PMS. The beekeeper should monitor their colony and be prepared to add a treatment immediately if the monitoring results dictate that it's time. Uncap perforated cells with your hive tool to see any issues that could be affecting the brood.

READ

How Is Hygienic Behavior Tested in Colonies?

A classic test for hygienic behavior is the freeze-kill brood technique. It consists in freezing a portion of capped brood in a colony using liquid nitrogen. This kills the brood. The frame is reinserted in the colony and left for 24 hours for the bees to detect the dead brood and clean out the cells. After 24 hours, the frame is removed and the proportion of removed pupae is calculated as a percentage. Typically, colonies with greater than 90% of the dead brood removed are classified as hygienic. Liquid nitrogen is applied to a portion of capped brood. Some brood becomes "bald" in the process. Twenty four hours later, Colony A did not do a great job of removing the freeze-killed brood. Although they uncapped many cells, this colony is not considered very hygienic. Photo: Bee Informed Partnership. Twenty four hours later, Colony B detected and removed 100% of freeze-killed brood. This colony is highly hygienic. Photo: Bee Informed Partnership. If a beekeeper wishes to test his or her own colonies for free, the "pinprick" method can also be used. In this method, capped brood is pricked with a sharp pin to kill the pupae inside. It is recommended that the beekeeper pricks at least 21 cells and that they apply a paint dot or smear to a few cells above the region they pricked to more easily identify that region later. After 24 hours, the beekeeper returns to the frame to count the proportion of pupae that were removed. If the beekeeper pricked 100 capped cells, and 24 hours later 92 of them were empty (had the pupae removed), that colony would display 92% hygienic behavior. Illustration of the pinprick method. In this example, the beekeeper pricked 30 cells (red dots) in two regions, for a total of 60 cells.

Yellow paint lines mark the two regions that were pricked so they can quickly be identified later. Two other assays to measure hygienic behavior in honey bees have been developed recently. They provide a faster, more precise, and less destructive alternative to the traditional freeze-killed brood or pin-killed brood tests described above. The UBO assay (Uniform Brood Odor Assay) measures the olfactory detection stage of hygienic behavior, meaning how bees respond to chemical cues that mimic diseased or parasitized brood. It is based on the finding that hygienic bees respond to specific brood pheromones and decay volatiles released by unhealthy or parasitized larvae. These odors act as olfactory cues that trigger workers to uncap and remove affected brood. This is how beekeepers can implement this assay in their colonies. Synthetic disease-related "brood odor" compounds (like β -ocimene and oleic acid) can be applied to healthy capped brood, and the hygienic response can be measured as the percentage of cells that worker bees uncap or remove 2–4 hours after application of the synthetic mix. That mix can be currently commercially obtained as "UbeeO" at <https://opterabees.com/>. The second recent assay developed is known as the capping/uncapping assay, which measures how often and how effectively worker bees open (uncap) and sometimes re-seal (recap) brood cells that contain abnormal brood, usually Varroa-infested pupae. It is based on the finding that when a Varroa mite reproduces inside a capped cell, the developing pupa releases subtle olfactory and vibrational cues that differ from healthy brood. Some bees, in particular from hygienic lines, detect these cues through the cappings and open the cells to investigate. If the pupa is severely parasitized or dead, they proceed to remove it. If it seems viable, they proceed to recap the cell. This behavior interrupts the mite's reproductive cycle, often killing or sterilizing the mites. These are the steps that beekeepers can use to implement this assay in their colonies. Step 1. Choose a frame with Varroa-infested sealed worker brood and mark a defined area of capped cells (~200–300 cells). Step 2. Reinsert the frame into the hive and leave it for 24–72 hours to allow worker bees to interact with it. Step 3. Examine the cells and record the following categories: capped cells (intact), uncapped cells (wax removed), recapped cells (new thin wax cap placed after reopening), removed brood. Step 4. Calculate the proportion of infested cells that were uncapped/removed versus those left intact.

WATCH

Genetic Stocks for Varroa Resistance

■ Soon after the Varroa mite jumped Key Points hosts to *Apis mellifera*, researchers began to breed for Varroa-resistant stock. The best way to give colonies a head start in controlling mite levels is the start with Varroa-resistant stocks. These are the best non-chemical tool to help control mite populations. While several stocks confer some degree of resistance, Varroa Sensitive Hygienic (VSH) has been the most effective in lowering mite populations. Hygienic, Minnesota Hygienic, surprisingly not many beekeepers use Russian, Purdue Ankle Biters, and these stocks in their own operation. Saskatraz. Share information about Varroa-resistant stock and where to order it to help spread the word at your local bee club. If entire communities begin to adopt Varroa-resistant stock, the region will have a good head start in combating the Varroa virus epidemic. There are currently five different Varroa-resistant stocks that are commercially available in the U.S.: Varroa Sensitive Hygienic (the newest lines are called Pol-line), Minnesota Hygienic, Russian, Purdue Ankle Biters, and Saskatraz. Some of these stocks are effective

at reducing Varroa infestation levels because they display hygienic behavior. Below you will learn more about the different Varroa-resistant stocks available in the U.S. Varroa Sensitive Hygienic This stock was developed by Dr. John Harbo and Dr. Jeffrey Harris at the USDA Bee Breeding Lab in Baton Rouge, Louisiana. Varroa Sensitive Hygienic bees kill and remove pupae from cells that contain reproductive mites. Immature mites die because they desiccate quickly once the cells are opened; in this way, mite reproduction is interrupted and the growth of the mite population is slowed. If there are many bees in a colony that perform hygienic behavior, this can significantly lower the mite population. This stock was originally called SMR stock (suppressed mite reproduction); with further research, however, it was realized that the impairment of mite reproduction was due to the bees' so-called hygienic behavior, so the stock was renamed Varroa Sensitive Hygienic (VSH). The VSH stock was developed in 1997 and has had many generations to test and refine this trait. These tests show that VSH colonies have significantly slower mite population growth than non-VSH colonies. In one of these tests, VSH queens were exchanged for non-VSH queens in colonies. When VSH queens were introduced to colonies previously headed by non-VSH queens with high Varroa mites, the mite population decreased. When non-VSH queens were introduced to colonies previously headed by VSH queens with low Varroa mites, the mite population increased. "Pure VSH" queens refer to queens that have been mated with drones that also carry genetics for the VSH trait. Some commercial beekeepers purchase pure VSH queens and raise daughter queens from them, allowing these daughters to naturally mate with random drones in the area. These second-generation queens are often advertised as "VSH" queens, but the trait is more diluted than it is in colonies with pure VSH queens. Some beekeepers may purchase pure VSH queens and mate their daughters with drones that have other traits of interest, such as gentleness, winter hardiness, or high honey production. By doing this, stocks of bees can be bred that carry several traits of interest to beekeepers. How Do VSH Colonies Compare to Other Colonies in Beekeeping Operations? When VSH queens have been introduced to commercial colonies in the U.S., they have helped significantly reduce mite levels. One study found that colonies headed by pure VSH queens reduced mite infestation levels by 76% and daughter VSH queens (that mate openly with random drones) reduced mite infestations by 44%, compared to colonies headed by non-VSH queens, which reduced mite infestations by only 7%. Another study comparing VSH colonies and Italian colonies in U.S. commercial apiaries found VSH colonies required fewer mite treatments during the study period: Only 12% of VSH colonies required mite treatments, while 40% of Italian colonies required mite treatments. Furthermore, both stocks produced similar amounts of honey. Minnesota Hygienic Originally developed to help colonies resist and recover from American foulbrood and chalkbrood, the Minnesota Hygienic line of bees was developed by Dr. Marla Spivak in 1993. Colonies that display 95 to 100% hygienic behavior are used as breeder colonies. This stock is now also advertised as helping reduce Varroa mites. Tests show that this line is not quite as effective at lowering mite population as VSH lines are, but it is more effective than non-hygienic stocks. In this video, Dr. Marla Spivak, Distinguished McKnight University Professor of Apiculture and Social Insects at the University of Minnesota, discusses the Minnesota Hygienic line that she developed. Video Transcript When I moved to the University of Minnesota in 1993, I started studying hygienic behavior of honey bees. This is the ability, it's a genetic trait, and it's the ability of some bees to detect and remove diseased and mite-infested brood. For a diseased brood, the bees do it very, very quickly. They're able to detect that the larva or the pupa is diseased. But they, when they're handling it, the larva or pupa has

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of colonies that die from Varroa mites. Purdue University distributed 102 queens to 39 beekeepers in 2014. Half of the queens were Purdue Ankle Biters, while the other half were of other commercial stock. Beekeepers received one queen of each and were asked to keep records on colony survival and honey production. The beekeepers did not know which queen was an Ankle Biter and which was not — it was a “blind” experiment. The beekeepers introduced the new queens in the summer of 2014, and by March 2015, 55% of Ankle Biter colonies were still alive, while only 22% of the non-Ankle Biter colonies were. Honey production was higher in Ankle Biter colonies: Ankle Biters produced an average of 52 pounds per colony, while non-Ankle Biters produced 11 pounds per colony. Varroa mite with missing legs. Photo credit: Bee Informed Partnership

Russian In 1997, the USDA-ARS program imported Russian bees and has continued breeding them for Varroa resistance. These bees have been naturally living with Varroa mites in eastern Russia for over 100 years and have developed some resistance through natural selection. USDA researchers studied the Russian bees in their own country before introducing them to the U.S. and found that they harbored lower levels of Varroa mites than U.S. colonies. In one study, untreated colonies in the U.S. had infestation rates of 33% and were collapsing from Parasitic Mite Syndrome, while untreated colonies of Russian bees in Russia only had infestation rates of 7% over the same time period. Trials in the U.S. echo these results: After three months without treatments, domestic colonies have a 17.3-fold increase in Varroa mite levels, while Russian colonies have only a 2.5-fold increase. Thus, Russian bees still exhibit a high degree of Varroa resistance, even when they are imported in the U.S. Furthermore, Russian bees in the U.S. produce equal or higher amounts of honey than domestic bees. Russian honey bees also demonstrate resistance to tracheal mites. Several traits of Russian bees better equip them to reduce Varroa mite levels. They have a high degree of VSH behavior, and they also cease brood production completely in times of food shortage (such as a dearth in the summer). This cease in brood production reduces the availability of brood cells for mites to reproduce inside, so their population decreases. After years of continuous selection in the U.S., Russian bees were made available in 2000. Since 2007, the Russian Honey Bee Breeders Association has bred and evaluated Russian stock available for purchase. As of 2017, 12 beekeepers are involved in maintaining and distributing this stock.

Grooming Behavior Grooming behavior is a type of social immunity that honey bees perform in the colony. Bees groom themselves or their nestmates to remove foreign debris, including Varroa mites. This behavior can help reduce Varroa mite levels in a colony. This effect can be more pronounced if the beekeeper also installs a screened bottom board with a sticky sheet beneath the colony. Mites that are groomed off the bees fall to the bottom of the hive and are unable to crawl back up to feed on other bees. The original host of the Varroa mite, *Apis cerana*, has more pronounced grooming behavior than *Apis mellifera*. It is thought that this behavior has contributed to *Apis cerana*'s ability to resist mites. *Apis cerana* workers regularly groom mites off of themselves. They have also been observed performing “grooming invitation dances” to stimulate their sisters to groom them. During this dance, a bee stands in one spot and vibrates its abdomen while intermittently grooming its body. Dr. Jennifer Tsuruda of the University of Tennessee is working to produce a stock of honey bees with increased grooming behavior to help reduce Varroa mite levels. Watch the video below (from Jennifer) to view two bees with mites on their bodies. The bee in the top frame does not exhibit the grooming behavior trait, while the bee in the lower frame does. Saskatraz

The Saskatraz breeding program started in Saskatchewan, Canada, in 2004. The goal was to develop a stock of bees that produced a lot of honey, overwintered well in cold climates, had a gentle

temperament, resisted brood diseases and viruses, and exhibited hygienic behavior. Colonies in this breeding program are tested every year for these traits and breeder queens are selected from the top performers. The program began commercial queen production in 2012. Saskatraz stock is available as either queen cells or pure Saskatraz queens from Saskatchewan, Canada, or as Saskatraz hybrids from Olivarez Honey Bees, Inc., in Northern California. For a list of reputable suppliers that sell pure VSH, Minnesota Hygienic, Ankle Biter, or Russian stock, [click here](#). How Do I Introduce New Stock to My Operation? After reading about the advancements in Varroa-resistant stocks, you may be eager to order some resistant queens, nucs, or packages for your own operation. If you only order a few queens to try in some of your colonies, place those colonies in a new yard. Having a colony with a Varroa-resistant colony near a Varroa-susceptible colony can diminish the effects. Even a colony that is better able to resist mites can become overwhelmed if the colony next door is infested and continuously introduces mites to it. Compare the resistant colonies to your other colonies in terms of Varroa mite levels and other traits in which you are interested; this will help you determine whether this stock is right for your operation. Incorporating resistant stock is only one part of IPM, and having resistant bees does not mean that you'll never need to treat your colonies again. Although these colonies generally have slower mite population growth, mites can still reach or surpass the treatment threshold over time. Always monitor your levels and intervene if necessary. Can I Select for Varroa Resistance in My Own Colonies? Yes! You can indeed begin selecting for more Varroa-resistant stock. Many beekeepers try to select for Varroa-resistant colonies by letting nature take its course. They don't treat their colonies, and every year lots of colonies die from mites. The few colonies that survive till spring may have some resistance. Queens can be raised from those survival colonies, and the process is repeated each year. The problem with this method is that your bees suffer, you risk spreading disease to your neighbors' colonies and the wild bees in the area, and you can waste a lot of money purchasing new bees to replace your losses. To avoid heartache while still working toward developing a resistant stock, you can select for Varroa resistance right in your own backyard. Dr. Meghan Milbrath from Michigan State University has developed a great method. In order to breed for any trait, you need to have more than one hive so that you can compare colonies. With this method, you also need to have a source for new queens during your first year. They can be from a local supplier, or perhaps a commercial stock that already has some degree of Varroa resistance. Once you are set up with two or more colonies, you are ready to start. Keep in mind this method works best if your apiary is isolated from other colonies that could potentially reintroduce Varroa mites to your colonies. Here are six steps to begin your breeding program: 1. Spread your colonies out in your yard and give each colony a unique landmark to minimize drift. 2. Monitor your mite levels using a reliable method each month and keep records on these levels. 3. Record any other behaviors that could influence mite populations (performing splits, queen supersedure, swarming, a dearth, robbing, etc.) 4. When a colony exceeds the treatment threshold, treat that colony and requeen it. This is a key step. Allowing mite levels to build up without treatments can create a situation that favors virulent Varroa mites instead of Varroa-resistant bees. Treating and requeening these colonies enables you to remove those nonresistant genetics from your apiary and prevent mite bombs from spreading to your other healthy colonies. This program requires very diligent monitoring and record keeping. Continue monitoring all colonies and treating individual colonies as needed. Prepare your colonies for winter at the end of the season. 5. The following spring, rear queens from the colonies with the lowest mite levels. In some

cases, you may have colonies that always remained below the treatment threshold. In other cases, all colonies may have required at least one treatment. In this instance, rear queens from the colonies that required the fewest treatments and had the lowest mite levels. 6. Repeat this process again during the second year. If you choose to requeen colonies in this year and following years, raise queens from your colonies with the lowest mite counts. Over the subsequent generations, you can continue to refine your stock.

WATCH

Genetic Stocks for Varroa Resistance Overview

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READ

Genetic Stocks for Varroa Resistance — Varroa Sensitive

Hygienic This stock was developed by Dr. John Harbo and Dr. Jeffrey Harris at the USDA Bee Breeding Lab in Baton Rouge, Louisiana. Varroa Sensitive Hygienic bees kill and remove pupae from cells that contain reproductive mites. Immature mites die because they desiccate quickly once the cells are opened; in this way, mite reproduction is interrupted and the growth of the mite population is slowed. If there are many bees in a colony that perform hygienic behavior, this can significantly lower the mite population. This stock was originally called SMR stock (suppressed mite reproduction); with further research, however, it was realized that the impairment of mite reproduction was due to the bees' so-called hygienic behavior, so the stock was renamed Varroa Sensitive Hygienic (VSH). The VSH stock was developed in 1997 and has had many generations to test and refine this trait. These tests show that VSH colonies have significantly slower mite population growth than non-VSH colonies. In one of these tests, VSH queens were exchanged for non-VSH queens in colonies. When VSH queens were introduced to colonies previously headed by non-VSH queens with high Varroa mites, the mite population decreased. When non-VSH queens were introduced to colonies previously headed by VSH queens with low Varroa mites, the mite population increased. "Pure VSH" queens refer to queens that have been mated with drones that also carry genetics for the VSH trait. Some commercial beekeepers purchase pure VSH queens and raise daughter queens from them, allowing these daughters to

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Genetic Stocks for Varroa Resistance — Minnesota

Hygienic Originally developed to help colonies resist and recover from American foulbrood and chalkbrood, the Minnesota Hygienic line of bees was developed by Dr. Marla Spivak in 1993. Colonies that display 95 to 100% hygienic behavior are used as breeder colonies. This stock is now also advertised as helping reduce Varroa mites. Tests show that this line is not quite as effective at lowering mite population as VSH lines are, but it is more effective than non-hygienic stocks. In this video, Dr. Marla Spivak, Distinguished McKnight University Professor of Apiculture and Social Insects at the University of Minnesota, discusses the Minnesota Hygienic line that she developed. Video Transcript When I moved to the University of Minnesota in 1993, I started studying hygienic behavior of honey bees. This is the ability, it's a genetic trait, and it's the ability of some bees to detect and remove diseased and mite-infested brood. For a diseased brood, the bees do it very, very quickly. They're able to detect that the larva or the pupa is diseased. But they, when they're handling it, the larva or pupa has not become infectious yet. The disease didn't reach the spore stage, so they can handle it without transmitting the disease throughout the colony. For mites, the bees detect that there is a pupa infested with a mite when the mite is reproducing, starts laying eggs. And so, it's not the mite they're detecting, it's something wrong with the pupae. And so, they open up the cell, remove the pupa, and it interrupts the mites' reproductive cycle. And so it limits mite population growth in the colony. So, it's good for diseases, American foulbrood and chalkbrood, and for Varroa mites. It helps reduce the mite and disease loads in the colony. We started selecting hygienic behavior using the freeze-killed brood test. So, this was either you cut a section of comb of pupae, you stick it in your freezer for 24 hours, and you replace it back where you found it in the brood nest very carefully. And then you put it back in the colony. And after 24 hours, you come and check the amount of the freeze-killed brood that was uncapped and removed. And colonies that remove the most diseased, the most freeze-killed brood in 24 hours, those that can do it the fastest, it's correlated with how quickly they would detect diseased brood and remove that from the nest. We moved from the sections of freeze-killed brood to using a liquid nitrogen assay. So, we freeze brood with liquid nitrogen, pouring the liquid nitrogen down a PVC tube. And it's easier for us, maybe not easier for small scale beekeepers, but we freeze-kill the brood

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WATCH

Genetic Stocks for Varroa Resistance — Grooming Behavior

Grooming behavior is a type of social immunity that honey bees perform in the colony. Bees groom themselves or their nestmates to remove foreign debris, including Varroa mites. This behavior can help reduce Varroa mite levels in a colony. This effect can be more pronounced if the beekeeper also installs a screened bottom board with a sticky sheet beneath the colony. Mites that are groomed off the bees fall to the bottom of the hive and are unable to crawl back up to feed on other bees. The original host of the Varroa mite, *Apis cerana*, has more pronounced grooming behavior than *Apis mellifera*. It is thought that this behavior has contributed to *Apis cerana*'s ability to resist mites. *Apis cerana* workers regularly groom mites off of themselves. They have also been observed performing “grooming invitation dances” to stimulate their sisters to groom them. During this dance, a bee stands in one spot and vibrates its abdomen while intermittently grooming its body. Dr. Jennifer Tsuruda of the University

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READ

How to Introduce New Stock and Select for Varroa

Resistance After reading about the advancements in Varroa-resistant stocks, you may be eager to order some resistant queens, nucs, or packages for your own operation. If you only order a few queens to try in some of your colonies, place those colonies in a new yard. Having a colony with a Varroa-resistant colony near a Varroa-susceptible colony can diminish the effects. Even a colony that is better able to resist mites can become overwhelmed if the colony next door is infested and continuously introduces mites to it. Compare the resistant colonies to your other colonies in terms of Varroa mite levels and other traits in which you are interested; this will help you determine whether this stock is right for your operation. Incorporating resistant stock is only one part of IPM, and having resistant bees does not mean that you'll never need to treat your colonies again. Although these colonies generally have slower mite population growth, mites can still reach or surpass the treatment threshold over time. Always monitor your levels and intervene if necessary. You can also begin selecting for more Varroa-resistant stock. Many beekeepers try to select for Varroa-resistant colonies by letting nature take its course. They don't treat their colonies, and every year lots of colonies die from mites. The few colonies that survive till spring may have some resistance. Queens can be raised from those survival colonies, and the process is repeated each year. The problem with this method is that your bees suffer, you risk spreading disease to your neighbors' colonies and the wild bees in the area, and you can waste a lot of money purchasing new bees to replace your losses. To avoid heartache while still working toward developing a resistant stock, you can select for Varroa resistance right in your own backyard. Dr. Meghan Milbrath from Michigan State University has developed a great method. In order to breed for any trait, you need to have more than one hive so that you can compare colonies. With this method, you also need to have a source for new queens during your first year. They can be from a local supplier, or perhaps a commercial stock that already has some degree of Varroa resistance. Once you are set up with two or more colonies, you are ready to start. Keep in mind this method works best if your apiary is isolated from other colonies that could potentially reintroduce Varroa mites to your colonies. Here are six steps to begin your breeding program: 1. Spread your colonies out in your yard and give each colony a unique landmark to minimize drift. 2. Monitor your mite levels using a reliable method each month and keep records on these levels. 3. Record any other behaviors that could influence mite populations (performing splits, queen supersedure, swarming, a dearth, robbing, etc.) 4. When a colony exceeds the treatment threshold, treat that colony and requeen it. This is a key step. Allowing mite levels to build up without treatments can create a situation that favors virulent Varroa mites instead of Varroa-resistant bees. Treating and requeening these colonies enables you to remove those nonresistant genetics from your apiary and prevent mite bombs from spreading to your other healthy colonies. This program requires very diligent monitoring and record keeping. Continue monitoring all colonies and treating individual colonies as needed. Prepare your colonies for winter at the end of the season. 5. The following spring, rear queens from the colonies with

the lowest mite levels. In some cases, you may have colonies that always remained below the treatment threshold. In other cases, all colonies may have required at least one treatment. In this instance, rear queens from the colonies that required the fewest treatments and had the lowest mite levels. 6. Repeat this process again during the second year. If you choose to requeen colonies in this year and following years, raise queens from your colonies with the lowest mite counts. Over the subsequent generations, you can continue to refine your stock.

READ

Breeding for Resistance in Your Own Operation

Some beekeepers really enjoy the stock they currently have but would like to incorporate some Varroa-resistant traits too. Great news - you can start selecting for Varroa resistance in your own operation. Many beekeepers try to select for Varroa-resistant colonies by letting nature take its course (a "survival of the fittest" philosophy). They don't treat their colonies, and every year lots of colonies die from mites. The few colonies that survive in spring may have some resistance. Queens can be raised from those survival colonies, and the process is repeated each year. There are many problems with this method: You risk spreading mites to your neighbors' colonies and feral colonies in the area. You risk spreading Varroa-transmitted viruses to wild bees in the area. You can waste a lot of money purchasing new bees to replace your losses. To avoid the heartache of losing bees and money, while still working toward developing a resistant stock, you can select for Varroa resistance right in your own backyard in a sustainable and responsible way. Dr. Meghan Milbrath from Michigan State University and Dr. Tom Seeley have both written about methods to select for mite resistance in a sustainable and responsible way. We'll outline a combination of their methods here. In order to breed for any trait, you need to have more than one hive so that you can compare colonies. The more colonies the better, as honey bee colonies can have a lot of natural variation to work with. With this method, you also need to have a source for new queens during your first year. They can be from swarms you caught in bait hives, a local supplier, or perhaps a commercial stock that already has some degree of Varroa resistance. Once you are set up with two or more colonies, you are ready to start. Keep in mind this method works best if your apiary is isolated from other colonies, as this limits the reintroduction of Varroa mites from nearby colonies. Here are six steps to begin your breeding program: 1. Spread your colonies out in your yard and give each colony a unique landmark to minimize drift. 2. Monitor your mite levels using a reliable method each month. Keep records on these levels. 3. Record any other behaviors that could influence mite populations (performing splits, queen supersedure, swarming, a dearth, robbing, etc.) 4. When a colony exceeds the treatment threshold, treat that colony and requeen it. If you don't want to use any chemicals whatsoever, you can also cull the colony with soapy water. This is a key step! Allowing mite levels to build up without treating or culling highly infested colonies can create a situation that favors virulent Varroa mites instead of Varroa-resistant bees. Treating and requeening these colonies enables you to 1) remove those non-resistant genetics from your apiary and 2) prevent mite bombs from spreading to your other healthy colonies. This program requires very diligent monitoring and record keeping. 5. Continue monitoring all colonies every month, treating and requeening or culling individual colonies that develop high mite levels as needed. Prepare your colonies for winter at the end of the season. 6. The following spring, rear queens from the colonies with the lowest mite levels. In some cases, you may have

colonies that always remained below the treatment threshold. These are excellent candidates for being breeders! In other cases, all colonies may have required at least one treatment. In this instance, rear queens from the colonies that required the fewest treatments and had the lowest mite levels. 7. Repeat this process again during the second year. If you choose to requeen colonies in this year and following years, raise queens from your colonies with the lowest mite counts. Over the subsequent generations, you can continue to refine your stock. If you continue this method for several years and your operation is isolated from other honey bee colonies, you will likely be able to have a sustainable treatment-free operation with colonies that convey some Varroa resistance. This more natural approach to beekeeping, sometimes called "Darwinian Beekeeping", is detailed nicely in Tom Seeley's Book "The Lives of Bees: The Untold Story of the Honey Bee in the Wild".

READ

Cultural Methods for Varroa Management Overview

■ Cultural methods are fairly simple non-chemical management practices that beekeepers use that take advantage of understanding honey bee and Varroa mite biology. Cultural controls take advantage of honey bee and Varroa mite understanding certain aspects of biology. A variety of different cultural methods have demonstrated varied success in reducing mite levels. None of these methods is effective at keeping levels below the economic threshold on their own, but combining multiple cultural methods as part of a greater Integrated Pest Management approach can be highly effective. We encourage beekeepers to incorporate as many cultural methods as they can, dictated by their beekeeping goals and their available time.

WATCH

Cultural Methods for Varroa Management — Drone Frame

How it works: One to two drone foundation frames are inserted into a hive at the edge of the brood nest (typically in the #2 or #3 position). The comb is built out and the queen lays drone eggs in the comb. After the eggs are laid, frames should then be removed between 10–24 days later, with an effort to remove the frame(s) with the maximum amount of capped cells on the frame. Frames should be frozen for 24 hours at 0°F (-18°C) to kill the drone brood and mites inside the cells. After freezing and thawing back to room temperature, the frames are reinserted into the colonies in the same locations. The bees clean out the cells and the process is repeated throughout spring and summer. After the first round, frames can easily be re-inserted and removed at approximately 24 day intervals, as it only takes a few days for the bees to clean the cells out and for the queen to start laying again. This method is most effective when the most drones are produced in colonies, from April until July in the Northeast US. Fewer drones are produced in late summer and autumn, making the drone frames an ineffective method later in the year. As drone production slows, the frames may be filled with honey instead. Simply scrape off the honey, rinse the frame, and use the frames again next spring. Colonies with low populations also do not produce many drones and may not use the drone frame. Watch this video to see how to use drone frames to help manage Varroa mite levels. Video Transcript This is a frame of drone comb foundation that I ordered from a bee supply company. And when you order these frames, they typically come in this bright green color, so it's very easy to pick out which one is the frame inside

your colony. And they're a plastic foundation with just a thin layer of beeswax. And we're comparing it to worker foundation. You can see that the cell sizes are much different. So, the worker foundation is much smaller than the drone foundation. And it's the size difference that's actually going to stimulate the queen to lay eggs that will develop into drones. So, to lay unfertilized eggs. So, now, let's go out into the colony and actually put this frame in. And we'll go through the process of using this for Varroa control with an integrated pest management approach. Okay. So, once we have our foundation, we're actually going to insert it into the deep that contains the largest portion of the brood nest. And, in this case, it's this bottom deep. And we can either put it in the number two position, which is what I'm doing here, or we can also put it in the number three position. You just like to have it on the periphery of the brood nest, if possible. So, when you first put in a frame of drone comb foundation, it might take a while for the colony to begin drawing it out with wax before the queen will lay an egg in there. And, remember, the timing starts from when the egg is laid. So, in the beginning, when you're just using foundation, you might need to check on it periodically to see when the queen begins laying an egg so that you can precisely time when you want to remove that frame. And once the egg is laid, it can be removed between 10 to 24 days, ideally more toward the end, just so that you're maximizing the amount of brood that's actually fully capped. When you're ready to remove your drone comb frame, it should look like this. This is ideal. Here, we have a frame that's just covered in drone pupa that are almost all capped. So, there's very few that have open larvae. And, in this way, we're actually maximizing the number of mites that we're killing. So, we're going to pop this frame directly into the freezer, leave it there overnight. The next day, let it thaw at room temperature. And we're ready to put it back in the hive and start over again. So, that's how to install drone comb foundation. It's a very quick, cost effective method to incorporate as part of your integrated pest management strategy for Varroa. Why it works: Mites are more likely to reproduce in drone brood, crawling into drone cells just before the cells are capped. Removing full frames of drone brood removes many reproducing mites, helping to drive down the population level. How effective is it? On their own, drone frames are not able to provide sufficient Varroa control for an entire season, but their use can delay the number of mite treatments needed. Several studies support drone frames as being an effective tool as part of an Integrated Pest Management approach for reducing mite levels. Research from Nicholas Calderone at the Dyce Lab at Cornell found that colonies from which drone frames are removed all season long have significantly lower levels of Varroa mites in autumn compared to colonies that do not receive any drone frame removal. Another study conducted by David Tarpy found similar results: drone frame removal kept Varroa mite populations below the treatment threshold until mid- to late summer, reducing the number of chemical treatments needed by the colonies. Throughout the entire study, the colonies from which drone frames were removed had significantly lower mites than colonies that did not. Studies in Europe find that using drone comb removal in combination with chemical treatments slows down the population growth of Varroa significantly more than colonies that received only chemical treatments. These studies also report there is no negative impact on colony development, colony strength, and honey production. Who shouldn't use it? If you rear queens and have drone-rearing colonies established for mating, this is not a good practice to use in those colonies, as a surplus of drones is needed for mating. Also, if you do not have the time or organization to remove the drone frames on schedule, you may inadvertently increase mite levels in the colony. Other considerations: Many beekeepers remove drone comb that has been constructed elsewhere in the colony. To do this, you can simply scrape off drone

brood with your hive tool while you are working your colonies. Colonies naturally tend to produce drone comb on the outside edges of the brood frames, making it easy to scrape off as you come across it. Unlike using drone frames, this method is opportunistic and is not timed to maximize the removal of mites.

WATCH

Cultural Methods for Varroa Management — Screened

Bottom Board How it works: A screened bottom board is installed below the lowest hive body, allowing it to sit beneath the hive in place of a solid bottom board. Screened bottom boards can be accompanied with a sticky board. If a sticky board is used, it must be cleaned or replaced regularly, as debris builds up. In the Northeast US, screened bottom boards are typically in place from May to October and then replaced with solid bottom boards for the winter months. In warmer regions, screened bottom boards may remain in place all year long. Screened bottom boards also help reduce debris build-up (which can host wax moth larvae, ants, and small hive beetles) and can also help reduce humidity and mold by improving ventilation. Watch this video to see how to use screened bottom boards. Video Transcript This is a screened bottom board that I purchased through a bee equipment supply company. I painted mine blue. This is going to sit directly underneath your hive. So, a hive body will sit on top of this, in place of a bottom board. As you can see, there's a screen on the top here that allows debris and mites to fall through. And, if you happen to have a sticky board attached underneath here, then it's going to stick to that sticky board so that the mites aren't able to crawl back up onto the bees. If we look underneath this board, we can see that it's not solid. So, I set mine on top of a regular bottom board, which I'll show you in a second, but you can also just put it directly on top of the stand, depending on what your highest stand is. On the back of the bottom board, we have this little tray that pulls out, and this is where you can attach your sticky board. So, you can purchase a sticky board from an equipment supply store or, if you are like me, you can make your own using card stock. And you simply draw a grid on there to make it easier for you when you're counting the mites. And you need to apply something that's going to make the mites stick to this. So, you can either use petroleum jelly or you can use cooking spray, and spray it on top of it. So, from there, you're going to put the sheet on top of your tray, and you want to make sure that you're using card stock that's long enough to take up most of the space on this tray. And then, from there, you're going to be able to insert it into the screened bottom board pretty easily, so that you're not having to open your colony or disturb your colony in any way. So, let's take a look at this outside and get an idea of how to use a screened bottom board as part of your Varroa mite management plan. So, here we have a screened bottom board that's actually installed beneath a hive. It's this top board, here, that's painted blue, and it's actually sitting on top of a bottom board that's turned the other way, that's also painted blue. And I just needed to put it on top of a solid surface because it's sitting on a cinder block, and otherwise, the cinder block would directly touch the tray that's in the screened bottom board. So, right now it's directly beneath these hive bodies so that when bees are landing to enter the hive, they're landing on top of the screened bottom board. And then the tray with the sticky sheet on it, we're actually going to install it into the back. So, we're at the back of the hive now and we have our sticky sheet that's thoroughly coated in petroleum jelly so that it will trap any mites that happen to land on it. And we have it sitting on top of this tray, which we're now just going to insert into the back of the screened bottom board. Some beekeepers will

leave these sticky trays in for about a month, and then after that, they'll swap them out for a new one. After a month, there's a lot of debris in there that might actually completely coat the surface, so that it's not really effectively capturing any other mites that are falling. But other beekeepers will actually use this to measure the number of mites that are dropping in a particular period of time, usually about 24 hours. And so what they'll do is they'll insert the tray, after 24 hours, they'll come back and just see how many mites have naturally dropped. Another way that you can use this is when you're putting on a mite treatment and you can insert the tray with a sticky sheet and you can get an idea of how many mites are actually dropping after you've put a treatment on. And while I can't really gauge quantitatively how effective that treatment is, it can give you an idea if lots of mites are dropping dead, then it could be working to some degree. But you definitely want to follow it up with a sugar shake, or an alcohol wash, or an ether roll just to actually measure what your Varroa levels are afterward. Let's come back after about 24 hours and take a look and see what kinds of things we'll find on the sticky sheet. If you're checking your screens after 24 hours, they usually don't have too much debris on them. And we can start looking through each of these squares to count how many mites we're seeing. So, one of the things that takes a little bit of learning when using these sticky boards is being able to decipher mites from other debris that can collect on these boards. So, sometimes you'll see little bits of wax flecks. Sometimes you'll see actual wax scales that have fallen off of the bees' bodies before they've been able to start shaping them. Sometimes you see bits of pollen or little bits of bees' bodies from dead bees. You might also see legs and wings. And you can also expect to sometimes see wax moth cocoons, especially if you have a big wax moth infestation. So, Varroa mites, once you train your eye to be able to recognize them, you can find them fairly easily as you're looking square by square on these sheets. So, in this particular instance, over a 24 hour period, we actually caught two Varroa mites. So, there's one mite found here, and then another mite found here. So, let's actually take a closer look at Varroa mites on sticky boards and get kind of used to seeing what they look like. So, in contrast with the board on the right, which is the one that I just pulled out of the hive, this is how much debris has built up in about 24 hours. We can look at the amount of debris that builds up after about one month. You can see it's considerably more, and it would be much harder to sift through and count the mites. And we would expect that there would be several more mites on this board that's been in for a much longer period of time, compared to the one that's only been in for 24 hours. So, after you have observed the mites on that sticky sheet, you can simply throw it away, add a new one, and repeat the process over and over again, so long as you have a screened bottom board beneath your hive. And there you have it, that is the step by step procedure of how to use a sticky board in conjunction with the screened bottom board as one part of your integrated pest management approach for Varroa mites. Why it works: Mites, particularly those that are chewed/injured by bees, may fall to the bottom of the hive. If they land on a solid bottom board, they can crawl back up onto the bees, but if they fall on a screened bottom board, they may fall out of the hive entirely and some may be unable to crawl back up. With sticky boards in place, mites will stick to the boards and die. How effective is it? Screened bottom boards are minimally effective in reducing mite populations and should not be used as the sole method of Varroa control. With them, mite populations may be reduced by up to 10%, but surveys conducted by the Bee Informed Partnership report no advantage of their use. Their ability to reduce mite levels likely varies seasonally. More mites naturally fall during broodless periods. Who shouldn't use it? Screened bottom boards can be used by beekeepers of all operation, sizes, and goals, but

cannot be used with top bar hives. Studies show there is no difference in weight gain or frames of bees between colonies with screened bottom boards versus those with solid bottom boards; in other words, screened bottom boards do not negatively impact honey bee colony growth.

READ

Cultural Methods for Varroa Management — Small Hives

How it works: Colonies are kept in a hive the size of one deep hive body with only 1-2 honey supers added during a honey flow. Why it works: Small colonies do not produce a lot of brood, so mite populations in a colony with 10,000–20,000 bees are not able to grow as large they would in a colony with 40,000–50,000 bees. Small colonies also swarm more frequently (usually at least once a year), taking about 30% of the mites with them as they leave and creating a break in brood rearing in both the parent colony and the new colony. How effective is it? Cornell bee researcher Tom Seeley has shown that colonies with brood nests housed in one deep hive body have significantly lower mite population growth compared to colonies kept in four deep hive bodies. Who shouldn't use it? Beekeepers whose main goal is honey production would not benefit from using small colonies, as they produce little to no surplus honey. Beekeepers who live in residential areas and are unable to catch the swarms should not use this method.

READ

Cultural Methods for Varroa Management — Colony Spacing

How it works: Colonies are spaced apart and provided with unique landmarks. Hive entrances face in different directions. Why it works: In most bee yards, colonies are crowded together. Typical commercial yards contain anywhere between 20–50 colonies, placed on pallets, in an acre or less. This promotes drifting and robbing among colonies, with bees transferring Varroa mites among different colonies. In nature, on the other hand, colonies are spaced about half a mile apart. Having colonies spread apart reduces drifting and robbing. Spacing colonies apart, using landmarks, and changing the directions of entrances helps foragers identify their home hive. How effective is it? Colonies spaced apart have reduced Varroa mite levels, reduced Varroa mite transmission, and increased overwinter survivorship. Tom Seeley has compared overwinter survivorship between colonies crowded together (8 colonies placed side-by-side on pallets) and colonies spread apart (9 colonies placed 30–80 yards away from each other). None of the colonies received any mite treatments. In his experiment, all 8 colonies that were crowded together died by mid-December, while 5 out of 9 colonies spaced apart survived the winter. The five surviving colonies had low mite infestation levels the following April (0.47 mites/100 bees). Similar experiments conducted by other researchers echo these results. Apiaries with colonies spaced at least 100 yards apart have significantly lower Varroa mite transmission from drift, compared to colonies that are next to each other or even 10 yards apart. Who shouldn't use it? Beekeepers that manage hundreds or thousands of colonies may not have the space available to place colonies far apart. Similarly, backyard beekeepers may not have the available space, either. Colonies placed in agricultural land for pollination services usually need to be placed on pallets for ease of transportation.

READ

Cultural Methods for Varroa Management — Brood

Interruption How it works: There are a variety of ways to introduce a brood interruption (also known as a brood break) into a colony: **Allowing the colony to swarm:** The remaining colony experiences a brood break as it waits for a new queen to emerge, mate, and begin laying. This brood break can last several weeks, especially if there are afterswarms. On the other hand, if you capture the swarm and put it in a hive where there is drawn comb available, it will take at least 9 days for capped brood to be present (if the queen begins laying right away). If no drawn comb is available, the brood break will be longer.

Performing splits: Strong colonies can be split into two smaller colonies. The colony without the original queen experiences a brood break. No new eggs will be laid for several weeks if the colony rears its own queen. In this case, there will be approximately 35 days without any capped brood present in the colony. Some beekeepers remove the queen entirely and have both colonies rear new queens.

Requeening: Requeening a colony can offer a small break in the brood cycle. This may not be enough to significantly lower the mite population if the newly introduced queen is mated and begins laying in a few days. In this case, the effects on the Varroa population are maximized if the colony is requeened with hygienic or Russian stock. The brood break lasts much longer if the colony rears its own queen (allowing approximately 35 days without any capped brood), or if a queen cell is introduced (allowing approximately 20 days with no capped brood). **Temporarily cage the queen:** Cage the queen for the duration of a brood cycle (25 days) to cause a brood break. If the queen is released after 25 days and begins laying eggs right away, this gives the colony approximately 10 days without any capped brood present. Release the queen after this time so that egg laying can resume. Keeping the caged queen in the colony allows the workers to detect the queen pheromone and prevents them from rearing a new queen. Use a cage where workers are still able to feed the queen.

Why it works: Mites require capped honey bee brood in order to reproduce. Without brood present in the colony, mite population growth momentarily ceases. Mites present on adult bees are also at risk to be chewed or fall off. Since adult mites can live for several months, the brood break is not a long enough period for the entire mite population to die off, but it can be reduced. Brood breaks are most effective when the queen is not permitted to lay for an entire brood period (at least 24 days). If a queen lays eggs the same day the brood interruption begins, 24 days ensures that the last eggs that were laid have had time to emerge as adults before she begins laying again. Longer brood breaks are more effective at reducing mite levels than shorter brood breaks. How effective is it? Brood interruption is mildly to moderately effective in reducing Varroa mite populations when used on its own. Studies where queens are caged inside colonies for 25 days find between 11%–24% mite mortality. One study found that mite levels before the queen was caged averaged 4.36 mites/100 bees, while after caging the queen for the duration of a brood cycle the mite levels averaged 3.24 mites/100 bees. Brood interruption can be even more effective when beekeepers exploit the broodless period to apply a chemical mite treatment. As most mite treatments only target phoretic mites, a brood break ensures that all the mites in the colony must be on the adults, providing an ideal opportunity to treat. A study published in 2016 evaluated the efficacy of the mite treatment Apiguard in combination with queen caging. Apiguard used alone killed 76% of mites in the colony and caging the queen for three weeks with no other treatment killed 40% of mites in the colony. But the combination of caging the queen and treating during the brood break killed 97% of mites in the colony. The researchers did not observe elevated rates of queen death. Similar studies with oxalic acid find Varroa mites are significantly reduced when application coincides with a

brood break from caging the queen. Who shouldn't use it? Longer brood break periods may reduce the population size of the colony, which may have negative implications on honey production. Beekeepers who live in residential areas and are unable or unwilling to catch the swarms that are cast should not use swarming as their brood break method of choice. Other considerations: Brood breaks are best conducted after the honey crop is removed and before the colony begins rearing winter bees (ideally around late July in New York). Brood breaks performed before the honey flow may reduce the overall forager population, while brood breaks conducted in early autumn may interrupt the colonies' production of winter bees. Also, performing splits in late summer or early fall might result in colonies too small to overwinter successfully. Finding queens to cage can be quite time consuming and unfeasible for large-scale operations. Some large-scale operations are having success by temporarily moving colonies into cold storage facilities to naturally induce brood breaks. Lastly, although there are benefits to using this method, it must be noted that forcing the entire mite population onto adult honey bees likely presents a risk to bee health, as their feeding damages fat bodies and transmits viruses. To reduce this risk and to maximize mite death, beekeepers can apply a Varroa treatment as soon as colonies become completely broodless.

READ

Ineffective Cultural Methods for Varroa Management

■ Varroa mites are more likely to reproduce in drone cells. While this is using small celled foundation does due to the developmental time of not reduce Varroa mite levels. Drones, some people thought that the size of drone cells may play a role as well. Drone cells are larger than worker cells. In small cell foundation, it was thought that mites may have difficulty reproducing, developing, and moving within the cell because there is limited space. For this reason, many people believe that using small cell foundation can reduce the numbers of Varroa mites in a colony... but does it really work? Past experiments have had conflicting results, but several more thorough studies published since 2001 have found that using small cell brood comb does not reduce Varroa infestations, and that mites enter small cells and regular-sized cells at the same rate. For this reason, small cell foundation should not be included as part of your IPM strategy to combat mites.

WATCH

Cultural Management Practices and Wild Honey Bee

Colonies ■ Tom Seeley has spent decades studying the wild honey bee colonies living in the Arnot Forest. He and his with Varroa mites have provided colleagues have observed how the insights into the cultural practices colonies have changed since the that beekeepers can use to better introduction of Varroa. Some of those manage mites in their own observations can help us manage our colonies in healthier ways. Watch this video to learn more. Video Transcript At this point, I wondered how are these wild colonies able to persist with Varroa? I mean, nobody's putting chemicals in them, helping them out. They're all on their own. Well, this was an interesting mystery. And we don't have it fully solved, but we're learning more every year. Part of the story is, of their survival, is that they have good genes. And part of their survival is that they have a good lifestyle, or a good ecology. Let's look first at their good gene side of the story. I don't have time to go through it in detail, but I'll emphasize one thing that is interesting about the genetic changes

that have occurred in these wild colonies as a result of natural selection, created by the Varroa, is that the wild bees now are smaller than they used to be. Back in 1978, if you measure their head width, for example, or the space between their wings, and you compared those 1978 bees with the 20 bees we collect from the wild in 2011, the bees are markedly smaller now than they were before. Now, why would that help the bees resist Varroa? It's because that smaller body size is probably a function of the bees having a shorter development time. We're not sure about that, but that's our working, current understanding. And by shaving a day off the development time, which is what they have done, that gives the mites one day less of the 21 days of development. It takes a day off the, a day less for the mites to mature before the bee climbs out of the cell. And that ends the mite's reproduction. So that's part of the story. Genetic changes changing development time, changing body size and other traits. The other side of the story is the lifestyle, or ecology of these wild colonies. And one thing I want to stress here is that a huge difference between how honey bees live in the wild versus how they live in our bee yards is the spacing of the colonies. We crowd colonies in our apiaries, whereas in the forest, they're widely spaced. On average, in the woods, they are a half mile apart between the colonies. And why is that so important for the bees' survival with the Varroa? It's important because when we put our colonies close together in an apiary, we're creating a situation where it's very easy for Varroa to get from one colony to the next. This can occur when worker bees drift carrying Varroa from one hive to the next. It can occur when bees rob from one colony. They can pull the Varroa out of a dying colony, for example, and bring those Varroa home with them. And it can happen also when beekeepers like ourselves move combs between colonies. Now, so Varroa can move from one colony to another in an apiary easily. And that means that in an apiary, the Varroa can be actually very virulent and very damaging to colonies because they can kill this colony, and by the time they've killed this colony, and this colony is collapsing, the Varroa, through drifting and the other mechanisms, has already moved off into the next colony. So, it doesn't kill, the Varroa doesn't kill itself if it kills its host. In the wild, it's a completely different situation, with these colonies a half a mile apart, living up in these trees in dense forest. If Varroa kills its colony, it kills itself because there's no good way for Varroa to get to another colony living in a tree that's half a mile away. And what does that mean? It means that in the wild, there's selection on Varroa itself to not be virulent, to contain its reproduction so it doesn't kill its host. So, colony spacing is a big part of the ecology that has changed the selection on the mites. And the mites have evolved their virulence. But it also, the colony spacing has another effect. It also means that there's just less drifting of the bees themselves from one colony to the next. There's probably zero drifting in the wild, whereas there's abundant drifting in an apiary. And we've measured this level of drift recently by doing a simple experiment. We set up two groups of 12 colonies, 12 colonies in each group. In one group, the colonies are crowded together like in a standard apiary. In the other, they were scattered. They weren't as widely spaced as the bee trees are in the wild. They were only 30 to 80 meters apart, or 30 to 80 yards apart. But when we measure the amount of drift of the drones between colonies in the crowded group versus the dispersed group, it was like night and day. The drones in the crowded group, if you looked at, we used genetic markers to measure this, but we found that one third of the drones going into a hive in the apiary setting, were produced from a different colony, from one of the other hives in that apiary. One third of the drones, that's a lot of drifting, a lot of mixing, and thus, a lot of opportunities for Varroa to get from one hive to another. In the scattered situation, it was 0% drift of the drones. All of the drones going into each hive came from that hive. So, it just shows you how

different it can be. And it also suggests that we beekeepers might want to think about spacing our hives out. They don't have to be a half mile apart, but if we can get them 30 yards apart with some trees and things like that in between, that can reduce the opportunities for Varroa to get around. There's another thing about colonies living in the wild that is helping them deal with Varroa, and it is that they're living in relatively small nest cavities. They're only about the size of one deep hive body. What does that mean? Why does that help? That helps because these colonies living in the wild swarm every year. And that helps them deal with Varroa because when a colony swarms, as the swarm leaves, it takes away about a third of the mites in the parent colony. So, it knocks down by one third the population of mites in a colony. But it also does more, because it creates a period of time in the colony when there's no sealed brood. And this period can be quite long. It can be several weeks long if there's after swarming going on. And when there's no sealed brood in a colony, the mites have to be out on the bees and on the combs. They can't be hiding away in the sealed cells. And when the mites are out on the bees and the combs, they're vulnerable to being groomed and chewed by the bees. And that knocks down their population as well. And we've studied this, again, with experiments. We've set up experiments. We have two groups of colonies. In one group, the colonies are living in hives managed for honey production. Four deep hive bodies. And those colonies have big brood nests and they make lots of honey, and they produce lots of mites. And then, the other treatment is colonies kept in one deep hive body. And those colonies don't make any honey, but they make lots of swarms, and they don't make many mites. And in the big colonies, it's what you would expect. The mite populations just grow exponentially and they kill the colonies. The colonies living in the small hives, the mite population stay under control and the colonies survive. So, these are some of the things that we're learning about how bees live in the wild and how we can use that, use what we're learning about how bees live in the wild to tune up our beekeeping practices if we want to keep bees without having to treat them heavily with acaricides to control the Varroa mites.

READ

Chemical Controls for Varroa Management Overview

■ When monitoring reveals that mite Key Points levels are at or above the treatment threshold, it is a good time to apply a treatment registered in NYS to chemical control. There are currently eight chemical treatments registered in NYS for Varroa control. Become familiar with the treatments for Varroa control in New York State. available in your state or province. Beekeepers in other states or countries Chemicals can be broken down should seek information about the into three classes: synthetic treatments registered in their regions. chemicals, essential oils, and The approved treatments in New York organic acids. Each chemical has include a variety of chemicals that can specific requirements be used under different circumstances, (temperature, frames of bees, depending on the temperature, the presence of honey supers, etc.) presence of brood, and the presence of that must be fulfilled in order to honey supers. Broadly, the chemicals use the product safely. can be broken down into two Always monitor colonies before categories: synthetic chemicals (also and after chemical treatments to known as hard chemicals) and natural evaluate the treatment's efficacy. chemicals (also known as soft chemicals). Natural chemicals can further be categorized as organic acids and essential oils. The organic acids and essential oils used for Varroa control are naturally occurring in plants. For instance, thymol comes from thyme plants and oxalic acid is produced

in rhubarb. The concentrations in the mite treatments, however, are much higher than what naturally occurs in plants. There are eight registered chemical treatments for Varroa in New York State: Apiguard® (active ingredient: thymol) Api Life Var® (active ingredients: thymol, eucalyptus oil, and menthol) Mite Away Quick Strips® and the similar new product Formic Pro® (active ingredient: formic acid) Oxalic acid (active ingredient: oxalic acid dihydrate) HopGuard III® (active ingredient: hops beta acids) Apivar® (active ingredient: amitraz). Note that widespread resistance to amitraz was found in the USA in 2024/2025. Apistan® (active ingredient: fluvalinate). We do not recommend using this product. CheckMite+® (active ingredient: coumaphos). We do not recommend using this product. The illustration below depicts how they are categorized by active ingredient.

WATCH

Chemical Controls for Varroa Management

Here, Emma provides directions on how to use many of the various chemical controls for Varroa mite management. Apivar The active ingredient in Apivar is amitraz. It has been the backbone of Varroa destructor control for over two decades but in 2019, the USDA documented amitraz resistance in New York State and other states in America. Then, in 2024/2025, nearly all mites screened by the USDA from USA-based collapsing colonies were resistant to amitraz, thus suggesting widespread resistance to this miticide, and this in turn is believed to be a major driver of the exceptionally high honey bee colony losses in the USA for 2024/2025. Signs of resistance have also been reported worldwide. Video Transcript The active ingredient and Apivar is Amitraz, which is a formamidine insecticide. The strips contain 3.33% Amitraz. The insecticide is released slowly from the strip. So, as bees touch the strip of Apivar, small amounts of insecticide transfer to their bodies. As bees move and interact within the hive, they spread the insecticide. And when it comes in contact with mites, it impairs nerve functioning in the mite, and the mite is killed. Before you apply Apivar, carefully read all instructions to ensure proper application and safety. Wear a long sleeve shirt, long pants, shoes and socks, and chemical-resistant gloves. This product is not temperature dependent. Apivar is best used in early spring before the honey flow has begun, and in autumn, after honey supers are removed. Don't use this treatment during winter or when honey supers are on. To use Apivar, first remove the strips from the pouch, and keep any unused strips in the original packaging. Hang the strips between the frames so that bees can come in. To hang the strips, insert a toothpick or a thin nail in the small hole in the top of the strip, and lay the ends of the toothpick on the frames. Make sure that you remove all honey supers that are on the hive first. Apivar cannot be used with honey supers, as residues can be found in the honey. The number of strips that you insert depends on the number of frames of bees in the hive. Use one strip for every five frames of bees, use two for every 6-10 frames of bees, use three strips for every 11-15 frames of bees, and use four strips for 16 frames or more of bees. Do not use more than four strips in one hive, and do not place more than two strips in each brood chamber. Place the strips within the bee cluster in the brood box. And if using two strips, place the two strips apart at least two frames from each other. Leave the strips in the hive for 42 days. If, after 42 days, you find that the bee cluster has moved and the bees are no longer in contact with the strips, reposition the strips into the bee cluster, and leave them there for an additional 14 days. Do not leave the strips in for longer than 56 days. After removing the treatment, wait two weeks before adding honey supers. Don't reuse the strips, and dispose of them according to the instructions on the label. Also, don't use Apivar more than twice a year. Apistan Video

Transcript The active ingredient in Apistan is tau-fluvalinate, which is a pyrethroid insecticide. The strip contains 10.25% fluvalinate. The insecticide is released slowly from the strip, and as bees touch the strip of Apistan, small amounts of insecticide transfer to their bodies. As bees move and interact within the hive, they spread the insecticide. When it comes in contact with mites, it impairs nerve functioning in the mites and the mite is killed. Before you apply Apistan, carefully read all instructions to ensure proper application and safety. Wear a long sleeve shirt, long pants, shoes and socks, and chemical-resistant gloves. Use this product when outside temperatures are warmer than 50 degrees Fahrenheit, or 10 degrees Celsius. This product is best used in early spring before the honey flow has begun, and in autumn after honey supers are removed. Don't use this treatment during winter or when honey supers are present. So, let's go through the steps of applying Apistan. First, we're going to remove Apistan strips from the pouch, and keep any unused strips in the original packaging. Remove honey supers that are on the hive. Apistan cannot be used when honey supers are on, as residues can accumulate in the honey. The strips will be hung between frames so that bees can come in contact with both sides of the strip. The bees will come in contact with the chemical and will pass it to other bees as they rub against each other in the hive. You can bend the top flaps of the strip to hang them effectively between the frames. After using this treatment, do not consume any honey or beeswax from the brood nest. The strips will begin wet, and they'll dry once they're placed inside of the hive. While they're wet, they can kill bees that come in direct contact with them. So, to minimize these risks, this treatment is best applied during times of reduced activity. So, this could be early in the morning, in the evening, or on a cloudy day. The number of strips that you insert depends on the number of frames of bees in the hive. So, you use one strip for every five frames, or fewer, of bees. This means a nuc or a weak colony with five or fewer frames of bees would receive one strip. A colony that has 10 frames of bees in a deep hive body would receive two strips. A colony with a brood chamber spanning two deep hive bodies and 20 frames of bees would receive four strips. Hang the strips so that they're within two combs of the edge of the bee cluster. The strips have to be in contact with the brood nest bees for the entire treatment period. For colonies in one deep hive body, hang the strips in the center of the brood nest. And for colonies with two deep hive bodies, hang the strips in alternating corners of the bee cluster. Leave the strips in for 42-56 days. This translates from 6-8 weeks. Do not leave the strips in for longer than 56 days. Once the treatment has finished, honey supers can be added immediately after. Don't re-use the strips, and make sure to dispose of any waste materials according to the label. There have been cases of resistance to Apistan. To minimize resistance in your colonies, rotate this treatment with other chemical treatments, and use an integrated pest management approach that includes cultural and genetic controls as well. Make sure to monitor for mites after the treatment is over to ensure the treatment was effective.

Api Guard Video Transcript The active ingredient and Apiguard is Thymol, an essential oil that's produced by thyme plants. Apiguard contains 25% thymol. The treatment comes in a gel formulation that provides a slow, controlled release of the vapors. In addition to the vapors dispersing throughout the hive, bees also pick up the gel to clean it out from the hive. And in doing so, they help spread it throughout the colony. Apiguard comes in two sizes. You can buy the single trays that we're showing in this video, or you can purchase it in a big tub that treats at least 30 colonies. When you purchase a big tub of it, you'll scoop the right amount onto foil trays that you place on the top bars. But in the single application method, no measuring is required. Before handling Apiguard, make sure to wear safety goggles, a long sleeve shirt and pants, socks and shoes, and

chemical-resistant gloves. And before applying Apiguard, remove all of your honey supers that are present. Apiguard can not be used during a honey flow. Use Apiguard in early spring or in autumn when the temperature requirements are met. So, it needs to be between 60 and 105 degrees Fahrenheit, or 16 to 41 degrees Celsius. Apply Apiguard by first opening the foil lid from the tray and peel it back. Leave one corner of the foil lid attached to the tray. Add a rim spacer or an empty super to the hives so that there is free space, at least a quarter of an inch, between the top of the tray and the inner cover. This will allow the vapors to dissipate in the hive and it will allow the bees to access the gel. Close the hive. After two weeks replace the tray with a new one. Leave the second tray in the hive for an additional two weeks. If there's still gel in the tray after that, leave the tray in the hive until it's empty. Remove the tray before putting honey supers on. If you purchase the tub of Apiguard and are measuring the dose to give your bees, be careful that you don't overdose the colony. Thymol overdoses can cause increased mortality, absconding, and an increase in defensive behavior. In the event that the colony is overdosed, remove the excess product from the colony promptly. After you're finished treating your colony, dispose of any packaging according to the label. [Api Life Var Video Transcript](#) Api Life Var is an essential oil treatment that contains 74% thymol, 16% eucalyptus oil, and 4.7% menthol. The treatment comes in the form of a tablet, and the essential oil vapors will evaporate from the tablet and dissipate throughout the colony. Before applying this treatment, wear long sleeves, long pants, shoes and socks, safety goggles, chemical-resistant gloves, and a filtering respirator. Api Life Var is also combustible, so keep it away from your smoker. This treatment cannot be used when honey supers are in place to prevent excessive residues in honey and wax. After the treatment is ended, wait 30 days before applying supers. And don't apply this treatment when bees are robbing the colony. The best time to use this treatment is in early spring and autumn, when temperature requirements are met. So, the temperature must be between 65 and 85 degrees Fahrenheit, or 18 and 30 degrees Celsius. Don't use this treatment more than two times a year. This treatment can only be used on Langstroth colonies that contain 8 to 20 frames of bees. All colonies that fulfill this requirement receive the same treatment dose. So, each package contains two tablets. First, open the package and remove one tablet. Break it into four even-sized pieces. The tablets break really easily and are quite fragile, so be careful handling them. Place the four tablets on the top frames of the brood nest at the edges of the brood nest. Don't put the tablets directly on top of the brood nest. Close the hive and keep the tablets in there for 7 to 10 days. After this time, replace the tablets with four fresh pieces and leave them in for another 7 to 10 days. After this time, replace the tablets again with another four fresh pieces and leave them in there for 12 days. After 12 days have passed, remove the tablets and dispose of them according to the label. So, in this way, the treatment of one colony requires one and a half packages of Api Life Var and four visits to the hive. Do not reuse tablets. Hop Guard II (Now HopGuard III) HopGuard III is a next-generation, more stable, and user-friendly formulation of the same natural hop beta acids used in HopGuard II. While the active ingredient is the same, the formulation, delivery system, and performance are substantially improved in HopGuard III thus offering greater consistency, less brood stress, and less cleaner handling. [Video Transcript](#) The active ingredient and Hop Guard II is potassium salt of hops beta acids. This ingredient comes from hops, the same flowers that are used to make beer. The treatment strips contain 16% hops beta acids. The strips are covered in liquid, and it's important not to remove any liquid from the strips before putting them in the colony. When applying this treatment, wear long shirt sleeves and pants, socks and shoes, safety goggles, and waterproof

gloves. Hop Guard II can be used a maximum of three times per year. It can also be used when honey supers are present. But it's less effective when brood is present. So, if you're using this treatment during the honey flow when brood is present, you may need to follow up with another treatment shortly afterward if your mite levels remain above the treatment threshold. Always monitor to check your efficacy. It is safe to consume honey from honey supers that are present during treatment application, but do not consume honey from the brood nest. This treatment can be used on colonies that have at least five frames of bees. Apply one strip per five deep frames of bees in the brood chamber. If the brood chamber has four or fewer frames of bees, do not use Hop Guard. If the brood chamber has five to nine frames of bees, use one strip. If it has 10 full frames of bees, use two strips. Bees in single brood boxes will use a maximum of two strips, while bees in double brood boxes will use a maximum of four strips. If using two strips, place them on different frames at least four inches apart from each other. And do not place any strips in honey supers. Once the strips are in, leave them in the colony for 30 days. The bees may chew up the strips in time, but if any remain after 30 days, make sure to remove them. Hop Guard II can also be used to treat package bees. In this case, beekeepers can cut strips in half and staple three half strips from the top of the package so that they're dangling inside the package and bees can come in contact with them. Keep the bees in contact with strips for 48 hours before installing them into a hive. Three half strips can be used for both two pound and three pound packages of bees.

Oxalic Acid Note that while there are no strict temperature requirements for using the fumigation method for oxalic acid, the product is most effective when used at temperatures above freezing (0°C/32°F). At temperatures below freezing, the tight clustering of the honey bees does not easily permit vapors to travel throughout the cluster and contact all the bees.

Video Transcript The active ingredient in oxalic acid is oxalic acid dihydrate. And this is an organic acid that naturally occurs in some plants, like rhubarb. The oxalic acid that you purchase for mite control contains 97 percent oxalic acid dihydrate. Trace levels of oxalic acid are naturally found in honey, but at higher levels, around 400-900 parts per million, it can alter the taste of honey. The current applications of oxalic acid cannot be used with honey supers on. Oxalic acid treatments can only be applied a maximum of two times a year. Do not give a colony oxalic acid treatments more than two times a year, as it can be harmful to the colony. The efficacy of this product is not temperature dependent. This treatment is only effective when there's no brood in the colony. This could be in early spring, late fall, or winter here in the Northeast, or if you're creating a broodless period, for example, by caging the queen. Do not use in colonies that are overwintered indoors, as the product can only be applied outside. Oxalic acid is approved for treating mites in three ways. First, it can be fumigated into a colony. Second, it can be dribbled in a colony. And third, it can be sprayed onto a package of bees. Let's first go over the fumigation method. When oxalic acid is fumigated into a colony, it begins as a solid, but it's heated up to become a gas. While you're applying the treatment, the gas is coating all the bees, phoretic mites, and the hive interior, so that everything will possess a thin film of oxalic acid crystals. With the appropriate dose, the oxalic acid film is not harmful to honey bees, but it is fatal to phoretic mites. When applying this miticide, wear a long sleeve shirt and long pants, shoes and socks, gloves, goggles, and most importantly, a half face respirator with a cartridge or particulate filter. It is very important not to breathe in the vapors, as it's very poisonous. And when inhaled, it can damage your lungs and other organs. Also, don't stand close to the hive while treatment is in progress, and take care to avoid any vapors that are released from the entrance. Do not use when honey supers are on the hive, as it can

contaminate honey. After treating a colony with oxalic acid, wait 14 days before adding honey supers. Close all upper hive entrances and seal any equipment cracks with tape to prevent oxalic acid vapor from escaping the hive. If you have a screen bottom board on your hive, replace it with a solid bottom board. Make sure to follow the instructions on your oxalic acid vaporizer. In this video, we're using the Varro vaporizer, but your vaporizer might have slightly different requirements. There are also some models that require a heat gun to use. Always follow the instructions that come with your specific device. So, let's get started. First, we're going to scoop oxalic acid into the pan of the vaporizer. The amount of oxalic acid you use depends on the configuration of your hive. Use one gram per brood chamber. If you have a colony housed in one brood box, use one gram of oxalic acid. If you have a colony housed in two brood boxes, use two grams of oxalic acid. And there will be a measuring device that comes with the oxalic acid when you purchase it. Scoop the oxalic acid into the pan and insert it fully into the hive entrance. After the vaporizer is inside, seal up the entrance with a strip of foam or a cloth so that the vapors don't escape. After the entrance is sealed, connect the vaporizer to a car battery. If you're using the Varro vaporizer, you'll need a 12 volt, 12 amp battery. Leave it inside the hive, connected to the car battery, for two and a half minutes. During this time, the pan is heating up, causing the solid oxalic acid crystals to vaporize into a gas. And this conversion of a solid into a gas is called sublimation. So, sometimes this is referred to as the Oxalic Acid Sublimation Method. If there are still some crystals left in the pan after this time, continue to heat for an additional half minute until all crystals have vaporized. And during this time, all the oxalic acid will sublime, and you may see fog coming out of the entrance of your hive. Disconnect the vaporizer from the battery and keep it inside the hive for another two minutes before pulling it out, to allow the oxalic acid vapor to settle. Remove the vaporizer and completely seal the entrance with a cloth or a strip of foam. And keep the entire entrance sealed for 10 minutes. Cool the vaporizer by dipping it in a bucket of water to ensure that the oxalic acid sublimation doesn't begin occurring before it's inserted into the next hive. The great thing about using oxalic acid in late fall is that you don't need to open up the colony in cool weather and you don't need to come back to the hive to remove any strips later. So, it's very convenient. It's really important to only use oxalic acid that's been formulated for treating Varroa mites. It will specify this on the label. Other oxalic acid, or wood bleach purchased from hardware stores, can contain additives that can potentially harm bees. Using non-registered forms of oxalic acid for mite treatment, or non-registered forms of any mite treatment for that matter, is against the law. Now, let's go over the dribble method. The dribble method of oxalic acid application requires you to dissolve oxalic acid into sugar syrup. So, dissolve 35 grams of oxalic acid dihydrate in one liter of 1-to-1 sugar syrup. This will make a 2.8 percent oxalic acid solution. You'll want to use warm water because this will completely dissolve the crystals and it won't chill the bees when you trickle it onto them. Using a syringe, trickle five milliliters of the solution directly onto the bees in each bee space in each brood box. The maximum dose for a colony is 50 milliliters, regardless of whether the bees are in a single brood box or double brood boxes. You can purchase a measuring syringe from your bee supply company when you purchase the oxalic acid. It's important to note that you shouldn't use the dribble method on cold autumn days or in the winter; any day less than 37 degrees Fahrenheit, or three degrees Celsius, as this can risk chilling the bees. In this environment, it is best to use the fumigation method instead. The last registered method for oxalic acid is the package method. The package method uses the same solution that we just discussed to spray onto a package of bees before you install them into a hive.

Before you spray your package, place it in a cool, dark area for 24 hours to encourage the bees to cluster together. First, you want to spray your package with 1-to-1 sugar syrup without any oxalic acid. This will encourage the bees to engorge themselves on syrup, so that they're not consuming a lot of oxalic acid when you give it to them later. Wait two hours after spraying them with sugar syrup. Then, you want to spray them with the oxalic acid solution. Apply three milliliters per 1000 bees, using a spray bottle. So, what does that translate to? Well, a two pound package of bees contains about 7000 bees, which requires 21 milliliters of solution. On the other hand, a three pound package of bees contains about 10,500 bees, which would require 31.5 milliliters of solution. Apply the solution evenly on both sides of the package, and then keep the package in a cool, dark place for another 72 hours before you put them in a hive. Some important things to note about oxalic acid is that it doesn't penetrate brood cappings, so it will not kill any mites that are inside the brood, only phoretic mites. In the US, oxalic acid is not registered to use when brood is present. However, in Europe, some beekeepers will use oxalic acid during spring or early summer, when brood is present in the hive. And they'll apply a treatment, wait a week for some of the brood to emerge, apply another treatment again, wait another week, and apply a third treatment. And some beekeepers in the U.S. have also experimented with this practice. But research shows that repeatedly using oxalic acid when brood is present isn't nearly as effective as using one application when brood is not present. Repeated applications during spring or summer gives an efficacy of only 37 to 73 percent, whereas applying oxalic acid to broodless colonies yields an efficacy of 82 to 99 percent. Furthermore, repeated applications of oxalic acid can be damaging to honey bee health and can cause increased winter mortality. Also, it can result in higher adult, brood, and queen mortality, and can reduce the amount of brood the queen lays for up to two months after the treatment ends. This most often occurs with the exposure to multiple applications using the dribble method, but not so much with multiple applications using the fumigation method. So, those are some considerations when using oxalic acid.

READ

Chemical Controls for Varroa Management — Mite Away

Quick Strips The active ingredient in Mite Away Quick Strips® is formic acid, an organic acid found in some plants. Formic acid is also naturally found in honey, sometimes in trace amounts and sometimes in concentrations as high as 2,000 parts per million. A Mite Away Quick Strip application increases the formic acid content in honey well below a hundred parts per million, so it does not exceed the levels that can naturally be found in honey. For this reason, it is safe to use when honey supers are present. Mite Away Quick Strips contain 46.7% formic acid. A Mite Away Quick Strip is a gel strip with a paper wrapping surrounding it to control the vapor release. The vapors are going to dissipate throughout the colony to kill mites. The product also has the ability to penetrate brood cappings to kill some mites that are reproducing in the brood. Because of its ability to use when brood and honey supers are present, it's a great option to use in summer, should you find your mite levels exceed the treatment threshold. Mite Away Quick Strips can also be used in spring and autumn when the temperature requirements are met. You want to make sure that you disturb your colony as little as possible before applying Mite Away Quick Strips. If you're inspecting or monitoring your colonies, wait at least three days before you apply the treatment. Applying the treatment right after disturbing the colony can result in increased queen loss and absconding. Formic acid can also affect queens and drones. Queen supersedure may be

observed and drone eggs are sometimes removed. Drone mortality is also sometimes observed. For these reasons, we don't recommend that you use Mite Away Quick Strips to treat colonies that are being used for queen-rearing or drone-rearing. There are two application options for Mite Away Quick Strips. Option 1 consists of a seven-day treatment; it is recommended for all critical treatments as well as for under-the-cap Varroa control. Option 2 consists of a 21-day treatment; it is recommended for control of Varroa mites in the dispersal (phoretic) phase. For both options, outside temperatures must be between 50 and 85 degrees Fahrenheit, or 10 to 29 degrees Celsius, for the product to be most effective. Hot temperatures above 92 degrees Fahrenheit, or 33 degrees Celsius, can cause excessive brood mortality and queen loss. You also want to ensure that colonies have plenty of ventilation during treatment. Remove any entrance reducers in the hive. Before applying the treatment, make sure that you're wearing long pants, a long-sleeved shirt, gloves, shoes and socks, safety goggles, and a respirator with an organic vapor-removing cartridge. For the seven-day treatment, remove the strips from the plastic wrapping but do not remove the paper wrap. Place two strips on top of the brood box, staggered so that they have about two inches between strips and about four inches between the ends of the hive body. For colonies in double brood boxes, place the two strips in the same manner, but put them in between the two brood boxes. If you have honey supers on top, place the strips beneath the queen excluder. Colonies will expand their cluster within the first three days as a response to this treatment, and it's normal to observe bearding during this time. For really strong colonies, you can add an empty honey super on top of the top brood box so that they have room to expand. It's important not to open the colonies for seven days after the treatment is applied because it needs to be vaporizing properly. For the 21-day treatment, apply one strip on Day 1 across frames in the center of the brood box then replace that strip with another strip on Day 14. If the application of the second strip is delayed by unsuitable temperatures, plan to apply that second strip as soon as weather conditions permit. Follow all recommendations listed above for the seven-day treatment (exception: removing and replacing the first strip on Day 14). After treatment ends, you can remove the strips and compost them, or you can allow the bees to chew them and remove them from the hive in time. Do not reuse the strips, and dispose of any packaging according to the label. Formic Pro® is a new product created by the same company that makes Mite Away Quick Strips. While the basic application and look of the product is the same as the Quick Strips, Formic Pro has a longer shelf life and the treatment duration is a longer period of time; two strips can be placed on hives for 14 days, or one strip can be placed on hives for 10 days and replaced with a fresh strip for an additional 10 days.

TOOL

One-Page Guide to Varroa Treatments

With so many different chemical treatments available, it can be overwhelming to determine which is the best. Here, we provide information on page guide to Varroa treatments. all the options available; it is up to individual beekeepers to decide which option makes the most sense for their colonies, depending on their goals, locale, time of year, etc.

READ

Safety and Biosecurity of Chemical Controls

Although mite treatments are designed to kill mites, some of them can also be harmful to bees and to humans. Always read the product label and use gloves, goggles, and ventilators when called for. It is a violation of federal law to use any pesticide product in a manner that is inconsistent with its labeling in the US. Too often, beekeepers ignore personal safety, putting themselves at risk for acute or chronic issues. Even natural chemical treatments, at the relatively small doses required to kill Varroa mites, can be poisonous to people. The beekeeper's safety is the number one priority. Be fully prepared when it is time to treat and train any employees, volunteers, or friends in personal protection safety when they are helping apply mite treatments. Oxalic acid vapor is particularly harmful and can be fatal if inhaled. A respirator face mask with a cartridge is required. Miticides can also be toxic to honey bees, particularly if they are not used according to the directions on the label. The risk to bee health for some chemicals is heightened if chemicals are applied when it is warmer than the upper threshold on the label or if they are applied more frequently than the maximum annual usage indicated on the label. Formic acid, Api Life Var, and Apivar can only be used a maximum of two times per year and can be harmful to honey bees if used more frequently than that. An overview of the documented symptoms that can be observed in bees is in the table below:

Treatment	Mortality	Mortality	Sperm	Laying	Development	Viability (Queen)	Apivar	Apistan
Apiguard	X	X	X	Api Life Var	X	Mite Away Quick	X	X
Strips	Oxalic Acid	X	Hop Guard	No adverse impacts reported yet; more research is needed.	III			

WATCH

The Controversy Surrounding Some Synthetic Chemical

Treatments For the past twenty years, the main chemical treatments for Varroa mites were Apistan (active ingredient fluvalinate) and CheckMite+ (active ingredient coumaphos). Fluvalinate is among the oldest treatment options and has been used to treat Varroa mites since their introduction into the US. For several years, this treatment worked very well to control mites; however, in 1998, the first resistance to fluvalinate was documented in the US. Worried about the inability to effectively control Varroa, the U.S. Environmental Protection Agency granted CheckMite+ an emergency approval. After only three years, resistance to CheckMite+ was also documented in the US. Apistan and CheckMite+ remain registered treatments for controlling Varroa mites, but they are almost never used by beekeepers as miticides (though sometimes CheckMite+ is used to control small hive beetles). At this time, the two most common treatments were unreliable, and yet Varroa mites were still posing a significant challenge to beekeepers. To make matters worse, research was beginning to uncover that fluvalinate and coumaphos sometimes had negative impacts on honey bee health. Specifically, these two chemicals can be harmful to queens and drones, interfering with their ability to reproduce. Studies showed that treatments of either chemical could lead to low birth weight of queens, high mortality rates of queens, and physical abnormalities of queens during development. Fluvalinate can interrupt a queen's ability to lay. Coumaphos can hinder the ability of a colony to rear queens, as fewer queens develop and are accepted. Coumaphos exposure can also cause low sperm counts in both queens (in their spermatheca) and in drones. As with many pesticides, fluvalinate and coumaphos are "lipophilic," meaning they are fat- soluble. For this reason, they persist in beeswax (which is made primarily of lipids) and are now ubiquitous in commercial combs and wax foundation. Several large-scale studies

have found these two miticides are the most frequently detected pesticides in beeswax. The largest survey of pesticide residues in wax examined 259 colonies across North America. The most prevalent pesticides that were present in these colonies were fluvalinate and coumaphos. 98% of colonies contained both of these miticides, and residues were present at higher levels than any other insecticide, herbicide, or fungicide. Coumaphos is the most persistent of the two chemicals, with a half-life of 115–346 days in beeswax. This means that the chemical can be present in the colony for years, well after application. The ability of these and other miticides to remain in wax in low levels can help contribute to mite resistance. Furthermore, coumaphos is not degraded by heat. Commercially recycling contaminated beeswax to create wax foundation sheets results in contaminated foundation. This has made it difficult for beekeepers to avoid introducing low levels of coumaphos into their colonies, even those who avoid using synthetic chemical treatments. The best way for beekeepers to reduce their bees' exposure to contaminated wax is by gradually recycling out their old combs. Plastic foundation likely has lower residue levels than wax foundation, although plastic frames do still contain a thin coating of beeswax. The most recently developed synthetic chemical, amitraz, breaks down quickly in wax and honey. Residues cannot be detected after several weeks, though its breakdown chemicals can be present in honey for up to two months. Watch this video to hear about the introduction of naturally-derived chemical treatments and some considerations for using any kind of chemical treatment in your hives. Video Transcript There's a lot to know about mite treatments, as controlling Varroa is the biggest challenge that beekeepers face today. So, I'd like to give you a brief overview of some different chemicals, and some considerations to keep in mind as you're using treatments to manage Varroa. So, the first mite treatments to come on the market were synthetic chemicals. And these chemicals were developed because they're much more toxic to mites than they are to bees. For instance, fluvalinate, the active ingredient in Apistan, is 800 to 1000 times more toxic to mites than it is to bees. This means that not many bees die during treatment. So, that was a really great development. Although researchers found that there can be other negative impacts on bee health from the synthetic chemicals of fluvalinate and coumaphos. So, there's more desire to use more natural compounds in hives. And this is especially true because of concerns with residues in honey and beeswax, and with the ability of Varroa mites to develop resistance to some of the synthetic chemicals that are on the market. So, for this reason, over 150 essential oils have been tested for Varroa mite control. And the most effective essential oils are the ones that can evaporate to produce vapors inside of the colony. And of the 150 that have been tested, only about 24 of them are able to evaporate quite readily. The next difficult thing about finding essential oils that work are that at concentrations needed to kill Varroa mites, many also kill honey bees. Unlike with synthetic chemicals, the specificity to target only mites is much lower in these natural treatments. So, for this reason, there's only a handful of essential oils that are toxic to mites, but safe for honey bees. The current registered treatments on the market include Thymol, eucalyptus oil, and menthol. It's important not to purchase your own essential oils from health food stores to make your own mite treatments because the products on the market are the right concentration and formulation to be safe for bees. They're also delivered in gels, packages, or tablets that allow them to vaporize at specific rates. Let's talk a little bit about treatment efficacy. So, the efficacy of treatments varies greatly depending on a number of environmental and colony factors. The amount of brood in your colony, for instance, affects treatment efficacy. So all treatments, except for Mite Away Quick Strips, only affect phoretic mites. Therefore, they're most effective when the colony

has little to no brood. This could be during autumn, or if you're creating a broodless scenario. If you're using mite treatments when brood is present, such as during spring, or early summer, or in the middle of summer, they tend to be most effective when mites have just exceeded the economic threshold. If you have very high mite populations well above the economic threshold, they might not be brought down significantly if you're applying a treatment during this time. Treatment efficacy can also be impacted by the environmental temperature. And this is especially important for treatments that are vaporizing in the colony. The rate that a treatment vaporizes changes with temperature. So, it might occur more quickly in hot temperatures, which can pose a threat to bee health and can reduce the efficacy of the treatment because it's not present for a long enough time. On the other hand, if it vaporizes too slowly at cold temperatures, then it can also cause the product to be less effective because the mites aren't being exposed to the right concentration at the right time. For this reason, it's very important to only apply chemicals during their appropriate temperature windows. Because treatment efficacy can vary, it's always important to monitor after the treatment is finished to determine whether or not it worked. You don't want to assume that your mites are under control simply because you applied a treatment. When you treat your colonies, it's also important not to open the colony while the treatment is in progress. So, you don't want to disturb the vaporization or anything else that's taking place during that time. It's also important to treat all the hives in an apiary at the same time. As soon as there's one colony in an apiary above the treatment threshold, all the colony should be treated because mites from highly infested colonies can easily be spread through robbing and drift. So for this reason, you want to treat on a yard-by-yard basis. With all these treatments available, beekeepers sometimes ask me, why are there still Varroa mites when we've been treating our colonies for decades? Well, there's a couple of reasons why. First, there's no treatment that kills 100 percent of mites on a colony. Even treatments with 99 percent efficacy still leave some mites present that will continue to reproduce in the colony, and perhaps spread to other colonies. The current treatment on the markets range in efficacy from 61 to 99 percent when used according to the label. So, there's always some mites that are still around. And second, not all beekeepers treat their colonies. The Bee and Farm Partnership surveys find that while most commercial beekeepers are treating their colonies, less than 50 percent of hobbyists manage Varroa mites in any way. So, any unmanaged colonies in an area can contribute mites to your colonies, even if you happen to treat them. So, until we have widespread adoption of honey bee stock that is resistant or tolerant to Varroa mites on their own, using chemical treatments when needed as part of a broader integrated pest management approach is our best strategy to keep mites at bay. So, I hope that some of these considerations help you when you're making decisions for mite management. Treating on an apiary basis is a great idea under these circumstances: 1. If the beekeeper doesn't know the mite levels of every colony in the yard but instead sampled a portion of them to understand the Varroa dynamics of the entire apiary 2. If the colonies in the yard are susceptible to drift because there are numerous hives in close proximity to each other 3. If the treatment will occur at a time of year when robbing is likely However, if robbing and drift is not a concern, and if beekeepers are aware of the mite levels in each individual colony, they can effectively control Varroa by treating on a colony basis (i.e. only treating colonies that exceed the mite treatment threshold).

WATCH

Varied Management Practices

Beekeepers use a variety of chemical managements. A Cornell University survey of 60 New York State beekeepers in 2016 show that chemical controls tend to vary by operation size. Hobby beekeepers (with fewer than 50 colonies) are least likely to use chemical treatments, and when they do, they most commonly use natural chemicals. Only 3% of hobby beekeepers in the survey used synthetic chemicals. On the other hand, synthetic chemicals are much more common among commercial beekeepers; 42% reported using synthetic chemicals as part of their Varroa control strategy. Both hobby and commercial beekeepers had similar levels of Varroa mites. Some beekeepers choose not to use synthetic chemicals in an effort to reduce the potential build-up of residues in the combs or to minimize the negative impacts on bee health. Other beekeepers may choose not to use any treatments at all. If a beekeeper chooses to be chemical-free, Integrated Pest Management methods should continue to be followed. Being a successful treatment-free beekeeper requires diligence and good record keeping. Colonies should be monitored on a monthly basis, genetic stock for Varroa resistance should be introduced, and cultural methods should be used. Colonies that exceed treatment thresholds should be selectively treated or culled. Without the use of chemical treatments, colonies may need to be managed more intensely and this management is likely to be more time consuming. As we mentioned earlier in this module, it is a risk to those and other honey bee colonies, including wild ones, to practice chemical-free beekeeping without the use of any cultural treatment options and without closely monitoring mite populations. Chemical-free beekeeping is more successful in isolated areas, where colonies that can effectively manage Varroa without treatments are protected from re-introduction by robbing from outside colonies. The combination of Varroa prevalence and its role in colony losses, the importance of effectively managing mite levels, and the wide variation of beekeepers' treatment preferences paves the way for disputes. During disagreements, it is very important to be respectful of other beekeepers' opinions and to try to understand the reasoning behind their choices. Master Beekeepers have an important role to play with regard to Varroa management. They must work to educate other beekeepers on Integrated Pest Management and help others be comfortable, knowledgeable, and confident in managing their colonies effectively. Watch these videos to hear honey bee experts Randy Oliver and Dr. Dennis vanEngelsdorp discuss the importance of mite management and the responsibility beekeepers have to their honey bees, their beekeeping community, and to nature. Randy Oliver Video Transcript Back in the day, beekeeping used to be much easier before the Varroa mite arrived. You could go in the Sears Roebuck catalog and order your hive, and it would be delivered by mail. You put the hive together, then you order your package bees, get your Starlines and Midnights, again through Sears Roebuck. They'd be delivered mail. You put your bees into the box, and that's the last thing you have to do. The bees would just take care of themselves. You go out once a year. You could take honey from the hive. And that hive would just survive for ten years, 20 years. No problem at all. No other input from the beekeeper. And I would hope someday, we get back to there, when bees evolve resistance to Varroa. But right now, that's not the case. It takes actual input to the beehive to practice bee husbandry to help them out. And the beekeeper has a responsibility to the bees that are in their care, and to the community of beekeepers, and to the feral population of bees. Taking a package of commercially-raised bee stock, which are bred for commercial production with lots of husbandry, and putting them in a hive, and then leaving them on their own is not fair to the beekeeping community, and it's not beneficial to the natural bee population because those

bees won't survive. They're going to crash from the Varroa virus complex, and then they're going to flood the surrounding community with extra mites. So, all you've become then is a Varroa producer to the detriment of the beekeeping community and the surrounding bee population. So, you have a responsibility to the community, and to nature, if you're going to keep bees, to manage them, to keep them from collapsing from the Varroa mite. Dennis vanEngelsdorp Video Transcript

The number one issue, I think, especially among backyard beekeepers, is Varroa mite. We consistently see, we sample a lot of colonies across the country. And we consistently see in September and October, when, that's the most critical time to get your mites under control because that's the time that you're going to produce your winter bees so they can live all year round or all winter. And so, when they're producing those winter bees, you want that colony to be as healthy as possible so that those bees are going to live as long as they can. And we consistently see that a majority of beekeepers have mite levels in September and October that are above what we think the colonies can withstand without hurting them. And so, we think that after three mites per hundred, you need a treatment. After five mites per hundred, your colonies are already going to suffer ill effects, even if you start to treat. We also notice, especially among backyard beekeepers, that a majority of backyard beekeepers are not using a known Varroa mite control product. And I think that one of the reasons they don't is actually very altruistic. I think it's a really good reason. Some people think, "Well, I can afford, I'm a backyard beekeeper, I'm not feeding my family on this hobby. I have 10 colonies. I can afford to lose nine colonies. If one colony survives and I can breed from that survivor and then I'll save the world." And who doesn't want to save the world? That's a great thing to do. I applaud it. It's actually a fantastic idea until you think about what happened to those last nine colonies. So, we've been doing this project called the Sentinel Apiary Project across the country, especially in Maryland. And what we find is there's three, so these Sentinel Apiary projects, what we do is we go in and we get the beekeeper to sample eight colonies once a month for six months. So, it's the same colonies and we know what's happening. And we find that there are three patterns of mite development. This is Kelly's work, my PhD student. The first is, is the people who don't treat. So, they start with low mites, and the mites go through the roof, and, indeed, are well above threshold in October, and some of them start to die. The other one is what you would expect, is you have, you see that grows slowly, mite populations double every month, you see they grow slowly, August or September they treat, mites go down, and the mite levels stay low. That's what you want. That's an effective, good treatment and mite levels are down. Then we have this other third of the population that treats, mites go down. And then a month later, the mite levels are three times higher than they possibly could be based on normal mite reproduction. And so, we think this is evidence of these mite bombs in the landscape. That these colonies that are dying from mites, those bees, those last bees and mites don't say, "Oh, too bad, I'm just going to die here." They take off and they fly away. As far as two kilometers away, we found these. They blow up and they spread these mites across all the beekeepers within a three kilometer radius. So, what's happening is these people who don't treat their colonies are basically letting those mites invade their other colonies. And you're never going to breed resistance to mites that way. Like, I mean, if you have all these mites invading, bees aren't going to be able to ever evolve resistance to that. And they're also infecting their neighbors. And so our message is: You need to have a Varroa management plan in place because it's irresponsible. We would not say that it was acceptable to breed a tick-resistant dog by letting our dogs die from ticks. That just would not be an acceptable thing. Why is it okay to do that with bees, especially if you're

invading and hurting your neighbor? And I think that's a really important question that we have to ask ourselves. I still think it's very noble to be breed persistence. Don't get me wrong. And a lot of people say, "Oh, what you mean?" I think what you have to do is you have to monitor your colonies. And you measure the mites. Your colonies that have the lowest mite levels, you breed from. And the other ones, you either treat those mites, or if you're really opposed to treating, kill those bees and those mites because they're going to die anyway. It's better to do that before you spread it to everybody else. So, I think there are ways of accommodating this very important and long-term solution to our mite problem is breeding resistance. But you can't do it just by letting them die and spreading mites everywhere. That's just irresponsible for everybody. And that's one of the biggest learning points that I've had, and that has surprised me over the last period of time.

WATCH

Developing Your Own Varroa Management Plan

The active beekeeping season is a busy time. Beekeepers are often making splits, trying to build their colonies up to maximize forager populations for the major honey flows, adding supers, extracting honey, and preparing their colonies for winter. Without planning a Varroa management calendar upfront, it's easy to let mite management fall by the wayside and then get blindsided with high mite counts. Watch this video to see how we plan our Varroa management calendar at the Dyce Lab and to gain some tips that you can use when creating your own plan. Video Transcript I think it's very helpful to sit down and actually plan out what your Varroa management plan is in a given year. And this is something that I started to do probably in my second year of beekeeping when I began monitoring for mites. And one year, in the middle of summer, I found that I had mite levels that were above the treatment threshold. And then I thought, oh my gosh, what do I do now? And I didn't know what my treatment options were, and I began just Googling all the information I could find on the internet. And I quickly realized that there was a lot of different treatments out there and they all have different specifications. So, some can be used on there's honey supers on, some can't. Some can only be used at a given temperature window. Some can only be used when the colony is broodless. And I was really unprepared for what to do when I found that my mite levels were high. So, by the time I finally decided what treatment would be my best option, I ordered it, it took awhile for it to come in the mail, and then when I went to treat my colony, it was actually dead. A few weeks had passed and I was too late. So, from there, I decided it was a really good idea to have a plan in advance now that we know that Varroa is so prevalent and can be so destructive to colonies. So, what I basically do is on a quiet winter day, I sit down and I decide what all my options would be for a given year and I physically take note or write out a calendar that looks something like this. So, we have all months here. So, it's currently December, so our calendar begins from January and runs all the way till next December. And then I have a column to look at my genetic options, my cultural options, and my chemical options. And I write these in here so that every month when I'm monitoring, if I do find that I'm above a given threshold, I know what my chemical options are and I already have a plan for what my genetic and cultural options are in advance. So, when I'm sitting down to do this plan, I also have a couple of printouts with me, the ones that are available in this course, and the one that really helps for the different chemical options is the Varroa mite control options throughout the year. And in this one, we have a month-to-month breakdown of the different natural chemicals and synthetic chemicals that you can use. So, this plan that I'm creating is

for the colonies here at the Dyce Lab. We have 20 colonies that we're managing and they're spread across two different yards, so 10 colonies in each yard. But you can adopt this plan for your own operation if you only have two colonies, to if you manage thousands of colonies, whatever works best for you. So, let's get started. So, the way that I do this, first and foremost, as I think about the genetic column, and really, there's only one way that we can help our colonies with regards to genetics. And I actually am planning on ordering some Varroa sensitive hygienic queens, and I spoke to the breeder yesterday and he anticipates that they will be ready by June, and that's when they'll arrive. So, I'm going to put in my June column VSH queens, and that's when I'm going to be installing them in some of my hives. And I'm already happy with the queen stock that I recently ordered in one yard. So, I'm actually just going to be putting these VSH queens in a different yard. So, that would be in yard 1. And in the other yard, I'm going to just perpetuate the line that I already have, and I'm going to do some splits and allow those colonies to requeen themselves. So, I tend to do splits in June here at the Dyce Lab. Some people can do that as early as May, or even April, depending on where they overwinter their colonies. So, doing splits is also a really great way to help with Varroa because it's going to allow a long brood break for the colony to begin rearing its own queen, for that queen to become sexually mature, for her to take her mating flight, and begin laying. So, the splits in yard 2, I'm going to allow them to just rear their own queens. So, now in addition to performing splits, there's also a couple other cultural options that I really like to do. And I have the time to do them while I'm here at the Dyce Lab because the yards are pretty close, and it's feasible for me to do inspections every 10 to 14 days so I can keep track of what's going on. The first thing that I really like to do is use the drone comb frames. And so, those, depending on how warm the year is, I typically install those at the end of April if we have a really warm April here in the Northeast. But usually, it happens to be in May. So, I use drone comb frames, and I continue to use these every month until the end of August. And colonies are typically rearing drones at that time, so it's not too effective to leave the drone comb frames in there. And then, in addition to using drones comb frames, there's a third cultural option that I like to do, and that's using a screened bottom board. So, the screened bottom board can sometimes go on at the end of April, again, depending on how warm it gets, but I typically like to do it in May because that's usually when the colonies are finished for winter and ready for the spring and summer season. And I leave screened bottom boards on until I winterize my colonies, and that typically happens at the end of October. And at the Dyce Lab, we don't leave the screened bottom boards on during the winter. And then for the winter months, there's not really any cultural options that we're doing to help with Varroa management. So, let's talk about the chemical column now. And I find that by planning which chemicals I'm going to be using throughout the year, it lets me have those chemicals on hand when I need it. And that prevents a similar catastrophe that happened to me in my second year of beekeeping. So, actually, right after I fill out this column, I'm going to go to the bee supply company website and I'm going to order all my chemicals to make sure that I have them on hand for when I need them next spring and summer. So, right now for where I am in the Northeast, we overwinter our colonies here. And during winter, we're not going into the colonies and monitoring or able to treat them. So, currently it's December, but I'm going to be leaving January and February and March blank. In April, we often have a couple of days that are warm enough for me to go in and do a quick sugar shake or alcohol wash to determine what our mite levels are. And so, April would be the first opportunity for me to use a chemical treatment if I find that the mite levels are above the treatment threshold. So, my preferred treatment in April is actually to use

Apivar, which is a synthetic chemical. And the reason why I really like to use this is because it's quite effective and it isn't temperature dependent. So, even if I put on the treatment and a couple days later it's going to get a bit cool, that's okay, I can still use that treatment. If you're not comfortable using a synthetic chemical, then you can try one of the thymol options. So, that would be Apiguard or Api Life Var. So, Apivar is a chemical that remains in there for 6 to 8 weeks, so it's a long, slow release chemical. So, I actually shouldn't see that I'm having high mite levels if this chemical is effective for me. But just in case, I want to make sure that I have options for every month. I don't want to be caught off guard if I find that I do do a sugar shake or an alcohol wash and find that my levels are high. So in May, if I find that I do have mite levels that are above the treatment threshold again, then my chemical of choice at that time would be to use Apiguard. And Apiguard is one of the thymol-based treatments. And the reason why I'm choosing that for May is because it tends to be a little bit warmer, and Apiguard does have a temperature window up between 60 and 105 degrees Fahrenheit. And also, it's not too long of a treatment. So, it's on there for about 28 days and I can super immediately after. And this is important for me because the time that I'm going to have my supers on is from June until September. So, once we get into June, I need to start thinking about different chemicals that I can use when I have honey supers on. So, let's say after my Apiguard treatment, or my Apivar treatment, whichever one I end up using, I am doing a sugar shake and I find that my mite levels are high. So, what are my options in June? My main option that I'm going to be using is Mite Away Quick Strips. I really like Mite Away Quick Strips because it's a very quick treatment. The pads are only on there for seven days. And the treatment is able to be used when you have honey supers on, and it can actually penetrate the cappings and impact some of the mites that are reproducing inside of the brood. So, Mite Away Quick Strips is my preferred option when I have honey supers present. If, after using that treatment, I'm doing my monitoring again in July and I find that my mites are high, then my option would be to do Hopguard. And Hopguard II is the only other chemical that you can use right now that has been approved for using with honey supers on. So, that would be my option for July. And then in August, I could use Mite Away Quick Strips or Hopguard II again. But I'm going to be rotating my treatments, so if I find that one time I'm using Mite Away, the next time I want to use Hopguard. So, honey supers are on until September, and then once we get into October, when I'm starting to winterize my colonies, I'm seeing that there's really not a whole lot of brood that's being produced. And so, October and also November are the key months when I would consider doing oxalic acid. And it's best used if there's no brood. So, the fewest amount of brood that's possible would be the time to do an oxalic acid application. In October, the weather is still usually warm enough here in the Northeast to use the dribble. But once we get into November, the colonies are pretty well winterized, and I don't want to be opening them up and exposing the brood to getting chilled and actually pouring some sugar syrup on them, which can get them to, or which can cause them to become chilled even further. So, in November, really the only option would be the oxalic acid vaporizer. So, now you're going to look at this sheet and you'll think, oh my gosh, she's planning on putting that many chemicals in her colonies? That is definitely not the case. I'll probably be treating twice, maybe three times, but only when my mite thresholds are being exceeded. So, these are just my options to have on hand. I may find that my mites are high only in May, and I can do an Apiguard treatment. And that keeps the levels down until September, when I need to do a quick Mite Away Quick Strip application. And then again in November, if the mite levels have crept up again, I can do an oxalic acid vaporization. But I find that it's helpful to write out all my

options so that I'm not caught off guard. If something happens and I find that there were mites that were reintroduced from other colonies, or if there's just a spike in the mite population that I wasn't anticipating, I'm now going to have the tools at my disposal to be able to deal with the problem promptly. And so, it really helps just to consider what your options are at the different times of year, so I know exactly what to do, should a problem arise. As we continue going through course material, we'll be able to add additional things to this calendar, like when am I planning to do my American foulbrood checks, when am I planning to sample for *Nosema* really quickly, and looking at my spore counts? All of those things can be put into this plan so that you're never caught off guard when you're beekeeping. And you can make sure that you have enough time dedicated toward managing pests and diseases. A lot of beekeepers find that the number of colonies they can manage depends on how many they can actually treat for diseases and inspect for diseases effectively. And we can continue to split colonies and grow as fast as we can, but if we don't have the time to monitor and treat those colonies as needed, then it's not going to be a great situation. So, being as prepared as possible is the best way to manage your colonies adequately for *Varroa* mites and other issues.

TOOL

Mite Management Calendars

To help you with your plans, we have  Download the Tool developed some example guides of Click here to download the past mite management calendars. In example calendars. these calendars, you will see the beekeepers' monthly mite monitoring results and see the decisions that they made. The first calendar is a hobby beekeeper's plan. In this plan, this beekeeper preferred to use only natural chemicals and had plenty of time to incorporate cultural methods since she was only managing a few colonies. The second calendar is a commercial beekeeper's plan. This commercial beekeeper was comfortable using all types of chemicals, but had little time to devote to the more time- consuming cultural practices.

MODULE WRAP-UP

Varroa Mites

Understanding *Varroa* mites — from their biology to their treatment — is fundamental to successful beekeeping. With this knowledge, beekeepers can better monitor their colonies and implement appropriate control measures to protect their bees from this devastating parasite.

Viruses

The study of honey bee viruses is a relatively new field that has grown in recent years with advances in molecular technologies. We now know that honey bees are host to a variety of different viruses, some that don't show many symptoms, and others that can be devastating to colonies. Of all the pests, parasites, and pathogens included in this course, viruses are by far the most difficult to manage. It is important for beekeepers to be aware of the potential viruses that can infect honey bees and what symptoms to look for, should any be presented. In this module, we will explore the most well-understood honey bee viruses and the impacts they can have on colony health.

READ

An Overview of Viruses

Viruses are small infectious particles that live and reproduce inside another organism (their host). Nearly all life forms, including plants, animals, and bacteria, can be infected by different kinds of viruses, and honey bees are no exception. *Apis mellifera* is susceptible to at least nineteen different viruses, with new species and strains continuously discovered. These viruses can have a range of impacts on honey bee health: individual bees may have no symptoms at all, individual bees might die while the colony survives, or the entire colony could be wiped out. Viruses are the most difficult diseases for beekeepers to diagnose for a variety of reasons: 1. Infected bees are asymptomatic at low levels and only begin showing symptoms at high levels 2. Symptoms may not be evident in all life stages of a honey bee 3. Many different viruses can exhibit the same symptoms 4. Several viruses can be present in a colony at once, making it difficult to interpret symptoms and distinguish the effects of one virus over another. Furthermore, viruses cannot be viewed with a simple microscope like the spores of American foulbrood or Nosema. Without access to facilities to analyze DNA, many beekeepers may have colonies with high rates of infection without knowing it. This can translate to infections spreading without the beekeeper realizing, and colonies dying without the beekeeper understanding why. Viruses are often in an inactive state or are asymptomatic, but under certain conditions they can be activated and become symptomatic. Some stress events, such as a bee experiencing other disease pressures (e.g., being exposed to certain pesticides or being chilled) can contribute to certain pathogens becoming “active.” At this point in time, we are aware of at least 7 viruses that are vectored by Varroa mites: deformed wing virus, Kashmir bee virus, acute bee paralysis virus, Israeli acute paralysis virus, slow bee paralysis virus, Varroa destructor virus-1, and Kakugo virus. This means that the mite can contract the virus in its own body from feeding on a honey bee and then transmit this virus to a different honey bee afterward. Often, these viruses do not show any symptoms until Varroa mites are present in the colony to exacerbate the infection. For these viruses, the best defense against high infection levels is to maintain low levels of Varroa mites for the entire year. If Varroa levels are not managed all summer long, but then a mite treatment is applied in autumn, the colony can often have low mite levels entering winter but still die from the viruses that have increased all season.

READ

Sacbrood Virus

■ Sacbrood virus is one of the most Key Points common honey bee viruses worldwide. Depending on the stage of the It was described in 1913, making it the infection, sacbrood virus can be first known honey bee virus. Sacbrood identified by the larva's curled- can infect brood and adults, but easily upward position and shrunken observed symptoms are only visible in head, by its fluid-filled sac-like brood. Once sacbrood is contracted in appearance, or by its removable a larva, it fails to shed its final skin to scale. become a pupa once the cell is Sacbrood is most common in capped. Instead, it remains in the larval spring or times of dearth. stage, curling upward and developing a “shrunken head” appearance, where its head looks smaller and pointier than normal. Over time, its skin becomes filled with fluid and turns pale-yellow and swollen. This fluid-filled “sac” appearance is the origin of the name

sacbrood. Many replicates of the virus are contained beneath the skin in this fluid, enough to infect many other bees if the skin is punctured and the fluid released. Eventually, the larva's head begins to turn brown and the body dries out and becomes brown or black, turning to a flat scale on the cell floor that can be scraped out with a toothpick or small twig. The virus can remain active in this state for several weeks. Adult bees infected with sacbrood virus don't have symptoms that are easily observed, but their behaviors can be altered. Infected adults usually transition from nurse bees to foragers earlier than expected. They become poor foragers, collecting less pollen than healthy bees. Progressive sacbrood infection. Credit: All photos from Bee Informed Partnership The virus can be transmitted from infected brood food and it is also associated with Varroa mites. Food becomes contaminated when nurse bees feed an infected larva and then move on and feed a susceptible larva. While Varroa has not yet been confirmed to vector this virus to honey bees, we do know that the mites can carry the virus. When inspecting your brood frame, look for any cells that have perforated cappings or the cappings entirely chewed off, an example that nurse bees are detecting the diseased brood. Open any perforated cappings with your hive tool and check for the characteristic “shrunk head” or sac-like larva. Sometimes, only one or two cells are affected. A brood comb with a heavy sacbrood infection appears spotty (due to hygienic behavior) and has many cells with perforations and symptomatic larvae. Sacbrood virus is most common in early spring and during periods of dearth. As with other viruses, sacbrood is primarily managed through proper nutrition and maintenance of colony vigor with a fertile queen. In mild cases of infection, bees can prevent a major outbreak by removing diseased larvae through hygienic behavior in the early stages of infection. If a colony experiences a bigger outbreak (with at least 5% of brood cells visibly infected), discard brood frames that contain many infected cells and requeen the colony (ideally with hygienic stock). Clean your hive tool and hands before working another colony. Sacbrood virus also infects bumble bees.

READ

Deformed Wing Virus

■ Deformed wing virus (DWV) is the Key Points most prevalent virus in North America, Deformed wing virus is the most and one of the most destructive to prevalent honey bee virus and has honey bees. High levels of this virus, been declared a worldwide combined with Varroa mites, are a epidemic. leading cause of over winter colony The most identifiable symptom is mortality. Every year, the USDA-APHIS deformed, dysfunctional wings. surveys find over 80% of honey bee colonies sampled across the U.S. have The virus occurs at low levels, often DWV. Cornell University's survey of over without symptoms when Varroa 200 colonies in 2016 found 96% of mites are not present. When mites them to be infected with DWV. are present, the virus can activate, Because of its prevalence, this virus has increase, and ultimately kill recently been classified as a worldwide colonies. epidemic. There are at least two different strains of DWV: strain A and strain B. Strain A is the historical strain of the virus distributed worldwide, but since the mid 2000s, a new, more virulent strain, strain B, has emerged in Africa, Asia, and Europe. Until recently, we were unaware whether or not it was present in the US. In early 2017, the USDA-ARS lab screened samples that were stored in a freezer and discovered strain B (also called Varroa destructor virus-1) is indeed present in the U.S. and was likely introduced sometime around 2010. The strain has since spread across much of the nation without our realizing, from a mere 2 colonies infected in 2010 to 64% of colonies infected in 2016. There is no indication yet that strain B

is more virulent than strain A in the US, but the USDA-ARS lab is continuing to investigate the virus, its spread, and how it relates to colony losses. The growth and spread of DWV coincides with the expansion of Varroa. Without Varroa mites present, this virus is transmitted inefficiently between bees and symptoms are rarely observed. But once Varroa mites become an issue in the colony, the virus can be transmitted rapidly. DWV replicates in mites just as in honey bees, increasing in the mite before it is passed to a bee. The more DWV there is present in a mite, the more likely the honey bee it is feeding on will develop deformed wings. Credit: Bee Informed Partnership

WATCH

Deformed Wing Virus Transmission and Symptoms

Although DWV is most often present asymptotically, when it does show clinical symptoms, they are distinct and striking. Its name comes from its telltale sign: deformed, dysfunctional wings. These wings are deformed from the pupal stage onward. Pupae often acquire DWV from Varroa mites while still in the cell. Infected pupae often fail to complete development and die inside their capped cells. If they do survive the pupal stage, these infected bees are usually unable to fly or perform regular tasks and quickly die. All castes can have deformed wings. Infected adults also have a shortened, bloated, and darkened abdomens. Watch this video to see the symptoms of DWV and learn how it can impact colonies. Video Transcript The name deformed wing virus comes from its telltale sign: deformed, dysfunctional wings. Infected adults may also have shortened, bloated, and darkened abdomens, and all castes can become infected with this virus. This is what a worker bee looks like with deformed wings. And here is what a queen bee looks like that has deformed wings. Now, this virus can be passed from bee to bee, through trophallaxis and cell cleaning. And it can even be passed from the queen to her daughters through the eggs that she lays. However, transmission from these routes mostly results in low levels of deformed wing virus, usually without any symptoms being observed. The introduction of Varroa mites has changed the epidemiology of the virus. High deformed wing virus infections are almost exclusively as a result of Varroa mite transmission. Without Varroa mites present, the virus is not a major problem. But with Varroa mites present, it becomes deadly. The symptom of deformed wings occurs when the bee becomes infected with high levels of the virus as a pupa. So, it's the mites that are reproducing inside the cell that feed on the pupa and infect it. The bee will emerge as an adult with these deformed wings, She'll never fly or be able to perform regular tasks, and she'll die at an early age. Many other pupae that become infected with this virus never emerge and die inside their cells. Varroa is a very effective vector of this virus. Because deformed wing virus levels are low in colonies without mites, maintaining low levels of Varroa is the best way to maintain low levels of deformed wing virus. After you apply a mite treatment, it will take about six weeks before the virus levels are lowered in a colony. This is because the lifespan of a worker is about six weeks long. So, the colony needs to experience a full turnover of the bee population before they're able to clear most of the virus infection. Even after Varroa is effectively controlled, colonies that were previously infected can still maintain low levels of the virus. So, while you're inspecting your colonies for disease, be on the lookout for deformed wings. If you see that symptom, monitor for mites to determine what your levels are and if it's time to treat. DWV can also infect bumble bees, carpenter bees, and orchard bees.

READ

Black Queen Cell Virus

■ Each year since 2009, the USDA-APHIS Key Points National Honey Bee Pests and Black queen cell virus is the Diseases Survey reports the national second most prevalent virus. trends of honey bee viruses. Black Despite how common it is, the queen cell virus has consistently been virus is most often asymptomatic. the second most prevalent virus, This virus infects all honey bee always found in over 80% of bee castes, but “black cells” are only colonies that are sampled. Aptly observed in queens and drones. named, when black queen cell virus (BQCV) infects a queen larva, the bee dies sealed inside the queen cell. After a few days, the body of the bee turns sac-like and yellow, and eventually dark brown or black. Often the cell wall also becomes dark brown or black. These symptoms are also observed in drone cells, though rarely. Queen cell with black queen cell virus. Credit: Bee Informed Partnership As with other viruses, BQCV can infect adult worker bees, but is most often asymptomatic. Infected worker pupae do not develop the characteristic black cell. Sometimes adult workers with BQCV have the symptom K-wing: their wings become disjointed (no longer connected to one another by the hamuli) and form the shape of the letter “K.” Unlike other viruses discussed here, BQCV has no clear association with Varroa mite infestations. It sometimes co-occurs with *Nosema apis*, and both are strongly evident in the spring. For this reason, BQCV may particularly be a problem for queen-producing operations. Because the virus is most often asymptomatic, it is unclear how much of a risk it poses to overall colony health. Watch this video clip to see a worker bee with K-wing. Management is primarily through good nutrition and proper health, maintaining sterile and clean grafting and hive tools, and by removing any infected BQCV queen or drone cells, if observed. Queen breeders should not re-use any cell builders or mating nucs in which they have observed symptoms of BQCV. Black queen cell virus can also infect bumble bees.

READ

Acute Bee Paralysis Virus Complex

■ Acute bee paralysis virus (ABPV), Key Points Kashmir bee virus (KBV), and Israeli The three viruses in this complex acute paralysis virus (IAPV) are closely are closely related and are related viruses from the same family transmitted by Varroa. that are best analyzed as a complex. All Symptoms of these viruses vary three viruses are distributed worldwide but include paralysis, hair loss, and and are fairly common. All of these acute death. viruses are transmitted by Varroa. The viruses generally persist asymptomatically at low levels, but if a bee experiences high infection levels, the virus or viruses will cause characteristic symptoms and rapid death. Once they reach their critical high level, these viruses can kill colonies so quickly that a beekeeper may never notice any symptoms of paralysis in the colony at all. Acute Bee Paralysis Virus Acute bee paralysis virus was first isolated in 1963. Until the population growth and spread of Varroa through the US, ABPV was not a serious problem. Now that Varroa is nearly ubiquitous, ABPV is much more common than it was in the past. The USDA-APHIS survey finds ABPV to be present in around 15–25% of colonies in the US, where it tends to peak in late summer. This virus infects both brood and adults. When symptoms are showing, bees may darken, lose hair, tremble, and be unable to fly. Bees with symptoms of APBV die within days, and at high infection levels, just one day (hence the name “acute”). The virus accumulates in the brain and hypopharyngeal glands of the adult bee. It is likely transmitted orally, either through contact with contaminated brood or feces, or by eating infected brood. This virus can also infect bumble bees. ■

Israeli Acute Paralysis Virus ■ Israeli acute paralysis virus was first discovered in 2004 in Israel when beekeepers experienced high numbers of colony losses. As it was isolated just before Colony Collapse Disorder became a widespread national concern, it was a natural suspect for the die-offs in the U.S. and some parts of Europe. Some studies found IAPV in U.S. colonies, and the presence of IAPV was strongly associated with colonies that died from CCD. However, more recent studies have shown that not all colonies with IAPV experience CCD, and not all colonies with CCD have IAPV, ruling out this virus as the cause of CCD. About 10–20% of colonies sampled across the U.S. are infected with IAPV. IAPV can be present in all life stages of honey bees and usually shows no observable symptoms. Bees that suffer from Varroa mite infestations and IAPV infections experience immune suppression. When symptoms are observed, bees display shivering wings and become disoriented. They may be paralyzed or crawl slowly, and they ultimately die (often just outside the hive). Their body often turns darker and the thorax and abdomen become hairless. Colonies infected with IAPV experience declining colony growth and reproduction, and in severe infections, death. Watch this video clip of a sick bee that has become hairless and shiny. Note: Sometimes bees that get nectar dripped on them during hive inspections can appear hairless and shiny. Sick bees can be differentiated from "wet" bees because they are not sticky to the touch and are often seen "crawling" like the bee in this clip. Kashmir Bee Virus Kashmir bee virus was discovered in 1974, and today there is 10–15% incidence in U.S. honey bee colonies. KBV infects brood and adults. It is transmitted by Varroa; when virus particles are fed to bees, the bees do not experience an infection. However, if the virus particles are injected directly into the hemolymph (which is what happens when Varroa mites feed on bees), the bees can develop the infection. The virus can be extremely virulent (destructive) following Varroa transmission. Once acting virulently, KBV can kill bees at any life stage and has no clearly identifiable symptoms—the trembling and hairless progression of ABPV and ■ IAPV does not occur with KBV. Infected bees die within three days. KBV tends to peak in autumn, later than ABPV. As with the other viruses infecting honey bees, KBV can only be managed by general considerations of nutrition and a strong queen. Most importantly, Varroa mite management is crucial for lowering viral loads. This virus can be transmitted from honey bees to bumble bees.

READ

Chronic Bee Paralysis

■ About 5 to 15% of colonies sampled Key Points across the U.S. have chronic bee There are two forms of symptoms paralysis virus (CBPV) infections. CBPV from CBPV: one causing “hairless causes the same symptoms as ABPV, black syndrome” and the other but bees take longer to die — often causing paralysis. several days compared one day for ABPV. Although current research suggests that Varroa mites do not act as a biological vector for CBPV because CBPV does not replicate within the mite, Varroa mites likely act as a mechanical or facilitating vector, worsening conditions that favor CBPV outbreaks. There are two forms of symptoms in adult bees. In the first syndrome, adults may turn dark and hairless (but are not paralyzed). This syndrome has been called “hairless black syndrome” in honey bees. In the other syndrome, bees are paralyzed or crawl, experience trembling wings and bodies, demonstrate K-wing, and do not darken or lose hair. Some bees develop bloated abdomens, and dysentery in and outside of the hive can be observed from the fluid buildup in their digestive tract. Both syndromes can be observed in different bees in the same colony at the same time.

Colonies with high levels of CBPV often have hundreds or thousands of bees crawling on the ground in front of the hive. If many trembling bees are present in a colony, it is more likely infected with CBPV than ABPV, as most bees from ABPV die before symptoms are observable. Because of the black, shiny appearance of some of these bees, they are occasionally mistaken for robber bees from other hives (who are stripped of their hairs from fighting). In addition, the trembling, flightless bees at the colony entrance have been mistakenly believed to be exhibiting symptoms of pesticide poisoning. Black hairless syndrome. Credit: Jennifer Tsuruda. This infection is spread by fecal-oral contact. It can also be spread through body contact, particularly in crowded colonies or colonies that are confined (e.g., during winter or transport).

READ

Other Viruses

In this module, we have only covered the main honey bee viruses that are regularly monitored and present in the US. Other viruses impact honey bee health as well, and even more viruses have been identified, but have not been well characterized and their genomes have not yet been sequenced, making it difficult to monitor their prevalence, distribution, and impact. For instance, the USDA-APHIS program is continuously monitoring for the introduction of slow bee paralysis virus (SBPV), which was discovered in England in 1974. To our knowledge, this virus is only present in Europe and Oceania. It is primarily an asymptomatic infection but can sometimes involve paralysis of the front two pairs of legs and is associated with the death of brood and adults. It is transmitted orally as well as by Varroa mites. In areas where this virus is present, viral loads peak midsummer. This virus also infects bumble bees. Additionally, one other virus is closely related to deformed wing virus (Kakugo virus). Collectively, DWV, Varroa destructor virus-1 (DWV strain B), and Kakugo virus are referred to as the Deformed Wing Virus Complex and all three are transmitted by the Varroa mite. While Kakugo virus and Varroa destructor virus-1 haven't been regularly screened for in the past, the latter will be routinely screened starting in 2018 after the discovery of its prevalence in the US. Scientists are working to create molecular screening methods to make it easier to track the prevalence and spread of these viruses.

READ

Managing Viruses

■ There are currently no treatments that Key Points target viruses in the honey bee colony. The only way to manage viruses is Keeping colonies otherwise healthy to keep colonies healthy with and free of other stresses is the best adequate nutrition, strong strategy to reduce the likelihood of populations, and few external virus levels increasing. Healthy colonies stressors. can be achieved by ensuring adequate Maintaining low levels of Varroa nutrition (which can help ensure mites all year long is key to optimum immune functioning), strong reducing infections of viruses colony populations, fertile queens, and transmitted by the mite. by limiting other stresses that can impair immune functioning or foster infection, including but not limited to pesticide exposure, moisture, or prolonged confinement. To limit the viruses transmitted by Varroa, maintain low mite populations all year. Frames with heavy infestations of viruses that impact brood (sacbrood or Black Queen Cell Virus) can be removed from the colony to reduce the infection load. To date, it is unclear how long wax or other equipment from a dead colony can retain live virus particles. New research in 2019 found viruses can be detected in

comb (and in honey and pollen stored in comb), and they can remain virulent for several months. Research is urgently needed to determine whether stored combs from dead colonies pose a significant risk to bees that are installed on them and, if they do, to develop effective sterilization methods. Reduce the transmission to other colonies by sterilizing hive tools and changing gloves/washing hands in between working colonies that have observable symptoms. Viruses can be present on your hands and tools for the short time period it takes to open a new colony.

MODULE WRAP-UP

Viruses

Our exploration of honey bee viruses has revealed the complexity of these pathogens and their varied effects on colony health. From seemingly harmless infections to those causing severe colony damage, we've learned to recognize key viral symptoms and understand their impacts. While these pathogens remain among the most challenging to manage, this knowledge equips beekeepers with the awareness needed to better monitor and respond to viral threats in their colonies.

Other Honey Bee Parasites

Honey bee parasites are organisms that live inside or outside a honey bee host and require the host to feed and/or reproduce. Parasites benefit at the host's expense. Honey bees are susceptible to a variety of internal and external parasites, and so far in this course we have described several serious enough to require their own modules: Varroa mites, Nosema apis, and Nosema ceranae. In addition to these three parasites, honey bees are susceptible to several others. This module explores the additional parasites that impact honey bee health, and discusses methods for identifying and managing them.

READ

History of Tracheal Mites

■ Tracheal mites are a parasite that, as Key Points the name suggests, reside in the The appearance of tracheal mites honey bees' trachea. The appearance in the 1980s caused an epidemic of of tracheal mites in the early 20th colony losses. Today they are a century was one of the largest honey minimal threat to colonies. bee disease epidemics in history, The Buckfast bee stock was bred wiping out thousands of colonies each for tracheal mite resistance. year in Europe, North America, and elsewhere until effective controls were developed. In this section, we will describe tracheal mite infestations and how beekeepers can manage infestations in their own colonies. Tracheal mites were first noticed on the Isle of Wight, an island just off the coast of the United Kingdom in 1904. Mysteriously and rather suddenly, honey bee colonies began collapsing on the island. Over the next 14 years, 90% of the colonies died. The problem spread to the rest of Great Britain, devastating the beekeeping industry and decimating feral honey bee colonies. The mysterious disease was named "Isle of Wight Disease" until, in 1921, scientists discovered the tracheal mite was likely one of the culprits of these colony deaths, along with Nosema apis and starvation. Shortly after the mite's discovery, the United States Honey Bee Act was enacted (1922) in an effort to keep the mite out of the U.S. by preventing the importation of honey bees from Europe. Brother Adam, a beekeeping monk from the Buckfast Abbey in England, bred

the Buckfast bee stock to be resistant to tracheal mites after 30 of the Abbey's 46 colonies died from the mite. This stock was adopted by many beekeepers, and is still in use today. Buckfast bee stock Despite the Honey Bee Act, tracheal mites were introduced into the U.S. in 1983, when it is believed that infested swarms crossed the border from Mexico. The movement of packages, queens, and nucs rapidly spread the infection across the U.S. and elsewhere. Importing bees from other countries where the mite was unknowingly present also contributed to its quick spread. Tracheal mites are now present in every continent except Australia. They are also not yet present in the province of Newfoundland in Canada. While they initially devastated honey bees, tracheal mites are not a major parasite threat today. The Beltsville Bee Lab has been tracking tracheal mites in the U.S. since their introduction. Based on samples submitted by beekeepers and state inspectors, their estimated incidence has decreased from a peak of 45% of colonies infected in the mid- 1980s, to a mere 2% in 2015. This reduction has likely resulted from 1) the development of resistant stocks, 2) natural resistance developed by honey bees that survived infestations, and 3) effective control strategies used by beekeepers once an infestation is confirmed. Nevertheless, infestations still do occur, so beekeepers should be knowledgeable about tracheal mites' impacts, symptoms, diagnosis, and treatments.

READ

Tracheal Mites Biology

■ The tracheal mite, *Acarapis woodi*, is a Key Points very small, white mite that measures Tracheal mites infest the trachea of about 150 micrometers in length (a honey bees, impairing their ability micrometer is one thousandth of a to breathe. millimeter). Their effect on honey bees With breathing impaired, infested is referred to as “Acarine disease.” They bees cannot fly and die at an infest adult honey bees of all castes earlier age. and do not affect brood. They reside exclusively in the prothoracic trachea— the tracheae located at the very front of the thorax, just at the honey bee's “neck.” To get to these tracheae, they crawl through the spiracle at the base of the wing of 1–2 day old adult bees, usually during the night when colony activity is low. After a few days in the tracheae, they begin to reproduce, laying eggs that will develop into adults. Their offspring mate with each other and mature inside the trachea. Tracheal mites only infest young bees, identifying them based on their odor. This allows the mites to complete their entire life cycle before the adult bee dies. Before the bee dies, the tracheal mites exit this bee's body through its spiracle and looks for other young bees to infest. Tracheal mite in honey bee trachea. Credit: Bee Informed Partnership. Inside the trachea, the mites feed on the bees' hemolymph by piercing through their tracheal walls with their sharp mouthparts. Honey bees experience reduced respiration as their breathing tubes are clogged with mites and become damaged and scarred. The bees also experience blood loss from the mites' feeding. Tracheal mite infestations tend to peak in autumn and winter in the Northeast US. When over 30% of bees in a colony are parasitized, the colony experiences a significant reduction in its adult population. As a result, the colony experiences decreased honey production and overwintering success. Colonies that survive the winter may build up slowly or die in early spring.

WATCH

Tracheal Mites Symptoms

Symptoms of tracheal mite infestations include the following: Adult bees crawling on the ground in front of the hive, unable to fly “K-wing” (in some cases) An inability to cluster in cool temperatures. When you open your colony on a cool autumn day, you expect to see a tight cluster of bees. With a tracheal mite infestation, you may instead see bees scattered on the frames, not in a cluster Dwindling adult population as infestation rates increase Watch this video clip of a bee in front of the hive, unable to fly. Bees that are parasitized by tracheal mites exhibit this behavior.

WATCH

Diagnosing Tracheal Mites

■ Tracheal mites can be diagnosed by Key Points microscopic examination, at home or With access to a dissecting at the Beltsville Bee Lab. microscope, diagnosing tracheal mites can be accomplished Tracheal mites can identified simply quickly and easily. and quickly using a dissecting microscope. These microscopes have low magnification, and you won't need slides. If you do not wish to purchase a microscope yourself, you can talk to your local bee club about a collective purchase. Alternatively, Cornell's Dyce Lab for Honey Bee Studies in Ithaca, New York, has a dissection microscope that you can use. Collect bees that are crawling on the ground, unable to fly, as these bees are the most likely to have tracheal mites. It is best to collect live bees, as diagnosis is difficult in bees that have died, especially if they have begun to decay. Keep collected bees in the freezer or in 70% alcohol until you are ready to dissect them. The dissection process involves removing the honey bee head and front two legs to expose the prothoracic trachea. You will be examining the trachea to look for the presence of 1) tracheal mites, and/or 2) scarring and discoloration that is caused by tracheal mites. Dissect around 20 different bees to test for tracheal mites. If you find 10-20% of the bees you dissect have tracheal mites, it is time to treat your colonies. Watch this video to see Dr. Christina Wahl conduct a tracheal mite dissection. Video Transcript So, I'm sitting here in Emma Mullen's office at Dyce Lab and we are going to be looking at some bees to see if they have tracheal mites. There was a colony here that had some bees crawling around on the ground in front of it, and it seemed like a weak colony. So, Emma grabbed some of those bees and froze them. And we're going just see, can we find out what's the matter? Now, if you are very happy with investigating your bees, you will want to try this yourself some time. And you need a little bit of equipment. The single most important thing you're going to need is a dissecting microscope. And unfortunately, those tend to be kind of pricey. You can borrow one, often, and you can find your bee club that maybe has one, or you can get some advice from somebody who knows a little bit about microscopes to get yourself a decent one. Don't spend money until you get some help choosing a microscope. That's important, in my opinion. But, everything else is cheap and easily acquired. So, what do you need? What you want to have is a little square of some kind of corkboard. You can get it, it doesn't matter what shape it is, but a little piece of corkboard that you can stick some pins into. And you want a hot glue there down under some kind of surface. This happens to be a plastic Petri dish, but it could be any other kind of a dish. Just not too high sides because you have to access it with your hands. And here's our bee we're going to dissect. And then we want a little pair of sharp scissors. These are regular dissecting style scissors that are reasonably good quality. So they are sharp. But, what I recommend you do is just get a good pair of personal grooming scissors, cuticle scissors, or something like that, because they tend to be nice and sharp and you want them to be small. And an eye dropper helps. This is a disposable pipettor, but an eyedropper works. And then we

have these little tools here, which are homemade. All you need to do is get yourself a couple of needles or pins, the smaller the better, the thinner the better, and some popsicle sticks. And you take the pin, if it's very sharp and thin already, you're all set. If it's kind of on blunt end side, hold it with a plier and use a blow torch on the tip to burn away some of that metal and make it nice and sharp for yourself. When it's sharp, then you take the blunt end, you shove it into the popsicle stick, and you hot glue it so it stays put. And you have a tool that's perfect for dissecting bee parts. And a couple of pairs of forceps. Here's a straight forcep, here's a curved forcep. I like the curved ones, you may or may not choose to use them, you could just use two straight ones. If you buy cheap versions, then they're going to be kind of blunt. And you'll need to sharpen them yourselves. And so, if you have any experience with this, any kind of knife sharpening, you'll have a sharpening block, or you can start with a, sort of a medium to fine sandpaper and get those tips as fine as possible. And then smooth them off as best as you can to nice, sharp tips. So, that can be done with cheap forceps, and you are ready, good to go. And I like to have some sort of a cap to cover my dissecting needles with because you can hurt yourself on those, and you can also break them and then you're stuck having to go back and make new ones. And so, I got a disposable pipettor tip here. But be creative, come up with something, and make your own. And we need water, and a honey bear works just great, and an eye dropper. So, those are the tools. Now, we're going to start doing our job, dissecting this bee. So, the object is going to be to take the abdomen and most of the wings off the bee. And you can usually do that in one fell swoop with your scissors. You just go behind the thorax, hold it with the forceps here, like this, and cut through the wings and the abdomen where it joins the thorax. So, we're not going to use the abdomen. If you've only collected a few bees and you also want to check for Nosema, you'll be saving those abdomens. But the part we're going to work with right now is this. So, I'm going to take that back over to our cork, here. And you're going to probably want to use your forceps to hold this bee down. It's going to want to jump around on you because the cuticle is pretty tough, and pins tend to bounce right off it. But, you're going to try and pin this thing. It may take a couple of trials here. You want to pin it so that you're not going through where the wings are because you're going to be looking at the tracheal tubes, which are right behind the wings. So, I'm pinning this one at about 12 o'clock, like that, into the cork. And then the other pin, I'm going to try and put at, not necessarily at six o'clock here, but a little off center. It seems to work pretty well. Oops. You'll get practiced with this, and like me, you are going to find that even though you've done a hundred of them, you'll wrestle with that bee. So, bee's head is straight up, her tongue is right here, looking at me. What we're going to do now is take that head off, and what we call the prothorax, which is attached to the front pair of legs. So, the thorax has parts to it, several different sections, some of them are fused, but the one right behind the head is not fused to the rest of the thorax. It's called the prothorax and it has the first pair of legs attached. And I'm going to be going under the head. And the first pair of legs here, and I'm going to grab it under those first pair of legs, and I'm not going to squeeze; I'm just closing the forceps enough to grab the head. And then I'm going to pull straight up, there. So, you can see here I got the two legs and head. If we did a good pull of the head, we won't have to do too much additional dissecting. We'll see in a minute. Here's where the water comes in handy. Because the bee is dead, all the interior organs have sort of collapsed. And we want to suspend them so that we can see that nice trachea. And so, now we're ready to look at the microscope. And make sure that it's on the lowest power magnification so that it's easy to find where the subject is. Get it into focus here. And I've got what I want. I'm looking sort of down the thorax

without the head into the body of the honey bee. Now, the object that we're trying to find is the main tracheal tube. That's where the air comes into the bee when she breathes. So, it's right behind the two pairs of wings. So, there's one pair of wings, second pair of wings, and then there's an opening for this big tracheal tube. When tracheal mites infect bees, that's where they crawl in, and that's the biggest of the windpipes in the bee. So, if there's a tracheal mite infection, you'll see them in there because they're congregating in there, they mate in there, and they do all their business in there. Which is why bees die, because they can't breathe very well with those things in the way. And most of the prothorax came off, but I see here, now, that I've still got a little bit on there. Here's a pair of the wings, what's left of them. This is on the animal's left side, and up above it is where the head was. And this little piece here that I'm pointing at is what's left of the prothorax, that collar right behind the head that I was trying to pull off. So, I didn't actually get the whole thing off. That happens, you just can't control it. So, I may have to take that off. And that's where these tools come in handy later. And because I'm going to have to do that, probably, I'm going to re-pin, this pin came out, I'm going to have to stick it back in again. So, it's more or less satisfactory. You'll find that your versions may vary. Your mileage may vary. I'm going to put a single drop of water on top of this hole now, and that really highlights everything in there much better than without it. So, it's an essential step. Don't forget your water, or you're really going to be struggling to identify anything. We focus the microscope, and I'm going to take my two little dissecting tools, and I'm going to go into the thorax and push away the glands here that are kind of obstructing the field of view. And right there, and I don't know if you can see that, but my right needle, here, that's the right needle, the one that's left, has just been poking at one of the big thoracic trachea. It's kind of silvery looking, and silvery generally means that there are no tracheal mites in that bee. If the trachea has a brownish cast or has black spots in it, then you have evidence of scarring, which is irritation, and that's caused often by these tracheal mites. Now there can be other things that cause tracheal scarring, but evidence of scarring is certainly suggestive of a tracheal mite infection in the honey bee, and even more, it suggests if you actually see the mites. And they will clog those two big trachea, so you can physically see them. And okay, there we go. So, I just peeled back the remnants of that prothorax, and I can see a great big tracheal tube just laying there. And here's where my two dissecting tools come in handy because now I'm going to go in to this opening that I've made and try and tease the tracheal tubes into a position that I can see them more clearly. And so, the back of the bee, top of her back is here, that would be 12 o'clock. Middle of her ventral midline, between the legs is down here, that'd be six o'clock. So, the tracheal tubes are at nine o'clock and at three o'clock. The big ones. Now, just to be thorough about this, I want to tell you that those tracheal tubes don't just stop the way we see them here, they get smaller and they branch, and they go through the whole bee and carry air everywhere, right up to the cells. And the air exits the bee through the abdomen. So it's a one-way street. The air goes in at the thoracic spiracle here, cycles through, and comes out of the abdomen. And this thoracic entrance is where most of the tracheal mites gain access to the bee. According to what has been written about this, the bee becomes infected by tracheal mites during her first couple of days of life. After that, she seems to get tougher, or her hair grows stiffer, or something, and they don't get in so easily. So, the infection happens after the young bee has emerged, while she's still hanging around in the hive. And they proliferate in there, and they can exit and infect another bee because they're crawling around on the surface of that bee, and then they'll jump off and get into some other bee when an opportunity comes along. So, that's pretty much it. This pair of trachea look normal, there's no

scarring, they have a slightly shiny appearance, they're transparent. I don't see any lumps in them. So, this bee did not have tracheal mites. Beltsville Bee Lab Test While dissecting honey bees yourself is the fastest method to diagnose tracheal mites, beekeepers may not have access to a microscope or confidence in their dissecting skills. In these cases, a good alternative method is to send a sample to the Beltsville Bee Lab. This is free of charge and beekeepers can expect their results in approximately two weeks. This link explains how adult bees should be shipped for tracheal mite analysis.

READ

Treating Tracheal Mites

There are several ways to treat and prevent tracheal mites infestations, such as by using resistant stock or employing chemical or non-chemical controls. Requeening with Resistant Stock Several stocks of honey bees are fairly resistant to tracheal mites, including New World Carniolans, Buckfast, and Russians. Using these stocks reduces the likelihood of a tracheal mite infestation developing in the first place. If you do experience an infestation, we recommend requeening your colony with a resistant stock after providing them with a treatment. Grease patties can help clear tracheal mite infestations. These patties are made by mixing 1 part vegetable oil with 3 parts powdered sugar to form a patty. Mix the ingredients very well until it has the consistency of Play-Doh®. Place it in between two sheets of wax paper, flattening it to the thickness of a thin pancake about ten inches in diameter. Place the patty on the top bars of the frames over the brood nest (beneath the queen excluder, if you are using one). As the bees consume and remove the patty, they become covered in grease, masking their odors. This interferes with the mites' ability to detect young bees to infest. Eventually, the old mites die without reinfesting new bees. Caution is required with this treatment, however, because it can attract small hive beetles and other pests. Menthol (under the trade name Mite-A-Thol™) is currently the only registered treatment for tracheal mites. The menthol chips arrive in a mesh bag. Place the entire bag on top of the brood nest (beneath the queen excluder, if you are using one) to prevent bees from removing the individual crystals, and leave it ■ Non-Chemical Control Using Grease Patties ■ Chemical Control ■ there for a minimum of 28 days. The temperature must be between 60–80°F (15–27°C) in order for the product to be effective. Temperatures above 80°F can lead to adult and queen mortality. As the menthol evaporates, it enters the bees' tracheae and kills the mites inside. Wear protective eyewear and gloves when applying this treatment. Menthol cannot be used when honey supers are present. Honey supers can be added 30 days after the treatment ends.

READ

Zombees: Honey Bees Parasitized by Phorid Flies

■ From 2010 to 2016, the media was Key Points abuzz with stories on “zombees”— Phorid flies parasitize bees, honey bees that left their hives at night altering their host's behavior so and behaved strangely. Beekeepers that they fly to light sources and scientists wondered if this new during night. ailment could explain Colony Collapse Phorid fly parasitism is uncommon Disorder, a phenomenon where bees in the U.S. and is not a major suddenly abandon their hive. As it concern for honey bee health. turns out, zombees are not a major issue for North American beekeepers; nor are they behind Colony Collapse Disorder. Nonetheless, it's important to understand this parasite in case it does become widespread and to field any questions from beekeepers or the public that have seen the media

stories. What Causes Honey Bees to Become Zombees? In 2010, researchers discovered that a species of fly, *Apocephalus borealis*, commonly called the phorid fly, was behind some odd behavior of honey bees. This phorid fly is native to North America and historically parasitized only paper wasps and bumble bees. Recently, it jumped hosts and is now able to parasitize honey bees. The phorid fly is a small white fly, 2 mm in length, that resembles a fruit fly. When a phorid fly parasitizes a honey bee, it inserts its ovipositor into the bee's abdomen and lays eggs. The eggs hatch and the larvae (maggots) crawl through the bee's body, feeding on it while the bee is still alive. Within a week, the bee dies from being consumed from the inside. After the bee dies, the larvae exit the bee's body; as many as 15 maggots can crawl out of one bee. They pupate outside of the bee's body, and adult flies emerge after 28 days, ready to mate and begin the cycle again. Phorid fly laying eggs in honey bee Photo from: Core A, Runckel C, Ivers J, Quock C, Siapno T, et al. (2012). "A new threat to honey bees, the parasitic phorid fly *Apocephalus borealis*" Symptoms Before the bee dies, its behavior is changed by the parasite. The main symptom is that infested bees abandon their hives at night. These bees tend to be attracted to light and can be observed flying around streetlights and outdoor house lights, even during rain or cool temperatures when they don't normally fly. The bees become stranded at these light sources and die. Disoriented bees may also walk in circles, or be unable to stand without falling over. Because of these strange behaviors, honey bees parasitized by these flies have earned the colloquial name of "zombees." It is still unknown why parasitized bees abandon their hives at night and are attracted to the light. It may be that this behavior benefits the phorid fly's reproduction in some manner, or it may be that abandoning the hive reduces the chances that the bees' nestmates also become infested. More research is needed to investigate this behavior. All bees that become parasitized die, and research has found that up to 18% of foragers in a colony can be infested. However, while it can stress a colony, entire colonies rarely, if ever, die from the infestation. More research is needed to understand what role the phorid fly plays in modern day colony losses, if any. There are currently no treatments for phorid fly infestations. Documenting Incidences The scientists who first discovered the phorid fly's parasitism of honey bees in the U.S. launched a citizen science project in 2012 to document cases of infestation in North America. Sightings are entered on the website <http://www.zombeewatch.org>. Thanks to these reports, we now know that the fly parasitizes bees in many U.S. states and Canadian provinces along the west coast and in Northeast North America. The U.S. states include Maine, Vermont, New York, Pennsylvania, Virginia, North Carolina, South Dakota, California, Oregon, and Washington. While the fly is widespread, reported cases are very low, with just under 100 confirmed incidences in the U.S. as of June 2017.

MODULE WRAP-UP

Other Honey Bee Parasites

Beyond the major parasites like Varroa mites and Nosema species, we've now examined the broader range of internal and external parasites that can affect honey bees. Understanding how to identify and manage these additional threats adds another important layer to our knowledge of honey bee health management, helping beekeepers maintain stronger, healthier colonies.

Brood Diseases

One fungal brood infection and two bacterial brood infections can pose a health risk to honey bees: chalkbrood, American foulbrood, and European foulbrood. These diseases infect and kill larvae and pupae, but do not impact adults. Chalkbrood is a fungal infection that can have minor to moderate implications on bee health. American foulbrood, on the other hand, is the most infectious and destructive honey bee bacterial infection. While foulbrood is arguably the oldest disease that beekeepers have been aware of, its prevention and management is changing due to new FDA regulations regarding antibiotics. In this module, we will describe these brood diseases and how to properly diagnose and manage them.

READ

Chalkbrood

■ *Ascosphaera apis*, commonly referred to as chalkbrood, is a fungal disease. Chalkbrood is caused by the fungus that infects honey bees in the larval stage. The fungus reproduces through brood turns hard and white, spores, and larvae consume these resembling a piece of chalk. spores in their brood food. The spores Fruiting bodies make the germinate in the hindgut of the larvae, mummies develop brown or black and once the cell is capped, they coloration. become vegetative and the active Chalkbrood mummies can be infection begins with visible observed in the brood cells, on the symptoms. Drone and worker larvae bottom board, or on the ground in are infected more often than queens. front of the hive. The genus *Ascosphaera* contains Chalkbrood can be managed by many species of fungal diseases that requeening and ensuring colonies infect bees. While honey bees are strong and housed in dry hives. infected by *Ascosphaera apis*, other species of *Ascosphaera* can infect bumble bees, leaf cutting bees, and mason bees with chalkbrood. In honey bees, the vegetative fungus resembles very small white threads. As it overcomes the larva's body, the larva becomes covered and eventually comes to resemble a piece of chalk. Hard, white larvae in this form are called chalkbrood "mummies." After the larva dies, the fungus develops fruiting bodies that eventually produce spores to reproduce. These fruiting bodies first appear as brown or black spots on the chalkbrood mummy and eventually, the mummy turns completely black as it becomes covered in fruiting bodies. Mummies can be seen inside their cells in the brood frame. Depending on the fungus' life stage, the mummies can appear white and chalky in appearance (with or without black spots), or can be completely black and hardened. Mummies can be removed from the brood frame by tapping the frame against a solid surface. This action helps differentiate chalk brood from other brood diseases—tapping the frame does not easily dislodge brood with other types of infections. Chalkbrood mummies are often removed by the bees through hygienic behavior and left on the bottom board, at the entrance of the hive, and/or on the ground. Like most fungi, the chalkbrood pathogen thrives in an environment that is cool and humid. Mummies are most often seen along the periphery of the brood nest where temperatures aren't as well-regulated. They are most common in spring, when brood may be exposed to cool temperatures, or in colonies with small populations that are unable to effectively maintain a warm brood nest. Moisture in the hive can also help promote the growth of chalkbrood. Honey bees can create a "social fever" in an effort to kill chalkbrood infections. They accomplish this by increasing the temperature of their brood nests by shivering their wings. Bees elevate the brood nest temperature just enough (around 0.9°F, or 0.5°C) to exceed the temperature for chalkbrood to develop, but within the normal developmental temperature window for bees. Social fevers can be initiated after

symptoms have begun developing or before symptoms have started. It is still not known how bees identify larvae infected with chalkbrood. Preventing Chalkbrood The key to avoiding chalkbrood is to maintain strong colony populations (with sufficient bees to keep the brood nest warm) and eliminate moisture in the hive. Avoid making splits or nucs in early spring without enough bees to warm the brood nest. More generally, ensure that your hives are not in cool, damp places, and provide ventilation in the hive bodies. It is recommended that the two oldest brood combs in each hive body be rotated out each year to remove any existing fungal spores that may be present. Treating Chalkbrood Chalkbrood is not regarded as a detrimental disease to the colony as infections usually clear up in the summer as the temperature rises. It rarely impacts the queen, and nurse bees can usually prevent or control it with hygienic behavior. There are no registered treatments for chalkbrood control. If infections do not clear up on their own, requeening (ideally with hygienic stock) may help a colony clear up the infection. If more than 10–20% of brood cells on a single frame are infected, remove those frames and replace them with frames of drawn comb or foundation.

READ

American Foulbrood: An Overview

■ American foulbrood (AFB) is a bacterial Key Points infection caused by the bacterium *Paenibacillus larvae*. The bacterium is the most infectious bacterial disease, with has two states: the vegetative state (when it actively causes infection and reproduces) and the spore state (when it does not cause infection). The vegetative bacterium reproduces by creating many spores, which can be found in honey and bee bread, as well as on wax and woodenware. AFB spores under a microscope. Credit: Bee Informed Partnership To contract the infection, young larvae consume spores that are fed to them by nurse bees, which they have picked up from food or from feeding an infected larva. Worker, drone, and queen larvae are only susceptible within the first three days after they hatch from the egg. Furthermore, as larvae age, they are less susceptible to AFB. Only one spore is needed to infect a one-day-old larva, while more are needed in two-day-old larvae, and even more in three-day-old larvae. Beyond that age, larvae do not typically develop the infection, even if they consume spores. Once a spore is ingested by a larva of the right age, the bacterium becomes vegetative. After one day it begins replicating in the larva's midgut. The infection continues to multiply as the bacteria feeds off the larva. Soon after the cell is sealed, the prepupa dies. The prepupa becomes a brown, mucous- like substance that lays flat on the bottom of the cell, eventually drying out into a black, brittle scale. The colony ultimately dies either because adults are no longer replaced or because there are too few of them to protect the colony against robbing. Spores can be present in the hive at any time of year and a colony can become infected whenever young larvae are present. Larvae laying flat in cell turning brown in color. Credit: Bee Informed Partnership What's the Big Deal? Although incidence is relatively low in New York State (less than 2% of colonies become infected each year) and elsewhere, the problem with this disease is that the bacteria remain alive in the spore state for over 40 years. The long-lived spore is resistant to changes in the weather, hot and cold temperatures, bactericides, and antibiotics. Colonies that are positively diagnosed need to be managed carefully and immediately to reduce the chances of the infection spreading. Most states in the U.S. and all provinces in Canada have inspection programs. with the top priority being to inspect for and control cases of American foulbrood. In New York State, colonies that

are leaving the state for pollination, as well as packages and nucs for sale, must be certified as AFB-free before their departure or sale. These inspection programs have been very successful in reducing AFB incidence and spread for over a century.

READ

American Foulbrood Symptoms

■ The early and late stages of a colony Key Points infection have different symptoms. In The telltale symptoms of American the early stages, you can expect to foulbrood are coffee-colored larvae observe the following: laying flat in their cells that rope when pulled. In some cases, the Spotty brood pattern with infection is associated with a foul sunken/concave or perforated odor. cappings. Caramel or coffee-colored prepupae. Prepupae laying flat on the lower cell wall with a mucous consistency. Prepupae stringing out into nearly inch-long "ropes" (2 cm) when pulled with a toothpick or small twig. (Note that prepupae in the very early stages of infection may not rope out.) If death occurs in the pupal stage, some pupae may have their tongues sticking out. Distinctive foul odor in some cases (though not all cases). The odor smells similar to old gym socks. It is worth noting that colonies can give off a foul odor in other circumstances, such as during infections of European foulbrood, during brood decay after prolonged colony death by any cause, and during the ripening of goldenrod honey. Ropy prepupa. Credit: Paul Cappy Black scale. Note we are looking down the face of the frame, and scale is on the bottom of the cells. Credit: Paul Cappy Heavily infected colony. Credit: Bee Informed Partnership Many infected "molten" prepupae Concave, moist cappings and molten prepupa Perforated and concave cappings Pupal tongues A lot of scale in stored comb In the late stages, usually after a colony has died or when the colony is nearing death, you can expect to observe the following: Dead prepupae become a thin, dark scale along the lower cell wall. In this stage, they are no longer "ropy." This scale does not pick out if you scrape it with a toothpick or small twig. Distinctive foul odor in some cases (though not all cases). Few adult bees present (toward the ultimate demise of the colony). Sunken cappings and black scale. Photo credit: Bee Informed Partnership

WATCH

Inspecting for an AFB Infection

■ Three times a year, do a Key Points comprehensive inspection for AFB Comprehensive inspections for (spring, post honey harvest, and before AFB should occur three times a wintering). This requires you to go year. through every brood frame in every hive, looking for the symptoms described on the previous page. To make sure you don't miss any cells, shake the bees off the frames as you check them. Notice whether the brood pattern is solid or spotty, then carefully examine the frame to look for cappings that are concave or perforated. Anytime you see a cell like this, use your hive tool or a small twig or toothpick to open up the cell. If you see a larva or pupa that is the distinct caramel color, it's time to do a test. Even if only one cell is affected, test it. It's best if you catch the infection in an early stage. Watch as Paul Cappy, New York State apiculturist, demonstrates the proper technique to inspect frames for American foulbrood. Video Transcript When you're looking at a frame of brood to determine if you've got American foulbrood, you first pull it out of the brood nest and shake off all the bees. And then, you hold the top bar toward you and the bottom bar away from you. This gives you the correct position, since the individual cells have a slight upward slope. And so, from the top bar you can look

down and look at the side, the lower side of the cells. And so, when you're holding it, you want to be able to rotate it in such a way that you will find the proper angle where you're actually looking down into the cell on the bottom side of the cell for the dead larva. It can be chalkbrood, Sacbrood, European, American, all of these, it can be any of those. And so, you practice by looking at the brood to start with. Unhealthy, but as you go along you'll start to pick up where cells haven't hatched out, they're sunken, or there seems to be something wrong, perforated, this type of thing. So, it's actually important to look and see the bad cells. And the good cells are always convex, they're oval. And the cells that will not hatch, they're usually sunken, or concave, and they're dark in color and can also be perforated. So, those are the kinds of cells that you're looking for. But it's very important to hold the frame, to get the right angle. So, you take a hold of the top bars and you rotate this around until you can see all the cells. There's about 2500 cells on the side of a comb. So, when you're looking, you should be able to look with a quick glance across the comb and be able to pick out the cells that did not hatch. And one of the last things is that when you're trying to do this, it's nice to do it on a sunny day, so the sun can be at your back and you can then see the sun that will come over your shoulder and look down in the cell and make it shine into the cells so you can see the disease quite nicely. And the final point is, when you're looking for American foulbrood, make that the only thing you're going to do when you're going through the colony. Just concentrate on looking for American foulbrood on four to six combs. And just look at all your colonies in a bee yard, concentrating, just looking for disease. Do not look for other things just concentrate mainly on looking for the disease so that you can easily pick out the disease.

WATCH

Tests to Diagnose an AFB Infection

■ There are four tests that you can use to Key Points diagnose AFB. The two most reliable The two most reliable ways to are the Vita AFB test kit and the diagnose AFB are to use the Vita laboratory test at the Beltsville Bee test kit and to send samples to the Lab. Additional tests, though less Beltsville Bee Lab. reliable, include the rosy test and the Holst milk test. Label any colony that is suspected to have AFB so that you can easily find that colony again later, should an inspector need to visit it. The Vita AFB Test The Vita AFB test is a small kit available from bee equipment supply companies. They cost about \$13 and are very convenient. The results are ready in about three minutes. At the Dyce Lab, we keep a couple of these test kits in our inspection tool box at all times. That way, if we see a suspicious brood cell we can take a sample right then and there. To use the kit, simply scoop out some of the suspicious brood and place it inside the small test tube with liquid in it. Shake the contents together and apply two or three droplets to the test kit. After two minutes, the results will be available. Watch this video to see how to use an AFB Vita test kit. Video Transcript The Vita AFB test kit is a really simple and convenient way to test whether or not your colony has American foulbrood. And it can be done right in your bee yard. Let's take a second to go through the steps of how to use this test. So, the test comes in this packaging and everything you need is inside of it. You'll see the expiration date in the bottom, left corner of the package, so make sure that you're always using a current test. When you open it up, the first thing you'll see are instructions. Make sure you read these so that you're actually following the protocol properly. Next, you'll also see the solution, which contains buffer and some of the antibodies and enzymes that are going to make this test useful. And the test kit itself is in this package, which we'll open a little bit later, once we're ready. You'll also see a plastic rod that we're going to use

to extract the larvae or pupae from their cells. And this plastic pipette that we can use to suck up some of the contents. So, we're in this colony that suspect might have American foulbrood, and we're going to remove at least one infected looking larva or pupa. And we're going to put it right in this buffer. You can extract more than one if you would like, but one is all that it takes to activate this test. You're then going to shake the contents together for at least 30 seconds. So next, we're going to open our package to reveal our test kit. And while we're doing this, we're going to allow our solution to sit for two minutes so that the chemical reaction can occur. So, inside the solution there are antibodies that are linked to enzymes. And if the antibodies bind to AFB bacteria, so if the bacteria is present in our sample, then the enzyme will give a color change. So, we're going to use the pipette to suck up some of the fluid from the bottle, and then we're going to apply two to three drops on this circle part of the test, right here. And then, it'll take about two minutes for the test to work. And, we're going to speed things up for you so you don't have to wait too long. You will start to see that this fluid moves across the test kit. So, in this example, we only see one line under the C, for control. There is no line under the T, for test. And this means that our colony does not have American foulbrood. If your colony does have American foulbrood, then you will see two lines: one under the T, and one under the C. Even if the line is very faint under the T, this is still a positive response. If your test kit shows that there are no lines, not a test line or a control line, then this means that the test was defective and you should use another one. And there you have it. That's how you use a Vita AFB test kit in your own bee yard.

Ropy Test Before test kits and laboratory testing, many inspectors in the field commonly diagnosed AFB based on the ropy test. With the ropy test, a toothpick is inserted into the cell, the goopy contents are stirred in a circular motion once or twice, and then the toothpick is slowly lifted out. If the contents create a rope about an inch long, it is a positive diagnosis. This test isn't always reliable, however, as AFB larvae do not always rope out and the presence of ropiness depends on the time of the infection. Furthermore, European foulbrood (another bacterial disease) can also rope out, confounding the diagnosis between these two infections. Doing a quick ropy test and finding that the infected brood does rope out is a strong indicator of AFB, but a Vita test and/or a Beltsville Bee Lab test should always confirm it.

Beltsville Bee Lab Test Another option — if you live in the U.S. — is to send a sample to the Bee Lab in Beltsville, Maryland, where they process samples free of charge. The Beltsville Bee Lab tests for the presence of American foulbrood, as well as antibiotic resistance. It is important to determine whether any confirmed infections are resistant to traditional antibiotics. Information on how to submit a sample to the Beltsville Bee Lab can be found [here](#). The results of the lab test are usually returned in two weeks. Because of this delay, it is best to use the Vita test kit while in the field and simultaneously take an additional sample to send to the Beltsville Bee Lab for diagnosis confirmation and resistance testing. If you have a positive result from the Vita test, you can follow the necessary procedures.

The Holst Milk Test To conduct the Holst milk test, scoop out the ropy larva with a twig and place it in a plastic or glass tube or jar. If you are testing scale, rip out the entire bottom of the brood cell using tweezers and place it in the tube. Add a small amount of water, enough to generously cover the contents in the tube or jar. Next, add a few drops of milk (skim is preferred) or a pinch of 1% skim milk powder. Enough drops are needed to make the water begin to look cloudy. Swirl the contents together and let the mixture sit for 30 to 40 minutes. If the milk solution becomes translucent and yellow (similar to the color of apple juice), it indicates the presence of AFB. If the brood is infected with European foulbrood, the solution may also change color slightly but remain cloudy. The reaction can be sped up by placing the container in hot

water (up to 165°F/74°C). This test is not always accurate and should be accompanied by a Vita test and/or a Beltsville Bee Lab test. The Holst milk test for American foulbrood. The vial on the right contains AFB, while the vial on the left does not.

READ

You Have a Positive AFB Result; Now What?

■ If you receive a positive result with the Key Points Vita test kit or from the Beltsville Bee In New York State, all AFB-infected Lab, report your result to your colonies must be reported to the state/provincial apiculturist. In New state apiculturist and destroyed by York State, it is the law to report all burning with the oversight of a cases of AFB. Find out the laws bee inspector. surrounding AFB reporting and treatment in your state/province/country. First and foremost, disinfect your equipment and wash your hands thoroughly. Do not open up another colony until everything has been cleaned. Scrape off all traces of propolis and wax from your hive tool, scrub it with a stainless steel wool pad, and soak it in a bleach solution. After it's dried, the hive tool can also be inserted into a smoker with an open flame for further disinfection. Use bleach wipes or bleach-soaked paper towels to clean the bellows of your smoker and any other items that may have come in contact with spores. Change your clothes. Bee jackets and leather gloves can be washed in the washing machine, but if leather gloves appear to remain contaminated with wax or propolis after washing, it is safer to discard them in the trash. Any toothpicks or twigs that were in contact with the infected larvae should not be thrown on the ground. Instead, they can be placed inside the affected hive or can be disposed of in a sealed garbage bag. Place all used test kit materials and tubes in a sealed garbage bag. Once you have a positive diagnosis, do not remove the hive from the apiary. The hive is considered to be in quarantine. In New York State, the state apiculturist will send an inspector to visit your operation. If a sample has not yet been sent to the Beltsville Bee Lab, they will send one to verify the results and test for antibiotic resistance. The inspector will inspect all of the hives in the affected yard to determine the extent of the infection. He or she will ask questions to help determine where the disease originated. If, for example, the colony came from a nearby apiary or nuc producer, inspectors may visit those sites to determine whether that was the source of the infection. Though most bees forage no more than two miles from the hive, some bees may forage over an area five to six miles or more in diameter, so tracking down a source of infection can be difficult. For this reason, we recommend keeping detailed records of where bees were purchased, which colonies came from swarms or building cut-outs, and where colonies have traveled. Beekeepers should also communicate with neighboring beekeepers to learn whether they are also monitoring their colonies for disease. Destroying Colonies In the state of New York, all confirmed colonies must be burned. Do not attempt to control the infection by discarding affected frames or treating with antibiotics—the entire colony must be burned with an inspector's oversight. The entrance to the colony is screened shut and the hive bodies stapled together to prevent bee escape. A hole is dug in the ground and the hive (including bees) are burned in it. After the burn is completed, the hole is covered up with enough dirt that an animal cannot uncover it. If the hole is in an agricultural field, the hive must be buried below plow depth. If the area is dry and fires are not permitted, the colonies can be buried deep in the ground on the beekeeper's property or in a landfill. Your county will issue a permit to bury it in a landfill. As with burning, the inspector must be present during the procedure. In some cases, the inspector may determine that not all the equipment has to be burned.

Some outer covers, inner covers, and bottom boards that do not have deep cracks or rotten wood may be able to be scorched with a blowtorch and used again. Not all states and countries have the same protocol as New York. It is important to determine the protocol for your region. A hole is dug in the ground. Credit: Jon Ryan Infected equipment is added and ignited. Credit: Jon Ryan Credit: Jon Ryan Credit: Jon Ryan The hole is covered up. Credit: Jon Ryan

WATCH

The History of Antibiotics

■ While you are not allowed to treat Key Points infected colonies with antibiotics, you While many beekeepers would can treat other uninfected colonies in treat colonies with antibiotics the yard to reduce their likelihood of prophylactically in the past, this contracting the disease. Antibiotics management practice is no longer can kill vegetative *Paenibacillus* recommended. larvae, but it cannot kill the spores. The three antibiotics available in New York are the following: Oxytetracycline hydrochloride (brand name: Terramycin) Tylosin (brand name: Tylan, Pharmasin, Tylomed-WS) Lincomycin (brand name: Lincomix) Historically, beekeepers applied these antibiotics prophylactically to their colonies in spring, fall, or sometimes both. This practice was especially common among commercial beekeepers, as a positive diagnosis prevents them from being certified to sell packages or nucs or transport their colonies out-of-state for pollination. An AFB diagnosis can translate to a considerable economic cost for some beekeepers. For this reason, beekeepers fed antibiotics to their colonies to reduce the chances of an outbreak occurring. Unfortunately, the overuse of tetracycline resulted in *Paenibacillus* larvae developing resistance to the drug, making it temporarily ineffective in preventing further outbreaks. It is not only that *Paenibacillus* larvae has developed resistance to tetracycline; many other bacteria have developed resistance to this antibiotic as well. Such antibiotics are used for honey bees as well as other livestock and humans. Consuming meat and other food products from animals, as well as coming in physical contact with animals that carry antibiotic-resistant bacteria, can cause antibiotic-resistant infections in humans. Once bacteria are resistant to a specific antibiotic, that antibiotic becomes ineffective at treating the infection. In serious cases, antibiotic-resistant bacterial infections in humans can be very difficult to control, putting a patient's life at risk and placing a considerable burden on the healthcare system. Escalating antibiotic resistance has been identified as one of the top health challenges for medicine, as two million people develop antibiotic-resistant infections and 23,000 die each year from these infections in the United States alone. Watch this video to hear Paul Cappy discuss the history of antibiotic use and resistance in New York State. Video Transcript In 2001, after a five-year gap of no bee inspection, general bee inspection being done, New York State brought on many bee inspectors to start to look for American foulbrood. At that point, Beltsville was showing that about five percent of the cases of American foulbrood were resistant to Terramycin. And as time went on to 2005, the resistance level grew to 30 percent of the colonies were actually resistant to Terramycin. And, at this point, in December of 2005, Tylan was registered, and in 2006, the commercial beekeepers started using Tylan aggressively. And as a result, they actually reduced the number of bee yards that were positive with American foulbrood by 91 percent in 2006. Along came 2007, and the sideliners actually started to use Tylan. And it took them about two years, but they dropped the number of apiaries that had American foulbrood by 88 percent in that two-year period. So, there was a tremendous drop of American foulbrood during that period of

time. In 2001, we had 8.6 percent, or over 400, cases of American foulbrood. And by 2009, we have 0.8 percent of cases of American foulbrood. And this was a substantial drop. But the drop was caused mainly by bee inspectors. But the other portion of the drop was really by having a new antibiotic, Tylan, that actually helped tremendously to lower the amount of foulbrood. And it's been a good tool in the toolbox ever since.

WATCH

New Antibiotic Regulations

■ Because of increased resistance to Key Points oxytetracycline and other antibiotics in As a result of bacterial resistance a range of human and livestock to tetracycline, new FDA infections (including *Paenibacillus* regulations require a veterinarian larvae), the FDA now requires a prescription or a veterinary feed veterinarian prescription or veterinary directive to obtain antibiotics. feed directive for antibiotics administered to food-producing animals (animals that provide meat, eggs, milk, honey, etc.). Bee inspectors are not allowed to write prescriptions. There must be a confirmed case of AFB in order to acquire a prescription or veterinary feed directive. Should you need a prescription for antibiotics, the veterinarian will explain how to feed it to your bees. It is most often mixed with powdered sugar and dusted on the colony. To reduce the likelihood of residues in honey, the treatment must be applied at least four weeks before supers are added. These regulations came into effect across the entire U.S. in January 2017. If you are in the US, you can find a list of veterinarians at the Honey Bee Veterinarian Consortium website. There has been a lot of concern from the U.S. beekeeping community around this new regulation. In several discussions surrounding the new laws, beekeepers have admitted to performing desperate behaviors to continue using antibiotics prophylactically. Dangerously, some of these antibiotics being purchased are prepared for other livestock and the concentrations/doses altered for bees. These actions might buy beekeepers some time to continue treating their colonies prophylactically, but the risks of antibiotic misuse impact more than just the beekeeping industry. It impacts the quality of our food and the quality of life in other animals and humans. The best way to minimize risk and infection spread is for all beekeepers to report all infections, and for all infectious colonies to be destroyed. It needs to be a large-scale, community-wide effort to ensure AFB levels continue to be kept low. Watch this video to hear Chris Cripps, Doctor of Veterinary Medicine and owner of BetterBee Inc, discuss the new regulations in more detail. Video Transcript So, January 1st of 2017, the FDA had, their new rules went into effect. So, now instead of being able to buy antibiotics over the counter for food-producing animals, and honey bees are food-producing animals, you need to have a veterinarian write a prescription or VFD in order to obtain those antibiotics, or to be able to use those antibiotics. What they're looking to do is try to keep control of all the antibiotics that are medically important for human beings. So, Oxytetracycline, and Tylosin, and Lincomycin are the three antibiotics that have U.S approval for honey bees. And those are the ones that are affected. So, Fumagilin is another antibiotic which is not medically important for humans, but it's not included in any of these regulations, and it's actually not legal for use in the U.S. However, the FDA has said we don't really care about that antibiotic because it's not important for residues, so they allow us to import it from Canada. So, it's not included, but it's not legal, either. So, VFD is Veterinary Feed Directive. And what the FDA has done is, when they were trying to come up with the rules for how are they going to regulate antibiotics in feed and make sure veterinarians are involved in approving their use and making

sure it seems appropriate, they made this group of antibiotics called Veterinary Feed Directive approvals. So, instead of having a pharmacist at a feed mill for each state, now the feed mills have licensing from the FDA that allows them to sell antibiotics that are made up in feed. So, that's been the big change. When they're writing the paperwork, for the most part, the information should be the same. So, there's some very specific things that a VFD needs, such as the concentration of the antibiotic in the feed, but typically on a prescription, you're going to give that, you know, how much is going to be mixed up in the feed, in the sugar, as well. And the difference is how the FDA has approved the drug. So, if you go to the FDA website, you can look up and see is this a VFD drug, or is this a prescription drug. And, you know, from there you decide what kind of, you know, then you know what kind of paperwork to fill out as a veterinarian. The veterinarian doesn't need to decide this is a VFD paperwork to use this or a prescription because the FDA has already done that. So, in New York State, the veterinarian can sell those drugs. There's online pharmacies where you can go to buy those drugs. It's going to be tough to go to your local big box store pharmacy and pick these up because a lot of times they don't carry these. So, most of the time it's going to be an online pharmacy that you can go to. Or if you're doing a VFD medication, you'll probably buy that from one of the bee suppliers. So, there are a couple out there that mix these up right now, and mix up the sugar and the antibiotics, and it's all ready to go when you get it in the mail. So, a VFD is good for six months, no more than six months. Once you use the VFD and you've fed the antibiotics, you can't get a refill on a VFD. So, that's a little bit different from a prescription. A prescription should be good for about a year. The first thing that you really want to do is contact the state inspector. So in New York State, the inspectors are in Albany, but they have different inspectors throughout the state, so they'll send somebody out to help you with that diagnosis. You could also call a veterinarian, and if their interest is in bees, they could help you make that diagnosis. Maybe they're available quicker, maybe they're not. You have to call them and see what's happening. But you need to get a good diagnosis. There are other diseases that look like American foulbrood. And if you have American foulbrood, a lot of states, including New York, require that you burn the hive. So, it's important that you get the diagnosis made, you we destroy what there is, but then, we're looking to control the spread. So, how do we keep our other hives from getting this? Because usually it's not every hive and in an apiary that gets it, but one, or two, or three, it's usually low numbers. So, if we want our other hives to keep from showing signs or developing the disease, we might feed it antibiotics. And that's when you need to have a veterinarian come and either write a prescription or VFD, so that you can obtain the antibiotics and be able to feed those. So, there is a group called the Honey Bee Veterinary Consortium who has a website at www.hbvc.org, and there's a list of veterinarians there that will see honey bees. So, you can look them up by their name, you can look them up by the state that they have a license, and be able to find who is available, what's their phone number, or send them an email. So, the reason that the FDA came up with all these rules is this was something that the president of the U.S. said, we need to make sure that these antibiotics always work. You know, what can we do to prolong their use? It seems like there's a lot of bacteria that are resistant to antibiotics, so how do we keep them from becoming resistant? And groups of scientists got together, they made recommendations. And one of the recommendations was that, if animals were being fed antibiotics, they should be doing so not just for growth promotion, just not to make more, you know, muscle tissue in a certain amount of time, but to, you know, be of therapeutic use. So this is something, we're actually trying to prevent disease, we're trying to control disease from spreading, or

we're trying to treat disease once we have it. So, those are the three things that are allowed. And they said, who should be in charge of this? The veterinarians are the logical choice because they know about diseases, they know about antibiotics, microbiology. So, rather than having a local police chief in charge, they said let's make veterinarians in charge. They have licenses. If they do it wrong, we can take their license. So, they already know how to work with us and they know that we have the knowledge to be able to use the antibiotics appropriately, and not just to overuse antibiotics. So hopefully, we're going to maintain that the antibiotics work for us well into the future. And one of the reasons that Tylan was approved, and Lincomycin, was there was a lot of resistance amongst foulbrood to Oxytetracycline. So, Oxytetracycline had been used for a number of years. Many beekeepers feed it all the time just because, and we need to kind of work through that and see if we really need to do that or not. Some beekeepers think they need to feed that twice a year, and if they don't, their whole operation is going to die of foulbrood. I don't know if that's truly the case, but there may be beekeepers out there like that. So, we'll have to see which camp all the beekeepers fall into. So, one of the things we're doing right now is we're training veterinarians to be able to work with bees. So, veterinarians in the U.S. have not had bees in their veterinary curriculum. Other countries, they have had bees in their curriculum, so that's a little bit different from a lot of the other vet schools in the world. When a veterinarian comes to you, or you have a veterinarian come to you, you need to establish what's called the veterinary client patient relationship, or VCPR. So in that relationship, the veterinarian says they're going to take responsibility for medical decisions, making sure you know what the medical choices are, and then helping you guide, decide which decision you want to make. They're going to have to be available for follow ups, so if something doesn't work, you know who to call in order to get something changed or fixed. And you, as a beekeeper, is going to have to agree to follow the directions of the veterinarian. So, it's a relationship. So, kind of a two-way street is what most people tell me about relationships, my ex-wife included. You know, but we need to establish that. And I think that as veterinarians learn more about bees, there are going to be useful for other things than just writing a prescription for American foulbrood control. So if we're looking at Varroa mites, if tracheal mites are an issue, European foulbrood, chalkbrood, if you have a hive that's queenless and you're thinking it's a disease or parasitic mite syndrome, let's get an accurate diagnosis, make sure we know how to prevent this disease next time. And that's a lot of what veterinarians do all the time. They know how to keep your dog and cat healthy, how to keep your dairy cows healthy, or your horses. And these are just a little bit of an extension for them for that. So, I think they're going to be well suited to provide all that information for you.

READ

How Does Resistance Develop?

Resistance occurs when the bacteria develop a mutation that permits them to escape the mechanism through which the antibiotic kills the bacteria. In a large bacterial population, there may be one or a few individuals that randomly develop this resistance, allowing them to survive in the presence of an antibiotic. The rest of the individuals that are susceptible to the antibiotic die during a regular treatment. Over time, the resistant organisms increase in population. The next time the antibiotic is applied, it might no longer work. An infection that used to be easy to control is now much more difficult. The misuse and overuse of antibiotics facilitate the development of this resistance. The illustration below

depicts bacterial resistance developing in a population. Long description ■ This image illustrates the process of antibiotic resistance development in bacterial populations through natural selection. Here's what it shows:

Initial State: The diagram starts with a mixed population of bacteria — orange cells (susceptible to antibiotics) and blue cells (resistant to antibiotics), with the orange cells being the majority.

Selection Process: When an antibiotic is applied, it kills most of the susceptible (orange) bacteria, while the resistant (blue) bacteria survive. Some susceptible bacteria may also survive if they didn't encounter the antibiotic.

Population Recovery: Between antibiotic treatments, the surviving bacteria multiply and repopulate. However, since more resistant bacteria survived the previous treatment, the proportion of resistant (blue) bacteria increases in each generation.

Repeated Cycles: With each subsequent antibiotic treatment, this selective pressure continues — more susceptible bacteria die while resistant bacteria survive and reproduce.

End Result: After multiple rounds of antibiotic treatment and bacterial reproduction, the population becomes predominantly resistant (mostly blue bacteria), with only a few susceptible (orange) bacteria remaining.

READ

Preventing AFB Infection in Your Colonies

There are several potential ways to protect your colonies from AFB infection. Ten Tips for Preventing AFB Infection in Your Colonies Significantly reduce the risk of infection in your colonies by following these ten essential preventive measures.

1. Inspect colonies for AFB three times a year: spring, right after removing the honey crop, and prior to overwintering.
2. Do not purchase bees or equipment from companies or beekeepers without an inspection certificate.
3. Do not purchase or use someone else's old equipment. When you acquire more equipment, always buy it new or build it yourself. With spores persisting for at least 40 years, there could be a risk of introducing infectious agents to your colonies.
4. Reduce drift in your own apiaries. Robbing and drifting bees can spread infection.
5. Always do a thorough inspection of the brood nest before transferring equipment, frames, or bees among your hives.
6. Sterilize your hive tool and wash your hands in between visiting apiaries or working in an infected colony.
7. Do not bring your gloves, hive tools, or other equipment to someone else's apiary. If you must, sterilize them thoroughly after the visit.
8. Swarms and cutouts can carry AFB. When acquiring bees of unknown origin, keep them isolated from other colonies for a few months to allow the colony to rear a few cycles of brood. Use separate equipment and monitor the colony closely.
9. When possible, order hygienic stock bees.
10. Do not feed colonies honey purchased from the grocery store, as it can contain AFB spores.

A "Vaccine" for Preventing AFB Infection in Your Colonies A novel "vaccine" against American foulbrood has been developed for honey bees. Rather than functioning like vertebrate vaccines, it exploits transgenerational immune priming (TGIP): By feeding killed *P. larvae* bacteria to worker bees, which then feed it to the queen, immune signals or fragments get passed into eggs. The offspring thus inherit enhanced resistance to *P. larvae*. The USDA has granted this vaccine a conditional license in 2023, making it the first insect vaccine of its kind in the U.S. It offers a promising preventive tool, but it is not a panacea because it cannot eliminate spores, it must be carefully administered, and its efficacy is contingent on maternal health and proper beekeeping practices. Beekeepers should apply the AFB vaccine preventively, ideally in early spring or when introducing new queens, so that immune signals are transferred to developing eggs. For best results, beekeepers should pair vaccination with the apiary practices outlined above in the "Ten Tips for Preventing AFB

Infection in Your Colonies." Note that this vaccine should not be used during active AFB outbreaks since the vaccine does not eliminate *P. larvae* spores. This means that infected colonies must still be destroyed under state regulations. It is also useless to use this vaccine late in the season (late fall/winter) since queens are not laying heavily by then, so immune priming won't reach many offspring.

READ

European Foulbrood Overview

European foulbrood disease (EFB) is another bacterial infection that can impact the health of honey bee colonies. Like AFB, this disease is found worldwide. This infection is caused by the bacteria *Melissococcus plutonius*. It does not reproduce using spores and is not nearly as damaging and difficult to control as AFB, but the bacteria can reside in old combs for several years. In addition to *Melissococcus plutonius*, other bacteria are often present in the dead larvae, which cause them to decay, change color, and develop a pungent odor. As with AFB, young larvae become infected with EFB when nurse bees inadvertently feed the bacteria to them. The larvae must be less than two days old to be susceptible to infection. Once they consume the bacteria, it begins reproducing in their midgut and competes with the bee for food. If larvae receive adequate food, they may be able to overcome the infection. During times of the year where there may be fewer nurse bees or inadequate nutrition, EFB can be evident in a colony. In the Northeast, EFB typically strikes in late spring, when many nurse bees are recruited to foraging with the onset of a nectar flow, leaving behind fewer bees to feed the brood. EFB can also strike when there is a dearth and food sources are scarce. Colonies sometimes seem to be more susceptible to developing EFB when they are nutritionally stressed providing pollination to nutrient-poor crops; this is most commonly seen following blueberry or cranberry pollination. This observation is not always consistent, however; some studies find a correlation between pollination and EFB, while others do not. All castes can be infected with EFB. Within a few days of contracting the infection, the larvae die. Unlike the effect of AFB, they most often die before they have been capped. They become yellow or brown and their tracheal system can sometimes be seen beneath their skin. Their body may become contorted into a twisted coil. Eventually, they decay into a goopy puddle in the bottom of the cell. In this state, they sometimes rope out when pulled with a toothpick (usually shorter than an AFB rope; much less than an inch long). The dead larvae eventually dry up into a scale that can be easily removed with a toothpick and feels rubbery to the touch.

READ

European Foulbrood Symptoms

■ The early and late stages of a colony Key Points infection have different symptoms. In The telltale symptom of European the early stages, you can expect to foulbrood is twisted brown larvae observe the following: with visible tracheal systems. Spotty brood pattern European foulbrood can Yellow or brown larvae, sometimes sometimes look similar to with a visible tracheal system American foulbrood. Always test Larvae twisted inside the brood cell suspicious larvae to confirm the Healthy-looking larvae in pools of infection. contaminated yellow jelly Melted-looking larvae with a mucous consistency Larvae that "ropes" (strings out to a length of around half an inch [~1.5 cm] when pulled with a toothpick or small twig) In the late stages, usually after a colony has died or is nearing death, you can expect to observe the following symptoms: Larvae become a thin, dark scale along the lower cell wall. This scale

can be picked out with a toothpick or small twig and is rubbery to the touch. Distinctive foul odor in some cases (but not all cases). The odor smells sour or fishy and differs depending on the bacterial community present in the decaying larvae. Few adult bees present (toward the final demise of the colony). The top six photos show early stages of the infection: Larvae are twisted and beginning to change color, or appear wet and swollen. In some cases, there is a yellow jelly. The middle three photos show the middle stage of the infection: Larvae have turned brown in color and have visible tracheal systems. The bottom three photos show the advanced signs of infection: Larvae become a hardened scale at the bottom of the cell. Just before this stage, larvae may be brown and goopy and "rope out" when pulled with a toothpick.

READ

Diagnosing and Treating EFB

■ During your usual colony inspections, Key Points look out for any indications of disease. European foulbrood colonies do not require burning. Colonies can be described here (especially be requeened and/or treated with twisted/discolored larvae or melted antibiotics. brown larvae), perform a test. In cases where symptoms overlap with AFB, testing is encouraged to accurately determine the bacterial species. European foulbrood infections can be reliably diagnosed two ways: The Vita EFB Test and the laboratory test at the Beltsville Bee lab. The Vita EFB Test The Vita EFB test works the same way as the Vita AFB test. This test kit costs about \$13 and is very convenient to use. The results are ready in about three minutes. Test larvae that are in the early stages of infection (twisted or curled in appearance and white or only slightly changing color), as these larvae will be infected with *Melissococcus plutonius* and will test positive. EFB larvae that are melted or scale may not test positive for EFB with this test kit, as by this stage it mainly consists of secondary bacteria. Beltsville Bee Lab Test The second option, if you live in the US, is to send a sample to the bee lab in Beltsville, Maryland. The Beltsville Bee Lab processes samples free of charge. They test for the presence of European foulbrood, as well as antibiotic resistance. It is important to determine whether any confirmed infections are resistant to traditional antibiotics. Information on how to submit a sample to the Beltsville Bee Lab can be found here. The results of the lab test are usually returned in two weeks. Because of this delay, it is best to use the Vita test kit while in the field, and simultaneously take an additional sample to send to the Beltsville Bee Lab for diagnosis confirmation and resistance testing. If you have a positive result from the Vita test, you can follow the necessary procedures. How to Treat the Disease In the US, European foulbrood does not need to be reported to the state apiculturist and colonies do not need to be burned. In some European countries, however, EFB colonies must be reported and destroyed. In many cases, a mild infection can clear up on its own through hygienic behavior of worker bees and adequate nutrition. To help a colony overcome a mild level of disease, the beekeeper can requeen it. For moderate to high level of infection, beekeepers can acquire antibiotic prescriptions from a veterinarian. EFB can be treated with oxytetracycline. The same measures taken to prevent and manage AFB apply to EFB.

READ

Idiopathic Brood Disease Syndrome

The foulbrood diseases and Parasitic Mite Syndrome can sometimes be confused for a different brood disease called idiopathic brood disease syndrome. This syndrome was first described in 1994 when a USDA researcher observed larvae of different ages that were sunken in the bottoms of their cells instead of healthy-looking larvae that are curled into “C” shapes and white in color. Some beekeepers refer to this syndrome as “snotty brood.” The unhealthy larvae can adopt a variety of appearances: They can be white and sunken in their cells as we sometimes see with Parasitic Mite Syndrome, they can be twisted or yellow in appearance like we see with European foulbrood, or they can have a brown “molten” liquefied appearance similar to the symptoms of American foulbrood. However, laboratory tests confirm that it is not a bacterial infection and the liquefied larvae do not rope out. This brood disease is not caused by Varroa mite infestations and is present without mites. Idiopathic brood disease syndrome samples analyzed in the lab are often infected with viruses, but the viruses are not the same in every case. The prevalence of idiopathic brood disease is not currently documented in the U.S. White circles indicate cells with idiopathic brood disease syndrome. To date, scientists do not know what causes this disease; hence, the name “idiopathic,” which means “the underlying cause is unknown.” We do know, however, that this disease increases the likelihood that a colony dies. Colonies with idiopathic brood disease are three times more likely to die within 50 days than colonies without this syndrome. It is therefore important to use diagnostic tests on all discolored, twisted, or molten larvae to determine whether infections are bacterial. Perform a sugar shake, alcohol wash, or ether roll to determine if it is parasitic mite syndrome. Pull off infected larvae from their cells and inspect the cells for mites or guanine crystals. If neither of these tests are positive, the colony has idiopathic brood disease syndrome. Unfortunately, until more is known about this issue, no treatments or prevention strategies are available to help clear and prevent it.

MODULE WRAP-UP

Brood Diseases

In this module, we’ve explored the three major brood diseases — chalkbrood, American foulbrood, and European foulbrood—and their varying impacts on colony health. From the relatively mild effects of fungal chalkbrood to the devastating consequences of American foulbrood, we now understand how to identify these diseases and implement appropriate management strategies. With recent changes in antibiotic regulations, this knowledge is particularly crucial for modern beekeepers to effectively protect their developing bees.

Bee Health—The Big Picture

Through this course, we’ve identified and discussed various pests, parasites, and pathogens that can impact our bees. The details are important for keeping your colonies healthy—but Master Beekeepers should also be able to take a step back and look at the bigger picture. What is the broader status of honey bee health in the US? What have been the effects of Colony Collapse Disorder? How can the health of your colonies impact the broader environment? It is important to be able to differentiate among different honey bee health concerns when you read media stories and scientific papers and to be able to communicate accurate information to new beekeepers and the public. This module explores the grander scheme of honey bee health.

READ

Honey Bee Colony Declines

■ You have no doubt heard that honey Key Points bee colonies are in decline. Since 2010, Honey bee colonies have been the Bee Informed Partnership (BIP) declining over the past decade in has reported total annual losses the U.S., Canada, and Europe. In ranging between approximately 30% the U.S., colony losses total 30 to and 45% of the total number of 45% each year. colonies in the U.S. every year. The 53 species of wild bees are also pests and diseases covered in this experiencing population declines course pose enormous challenges; in the Northeastern U.S. pesticides have affected honey bees since the 1870s, and the quantity and Despite decreases in honey bee quality (including diversity) of bee and several wild bee populations, forage has declined in most areas of the demand for crop pollination the U.S. Each of these issues, along continues to grow. with others, interact to threaten honey Multiple factors contribute to bee colonies even further. In light of pollinator health declines. these challenges, beekeeping requires ever-more intensive management as beekeepers must purchase more nucs and packages, split more colonies, and manage smaller operations. They also have to spend more time and money monitoring and managing disease and pest pressures. U.S. national colony losses from 2006 to 2020. In 2010, Bee Informed Partnership began collecting annual loss data (losses from April of one year to March of the next year) in addition to winter loss data. Credit: Bee Informed Partnership. Shared with permission. These stresses shape annual losses, but changes in the number of managed hives domestically and globally reflect economic factors as much as ecological ones. From year to year, colony numbers fluctuate based in part on the price of honey or the demand for wax, subsidies, and pollination contracts (particularly almonds). Competition from imported honey and cheap sugar and sugar substitutes also drive declines in the number of colonies in industrialized countries. Even social and political events shape the number of colonies: For example, the collapse of the Soviet Union halved the number of colonies in Eastern Europe in the 1990s. In short, the number of managed colonies in U.S. is largely driven by economic factors. In 2006, Colony Collapse Disorder (CCD) made headlines in the U.S. Unfortunately, media stories have often tended to conflate CCD with colony deaths more generally and long- term trends in colony numbers, even though these are three very different things. Master Beekeepers should be sure not to make the same mistake. Today, many beekeepers are losing colonies for a variety of reasons; some relatively recent and others longstanding. More to the point, most colony losses in the U.S. and elsewhere are not attributable to CCD. Although CCD shook the industry and set into motion a great deal of research, it was a short-lived phenomenon and very few incidents have been confirmed since 2010. Wild bee species have also experienced population declines throughout North America, and by 2017 eight species of wild bees were officially listed as “endangered” under the U.S. Endangered Species Act. Of the 416 documented species of wild bees in the Northeastern U.S., 53 of them are currently experiencing population declines, and 35 of these species are now considered rare. Wild bees face some of the same pressures as honey bees as well as very different ones: Habitat loss, parasites, pathogens, and climate change are among the leading factors responsible for wild bee declines. At the same time, the number of pollinator-dependent crops planted in the U.S. is increasing. One oft-repeated concern is that food production might be impacted should there be too few bees to pollinate the flowering crops. Honey bees are the most common pollinator commercially used in agriculture, pollinating over 100 fruit,

nut, and seed crops in the U.S. Indeed, in 2005, there were not enough honey bee colonies available in the U.S. to pollinate over a half-million acres of almonds, and colonies were imported from Australia to ensure that there would be a sufficient almond crop that year. But almonds are unique: In the last few decades, the number of acres of almond orchards in California has doubled and they are pollinated almost exclusively by honey bees. Almonds make up a disproportionate part of the pollination “demand,” affecting the “supply” of honey bee colonies and pollination contract prices. Most staple crops do not depend on insect pollination, but many of our favorite and most nutritious fruits, vegetables, and nuts/seeds do. In other words, concerns about “food production” must be specific about which crops are threatened or made more expensive by declines in honey bees and other pollinators, where those crops are, their degrees of dependence on insect pollination, and other details. In 2015, to address rising annual losses of honey bee colonies and the decline of some wild pollinators in the U.S., President Barack Obama announced the development of a federal strategy to promote pollinator health. Each state was required to create an action plan to help improve the health of pollinators and reduce colony losses. Thanks to funding through these plans, in addition to other government, organization, and academic funding, pollinator health research and conservation is an active area of work in recent years. Taken together, thousands of studies have identified that there is not a single cause for pollinator health problems; instead, multiple overlapping factors interact and contribute to both honey bee and wild bee health problems. For honey bees, parasites, pathogens, pesticides, beekeeper management, climate, genetics, and nutrition all influence bee health. Beekeepers must therefore think about bees holistically and consider all levels of the honey bee system (the bee, the colony, the apiary, the operation, the beekeeper, the community, and the environment), along with all the inputs that can affect it (disease, environment, management, genetics, as well as social, physical, and financial factors). Thinking about bees in this big-picture way can help us, as Master Beekeepers, to identify and take action on many aspects of bee health, pollinator health, and the broader health of agricultural and natural ecosystems.

READ

Colony Collapse Disorder

■ In 2006, a phenomenon appeared in Key Points the U.S. (and later in Switzerland) that Colony Collapse Disorder is a became known as Colony Collapse syndrome with specific symptoms. Disorder (CCD). Beekeepers found that The term is not synonymous with their recently healthy-looking colonies the general “colony declines” that had collapsed: there were little to no are observed each year. adult bees present, resulting in sudden A single cause for CCD was never colony death. These colonies did not discovered. Instead, several factors exhibit the recognizable signs of death likely contributed to CCD. and instead had the following symptoms: A rapid decline in colony population, with the majority of the adult bees disappearing, leaving brood poorly attended and the queen with only a few young nurse bees Few or no dead bodies in or around the hive Sufficient stored food (honey and pollen) A lack of robbing by bees from neighboring colonies or pests such as wax moths or small hive beetles Levels of Varroa and Nosema that are below the treatment threshold CCD was most frequently reported by migratory commercial beekeepers who wintered their colonies in warmer states, but also by some beekeepers who kept their hives stationary in one region. Beekeepers of all operation sizes were affected. These beekeepers experienced a range in losses from this mysterious phenomenon, from

30% of their colonies to over 90%. The scientific community has tried to determine what was behind this disorder, but to this day no single causal factor has been found. Over the winter of 2006–2007, 23% of beekeepers lost colonies to CCD. Those affected lost an average of 45% of their colonies. The condition was widespread, affecting beekeepers in 22 states. This syndrome was never observed in Canada or most of Europe. CCD was continuously reported from 2006 until about 2010 in the US. Today, CCD is not as prevalent, with only a few unconfirmed cases reported in the past few years. For instance, only 6% of beekeepers reported losing colonies to CCD over 2013–2014. That number is likely an overestimation, as some beekeepers may report that their colonies died from CCD if they don't know what the true cause was (and if they don't know the specific criteria for CCD diagnosis). Dr. Dennis vanEngelsdorp and his colleagues at the University of Maryland have been leaders in trying to determine the causes of CCD. In 2009, they conducted a comprehensive study examining nearly all known stress factors to honey bees for their potential involvement in the disorder. They inspected and sampled colonies that exhibited CCD characteristics to compare to healthy colonies, looking for one or a few variables that would point to an underlying cause. In this study, they compared 61 variables, including body mass, protein content in bees (an indicator of nutritional stress), genetic origin of bees, bee body symmetry (an indicator of healthy development), brood pattern, Varroa mite levels, Nosema levels, chalkbrood, tracheal mites, virus levels, European foulbrood, and levels of a variety of pesticides (including fungicides, miticides, insecticides, and herbicides). No single factor was proven to be responsible for CCD. Instead, researchers found that colonies in the throes of CCD were less healthy overall. These colonies were more likely to 1) be infected with multiple viruses, at higher levels, and 2) be infected with both species of Nosema simultaneously. The pesticide levels in colonies with and without CCD were the same. These findings suggest that either colonies were collapsing because they were faced with several stressors, or they had compromised immune systems from some unknown, undiscovered cause, making them vulnerable to viruses and Nosema. Unlike a previous study that had found Israeli Acute Paralysis Virus (IAPV) to be present almost exclusively in CCD colonies, the vanEngelsdorp study (which sampled beekeeping operations more widely) did not find IAPV to be exclusive to CCD colonies. Instead, IAPV was present in both CCD and non-CCD colonies. At this point, the scientific community has come to the consensus that several factors likely contributed to CCD. Sudden, widespread colony losses are not uncommon throughout history. Indeed, there have been many instances of colonies collapsing for unknown reasons, many of which exhibit similar symptoms that are observed in CCD (sudden adult population loss, leaving behind a queen with some young bees, food, and brood). There have been 18 known documented cases of mysterious large-scale colony losses since 1869, with these syndromes given names like “disappearing disease,” “fall dwindle disease,” and “disappearing syndrome.”

WATCH

Colony Collapse Disorder Investigation

Watch this video to hear Dr. Dennis vanEngelsdorp discuss his investigations into CCD. Video Transcript So, the story of colony collapse disorder, in its current, in the last 10 years, what we discovered. So, 10 years ago, I was state apiarist in Pennsylvania, and the largest beekeeper, who has given his permission for me to tell this story, the largest beekeeper in Pennsylvania, David Hackenburg, I'd seen him at the October state meeting, and he was bragging about how great his bees looked. They

were fantastic. And then, he went down to Florida a month later, as he does all the time, he moved all his bees down there, and he went back a month later. And he describes getting out of the truck and it being dead silent. And as you know, if you're a beekeeper, one of the great joys is that big, loud buzz when you get of the truck, and that was gone. And so, he didn't really know what was going on. And so, he describes looking around and noticing just nothing was moving. Nothing was there. He got his hands and his knees, and looked in the entrance. And then, there was just nothing. There were no dead bees. Just everything had disappeared. And this was peculiar. There was honey on the colonies. And so, he knew something was different here. And so, he picked up a couple of these colonies and some that were just in the last stages of collapse. And he shipped them up to Pennsylvania, where I received some of these samples. And at the time, we didn't call it colony collapse disorder. We called it fall dwindle disease, because what we did was we looked through the literature, and we said, did this happen before? And what should we call it? And so, we sent out a report online to do that. And then it became really clear that we need to go to Florida and sample other colonies. And so, we did. We found that a lot of other operations were suffering this very strange condition. And that strange condition we define very, very carefully. We define it as the rapid loss of adult bees. And that's important. Rapid, as in, like, two weeks, that they really quickly, all the adult bees left. And we knew that this happened in Florida, because we saw big sheets of brood. And we know that they could not have laid all these eggs and maintained that brood with such a small cluster size. We also knew that if there were bees left, they were young bees, so just emerging bees from the brood, presumably, and the queen was there. There was enough honey stores in colonies that had completely collapsed. Those stores weren't getting robbed out. So, that was peculiar. Until we cracked it open. And if we cracked it open, the next day they might get robbed out. So, there were these really clear set of characteristics that describe these dead colonies. And so, that's when we got on a conference call with a whole bunch of other people who had been sampling. And we coined this new term, colony collapse disorder. And so, we termed that to describe, very specifically, the set of symptoms, the rapid loss of adult bees, the queen is present, there were food stores, and, really importantly, the amount of the Varroa mite and Nosema in these colonies could not explain that loss. And that's an important distinction because Varroa can give you these symptoms in the late summer. If your strongest colonies, I mean it's famous, you know, the colonies that had the most honey supers you could go to, and suddenly there's no bees left. And that's Varroa mediated. You can find the Varroa in the brood capping. If you had done monitoring, you would have known it was coming. But in the case of CCD, this was not the case. Now, I do want to say that, in general, the new term, colony collapse disorder, really captured the public's imagination. And so suddenly, every dead colony was colony collapse disorder. This was never our intent when we came up with that phrase. That phrase was meant to be very specific to that case definition. But you really can't fight the media. And so, that's become, I think, a very common word to describe unusual losses. But that use of the word is not what we originally intended. Our original intent was to describe this very specific set of symptoms. What's really quite amazing about this is that if you look back in the literature, there's some evidence from 900 years in the current era. So. In Ireland, where they described something very, very similar. Just the other day, I have a friend who works at the National Agricultural Library. And he gave me an article from 1862, the South Carolina Farmer. And it talks about a symptom, and these beekeepers who describe exactly what we describe as CCD. And so, it seems like this condition has occurred for apparently thousands of years, and it pops up once in a

while. And they called it different things at different times. But this symptom has, these symptoms, this group of symptoms, seems to be consistent over time. So, what's going on? Well, when we first did this study, we took all these samples, and we wanted to do descriptive study. So, wanted to describe what was happening. So, we map the yards where this were. And what we found was, that colonies that were sick were more likely to be close to sick colonies than colonies that weren't sick. Which just seems obvious if it's a contagious condition. And so, that suggests there was a contagious condition. We looked at Varroa and Nosema, and that didn't explain differences. We looked at viruses and we found that, in fact, the CCD colonies had much higher levels of viruses than the non-CCDs. But It wasn't a consistent virus. At first, we thought it was just Israeli acute paralysis virus. But then with more analysis, and new tools, and a bigger data set, it was clear it wasn't just one virus. It was several viruses. And we think that it's actually those viral infections that caused the symptom of CCD. And bees are social organisms. And other social organisms, that means you don't look at a bee as an individual, you look at the colony as the super organism. And so, super organisms, these collective units of living things, have evolved a lot of different social ways of protecting themselves against damage. And one of those is that if individuals get sick, they let other individuals know, don't touch me, and then they fly away and leave the colony. You see this in termites, for instance. They don't find away, but what they do is they get a fungal infection, and as they walk out of their nest, they do something called the leper's walk, where they pound on the ground with a steady beat, saying don't touch me because I've got a fungal infection, and you'll get sick, too. And then they walk out of the colony and die. And so, some really good work out of North Carolina. Dr. Rueppell showed that they could imitate this altruistic suicide with pesticide exposure. And so, we think what happens is, is that, for some reason, these viruses have taken a hold in this colony, spread really rapidly. And so, all these bees are leaving collectively because they're sick and dying away from the colony, trying to prevent the colony from dying. But since everyone left at one time, the colony collapses. And so, we think that's the driver of those symptoms, is these heavy viral infections inducing this mass abandoning the colony. It's sort of a bad consequence of this behavior. So, we started seeing these symptoms first in the fall of 2006. Most of our samples got taken in California and Florida in the winter of 2007. And that's when we came up with the term, colony collapse disorder. We found it in lots of different operations. How widespread? I don't know. Because one of the things we do, of course, is survey. And ask people, did you have colonies die without bees? Now, we still see that symptom come up, and often people say, well that's CCD. But, of course, there's other symptoms involved. And so, Varroa can cause this, too. And so, it's a little bit leery just defining something CCD on that one symptom. Although that tends to be what happens. So, we think that these colonies are dying from the symptoms of these heavy viral infections. What we don't know is, why did they become so susceptible to these viruses? One possibility is that there were these lethal strains, and at that time, we did not have the technology that we could have differentiated that. The other, of course, is that they were exposed to other things that predisposed them to getting ill. And one of those is pesticides. We hear pesticides a lot. For the CCD samples we sampled in 2007, one of the clearest distinctions between sick and healthy colonies was the healthy colonies have more pesticides, which is not a very easy data point to reconcile. But that clearly suggests that it wasn't pesticides that underlay these conditions. Also, the fact that we've seen these conditions occur in time, well before we started using pesticides, suggests it's not pesticides. And so, do we know the answer of what caused CCD? No. We think it's viral related, and we think it's just

maybe a very highly virulent strain of the virus that's sort of wiping through the population and then spreading through time. I do think that that event, colony collapse disorder, even though we don't see very much of it all, I have not seen what I would say is a classic case of CCD in the last four years, and I looked. I've been looking pretty actively. Not so much this last year, but I've been looking, and we haven't seen it. But what it did highlight, I think, was the fact that we were losing a lot of colonies, year in and year out, and beekeepers weren't talking about it because they were sort of embarrassed. And we come from a time that, well, I did something wrong, and so we're not going to talk about it. I think this sort of opened that door, and it showed that colonies were dying from a whole host of factors.

READ

Honey Bee Health in History

■ Honey bee health has faced periodic Key Points problems, and the rates of annual Recent honey bee colony losses as well as the number of are not unique in history. There colonies in the U.S. in general has risen have been several periods in the and fallen over time. past where large-scale losses were observed. Between April 2016 and March 2017, a third of the colonies in the U.S. Honey bee colony numbers are perished, according to the Bee influenced by economic, social, Informed Partnership, the gold and environmental factors. standard for measuring honey bee colony numbers in the US. This is worrying, although it represents a slight improvement from recent years, where as many as 45% of colonies perished. More importantly, the BIP survey is a relatively recent snapshot. Information on annual losses has only been collected for seven years—before 2010–11, the surveys only covered winter losses, and before 2006–07, there were no national loss surveys at all. And some practices have changed dramatically in the last century, making comparisons unreliable. For instance, before the advent of modern beekeeping and the Langstroth hive, some beekeepers killed their strongest and weakest hives to harvest the honey. The logic was that the strongest hives usually had the most honey and that the weakest were least likely to survive the winter anyway. This makes winter loss rates seem as though they used to be much lower, when, in fact, beekeepers used to simply “take their losses” earlier, although we don't know how much this affects the numbers. In short, it is difficult, if not impossible, to make meaningful comparisons to past honey bee colony losses since methods for gathering data, counting losses, harvesting honey, and other indicators have changed over time. Lessons from the long-term trends in the number of colonies across the U.S. are also limited. The number of honey bee colonies has been monitored by the National Agriculture Statistics Service since the 1940s; earlier records are just estimates based on historical data about honey production. These statistics tell us that the number of honey bee colonies has been declining for the past 70 years: in 1947, there were 5.5 million colonies, and by 2006, there were 2.4 million—a 56% reduction. But these declines need to be understood in their economic and historical context: most colonies in the U.S. are managed as livestock and are not wild creatures. And the number of colonies today is similar to the number (estimated) a century ago; the figure from 1947 is the high point, as World War II made beekeeping more profitable and popular. A historical perspective is therefore helpful. Here, we describe some of the factors that have affected honey bee health historically or that continue to shape colony numbers in the US. Foulbrood In the late 1800s and early 1900s, a scourge of foulbrood spread across the US, seriously depressing honey production for decades (vanEngelsdorp and Meixner, 2010). The science was still being worked out: American and European foulbrood were often conflated well into the

1900s, and for a while, foulbrood was known as “New York bee disease,” to the consternation of beekeepers in the Empire State (White, 1906). Ironically, disease was spread further with the adoption of modern equipment and related practices (e.g., re-using comb in movable frame hives, year after year), as well as the growth of the industry. Efforts to prevent and treat foulbrood gave rise to new laws requiring the destruction of infected colonies, government inspection programs, and education through extension programs, which together have made foulbrood a more minor concern today.

Beekeeping in WWI and WWII: During both world wars (1910s and 1940s), the rationing of sugar, the jump in prices of sweeteners, and the many uses of wax pushed honey prices up, making the number of beekeepers and honey bee colonies in the U.S. soar. Indeed, there were approximately 6 million colonies in the U.S. immediately following WWII, which is still the highest number of colonies ever in U.S. history. While the landscape (both agricultural and urban) was different then, these numbers are also a helpful reminder that changes in the number of honey bee colonies upward or downward generally reflect economic factors as much as or more than ecological ones. In other words, ■ markets drive the number of honey bee colonies as much as disease, or pest pressures, or forage availability.

Isle of Wight Disease In the early 1900s, bees in England perished in large numbers. The malady was traced back to, or at least named for, the Isle of Wight (an island south of England, in the English Channel), where 90% of the honey bees died, in three separate years. To this day, there is no complete agreement on the cause: the deaths are generally attributed to tracheal mites, but it has also been suggested that *Nosema apis* was at fault, or simply overcrowding and beekeeper mismanagement that led to starvation. Today, there is much discussion about the insidious, chronic, and sublethal effects of new classes of pesticides. Pesticides have caused problems for honey bees since the 1870s (Johansen and Mayer, 1990), when arsenical sprays on fruit trees led to many losses. In the 1920s, aerial spraying of cotton fields killed many bees as well. In the 1950s, the insecticide Sevin was discovered to have the ability to damage colonies even one season after it was applied if it contaminated stored pollen. In 1967, it is estimated that half a million colonies were poisoned by pesticides, a significant fraction of the total number in the U.S. (Johansen and Mayer, 1990). Today, newer generations of pesticides can be used at lower dosages, and there are increasingly stringent rules about their application. However, there is still much to be learned about their possible effects on pollinators, including synergies between different agricultural inputs, particularly those initially thought to be harmless to bees (e.g., fungicides and adjuvants).

■ **Pesticides** ■ **Environmental Factors** The quality and quantity of good bee pasture available in the U.S. has declined over the last several decades because of more intensive agriculture. This has included less cover cropping, a reduction in forage in field margins and in understories, and regional changes in agriculture (e.g., the loss of alfalfa in New York State as the dairy industry has declined). Depending on the habitat surrounding colonies and the amount of competition in the area, bees may be at a higher risk of malnutrition and may require more intensive feeding. Ironically, some invasive species such as purple loosestrife and Japanese knotweed have been a boon to honey bees, making environmental conservation programs to eradicate them less straightforward! Finally, while the effects are so far limited, climate change may soon begin to impact honey bees (it is currently only documented to affect bumble bees). Indirectly, honey bees will be most affected by changes in forage. The flowering of plants is affected generally by water availability, temperature, and photoperiod. In the Northeast, nights and days will both likely be warmer, with more frost-free days, and the weather wetter, so the spring greening will probably come earlier,

with mixed effects for honey bees. The profitability of beekeeping wanes and waxes based on many factors, including the price of honey, pollination contracts, and the cost of treatments for various pests and diseases. These factors contribute to the total number of hives in the U.S. or specific regions. Shifts in international trade in bees and the movement of bees within countries have contributed to the genetic diversity of honey bees in the US, but also inadvertently introduced and spread pests and pathogens. More broadly, major political events and social change can affect the beekeeping industry. For instance, in ■ Social and Economic Factors ■ the early 1990s in Europe, the number of colonies declined quickly, largely due to the collapse of the Soviet Union. Photo credit: Bee Informed Partnership

READ

Tying It All Together

■ As the primary caretaker of our Key Points colonies, we also serve as the "primary Performing colony inspections, care physician" to our bees. We look for monitoring, and communicating signs of unhealthy colonies, identify with other beekeepers can the set of symptoms we see, interpret increase your opportunity to those symptoms, and diagnose a identify issues along with your problem. From there, we weigh our confidence to diagnose them. management options on how to Follow a systematic routine for restore the colony back to health, then identifying abnormalities in your choose the best option for the colonies. situation. Refer to the symptom chart to In this course, we presented you with help diagnose diseases. When in different pests, parasites, and doubt, send a sample to the pathogens that can impact bee health. Beltsville Bee Lab or reach out to There's a lot of information to navigate: your local extension associate. Some ailments impact adult bees, while some impact brood; some are more common in the spring, while others are more common in summer and autumn; and sometimes colonies are afflicted with more than one ailment at once, as fighting one infection may make a colony more susceptible to developing another. Perhaps the most difficult aspect of diagnosing bee diseases is that many of the symptoms overlap or seem vague (spotty brood, K-wing, etc.). Diseases appear different in different cases. The photographs here may not look identical to what you're seeing in your own colonies. All of this taken together can make it difficult to feel confident in a diagnosis and choose a treatment option. To gain more experience and confidence in diagnosing bee diseases, be sure to perform regular colony inspections. If you have friends in your bee club that are experiencing issues, ask if you can observe them (but remember the guidelines about using your equipment, such as hive tools, in other bee yards). Attend workshops, conferences, and meetings that allow you to observe more case studies in specific diseases. The more exposure you have in observing honey bee diseases, the better you will recognize them in your own colonies. Make time to inspect your colonies every 10 to 14 days. Observe the outside and entrance of the hive for dead, disoriented, or twitching bees. When you examine the brood nest, look at three or four frames of brood to get an idea of what's going on in the colony. Keep your eye out for common symptoms such as reduced adult population, spotty brood pattern, supersedure cells, and perforations in capped brood. These symptoms are common for a variety of diseases; if you observe any of these symptoms, a closer look is warranted. Carefully examine the brood cells. Look for discolored brood and larvae that aren't in the normal "C" shape. If you see perforated cappings, open them with your hive tool and examine the larvae or pupae inside. Take a look more broadly at the adults on the frame. How does the population size seem? Is it normal

for that time of year? Do you see any adult bees with deformed wings, hairless bodies, or K-wing? In addition to these regular, brief colony inspections, make time in your beekeeping schedule to perform routine monitoring for the most common and most serious diseases. Monitor for Varroa mites once a month from March to October. Monitor for Nosema when you see symptoms by sending in a free sample to Beltsville or doing a spore count under a microscope. Monitor for American foulbrood three times a year (spring, post honey harvest, and pre-wintering) by looking for symptoms on every frame in the brood nest. Take advantage of tools and resources that help take the guesswork out of diagnosing diseases. The Beltsville Bee Lab provides free diagnostic services for Nosema, tracheal mites, chalkbrood, European foulbrood, American foulbrood, and small hive beetles. In addition, you can purchase Vita test kits for AFB and EFB for quick and convenient results. Regular inspections, scheduled monitoring, and incorporation of diagnostic tools give you the best chance at finding and addressing diseases early.

TOOL

Honey Bee Symptom Chart

We've created a symptom chart to help **■** Download the Tool take the guesswork out of diagnosing diseases. When you observe symptoms Symptom Chart. in your colony, refer to this chart to narrow down your disease options. It is worth emphasizing that anytime you observe larvae or pupae that are brown and mucousy, test them immediately for AFB. If you do come across symptoms that you can't diagnose, the Dyce Lab is here to help. If you are located outside New York, contact your local extension associate.

READ

Pathogen Spillover to Wild Bees

■ Throughout this course, we've Key Points mentioned pathogens and parasites Some parasites and viruses are that are shared between honey bees shared among different bee and wild bees, such as Nosema species. ceranae, deformed wing virus, black queen cell virus, acute bee paralysis virus, sacbrood virus, and slow bee paralysis virus. Because these pathogens share some bee hosts, there is the possibility for the transmission from sick honey bees to healthy wild bees. Pathogen spillover refers to the transmission of pathogens from a reservoir population (with high infection prevalence) to a new host population. In this module, we are referring to pathogens spilling over from infected honey bee colonies to wild bee populations, and vice versa. Research has shown that honey bees can and do spread infections to the broader pollinator community. One study tested honey bees and bumble bees for disease across Great Britain and found that the prevalence of honey bee deformed wing virus and Nosema ceranae was strongly linked to the prevalence of these infections in bumble bees: Areas with many infected honey bees also contain many infected bumble bees. Infection rates were higher in honey bees, suggesting that these pathogens are “spilling over” from the honey bee population to the wild bee population. Another study from Vermont in 2019 found DWV and BQCV infections were more prevalent in bumble bees that were caught near apiaries. When honey bees in the area had high infection levels, so did nearby bumble bees. And when there were no apiaries near bumble bee communities, there were no DWV infections in bumble bees. These findings are a staggering reminder of the importance of being responsible beekeepers by maintaining healthy honey bees.

WATCH

How Do Wild Bees Catch These Infections?

■ Honey bees are kept in nearly every Key Points country that is inhabited by humans, Honey bees can spread diseases to with numerous colonies distributed wild bees by visiting the same throughout the landscape. They are flowers and sharing other transported to a variety of agricultural resources. areas for crop pollination and they are As stewards of the environment, it also scattered in natural and urban is critical that beekeepers settings as backyard beekeeping maintain healthy honey bee continues to increase in popularity. As colonies to protect wild bee viruses and Nosema are so prevalent in communities. honey bees, and honey bee colonies are so numerous and widely distributed, the risk of pathogen transmission to wild bee communities can be quite high. Many of these pathogens are picked up from flowers. Honey bees, bumble bees, and many other wild bees are generalist pollinators, meaning they collect food from many different types of flowers. Unfortunately, it also means they can deposit and pick up pathogens from a variety of flowers. Pathogens can be deposited and collected from contaminated nectar and pollen, and they can also be deposited through infected feces on plants. Simply put, if an infected honey bee visits a flower and inadvertently deposits pathogens on that flower, the next wild bee that visits may pick up that pathogen and become infected. Watch this video to hear Dr. Scott McArt, Professor of Pollinator Health at Cornell University, discuss pathogen spillover between wild and managed bees. Video Transcript So, a lot of pathogens that you have learned about in this course are not specific to honey bees. So, those pathogens can infect multiple bee species, or at least be present in multiple bee species. Sometimes we find that these pathogens, such as *Nosema ceranae*, can actually cause active infections in wild bee species. Other times, we find that there are bits of DNA of the *Nosema ceranae* in these wild bee species, but they're not necessarily causing an active infection. As you well know from 416 species here in New York of wild bees and about 20,000 species worldwide, it's not always going to be the case that every single one of those species is going to develop an active infection from a pathogen from honey bees. But some of those bee species will develop an active infection and have negative consequences as a result of that. So we're finding out more, this is definitely an emerging field. We're finding out more and more about what types of pathogens are present in different bee species on an ecological time scale, and which pathogens are actually having a negative impact. So, it's not just, that the pathogen was found, a little bit of DNA was found, but it's actually is causing an active infection and negative impacts on bees. So, some of our current research is actually looking into the transmission biology of these pathogens. So, any time we talk about potential for parasite spillover from managed bees into wild bees, or either just transmission from wild bee to wild bee, or managed bee to managed bee, the key to controlling these, these types of interactions and trying to limit parasite spread, is understanding how transmission is actually taking place. And there's been an assumption in the literature for probably the last five or ten years or so, that transmission has been taking place at flowers. But this was largely untested up until just a year or two ago. So now, there is one study out by Pete Graystock, who is a post-doc in our lab here at Cornell, actually showing in a flight cage that particles of both know *Nosema ceranae* and *Nosema bombi* can actually be transmitted in flowers between honey bees and wild species of bumblebees, and that those can lead to active infections once they are transmitted at flowers. One of the more surprising results that we found scientifically in the past five or ten years or so, has been a

surprising level of bee pathogens that we have found on flowers. So, in about 36 percent of the 200 flowers or so that we have screened for honey bee pathogens, we found detectable levels of those pathogens. So, about a third of flowers that we have screened actually have detectable levels of pathogens on them. Now, this is significant in the amount of pathogen that we're finding, the prevalence of pathogens on flowers, but it should be noted that we don't know whether those levels are actually at sufficient quantities to be causing transmission and active infections in bees. We are currently following up with many studies as you might suspect, to try and understand what predicts a transmission event. What level is actually required on flowers to result in an active infection and a bee that has this pathogen transmitted to it. That said, it's sort of staggering to me that a third of flowers out there actually do have detectable levels of bee pathogens on them. So, as a bee interacts with a flower, it is obviously interacting quite a bit with the anthers, where it's trying to grab pollen, and also the nectary to try and grab nectar. As it's doing so, it's rubbing itself on various parts of the flowers. So, transmission could be taking place if the bee actually has pathogen particles on the surface of its body. So, just simple contact with the body surface of those bees on the flower could be transmitting pathogens. The other main venue that people have really looked into is fecal-oral contact. So, if a bee is defecating on the flowers, *Nosema* particles, in particular, could be being deposited on the flower surface. As the bee is lapping up nectar, or lapping up pollen, it is potentially coming into contact with some of those fecally transmitted diseases via that oral contact as well. So what you can see here is the common eastern bumblebee. So, this is *Bombus impatiens*, our most abundant wild bee here in New York. It's interacting with a lot of different golden rod flowers right here. And what we found is on this particular species, here's a honey bee right here, actually, interacting on another part of this flower. But what we found with, here in New York, is this particular species, *Solidago altissima*, what we found is that the vast majority of these flowers on *Solidago* have detectable levels of honey bee pathogens on them, some of which are known to be transmitted to this particular bee species, *Bombus impatiens*. So, one of our best studies that exists to date on the question of whether management practices and managed honey bees can be having impacts on wild bee species comes out of the United Kingdom. Where last year, so in 2016, a study was published in *Nature*, showing that honey bee viruses, in particular, deformed winged virus, is shared among wild bumblebee species and managed honey bees. And what they found is that, in those, in the places with a large proportion of managed honey bees and potentially bees that were not necessarily being controlled very well for *Varroa*, and therefore have large quantities of deformed wing virus, there was a higher proportion of bumblebees. And this was across three different bumblebee species that had higher levels of deformed wing virus. They took those results into the lab and actually did bioassays with those bumblebee species and those particular strains of deformed wing virus, and they showed that, lo and behold, that particular strain was highly virulent in those bumblebee species and caused accelerated mortality. So that, in combination with what we know about bumblebee species declining globally, I would say, is two independent data sets suggesting that viruses and other pathogens can be having impacts on bumblebee species, and here's one case example where we actually showed a bit of cause and effect. Whether that can be extrapolated to all the other declines that might be occurring for bumblebee species, most likely not. But it would be surprising to me if that was the only isolated incident where pathogen spillover from managed honey bees to wild bees was actually having an impact on those wild bees. As someone who's keeping managed bees, just as someone who is actually keeping any managed animals, such as

cows, or pigs, or anything else, the best thing that we can do to protect our wild bees is to keep healthy managed bees. So this involves actively looking for *Nosema* infections and treating if you do have large problems with *Nosema*. And probably the biggest thing you can do, and the simplest thing you can do, is what most beekeepers are concerned about, and that is *Varroa* management. So, as you know *Varroa* transmits many of the viruses, such as deformed wing virus and other virus such as Kashmir bee virus, Israeli Acute Paralysis Virus, slow bee paralysis virus, lots of different viruses. *Nosema ceranae*, especially, and also those viruses are known to be prevalent and causing active infections in many of the wild bee species that we look for. So, simply managing and keeping, managing for those diseases or parasites and keeping your bees healthy can potentially really help wild bees out there, in terms of minimizing pathogen spillover from your managed bees into wild bees communities. As stewards of the environment, it is critical that beekeepers actively maintain healthy honey bee colonies. It's worth remembering that if wild bee colonies become sick, they cannot receive any treatment to help them. On the other hand, beekeepers can intervene to help sick honey bee colonies. They can control *Varroa* mites with a variety of chemical and cultural methods to reduce viruses. They can supplement colony nutrition and requeen colonies that are struggling. And if colonies do die from disease, beekeepers can purchase more packages and nucs to replenish their stocks. Wild bees do not have these same opportunities. If they are sick, no one provides them with treatments. If they die, their populations are not replaced. Many species of wild bees are already facing declines from challenges including diseases, pesticides, climate change, and nutrition. It is our responsibility to avoid contributing additional challenges. Keeping our honey bees healthy can help ensure we are keeping our surrounding pollinator population healthy.

MODULE WRAP-UP

Bee Health — The Big Picture

Throughout this course, we've delved into specific threats to honey bee health, but now we've examined the broader context of these challenges. Understanding Colony Collapse Disorder, current bee health trends in the U.S., and the environmental implications of colony health helps us become more informed beekeepers. This broader perspective enables us to better evaluate media coverage, interpret scientific findings, and effectively communicate accurate information about honey bee health to both new beekeepers and the public.

[Back to Table of Contents](#) [Thank You and Farewell](#)

Congratulations on completing "Master Beekeeping Part III: Managing Pests and Diseases." I hope that you now have a better grasp of all the different pests, parasites, and pathogens that interfere with honey bee health and productivity. The material covered in this course provided you with Emma Walters the ability to monitor, diagnose, and control specific problems in Honey Bee Extension Associate Dyce Lab of Honey Bee Studies your colonies to be proactive and take charge of the health of your Cornell University bee operation. I hope it has met your expectations as you support your colonies' survival and success. From all of us at Cornell University and eCornell, thank you for participating in this course. Sincerely, Emma Walters

[Downloadable Resources](#) You may wish to download and save these files for future reference.

- [Varroa chemical controls 2017/2018](#)
- [Reputable Varroa Resistant Bee Breeders in the US](#)
- [Guide to Varroa Treatments](#)
- [Example Varroa Mite Monitoring and Treatment Calendar](#)
- [Honey Bee Disease Symptom Chart](#)
- [Recommended Books](#)

There are more books written about honey bees than any other animal on the planet, besides humans. With so many to

choose from, we've compiled a list to help guide you in your readings. There are no required readings for this course; all examination questions will be based on the online material only. However, if you are interested in delving deeper into topics we've discussed, here are some great resources: Honey Bee Pests, Parasites, and Pathogens "The Hive and the Honey Bee" (2015), by L.L. Langstroth, et al. "Honey Bee Biology and Beekeeping" (2013), by Dewey Caron and Lawrence Connor. "Diagnosis and Treatment of Common Honey Bee Diseases" (2014), by Benn Sallmann and Rob Snyder. "The Beekeeper's Handbook" (2006), by Diana Sammataro and Alphonse Avitabile. "Honey Bee Pests, Predators, and Diseases" (1997), by Roger Morse and Kim Flottom. "Tools for Varroa Management: A Guide to Effective Varroa Sampling and Control," sixth edition (2017), by the Honey Bee Health Coalition. "A Field Guide to Honey Bees and Their Maladies" (2011), by Penn State University. "Honey Bee Diseases and Pests" (2016), by Marla Spivak and Gary S. Reuter. "Diagnosis of Honey Bee Diseases" (2000), by Hachiro Shimanuki and David A. Knox. United States Department of Agriculture. Handbook 690. Natural Beekeeping "The Lives of Bees: The Untold Story of the Honey Bee in the Wild" (2019), by Tom Seeley. Princeton University Press.

Part IV: The Rewards and Contributions of Beekeeping

Honey

Honey is a historically important food, valued by ancient populations as well as contemporary cultures around the world. Humans have consumed honey for some 10,000 years, as depicted by cave drawings of honey hunters in Spain from the Stone Age (6,000–8,000 B.C.). To this day, Indigenous honey hunters and beekeepers alike are revered for their ability to provide honey and beeswax. In this module, we will explore honey: how it is made, what it is made of, and the different ways in which beekeepers process and pack it.

READ

From Nectar to Liquid Gold

■ The main source of energy for a bee Key Points colony, honey is comprised of Honey is the main energy source carbohydrates and eaten by the bees for honey bees; it is comprised to provide energy for activities such as mainly of fructose and glucose. flying, fanning, and heat production, or Bees ripen nectar into honey by it can be converted to fat for later use. adding enzymes and lowering its Storing honey in the hive offers a food level of moisture. source during times of poor weather or a nectar dearth. With some exceptions (e.g., honey produced from the honeydew of aphids), honey is derived from sugary floral nectar that has been manipulated and dried by bees of the genus *Apis*. Bumble bees and other honey bee species within the *Apidae* family make variations of honey, but we will focus on what is made by *Apis mellifera*, the typical honey bee cultivated by beekeepers. Honey is a saturated solution of sugars — primarily fructose and glucose — along with numerous other minor components. Its composition varies by the different sources of nectar and with regard to the environmental and seasonal conditions during its production. Under the right circumstances, a colony of *Apis mellifera* bees produces more honey than is necessary for winter survival; indeed, it is the only bee species that produces enough to allow for harvest in North America. The beekeeper's challenge is to manage the colony to maximize the honey production with a surplus greater than the bees' needs for winter. Nectar Sources Influence the Classification of Honey Honey can be polyfloral (or multifloral), monofloral (or varietal), or honeydew honey. Most honey is polyfloral, from a wide variety of honey plants in bloom, with no main nectar source dominating the flavor or other characteristics. Such honey might be labeled as wildflower honey, spring honey, summer blossom honey, fall honey, or meadow honey. Monofloral honey is harvested from a distinct source of nectar. Though there are typically other plants in ■ bloom simultaneously, and most monofloral honey labeled as such probably includes nectar from several different plants, beekeepers identify these unique honey varieties based on the heavy-flow sources like black locust, clover, basswood, or goldenrod, to name a few of the most common wild sources here in the Northeast. Monofloral honey may also result when colonies are near large agricultural plantings of alfalfa, sunflowers, apples, oranges, strawberries, blueberries, soybeans, buckwheat, and numerous

other crops. Honeydew honey, sometimes called forest honey, is a rare variety produced from the sugary excrement of sap-sucking insects such as aphids, leafhoppers, and psyllids. Honey bee on orange blossom flower. Collecting Nectar Nectar is a mix of chemically altered waterborne nutrients that are secreted into plant nectaries. Nectaries are typically located inside the blossoms, although some are extrafloral (meaning that they are found on leaves, stems, and other plant parts). The sweetness (concentration of sugar) of nectar varies, both in plant species and individuals. Honey bee foragers prefer the highest sugar content available. Pear blossom nectar, for example, is typically about 5- 12% sugar and therefore often bypassed for other flowers that may yield as much as 60%. ■ In one foraging session, a honey bee typically gathers 25–40 mg of nectar by visiting many flowers. To gather this much nectar, she visits up to 500 flowers in a foraging session, nearly always visiting the same type of flower. On a good day, foragers embark on about ten foraging sessions. If nectar is plentiful, bees can carry loads as great as 70 mg — about 85% of their body weight! As nectar travels from a forager's mouth to her honey stomach for temporary storage, enzymes from her hypopharyngeal and salivary glands are added. This initiates the process of the nectar becoming honey. These enzymes — namely invertase and diastase — break down the large sucrose molecules of nectar into the smaller sugars of fructose and glucose. From Nectar to Honey Upon her arrival at the hive to unload the nectar, the forager is greeted by receiver bees, who add a variety of additional enzymes to the nectar. When ripe (fully converted from nectar), most honey has less than 5% sucrose. The bees also add glucose oxidase, converting small amounts of glucose into gluconic acid and hydrogen peroxide. This increases the acidity of the honey, thus lowering the pH. The nectar is passed from bee to bee, with each bee adding more enzymes (as well as the amino acid proline) to the liquid. With all of these additions, the chemical properties of the nectar change to become more like honey. Finally, after being passed along by several bees, the unripened honey is deposited in a honey cell. ■ Once in the cell, the unripened honey must have its moisture content reduced. Nectar receivers smear it on the walls of the honey cells, and other house bees fan their wings to circulate air throughout the hive and promote evaporation. When the water content is reduced to approximately 18%, usually after a few days of fanning and handling, the honey is fully ripened. The time devoted to the process and eventual moisture content depends on the ambient humidity and temperature, the characteristics of the nectar (in other words, the mix of flowers from which it is derived), and the colony population and its other activities. When the transformation of nectar into honey is complete, bees cap the honey cell with beeswax. The properties of honey give it a very long shelf life; indeed, it is one of the only foods with no expiration date. Specifically, the high sugar content, the low pH, and the presence of hydrogen peroxide and various enzymes make it an inhospitable environment for most bacteria and other microorganisms.

READ

Medicinal Benefits of Honey

■ Honey has been revered for centuries Key Points for its medicinal value. Ancient medical Honey is effective in managing philosophers such as Hippocrates and coughs and wounds. Aristotle praised and promoted honey for a variety of internal and external ailments and injuries. In ancient times, honey was used to “cure” everything from intestinal distress, cough, insomnia, wounds, heart palpitations, anemia, poor eyesight, gout, nervous disorders, fever, contraception, baldness, and tuberculosis. It was a major

ingredient in medicines in ancient Egypt, Greece, India, China, Babylonia, and Rome, and there are references to honey's healing properties in the Bible, the Qur'an, and the Torah. Since the introduction of antibiotics in the early 1900s, the use of honey as a medicine has largely fallen by the wayside, although many websites and naturopathic doctors continue to promote honey as medicine and cure for many diseases. But few of the claims about the diseases and ailments that honey is advertised to cure have been proven scientifically. In order to be considered medicine and recommended for use in hospitals and doctors' offices, drugs and other treatments must be rigorously tested in randomized clinical trials with a large number of human patients. Today, there are only two conditions for which honey has been scientifically shown to be medically effective: treating coughs and treating wounds. Clinical trials have shown that honey can be an effective cough suppressant in children. It works better than taking no cough suppressant and can be as effective at reducing cough frequency as dextromethorphan (the active ingredient in most over-the-counter cough suppressants). Honey helps to soothe and coat the throat, and a teaspoon and a half before bed can help children sleep longer through the night with fewer coughing episodes. To treat wounds, honey is applied to a dressing and the dressing is placed on the wound. This accelerates skin healing and kills and prevents bacterial growth on the skin's surface. Some companies sell prepackaged honey-coated wound dressings. Of course, not all honey is created equal. Honeys with higher peroxide concentrations exhibit stronger antibacterial effects. Honey dressings have also been shown to accelerate the healing of burns as compared to other types of wound dressings, with patients who receive honey dressings healing four to five days sooner. Some evidence suggests honey can heal postoperative wounds, pressure ulcers, and diabetic foot ulcers faster than other antiseptic treatments and with fewer adverse effects. The beneficial effects of honey as a wound dressing are primarily due to its antimicrobial activities. It helps kill bacteria at the wound site, keeping the area clean while the wound heals. Some antioxidants present at low levels in honey (particularly dark honeys) can help reduce inflammation and improve healing.

READ

Manuka Honey: The Most Antibacterial Honey

■ In the 1980s, manuka honey was Key Points discovered to have unusually strong Manuka honey has exceptional antibacterial effects; more so than antibacterial properties due to the other types of honey. Manuka honey is presence of methylglyoxal (MGO). derived from the manuka bush native to New Zealand and Australia. Methylglyoxal (MGO) is the chemical responsible for these enhanced antibacterial effects. The higher the level of MGO, the greater the honey's antibacterial effects. Manuka honey is primarily used as a topical ointment for wounds and burns. Some hospitals use manuka honey dressings on patients. Studies in the laboratory exposing bacteria to manuka honey in petri dishes have found that manuka honey can kill 58 different species of bacteria, including *Helicobacter pylori* (responsible for stomach ulcers) and methicillin-resistant *Staphylococcus aureus* (a type of antibiotic-resistant infection sometimes found in hospitals). This laboratory research is promising, but scientists have not yet tested its use to control these particular infections in human patients. Just like other honeys, not all manuka is created the same. Some manuka honey has less- pronounced antibacterial properties. It believed that climate and soil type play a role in the level of chemicals in manuka bush flowers. In New Zealand, manuka honey is tested for its level of MGO and assigned a

number called a Unique Manuka Factor (UMF). Honey with a value of at least 10 UMF is considered therapeutic; it can be marketed as “Active Manuka Honey” or “UMF Manuka Honey” and sold for a premium price. An 8-ounce jar of 10 UMF manuka honey retails for \$40–\$50, and an 8-ounce jar of 20 UMF manuka honey retails for \$100–\$200. Given these prices and the fact that New Zealand's manuka honey market has grown substantially over the past ten years, it is sought after by many beekeepers. Some follow the manuka honey flow across the country, flying their hives into manuka-rich areas by helicopter.

READ

Composition of Honey

■ Sugars comprise 98% of the solids Key Points found in honey, making it a great In addition to sugar and water, natural sweetener. The major sugars honey contains acids, vitamins, present in honey are fructose and minerals, and some glucose, while the minor sugars are microorganisms. sucrose and maltose. The chart below The Clostridium botulinum spores shows the typical composition of naturally present in some honey honey, based on 490 samples tested in are a risk to infants. the U.S. Honey Composition Component Average Range Moisture 17.2% 13.4–22.9% Fructose 38.19% 30.9–44.3% Glucose 30.3% 22.9–40.7% Sucrose 1.3% 0.2–7.6% Maltose 7.3% 2.74–15.98% pH 3.91 3.42–6.10 This chart is based on White, J.W., M.L. Riethof, et al. 1962. Composition of Americans honeys. U.S. Department of Agriculture, Technical Bulletin 1261:124 Acids Although honey has a sweet taste, it is actually acidic. Much of the acidity is contributed by gluconic acid but other acids are naturally present in honey as well, such as formic acid, oxalic acid, acetic acid, and lactic acid. Vitamins and Minerals Honey contains trace amounts of vitamins and minerals. Darker honeys and honeydew honeys are highest in vitamins and minerals. These include Vitamins B1, B2, B3, B6, and C, along with the minerals sodium, potassium, magnesium, calcium, copper, iron, manganese, phosphorus, chlorine, and sulfur. The levels of vitamins and minerals are extremely low — only about 0.17% of honey — and make up a tiny fraction of a person's recommended daily intake. For this reason, honey should not be relied upon as a source of nutrients for the human diet. Microorganisms Most bacteria do not grow in super-saturated sugar solutions such as honey. Nonetheless, some species of bacteria and yeasts can survive in this environment. Sugar-tolerant yeast spores are often present in honey, but the low moisture content prevents their activation. Should the moisture content increase, yeast spores can activate and the yeast can multiply, putting honey at risk of fermentation. The most notable bacterium is Clostridium botulinum. While the active bacteria cannot survive in honey, its spores can. A test of over 2,000 honey samples found that 5% contained Clostridium botulinum spores. If spores are consumed, they can become active and cause an infection of botulism. The bacteria produce a neurotoxin that can cause paralysis and death. Infants under the age of 12 months are at risk because they have not yet developed the healthy gut microbiome that outcompetes the bacteria. (Adults, on the other hand, are not affected by the spores.) For this reason, honey should never be given to infants under a year old. Many U.S. beekeepers therefore include cautionary language about feeding honey to infants on their labels.

READ

Changes in Honey Over Time — Crystallization

■ In time, honey can undergo some Key Points changes. Most honeys will eventually Crystallization occurs when crystallize, and some ferment. glucose precipitates out of the water in honey. Crystallization Honey is a super-saturated sugar The rate of honey crystallization solution. This means that the water has depends on the presence of absorbed more sugar than it can foreign particles, the nectar under normal circumstances. This source, and the temperature at state is not very stable, and some of which it is stored. the glucose naturally precipitates out of the water in time, forming solid crystals. This is called “crystallization.” Once a crystal forms, it is a platform for more crystals to join, meaning that once the crystallization process begins, it continues until the entire jar of honey is crystallized. Sugar crystals are also more likely to form around other foreign particles in the honey, such as granules of pollen, wax, propolis, dust, or even air bubbles. Crystallization is also more likely to occur when honey is stored in cool temperatures (typically below 68°F [20°C]), as water absorbs more sugar in warm temperatures (such as the warm temperatures in a bee hive). Above all, the floral source of the honey influences the rate of crystallization. Some honeys are notorious for crystallizing quickly, such as goldenrod, alfalfa, clover, apple, canola, raspberry, and dandelion, which can crystallize in several weeks or months. Honeys that are slower to crystallize include acacia, black locust, sage, buckwheat, orange blossom, and tupelo. These honeys may not crystallize for many months or years. Honeys with a low moisture content tend to develop finer crystals (for a texture similar to creamed honey), while honeys with a higher moisture content develop coarser crystals. The best way to liquefy or soften crystallized honey is to place it in a warm water bath (95– 104°F [35–40°C]), with the water reaching the level of honey in the jar. The heat slowly dissolves the sugar crystals to make the honey liquid again. If you sell honey, you may want to include these instructions to your customers and explain to them that crystallization is a natural process. Some customers think that crystallized honey has gone bad or has additional sugar added to it, since they are more accustomed to seeing filtered honey on grocery store shelves. Delaying Crystallization Without Pasteurization To delay crystallization, strain the honey through a fine nylon cloth before storing or bottling it. This removes most of the foreign particles, such as flecks of wax or pollen grains, that can initiate crystallization. Crystallization does not occur below 32°F (0°C) and occurs very slowly below 50°F (10°C), so honey can be stored at cold temperatures until it is ready to sell. Crystallization can also be further delayed by allowing bees to clean extracted “wet” honey supers before the equipment is stored for winter. If extracted supers are stored wet, the honey can form crystals over the winter. When you place the supers on colonies next spring, last year’s crystals may be incorporated into this year’s crop and contribute to earlier crystallization. To allow the bees to “clean up” the wet honey on the frames, place the extracted supers on top of their hives (above the inner cover but below the outer cover) for a couple of days.

READ

Changes in Honey Over Time — Fermentation

■ The sugar-tolerant yeast spores Key Points present in honey can become active if Under some conditions, natural the moisture content exceeds 18%. yeasts can consume the sugar in Honey with a moisture content above honey, transforming it into alcohol 19% will likely ferment within one year. and carbon dioxide. Usually, when honey is sufficiently ripened and capped by bees, the moisture is low enough to prevent yeast growth. But sometimes during strong nectar flows, such as orange blossom

flows in the South, bees bring in nectar so quickly and in such great quantity that they do not ripen it sufficiently before capping. This honey can have high levels of moisture and is more susceptible to fermentation. Most honey fermentation occurs when honey becomes crystallized. Water is freed when sugar crystals precipitate, leaving a jar of honey with crystallized sugar on the bottom and liquid on the top. This liquid is susceptible to yeast growth. When yeast begins to grow in honey, it consumes sugar, transforming it into carbon dioxide and alcohol in a process called fermentation. Honey in the early stages of fermentation tastes fruity but, in time, develops a yeasty, unpleasant taste. It may also look foamy or bubbly. Preventing Fermentation Without Pasteurization Before Honey Is Extracted Honey with a moisture content below 18% is safe from the risk of fermentation, while any honey with a moisture content above 18% has the potential to ferment. To reduce the risk of fermentation, only frames that are at least 80% capped should be extracted. Beekeepers can test their honey before extraction to determine the moisture content; this can be accomplished with a honey refractometer. Test small samples from a few cells on a frame to get an idea of the moisture content for the entire frame of honey. If the moisture content is below 18%, you can extract it. If it is above 18%, you must reduce the moisture level first. To do so, honey supers can be temporarily stored in a warm room with a dehumidifier or a warm fan blowing. In several days (depending on the initial level), it can be brought down to an acceptable range. Many beekeepers have drying rooms (called hot rooms) in their honey houses for this purpose, where warm, dry air (90–95°F or 32–25°C) circulates. With this setup, 1–3% moisture can be removed in a 24-hour period. Alternatively, honey with a higher moisture content can be blended with honey of a lower moisture content to create a large batch of honey with an overall moisture content below 18%. After Honey Is Extracted Straining honey immediately after extraction with a fine mesh or nylon removes particles that can initiate crystallization. After the honey is placed in bottles, barrels, or Fermented honey on the buckets, ensure these A refractometer. frame. containers are airtight. Photo Credit: Betterbee Honey is a humectant, meaning it draws in moisture from the environment. If it is in contact with the air, moisture can accumulate on the surface of the honey, and fermentation can begin. Honey can also be stored at cool temperatures below 52°F (11°C) until it is sold. At these temperatures, yeast cannot grow.

READ

Processing Liquid Honey

■ After honey comes out of the extractor, Key Points it can be processed in a variety of ways. Straining, filtering, and ultra- filtering remove varying amounts For most backyard beekeepers, honey of particles from honey, including is consumed or sold raw. The U.S. pollen. National Honey Board defines raw honey as "honey as it exists in the Pasteurization delays beehive or as obtained by extraction, crystallization and fermentation, settling, or straining without adding but it breaks down the enzymes heat." Today's consumers are very that are naturally present in honey. interested in raw honey, which is convenient for small-scale beekeepers as they can extract and package it with minimal steps. These beekeepers may not process honey at all, opting to bottle it with all its pollen and propolis, along with any small bee body parts that happened to fall into it. Raw honey must be packed airtight. Nearly all raw honey eventually crystallizes. Beekeepers can remind their customers that raw honey is a superior product — the crystallization marks it as such — and that crystallized honey can be re-liquefied through slow, gentle warming. Some beekeepers process their honey by removing some of

the particles that help initiate crystallization, and others even heat their honey. Below, explore how honey can be processed once it is out of the comb but before it is in the bottle.

Straining Before being stored in containers, honey can be strained to remove large particles such as wax, propolis, bee body parts, dirt, wood splinters, and metal flakes from equipment. Strainers tend to be small enough to stop these products from entering the container but large enough to permit some pollen, propolis, and beeswax to flow through (200 micrometers or larger). The resulting honey is clear and attractive to consumers. A range of fabric or wire strainers are available: Cheesecloth can be used, or other metal or nylon fabric filtering bags and bucket strainers that are 200, 400, or 600 micrometers can be purchased through bee equipment supply companies. For large-scale beekeepers, commercial honey strainers are also available for purchase through bee equipment supply companies. Honey straining through a nylon filter basket. Beekeepers may wish to filter their honey after straining it to improve the clarity even further. Filtering also prolongs the time it takes for honey to crystallize, as it removes most of the pollen grains and dust that remain in the honey after straining. Diatomaceous earth, a very fine-powdered sedimentary rock, is usually added to the honey before it passes through the filter; this process removes nearly all the particles in the honey. Honey filtered in this manner — with little to no pollen remaining — is still honey. Ultra-filtration is a different type of filtering that is rare in North America. In ultra-filtration, honey is forced through an extremely small filter (one-tenth of a micrometer). Every last particle is removed, and the final product is considered a “sweetener” — rather than honey — by the U.S. Food and Drug Administration. The U.S. honey qualifications grade honey on the basis of clarity. Grade A honey (the highest score) is filtered to remove nearly all particles from honey. Grade A honey may contain trace amounts of pollen grains and other fine particles that do not affect the overall clear appearance of honey. Grade B and C honey contains more of these particles, and the honey is less clear. Honey should contain some pollen grains. The presence of pollen grains is used to identify the source(s) of the honey. While bees collect nectar from a plant, they often incidentally collect some pollen that makes its way into the honey. This can verify that a particular honey is a specific varietal (for example, honey marketed as “orange blossom” should contain orange pollen), and it can identify the country or region from which the honey sources have come. With an estimated 91 million tons of illegal honey entering the U.S. each year, being able to identify the source of a honey based on the pollen it contains is very useful.

Warming Honey that is warmed is less viscous; it flows freely from combs and through machinery, making extracting and bottling more efficient. Warming honey also prevents crystals from developing on equipment, which can contribute to crystallized honey. If honey is not heated above 105°F (40°C), it remains raw and the nutritional and medicinal values are preserved, as well as the natural aromas and flavors. Warmed to about 90°F–105°F (32–40.6°C), the equivalent of the temperature in an active hive, the honey is liquefied enough for faster extracting and bottling. At this temperature, the enzymes in honey remain intact and the honey is not permanently changed in any way. Honey can be warmed on the frame, with supers placed in hot rooms prior to extraction, or warmed at various stages of the extraction process. It may be warmed when it is passing through a sump (a temporary holding tank that receives honey from the extractor before it is pumped to other processing tanks), while it is in settling tanks, and while it is being bottled. This heated bottling tank warms honey.

Settling Commercial beekeepers who sell their honey in bulk (usually in barrels) often do not filter or bottle their own honey; instead, they remove most of the debris with settling tanks. A settling tank is a large tank that holds honey for two or three days. While

the honey settles, most of the debris floats to the top. The clean honey is then piped out of the bottom of the tank into barrels, leaving most of the debris behind in the tank. Often, tanks are heated with a water jacket to aid in settling and reduce crystallization. The honey may be filtered further before bottling. Small settling tank. Pasteurizing Pasteurization is the process of heating food to kill microorganisms. In honey, pasteurization destroys the yeasts responsible for fermentation and delays crystallization by dissolving any crystals already present. It does not, however, destroy *Clostridium botulinum* spores. Honey can be pasteurized by raising the temperature to 145–155°F (63–68°C) for 30 minutes, or through flash heating that brings the honey to 160°F (71°C) then immediately cools it. Most grocery store honey is pasteurized so it will remain clear and liquid. Honey that remains liquid for long periods of time is easier to handle at the table, as it pours easily. For hobby beekeepers, pasteurization can occur in the kitchen, heating the honey in a pot on the stove. Keep a food thermometer in the honey to monitor the temperature and stir it to ensure even heating. For larger commercial beekeepers, honey-pasteurizing equipment is available through some bee equipment supply companies or food processing companies. The disadvantage of pasteurized honey is that heat breaks down the enzymes naturally present in honey, such as invertase and glucose oxidase. The higher the temperature, the faster these enzymes break down. For instance, the half-life of invertase at 176°F (80°C) is 8.6 minutes. The breakdown of glucose oxidase diminishes some of honey's healing properties when applied to wounds; without glucose oxidase, the production of hydrogen peroxide halts. Heating at too-high temperatures can also burn the honey, causing it to caramelize, especially at the higher-end temperatures used in pasteurization. This changes the color, aroma, and flavor of the honey, thus reducing its grade and marketability.

READ

Components and Design of a Honey House

■ By now, students in this course are Key Points likely familiar with basic, small-scale A honey house should be carefully honey extraction (while some are designed to maximize productivity familiar with large-scale extraction). If and efficiency. beekeepers manage a few to a few dozen colonies, honey extraction can take place in their own kitchen or garage. Beekeepers who manage more than 50 colonies likely want to construct or purchase a honey house to process honey more efficiently and store honey and equipment. Let's discuss some of the main components of a honey house. General Considerations It is best for beekeepers to locate their honey house on a separate piece of land from their own home, as sharing a property can affect the retail value of the home. Keeping these buildings separate also allows beekeepers to sell each place separately if needed. The honey house should have access to electricity, water, and good drainage. Beekeepers need to maintain a clean honey house, for example, by mopping the floors. The honey house should have floor drains and a hose spigot for the convenience of hosing floors. It's also useful to have a sink for hand washing and a cupboard to store cleaning supplies. The honey house should be easily accessible by truck. Single-story honey houses on level land will need pumps to push the honey from the extractor to strainers or tanks. With a two-story honey house, extraction can take place on the top floor and bottling on the lower floor, allowing the honey to flow by gravity from the extractor to the strainer, settling tank, or bottling tank. Two-story honey houses should have truck access on both floors to eliminate the need to move equipment up and down stairs. A freight elevator will also work. Carefully plan the layout of the

honey house to make it efficient and convenient. Consider the logistics of items moving into and out of the building: full supers coming into the building; empty supers going out of the ■ building (or being stored in the building); and barrels or bottles being filled, stored, and shipped. Maximize the flow in the rooms so people are not bumping into each other and so that heavy items do not have to be moved long distances. Consider how many people will be in the rooms at one time and plan accordingly for space. Visit other beekeepers' honey houses to learn how they have been constructed, what aspects they like most about them, and what aspects they would change. This can help beekeepers design a more optimal honey house for themselves.

Loading Dock The loading dock is the area where honey supers are removed from the truck and placed in the honey house (often the hot room). The beekeeper can use a forklift to easily remove supers from the truck and bring them into the honey house. If the beekeeper does not have a forklift, the loading dock should be level with the truck bed to minimize lifting. The loading dock is also the place where bottling and bee yard supplies are accepted, so it is wise to consider the height (or heights) needed for freight truck deliveries in addition to the level required for bee yard truck offloading. Many large trucks will not deliver to the honey house unless there is a forklift or loading dock available. The hot room is where supers can be temporarily placed to be warmed and should be in close proximity to the loading dock. Its temperature is usually set to 80–100°F (26–38°C), and relative humidity is kept to 0–20%. While in the hot room, honey will become less viscous and easier to extract. If honey moisture is too high, warm air and/or a dehumidifier in this room can help to reduce water content before the supers are extracted. The hot room should contain a thermometer and a humidity gauge.

■ **Hot Room** ■ Beekeepers without space for a hot room can consider purchasing a honey drying machine from Dadant. These machines lower moisture content very quickly; 1–2% moisture is removed from two barrels' worth of honey every hour.

Extracting Area Once honey is warm and at the right moisture level, it can be moved into the extracting area. In this area of the honey house, honey combs are uncapped and placed in an extractor. The size of the beekeeper's operation influences the type of extractor they use, which in turn influences the setup of the honey house. Equipment should be thoroughly researched ahead of constructing the honey house. It is also wise to plan around potential equipment upgrades, additions, and increased power needs. While hobby beekeepers tend to use heated uncapping knives and forks to remove beeswax cappings by hand, sideline beekeepers increase efficiency by using uncapping machines to do the process for them. Various versions of "sideliner uncappers" exist, but in most, the equipment calls for beekeepers to feed frames one at a time between two metal rollers with circular cutting knives on them. The circular knives slice holes in the wax cappings on both sides of the frame at once. They do not employ electricity or heat. These sideliner uncappers are resting on top of uncapping tanks to catch cappings and honey drippings. Frames are then lifted manually into the uncapper and into the extractor. Automatic electric uncappers also exist. They function by using heated, oscillating knives to cut cappings from the honey comb. Frames are loaded onto a short conveyer belt chain, which feeds the frames one by one into the uncapper. One frame at a time is guided downward between two heated, oscillating knives. The cappings and honey drippings collect in an attached cappings tank. Commercial beekeepers often purchase large pieces of equipment that form a big assembly-line system for uncapping and extracting. These systems require minimal labor to load the automatic uncapper (again, functioning by using heated, oscillating knives) and the extractor. There are even machines that can lift honey frames to feed them into the uncapper, preventing the need for the ■ beekeeper to bend down

to grab and load frames. These commercial systems have the potential to extract honey from hundreds of supers in a day. After the wax cappings are removed, the honey frames are placed in extractors. Different types of extractors will be explored later in the module.

Sump In small hobby operations, honey can flow directly from the extractor into buckets. Some beekeepers may place filters over buckets to strain the honey. In larger operations, however, honey flows out of the extractor into a sump, a temporary storage tank that begins straining the honey. Sumps are placed below extractors so honey can flow into them by gravity. Any honey collected in uncapping tanks often flows to sumps as well. Sumps are often gently heated with water jackets so the honey can be strained more efficiently and wax separated from it more easily. Screened baffles help separate wax and particles out of the honey as it runs through, helping it flow through the pump without clogging it. Many sumps have floats to detect whether honey is about to overflow. Pumps connected to the sump can be automatically turned on to allow honey to drain out and prevent spills.

■ **Sump with three screened baffles for honey to pass through.** Pump If a beekeeper's honey house is located on one floor, the honey will need to be pumped from the sump to the strainer — assuming the beekeeper wants to strain their honey more finely than what the sump accomplishes — or to the settling tank if they simply want to separate any remaining wax and particles by settling. All pipes must be "food grade"; i.e., rated for use with food products. Large-scale operations use commercial strainers or settling tanks to further process honey. Typically when beekeepers plan to bottle and sell their own

■ **Strainer or Settling Tank** ■ honey, they strain it first. If beekeepers sell wholesale, they often use a settling tank instead. Straining ultimately produces a higher-clarity product. If a beekeeper strains their honey through a fine mesh, a sump that removes the largest, coarsest particles ensures that the mesh does not clog quickly. A commercial-sized strainer uses a mesh made of nylon or a metal screen. This mesh is located at the bottom of a tank, submerged underneath the honey. In this way, minimal air bubbles become incorporated in the honey as it passes through the mesh. Honey is strained at a warm temperature of around 100°F (38°C) to reduce the viscosity and increase the flow through the strainer. Honey should not be strained at a higher temperature, however, because any wax will melt and pass through the mesh, making the honey cloudy when it cools to room temperature. Mesh straining bags can be replaced as needed. After passing through the strainer, honey is bottled. Beekeepers who do not finely strain their own honey use settling tanks as the last processing step before they run their honey into barrels for sale. Honey exits the pump into large tanks (often made of stainless steel) to sit for two to three days. Beekeepers who market different varieties of honey may have several separate tanks.

Packing Honey Beekeepers who use barrels simply sell the barrels to other companies that bottle the honey. These barrels must be made from or coated with food-grade materials. Sideline and commercial beekeepers who bottle their own honey often use automatic bottling machines. These machines ensure that the bottles receive a consistent and accurate volume, efficiently and with minimal spills. Labeling machines attach labels to the bottles. Small-scale beekeepers usually bottle and add labels to their honey by hand. Designated areas for storage are helpful in the honey house. Beekeepers can temporarily store extracted supers before returning them to colonies, as well

■ **Storage** ■ as barrels and packages of honey before transport or sale.

READ

Types of Extractors

Let's explore the three different types of extractors, from smallest to largest. Tangential extractor: These extractors hold between two and 16 frames at a time and can be hand cranked or electrically powered. The frames sit in a metal basket, and only one side of the frames is extracted at once. After extracting one side of the frames, the frames are flipped and the other side is extracted. Small models require the beekeeper to flip the frames by hand, but other models may rotate the metal basket inside for the beekeeper so the machine does not need to be opened again until the full extraction is finished. These extractors are usually used by small-scale beekeepers who manage fewer than 50 colonies. In the U.S., equipment of this scale is typically purchased through the bee equipment companies Maxant, Dadant, and Mann Lake. We encourage beekeepers to explore current catalogs to familiarize themselves with honey extraction products. Radial extractor: A radial extractor can hold from 10 to over 100 frames and is electrically powered. The frames are placed in the extractor like spokes in a wheel, and both sides of the frame are extracted simultaneously. These extractors are typically used by beekeepers who manage between 50 and 500 colonies. In the U.S., equipment of this scale is typically purchased through the bee equipment companies Maxant, Dadant, and Mann Lake. Parallel radial extractor: Parallel radial extractors are radial extractors that are tipped horizontally. This allows frames to be loaded in a parallel fashion by pushing them along a ramp directly into the extractor. In this way, no lifting is required by the beekeeper. They are commonly used in large sideler or commercial operations, for beekeepers managing hundreds or thousands of colonies. The largest extractors in the U.S. can handle 240 frames at once! These extractors are often part of a large extraction system with an automatic electric uncapper, a conveyor belt, and a powered extractor. In the U.S., these large systems can be purchased through Cowan Manufacturing or Cook & Beals.

READ

Other Honey House Considerations

When thinking about honey houses, here are a few further considerations. Honey houses should be bee-proof, so no bees or wasps can enter from the outside. Any windows that may be left open should have screens. There should be a screened window in each room of the honey house with a small bee escape. Any bees that enter with the supers will be drawn to the window and find their way out. Use concrete or tile floors with drains in each room. These floors are easy to mop with water to promptly clean up honey spills. Floors should be treated with paint, acrylic, or another treatment to protect them against acid wear from honey. Honey and water conduct electricity. Electrical wires and outlets should be protected with waterproof fittings and coverings to reduce the chance of shock. Honey extracting and processing equipment should be made with stainless steel to ease cleaning and to last a long time. Iron and steel equipment can rust over time and can darken honey if they remain in contact for long periods of time. Below is a design for a sideler-scale honey house. In this particular example, the facility is located on one floor and uses a parallel extractor. Long description ■ A floor plan of a honey processing facility. Storage tank with strainer and bottling valve is located in the upper right corner, with sink and counters adjacent. A wax melter sits below these. Two loading docks line the left wall. In the center, a sump and pump is installed below floor level. A wax spinner is positioned in the middle right section, with a hot room beside it. An uncapping tank is situated beneath the hot room, with an extractor at the bottom right of the facility. Storage areas occupy the upper portion of the layout.

WATCH

Virtual Field Trip to Honey Houses

Now that we're familiar with the general organization and components of a honey house, let's take a field trip to explore honey extraction in two operations. Ryan Apiaries Jon Ryan is a small commercial beekeeper who manages 300 colonies. His honey house is two stories, conveniently located on a hill. His extractor is on the top floor, while his bottling tank is on the bottom floor. Gravity allows the honey to flow from the top-floor extractor to the bottom-floor bottling tank; because of this, he does not require any pumps. Jon Ryan sells his honey to a company in its rawest form. The honey is not strained at all and contains beeswax and other debris. It is not pasteurized. The honey is allowed to crystallize. This is exactly how the company (and their customers) wants the honey; they advertise it as "raw" to their consumers. In this video, you will accompany Jon, owner and operator of Ryan Apiaries, as he extracts honey in his honey house. Video Transcript Hi. I'm John Ryan and I'm a commercial beekeeper. I say commercial, but running about 300 colonies, which, by commercial standards, is pretty small. But I have a pretty good market with really raw honey. I will give you a tour of my honey house. Again, keep in mind that this is a unique situation, any honey house can be different. I think a lot of the things that you'll see here are good ideas that can be implemented in different ways. I will say that this building was built in 1935-36. It was right in middle of the Depression and the fact that a beekeeper could build this, considering those economic times, I think is pretty impressive. And I like this building very much. Welcome to my honey house. Come right on in. The first part that you're going to see is basically storage, where all the supers come in from various yards and are all stored here. Right now, it's pretty empty, believe it or not. It needs a lot of organizing. But this will be filled to the gills by the time I'm done. I guess I'll give you a path of the honey, as to how it gets here. You'll notice that the floor of the honey house is about even with the bed of my truck, I don't know if you can see that, which makes it very handy for unloading of the supers, which I'll show you how I do that in a second. Okay. The honey is all taken off the colonies, out in the yards. I have a boom, I don't know if you saw that in the back of the truck, but that's made my life a lot easier. So, the honey is taken off, often two or three supers at a time, placed on pallets, like this, right on the truck. From that point, I'll take my hand cart. And because my truck bed is level with the floor, you just unload them at the handcart, and bring them into the honey house. Now, a consideration to keep here, which I think is very significant, and was very clever of the Stevens when they built this building, was that they built this honey house on a fairly gentle slope. So, I have a loading dock on one level on the ground floor, but I also have a loading dock on the downside downstairs. It's truck-bed height for a tractor trailer. So, again, I have two loading docks. One on two different levels. Also, by being on a hillside, as you will soon see, my honey is all extracted on this floor, and it all flows by gravity downstairs, where it is finally either put in barrels or jars. But, I do not need a pump of any kind, which that means eliminating the sump. In other words, an intermediate storage container for the pump in and then pump out of and pump to a tank or whatever. If you can somehow incorporate that into a design of any honey house, I think it would be well worth doing. From the truck, again, the honey is all put on pallets. It is then wheeled in on my hand cart, and then it comes into this room right here, which I call a hot room. The hot room accomplishes several things. First of all, once honey is extracted, it's very hard to do anything about the moisture. And moisture is critical. It has to be 18.6 or less. Though fermentation really can take place at basically any moisture level. But to be grade A, you need to be at 18.6. And again, once the honey is extracted, it is very hard to remove any moisture. By putting your honey in the supers, before they are extracted, in a warm, dry environment,

you can actually lower the moisture content. I tried putting a dehumidifier in here one time and it really was not all that productive. It was already so warm and dry that it took quite of the moisture out anyway. But even if you don't have a whole hot room, just some sort of smaller room, or even a closet-like that you can warm the honey, I think it's worth doing, just for the moisture content. The other thing that it does by having a hot room here is storage for a rainy day. The other thing that is very significant about this room is you will see at one end is a window. And no matter what method you use for removing bees from the supers, you're probably going to have a few bees left over. Like people, there's always 10 percent that don't get the word, or maybe a little bit less than that. But at any rate, I don't like having bees in the extracting room. And by bringing it into the hot room here, with one window down there, in a few hours or even a day sometimes, the bees usually work their way up, out of the supers, the stack of supers and go down to the window. From there I have a shop vac. And I can suck the bees up. And I don't think it hurts them too bad, and take them back out to their original bee yard and dump them. Or if you have a nuc or something out in your front yard here, you can dump your swarm, your hot room bees there. At one time, I used to fill this hot room completely and it would hold about eight tons of honey, if the supers were reasonably well filled. However, since the introduction of the small hive beetle, I can't do that anymore. I have to, from the time I escape, and that's how I get the bees out of the supers. From the time that the bees are removed from the supers, to the time you extract it, depending on the temperature, but let's say about five days, after that time, the beetle larva is going to emerge and they can be extremely destructive, not to mention disgusting. So, now I only fill my hot room partway because I'm fairly slow at extracting. Though, that's, again, my limitation. But at any rate, I don't keep the temperature all that warm in here. I think low 70s is good enough for what I'm trying to accomplish. I know some beekeepers round it up to 80, 90 degrees, but that may be because they're trying to run the honey out more quickly or pass it through the hot room. So, from here, now that the honey has been in the hot room for a day or two, hopefully most of the bees have vented. Hopefully, the moisture content is reasonable. And we're going to extract today. So, the next step is to move these stacks over into the next room, which is the extracting room. And to do that, again, I use my hand cart. And we'll step out over into the next room. So, the next step in this process is to scrape off the burr comb, both the top bars and the bottom bars. And to do this, I made a set of rollers, which, again, is a copy, basically, from some other beekeepers have had this idea. I somehow need to get from here to down to where it's actually going to be uncapped. And at the same time, it's nice to have a place to actually keep your scrapings because they do make a mess. When it hits that, it shuts the lift off. So, the supers, basically, are always maintained at this well, which not a big deal, but if you get tired of bending over to pick up an 85-pound box, after the end of the day, you will appreciate having something like this. I think it's a great idea. Okay. I'm just getting a little more scraping down here. And once the line is still, I will go to uncapping. To really run this equipment efficiently, it's really a two-person operation. Somebody to be running this end and loading the extractor, and somebody to be, basically, doing nothing but uncapping. And that's how you get most production done. There is a commercial, deboxer, it's called. You need to somehow now get these frames up out of the super and over to the uncapper. I stole this idea from another beekeeper down in Spencer. Basically, all you're doing is putting the super on that raised rim. And, this is a well filled super, so these are pretty tight. But when you push down, it pops the frames out. Which doesn't seem like a big deal, but it sure makes it a lot easier for getting the combs out of the super without having to dig each one out. So from this point,

we're now going to go to the uncapper. Regardless of the size of the operation you have, you somehow need to uncap the honey. There are basically two ways of doing it. Either with a knife, or knives, or with a flail like a Dakota, which basically is chains that just beat the cappings off. I prefer the knives. I think it was a cleaner job. Though, to be perfectly honest, I've never used a Dakota. I know some beekeepers do like it. It is fast, but I think it's hard on the wooden part of the frame, too. So from here, that's a reasonably capped, but not great. So from here, we will now uncap it. And this is a Cowen uncapper. In fact, my whole extracting system is Cowen out of Parowan, Utah. I think they make a very nice setup. So, you now uncap. This will do about eight frames per minute, which is about as fast as I can feed them up there. Okay, we now have 60 frames uncapped. And the next step is to actually load them into the extractor. These frames were extracted yesterday. They are now empty. So to load this, again, we have 60 frames here, I have to then open the gate between the extractor and the trough, or whatever you call it. At the same time, the frames that were extracted yesterday will be pushed out as the new frames, the frames that have not been extracted yet, are pushed in. Again, Cowen makes a very clever thing. Just the way they join the extractor with the ramp here, I think is pretty clever. But another very simple and smart thing to do, this takes basically 30 frames. Now, I can sit here and count out 30 and that would take me forever, but they just made a simple little rail. And when that drops off the rail, there's 30. So I don't have to think. And I don't like being encumbered by the thought process. So, we will now proceed to load these and empty these all at the same time. And now there're 30 frames right there, which is half a load. And at the same time, you can see we pushed out 30 frames that were extracted before. Now, we just switch the reel over to the other side of the reel, and also advance the other 30. There are 30 in and 30 out. Okay, this is the parallel radial extractor. And as you could see, I loaded 30 frames in two different sides of the reel. I don't know why anybody would buy a large radial, regular old radial extractor, where each frame sits in as a radii. The only time I have to touch the frame is when I load it to uncap it. And some of the beekeepers even have a mechanism for loading the uncapper. But that's the only time I touch the frame, as you'll see when I unload these. I load nine at a time back in the box, I don't touch individual frames. This is so much faster. It does a very nice job. It's very efficient, and it gets the job done. This is a 60 frame, it holds 2 batches of 30 frames. I believe Cowen makes a smaller one, maybe for hobby size. And I think they also make 120, a really big one. I know some of the commercial beekeepers will run two of these parallel to each other. They will have two uncappers and two extractors all at one time. So, I give it a little pull just to help it along. There it goes. And I can't tell you the RPM, but that gets going pretty fast. So, the next step will be then to put these frames here, that are now empty, back in their original boxes. So, rather than carry this over I can just lift it off. Usually, the whole line is filled with about 6 of these, one full load, extracted load. Supers are then scraped. It's just a couple of boxes stacked up, nailed together. The scrapings all fall down into there and, you know, occasionally, I'll pick them, put them in a box. I think by scraping up the supers, again, it just makes it easier come next spring to space out your frame. This stack right here is the frames that are now in the extractor and have already been spun. This box is going to be the first box in the next load. That will go down here, on this pallet. The first frames into the extractor will be that super. The last frames into the extractor will be the top box. And when they come out of the extractor, the same frames go back into the same box. The same frames are basically kept together. It may be kind of a minor point but by not mixing your stuff, as little as possible, anyway, I think if there is a disease situation, that's well worth doing. However, to be really conscientious about American

foulbrood, you should be inspecting your bees before you remove any honey because once it goes into the honey house, into the extracting system, it's easy to spread it around. The honey from the extractor and the honey from the spinner are plumbed back together again, as you can see from these pipes right here. And then goes downstairs. And, again, one of the beauties of this building is that there is no sump, there's no pump. The honey, all by gravity, flows out the extractor, out of the spinner. And goes downstairs, hopefully to where we want it. As you can see, the honey is coming down this pipe right here, along and around the corner, and over into that kettle, where it will then be bottled. In the past, I put most of my crop into barrels. Right now, the price of honey in barrels is not great. But by packing for this company really raw honey, I get a significantly greater price. And in the past, I know we beekeepers often would fight over half cents. And I'm talking dollars difference here between barrels and putting it into jars. The problem is this is very labor intensive. But the process is this: The honey runs into this kettle right here. And because I'm doing a product that they want wax in, I've got an auger also in this kettle that keeps the honey and the wax cappings mixed up into, hopefully, a homogeneous mixture so that they do get an equal amount, or a proportional amount, of cappings in each jar of honey. On this tank I have a honey pump. There are several different brands. I like this one. It's pretty good. You put in a number, I don't know what it is, I might be RPM, I'm not sure what it's counting. But when it gets to that number it will shut off and, hopefully, that is exactly one pound. However, because a jar has to be filled to exactly one pound or more, we weigh each jar. Well, we don't weigh each jar. We spot check about every fifth jar so to make sure that they are properly filled. Because as the honey to wax ratio changes, that will also change the weight. There's nothing really special about this pump other than, again, it will meter out. It's got a foot shut off right here. Again, you dial in a certain number. And when I hit this pedal, there it goes. And, hopefully, it will shut off. Now, see I have to readjust the amount because it's just started. But that's the process. From there, the jars are then all put on these pieces of plywood. We go three tiers tall. And because this is the raw product with cappings that have not been filtered, not been strained, nothing, there could be dead bees, even a piece of wood or something in there. So, each jar then has to be checked. If you have a little stick, you have to go through and check on the top to see if there's anything that looks yucky to the general consumer. Pick it out. Caps are then put on all the jars. And once there's a cap on it, each jar is checked on the bottom to make sure there's not a nail or something that is heavy, like even a paint chip. And to do that, rather than look at the bottom of each jar, we just have a mirror we can look into and, very clearly, it will show up if there is any foreign particles in it that should not be. And that's the end. Kutik's Honey Farm Chuck Kutik runs Kutik's Honey Farm, the largest commercial beekeeping operation based in New York State; he manages 8,000 colonies. Chuck's honey extraction facility is located all on one level, so it requires pumps and sumps to move honey through the system. He runs two parallel radial extractors to increase the efficiency in his operation. Chuck sells a lot of honey wholesale in barrels, as demonstrated by this video in which you explore his honey extraction facility. Video Transcript Let's take a trip through Kutik's Honey Farm's extracting facility. I'm going to narrate this video because the extracting facility is quite loud, and the audio is difficult to capture. First, we start outside of the honey house. Chuck Kutik transports all of his supers on large flatbed trucks, and he parks the trucks just outside of the loading dock to the hot room. This garage door is the entrance to the hot room. He uses skidders to lift the pallets full of supers from the truck into the hot room. The hot room is a dark room that could hold many pallets of supers. This room is heated to between 90 to 100 degrees Fahrenheit,

and the temperature and humidity can be controlled in this room. Also, ceiling fans help circulate the warm air. After two days of sitting inside the hot room, the warm supers are now ready to extract. 48 supers are moved in on electric pallet jacks at a time into the honey extraction area. The honey house has two Cowen extracting lines really similar to the ones that we saw in John Ryan's facility. Each extractor holds 60 frames at once, so Kutik's honey house is able to extract 120 frames in one cycle. He has two employees that are loading the frames into the uncappers, and the whole setup minimizes handling time. Full supers are first placed on unboxers, where an electric lift lifts the frames out of the box and onto a short conveyor belt. Burr comb is scraped off of the top of the frames. The frames are then fed into the uncapper, one by one, and they move down the ramp, again, with a long conveyor belt, and are pulled toward the extractor. Once ready, 60 frames are pushed into the parallel radial extractor, and the empty super boxes are then slid down a ramp with wheels. All of the honey and wax from the uncapping tank and the extractor are pushed with an auger into sumps that are located beneath the extracting line. The sumps have automatic switches with floats to ensure that they don't overflow. From the sumps, all of the honey and wax that comes out of the uncapper and the extractor is actively pumped through large pipes that we're following up here, and it makes its way into a Cook & Bea's heat exchange unit that you can see here on the wall. It's the 12-foot stainless steel piping equipment. And the heat exchange unit heats the honey and wax to a temperature of between 95 to 105 degrees Fahrenheit, and it's able to heat about 3,000 pounds of honey per hour. The way that it works is it has this one central pipe that's filled with hot oil, and several smaller tubes that carry the honey and wax mixture that travel through this large pipe. So, as the honey mixture is traveling through the small tubes, the circulating hot oil transfers heat to it. Having honey travel through multiple small pipes is beneficial in the event that one of the pipes gets blocked, the process isn't interrupted. From the heat exchange unit, the hot slurry gets dumped into the Cook & Beals wax separator. The honey and wax flow into the separator, which is this big revolving drum, or a centrifuge. The mixture first lands on top of this rotating disk plate that's sitting on top of the drum, and it causes the mixture to fling out to the sides of the drum. The wax sticks to the sides, while the honey flows downwards and drops into the outer part of the drum. So the honey flows downward and drops into this outer part of the drum, and then it enters another sump, and it gets pumped into a settling tank. The wax, on the other hand, is cut out by a rotary cutter, and it drops out of the center of the centrifuge, which we can see here. The wax separation is continuous. The separator remains turned on during the whole process of extracting, and it's not a bottleneck in any way. And the Cook & Beals system is actually really effective at separating the wax from the honey. So the honey that's flowed into the settling tank remains there for a while to allow any remaining wax to float to the top of the honey so that it's not being run out into barrels. And after the honey settled, the gate is opened, and it's allowed to flow into these barrels and fill them up. Once a barrel is filled, it's sealed with a ring, and it's carted into the storage area of the warehouse. This storage area is really handy because it holds barrels until they're ready to be shipped out, and it also temporarily holds these wet supers that have been recently extracted until they're ready to be brought outside for the bees to clean. The barrels are eventually shipped out to a bottling facility, where the honey is further strained, and it's run into glass jars for retail. The wax cappings get shoveled into barrels. And Kutik's Honey Farm doesn't sell much of their own wax because they use it themselves. They melt it down into blocks and paint it onto their foundation frames that they're going to use in their nucs. And painting the plastic foundation with additional wax gives the bees a good head start to draw

out comb really quickly. Now, let's get back to those empty supers. After the honey frames are removed, the empty super boxes are slid down this ramp with wheels, and an employee at the other end picks up these boxes and scrapes off burr comb and propolis. After that, she then inserts three frames of fresh foundation into every box. And once the extracted frames come out of the extractor, six of them are placed inside a box, along with the three foundation frames in the middle. And this is a great way to rotate out old frames and introduce new foundation every year. So here we can see the extracted frames being removed from the extractor and getting loaded back into the supers. And these wet supers are temporarily stored in that storage area until they're ready to be brought outside. In an average year, this honey house produces between 600 and 700 hundred barrels of honey, which translates to just under half a million pounds. On an average day, the team of five employees extract about 528 supers, which translates to about 20 barrels in a day, which equals about 14,000 pounds. It's quite remarkable, and I hope that you had a lot of fun watching this whole process happen.

WATCH

Selling Local Honey as a Small-Scale Beekeeper

■ Hobby beekeepers and their honey Key Points crops are no doubt appreciated by Small-scale beekeepers can sell their family, friends, and neighbors; honey through a variety of retail having a honey hookup is a true perk avenues depending on their time of knowing a beekeeper. But availability, interests, and goals. beekeepers also experience a constant All honey being sold must adhere flow of expenses, and they are to regional and national food label interested in recouping those laws. expenses on every scale. After family, friends, and neighbors are supplied, many beekeepers consider selling their honey. With effective marketing and practical management in both the apiaries and the honey house, beekeepers can anticipate a profit as well. This Cornell Cooperative Extension publication "From Honeycomb to Consumer: Marketing Local Honey in New York State" features some tips and considerations for selling local honey on a small scale. The document is tailored toward marketing honey in New York State but includes considerations that are important in many regions. Small operators and hobbyists can do well with a focus on retail sales; they can attest to great care of the crop, stressing a high-quality product and a finite harvest. The "handcrafted" product adds value, as does making creamed honey or providing raw honey or varietal honeys. Hobbyist beekeepers are often active with individual sales to friends and acquaintances at work, school, and other organizations to which they belong. At the hobbyist level, it is said that honey sells itself. Indeed, word of mouth goes a long way over time, but there are a few other venues for selling local honey as well. Explore the tabs to learn about the main venues used by hobbyists to sell honey. Roadside Stand A simple honey stand can be lucrative, especially in a location on a well-traveled road with convenient parking. Stands vary from simple folding tables to elaborate, permanent wooden structures. Beekeepers can put up easy-to-read signs close to their stand to draw in customers. Honey stands of this nature are often self-service operations, with the prices clearly marked and a money slot. Beekeepers generally report very little theft. Setting up a stand in a beekeeper's home driveway is also a common practice, as the stand can be attended by the beekeeper when a customer drives up. Many beekeepers post a simple sign inviting people to knock on the porch door to buy a jar of honey. Farmers' Markets and Festivals With a simple tabletop display, a beekeeper can make sales at festivals and farmers' markets. This marketing venue provides more foot traffic than a roadside stand, and it allows beekeepers to develop

good relationships with their customers. Conversations with customers offer opportunities for the beekeeper to educate them on the lives and behaviors of honey bees, the use of honey for cooking and health, and the important pollination services that bees provide to gardens and farm crops. Customers generally come to appreciate local honey when beekeepers explain to them how it is a snapshot of the landscape within a two-mile radius. Beekeepers can discuss the difference between raw honey and pasteurized honey and offer samples for folks to taste the differences among several varieties. These conversations help create a relationship that gives the customer confidence both in the product and in the beekeeper selling it. With provided contact information, the customer is likely to call for restocking. Offering honey samples is recommended. Most often in marketing, the longer a customer spends at a booth, the higher the likelihood that they will make a purchase. Some beekeepers find that encouraging customers to linger by providing a bench beside their booth or some shade can offer the opportunity to engage the customer in a conversation and make a sale. An observation hive may also encourage people to come up and learn about bees. Listening to customers at farmers' markets and festivals can provide beekeepers with valuable insight. They will learn when customers think a product is priced too high or when they think the packaging is inconvenient. Beekeepers should make an effort to learn what will help increase sales and improve packaging and marketing as opportunities arise. Beekeepers should also keep records of the most popular items they sell and what customer demographics they tend to attract, which will help them cater to those customer bases in the future. Checking out other booths at farmers' markets or festivals will let them see which ones attract the most customers. What do their booths look like? How are their products displayed? What other visual interest first attracts customers and makes that booth stand out from the rest? Beekeepers can mimic some of these qualities with their own stands. Festivals provide a lot of opportunity to get creative. Any festival with the theme of local food, health and wellness, cheese, wine, Earth Day, the environment, pollinators, or gardens are great avenues in which to participate. Some beekeepers tailor their honey labels and packaging to specific festival themes. Farmers' markets and festivals can be costly, both in terms of time and money. Nearly all venues charge a fee to set up a booth, and liability insurance may be required if beekeepers are providing honey samples or have an observation hive (in the event of allergies). A food vending permit may also be required by the local health department if the beekeeper is providing honey samples. Beekeepers should consider the value of their time attending a booth for several hours, plus the time it takes to travel to and from the event, set up, and clean up. If a beekeeper is not making enough money at the market for this avenue to be worth their time, they should consider marketing the honey in other ways. In this video, Bob and Vin Finch, owners and operators of Berkshire Hills Honey Bee Farm, discuss their experiences selling honey at farmers' markets and festivals. Video Transcript I've been a beekeeper for 47 years now, and I'm a fifth generation beekeeper. So, I guess I had to be a beekeeper after that many generations. I feel like I would let my relatives down, I guess, if I didn't become a beekeeper. But I thoroughly enjoy it. It's a passion of mine, beekeeping. So, I did it as a hobby for 47 years, and then, when I got near retirement, we thought about starting a small business. And we have. I got my d.b.a. in 2011. So, we've been doing, how are we going to make a living at it? How we were going to make it pay for itself. And so, for seven years now, we've been doing sales through farmers markets, festivals, and some sales out of our home. So I've been doing it for seven years, but five of those years, we have been at the Ithaca farmers market, which is our main source of income. We still do festivals and small markets. So, that's how we have generated the income to

maintain our bees and buy equipment. And beekeeping is not a cheap hobby, even, it's very expensive. But thus far, we have been able to pay for our business and operation all through the sales of farmers markets and customers. And when you do the farmers markets, there's general fees that you have to pay, there's a structure you have to follow, you have to bring your own tables, your own tents and set up. There's usually a requirement for insurance, you have to have an insurance certificate proving that you have insurance. Some customers come to the market and they may run out of cash or they may not carry cash and they want to pay with their credit cards, so it's good to have Square so that you can take credit card payments. And Square is a really neat digital, electronic way of paying for stuff. And also with Square, a lot of people request it today, but with Square also, you can carry your inventory, you can have all everything inventoried and at the year end they calculate your tax and sales and everything. So it's really handy tool if you're going to be doing any amount of vending. It does take a lot of time to do farmers markets and festivals, more so probably than when we jumped in. It's eye opening and it's a learning experience, it's challenging. And there's a lot of work behind the few hours. Most farmers markets are four hours to six hours long, but you have all of the behind scene. Not only managing bees and the business, but bottling, packaging, marketing. How do you market? I do market through not only a website, but Facebook and personal one-on-one sales at the market, which is what I really enjoy the most. I didn't think we would enjoy that, I'm somewhat of an introvert and didn't think that I would ever want to do sales, but take an observation hive and doing personal tastings with people and talking, we've now discovered that really has become a passion to get out there and talk to other people, meet people. It's a great way to network, even with other beekeepers, and entomologists, veterinarians. I talked to all kinds of people and it's just a great source to become interactive with. We always engage people, people walking by, we always say hello, hi, how are you. Try to start a conversation. Don't just sit behind your table and watch them go by. I think it creates a better atmosphere as far as sales and getting to engage people, the observation hive which we take is a great tool because it creates a lot of questions. It might even startle people, seeing that there's live bees there. And they're startled by it, but generally, they will start asking questions and because colony collapse disorder made the news in 2005, it does, it creates a platform where you can discuss your bees and the health of the bees today and the industry as well as bring it down to your own personal operation and offer them a taste of honey. Generally, people taste honey, they are pretty interested in it and will buy it. Another thing that attracts them is the different colors of honey. They always ask, why are they different? What makes them different? So you can explain that to them. Yeah, the source of the nectar always creates, the floral source, always creates the color and the flavor in the honey. So it's just a nice way to engage people. But we really enjoy engaging people at the market. Online Today, online shopping is more common than ever. Selling honey online can substantially widen a beekeeper's customer base. There are also several companies that make creating modern, sleek, attractive websites easy for anyone to accomplish. Some of these websites that permit people to sell products include WordPress, Squarespace, Wix, Weebly, and Shopify. They can be set up to conveniently use PayPal, a service that enables customers to purchase goods with credit cards and notifies beekeepers as soon as purchases are made so they can promptly ship goods. Effective websites can be used to educate customers about a beekeeper's region, their specific beekeeping practices, and honey bee biology and health. High-quality photos of a beekeeper's honey house and hives give customers the feeling that they know exactly where their food is coming from as well as the

pride the beekeeper takes in their craft. An "About" section with information about the beekeeper or company history can help build trust with customers. Some websites will also include blog posts about the latest happenings in the bee yard, the operation, or recent scientific discoveries. Customers can sign up for listservs to get email notifications of sales, discount codes, new arrivals, and monthly specials. Having a website linked to social media pages such as Facebook, X, or Instagram can further connect a beekeeper with customers and create a brand for their business. These features allow the beekeeper to stay connected to their customers, sending them notifications, emails, and news whenever new information or products become available. Staying up to date on honey marketing research helps beekeepers sell their products. For instance, consumer surveys show that 57% of honey is consumed at breakfast. Providing pamphlets to customers that include breakfast recipes using honey, as well as posting photos of honey paired with various breakfast foods and tea on social media sites, can inspire consumers to purchase honey for this purpose. Honey is also a common ingredient in baking, which surveys show is most commonly done throughout the winter holiday season as well as at Easter. Promoting honey as a holiday season staple can help boost sales during these time periods.

WATCH

Packaging and Labeling the Product

■ There are a variety of ways honey can be packaged and labeled for sale, and Customer research can provide beekeepers with valuable information about using a package and label that best reflects their brand. In addition to creating a visually appealing product, there are very specific criteria that must be included on the honey label. This video addresses how to properly label honey for sale along with some of the factors to consider when packaging and labeling the honey. Video Transcript Labels are the main identifiers that customers are going to use when they decide to purchase your honey. There's a lot of freedom in packaging honey, and beekeepers are encouraged to get creative and make their honey stand out from the rest. However, the label on the jar or bottle must be created in accordance with the US Food and Drug Administration laws, as well as laws in your state. You need to have a legal label in order to sell at supermarkets, farmers markets, roadside stands, fairs, and festivals. Here, we'll focus on the label laws according to the US Food and Drug Administration. The first thing you need to make sure of is that the word "honey" appears in large, legible type font on the front of the container. The product name could be clover honey, summer blossom honey, or wildflower honey, as long as the word "honey" is prominent. All packages of honey for sale must include the net weight of the product. This is the weight of the honey itself, not including the weight of the packaging, and this must appear on the lower third of the front label. The weight of honey needs to be displayed in pounds or ounces and also in metric units. For example, the label could read, "Net weight 16 ounce," "454 grams" in brackets. Some beekeepers may include other words on their label to provide a marketing edge to their product. These words could be "pure," "natural," "raw," or "local." And none of these four words are regulated as there's no legal definitions for them. We encourage beekeepers to include "pure honey" and "natural honey" on their labels as they wish. However, they should be honest about whether their honey is truly local, produced in the region they're selling it in, or if it's raw, so not pasteurized. Other words are regulated and can only be used under certain circumstances. For instance, the word "organic" does have a specific

meaning, and beekeepers can only place this word on their label if they receive certification. Honey must be certified through an agricultural marketing service accredited certifying agent. There's 48 agents based in the US., and 32 based in foreign countries. All of them are authorized to certify operations to the USDA Organic Standards, allowing you to place a USDA Organic label on your honey bottle. You also need to be careful not to include any health claims on your label. Health claims are regulated by the US Food and Drug Administration, and at this time, there's no approved health claims that can be made for honey. Beekeepers, if they wish, can denote the variety of honey on their label as well. So, if you like to specify the dominant nectar source, whether it's clover honey, buckwheat, or orange blossom honey, you can include it on the label. However, the National Honey Board does recommend beekeepers be prepared to show proof of the main nectar source if they're asked by an inspector. And this is more likely to happen to beekeepers that are selling their honey in large venues like grocery stores. If beekeepers are interested in testing their honey sources, they can be tested for a fee through the Texas A&M pollen identification laboratory, where small samples can be milled to determine what the main honey is that you're advertising. It's great to keep these records nearby in the event that you get asked for them. A beekeeper's contact information is also required and must be located on the front label in a font that's at least one sixteenth of an inch tall. If all the honey that a beekeeper is selling is from their own colonies, they can say "produced by" on the label. For example, "produced by the Dyce Lab." However, if any of the honey that they're selling is from other beekeepers, they would use the words "packaged by" or "distributed by" the Dyce Lab. You can use your own name or your business or apiary name as long as there's a name on the bottle, as well as a complete physical address and a phone number to accompany it. Let's talk about nutrition labels. Not all beekeepers who sell honey need to include a nutrition label on their package. It depends on how much honey they're selling. Beekeepers who are selling more than \$50,000 of honey, or 100,000 units of honey in a year, must include a nutrition label on their jar. These beekeepers can purchase nutrition labels through a bee equipment supply company. However, if you're selling more than \$50,000 dollars of honey but less than 100,000 units, you don't need to include a nutrition label, but you must apply for an exemption with the FDA every year. Lastly, if you're \$50,000 dollars of honey, you don't need to list the nutrition on the jar and you don't need to file for an exemption with at the FDA. Nearly all small scale beekeepers will fall into this category. Some small scale beekeepers like to put nutrition labels on their jars anyway because it's good practice, it looks professional, and it provides transparency to their customers. Ingredients. If you only have honey in the jar, you don't need to include a list of ingredients. However, if you do have additional ingredients you need to include these on your label in decreasing order, by weight. So for instance, if you're adding spices or flavors such as vanilla beans, cinnamon, or cocoa powder, this must be listed in the list of ingredients. And if you're selling enough honey to require a nutrition label, the label must be updated to reflect the nutritional value and all the ingredients in the product. If you're selling honey to a grocery store, beekeepers may also need to require a universal product code. This is the barcode that can be scanned by the cashier to keep inventory for store sales. It's up to the beekeeper to visit a universal product code website to obtain their own barcode. Beyond these laws, beekeepers should make their labels stand out from other honey being sold on the shelf. So before designing your honey label, stroll through grocery stores and farmers markets and check out websites to see what other people have designed on their labels and to take note of what makes their labels stand out. Some beekeepers design their own labels, while others hire graphic designers to help

them make it look more professional. Regardless of who designs your label, there's a few things to keep in mind. First, consider having your labels printed with waterproof ink on waterproof paper. If customers are wiping sticky bottles with a wet cloth, or if they place the jar in a water bath to liquefy crystalized honey, you don't want the label to disintegrate on them. Next, most customers want to see honey in the bottle. So make your label large enough that they can easily read the text, but small enough so that they can clearly see what the honey looks like inside. Also, if you're using transparent labels, be thoughtful to the color of honey that's inside the bottle. If you're selling a dark buckwheat honey and your label has black ink, it's likely going to disappear against the background, and white ink might be better. So test out sample labels on your filled jars before you print a big batch just so that you can see how the overall product is going to look. And lastly, check out customer preferences regarding honey purchases. This can help you come up with ideas to design your label and also your overall packaging, too. Many customer surveys can be found online through the National Honey Board website. This research investigates what packaging qualities influence customer purchases. For instance, the National Honey Board attitude survey found 83% of consumers will pay more for a product that promotes bee-friendly causes. They also found that millennials, aged 18 to 34, are the main purchasers of honey, followed by people over the age of 65. So check out other product labels and advertisements that are targeted for these age groups to get some ideas for your own product. The 2016 Nielsen honey category overview found the most frequently purchased bottle size is 12 ounces. More so than any other size on the market. Also, glass bottles commanded a higher price than plastic bottles, giving a high quality look. The preferred format for purchasing honey was in a bottle instead of a barrel or a tub. This survey also found that liquid honey is purchased more frequently in the US, followed by creamed honey, and then comb honey. So do your research and use it to your advantage to gain a marketing edge. I hope that these considerations and laws help you when it comes time to design the perfect label for your honey products.

READ

Selling Honey as a Sideliner or Commercial Beekeeper

■ Sideliner and commercial beekeepers Key Points may continue to sell retail (directly to Many large-scale beekeepers sell customers) and choose to do so partly honey wholesale to packers or by including a sales outlet as a room in brokers. their honey house, thus encouraging The value of honey fluctuates over the public to buy jars of honey right on time as a result of supply of and site. Some also offer tours of the honey demand for domestic and house for a memorable agritourism imported honeys. experience. Large-scale operations do well with retail sales but usually have the capacity and need to address the option of selling larger quantities at wholesale rates. "Wholesale" refers to bulk honey sales to packers, distributors, or grocery stores; the beekeeper does not interact directly with customers. Wholesale honey can be sold as cases of jarred and labeled honey to distributors or grocery stores. Alternatively, beekeepers can sell their honey without labels, simply packaging it in larger containers. Some sell 5-gallon buckets (~60 pounds of honey), but large-scale beekeepers most commonly sell their honey in 55-gallon drums (barrels holding 650 pounds of honey). Drums of honey are typically sold to honey packers (who strain, bottle, and label the honey) or to honey brokers (who, in turn, sell it to packers or sell it internationally). Many big honey companies in the U.S. purchase other beekeepers' honey, process it, label it, and sell it. Two of the largest and best known are Sue Bee and

Dutch Gold. If a beekeeper sells honey to a broker who is, in turn, exporting it to another country, the beekeeper must be knowledgeable on the laws, standards, and requirements of honey in that country.

Honey in 55-gallon drums. The value of honey, both on retail and wholesale scales, fluctuates monthly. Beekeepers who are selling wholesale should familiarize themselves with evolving honey prices over time. Accurate sources of this data can be found on the National Honey Board website, the National Agricultural Statistics Service website, or in every monthly issue of Bee Culture Magazine. The price of honey varies each month and within every region of the U.S. It also varies depending on the color grade of the honey, with lighter-colored honeys typically priced higher than darker-colored honeys. Generally speaking, the smaller the package, the higher the price per pound. This is due to the added costs of bottles and labels as well as the added labor by the beekeeper to package the honey. Below is an example of the prices for selected package sizes of U.S. honey in September 2020, based on the regional honey report in Bee Culture Magazine.

Package	Average Price
55-gallon drum (light)	\$2.13
55-gallon drum (amber)	\$2.06
Case of 12 pint jars	\$5.31
Pint	\$7.29
12-ounce plastic jar	\$8.26

The graph below shows how U.S. honey prices have increased in value over the years, from 1987 to 2016. These data are from the National Agricultural Statistics Service.

Long description ■ A graph showing US honey prices from 1987 to 2006. In 1987 the average price per pound hovered around 60 cents. Starting around 1994 honey prices began to rise. After a dip in 2005, the chart continues to increase and peaks in 2014 at \$2.20 per pound.

What Influences Honey Prices? The U.S. and international honey markets are complex, and the intricate details are not needed for the Cornell Master Beekeeper Program final exam series. One of the many factors contributing to the increased price of U.S. honey in the past 30 years is the drop in honey produced domestically. In 1987, the National Agricultural Statistics Service estimates that 3.3 million colonies produced honey in the U.S.; in 2019, this number was reduced by 15%, to 2.8 million. This is partially reflected in honey production: In 1987, 235 million pounds of honey were produced in the U.S., whereas the amount was 33% lower (157 million pounds) in 2019. At the same time, colonies that are producing honey produce less than they did in the past. In 1987, the average colony produced 71 pounds of honey; in 2019, it produced 56 pounds. The decrease in honey production is related to a variety of factors, including parasites, disease, pesticides, management practices, and shifts from honey-focused operations to pollination-focused operations. Each year, U.S. citizens are steadily increasing their honey consumption. In 2016, the average U.S. citizen consumed 1.60 pounds of honey (up from 1.20 in 2010), whether in the form of table honey, baked goods, processed foods, or candies. To fill the market's supply for honey, the U.S. imports it from a variety of different countries; in fact, the majority of honey consumed in the U.S. is of international origin! In 2009, the U.S. produced \$208 million in honey, while the value of honey imported into the U.S. from other countries was \$220 million. The countries that export their honey to the U.S. change slightly each year, but the biggest in 2016 were Vietnam, Argentina, India, Brazil, Canada, and Mexico, respectively. Most imported honey is extra-light amber and light amber in color. Nearly all organic honey sold in the U.S. is imported from Brazil. The amount of honey imported and exported from the U.S., as well as the price of these honeys, also influences the changes in price of U.S. honey. Until 1995, the majority of honey imported into the U.S. was from China. China is a leader in honey production, largely due to its vast land area, the climate, and its floral diversity. For many years, China was producing a big surplus of honey and selling it to the U.S. at very low prices — lower than what they would sell it for in their own country, and lower than U.S. honey

prices. Selling honey for this low price began to create an unfair competition for other honeys being sold in the U.S. To oppose this, the U.S. International Trade Commission enacted antidumping laws, preventing China from simply "dumping" their excess honey on the U.S. for low prices to gain an unfair advantage in the market. Tariffs were placed on any honey that was imported from China, and they restricted the volume of Chinese honey that was allowed to enter the U.S. With these tariffs in place, China now exports very little honey to the U.S. Bee hives placed in canola fields in Luoping, China. This situation made the news several times over recent years, with a few incidences of Chinese honey being illegally brought into the U.S. Some Chinese suppliers would circumvent the expensive tariffs by first shipping their honey to other countries before selling it to the U.S., claiming the honey originated in another country. The honey could be processed to remove nearly all pollen grains (and in some cases, antibiotics) that would identify the country of origin of the honey. Efforts by the U.S. Customs and Border Control, U.S. Department of Homeland Security, U.S. Immigration and Customs Enforcement, and the U.S. Food and Drug Administration are actively helping to prevent illegal honey from entering the U.S. Many beekeepers in the U.S. feel the competition with international markets, as imported honey is often priced lower than local honey. As Master Beekeepers, it is critical that we educate consumers on the importance and quality of local honey, so as to help U.S. beekeepers remain competitive in this market.

READ

Selling Honey as a Sideliner or Commercial Beekeeper —

Influences on Market Price ■ The U.S. and international honey Key Points markets are complex, and the intricate details are not needed for the Cornell time as a result of supply of and Master Beekeeper Program final exam demand for domestic and series. One of the many factors imported honeys contributing to the increased price of U.S. honey in the past 30 years is the drop in honey produced domestically. In 1987, the National Agricultural Statistics Service estimates that 3.3 million colonies produced honey in the U.S.; in 2019, this number was reduced by 15%, to 2.8 million. This is partially reflected in honey production: In 1987, 235 million pounds of honey were produced in the U.S., whereas the amount was 33% lower (157 million pounds) in 2019. At the same time, colonies that are producing honey produce less than they did in the past. In 1987, the average colony produced 71 pounds of honey; in 2019, it produced 56 pounds. The decrease in honey production is related to a variety of factors, including parasites, disease, pesticides, management practices, and shifts from honey-focused operations to pollination-focused operations. Each year, U.S. citizens are steadily increasing their honey consumption. In 2016, the average U.S. citizen consumed 1.60 pounds of honey (up from 1.20 in 2010), whether in the form of table honey, baked goods, processed foods, or candies. To fill the market's supply for honey, the U.S. imports it from a variety of different countries; in fact, the majority of honey consumed in the U.S. is of international origin! In 2009, the U.S. produced \$208 million in honey, while the value of honey imported into the U.S. from other countries was \$220 million. The countries that export their honey to the U.S. change slightly each year, but the biggest in 2016 were Vietnam, Argentina, India, Brazil, Canada, and Mexico, respectively. Most imported honey is extra-light amber and light amber in color. Nearly all organic honey sold in the U.S. is imported from Brazil. The amount of honey imported and exported from the U.S., as well as the price of these honeys, also influences the changes in price of U.S. honey. Until 1995, the majority of honey

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Honey Grades and Standards ■ Currently, there is no official definition Key Points or standard for honey in the United States; honey is covered under the USDA guidelines grade and States; honey is covered under the classify honey based on color and same general regulations as all other clarity. foods. However, voluntary honey grade standards have been developed by the USDA to designate the quality of bottled honey for sale. Beekeepers can order a grading kit to assist them in grading their honey, then label their honey for sale accordingly. The person who packages the honey determines the grade that appears on the label. This grading system is voluntary; it is not enforced or verified by the USDA. The grading system is based on points accumulated for moisture content, clarity, defects, flavor, and aroma. It does not take into account the presence of pesticides or antibiotics, nor does it consider purity (which can be reduced with the presence of added sweeteners like high-fructose corn syrup). The total score after adding points in each of these categories determines the grade of the honey. Grade A honey scores a minimum of 90 points, Grade B honey scores a minimum of 80 points, Grade C scores a minimum of 70 points, and any honey with fewer than 70 points is considered substandard. The USDA's honey grading system was developed in 1985; it is helpful to become familiar with the criteria that it assesses. Honey in a variety of colors and grades. Color Classifications The range of color and flavor of honeys depends on the source of the nectar gathered by the honey bees. There is a large range of variation — the National Honey Board states that there are more than 300 unique types of honey available in the U.S., each originating from a unique floral source. With colonies typically gathering nectar from a variety of sources each day, the number of potential honey blends is essentially infinite! Honey color ranges from

nearly colorless to dark brown, and its flavor varies from delectably mild to distinctively bold. As a general rule, light-colored honey is milder in taste and dark-colored honey is stronger, but there are exceptions. Honey is produced in every U.S. state, but, depending on the location of the flora, certain types of honey are produced only in a few regions. The color of honey does not influence its grade, but there are specific color classifications that should be included on the label. The color designations include the following, from lightest to darkest: water white extra-white white extra-light amber light amber amber dark amber Honey color can be classified in two ways: using the Pfund scale or using the optical density scale. The Pfund scale uses visual observations of a honey sample placed in an instrument. The honey color is matched to a specific color on a color spectrum chart that ranges from light to dark. The color chart is 140 millimeters long. The score is determined based on how many millimeters the color of the honey sample is from the beginning of the color chart. A color close to zero is water white, while a color close to 140 is dark amber. There is a bit of subjectivity to using this method; the Pfund scale is not the approved method for color classification by the USDA. More modern and precise technology involves various instruments that measure the optical density of the honey. These instruments provide a reading based on the degree of light intensity loss that occurs when a light is passed through the honey. Lower numbers, close to zero, are seen in water-white honey, whereas numbers above 3 are seen in amber honeys.

MODULE WRAP-UP

Honey

In this module, you explored all different aspects of honey: how it is made, what it is made of, and the different ways in which beekeepers process and pack it.

Additional Hive Products

Honey is only the tip of the iceberg when it comes to the products that can be harvested from the hive! Whether beekeepers are looking to produce value-added products for their businesses, or to simply have fun while developing new skills along the way, the number of products from the hive and ways to incorporate them into our lifestyles are nearly endless. In this module, we will explore the most common products harvested from the hive and evaluate the scientific evidence behind using them for medicinal purposes.

READ

Beyond Honey

■ Honey is far from the only hive product Key Points that is useful and valuable to humans. Humans utilize many products Following honey, the most common is from the bee hive. beeswax, which has hundreds of uses The use of bee hive products in and, although it has been superseded human health and medicine is by synthetic materials in many called apitherapy. applications, is still the best medium for candles as well as an ingredient in a range of beauty products. Wax is typically a part of honey competitions at fairs and conferences. Less-common products include the pollen pellets collected by bees, propolis, royal jelly, and bee venom. These, along with honey, are marketed for their health and

medicinal benefits, collectively known as apitherapy. However, these products tend to enjoy an outsized reputation, which has paradoxically eroded many people's faith in what these products actually are and what they can do. Master Beekeepers have a role to play in helping consumers separate the science from the myths. Hive products clockwise from top: pollen, honey, propolis capsules, beeswax candle. The scientific research on hive products is limited for a number of reasons. There are the usual caveats about the placebo effect; the limitations of testimonials; and the necessity of large-scale, rigorous scientific studies. It is also difficult to ensure uniformity in hive products, as they vary widely based on environmental conditions and seasonality, as well as other differences among colonies. Some researchers point to the difficulty that the pharmaceutical industry would have in patenting and making profitable bee products (relative to synthesized compounds), which tends to suppress lines of research. In short, further study is needed to more conclusively prove which effects are consistent and which are not. Importantly, although some products may be beneficial for some people, they may also be dangerous: Bee venom can be lethal at high doses, and ingesting hive products may trigger allergies (although this is statistically less likely than the most common food allergies). Less dramatic but more common is the misconception that these products are natural and therefore organic or free of pesticides or other contaminants. Even if a colony is managed for pests and diseases without chemical treatments, it is exposed to whatever is being sprayed or planted in the two- to three-mile radius around the hives; there are vanishingly few places in the U.S. without any such chemicals. Collecting these materials from the hive requires attentive care, good nutrition, and careful timing — the more of these products that are harvested, the more careful management needs to be in order not to negatively impact the health of the hive. Some subspecies of *Apis mellifera* collect and produce more propolis, or can be bred to produce more royal jelly or collect more pollen, so a small business with these hive products would also be aided by keeping the most productive subspecies or stocks.

READ

Rendering Beeswax

When extracting honey, the beeswax cappings can be saved, cleaned, and melted into wax blocks, which can be used by the beekeeper or sold. The process of removing and processing beeswax is called rendering beeswax. Fresh wax cappings are the cleanest wax in the hive. Older wax from other parts of the hive, particularly the brood nest, is darker and filled with debris such as larval feces and cocoons. Nonetheless, with lots of processing, some beekeepers do clean it and use it. No matter the origins of the wax, rendering and filtering it is important for beauty products, competitions, and candles. It should also be processed soon after removing it from the hive: While wax moths and other pests may ruin combs, they do not go after solid blocks of wax. Removing uncapped honey/nectar and other perishable debris before they spoil also helps preserve the wax's color and aroma. Wax is best processed in stainless-steel, aluminum, tin, enamel, or ceramic containers; iron, brass, steel, or copper may stain the wax. Care should be taken with temperature. Wax heated over 185°F (85°C) may discolor. It should never be heated directly; only with a double boiler or other indirect setup. It is highly flammable and may ignite if it gets too hot or is splattered onto an open flame. Also, wax that cools too quickly may crack, bubble, or shrink.

READ

Methods of Rendering Beeswax for Hobby Operations

■ For beekeepers with only a few hives, Key Points wax rendering is simple and requires The equipment used to render little equipment. The cappings are beeswax depends on the collected in a small tank (either plastic operation size. or stainless steel) and allowed to sit for Wax melts and flows freely at 143– a few hours, as they still contain a 149°F but can discolor above 185°F. substantial amount of honey. The uncapping tank has a metal grid that Slumgum is the name for the allows the wax to remain on top and debris filtered out of beeswax. honey to drip out and gather below. The honey is collected out of a spout at the bottom of the tank. The wax cappings can then be gathered and rinsed or soaked in water to remove any remaining honey. From there, they can be melted using the stovetop or placed in a solar wax melter. Stovetop Melting The simplest technique to process the wax is to wrap the cappings in tightly tied cheesecloth and filter them over a stovetop in a big metal pot. The pot is filled with water and the cheesecloth with the wax is placed in the water. The water is heated on low heat and the wax melts through the cheesecloth, leaving any debris inside the cloth. The clean wax floats on the water as it hardens and can easily be removed after it has cooled. If any debris enters the water, it either sinks to the bottom of the pot or aggregates on the underside of the wax. It can be scraped off when the wax re-hardens. The remaining debris in the cheesecloth is called slumgum; it is a brown-black sludge that contains bee body parts, honey bee and wax moth cocoons, and other debris. This slumgum can be discarded or used in small pieces (or dirty filters) for lighting smokers. Slumgum may contain disease (AFB) spores and should be kept away from bees. ■ Solar Wax Melting Many beekeepers use solar wax melters, which can be made at home or purchased from bee supply companies. Specific designs vary widely, but in general, these are shallow, sealed boxes with glass or clear plastic lids that concentrate the sun's heat to melt the wax over several hours. The melted wax drips down a metal tray and through a filter (if one is added), collecting in a small container. Once the wax is hardened, the beekeeper can easily remove the wax blocks. Slumgum remains trapped by the filter or on the tray and can be easily scraped off and discarded. While simple and cheap, solar wax melters are inefficient in terms of the percentage of clean wax rendered from the cappings (meaning that the wax content of the remaining slumgum is high). They are also more difficult to control the heat and need to be monitored so that they do not reach temperatures high enough to discolor the wax. Left: Wax placed in a solar Right: The melter with the wax melter without the solar screen in place. screen in place. As wax melts, it drips down the metal tray into the collection tin on the left side. In large-scale operations that routinely uncap hundreds or thousands of frames, a large amount of cappings builds up quickly in the uncapping tank ■ Methods for Large-Scale Operations ■ and must be processed efficiently. This honey-wax slurry can be pushed along a pipe with an auger into a machine called a wax spinner or a honey/wax separator. The wax spinner is a large cylindrical device that works similarly to a washing machine on the spin cycle. The wax slurry is deposited on top of a stainless-steel spinning reel with very small perforations. The honey is flung out of the perforations as the reel spins, collecting at the base of the spinner and eventually flowing through a pipe to a sump or tank (settling tank, filtering tank, or bottling tank, depending on the operation). The wax cappings too big to fit through the perforations collect on the reel and can later be scraped off. The cappings are fairly dry after being spun. The honey/wax separator is a large centrifuge that is also very effective at separating honey from wax. The honey-wax mixture drops into the centrifuge and is flung onto the inner walls. Wax sticks to the walls and is later removed with rotary knives to drop out of the bottom of the centrifuge, where it is

quite dry. The honey flows downward into an outer collection rim of the drum and is pumped to a sump, strainer, or settling tank. The wax cappings can then be transferred to a wax melting tank. The wax is slowly heated (via a water-jacket system or radiant heat) to 143–149°F (62–65°C) until it melts and flows through a series of stainless-steel filters to remove any particles. When the wax melts, any remaining honey and debris are separated from it. The wax floats above the honey and can be drained as a liquid into containers through a spout. Honey (often caramelized, darkened, and unusable after the heat) flows out a separate spout. In a few minutes, the wax begins to harden into a wax block. The wax blocks can be sold as is, or the beekeeper can further process them to create candles or other value-added products. The remaining slumgum, in large enough amounts, can be sold to a more efficient wax-rendering operation or discarded.

WATCH

Processing Beeswax Example

In this video, Jon Ryan, owner of Ryan Apiaries, processes beeswax. Video Transcript So, once the cappings have come off your knife, once you have uncapped the combs, you are going to have to somehow separate the honey from the wax. There are many different methods of doing this, there are different pieces of equipment that's out there on the market right now for doing this. The size of operation that you have is going to determine, to some extent, what equipment you are going to use. But again, in the end, the principle is all the same. You somehow are going to have to separate the wax from the honey. I will in a few minutes show you how I do it. And again, you are subjected to my opinion, but I think for my size operation, it is the way to go. For a larger, commercial operation, which I think you will see later on, it is called the Cook & Beals system in which the whole crop, in other words, both the wax, the cappings, honey, everything goes through the heat exchange and then through a spinner. In my opinion, spinning is the best way to separate the honey from the wax. And the thing that I like most about it is that by separating the wax from the honey before any heat is put on, I don't think you do as much damage to either the wax or the honey. Okay, the first step in this whole process of obtaining beeswax is, and regardless of the size of operation you have, you are still going to have to somehow uncap the combs. For my operation, and for the size operation that I have, I very much like this Cowen system. Here are the knives are right here. The knives, obviously, are going to uncap the cappings, and the cappings will now drop down into this hopper. On a good super, I have determined that approximately 40 percent of the honey in a super is actually uncapped by these knives and drops into the hopper. So, it is very important to have a good recovery system so you don't lose all that honey. At the same time, a recovery system that does not damage the honey, I think is very important. And I think that is what this system will do. And in the bottom of the hopper is an auger, and that auger will then move the cappings, which will basically be a slurry of both wax and honey over to the capping spinner, also made by Cowen. The auger ends about there, where the metal straps are, and from that point on, it is just pushing the cappings, which have a consistency of about like oatmeal, I would say. So, the cappings are being pushed, and then dropped down to this reel, which is spinning at the time that the cappings are actually dropping. The reel is stainless steel, it has many perforations in it so the honey can go through those perforations. The spinner basically works like your washing machine on the spin cycle. It's just spinning, the liquid honey is going through the perforations and then out to a tube at the back of the spinner. The cappings themselves are being plastered on to the reel. And then,

once it has been spun and the cappings have dried off, they're shaved off, basically, as you would use a lathe, a wood lathe or a metal lathe. This chisel here then gauges the reel. And by turning this, we move the chisel across the face of the reel, shaving off the cappings. So, you have gone from something that is, you know, the consistency of oatmeal, to cappings that are fairly dry. Most of the honey has been removed. I can't tell you exactly what percent. The cappings are then switched over to another tank and when they're melted, about three of these muck buckets will yield maybe two gallons of honey, which, by the way, has basically been destroyed. It is black. That honey is really marginally salvageable. But most of the honey has been completely unheated, undamaged, and retains all its goodness. From here, the muck bucket, we'll go over and dump in this wax chest. This wax chest will hold about three and a half of these muck buckets full of cappings. This is a homemade wax melter, wax chest. The way it works is, it is water jacketed. In the bottom of this, there are four courses of copper tubing, which steam runs through, so I can heat this, basically, to any temperature I want. However, I also have the option with an immersion heater to heat the water. The way this works is the cappings, the wax with honey put in here. The temperature is such that it will melt the wax. The whole liquid mess will then come down to this end. This dam, if you will, will keep the unmelted portion from sliding in to the front part. The critical thing about this, which really makes this work and work so nicely, is this drop off right here. The honey, being heavier than the wax, is going to drop off into the drop off. The honey will then baffle over to this right here. There is a hole in the bottom of this so that the honey will slide over into the baffle and then, being a fluid, it will seek the level of whatever the total fluid is. So, the honey can then be baffled out through this spout. The wax will come out this spout right here. If you look in front of the tank, you can see. So, the wax will come out here and mold it into whatever mold you wish to use. We find dish pans pretty convenient. As the cappings are melted, it is going to separate the honey from the wax. And the honey will come out here. And as you can see, that honey has been heated enough to actually burn it. It will have a burned flavor. And that was nice, light honey. But again, it is a fairly small proportion, and the rest of the honey is completely unaffected. So, this is the sacrifice I made. I think it is about the best system for rendering cappings that I know of. And here is nice, yellow wax. I love watching the wax spread over the water. So, we now take quality beeswax. And what we'd like to do is show you a few things that you as a beekeeper can do to add value to your product. In other words, make a quality beeswax and quality beeswax products. To obtain quality beeswax, the best, nice, lemon-yellow beeswax is going to come from cappings that have been produced off of combs that have been used for extracting. And the best cappings are going to come from combs that have been used exclusively for honey. In other words, if you've got brood combs mixed in with your extracting combs, you are going to get a slightly darker wax. So the cleaner your beeswax can start with initially, the more lemon yellow, the higher quality beeswax you are going to get as a final product. Once you've obtained quality beeswax, a nice lemon yellow as I mentioned, you can make products such as this, or this. And again, it is pretty simple, but there is a significant market for people that want beeswax, for who knows what, perhaps soap making, or candles, or whatever. But there is one thing you can do to add little value. And then, of course, there is an array of things that you can do with the beeswax, again, to add value. You can get molds, such as this, or the old fashioned tin molds, which they're kind of a pain in the neck. I think that the dip candles are far superior in both quality and in production. Also little things you can do. There is this for putting the ends, a fluted end on the candles. We have not been doing that. We have thought that they actually looked better with a flat

end. But again, we will get to that later. You can also use some of these porcelain molds for making figures such as this, or this. And we also made a mold for making Christmas candles. By Christmas, I mean these go on the tree. We haven't burned our house down yet. Or, if you are really ambitious, you can make little tiny candles for birthday cakes. So, again, your imagination is the limiting factor as to what you can do with the beeswax. But, again, a quality beeswax, I think, is important. So, one final thing that I would like to tell you about as far as our beeswax show and tell, is how do you clean up, because inevitably, spills and cleaning up is necessary. This wax chest, which you show, which I showed you earlier, that has been used for probably at least twenty years. And if you noticed earlier, it is absolutely spotless. And it did not take all that much effort to get it that way by using good, old lye. Dadant, I believe, sells this. It is not as easy to find as it once was in hardware stores or grocery shelves. Soap making supplies may also have lye. It is great stuff, you need to put it with heat. In other words, when I clean up the wax chest I have heat on it. It is also great for cleaning up excluders. Once they become caked with wax from using one of these, I run it through a very hot lye bath, hot water lye bath, and that cleans the excluders right to brand new. So, it is great stuff. It is also good for disposing of your murder victims in your bathtub. I have a wooden floor, and I guess, depending on the surface that you have where you are doing on wax melting, your wax processing. Occasionally, I spill some wax on the floor. In fact, sometimes I have whole buckets run over. And what you do is nothing. You just want to sit there and cool. And then it will just peel right up off the floor. And you get things like this, which I think that is a calf, but I get all kinds of great shapes without any effort. All I have to do is let the spout overrun. Another thing I should say as far as clean up is concerned. If you get wax on your clothing, the best thing to do is take a fairly warm iron, put a paper towel over your spilled garment, and then iron it. And usually, the wax will wick up into the paper towel and nobody will know that you did it.

READ

The Many Uses of Beeswax

■ Beeswax has a long and storied history. Key Points Its uses were manifold: sealing Beeswax can be used for a variety envelopes, waterproofing and of purposes including candles, caulking, embalming, etc. The soaring beauty products, lubricant, food popularity of beekeeping during WWII covering, and furniture polish. had much to do with the use of wax in Over time, beeswax products can the war effort (as well as the high price develop “bloom” that can be of honey). Today, beeswax has been removed with a soft cloth. replaced in many applications by synthetic and petroleum-based waxes (such as paraffin), but it is used for a range of beauty products, including lip balms, hand creams and cold creams, makeup, and soaps. It can also be used as a furniture polish; for lubricating stuck zippers or skis; in art; or as hive foundation, crayons, cheese coatings, coatings on fruits, and more. Beeswax is often shown competitively at fairs and conferences. Different categories include wax in blocks, carved/molded artistic figures, candles, and products. Entries of wax are judged based on their cleanliness, uniformity — including an absence of cracks — and color. Sometimes aroma and artistic design are also criteria. Wax can easily become contaminated by in-hive or environmental toxins, including agricultural pesticides and miticides in the hive. As a lipophilic (fat-loving) substance, it absorbs fat-soluble toxins (more than honey or adult bees). This includes wax purchased as comb foundation from bee supply companies. Note that the cloudy, whitish “bloom” on wax that develops over months or years is normal (and, in fact, indicates that a product is made all or in part from pure

beeswax) and can be rubbed off with a soft cloth.

WATCH

The Many Uses of Beeswax — Making Candles

■ Candles are probably the most iconic Key Points use for beeswax. Compared to candles Pure beeswax candles burn longer made from paraffin and and produce less smoke than vegetable/palm waxes, pure beeswax paraffin and vegetable wax candles burn longer and create less candles. smoke. There are a variety of candles that can be made with beeswax, such as votive candles, pillar candles, and candles poured into glass jars or molds. Here we will explore the traditional form of beeswax candles: dipped taper candles. Explore the process in this video with Ann Sill, an award-winning candlemaker for Ryan Apiaries in New York State. Video Transcript We're now going to make dipped beeswax candles, which, in my mind, are the perfect use of beeswax, and the perfect candle. So what we have here is a homemade candle melter. It is a water jacketed tank, I can't tell you how many gallons, but it holds probably about 80 pounds of beeswax, would you say? Is that right? I think that's right. About 80 pounds of beeswax. So the first step that we're going to do here is to re-filter the beeswax. It's very important to get your wax really well filtered because any little impurities, little specks of dirt, lint, that sort of thing will show up on your finished candle, which, obviously, is unacceptable. We have found that the best filtering material is something that I would call an organdy cloth, it's a very fine nylon mesh. I don't know if the organdy cloth is the correct terminology or not, but if you go to a good fabric store you'll be able to find what you're looking for. This is a well-used bag, but still quite functional. And, again, the way we've filter it is just we've made a bag, the block of wax goes inside the bag, and as the wax melts, it comes out of the bag and is now filtered. So, here we have the filtered wax. Again, the block of beeswax goes into this bag and as it comes out. the wax is then filtered. That's good. So, on a particular day if I'm going to decide if I'm dipping, I'm going to do dip candles, I have two different sizes that are easy, and we have frames for them. John has made the frames. And it's important to keep, instead of, I know a lot of companies have ways of weighting down your wick, but we have found that if you can tie your wick onto these stainless steel pegs, and then I just start wrapping around, trying to keep it, these are spaced for the first dip of making dipped candles. It's important to keep the wicking extremely taut so you get a straight candle. When you go to purchase your wick, we buy pure cotton wicking. Size depends on what size candle you're going to end up with as far as your end product. If you're making pillars, there's a diameter, and most of the companies will have a chart for you to look at to decide what size you need. This is for dinner candles, it's a spool. We have purchased a large quantity of it just because it's cheaper to do that. Another thing to consider is that when you're buying wicking, make sure that it is considered for beeswax, not paraffin, because paraffin burns at a lower temperature. And so, there is a difference in wicking requirements, whether it be beeswax or paraffin. Okay, so once you have strung you wick and made sure it's reasonably tight, it's very important to keep it taut, you can have any size frame to do it on. I use these because it works for my cooling batch. So that's what I use. And then you can, at some point, you need to mark on your top of your wick the center of the wicking. And I will explain later as to why that's important. Temperature is where I want it. I will go down, and for the first dip, you will put it in and you hold it there. You will see bubbles coming up and out of the wax. But all that is your holding it there so the wick can get, absorb the hot wax. And I can't tell you a time, but when it quits bubbling, then you have done that. We'll let

that frame cool. In between the dips, it is important to let it cool slightly because if you have it too warm and re-dip into the hot wax, you'll never get it to build up. It just will keep melting on your wick. After it has cooled slightly, three to five, or whatever your schedule allows. This will be the second dip of the first phase. And again, it's important to try to keep it, when you go back down into the hot wax to keep it continuously without stopping. If you stop, sometimes it'll leave a ring. So we would go down. Now this time, I'm going to hold it in the wax for approximately four seconds, and then come back up and out. And in between each dip I do turn the frame, so that when I get to the cooling table it will be ready to pick up. Turning it, I believe, helps keep the bottoms from wanting to curl. This will be the fifth dip. From here on in, all the dips will be a continuous motion. Important to keep moving in the hot wax and not pause. This will establish, kind of, how your tip at the top will look with the finished product. After 13 to 15 dips of going in and coming right back out, it will get to this size. My limiting factor with my frame is that if I make it dip anymore, it will either stick to the side of my frame or they'll stick to one another, which is not good. So what I do is I let them cool over night. It makes them easier for the next part, where I have to cut them from the frame. Cutting them off, which we just did. This is the start of phase two. Why do I have to transfer them? This is a wider board, and for the second dip, you're going to add more diameter to your candle, so you need to separate them, or they will stick together. Again, you will see that now we have that mark that we put on our frame for phase one. I now have a line on my large boards down the center. And I just match my wick blue line, red line, whatever color you choose. And that just rests on the line of my wider boards. So, once I have transferred all of them off my first phase dipping board, and I would be ready to now dip for my second phase dipping to complete the process of the dinner candles. So, with the wider boards, candles centered with our marks, we are starting the first dip of the second phase. Now on this one it's important, again, especially if it's been more than 24 hours, you're going to go down into the melted wax, and you're just going to, for approximately four or five seconds, you're warming up the cooled-off wax. And then coming back up. It's a continuous down and back, except for you're, basically, on the first four dips of the second phase. You're just trying to warm the candle back up so it will accept the hot wax. Again, on the second dip of the second phase, I put them in for about three seconds. And then the next dip, I'll do two seconds, and then finally, it'll just be down and up. And at this part you can see that they start to drift at the bottom, but that is what you want. After approximately 8 to 9 dips on the wider board in the second phase, I'm getting close to the size of the candle that I want, but I'm not really sure. Now you can take a pair of the candles. I break up the tip because I'm going to put it into a stick to size it. And I just approximately say, yeah, that's going to work in a stick. They do a shrink a little bit. So if it's a little bit loose at this point, you probably should dip another one or two times because you almost want it too big for, this one is cool, so it has shrunk down, but it's now right. And this was after nine dips of the second phase. So once I have completed second phase my candles are now ready. I will inspect each one as I take them off the wide board to pack them in the box. I am now looking for candle show candles. I'm looking for major flaws, and they can be at this point a crack, or a blemish, dirt, anything that would make it go from being a number one candle to second, what we call seconds. Doesn't usually affect its burning, but it's got to look good for me to put it in a package. Once you have put your base on your candle, it can be fluted, it can be flat. Ours end up being flat. We have bought boxes from a bee supply. And we put in a little blurb of do's and don'ts. And talk about bloom and that. I just put that in. And they're ready for packaging and selling. Over time, the bloom on a candle is almost inevitable. You can see that it has kind of a white, powdery,

glucose finish to it. That is just supposedly the more volatile waxes within what comprises beeswax coming to the surface. Some people might prefer that effect. It has kind of a nice aesthetics to it. But if you want to remove it, you can either buff it with a soft towel, which I think is a lot of work. And more easily done, you just run it under fairly warm water. I wouldn't make it too hot, but warm water will take that bloom right off and it'll shine the candle right back up to bright yellow.

READ

Pollen

■ Pollen is a high-value hive product that Key Points can be collected and sold by Pollen is a protein-rich food. beekeepers. Its most common uses are as nutritional supplements for humans, feed for bees during times of low protein availability, and supplements for other animals such as horses. It is normally sold in small pellets — the tiny, intact balls of pollen that the bees bring in on their corbiculae — but can also be added to other foods or mixed with honey. The pollen collected by bees comes from a variety of plants in a range of environmental conditions and is therefore enormously variable. Indeed, bees prefer a mixture of pollen from multiple sources to a uniform source. Even in the same place and at the same time of year, two neighboring colonies may find different sources of pollen and bring back a very different composition of pollen to the hive. Pollen is therefore difficult to characterize in definite terms, making scientific research about its medicinal effects in humans less conclusive. However, its general composition has been established as a mix of protein, carbohydrates, starches, fats, vitamins, and minerals. The table below outlines the breakdown of pollen's nutrients. Pollen Composition

Component	Average Proportion
Protein	23.7%
Carbohydrates	27%
Lipids	4.8%
Vitamins and Minerals	4.6%

Pollen is considered a protein-rich food. For comparison, a 3.5-ounce serving of pollen contains 24 grams of protein, while a piece of chicken of the same weight contains 27 grams of protein (mind you, no one consumes that much pollen in one sitting). It has trace amounts of the B vitamins and Vitamin A, as well as the minerals calcium, copper, iron, magnesium, manganese, potassium, and zinc. In addition, it has low levels of sodium. Pollen's levels of Vitamins D, E, and K, however, are low or entirely lacking.

READ

Medicinal Properties of Pollen

■ Some people who practice alternative Key Points medicine recommend consuming There is no clinical evidence to honey bee pollen for treating various suggest consuming pollen can conditions, including gastrointestinal help treat medical conditions. issues, joint pain, cancer, cancer treatment side effects, premenstrual syndrome, aging, and as an athletic performance enhancement. The most commonly invoked benefit of pollen (whether consumed straight or in a local honey) is its potential to help relieve or reduce allergies to airborne pollen, or hay fever. The recommended dosages vary, but are usually around one to two tablespoons of pollen per day. To date, there is insufficient evidence to support using honey bee pollen for any of these conditions. Although it is rare, consuming pollen can cause serious allergic reactions in people who have pollen allergies. While most bee pollen is from insect-pollinated plants, it can contain small amounts from wind-pollinated plant as well (the variety that causes most human allergies). Several cases of anaphylactic shock have occurred from people consuming bee pollen in an effort to treat their seasonal allergies. Despite the lacking science, there is

a good market for bee pollen at farmers' markets and health food stores. Beekeepers should be careful to never advertise their pollen for sale to be used as a treatment for allergies, and they should be ready to openly discuss the risk of allergic reactions to those who wish to purchase it from them for this purpose. The most common side effect of ingesting pollen is stomach upset and gastrointestinal problems. Should anyone wish to consume pollen, they should ease their way into the dosage to minimize these reactions. Start with a tiny pinch and increase the dose slowly until able to consume one or two tablespoons. Anyone who plans on consuming bee pollen should speak to a physician beforehand.

READ

Harvesting Pollen

■ Pollen can be harvested from hives Key Points using pollen traps (available from most Pollen traps should only be bee equipment companies). There are engaged temporarily so as to not various pollen traps on the market, interfere with colony nutrition and including ones mounted beneath a brood rearing. hive, on top of a hive, or directly over Freezing pollen can preserve its the entrance. Each of these traps either nutritional content. reroute the bees through a new entrance into the hive or partially block the main entrance, forcing incoming foragers to squeeze through wire mesh screens large enough to allow bees through but small enough to strip off the foragers' pollen loads. In this way, pollen traps collect up to 70% of the pollen that comes into the hive. Pollen traps come with switches or parts that allow the trap to be engaged (allowing pollen to be trapped) or disengaged (allowing bees to enter freely and preventing pollen from being trapped). When the trap is engaged, the pollen pellets accumulate in the tray, which needs to be emptied every two days to reduce mold growth and pests, and daily if conditions are damp. Pollen collected in the tray of a pollen trap. The amount of pollen that can be collected in a trap varies greatly and is generally between two and six pounds per colony per year. They should only be used during heavy pollen flows, as colonies require nearly all the pollen they collect during scarcer conditions. Furthermore, pollen traps should not be placed on weak colonies, as pollen is needed to grow the population, or on colonies while they receive mite treatments. Pollen traps placed on hives in pollination or adjacent to agricultural fields may collect pollen with more pesticides than those placed on hives surrounded by unmanaged natural areas. Limiting the amount of pollen coming into a hive stresses a colony and can impact brood rearing. There are no specific recommendations for how long a trap should be engaged. Cornell University suggests keeping a trap engaged for no longer than seven to ten days at a time to allow enough nutrition for the colony. If a trap is going to be activated for longer than that, beekeepers can provide the colony with a pollen substitute to help with brood rearing. As soon as pollen is collected from the tray, it should be cleaned. This is accomplished by picking out all the pieces of bee body parts and other debris by hand or sift. After cleaning the pollen, place it immediately in the freezer to retain its nutrients and kill any potential pest insect eggs that could be present. If pollen is left unfrozen and sitting at room temperature, it will lose 76% of its nutritional content after one year. Pollen can be removed from the freezer the night before it is packaged or consumed. To maintain freshness, beekeepers should only package the amount of pollen they can expect to sell fairly quickly. Encourage customers to keep the pollen in the freezer and to thaw small quantities at a time when they are ready to consume some. Some beekeepers choose to dry pollen instead, placing it in a food dehydrator until much of the moisture is removed. Drying pollen

in this way can cause it to become crunchier and less able to dissolve in other foods. The method of preserving pollen depends on the beekeeper's preferences. Debris (as shown here) should be removed from pollen before it is stored or sold. After pollen is processed in this way, it can be placed in clean glass jars for sale. Only intact pollen pellets should be in the jar, as an excess of pollen powder is generally not attractive to consumers and can affect the marketability of the product.

READ

Propolis

■ Propolis has long endured a bad Key Points reputation among beekeepers as the Propolis has antibacterial and sticky stuff that gums up frames and antifungal properties. other hive equipment. Indeed, beekeepers have over the centuries consciously or unconsciously selected for bees that collect and produce relatively little propolis. But propolis has an important function: It is what the bees use — a mixture of resins collected from trees and wax — to seal holes, line the interior of the hive, and entomb intruders living or dead (e.g., a “prison” for small hive beetles). New research by Dr. Marla Spivak’s team at the University of Minnesota has shown the benefits of propolis to honey bees in protecting their hives from chalkbrood and AFB, thanks to its antibacterial and antifungal properties. This stands to reason, given the origin of the resin in propolis as a plant defense of its young buds and wounds. Like pollen, the chemical composition of propolis within a hive or an apiary varies widely, depending on from which plants the bees have foraged the resins. Propolis has properties that make it an antimicrobial and anti-inflammatory agent. It is used in alternative medicine for treating wounds and burns, sore throats, and stomach ulcers. Topical applications, including dental ones, are the most promising, but as some people may be allergic to propolis or other hive components that are mixed in, care should be taken when ingesting or topically using a propolis product. The most well- researched avenue for propolis in human medicine is its use in oral health. In laboratory studies, propolis has been shown to kill several species of gram-positive bacteria and microorganisms common in human mouths and has therefore been incorporated in some toothpastes and oral rinses. Small clinical trials with humans, however, fail to show strong support for its use to prevent cavities or help cure oral infections. Larger and more robust clinical trials are needed before propolis can be recommended as a definitively helpful product in human medicine. Propolis can also be marketed for beekeepers to use as a stain for their hives, supplementing the natural propensity of their bees to propolize the interior and lowering demands on the bees’ immune systems. Thus, propolis can be sold fresh or powdered, in a tincture, or as an ingredient in a range of products (ointments, creams, and wood finishes) to be applied topically to humans or applied to wood.

READ

Harvesting Propolis

■ For most home uses, chips of propolis Key Points can simply be scraped off from Harvesting propolis is wherever it has accumulated in the accomplished by scraping it from hive, although this propolis is usually the hive or by using a propolis trap. mixed with wax, some wood particles, and other debris and may need to be cleaned before being used. These propolis scrapings can be soaked in warm water several times to help dislodge any debris inside. Propolis pieces sink, while other debris tends to float. Larger amounts of pure propolis can be collected with a propolis trap laid across the top bars of a hive

(replacing the inner cover) at any time during the active season. These traps look like queen excluders, but the openings are smaller than bee space, meaning that the worker bees cannot fit through them and they are stimulated to fill the holes with propolis. Once removed, the trap is frozen to kill any pest insect eggs and make the propolis brittle so that it cracks off the trap when the trap is flexed or rapped sharply against another surface. If beekeepers plan to sell propolis, they should use a propolis trap for a cleaner product, rather than simply using the scrapings off various hive equipment.

READ

Preparing Propolis

■ Some beekeepers sell dried propolis to Key Points healthcare suppliers to create propolis. Propolis can be incorporated into capsules to be consumed. These are capsules or tinctures for human made by grinding frozen propolis with consumption. a pestle and mortar to a fine dust. More commonly, beekeepers produce propolis extracts to sell as a specialty item. A propolis extract is created by adding propolis to high-proof grain alcohol (not to be confused with rubbing or other kinds of alcohol that can be toxic for human consumption!) to extract and dissolve compounds from the propolis. There are many recipes online and in books, but commonly a 5–30% concentration propolis solution (by weight) is created by adding it to grain alcohol and placing it in a closed container. This solution is allowed to sit for two weeks in the dark at room temperature, until the solution has taken on the color of the propolis (often brown or red). After this time, the solution is filtered through fine filter paper to remove any chunks or debris and is bottled in small, dark glass dropper bottles. The alcohol preserves the propolis extract for over one year. On its own, a few drops of this extract can be consumed as a tincture.

READ

Royal Jelly

■ Royal jelly is a complex and nutritious Key Points blend of sugars, proteins, fatty acids, The most common uses for royal amino acids, vitamins, and salts. While jelly are in nutrition, cosmetics, royal jelly helps queens live an order of and queen rearing. magnitude longer than their worker- Royal jelly is the hive product that sisters, its effects in humans are much commands the highest sale price. less well understood and not nearly so dramatic. Still, the product commands Royal jelly is harvested from queen a high price as a nutritional larvae cups. supplement and an ingredient in cosmetics (e.g., creams and masks). Of course, beekeepers use it to rear queens by priming cells before grafting young larvae. Nutritional Supplement As a nutritional supplement, royal jelly is commonly consumed as dried powder inside capsules, or a small spoonful of fresh jelly is dissolved under the tongue. While royal jelly does display some antimicrobial and anti-inflammatory properties in laboratory studies (likely from the fatty acids), there are no large-scale human clinical studies that have investigated any medical benefits. Small-scale clinical studies are exploring royal jelly's ability to help regulate blood glucose in patients with diabetes. Findings are conflicting, and more robust studies are needed before recommendations can be made. Royal jelly nutritional capsules. Cosmetics Because royal jelly consumption stimulates cellular pathways that enable queen bees to live so long, many people believe that it has anti-aging properties that also work on humans. Royal jelly is included in face masks and moisturizes with the idea that it can help skin remain youthful. No scientific studies support this notion, yet the royal jelly cosmetic industry remains huge, especially in China and other Asian countries. China

produces the most royal jelly in the world, but there is also a market in the U.S. Royal Jelly is the hive product that carries the highest value; in the U.S., one ounce of royal jelly can typically be sold for around \$15. Collecting Royal Jelly Collecting royal jelly is time consuming, although its logistics make it relatively easy to combine with a queen-rearing operation. Indeed, the steps for jelly collection are largely the same as the first few steps of queen rearing, namely, creating queenless, populous colonies with lots of nurse bees, supported by protein and carbohydrate resources to create a "starter colony." One-day-old queen larvae are grafted into queen cups along cell bars and are placed in the starter colony. When the larvae are three to four days old, the cell bars are removed. The larvae are picked out with tweezers (and are either disposed of or used in animal feed), and the royal jelly is spooned out with a small spatula or sucked out with a small vacuum device. Three- or four-day-old larvae have the most royal jelly in their cups; harvesting jelly from younger or older larvae give smaller yields. Seasonality is also important: Queen rearing — and, by extension, the production of royal jelly — is easiest in the spring swarm season. Royal jelly production requires some specialized equipment over and beyond what is necessary for queen rearing, particularly items that can handle small quantities with exact measurements: an accurate scale, small storage containers, extraction tools, and a freezer. The jelly should be frozen or refrigerated immediately. When frozen, it retains its nutrients for one year. When refrigerated, it retains its nutrients for four months.

WATCH

Venom

■ Most beekeepers take measures to Key Points avoid being stung. But some Bee venom injections are used to beekeepers specialize in the treat bee sting allergies. production of bee venom for human Venom is most commonly medicine. collected through the Many studies with animals have electroshock method. demonstrated bee venom has anti- inflammatory and pain-relieving effects. This is true of extracted venom (purified and usually diluted) injected into a patient, as well as live bee stings, but not whole-body extract injections (bees crushed, blended, and clarified), which are ineffective. The clearest benefit of bee venom is, unsurprisingly, in treating bee sting allergies (immunotherapy). This is usually done by injecting patients with very low doses of venom and gradually increasing the dosage over a period of days or weeks to desensitize them, followed by maintenance injections for several years. This is expensive, complicated, and labor intensive to do correctly for individuals who suffer from life-threatening allergic reactions to bee stings, but clinical studies have found that venom injections are effective. Because of the risk of fatal allergic reactions, immunotherapy should only be conducted under the care of a physician. Aside from immunotherapy, the most thoroughly researched applications for bee venom are for rheumatoid arthritis. Because venom has been shown to have anti-inflammatory properties in animals, there is the potential that it can help with inflammatory diseases. To date, only a few small-scale clinical trials have been conducted, but these have yielded promising results. More robust large-scale clinical studies are needed to verify these results before recommendations can be made. As always, caution should be exercised when using venom. Medical professionals should oversee treatments. Allergic reactions can be fatal. Venom extraction from a colony. Extracting Venom Since venom collection began in the 1930s, a variety of methods have been developed. Perhaps the most painstaking was by carefully squeezing several bees, one at a time, until a single drop of venom collected at their sting tips that could be gathered.

Today, the most common method is to use the electroshock method. In this method, a tray-like device containing several wires placed in parallel lines over a glass plate is positioned at the entrance of the hive. Electricity is turned on to run through the wires, which agitates any bees that leave the hive and come in contact with it. Momentarily, many bees can be stinging these wires, as alarm pheromone attracts more defenders. The bees do not lose their stinger during this process and therefore survive. With each sting they dispense a portion of their venom, which collects on a glass plate beneath the wires. The electricity is run for about 15 minutes at a time (for up to three rounds, totaling 45 minutes), and then the tray is removed and the venom is allowed to dry into crystals. The crystals are collected and are often stored in the freezer to maintain freshness. It is estimated that it takes about one million bee stings to produce one gram of dry venom. Venom collection is not very common, and there is not a large market for it in the U.S. It is currently used mainly for human medicine. In this YouTube video, Timothy Ferris, owner of Extract and Box Ltd, describes the process of venom collection.

READ

Mead

■ Mead is an ancient drink that is Key Points underappreciated! A simple fermented Mead is an alcoholic beverage with optional flavoring from made with fermented honey, fruits, spices, and even grains, a fine mead is as good as any wine. Mead can be dry or sweet based on the brewing and bottling processes, strongly or subtly alcoholic, and sparkling or still. Its flavor is infinitely variable, based on the assortment of monofloral honeys used. Commercial meads are still relatively rare; a majority of mead lovers brew their own small batches at home. Mead making is not a quick endeavor — most require at least six months to ferment — but it is simple, and there is much room for variation once the basics have been perfected. In essence, mead is simply spoiled honey. Being hygroscopic, exposed honey attracts water from the surrounding atmosphere. When the water content of honey increases beyond 18%, it becomes susceptible to microbial growth from bacteria, fungi, yeasts, and other microorganisms. Yeasts are the primary microbes responsible for converting sugars into alcohol; however, they require nutrients as well as the proper conditions in order to do so. There are many kinds of yeasts in the environment that foragers are exposed to and bring back to colony. These yeasts are incorporated or trapped in the honey as it is ripened. When water is added to raw honey, mead is produced as the sugars present in honey are converted to alcohol by yeasts that have been activated in the honey. Since raw and minimally processed honey already contain yeast and sugars, it is thought that mead may be the oldest alcoholic beverage.

READ

Making Mead — The Basics

Mead making is simple — or at least it can be. Raw honey already comes prepackaged with two of the three main ingredients: honey (sugars) and yeast. To make a basic mead, beekeepers can just extract their honey and add water; it's as easy as that. However, if beekeepers want to make mead that is even more palatable, there are additional things to consider, such as honey, yeast, equipment, and good fermentation practices. Honey First, not all honey is created equal. Although every beekeeper knows honey comes from nectar, it takes time and experience to learn which flowers contribute to different honey flows and when. In central New York, the major honey flows come from clovers, black

locust trees, basswood/linden trees, knotweed, goldenrod, and aster. All of these honeys produce distinctly flavored meads, provided they are treated properly throughout fermentation. Linden is a prized mead-making honey in much of Europe, as its pronounced minty flavors tend to carry through to the beverage. Goldenrod honey is a regional favorite with which to ferment, as it yields fruity and malty notes in its resulting mead. Second, not all yeasts are equal in the eyes of a mead maker. Thanks to the wine and beer industries, many strains of yeast are available that have distinct fermentation characteristics that factor into the production and, ultimately, the taste of the beverage. For instance, temperature influences the fermentation speed and final flavor profile of the mead; most yeast strains have a recommended temperature range for fermentation. In addition, the alcohol and sugar tolerance determine how much alcohol and sugar any particular strain of yeast can handle before they either stop or start fermenting. Most people use these tolerances to understand the upper alcohol limit for a particular mead recipe; for example, many beer yeasts stop fermentation after 7–10% alcohol by volume in solution. ■ Yeast ■ Different yeasts also vary in their “congener profiles.” Congeners are compounds other than alcohol and carbon dioxide that yeast produce during fermentation, and they can affect the flavor, mouthfeel, color, and overall complexity of any fermented beverage. Some of the most recognizable congeners are flavor compounds such as diacetyl (buttered popcorn), acetaldehyde (green apple), 4-vinyl-guaiacol (clove), and isoamyl acetate (banana). The same congener can be perceived as either desirable or a fault in the beverage, depending on personal preference. Lastly, nutrient requirement is of the utmost importance when making mead. Other than sugar, nitrogen is the most important nutrient needed for fermentation, and the right amount of nitrogen is needed to feed the yeast. Because honey is low in nitrogen-containing compounds, the mead maker must add some in a specific dose in the early stages of fermentation. Protocols detail the doses and proper times to add the nitrogen. Taking good care of the yeast requires being mindful of all of these factors. However difficult it may be to master, learning to achieve a balance with the yeast will bring the mead to new levels and audiences. Yeast (left) and yeast nutrient (right). Equipment ■ Basic meads do not require much equipment; a fermenter, airlock, and mixing implement are the three main pieces of equipment necessary for a small batch of mead. However, it is helpful to have a kitchen scale, a spray bottle, and a racking cane (a tube used for siphoning liquids) as well. These items allow mead makers to better implement and record recipes, sanitize equipment, and transfer mead to another container after primary fermentation as well as to the final bottle. Procedure Honey and water are mixed together in the fermenter; in a separate bowl, yeast is activated by providing it with warm water and some nutrients. After the yeast is activated, it can be combined with the honey-water mixture and sealed with an airlock. Over the course of a few days, initial nutrients can be added according to the specific recipe the mead maker is following. After the fermentation stops and the yeast settles, the mead can be transferred to another container, where the yeast is allowed to fully settle out of the solution. After this step, the mead maker conditions and ages the mead however they choose. Additions can be made here, such as oak, spices, fruits, or clarifying agents. When the mead has been sufficiently conditioned, the bottles are filled and stored. The mead should be allowed to sit and age in the bottle for one month to one year or longer. Aging time is determined by frequent tasting; when the mead maker is satisfied with its flavor, the mead is declared “ready to drink.” ■ Mead fermenter (left), floating airlock (middle), and mixing implements (right) for making mead. Mead Styles Since mead has been around for so long, it is no wonder that humans have experimented with putting almost everything imaginable into it! In

most cases there are names associated with various additives or styles: cyser (apples), pyment (grapes), acerglyn (maple syrup), metheglin (herbs/spices), melomel (fruits), rhodomel (rose), braggot (barley/grains), poltorak (Polish mead made with a honey:water ratio of 2:1), bochet (caramelized honey), etc. Conversely, traditional or “show meads” are simple meads that do not feature any additives; they consist of water, honey, and yeast. In addition to the limitless possibilities for additives, there are many stylistic choices that can affect the final product. Meads can be clarified and filtered to be brilliantly clear, or they can be left unfiltered and hazy for a rustic, or saison, style. Mead can be fortified with spirits for a port style, or aged in a solera system to achieve a sherry-like flavor profile. It can be made sweet, semi-sweet, or dry, and carbonated or still. One stylistic choice that is distinctively tied to mead is aging. Short meads, much like beers, are to be consumed almost immediately after production. Most meads are similar to wines in that they can be aged past two years and grow in complexity. Mead is distinct in that ■ there is a category called grand meads, which are meant to be aged for 12 to 20 years before reaching peak flavor and complexity! Many meads take on more nutty, caramel, and pleasant earthy aromas in this time.

MODULE WRAP-UP

Additional Hive Products

In this module, you discovered there are many more products than just honey that can be harvested from the hive. We explored the most common products and evaluated the scientific evidence behind using them for medicinal purposes.

Pollination Services

Bees are the main insect pollinators throughout the world, and their services are critical to our agricultural systems. Wild and managed bees are estimated to contribute \$20 billion to the U.S. agricultural economy and over \$217 billion worldwide. Honey bees are the most common bee species used in pollination, and colonies are rented to pollinate at least 70 crops in the U.S. While many other bee species are excellent pollinators and forage in rainy and cool weather, it is very easy and convenient to bring honey bees into a crop system. Hives can be moved on trucks and placed wherever they are needed, and each hive has around 20,000 foragers that can be immediately introduced into a field. Some other bees are also managed for pollination in the U.S., though to a lesser extent than honey bees. Bumble bees are commonly used in greenhouses, especially for tomato pollination; alfalfa leafcutter bees are used to pollinate alfalfa in western U.S.; and two species of orchard bees are used to pollinate early-flowering orchard fruits.

WATCH

Pollination: The Birds and the Bees

Pollination facilitates the sexual reproduction of plants. When an ovule becomes fertilized by pollen, it develops into a seed to generate a new plant. Many plants rely on wind or water to transfer pollen. Some examples of wind-pollinated plants include wheat, corn, rice, oats, and barley. These plants do not produce bright, attractive, fragrant flowers because they do not need to advertise for animal visitors.

When you see pollen in the air and experience pollen allergies, this is mostly from wind-pollinated plants. Many water weeds and pond weeds use water pollination, where pollen grains are released into the water and float on the surface until they contact a flower. Most of the flowering plants in the world, however, rely on animals to pollinate; in fact, around 90% of the world's flowering plants require animal pollination in order to reproduce. Flowers and bees have co-evolved over billions of years. Flowers attract bees through color, light patterns, and fragrance, advertising their pollen and sugary nectar. Bees, in turn, have evolved specialized anatomy to collect pollen. This video delves into flower pollination and how honey bees transfer pollen efficiently. Video Transcript Pollination is the sexual reproduction of a plant. It's the process where pollen from the male part of the flower is transferred to the female part of the flower. And this results in fertilization, where a plant sperm fertilizes a plant's egg and gives rise to the production of a seed. The male part of a flower is collectively called the stamen, and it consists of the filament, which is long stalk, and the anther, containing the pollen grains. The female part of a flower is collectively called the pistil. It consists of the stigma, which is the receptacle that receives the pollen, the ovary, and the style, which is the stock that connects the two together. The pollen grains, which contain a plant sperm, are located on the anthers of a plant, while the egg, or ovule, is located in the ovary. During pollination, bees transfer pollen grains onto the plant's stigma, where it can grow a pollen tube that travels down the style of the plant, ultimately reaching the ovary and the ovule it contains. The sperm fertilizes the ovule, and a seed is eventually produced. After pollination, the petals fall off the flower. The seed begins to form and the plant's ovary begins to swell, turning into a fruit such as an apple, pepper, cucumber, or into a pod. Some plant species can reproduce through self pollination or wind pollination, where they don't need bees' help to generate seeds. Corn, wheat, rice, soybeans, and sorghum are all examples of plants that do not need pollinators. The majority of flowering plants, however, need pollinating animals in order to help them transfer pollen grains to the stigmas of flowers on other plants of the same species. And without these pollinators, the plants would not be able to reproduce. Flowering plants have been evolving with bees for millions of years, and have developed specialized structures, colors, sizes, odors, and carbohydrate-rich nectar rewards to attract bees and other animals. In turn, bees have developed specialized tongues, hairs, and pollen carrying structures that enable them to collect pollen efficiently. And this evolved dependence of bees and flowers is truly one of the most remarkable and beautiful relationships on the planet. And honey bees have evolved to be excellent pollinators. The first behavior that makes them such great pollinators is their communication of food sources. Scout bees are the first to discover a pollen or nectar rich plant in bloom, and they return back to the hive to tell their sisters all about the food source, through the waggle dance, through the plant's odors that they carry back on their bodies, and by sharing small samples of nectar with them. In a short period of time, this communication can result in a plant becoming covered with bees, foraging on its nectar and pollen, and visiting nearby plants of the same species. This helps the plant by greatly enhancing the likelihood of pollination. It has a lot of helpers. Scouts are excellent at locating the most abundant and sweetest nectar sources near the hive. Next, bees' behavior to actively collect pollen, as well as their hairy bodies, helps them to transfer pollen from plant to plant. While collecting pollen, the bee's body can become dusted with it, making it really easy to transfer to the stigmas of other flowers they visit. Bees visit multiple flowers in a single foraging trip, and in this way, individual bees can help pollinate flowers before returning back to the hive. Even after a forager's trip is over and she returns back to the hive,

the colony is still aiding in pollination. Studies find that flower pollination is enhanced by pollen transfer within the hive. So a bee that's recently returned from foraging will come in contact with many of her nestmates in the crowded hive as she walks toward a cell to deposit her pollen. In doing so, her nestmates get dusted with small amounts of pollen. And when these bees fly out to visit flowers, they can effectively pollinate them. Lastly, honey bees display floral fidelity. This means that individual bees tend to visit the same species of flower for an entire foraging trip. If the flower species is abundant, the same bee will continue visiting that same species throughout the entire day. This benefits the plant tremendously. It means that honey bees are transferring pollen to the same species of plant, increasing the likelihood that ovules will get fertilized. For instance, it would not be beneficial for an apple tree's reproductive success if after a bee collected its pollen, it then went to visit a dandelion. Instead, traveling from apple tree to apple tree increases its chances of reproduction. By scouting out food sources that are rich pollen nectar sources, by communicating beneficial food sources to other foragers in the hive, by trapping and passing pollen in their sticky hairs, and by being loyal to a single type of flower during a foraging trip, honey bees can optimize their own food collection while inadvertently optimizing a plant's chances for reproduction. There are a variety of animals that help plants pollinate: bees, some flies, butterflies, some wasps, some beetles, and hummingbirds. Of these, bees are the superior pollinators and very efficient at collecting pollen. While butterflies and hummingbirds are iconic pollinators, they only actively collect nectar from plants; some pollen incidentally lands on their bodies and gets transferred from plant to plant while they feed on nectar.

READ

Preparing Colonies for Pollination

■ A colony ready for pollination should have a strong population and should be growing in size. Colonies with little brood should contain eight frames of brood production and four frames of brood. and may not produce a lot of foragers. Preparing strong colonies for to be available for pollination. Spring pollination begins the Strong colonies are also good for previous autumn. business, as growers who can see lots of honey bees on their flowers and who experience good crop yields of high-quality fruits and vegetables will want to continue renting colonies from a beekeeper year after year. Colonies for pollination are typically organized into two double-deep Langstroth hive bodies. If the crop tends to produce enough nectar to yield a honey crop (like oranges or blueberries), beekeepers may add supers on top. Pollination colonies should contain at least eight frames of bees and at least four frames of brood when they enter pollination; however, the contract between the beekeeper and grower may vary. Below, you will explore the requirements of a colony that is ready for pollination. Bees The eight frames of bees should contain individuals of various ages: Older bees can forage immediately on the crop. Middle-aged bees will soon transition to foraging and can receive nectar from incoming foragers and build comb for storage. Younger bees can tend to the brood and keep the colony growing. If frames of bees cannot easily be counted because it is favorable weather and most of the foragers are outside, an exiting rate of 60 bees per minute with at least one-third of returning bees carrying pollen is a good indicator of a strong colony. Brood The four frames of brood serve two purposes: Brood emits pheromone cues to stimulate foragers to collect pollen. If colonies have little brood, some beekeepers add a strip of synthetic brood pheromone to the hive to stimulate pollen collection. The product SuperBoost® is a commercially

available synthetic brood pheromone product that comes in a pouch to be placed between brood frames. Brood keeps a colony growing and replaces the cohort of bees that die. Drawn Comb Several frames of drawn comb should be available: The queen should have room to lay to prevent swarming and to continue brood production. Foragers reduce pollen collecting when the brood population decreases. There should be room for nectar and pollen storage coming into the hive. Foragers reduce food collection when they are honey- or pollen-bound. Queen A strong, fertile queen should be present in the hive: She can continue brood production, contributing to a growing colony and providing impetus for foragers to collect pollen. Weak queens may be superseded, which can reduce colony productivity and growth for a short while. Strong queens laying lots of brood often result in strong colonies, better equipped to overcome pests and some infections. Food Stores Food stores should be in the hive: The colony should have enough honey and pollen stores to fulfill their nutrition and energy needs while it is in pollination. Colonies must not be nutritionally stressed. Thirty pounds of stored food is typically expected, but the grower and beekeeper can negotiate this amount. Hives in an almond orchard. Building strong colonies for spring or summer pollination begins in the previous autumn. Ensuring colonies have enough protein (pollen or pollen substitute) and carbohydrates (honey or syrup) to survive winter and build up quickly in the spring is important. The winter cluster should be comprised of a large number of young bees. These bees require adequate protein and fats for their development in September and October so that they are able to raise brood in late winter. In addition to fall feeding, beekeepers who prepare their colonies for pollination often begin feeding dry sugar in January and pollen patties in February. Sugar syrup feeding begins in March. These carbohydrate and protein supplements are continuously fed until the first nectar and pollen flows occur. This intensive nutritional management ensures that colonies are active and strong, with a large forager population coinciding with crop bloom. Colonies that exit winter weak but are otherwise healthy can be merged to boost the population strength. In addition to healthy, populous colonies, hive equipment should be clean and in good condition when it is brought to someone's farmland. Colonies should be free of any observable disease and should have Varroa mite levels below the treatment threshold for the particular time of year. Colonies should be placed in farmland as soon as some of the crop flowers are already in bloom. The presence of flowers attracts foragers to that crop. If colonies are moved into a field before the crop is in bloom, the bees look for other food sources and may not visit the crop as readily once it does bloom. Colonies are typically removed just before flowering finishes or immediately after flowering finishes. The grower should inform the beekeeper when this occurs. Colonies should be removed before any post-bloom pesticides are applied to the crop.

READ

Hive Stocking Rates

■ Extensive research has been done to Key Points determine the recommended hive Hive stocking rates have been stocking rates for each crop. Some established for the main crops require more honey bee hives to pollination crops in the U.S. produce a high quantity and quality yield as the flowers may be less attractive to bees, either because they have lower sugar concentrations in the nectar or because they are physically deterring (such as alfalfa, which thumps honey bees in the face when they collect pollen from it). Alternatively, fewer hives are needed if a crop area is small and surrounded by natural vegetation that promotes many native pollinators. Beekeepers can recommend hive stocking

rates to use as a benchmark, but the number of hives to be rented is ultimately determined by the grower. The table below highlights general hive stocking rates for crops produced in the U.S.; you can also download the Hive Stocking Rate Chart for future reference.

Hive Stocking Rate	Crop (Hives per Hectare)
2.5–8	Cucumbers
2.5	Honeydew
2.5	Cantaloupe
1–2.5	Watermelon
1–8	Squash
1–8	Pumpkin
1–8	Zucchini
2–5	Apples
2–5	Pears
2.5	Plums
1.5–5	Cherries
2.5	Apricot
1–2.5	Peaches
1–2.5	Nectarines
4	Red Currant
2.5–5	Black Currant
1–2.5	Raspberry
1–2.5	Blackberry
0.5–20	Strawberry*
1–7.5	Blueberry
2.5–7.5	Cranberry
12	Alfalfa
2.5–8	Alsike clover
1–8	White clover
1–15	Canola*
1–2.5	Sunflower
3	Buckwheat
8–10	Kiwi
5–6	Almond
1	Orange
5–8	Avocado

*More research on necessary stocking rates is needed for this crop.

Pollination Opportunities in the U.S.

Many beekeepers pollinate a variety of crops during the year, transporting their colony from state to state to follow the timing of the blooms. The earliest U.S. flowering crop is almonds, beginning in mid-February, followed by cherries, apples, and plums, which are often in March and April. After these fruits finish flowering, colonies can be moved to pollinate blueberries and cranberries around May and June, and, finally, they can pollinate melons, squash, pumpkins, and sunflowers in July and August.

The map below shows the top crops pollinated in various U.S. states. Long description ■ The graphic shows the United States and the different pollination opportunities in the U.S. Maine, New Hampshire, Massachusetts, Rhode Island, Connecticut, New Jersey, Vermont, New York, Pennsylvania, Ohio, Indiana, Michigan, Illinois, Wisconsin, Iowa, Nebraska, and Kansas are highlighted blue. This indicates they have the following plants for pollination opportunities: apple, cherry, peach, watermelon, blueberry, cranberry, pumpkin, and squash. Delaware, Maryland, West Virginia, Virginia, North and South Carolina, Kentucky, Tennessee, Georgia and Alabama are highlighted in gray. This indicates they have the following plans for pollination opportunities: apple, cantaloupe, watermelon, blueberry, cucumber, pumpkin, and squash. Florida, Mississippi, Louisiana, Arkansas, Missouri, Oklahoma, Texas and New Mexico are highlighted in orange. This indicates they have the following plants for pollination opportunities: other tree fruit, orange, cantaloupe, watermelon, blueberry, cucumber, pumpkin, squash. Montana, North and South Dakota, Minnesota, Wyoming, Colorado, Utah and Nevada are highlighted in green. This means they have the following plants for pollination opportunities: apple, cherry, and pumpkin. Alaska, Oregon, Washington and Idaho are highlighted in yellow. This indicates they have the following plants for pollination opportunities: apple, cherry, peach, pear, watermelon, blueberry, cranberry, raspberry and clover. Finally, California, Arizona and Hawaii are highlighted in red. This indicates they have the following plants for pollination opportunities: almond, apple, avocado, cherry, plum, other tree fruit, orange, kiwi, cantaloupe, honey dew, watermelon, blueberry, cranberry, raspberry, cucumber, squash, alfalfa and sunflower.

READ

Renting Hives — Costs

■ Beekeepers have historically made the Key Points majority of their profits through selling Pollination services have shifted to honey. In recent decades, however, become a major source of profits there has been a shift in the for beekeepers in recent years. beekeeping industry toward providing Hive rental rates have increased pollination services over honey dramatically over the past few production. In fact, many commercial decades. beekeepers focus primarily on pollination, selling honey and wax as a Providing pollination services can small part of their business. With hive be stressful to colonies.

rental rates having increased so dramatically over the past few decades, beekeepers are able to make good money renting colonies for pollination. Furthermore, this money can be made in a short period of time, as colonies may only need to be placed in a field for two weeks or so depending on the bloom period of the crop.

Cost: Stress to Colonies Multiple pollination events are stressful to honey bee colonies. Beekeepers are wise to prioritize colony health, since losing a substantial number of colonies comes at a big cost. Transporting colonies can be stressful, as bees are exposed to temperature fluctuations, vibrations, and confinement for days at a time. Maintaining the brood nest temperature can be particularly difficult for bees while in transport. During almond pollination, bees may still be clustering and trying to stay warm while moving at high speeds on a truck. During summer months, colonies may have a difficult time keeping cool, unable to leave their colonies to beard or collect water. Queen losses during transport can also occur. Once placed for pollination, bees can experience nutritional deficiencies if their diet is comprised nearly entirely of one pollen source. These nutritional deficiencies are more common in large monoculture fields and less common in farmland surrounded by natural vegetation. Beekeepers can provide pollen patties during pollination to supplement bee diet. If the grower has rented hives from several beekeepers, a colony can become exposed to Varroa mites, foulbrood, viruses, and pests from its neighboring colonies. Hives are usually placed on pallets in or adjacent to fields, with four or six colonies in close proximity. Sharing flowers, drifting, and robbing can occur, facilitating disease spread. Colonies should be carefully inspected before and after pollination to promptly manage parasites, pathogens, and pests. In addition to potential stress from disease and pests, colonies can be exposed to many pesticides, and the number of pesticides to which they are exposed grows with increasing pollination events. Pollination contracts should clearly define which pesticides can be used during pollination and whether or not the grower will compensate any colonies lost from acute pesticide poisoning events. All pesticide poisoning events should be reported to both the state apiculturist and the appropriate state department (usually the Department of Environmental Conservation [DEC] or Department of Plant Industry). Beekeepers can find the appropriate office to call for their state by consulting this [Quick Guide to Reporting a Pesticide-Related Bee Kill Incident](#). To circumvent nutrition stress, many beekeepers place their colonies in wildflower meadows to recover between and following pollination events. There is no scientific research that has examined how many crops a colony can tolerate pollinating in one year and still overwinter healthy and strong. Many commercial beekeepers relieve overwintering stress on bees by placing them in warm winter locations where they can continue to fly and where they can be fed and re-queened as necessary. It is up to the beekeeper to judge the health of a colony and determine whether or not the colony should continue entering new crops for pollination.

READ

Renting Hives — Benefits

A large benefit of renting hives is the profit you'll derive from it. Determining the price to rent hives takes many factors into account. First, it must be profitable for the beekeeper. Before entering a pollination contract, beekeepers should think about how much it costs them to perform the service. The following factors should be tallied: Estimations of mileage, gas, meals, and hotel stays to move colonies to and from the crop. Price of labor if help is needed. Price of transportation if trucks are to be rented. Price of syrup and pollen patties to be fed to colonies before, during, and sometimes after pollination. The

pollen from some crops, such as alfalfa and blueberry, are not very nutritious for bees, and the bees can exit these crops nutritionally stressed or with limited population growth, requiring more attention from the beekeeper. Cost of miticides or other treatments that might be necessary after exiting pollination. The beekeeper's hourly rate and number of hours they will spend preparing, moving, and placing colonies. This last point is very important: Too often beekeepers do not include the value of their own labor in the price. To determine the minimum price for a hive rental, the value that is determined from the exercise above can be divided by the number of colonies that will be rented for pollination. This number can then be compared to average rental prices in the beekeeper's state, and both can serve as a helpful reference for price negotiations between beekeeper and grower. The National Agriculture Statistics Service (NASS) collected hive rental data for the first time in 2016. This makes it possible to know how much beekeepers were paid to rent their hives for pollinating different crops, how many acres of each crop were pollinated, and how many colonies were rented.

State-specific rental rates for a particular year can be found on the NASS website, but here we present a table summarizing hive rental information for the entire U.S. Take some time to become familiar with which crops rented the most hives and which crops paid the highest hive rental prices. You can download this Crop Pollination in the U.S. Chart (displayed below) for future reference. Crop Pollination in the United States Average Number of Number of Acres Crop Price per Colonies Month Pollinated*

Pollinated Colony	Rented	Kiwi	\$205.00	2,800	4,800	May–June	Almond	\$167.00	971,400	1,680,000
February–March	Cranberry	\$74.57	39,300	88,000	June	Pumpkin	\$73.53	22,500	20,500	
July–September	Blueberry	\$66.54	73,800	147,000	March–June	Raspberry	\$66.15	11,500	22,000	
May–August	Plum	\$64.40	29,700	46,000	March–May	Alfalfa	\$63.60	42,700	76,900	
May–July	Orange	\$59.70	9,400	49,000	March	Watermelon	\$59.42	64,850	76,900	
July–August	Cucumber	\$55.13	44,100	31,500	June–September	Peach	\$53.30	5,050	4,700	
March–April	Pear	\$53.10	24,300	30,000	March–May	Apple	\$52.64	129,600	183,400	
March–May	Cherry	\$51.80	99,400	134,100	February–April	Cantaloupe	\$51.73	42,750	46,600	
July–August	Honeydew	\$51.30	9,300	10,000	July–August	Squash	\$47.75	21,500	27,000	
July–August	Avocado	\$40.80	12,800	69,000	February–March	Clover	\$33.60	8,300	10,500	
April–September	Sunflower	\$28.80	22,800	30,000	May–September					

*Flowering times differ depending on the species (e.g., some plum species bloom in early spring, while others bloom in late spring) and may vary from year to year due to climate variations. Average price per colony is based on 2016 NASS data.

What Influences the Price of Hive Rentals? Hive rental price is dictated in part by the necessity for honey bee pollination. For instance, almond pollination in California is completely dependent on honey bees. Orchards are so vast and separated from other natural vegetation that very few wild bees visit their flowers, and the number is not sufficient to produce an almond crop. Because almond growers are so dependent on honey bees, they are willing to pay premium dollars for them. Other crops may be surrounded by natural vegetation and thus might support a robust wild bee community. In these cases, honey bees supplement the natural pollination services already occurring, and growers may pay less for honey bee hives. Pollination rates have increased drastically over the past few decades. This is most notable with almonds. The acreage of almond orchards in California has grown significantly over the years; in 1996 there were 430,000 acres, while in 2016 there were 971,000 acres! At the same time that the demand for almond pollination services was increasing, the supply of honey bee colonies available for pollination was decreasing. Beekeepers across the U.S. have been experiencing higher colony losses than normal due to a variety

of factors, including parasites, pathogens, pesticides, and management practices. The dynamics between supply and demand resulted in skyrocketing rental prices. For example, in 2004 the average cost to rent one bee hive for almonds was \$54, while in 2016 the average rental price was \$167! This increase in price for almond pollination also helped to drive rising prices in pollination of other crops. Additionally, the percent of agricultural land since the 1960s has increased faster than the availability of honey bee colonies; this has driven growers to pay a premium price for the limited number of colonies available.

TOOL

Creating a Pollination Contract

Like any contract, a pollination contract is a legally binding document. You can download this example for both the grower and the pollination services contract. It outlines the expectations Beekeepers may use this template of both parties and includes the as a starting point for their own following information: grower negotiations and edit it to How many hives will be rented. suit their needs. The rental price for each hive. The date the rental fee is payable. The status of those hives upon entering the crop (number of frames of bees, brood, and amount of stored food). Any cost penalties from deviations of the minimum colony status should be clearly outlined. The dates the hives will be placed and removed. Some growers do not provide specific dates, as crop flowering can vary from year to year and it can be difficult to precisely time flowering. In these cases, the grower should carefully monitor crop bloom and provide beekeepers with a notice period (for example, 48 hours) for when the hives should be delivered and removed. The application timing of specific pesticides. Whether or not the grower assumes liability if hives are damaged or stolen while on their property. The contract should outline the compensation cost of a hive that was damaged or lost in these circumstances. Additional agreements can be negotiated by the beekeeper and grower and outlined in the contract. For instance, if a hive has fewer frames of bees or brood than expected, a specific previously agreed-upon discounted rate can be given to the grower. On the other hand, if a colony is stronger than the minimum colony status, an increase in fee could be agreed upon. If one party does not fulfill their obligations as outlined in the pollination contract, the dispute can be settled in court. In the past, many crop pollination services were done simply with a "handshake" contract. Nothing was written on paper or signed; the parties simply shook hands in agreement. The issue with these types of contracts is that individuals can misremember specific details of what was discussed. Writing down a contract ensures both individuals think precisely about specific details. Drafts can be exchanged before the contract is signed to ensure all factors are considered before the colonies are moved. With oral handshake contracts, some details might be missing that need to be resolved later. As pollination services are a business relationship between a grower and a beekeeper, we recommend that written contracts be utilized on every occasion. Witnesses should also be present to sign the contract. It is best to begin making pollination service agreements with growers before winter so the beekeeper has sufficient time to prepare colonies to be strong and healthy for spring pollination.

READ

Preparing to Transporting Hives

■ Transporting hives for pollination is a Key Points big endeavor that requires careful Beekeepers must apply for the preparation and planning. While this proper permits and certifications section is written from the context of to transport bees into and out of pollination services, these certain states and provinces. recommendations can be used when moving colonies for any purpose, whether tracking a honey flow, changing apiaries, or moving to a new wintering site. With travel dates approaching, the beekeeper should assess the needs of the colonies and prepare the bees for travel. It is necessary that the colonies are well populated and free of disease. Their stores should be assessed and the bees should be provided supplemental sugar feed and pollen substitute if necessary. Weak colonies should be united with strong colonies a few weeks in advance to give them enough time to organize themselves. There are numerous hive configurations that are good for transportation. Typically, most are moved in two deep hive bodies, but sometimes they are shipped in a single deep, a deep and a medium, or either of these configurations with a queen excluder and a super. Hives taller than that are seldom seen in a load for travel. More importantly, uniform hive heights make the most secure load. Allowing plenty of time to prepare colonies and being thorough improves the chance for success. The colonies should have enough time to work propolis to secure the boxes, lids, and combs before loading. This reduces comb slapping and bee crunching that can occur while taking corners on the road. The risks associated with transportation of bees for pollination require vigilant monitoring and assessment. For the collective health of honey bees and preventing the spread of contagious diseases, it is the responsibility of the apiarist to be compliant with pertinent laws. At least two weeks in advance of an interstate move, the beekeeper must apply for a Certificate of Apiary Inspection for Interstate Movement of Honey Bee Colonies and Equipment through the State Department of Agriculture of the origin state. In New York State, this is acquired by calling the NYS Apiculturist. Truck drivers must always have this certificate on hand so that they can show any police officials that they are transporting a legal load. Some states, like Florida, have an agricultural station that asks for the certificate at the border. Though the regulations address the same purpose, there is a lack of uniformity with various requirements from state to state. In California, for instance, bees must be registered with the Department of Agriculture in the destination county. Some states like Texas require numerous permits, like the Importation Permit, Exportation Permit, Bee Removal Transportation Permit, Intrastate Travel Permit, and Certificates of Health for Queen Breeders. Maryland requires the beekeeper to show a health certificate from both the beekeeper's home state and an entry permit for Maryland in order to enter. It is the responsibility of beekeepers to know the laws of the states and obtain the required permits and certificates. To ease any inspections that may occur, hive bodies and the pallets on which they are sitting should be clean, with no mud or vegetation. Any pest insects found by border guards or police officials on hitchhiking plants or mud can considerably slow (or prevent) a beekeeper's travel through a state. Upon arrival at the pollination site, the bees should be in prime condition for inspection and for pollination.

WATCH

Transporting Hives — Loading the Truck

■ Honey bees are moved on all sizes of Key Points trucks, from small pickup trucks to Colonies should be secured to large semi trucks that carry just under trucks with ropes and checked on 500 hives. Whatever the size of truck or frequently; certain conditions also load of colonies, the considerations for

require providing water and transporting honey bees remain the traveling screens. same. All hive entrances must be screened to prevent bees from exiting while in transit. If the weather is hot, it is prudent to replace the hive covers with traveling screens to increase ventilation (no inner cover or outer cover is placed on top of the traveling screen). Each hive receives a strap to hold together the bottom board, hive bodies, and covers. Many beekeepers staple these hive pieces together for added security. Once each hive receives a strap, it is loaded onto a truck. Depending on how many hives a beekeeper is transporting, they can be loaded by hand (with a two-person hive carrier it is possible to hand-load heavy colonies), with a boom, or with a forklift. Some beekeepers recommend leaving the engine of the vehicle running while the bees are being loaded and unloaded; they claim the vibrations seem to either stun or calm the bees, keeping them quieter than they would be with the motor off. This clip demonstrates how hives can be loaded onto a truck using a boom. Video Transcript There is no audible dialogue in this presentation. Conversation and machinery sounds are heard in the background. Two people work together in a grassy field, using a hoist to lift hives from the field onto a flatbed truck. If possible, the hives should be arranged so the combs are oriented to the direction of travel. The hive entrances face forward or backward (towards the driver or tail lights), while the broad faces of the combs should face the sides of the truck to avoid the bee-killing “comb slap” that happens if they are loaded sideways and sudden stops occur. This may be more difficult with big loads containing hundreds of hives but is easily accomplished with smaller loads. The hives are roped or strapped to the truck or trailer for the trip. It is always good practice to first drive a few miles, then stop to retighten the straps or ropes before proceeding into the bulk of the trip. Ropes and straps should then be checked every couple hours or about every 100 miles. With big loads, additional mesh netting is thrown over all the colonies to further reduce the chance of bees escaping. The netting should be checked so it remains taut and bee tight around the bottom. This mesh should also be hosed with water occasionally to keep bees cool. Bees normally collect water when they are hot, but they do not have access to it when they are confined to their hives on trucks. With water passing through the mesh and coming in contact with the hive entrance screens, bees are able to collect water to help with thermoregulation. These tasks should be regularly and thoroughly attended while assessing the bees' needs for water. Larger operations do better by loading palletized bee hives (four or six at a time), often with the hive entrances left open. Loaded at night when all the field bees are in the hives, these colonies are then covered with a net. Even though the bees can crawl out of their hives, the bulk remain in the hive and the net prevents most escapees. Many bees become disoriented during the loading and netting. Whether the load is large or small, loading should always be done during the night. It is good practice for a team to work together to make loading bees as quick and smooth as possible. The team must dress with fully secured clothing since the bees at night are “crawly” and find their way into poorly secured coveralls and veils that are sufficient during the day. Smokers must be used liberally and kept handy. It is best to use headlamps to keep your hands free. On a tractor-trailer flatbed, a pair of beekeepers can unfold the net over the load as the forklift operator positions the bees from front to back. Once the net covers the entire load and is tucked in all around, it can be strapped down with the load. This keeps the net from blowing and flapping, while keeping the stacks of bees solidly anchored to the truck.

READ

Transporting Hives — Travel Considerations

■ Monitoring and watering the load on Key Points the road is an important consideration. Some beekeepers transport bees. Appropriate parking is also constant in refrigerated vans. A challenge for a driver en route with a load of bees. A collection of “hitchhiker” honey bees typically accumulates on the outside of the load. When the truck stops during daylight hours, these disoriented bees fly about and can become a nuisance if the truck is parked too near human activity. This is a challenge for fuel stops close to a place of business. A conscientious driver is constantly on the lookout for low activity and suitable facilities to top off the fuel tanks or perform other necessary maintenance on the truck. A load of bees on a flatbed trailer or truck, even in the open air, generates a lot of concentrated heat, which is hard on bees in transit, especially on hot days in full sun. To help bees stay cool in such conditions, it is necessary for the driver to park in the shade to check the bees, observe them for a nervous clamorous buzz, and provide water as needed to cool the hives. The bees should be misted and not soaked. It is helpful to have a thermometer with a probe in the center of the load. Temperatures over 100°F indicate that the bees need assistance cooling the hives. While spraying water on the hives, the driver can generally see the bees relax. Through these experiences, a beekeeper gains a sense of how to best keep them adequately supplied and calm. It is handy to carry a container of water, a hose, and a pump to spray water on the load while parked in an isolated location. On a semi, this might be a 55-gallon barrel with an electric pump. On small loads, a garden-sized sprayer with a hand pump is adequate. However, especially with tractor-trailer loads, watering is usually done at truck stops or other facilities that can provide water and a hose near appropriate parking. Grain elevators and farm supply stores in small rural towns are often good places for watering. The driver uses such isolated locations to check tires, the tautness of the straps, and the seal of the netting that prevents bees from escaping through voids. For trips longer than 500 miles, it is generally best to have a team of drivers who can switch off while continuing to travel, sleeping en route. The beekeeper-drivers should put fuel in the vehicle before loading the bees. The vehicle should be roadworthy and current with all maintenance needs before loading. Roadside mechanical problems are challenging, and mechanics are reluctant to assist vehicles loaded with bees. Stopping en route is best done routinely and deliberately to avoid problems. The route should be thoroughly planned, including the location where the colonies are to be unloaded. The drivers should be prepared with equipment and knowledge to handle bees in case of an unpredicted weather event, mechanical problem, or traffic accident. Coveralls, smoker, and veils should be part of the general on-board tool kit.

Hauling Bees With Refrigerated Trucks Many beekeepers haul hives in refrigerated trucks (also known as “reefer trucks”). The cooler temperatures help to keep bees less active, and the load is more reliably contained than with netting on a flatbed. Closing the doors to keep bees contained is also simpler than netting a flatbed load. Watering the bees in reefers is unnecessary. While it seems that the temperature of a reefer can be set to keep the bees in a constant loose cluster, the reality is that typical refrigerated vans cannot compete with the heat generated by the colonies. Colonies can increase the temperature to the point where bees are active inside the van. The bees also need fresh air and ventilation, which is not provided with most reefers. Some beekeepers have retrofitted refrigerated vans with adjustable ventilation systems and powerful refrigerator units. Their success indicates that this technology will likely be further developed and made more available in the future. Refrigerator vans need a loading dock for forklift loading and unloading. Beekeepers and drivers using reefers need to make special plans and arrangements to load and

deliver their colonies. **Special Transport Considerations: Hauling Nucleus Colonies and Packages**

Aside from providing pollination services, beekeepers may move their colonies for a variety of other reasons (tracking nectar flows, moving to wintering sites, etc.). Beekeepers who raise queens and produce nucleus and package bees have numerous occasions to haul bees to various apiaries and customers. Many of those customers also make trips to the breeder/production farms to pick up their own colonies. All of these people need to be aware of the special needs of these bees in transit. Nucleus colonies and packages are generally less stable than established colonies of bees that have fully occupied one or two deep hive bodies. Many of these bees are subject to long trips as well, such as bringing packages from southern Georgia into central New York. In such situations, nucs are handled like full colonies (as described above), keeping a keen eye on the heat. Nucleus hives and package bees can be sprayed with cool water to relieve overheating. While in transit and well watered, they typically arrange themselves in a cluster. If they are thirsty, they break out of the cluster into a nervous clamor. When they are misted with water, they should quiet down. Package bees can be vulnerable to cold, wind, and rain; in such conditions, they should have protective side panels strapped onto the load. If the weather is severe, the driver will need to offer some protection on the top of the package as well. Such protection can be provided with plywood or cardboard panels strapped onto the load. If the weather is warm and sunny, they can be less protected but may need to be watered.

MODULE WRAP-UP

Pollination Services

In this module, you dove into the world of pollination services. You explored how honey bees can be rented out to contribute to pollination in other states as well as how these bees are invaluable to the agricultural economy.

Confidence in Communicating About

Pesticides The discussion of pesticides is so common among members of the beekeeping community that it warrants its own module. Beekeepers are frequently asked about the impacts of pesticides on bees and may not feel equipped to properly answer the question. This is further complicated by the fact that research on pesticides is vast and continues to update each day, making it difficult to stay on top of the latest findings. In this module, we will discuss the research on various pesticides and how they affect bees. We will also discuss why there exists such a strong controversy about neonicotinoid pesticides as well as what recommendations you can give the public and other beekeepers to help reduce pesticide use and exposure to bees.

READ

An Overview of Pesticides — What's in a Pesticide?

■ Pesticides are an integral part of Key Points conventional agriculture. Insecticides, Pesticide products contain active herbicides, and fungicides help ingredients and inert ingredients; maximize crop quantity and quality by both can impact honey bee health. reducing the pests and diseases that cause damage. Especially in today's world of large monoculture fields, pesticides are needed to keep

out the various fungal, insect, and weed pests that can flourish in these expansive systems. While nearly all pesticide research investigates products used in agricultural areas, their use is not restricted to farms. Many pesticides are also applied to urban and suburban gardens and lawns. “Pesticide” is an umbrella term that applies to insecticides (intended to kill insects), herbicides (intended to kill weeds), fungicides (intended to kill fungi), and even miticides (intended to kill mites). Pesticide products contain many ingredients, classified as active ingredients (the ingredient that causes the pest's death) and inert or inactive ingredients. The active ingredients must be listed on the product label. Often, it comprises a very small amount (often around 5%) of the pesticide product. The majority of our understanding of how pesticide products impact honey bee health is by studying active ingredients, and we will explore these impacts in detail over the next few pages. The remaining 95% or so of pesticide formulations are inert ingredients. In the U.S., inert ingredients are not required to be listed on pesticide labels, as pesticide manufacturers consider them “trade secrets.” They are various chemicals that help deliver the pesticide and increase the efficacy of the pesticide when applied to a plant or soil. Adjuvants are a common type of inert chemical in pesticide formulations, and their function is to improve the performance of the active ingredient. The most common adjuvants are surfactants, used in products sprayed onto plants. They help the active ingredient stay on the plant instead of quickly rolling off as droplets. Other adjuvants help the active ingredient penetrate the tissue of the pest, some reduce degradation of the active ingredient from UV light, and others help thicken the formulation to reduce pesticide drift into the environment. Little research has investigated the impact of adjuvants and other so-called inert chemicals on honey bee health. However, in recent years, some scientists are beginning to find that not all of them are inert for bees. In 2012, researchers at Penn State University found bees that ingested organosilicone surfactants — the most frequently used adjuvant in almond orchard pesticides — experienced impaired learning. In early 2017 the team found that these same adjuvants could lead to worsened virus infections. The impact of inert ingredients on the environment is an active field in science.

READ

Pesticides in the Environment and in the Hive

■ Close to 1 billion pounds of pesticides are applied to U.S. agricultural lands each year. Pesticides are common in the environment and in bee hives. Herbicides make up most of the usage, followed by insecticides, and then fungicides. Pesticide applicators are required by law to report pesticide use data to the Department of Environmental Conservation. The U.S. Geological Survey compiles the data for pesticide usage each year, and that data can be viewed online through the work of the Pesticide National Synthesis Project; simply select the pesticide and year to view a map showing the areas it was used and the crops to which it was applied. Due to the sheer quantity of pesticides applied to land in the U.S., they are very common in the environment. Low levels of pesticides can be found in plants, soil, water systems, and a variety of animals. For this reason, all registered pesticides must undergo human health and environmental evaluations to determine whether they pose a risk to the environment or to people. Laboratory tests with honey bees are included as part of the environmental evaluations by the U.S. Environmental Protection Agency (EPA). Pesticides are common in bee hives. They come into the hive on pollen and nectar from plants that were treated with pesticides, in water that was collected from contaminated puddles or guttation droplets on plants, or

directly on bees' bodies after they have been foraging in treated landscapes or walking on contaminated soil. Pesticides are even placed in the hive by beekeepers trying to manage hive pests such as Varroa mites and hive beetles. Extensive studies across the U.S. have found that there are, on average, between three and seven different pesticides in a bee hive at any given time. Of course, this varies depending on how close the hive is to pesticide-treated lands, how often it is used in pollination services, and how frequently beekeepers apply mite treatments. The science is clear that pesticides are common in bee hives, and that fact alone might be alarming to people. The real question, however, is whether they are at levels that cause harm to bees. The simple fact that they are present does not give us information as to whether they are partially responsible for the widespread colony health problems that have been experienced in recent years. To give this some context, think about your dog's flea or tick medication. Prophylactic treatment of the animal is meant to reduce the incidence of illness that occurs when pests are left unchecked. Keeping the treatment below the threshold where the treatment itself becomes a problem is the balancing act.

READ

Measuring Harm to Honey Bees

■ There are a variety of ways to measure whether or not a pesticide harms honey bees; these ways can be broken down into four categories: residual, and chronic toxicity, as well as sublethal and synergistic Acute Toxicity effects. A common way to measure whether a pesticide is toxic to bees is by The LD is the dose of pesticide 50 measuring acute toxicity. Acute toxicity that kills half a honey bee determines the dose of a pesticide that population in a short period of is needed to kill a bee in a short time time. period (often 24 hours, but sometimes 48, 72, or 96 hours). To determine this amount, many individual bees are exposed to various doses of the active ingredient in a laboratory. Bees can be fed these active ingredients to determine the ingredient's oral toxicity, or scientists can apply small droplets to bees' bodies to determine its contact toxicity. The dose that kills 50% of the bees in this acute time period is known as the Lethal Dose 50 (or the LD). Similarly, the 50 concentration that kills 50% of bees in this time period is known as the Lethal Concentration 50 (or the LC). 50 The lower the LD , the more toxic the pesticide. For example, a pesticide with an LD of 1 50 50 microgram per bee is more toxic than a pesticide that has an LD of 8 micrograms per 50 bee, because it takes a much smaller dose to kill half the bees. The LD and LC values for honey bees have been determined for pesticide active 50 50 ingredients. Determining these values is part of the environmental evaluations conducted by the EPA before a pesticide is registered. This information is important because it tells us whether there is a risk of bees dying when they are foraging during or shortly after pesticides have been applied. If it is determined that the pesticide poses a risk of acute toxicity to bees, there is language included on the product label that prohibits applicators from applying it during bloom, when bees are actively foraging. When beekeepers observe pesticide poisoning events shortly after a pesticide was applied or after the planting of seed-treated crops, they are observing acute toxicity of the pesticide. The EPA classifies the acute toxicity of pesticides into "highly toxic," "moderately toxic," and "practically non-toxic" based on their LD values. A value of <2 micrograms/bee is 50 considered highly toxic, a value of 2-10.99 micrograms/bee is moderately toxic, and a value of 11 or greater micrograms/bee is practically non-toxic. Below is a list of contact and oral LD values for various pesticides. This is only a small 50

subset of the most commonly used pesticides in the U.S., as there are nearly a thousand active ingredients used today. There is no need to memorize any of these numbers, but students should familiarize themselves with how the data is presented. Which pesticides are highly toxic to bees, and which are practically non-toxic? Which are more toxic when consumed by bees, and which are more toxic when they are applied to a bee's cuticle? Notice that insecticides are generally the most toxic substances to honey bees, relative to fungicides or herbicides. The LD values are much lower for insecticides; often it takes 50 less than 1 microgram per bee to kill half those tested. This makes sense, as insecticides are targeted to kill insects, and honey bees are insects. In this list, the top three most toxic insecticides ordered by oral LD are clothianidin, 50 imidacloprid, and thiamethoxam — all neonicotinoid insecticides. By examining acute toxicity data, we can understand which pesticides pose the greatest risk to foraging bees when they are used. LD values in this table are from McArt, et al., 2017 and the references therein.

50 Long description ■ This image shows a data table listing various pesticide active ingredients and their toxicity measurements for bees. The table has four columns: Pesticide Active Ingredient, Type of Pesticide, Oral LD50, and Contact LD50 (measured in microgram/bee). The pesticides are categorized into three types: insecticides, fungicides, and herbicides. The table includes 24 different pesticides, with insecticides generally showing lower LD50 values (meaning higher toxicity) compared to fungicides and herbicides. Clothianidin and Imidacloprid show the lowest oral LD50 values at 0.004 microgram/bee, while Penthiopyrad has the highest known oral LD50 at 385 microgram/bee. The contact LD50 values range from 0.003 (Spinosyn) to 312 (Penthiopyrad) microgram/bee. The table uses alternating white and light blue rows for easier reading, and the header row is displayed with a teal background. One entry (Atrazine) has an "unknown" oral LD50 value. All measurements are precisely listed to up to three decimal places where applicable. The data is organized in roughly ascending order of oral LD50 values within each pesticide type category, making it easy to compare relative toxicity levels among different pesticides.

Residual Toxicity Residual toxicity is the time required to reduce the acute toxicity of a pesticide (and its residues) so that bee mortality is lowered to 25%. This means that when one in four bees that visit the flowers die, the pesticide has reached "low residual toxicity." The residual toxicity of a pesticide influences when growers are allowed to apply the pesticide to blooming plants. Pesticides with extended residual toxicity remain highly toxic on the plant for at least eight hours after application (and with some pesticides, over one week!), whereas pesticides with low residual toxicity remain highly toxic for fewer than eight hours. Most pesticides with high residual toxicity cannot be applied to blooms because the risk to beneficial insects that visit the flowers is too high. The EPA includes language on the product label prohibiting high residual toxicity pesticides from being applied when bees are visiting the area being treated. Low residual toxicity pesticides, however, can be applied at night when bees do not visit the flowers. By morning, the pesticide no longer presents an acute risk.

Chronic Toxicity Acute toxicity only captures the lethal effects within a very short time frame. However, pesticides often remain in the hive for longer than one to four days. They can be brought in with pollen and stored as bee bread. They can also accumulate in wax for days, months, or even years, depending on the pesticide. The chronic toxicity of a pesticide — the adverse effects after long-term exposure — have only been calculated for a handful of pesticides. A pesticide that causes chronic toxicity does not kill bees immediately, but it may kill bees at a later time or exhibit other negative impacts on bee health or behavior. The chronic effects of pesticides is being intensively studied at present.

Sublethal Effects Health impacts other than

mortality are considered “sublethal” effects. Investigating the sublethal impacts of pesticides is a fairly recent area of research and has revealed that many pesticides harm bees without killing them outright. Many sublethal impacts can affect individual bees, through impacts on learning, memory, locomotion (movement), feeding, reproduction, and foraging behavior. They also include cellular and physiological effects on bees. These may be changes in nerve transmission, changes in which genes are turned “on” or “off,” alterations to the gut microbiome, and the death of certain cells. Sublethal impacts can also be explored at the colony level. These studies investigate whether the consumption of or contact with pesticides by individuals can affect the entire colony. Some outcomes that can be measured include colony growth, overwintering survival, honey production, and brood production. Sublethal effects have been explored in fairly great detail for neonicotinoids and some miticides but remain largely unexplored for most other classes of pesticides. An example of a sublethal impact would be a reduction in foraging behavior. Synergistic Effects Traditionally, when researchers investigate the impact of pesticides on bees, they study how a single pesticide affects bees. With our growing understanding of the multitude of pesticides with which bees come in contact at any given moment, we now know that in order to understand the overall risk posed to bees in the environment, we need to investigate how those pesticides may be interacting with one another. Two pesticides can interact synergistically, meaning they are more toxic when occurring together than the sum of both pesticide’s toxicity levels on their own. Through these next few pages, we will explore the current knowledge of insecticides, fungicides, and herbicides on honey bee health. New science is continually being published on this topic, so students are encouraged to stay up to date on the latest research if they are ever in a position to speak or write on this subject. The final exam series for the Master Beekeeping Program will only ask questions based on the information we cover in this course.

READ

An Overview of Insecticides

■ Insecticides are generally the most Key Points toxic substances with which honey In general, insecticides are the bees can come in contact. While there most toxic pesticides to honey were some insecticides created before bees. World War II, the development of most insecticides, as well as the widespread adoption of their use, occurred after 1947. With respect to honey bee evolutionary time, this is a fairly new stress for them to face. Until the development of modern chemistry techniques that allow the detection of low levels of pesticides in bee products, beekeepers were unaware that there were often low levels of pesticides in bee hives. Now we understand that insecticides are fairly common in bee hives, and many can exhibit chronic and sublethal effects on bee health. Nearly all pesticide poisoning events reported by beekeepers are from applications of toxic insecticides. Rarely do fungicide or herbicide applications cause acute bee death. If insecticides are sprayed on plants when honey bees are actively foraging, they can come in contact with toxic doses. If insecticides have extended residual toxicity, acute bee kills can occur when bees are not present during spraying but visit flowers in the days afterward. Reducing unnecessary acute exposure of honey bees to toxic insecticides is one of the best ways to mitigate the risk of insecticide poisoning. One outcome of the U.S. Pollinator Protection Plan addresses this exact issue. Beginning in 2017, the EPA implemented new label recommendations to reduce the risk of acute toxicity. Any pesticide that (1) exceeds the EPA’s acute risk level of concern to honey bees; (2) is applied to foliage as a liquid or dust (a

formulation more dangerous to bees); and (3) is applied to crops that use bees for pollination services should include updated label language. The label will now clearly state which pesticides are toxic to bees and outline specific restrictions for applying the product. Some restrictions include the following: The pesticide cannot be applied during bloom. The pesticide can only be applied in a specific temperature window (avoiding temperatures when bees are active). The pesticide cannot be applied if bees are on site. The pesticide should be applied in a manner that reduces drift. The rate of application on a site should be below the EPA's level of concern for bees. As a Master Beekeeper, there is no need to memorize extensive lists of insecticides, nor be a specialist in chemistry or honey bee toxicology. Instead, Master Beekeepers should feel confident in discussing pesticide issues with other beekeepers, the public, and the media, and they should be aware of the current understanding of the health impact on bees. To help beekeepers build knowledge in this topic, we will cover the most common classes of insecticides — particularly those known to be highly toxic to honey bees — and discuss the current status of honey bee research in this field.

READ

Insecticides

■ There are tens of thousands of Key Points pesticide products registered in the U.S.; most of these products can be the nervous system in insects. classified into the five major classes of All classes of insecticides vary in synthetic insecticides: their risk to honey bee health. Organophosphates Some are practically non-toxic to bees, while others are highly toxic. Pyrethroids Insect Growth Regulators Neonicotinoids (covered on the next pages) There are several other classes of insecticides that are less commonly used and therefore not covered in this course. In addition to synthetic pesticides, there are also a variety of natural pesticides derived from living organisms (such as bacteria) that are used in organic farming. Their effects on bees are not well studied; in fact, very little research has investigated the sublethal impacts of insecticides other than neonicotinoids on honey bee health, despite the fact that many other insecticides are used widely in agriculture. Organophosphates Organophosphates are chemicals that inhibit the enzyme acetylcholinesterase, an important molecule needed for normal nerve functioning in insects (as well as most animals and humans). When the pesticide inhibits this enzyme, it disrupts nerve signaling and causes nerves to fire continuously. At doses high enough to cause acute toxicity, pest insects experience paralysis, twitching, and death. At low levels, it can cause sublethal impacts such as disorientation and impaired learning and memory. Organophosphates vary widely in their toxicity to honey bees. Some are practically non-toxic, while others are highly toxic. The table at the bottom of the page lists the organophosphates that are most toxic to honey bees. Shortly after they were developed, organophosphates became widely used pesticides. Their use is declining in recent years, however, as some have been shown to be toxic to a variety of organisms (including humans), and many pests have developed resistance to some of the active ingredients. Nonetheless, some organophosphates continue to be used in conventional agriculture. The most commonly applied organophosphate in the world is chlorpyrifos. This insecticide is highly toxic to honey bees and is unfortunately commonly detected in hive products: A large-scale analysis of pollen and honey samples from North America, South America, and Europe found it in 15-20% of hives. The study also found that phosmet (another organophosphate) was commonly detected. The levels of these two organophosphates detected in

pollen and honey were high enough to classify them as “posing a high risk to honey bee health.” In fact, of the 161 pesticides detected in hive products during this study, these two organophosphates posed the second- and third-highest risks to bees, respectively, after thiamethoxam (a neonicotinoid). Experimental studies have found chlorpyrifos can impair learning and memory in honey bees, and bees show an increased susceptibility to deformed wing virus and black queen cell virus following exposure to the insecticide. Furthermore, field-realistic doses of chlorpyrifos-spiked pollen fed to colonies have been shown to interfere with queen development. Other organophosphates have shown sublethal impacts on bees. For instance, methyl parathion can impair foraging behavior, and diazinon can impact learning and memory. Carbamates Carbamates act on the acetylcholinesterase enzyme in the same manner as organophosphates. High levels kill pest insects through disrupted nerve functioning. Similar acute symptoms of paralysis and sublethal symptoms of learning and memory impairment are observed as with organophosphates. Carbamates range from practically non-toxic to honey bees to highly toxic. The sublethal impacts of carbamates are not well known; only one study examined impacts on adult bees and found the carbamate carbaryl reduced bees' lifespans and delayed the age they transitioned to become foragers. Pyrethroids The passage of sodium through sodium channel proteins allows nerves to communicate with one another. Pyrethroids bind to these sodium channels and hold them open, causing nerves to fire repeatedly. At high doses, twitching, paralysis, and death are observed. Pyrethroid insecticides are considered broad- spectrum insecticides, meaning they kill a wide variety of insects instead of selectively targeting specific pest species. For this reason, many are highly toxic to honey bees and other beneficial insects. In general, pyrethroids have low residual toxicity, so the residues dissipate quickly. At high sublethal levels, the pyrethroids esfenvalerate and permethrin have been shown to reduce honey bee locomotion and social interactions among nestmates. Low doses of deltamethrin can impair development, reduce foraging behavior, and hinder foragers' ability to find their way home, and low doses of cyfluthrin, permethrin, cypermethrin, and tetramethrin have all been shown to impair motor functioning in bees in various studies. An experiment in 2016 exposed young adult bees to pollen contaminated with field- realistic levels of cyhalothrin and found that the young bees consumed less pollen (as if in an effort to avoid it). These bees experienced shorter lifespans and poor nutrition. Insect Growth Regulators Insect growth regulators (IGRs) are chemicals that disrupt the growth of immature insects using a variety of mechanisms. This makes them toxic to insects in the larval stage, but not to adults. Some IGRs work by altering juvenile hormone levels in pest larvae, which prevents these insects from developing into adults (in honey bees, juvenile hormone has several roles). Other IGRs act by altering the ability for an insect to molt or by preventing the formation of the hard outer cuticle on an insect. All these mechanisms result in eventual larval death. While IGRs do not pose a risk of acute toxicity to honey bee adults, several experiments have documented that the larval-killing effects of IGRs occur in honey bee larvae as well; these larvae die, develop malformed, or are removed by their nestmates. Those who do survive can have altered foraging behavior and are more likely to be rejected from the colony by their nestmates. The table below lists some common insecticides that are known to be highly toxic to honey bees. Students can refer to this list in the event that they are asked by the public, growers, or other beekeepers about the risk of specific pesticide products to honey bees.

Highly Toxic Insecticides to Honey Bees

Organophosphates	Pyrethroids	Carbamates	Neonicotinoids	Acephate	Cypermethrin	Aldicarb
Clothianidin	Azinphos-methyl	Cyfluthrin	Carbaryl	b	Dinotefuran	Chlorpyrifos
Bifenthrin	Methomyl					

Imidacloprid Diazinon Deltamethrin Oxamyl Thiamethoxam Dimethoate Esfenvalerate Malathion Fenpropathrin Methidathion Cyhalothrin Naleda Permethrin Oxydemeton-methyl Phosmet a In South Carolina in 2016, Naled aerial sprays for killing mosquitos as a Zika virus prevention measure caused large-scale, widespread honey bee deaths. b Carbaryl sprays on crops in the 1960s-1980s in the U.S. caused many large-scale honey bee poisoning events. Residues continue to kill bees up to 12 days after application. Carbaryl is the active ingredient in the product Sevin®. Information for this chart is from Hooven, et al. 2013. How to reduce bee poisoning from pesticides. Oregon State University.

READ

A Special Class of Insecticides: Neonicotinoids

■ Neonicotinoid insecticides were Key Points developed in the 1980s in response to In recent years, neonicotinoids an increase in pest resistance to other have become the most used common pesticides on the market. insecticide worldwide. They were commercially introduced in the 1990s, making them a fairly new class of pesticide. In a short period of time, they have now become the most-used insecticide worldwide, applied in 120 countries and on over 1,000 different crop types. Their surge in popularity was due to the fact that they are very safe for humans and are able to target and kill many different pest species. There are seven neonicotinoids commercially available: imidacloprid, thiacloprid, clothianidin, thiamethoxam, acetamiprid, nitenpyram, and dinotefuran. Neonicotinoids, like the other insecticides we have presented, interfere with the nervous system, including the insect brain. They exhibit a similar mode of action as nicotine by binding to acetylcholine receptors and overestimating nerves. At high-enough levels, they cause paralysis, twitching, and death. Humans and other animals also have these receptors in their nervous systems, but neonicotinoids bind with greater selectivity to insect receptors than mammal receptors, which makes them much less toxic to humans. Bee nerves also contain acetylcholine receptors, so they have the potential to experience the toxic effects of neonicotinoids. If exposed to high levels, bees die; at low levels, they can experience a variety of sublethal impacts. For instance, foraging is reduced, locomotion is affected, and bees become disoriented and have trouble finding their way back home. Because many acetylcholine receptors are found in the honey bee mushroom body, neonicotinoid exposure has also been shown to impair learning and memory in bees. With locomotion, orientation, learning, and memory so crucial to honey bee foraging and colony success, concerns arose that neonicotinoid exposure could be involved in widespread honey bee health challenges. Neonicotinoids act on acetylcholine receptors throughout the bee brain, including the mushroom body (highlighted here). Neonicotinoids Are Systemic Pesticides Neonicotinoids are applied to plants in a variety of ways. They can be sprayed onto plants directly, injected into tree trunks, drenched into soils, and coated on seeds. This last method — termed “seed treatments” — is the predominant mode of neonicotinoid application. With this method, seeds (usually corn and soy) are coated in neonicotinoids and planted into the soil. As the plant grows, all its tissues take up small amounts of this pesticide. Neonicotinoids are effective pesticides because they are distributed throughout the plant’s tissues to protect the whole plant. This action classifies neonicotinoids as “systemic pesticides” because they protect the entire plant system instead of one local, isolated area. The petals, leaves, stem, and roots all take up the pesticide; in this way, the plant is protected from a variety of pests that feed on or damage different parts of the plant. Unfortunately, this also means that pollen and nectar can contain neonicotinoids, which can put pollinators at risk. In

the U.S., 79–100% of corn hectares and 34–44% of soybean hectares are treated with neonicotinoids before planting. This type of pest control is a step backward from integrated pest management: Instead of applying pesticides only when treatment thresholds indicate to do so, these pesticides are applied prophylactically — regardless of whether or not they are needed. As a result, neonicotinoids are present in the soil (where they can remain for up to a few years), water, and nearby vegetation. Neonicotinoid treated corn seeds. Exposure Bees are exposed to neonicotinoids through a variety of avenues. Often, they collect contaminated pollen and nectar from flowers. This food is brought back to the colony and fed to larvae and adults. Studies show that bees do not only collect pesticide-contaminated pollen from the crops that have been sprayed, but they also collect it from nearby wildflowers exposed to these pesticides through application drift and from soil uptake into the roots. Neonicotinoids leach from plants and treated seeds into the soil and waterways, presenting additional routes of exposure to bees and other wildlife. From these routes of exposure, bees come in contact with low levels of neonicotinoids that do not often cause acute pesticide poisoning but may exert sublethal effects, chronic effects, or no effects at all. Lastly, when treated seeds are planted in the soil, the planting machines release contaminated dust into the environment. Any bees that are foraging in close proximity to seed planting (usually within one mile, but up to three miles) can come in contact with this dust and become exposed to high doses of neonicotinoids. Exposure through this route can cause acute pesticide poisoning. In these circumstances, beekeepers may observe their bees twitching, acting disoriented, extending their proboscises, and dying in large heaps at the entrance of their hive and nearby. Planting corn seeds releases dust that drifts in the environment.

WATCH

Toxicity of Neonicotinoids

■ Neonicotinoids range from practically non-toxic to highly toxic to honey bees. Nitro-containing neonicotinoids The variation of different are more toxic to bees than are neonicotinoids in toxicity is based on cyano-containing neonicotinoids. their chemical structure. Nitro- Neonicotinoids do pose a risk to containing neonicotinoids honey bee health in some regions (imidacloprid, clothianidin, and under some conditions. thiamethoxam, nitenpyram, and dinotefuran) are more toxic to bees than cyano-containing neonicotinoids (acetamiprid and thiacloprid). The nitro-containing neonicotinoids have LD values under 2 micrograms/bee, classifying them as highly toxic. 50 There is a high likelihood that bees die after consuming and coming in contact with neonicotinoids at these doses. On the other hand, the cyano-containing neonicotinoids have higher LD values that classify them as moderately toxic to practically non-toxic. 50 The majority of studies investigating neonicotinoid toxicity have been laboratory studies. These experiments expose bees to various doses and concentrations of neonicotinoids to see if any negative impacts result. Pesticides are applied directly to the bees' bodies or are fed to them through neonicotinoid-spiked sugar syrup or pollen. Scientists then observe what happens over a period of time. Collectively, these studies have demonstrated that low doses of neonicotinoids can cause a variety of sublethal impacts on bees, including impaired learning and memory, locomotion, feeding behavior, foraging behavior, altered gene expression, and reduced immunity. These studies are beneficial because they allow us to see exactly how the pesticides alone impact bee health, without the effects being compounded by the additional factors often experienced by bees, such as climate, diseases, or nutrition. There are several limitations to these types of studies,

however. First, they only evaluate the effects at the individual level, eliminating our understanding of colony-level effects. The bees are also forced to encounter neonicotinoids, since they are contained in all the food they are given. As such, the risk from these studies does not reflect the risk to bees foraging naturally in the world around them. Lastly, there has been criticism of some studies that expose bees to high concentrations or doses of neonicotinoids — much higher than what bees would encounter in the real world. To understand how these pesticides can impact colonies, scientists conduct field studies in which nucleus colonies or full-sized colonies are placed outside. Bees are fed field- realistic concentrations of neonicotinoids in sugar syrup or protein patties, or they are placed adjacent to neonicotinoid-treated crop fields. Colonies are followed for periods of time that usually range from two weeks to years. Studies of this nature explore what bees experience in the real world. They can show effects on entire colonies and for longer periods of time. If neonicotinoids are partially responsible for the widespread colony health issues documented in recent years, we would expect to see some negative effects through these types of studies. It is these studies that tend to produce the most controversy on the topic of neonicotinoids. Some field studies show negative impacts to colony health, while others do not. The results have been inconsistent and have rarely demonstrated that colonies die after being exposed. One study in particular found that while individual bees experience health issues and sometimes die, strong colonies are able to compensate for the losses of some bees and continue to grow and be productive. Of course, producing strong colonies requires a range of suitable environmental factors. Since these are not uniformly available, we again note that "all beekeeping is local," therefore field study results cannot be universally applied. With all the controversy, there has been a priority for these studies to be well designed, well replicated, and as realistic as possible to reflect the risk posed in nature. Some excellent studies have been published in recent years, and they are starting to shed some much-needed light on the subject. This video provides a summary of some of these studies. Video Transcript Studying the sublethal impacts of neonicotinoids on bees is challenging. Because there's so much natural honey bee colonies, in order for results to be statistically significant in experiments, it requires that researchers have access to many colonies, ideally close to a hundred, and lots of plots of land in the right places. These plots of land either need to be in close proximity to neonic-treated fields or really far away from neonic-treated fields. It also requires a lot of manpower because these colonies need to be regularly monitored and outcomes need to be measured. You need teams of people to accomplish this. So, by nature, it's tricky and expensive to accomplish all of these factors. Nonetheless, there's been several studies that have investigated the impacts of neonicotinoids in the field. Five studies that were conducted before 2014 reported no negative health impacts on honey bees as a result of being placed near neonicotinoid- treated canola or sunflower seeds. These studies looked at colony mortality, brood production, colony weight, worker lifespan, various health outcomes, honey production, and foraging behavior. And these researchers didn't dose or feed their colonies neonicotinoids; they simply placed them next to small fields that were either treated or not treated, and the honey bees foraged as they may. The results from these studies raised a lot of controversy right off the bat, as these findings conflicted with what researchers were finding in the laboratory. In the lab, when we're feeding honey bees neonics, we're seeing that this can lead to impairments in learning and memory, locomotion, feeding, foraging, the expression of certain genes, and immunity. So why weren't these outcomes being observed in field studies? Part of the reason could lie in the limitations of the studies. One of the biggest limitations is that treated fields are

really small. So honey bees forage two to three miles around their colonies, far beyond the application areas, meaning that there is no guarantee that a colony is going to primarily feed on that one treated plot. A two hectare research plot, for instance, encompasses only 0.025% of a honey bee's foraging area. In another one of these studies the untreated and treated canola fields were about 300 meters apart. So colonies could easily visit both fields, making it really tricky to detect differences between the two groups. Also, many of these studies only place honey bee colonies in treated fields for short periods of time, failing to capture the effects of chronic exposure to these chemicals. So, as I mentioned before, the constraints of having access to enough land in the right places and the time and manpower to measure outcomes for extended periods of time can make it challenging to study these effects in a meaningful way. Here's a summary of the field studies that were conducted up until 2015. It's easy to see where the conflicting results lie and understand why there's been so much controversy and confusion surrounding neonicotinoids. In this graph, the number of studies are found on the y-axis, whereas the outcomes are measured on the x-axis. So, these studies are looking at mortality, colony growth, brood production, honey production, foraging, overwintering, acetylcholinesterase activity in the brain, and virus infections. The blue parts of the bar represent studies that are not finding any impacts from neonicotinoids. Whereas the red are studies that find negative impacts from neonicotinoids on these outcomes. So, right off the bat, we can see that a lot of factors have no effects. Colony growth, brood production, honey production and overwintering are all not affected in these studies by neonicotinoids. Whereas acetylcholinesterase activity and the susceptibility to developing virus infections are impacted by neonicotinoids. And in some areas there's conflicting results. Six studies find that mortality is impacted, whereas two don't. And one study finds that foraging is impacted, where another study doesn't. So it can be really hard to interpret what this means. Some scientists wanted to make sure that honey bees were, in fact, consuming these low doses of neonics, and that these studies weren't limited by the amount of land that they had access to. So, instead of placing colonies near small fields that were either treated or untreated, the scientists would spike pollen or sugar syrup with pesticides to colonies and observe what happened with the bees. There were a lot of early field studies, however, that were actually dosing colonies with high levels of neonics that were being criticized for not reflecting realistic exposure. So recent field studies are now emphasizing the use of field-realistic levels of neonics to evaluate the real world risk that these pesticides are posing on bee health. And before we dive into looking at these studies, let's take a second to talk about what doses we tend to find in hive products. Real world doses of neonicotinoids differ depending on the application method. Seed treated applications, which is what we have with corn and soy, results in levels and hive products that average around 0.8 to 3.9 parts per billion. Levels in hive products tend to be higher if neonicotinoids are applied by soil drenching. They tend to be in the realm of 12 to 131 parts per billion. And lastly, foliar sprays are associated with the highest level of neonics in hive products, often between 5 and 208 parts per billion. For this reason, a lot of foliar sprays are not allowed to be applied when flowers are actually blooming. Lastly, applications to ornamental plants are applied at higher doses than applications to agricultural plants. So this can translate to higher doses in hive products. More recent studies have done a great job to thoroughly investigate the impact of neonics on bees in the field and they're helping to clear up some of the confusion. Let's go through two that are particularly well designed, and then we'll recap what we can say when we take everything together that we've discussed so far. The first study is one that was conducted in Ontario, Canada, and published in

2017. It studied neonicotinoid exposure in 55 hives that were distributed across five apiaries, either very close to cornfields or very far away from cornfields. They sampled the bees, pollen, and nectar from the hive at various time points during this study. What the researchers found is that honeybees near cornfields in Ontario are exposed to significantly more neonicotinoids than colonies far from cornfields. They also found neonics were present in the hive products for three to four months after corn seeds were being planted, demonstrating that there was a chronic exposure of these pesticides for most of the colonies' active foraging season. The researchers then took the average dose of neonics that they were finding in pollen and dosed colonies in a separate experiment. They found that this exposure to chronic low doses of neonics, exactly what the colonies were experiencing on Ontario, does have sublethal health impacts. The long term doses caused a significant increase in worker mortality, a decrease in hygienic behavior, and an increase in queen failure. The largest study conducted to date was done in Europe and was also published in 2017. This study was a massive undertaking. It followed a 198 colonies in 33 sites across the United Kingdom, Hungary, and Germany for two years. At each of these sites the researchers had plots that contained seed treated oil seed rape and large pots that contained untreated oil seed rape. They measured impacts on colony health during the crop flowering period, and also checked in after winter to measure the chronic effects on colony survival. And the authors found some interesting results. The biggest finding was that the effects differed depending on the country the bees were in. For instance, in the UK and in Hungary, colonies near the treated fields experienced population decreases. While in Germany, colonies in treated fields actually experienced slight population increases. Another example is that in Hungary, the negative effects of neonics persisted into the following spring. Colonies near treated fields were 24% smaller compared to colonies near control fields that received no treatments. And these effects were not seen in the other two countries. So how on earth do we interpret this? It's really easy to see where the controversy lies. It's clear that in some places at some times with some types of neonicotinoids, bee health is negatively impacted. But this relationship is not observed all the time. This makes it really easy for pesticide companies to interpret the results to say that there is no hard consensus that neonicotinoids harm bees. Researchers, on the other hand, say yes, sometimes neonics can harm bees. The take home message is that neonics can be harmful. However, whether or not a colony is affected depends on a variety of interacting factors. The first is how strong and healthy the colony was to begin with. Strong, healthy colonies may be less impacted by neonicotinoids. Colonies that are experiencing other stress factors such as disease may be more susceptible to the stress of pesticides. I already mentioned the correlations between neonicotinoids and virus levels. And studies also find interactive effects with the neonics and Nosema infections as well. Similarly, colonies that are weak with few foragers to begin with may not be able to recuperate from the adult population losses that sometimes happen from neonicotinoid exposure. Second, how close the colony is located to an area that's treated with neonics. This varies by time and by region. Third, the dose of neonics that are present on the bees, in the pollen, in the wax and in the nectar of the hive. Low doses below sublethal impacts may not affect honey bee colonies, while higher doses can. The Xerces society has put together a table that shows the doses that cause sublethal and chronic impacts on honey bees. Depending on the neonicotinoid, it can take between 0.25 parts per billion to 12,500 parts per billion before negative impacts are observed. You can download this table at the end of the course in the downloadable resources section. So it's easy to see that in some cases, colonies are at risk of effects

from neonics more so than in other cases. Studies in the field have also documented that neonics can be quite persistent in the environment. They can be found in plants and soil for several months, or even years after being applied. Neonics that have leached from plants or seeds in the soil can even be taken up by other plants, including wildflowers that were never sprayed or treated to begin with. So, neonics and other pesticides are quite prevalent in the environment, though most often in low levels. And although we've been studying these pesticides for many years now, we're still in the early stages of understanding exactly how their presence fully impacts honey bee colonies, but we know that it is a complex and interacting thing.

Bees Often Encounter Neonicotinoid Concentrations at Levels That Cause Sublethal and Chronic Effects

In 2020, Cornell University published an assessment of neonicotinoid risk to pollinators, summarizing the findings of field trials from over 400 published scientific studies across North America. The report, written by Grout, et al., sheds light on how frequently bees are exposed to neonicotinoid concentrations known to cause sublethal impacts in different agricultural landscapes. The authors estimated risk by using a metric called "lowest observed effect concentration," or LOEC. The LOEC is the lowest concentration of a chemical where scientists observe sublethal or chronic effects on an organism. Scientists have determined the LOECs of different neonicotinoids for a variety of responses in honey bees, such as impacts to physiology, behavior, and reproduction. By comparing the concentrations that bees are exposed to in the environment to their respective LOECs, the report authors determined the risks that different landscapes pose. The analysis found that honey bees are commonly exposed to neonicotinoid concentrations that can cause sublethal and chronic impacts. The most evidence exists for field crops (corn and soy). Data from 96 field crop assessments indicate that 74% of neonicotinoid exposures are at concentrations above those known to impact honey bee physiology, 58% are above those known to impact honey bee behavior, and 37% are above those known to impact honey bee reproduction. This means that the majority of exposures to neonicotinoids in corn and soy fields are risky; the usage of seed treatments are likely to impact bee physiology in three-quarters of cases, behavior in over half of cases, and reproduction in over one-third of cases. The risks for fruit crops, vegetables, and landscape management were also calculated; they are included in the table below. You'll notice that risks are higher in vegetable crops and landscape plants but lower in fruit crops. The take-home message from this report is that honey bees are frequently exposed to concentrations of neonicotinoids that can result in sublethal and chronic impacts, though the risk varies depending on the agricultural landscape.

Neonicotinoid exposures to bees in different agricultural landscapes compared to LOECs for sublethal and chronic effects:

Exposures	Percent of landscape above the LOEC for physiology	Percent of landscape above the LOEC for behavior	Percent of landscape above the LOEC for reproduction
Field crops	74%	58%	37%
Vegetable crops	88%	75%	54%
Ornamentals, turf and landscape management	92%	88%	72%
Fruit crops	50%	38%	17%

WATCH

Neonicotinoid Encounters Cause Sublethal and Chronic

Effects ■ In 2020, Cornell University published *Key Points* an assessment of neonicotinoid risk to pollinators, summarizing the findings honey bee health in some regions of field trials from over 400 published and under some conditions. scientific studies across North America. The report, written by Grout, et al., sheds light on how frequently bees are exposed to

neonicotinoid concentrations known to cause sublethal impacts in different agricultural landscapes. The authors estimated risk by using a metric called “lowest observed effect concentration,” or LOEC. The LOEC is the lowest concentration of a chemical where scientists observe sublethal or chronic effects on an organism. Scientists have determined the LOECs of different neonicotinoids for a variety of responses in honey bees, such as impacts to physiology, behavior, and reproduction. By comparing the concentrations that bees are exposed to in the environment to their respective LOECs, the report authors determined the risks that different landscapes pose. The analysis found that honey bees are commonly exposed to neonicotinoid concentrations that can cause sublethal and chronic impacts. The most evidence exists for field crops (corn and soy). Data from 96 field crop assessments indicate that 74% of neonicotinoid exposures are at concentrations above those known to impact honey bee physiology, 58% are above those known to impact honey bee behavior, and 37% are above those known to impact honey bee reproduction. This means that the majority of exposures to neonicotinoids in corn and soy fields are risky; the usage of seed treatments are likely to impact bee physiology in three-quarters of cases, behavior in over half of cases, and reproduction in over one-third of cases. The risks for fruit crops, vegetables, and landscape management were also calculated; they are included in the table below. You’ll notice that risks are higher in vegetable crops and landscape plants but lower in fruit crops. The take-home message from this report is that honey bees are frequently exposed to concentrations of neonicotinoids that can result in sublethal and chronic impacts, though the risk varies depending on the agricultural landscape. Neonicotinoid exposures to bees in different agricultural landscapes compared to LOECs for sublethal and chronic effects: Neonicotinoid Exposures to Bees in Different Agricultural Landscapes Percent of Percent of Percent of Agricultural exposures exposures exposures landscape above the LOEC above the LOEC above the LOEC for physiology for behavior for reproduction Field crops 74% 58% 37% Vegetable crops 88% 75% 54% Ornamentals, turf and landscape 92% 88% 72% management Fruit crops 50% 38% 17%

READ

Herbicides

■ Herbicides are the most commonly used class of pesticide in conventional agriculture. In general, most herbicides do not pose a major risk to honey bees. By far, the most frequently applied herbicide throughout the world is glyphosate (the active ingredient in Roundup). Glyphosate can indirectly impact bees by limiting floral resources. Glyphosate became commercially available in 1974 and is a very effective weed killer. In fact, it kills all vegetation to which it is applied, so it was originally only applied to areas where growers and landowners wanted to eliminate all plants. In 1996, Roundup Ready glyphosate-tolerant crops were developed. Since then, glyphosate could be applied onto these crops; weeds are killed, but the crops are genetically engineered to be unaffected by the herbicide. Roundup Ready cotton, corn, soy, and canola became common worldwide, and as a result glyphosate use skyrocketed. In 1994, the U.S. used 124 million pounds of glyphosate. By 2014, the usage increased to reach 1.82 billion pounds! Glyphosate has been detected in air, soil, water, and food products, including honey. Concerns have arisen about the prevalence of this pesticide in the environment and in human food; in 2015, the World Health Organization classified it as a “probable human carcinogen.” Not much research has been conducted to understand the risk of herbicides to honey bees. Of all herbicides, glyphosate is the most researched, likely because of its prevalence.

Overall, this herbicide is classified as practically non-toxic to honey bees. The LD values 50 are greater than 100 micrograms/bee, and while glyphosate is very prevalent in the environment, levels this high are not commonly detected in bee hives. Of the very few studies that have investigated the sublethal impacts of glyphosate on bees, these studies find conflicting results. One small-scale study determined that sublethal levels of glyphosate could impair bees' ability to return home after foraging trips, while another study found no effect on bees' ability to forage but demonstrated that learning can be slightly impaired. The latest paper published in 2018 reported that glyphosate can alter adult bees' beneficial gut microbiome. On the other hand, several other studies report no negative impacts of glyphosate exposure on survival or development. The actual concentrations and doses of glyphosate bees are exposed to in the field is not well documented. Glyphosate testing is not commonly done on honey bees because its analysis requires a different (and expensive) machine as compared to most other pesticides. Without more research to determine the levels that honey bees experience in the real world and more studies investigating the sublethal impacts of these realistic doses, it is too soon to say with confidence whether exposure to low levels of glyphosate in the field can cause effects at the colony level. Most other herbicides are classified as non-toxic to honey bees as well. After glyphosate, atrazine is the second-most commonly used herbicide in the U.S., followed by metachlor. LD values are high for these two pesticides and, as such, they do not pose a risk to honey 50 bee mortality. These herbicides frequently turn up in honey bee hive products. Research conducted by the NYS Beekeeper Tech Team in 2016 found these herbicides to be common at very low levels in honey bee wax; atrazine was detected in 17% of colonies sampled, while metachlor was detected in 58% of colonies. Another study conducted by Dr. Scott McArt at Cornell University found similar results: Low levels of atrazine were present in the bee bread of 20% of colonies tested. Despite their environmental prevalence and their occurrences in bee hives, these particular herbicides have not been studied with regard to their potential sublethal impacts on bee health. Some less-common herbicides are highly toxic and include language on the product label that emphasizes not to apply them when honey bees are foraging. If honey bees come in direct contact with these herbicides when wet, they can be toxic. However, after the product has dried and no longer presents a risk of residual toxicity, there is low risk to honey bee health. Indirect Effects on Bees While there is not strong evidence to suggest that exposure to herbicides directly harms bees, herbicide use can affect resource availability for honey bees and wild bees. Weeds and wildflowers surrounding agricultural, urban, suburban, and commercial land provide valuable food sources for nearly all species of bees and nesting sites for wild bees. While no study has investigated whether the increased use of specific herbicides correlates with a reduction in certain bee populations, habitat loss has been identified as a primary driver for native bee declines worldwide. Furthermore, the increased use of glyphosate is suggested to be a contributing factor to the decline in monarch butterflies, as its use is correlated with decreased milkweed abundance in agricultural fields in recent years.

WATCH

Fungicides

■ Fungicides control fungal diseases that Key Points infect a variety of vegetables, fruits, and EBI fungicides can harm bees by grains. Because several fungal diseases increasing the toxicity of other of crops commonly occur during pesticides. bloom, many fungicides can be sprayed while these plants

are flowering. Bees in these crop systems can easily carry fungicides back to the hive. In fact, next to miticides applied by beekeepers to control Varroa, residues in hives are predominantly fungicides. These residues have been detected in pollen, bee bread, wax, and honey. Despite the prevalence of fungicide use in conventional agriculture, scientific research is only beginning to uncover how they may affect bees. Most fungicides are not acutely toxic to honey bees. Often, adult bees that come in contact with fungicides do not die, unless levels well exceed those normally found in landscapes. There are a few, however, that can be toxic to other life stages of honey bees. For instance, larvae fed field-realistic doses of the fungicides captan, iprodione, and ziram all fail to develop to adulthood in the laboratory. Honey bee colonies used to provide pollination services for multiple crops are at the biggest risk of exposure to fungicides, simply because they are transported from one crop to another, usually at times when fungicides have been recently applied. One study conducted in California found this exposure can affect the quality of bee bread fed to larvae. The researchers sampled bee bread from colonies that provided both almond and apple pollination in the same year, and they found that bees incorporate fungicide residues into bee bread at high enough levels to alter its composition. Specifically, the presence of fungicides kill the beneficial fungi that are naturally present in bee bread and that help in the fermentation process. The downstream effects of larvae consuming this altered bee bread has not yet been studied. Another interesting finding that deserves further research is that a positive correlation exists between the presence of fungicide residues in honey bee hives and the occurrence of colony disorders (population weakness, queen loss, and brood problems); colonies that experience these disorders tend to have more fungicide residues in their hives than healthy colonies. The exact reason why is not yet known. Watch Scott McArt — assistant professor of pollinator health in the Department of Entomology at Cornell University — discuss how his lab has been studying fungicides for the past few years. Video Transcript My lab has been studying the impact of fungicides on bees for the last three to four years now. And what I'm going to share with you over the next five minutes or so are two of the studies that we have been pursuing in our lab. So, the first study, I think, really encapsulates what a lot of other people are finding, other researchers are finding and in terms of what the bees are actually bringing back to their colonies. So, what we did for this particular study is we, this was motivated by beekeepers who were doing pollination services, really throughout the US, but mostly from New York beekeepers. And most of those beekeepers were coming to us and saying, "You know, when we put our bees in to do pollination, they really seem to be impacted by pesticides." But not only that, most beekeepers, recently who I've been talking to say, "Listen, we're really concerned about fungicides in particular. We think that the bees are getting exposed to a lot of fungicides, and that might be having some impacts on them." So, for this particular study, what we did is we took 120 experimental honey bee colonies, and we put them into 30 different apple orchards here in New York. So, apple is our major pollination-dependent crop here, and where a lot of the beekeepers are providing those pollination services. We used 30 different orchards because we really wanted to get a good picture, not just of a few different orchards, but what about the breadth of all, you know, the pesticide exposure that a lot of beekeepers might actually be experiencing. The major result from this study, along the fungicide lines of things, is that 94 percent of the pesticide residues actually were fungicides. So this is probably not surprising for a lot of people who may be putting their bees into these crops for pollination, especially apple. Apple scab is the major pest of, the major fungal pathogen of apple, and growers are actually spraying fungicides during the bloom period. So when the bees are

actually out there, not surprisingly, the bees are bringing a lot of fungicides back in as a result of this. Okay, so the next study, along the lines of fungicides that I'll summarize is something that we did recently related to four declining species of bumblebees across the United States. So, this is a really impressive data set. Luckily, I didn't actually take the data on, I was able to borrow some data from someone. But what these researchers did is they went out to 284 sampling locations across the United States, sampled 10,000 bumblebees, and this included four species that are known to be in decline. So these are experiencing range contractions, and this includes the rusty patched bumblebee, which is now on the federal endangered species list. So across all of those different sampling sites and all of those different bees, believe it or not, people actually dissected all 10,000 of these bees and found out whether they had *Nosema bombi*. It's a related pathogen to *Nosema ceranae*, which we find in a lot of honey bees. So, what we were able to do from this data set is we were able to say, "Okay, here's the prevalence. So how many bees are actually sick?" But because we had all of these different sampling locations across the United States, we could say, well, what are the different attributes of those locations which may have actually caused the bees to become sick and potentially caused them to be lost from sites where they used to be. So, over particular regions where we know that the bumblebees used to be, but now they no longer exist in those particular areas. So what we did is we looked at lots of different factors. We looked at things like habitat fragmentation, things like latitude, longitude, elevation, urbanization, the human population density around those areas, the type of agriculture that was actually used in those areas. And then we also used a new database where we can actually make good predictions of pesticide usage across the whole United States. And that was applicable to every single one of these sites as well. So we can get good estimates of all 456 compounds of pesticides that were potentially sprayed across these locations. So as you can imagine, this was a large analysis with a lot of data. So we used some sort of fancy statistical techniques to look at this. But what we found in the end, is that the best predictor of *Nosema* prevalence, how many bees were actually getting sick at these sites, was not percent agriculture, it was not neonics, it was not human population density. It turns out it was the single most used fungicide in the US, chlorothalonil. So chlorothalonil was the most predictive of whether the bees were getting *Nosema* in them or not. We also know that in these four declining species they have elevated levels of *Nosema* compared to the species that are not necessarily doing so bad. Whether that's causal or not we don't know, but we do see that pattern. So the bees are getting sick where there is a lot of this fungicide chlorothalonil sprayed and there are more sicknesses in these declining species. The next thing we looked at was what was the best predictor of range contraction. So, whether the bees were actually lost from those sites, or whether they were still present in those sites. And, low and behold, what we find after we look at all these different factors again, in terms of what's predictive of loss of the bees, those declined species from the sites, is total fungicide usage in the United States. So again, I think if we were sort of to tie all of this data together, we find with the honey bees in apple orchards and a lot of other studies looking at similar things as we did that fungicides are the main things that our bees are bringing back to their hives. This is true for honey bees, this is true for bumblebees and we actually know for some solitary bees as well. The fungicides are the main things that they're being exposed to. In addition, now we have some data showing that fungicides may not necessarily be as benign as we think. There could be interactions with insecticides that could increase the toxicity of those insecticides, or there could be interactions with pathogens. So specifically, this *Nosema* pathogen that we found was at elevated levels and related to

fungicide exposure. We also have data, manipulative experiment data, in the lab with honey bees. So, Jeff Pettis showed several years ago that if you expose bees to, or you feed bees pollen that contains chlorothalonil versus control pollen that does not contain chlorothalonil, those honey bees that are exposed to the chlorothalonil are more susceptible to Nosema. Why exactly that is the case, we don't know. But it's something that we're looking into and hopefully something that we're going to be able to pass on recommendations to hopefully improve this situation with risk about fungicides in the future.

READ

Fungicides — Synergistic Effects

■ Laboratory studies have shown that a Key Points certain class of fungicides can be quite Entombed pollen is associated dangerous to honey bees when found with high levels of fungicides. in combination with other pesticides. This class is the ergosterol biosynthesis-inhibiting (EBI) fungicides. Several studies have found that these fungicides exhibit synergistic effects when found in combination with pyrethroid insecticides, neonicotinoids, and even some Varroa miticides. The most well- studied EBI fungicide in this regard is prochloraz. This particular fungicide is not registered in the U.S., but there are several other EBI fungicides that are. Laboratory studies find prochloraz can substantially increase the toxicity of the pyrethroid insecticide deltamethrin. When both pesticides are fed to honey bees at sublethal doses, mortality is higher than when bees are fed either pesticide alone. The combination is not always fatal but can have other negative impacts on bee health as well. For example, the joint exposure of prochloraz with pyrethroids deltamethrin or difenoconazole impairs bees' ability to regulate their body temperature. If this exposure occurs in cool climates, bees may be susceptible to hypothermia. In addition to pyrethroids, EBI fungicides increase the toxicity of miticides. Prochloraz elevates the toxicity of tau-fluvalinate (Apistan®), coumaphos (CheckMite+®), and fenpyroximate (a miticide registered for use on some crops but not for Varroa control). This effect was not seen with amitraz (Apivar®) or thymol (Apiguard® and Api Life Var®). These synergistic effects are especially concerning given the prevalence of miticides and fungicides together in honey bee hives. Lastly, EBI fungicides can also increase the toxicity of neonicotinoids. Imidacloprid and acetamiprid become especially deadly when found in combination with the EBI fungicide fenbuconazole. The neonicotinoid thiacloprid increases greatly in toxicity when bees are also exposed to the EBI fungicides prochloraz and tebuconazole. These observations are from laboratory studies but have not yet been documented in the field. For instance, one study investigated colony effects due to the combination of thiacloprid and tebuconazole and found no impacts on mortality, foraging, colony growth, brood production, or honey production. More field studies examining potential synergistic effects are critically needed. Studying pesticide interactions is very important because it is rare for honey bees to encounter only one chemical at a time. The environment, and therefore bees' hives, tend to test positive for several different pesticides at once. Why Do EBI Fungicides Increase the Toxicity of These Other Pesticides? When pyrethroids and neonicotinoids enter the honey bee body, they are broken down in part by an enzyme called cytochrome P450-monoxygenase. This enzyme is important for honey bees to detoxify chemicals from their bodies. EBI fungicides work by stopping the action of cytochrome P450-monoxygenase in fungi. When these fungicides enter honey bee bodies through contact or consumption, the same enzyme is inhibited in the bees' bodies! Now, when bees are exposed to pyrethroids and neonicotinoids, they are unable to break them down and clear them from their bodies.

In this way, low levels of these insecticides can build up and become very toxic. Entombed Pollen In 2009, Dennis vanEngelsdorp's lab discovered a strange phenomenon in some U.S. honey bee colonies: entombed pollen. Entombed pollen has a dark, sunken, wax and propolis capping. Often there are several cells of entombed pollen scattered among normal-looking bee bread and brood cells. When the researchers scraped off the capping to reveal what was beneath, the pollen was brick-red in color. Laboratory analyses found that every sample of entombed pollen had high levels of the fungicide chlorothalonil (on average, 40 times higher than in regular pollen!). This fungicide is commonly detected in honey bee colonies. After following these colonies over time, they discovered those with entombed pollen died at significantly higher rates than those that did not. It appears the bees' sealed off the pollen, perhaps due to the high fungicide levels, to avoid it. Entombed pollen tends to be more commonly found in older wax combs, possibly because old wax tends to accumulate more pesticide residues than younger wax. Left photo shows a close up view of cells containing entombed pollen; middle photo shows a frame containing many affected cells; and right photo shows an entombed pollen cell with capping scraped, exposing brick red pollen. Photo credit: Bee Informed Partnership Entombed pollen is fairly common today. In spring 2017, the NYS Beekeeper Tech Team found entombed pollen in 11.5% of colonies they sampled across the state (37 out of 321 hives). This entombed pollen was a variety of colors. Beekeepers should keep toothpicks in their inspection kits to scrape off suspicious wax and propolis cappings to determine the presence of entombed pollen. If beekeepers observe many cells of entombed pollen in their hives (5–10% of cells), it is recommended that they discard the affected frames and replace them with frames of foundation or newly drawn comb.

READ

Reducing Bees' Exposure to Pesticides

■ Throughout this module, we have Key Points examined how pesticides are prevalent. Ultimately, reducing pesticide in the environment. Low doses of some exposure in honey bee colonies pesticides have the potential to cause requires collaboration among sublethal impacts on honey bee beekeepers, growers, pesticide health, and the interactions among applicators, the public, and policy pesticides common in bee hives have makers. been largely unexplored. Acute Beekeepers can reduce miticides pesticide poisoning events occur when in their colonies by using honey bees are in areas where integrated pest management. pesticides are being applied or are foraging on recently treated plants Communicating with landowners within the residual toxicity time period. in the area and rotating old combs are good practices to reduce Beekeepers may feel powerless when pesticide exposure. it comes to reducing pesticide exposure. They may have little control over the proximity of their apiary sites to agricultural fields, and providing pollination services may be a focal part of their beekeeping business. Losing bees from acute pesticide poisoning events is a devastating experience for everyone. While it is nearly impossible to prevent bees from being exposed to pesticides, there are several measures beekeepers can take to reduce the degree of exposure. Protecting pollinators from pesticide exposure involves beekeepers' cooperation with a variety of groups of people, including pesticide applicators, growers, and the public. The importance of communication among these groups cannot be overstated. Combs Many pesticides and their residues (including some Varroa miticides) accumulate in wax over many years. Residues can be slow to break down, causing exposure to bees for months or even years after the pesticide was first brought into the hive. To reduce contaminants in hives,

beekeepers should rotate out two of the oldest frames from each hive body every year. If beekeepers experience a pesticide poisoning event whereby brood, nurse bees, and/or the queen is affected, it is likely that contaminated pollen was brought into the colony (because the foragers did not die immediately in the field). In these cases, beekeepers may want to promptly remove all comb with bee bread/pollen and replace it with drawn comb. Because pollen usually surrounds brood, brood may be discarded in this process. Beekeepers can make this decision based on the severity of the pesticide poisoning. In severe cases, where hundreds of bees continue to die each day and the queen does not resume egg laying after several days, it may be worth it to discard the frames, brood and all. In mild cases where only some adults and brood are affected, but the effects are not continuously observed over consecutive days, it may not be necessary.

Miticides Nearly all studies that quantify pesticide levels in beeswax and pollen find that *Varroa* miticides are the most frequently detected and contribute the highest residue levels in hives. While most beekeepers need to apply miticides to their colonies to control *Varroa*, practicing integrated pest management (IPM) is the best way to minimize miticide residues. By incorporating genetic and cultural controls, fewer pesticide applications are needed. IPM does not encourage the prophylactic use of miticides; instead, they are only applied when mites exceed a treatment threshold. Beekeepers can further aim to reduce pesticide residues by using those with short residual periods in beeswax first (essential oils, organic acids, and amitraz) and by limiting (or eliminating) their use of fluvalinate or coumaphos- based chemicals, which can remain in comb for years.

Pollination It is very likely that the risk of pesticide exposure is higher for colonies that pollinate multiple crops in a single year as compared to those that pollinate no crops or only a single crop, though no studies have specifically investigated this topic. The idea is that with each new crop, bees are exposed to a new cocktail of chemicals. In 2019, the NYS Beekeeper Tech Team found higher pesticide residues in commercial colonies compared to hobbyist colonies, though it is not clear whether this is driven by pollination services, comb age, or other management practices. To reduce exposure to the multitude of pesticides bees face when pollinating a variety of crops, beekeepers may wish to be strategic about which colonies they use for pollination and how many pollination events they work. For instance, beekeepers could limit pollination service events for individual colonies. If they manage 100 colonies for pollination, 50 could be used to pollinate apples, while the other 50 could be used to pollinate squash. Of course, to determine the strategy that works best for their operation and goals, the beekeeper will want to weigh the costs and benefits of renting fewer colonies with the stress the colonies can incur and with the income they will generate through providing these services. An alternative scenario could be to specifically manage colonies for a certain purpose. Beekeepers could achieve this by keeping a group of colonies that are maintained in low-pesticide environments for honey production or for reserves (building comb or rearing queens for later use) and have another group of colonies that are used for the riskier task of pollination services. Pulling resources from the healthy colonies to use in the pollination colonies can help mitigate losses. This method of avoiding “putting all your eggs in one basket” may help reduce the overall operation’s pesticide exposure. If colonies do experience pesticide stress or die from acute poisoning events, beekeepers have a reserve of strong, healthy colonies from which they can recuperate losses. Colonies that are used for pollination should be strong and healthy, and they should receive nutritional supplements (carbohydrates and/or protein) before, during, and after, as needed. In this manner, pollen substitute and sugar syrup can provide a clean, uncontaminated food option for bees to consume. Colonies that

come out of pollination can be placed in wildflower fields for the remainder of the year to have the opportunity to forage on a variety of plants with minimal pesticide exposure and recover. Contracts Pollination service contracts are a great way to communicate expectations between beekeepers and growers. They allow beekeepers and growers to determine up front the pesticide risk to honey bees, as well as what would happen if a poisoning event were to occur. Beekeepers and growers can negotiate which pesticides can and cannot be applied to crops while bees are on site, how much notice beekeepers require before pesticide application, and how much a grower will compensate a beekeeper for pesticide poisoning events. These contracts can reduce pesticide exposure to honey bees, help form trusting relationships between growers and beekeepers, and put both parties at ease before any services are performed. Reporting Beekeepers are encouraged to report all pesticide poisoning events to the state apiculturist as soon as they are experienced. They should take photos and videos of the bees at the entrance of the hive, as well as photos of any affected brood. Additionally, they should photograph and document the crops that are blooming in the area at the time of the event.

Investigators will interview the beekeeper and sample dead or dying bees from the affected colonies for pesticide residues. These bees must be as fresh as possible and not yet experiencing any decay. Remember, many pesticide residues can break down after several days, so the quicker the beekeeper acts to report the event, the better. Beekeepers are discouraged from taking any bee samples themselves before reporting the incident. Technicians will advise them whether or not to collect their own samples to store before the investigation occurs. In addition to dead bees, the inspector may also take samples of pollen, wax, or any pesticide dust at the hive entrance. The inspector will then explore nearby sites of possible pesticide use (such as nearby agricultural fields, ornamental gardens, utility lines, and roadside drainage ditches) in an effort to determine whether pesticides were applied illegally (i.e., not in accordance with the product label). The inspector may interview landowners and take samples of vegetation, soil, or water for pesticide analysis. Pesticide samples taken by the investigators are analyzed free of charge, but it may take several months before results are provided. Alternatively, beekeepers can send in their own samples for a fee. The Bee Informed Partnership offers this Emergency Response Kit that includes testing 16 hives for over 170 different pesticides. The cost for this kit is \$1,229. The turnaround time for results is much quicker than waiting for the state government results. The U.S. used to have programs to compensate beekeepers for colony losses from pesticide poisoning events, but no such programs exist today. Unfortunately, beekeepers are the ones who bear the cost for these tragic situations, even though the incidents are nearly always beyond their control. However, there is still a benefit to reporting pesticide kills. Records of pesticide poisoning events can help uncover trends as to which pesticides are causing the most acute harm to bees. Ontario, Canada's 2015 decision to regulate neonicotinoid seed-treated corn and soy was partially as a result of the number of acute pesticide poisoning events that were reported after seed planting. In 2012, pesticide kills in 240 yards were reported. Investigations found 70% of dead bees tested positive for neonicotinoid residues. In 2013, a total of 340 yards were reported, with 75% of dead bees testing positive for neonicotinoids. Without these reports, there might not have been a strong enough case for the new regulations. Pesticide poisoning events are underreported in the U.S.. Some beekeepers feel that without government compensation for their losses, there is no point to starting an investigation. However, reporting losses and investigating the pesticides responsible can lead to changes in pesticide regulations. Talking to Growers In addition to drafting pollination contracts with

growers, simply communicating with growers in the area can make a big impact. Beekeepers should be friendly and open with farmers and let them know where their bee hives are located. Growers and pesticide applicators cannot be expected to avoid spraying pesticides near honey bee colonies if they do not know where they are. Some states and countries have mandatory registration where beekeepers can be contacted and notified before pesticide applications, and others use an online program called FieldWatch®, where growers are able to get in touch with beekeepers prior to applying pesticides. In states and countries without such programs (like New York), it is the beekeeper's responsibility to make their colonies known. Beekeepers should provide nearby landowners with their contact information and print their contact information (in big letters) on signs near hives in high-risk areas so that they can be seen from a distance. Bringing nearby growers a jar of honey with their contact information on it is a great way for a beekeeper to start the discussion. Generally, growers want to protect bees. They are aware of the benefits they bring ecosystems as well as the crop quality and yield improvements that result from bee pollination. Ask them what pesticides they plan on applying this year and whether any of the labels include precautionary statements for bees. Many are willing to adopt integrated pest management practices to reduce pesticide use in their fields. Others are willing to use alternative pesticides that are delivered in a formulation safer for bees or that have shorter residual toxicity periods. Developing friendly relations and regularly touching base with growers helps to facilitate these open conversations. Growers may be more likely to keep honey bees in the backs of their minds when planning for their next growing season. Moving Hives Moving colonies out of agricultural areas to accommodate pesticide applications is a contentious issue. Many beekeepers feel that having to move their colonies puts the onus of pollinator health on beekeepers instead of on growers, pesticide applicators, or landowners. Often the notification period for informing beekeepers of a pesticide application is 48 hours. This can be a very short time period for beekeepers to pick up their colonies and move them, especially if they are in another state or province (which is sometimes the case, especially among commercial beekeepers). It can also be incredibly tricky for beekeepers to find alternative safe land on which to temporarily move their colonies. These barriers make it difficult to assume every beekeeper can move their colonies out of pesticide areas. Some beekeepers, however, may be able to move their colonies; if they are able to, they should. Moving colonies at least four miles from the area that will be treated can make a difference with regard to a pesticide poisoning event. Beekeepers should not return the colonies until at least 72 hours after the pesticide has been applied (or longer if the grower informs them of specific label language for residual toxicity for bees). Beekeepers should consider establishing "holding apiaries" for bees in advance so that they have a location to move colonies before a pesticide risk event occurs. Talking to the Public Today the public is more aware of pollinator health than ever. Many homeowners actively plant pollinator gardens and want to do their part to help the bees. Even home and garden stores have felt the pressure from the public to sell plants that are free of certain pesticides. Home Depot, Inc., for instance, provides tags on their plants to alert customers as to which ones have been treated with neonicotinoids and plans to phase them out entirely. This is an excellent first step that demonstrates that the public cares about pollinators and has the power to drive change. Beekeepers can also inform the public where to purchase plants. Studies from Europe, the U.S., and Canada find that most plants in garden stores labeled as "bee friendly" are actually sprayed with insecticides and fungicides — the label simply means that they are a flower that bees often visit and from which they feed. Instead of purchasing

these plants, beekeepers can inform people of the benefits of purchasing plants from an organic nursery. Beekeepers can even help change the public's perspective on weeds to regard them as valuable bee forage instead of an eyesore. Through these interactions with neighbors, customers, and local gardeners, beekeepers can make a difference in whether or not the public applies pesticides to their lawns and gardens.

MODULE WRAP-UP

Confidence in Communicating About Pesticides

In this module, you explored pesticides and how they impact honey bees. You also examined the controversy about neonicotinoid pesticides as well as what recommendations you can give the public and other beekeepers to help reduce pesticide use and exposure to bees.

Turning Bees Into Business

As a beekeeper, you have by now deepened your knowledge and understanding of honey bees and explored management strategies to maximize their health and productivity. As a Master Beekeeper student, you have likely also successfully implemented much of your learning into good practice and have possibly seen an increase in colony survival. At this point, many hobby beekeepers are satisfied to sustain their colony numbers without having to purchase bees each year. They produce enough honey for their bees, for themselves, and for gifts to friends and family. Some hobbyists even produce enough to sell their honey to other local beekeepers' operations or have ventured into small-scale honey sales of their own. Other beekeepers are ready to grow beyond the hobbyist realm and into a legitimate business where the bees and their products are paying for themselves and making a profit for the beekeeper. The transition from hobbyist to business owner requires a shift in mindset about time management, expenses, and hive/yard management. These beekeepers work for their bees as well as for themselves. In this module, we will explore the factors beekeepers can consider when growing their operation and beginning a beekeeping business.

READ

Considerations When Growing Your Operation

■ Growing a beekeeping operation can Key Points certainly happen organically. Many Before starting a business, beekeepers have found themselves beekeepers should develop a clear starting out with a couple of colonies, goal and plan to reach it. and within three to six years having 20 Increasing a beekeeping operation colonies or more. When growing the slowly is less risky than purchasing operation, especially for a self- many new colonies at once. sustaining business, it is important to have a firm goal in mind. It can take years for a beekeeper to develop a sense of what they would like to do with their bees to turn a profit, and that's fine. However, when the decision is made to grow, there needs to be a goal. Without a goal, a plan forward cannot be formed, and the operation may be ill-equipped to handle issues that can arise in a bad year, such as low honey and wax yields, slow population growth, inclement weather in mating season, unexpected colony losses, and particularly harsh winter and spring conditions. Identifying the main goal(s) will help beekeepers to focus their

business on the facet of beekeeping that is most appealing to them. A beekeeper's business goal is often centered around the main product or service that sets the operation apart from others. Will the beekeeper be producing bees for sale; focus on queen rearing; operate as a contract pollinator; offer bee retrieval and cutout services; produce specialty honey for a specific market; sell bulk hive products like wax and honey; or create value-added products such as soaps, balms, and mead? Choosing one or a few products is not meant to limit the operation and prevent the beekeeper from offering all of these products or services someday, but making this choice is the start of understanding how to structure the business around a central product so that it can generate enough revenue to sustain the business. This, in turn, allows the beekeeper the time and budget to expand their product lines and services as the business grows. With larger operations, time management is a critical skill to develop. Planning business and apiary tasks is best done in advance, and many beekeepers find it easiest to plan in the quieter winter months. Falling behind in management tasks one month can carry through to the next. For instance, not getting the colonies off to a good start in the spring affects management for the summer and fall. Not managing bees for optimized health going into winter sets the beekeeper's operation up for a difficult season in the year to come. So, although beekeeping seems to consist of discrete, year-to-year events, success each year is dependent on the year before. Beekeepers are encouraged to plan thoroughly and to schedule time points for conducting spring and autumn colony assessments, feeding, mite monitoring (and treating), honey extraction, checking inventory, equipment assembly, equipment ordering, and organizing records. Two main modes of business startup and growth exist: (1) organic/cumulative; or (2) aggressive expansion/fast track. If beekeepers are new to business ownership and relatively new to beekeeping, organic growth is the way to go, as it affords more room for error and the inevitable steep learning curve the beekeeper needs to overcome for success. Aggressive expansion is possible, but the margin for error is slim, and aggressive expansion requires ample amounts of funds as well as the realization that the business may be in debt for the first several years while startup costs are being paid off. The fast-track pathway should only be considered by entrepreneurs who are familiar with the ins and outs of business financing and operations. It requires extensive planning and a five- to ten-year business model that is written down and can be reviewed by potential investors and financial institutions. These facets of the fast-track pathway are outside of the scope of this course and require extensive business coursework, training, and field experience that can inform the process. We will instead focus on the organic growth model, where a beekeeper can learn good business practices as the business grows.

WATCH

Considerations When Growing Your Operation — Record

Keeping Just as keeping colony records is important for understanding and tracking the health of bees, keeping financial records is essential to understanding and tracking the health of the business. In this video, Mary Kate Wheeler, NYS Beekeeper Tech Team Agricultural Economist, discusses the financial records beekeepers should keep and offers advice on ways to make record keeping convenient. Video Transcript Keeping good financial records is super important for beekeeping businesses. A lot of beekeepers put a lot of time and energy into keeping really good production records, which help them to understand the health and productivity of their colonies, and it's just as important to keep really good financial records, because that allows you to understand the health and the viability of your beekeeping

business. Good record keeping also helps beekeepers to make better decisions about growth opportunities and other management decisions in their business. And it also helps to communicate the financial position of your business to other people, like lenders, accountants, and other service providers. So, there are four different types of financial records that you're going to want to keep in your in your beekeeping business. The first two, most people are pretty familiar with, and often they do a good job of tracking these. So, these are your revenues, business revenues and expenses. Revenues reflect all of the income that your business generates through your normal operations, so, through your sales of products or services, that money coming in is the revenue of your business. And expenses, on the other hand, reflect all the costs of doing business. So, when you're purchasing supplies and materials, or when you're purchasing larger pieces of equipment that your business needs, those are all examples of expenses. And most people do a great job tracking these because it's actually required at the end of the year by the IRS when you're filing your tax return. So the difference between your business revenues and your expenses is the profit that your business earns. And that is typically calculated over a period of time. So, normally people are looking at their business profits over the course of a calendar year. And that's what gets reported to the IRS when you file your tax returns. So, that's revenues and expenses. And the other two types of financial records are called assets and liabilities. So, assets reflects, well, assets are resources. They're productive resources that the business actually owns and uses to generate value or to generate income through the course of your operations. So, an asset, livestock is a type of asset, so your honey bees are actually an asset of your business. Supplies that you have on hand are an example of an asset. The cash balance of your checking account and your business savings account, those are assets. As well as buildings, and property, and larger pieces of equipment and machinery. So these are all different types of assets that your business is using to be productive, to produce honey or bees for sale. So, on the other hand, your liabilities are obligations, like debts or financial obligations that the business has that have to be paid in the future or have to be met in the future. So, examples would be a mortgage, loans like a vehicle loan or an operating loan. Even your credit card balance is a short term liability. And another type of liability happens if your customer pays you for a product or service that you haven't delivered yet. So that also becomes a liability because it's an obligation that your business has to meet in the future. So when we think about that relationship between assets and liabilities, we think about another accounting concept called equity. So, equity is the share of the business that actually belongs to the owner. And the relationship there is your equity is equal to the total business assets minus the liabilities. So that's like saying all of the resources that the business owns, minus all of its financial obligations. And you definitely want to see over time your equity increase in the business. So it's important to track these things consistently and to track them over time. Equity is a little bit different from profit, which we talked about, because equity is measured at a specific moment in time. We looked at profit over the course of a year, but your equity is typically measured at least annually, and it's often done on January first because that represents the ending of your last year's financial cycle and the beginning of the new cycle. So what we recommend is doing an inventory every year right around January first, where you actually go through all of the assets that your business owns, you track, you count those things and figure out their value, and then at that time you also take stock of the business liabilities, the financial obligations. And taking the difference between those two things gives you your equity at that January first date. So tracking these things over time and keeping your financial records is going to help you to

make better decisions about taking advantage of opportunities and growing your business in the ways that work best for you. So we've talked to all the different kinds of financial records you're going to want to keep, but then the next question is, "Well, how I actually track these things," right? And so there's a couple different ways to track financial records. It's important to pick a system that's going to work for you. You don't want to pick a system that's going to be difficult for you to use because then you're going to, it's going to be really easy to fall behind in your record keeping. So, some people use a really simple system, pencil and paper. You can find these accounting booklets that are structured to provide room for you to record all your business transactions, so all your purchases and sales over the course of the year. So if you want to, you can keep records that way. Some people use Excel and actually generate spreadsheets for tracking all the business transactions and assets and liabilities. And the benefit there is that Excel can do a lot of the math for you, it can actually calculate your totals at the end of the year, and it's a little bit more powerful tool for doing analysis along the way. And then a lot of folks also use QuickBooks, which is an accounting software that's very powerful, it's much more sophisticated, and it can also be used to track your transactions throughout the year. And a benefit of QuickBooks is that it can be, make it really easy to share your financial data with accountants and other service providers that are going to help you either put together your reports or do financial analysis. So I hope that helps you, gives you some things to think about in terms of what to track, how to track it, and how that investment, investing your time in record keeping is going to help you to be a better beekeeper and make better business decisions. To assist beekeepers with record keeping, Mary Kate Wheeler developed this step-by-step Inventory & Financial Analysis Handbook for owners and managers to inventory their business and construct financial statements; feel free to download it for your own use.

TOOL

Inventory and Financial Analysis Handbook

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READ

Building Your Business

Business growth happens in three main phases: startup, market establishment, and stabilization. Let's explore each in detail. Startup The startup phase consists of planning the business: exploring the market for the main product, researching competition, developing the business strategy, differentiating products from competitors, and formation of the business as a legal and taxable entity. In other words, the startup phase requires a lot of research and even more decision making. It is a very daunting phase; however, there are ample resources available to the public that make starting a business an achievable dream. For starters, the U.S. Small Business Administration offers resources, staff, mentors, coursework, and workshops to those looking to start a business. The USDA offers online resources for farmers and business owners in agricultural trades. In New York State, the Department of Agriculture and Markets is committed to aiding small agriculture businesses and farms, and can offer

protections and guidance to startups that are located within an "Agricultural District" (land that has been designated to be preferentially developed or maintained for agricultural use). New York also houses the organizations Groundswell (a farm/business incubator) and Cornell Cooperative Extension (CCE); both are dedicated to serving agribusinesses, especially those in the startup phase. For example, CCE in Tompkins County has partnered with local marketplaces to offer spaces to host small farm market events for farms that are not featured in established farmers' markets, which can have membership fees and practices that are high barriers of entry for startups. Beekeepers should explore business-supporting organizations and programs in their area before they start out, and they can reach out to agricultural extension personnel for resources. ■ In the startup phase, beekeepers should consider a growth rate that is scalable with the product. For instance, if the beekeeper is planning on gaining profits through honey sales, they should determine how much honey — and at what price — results in high-enough revenue for the beekeeper to pay themselves and to purchase more equipment and bees for growth the following year. It is suggested to only add 10–30 new colonies per year for the first two or three years. This growth rate allows for recovery of colony losses, low crop yields, and growing/grooming the market share or consumer base. This also enables the beekeeper to transition into new business and yard management strategies that stem from time limitations. There are many factors that go into placing a product in the market, but one key to successful deployment is packaging that is appropriate and appealing to the demographic the beekeeper is targeting. For instance, a commercial operation may only sell honey in bulk containers to other businesses, and therefore packaging costs can be limited to containers and simple labels. However, small honey producers in a competitive marketplace, such as a farmers' market, require an edge over other competitors and need to invest in packaging that both differentiates their product and reflects their brand and values (e.g., glass jars, labels that are informative and attractive, ribbons and accessory packaging such as bundle packs and baskets). The small business must be aware of how the buyer experiences the product, as well as the quality of the product itself. For a small business, marketing is akin to telling a story and inviting the consumer into the experience.

Market Establishment The second phase of business growth is market establishment. This is the experimental phase. No matter how much thought and planning goes into the startup phase, establishing a business and selling products in the market takes time. During this phase, beekeepers should seek ample feedback from consumers and businesses that are buying the product. Market events are invaluable in this sense, as they allow business owners to directly interact with consumers and solicit immediate feedback about their products. Events also allow owners to observe how consumers interact with the product and how they respond to the overall story. This phase is the best time for the business to react to customer feedback by refining packaging and storytelling, and ultimately establishing a successful and cohesive brand. This phase is both nerve wracking and rewarding. Often expenditures are on par with revenue, but mindfully investing in the business while refining the targeted market should yield steady growth in the following years. This is the time when pivoting is easiest and most frequent. Pivoting is making changes to the business model in reaction to the reality of what the business experiences in the marketplace as opposed to what was expected to happen. A business that does not react to the market, or adheres too rigidly to a plan, will have a difficult time entering the final phase in the growth cycle, or may fail outright and shut down. The final phase of the business growth cycle is stabilization. This phase is marked by steady adoption of product into the marketplace, climbing sales, ■ **Stabilization** ■ and a stable or consistent pattern of

revenue on a yearly basis. An indication that the business, brand, or product is entering the stabilization phase is the occurrence of repeat customers and orders. Word-of-mouth feedback (referral sales) is another indicator. These are exciting developments for any business, especially for a new business owner; however, they do not indicate that the work is done. The stabilization phase indicates the time when the beekeeper may be able to divert some time and effort from active sales outlets (marketplaces, events, etc.) to passive sales outlets (online, storefront, etc.), as well as considering whether or not it is time to offer more product lines. It is not advisable to consider diversification of products before the stabilization of the core business. More products equals more time spent away from the bees and more processing equipment and packaging in which to invest. If your business reaches this phase, give yourself a good pat on the back — you've earned it!

WATCH

Mindset and Management Changes

■ As beekeepers begin the path to Key Points business ownership, it is important to Adjusting from a hobby operation understand the changes in time to a large-scale operation requires management, expenses, and mindset changes in time management and that accompany growing the expense management. beekeeping operation. The saying Large equipment purchases are “time is money” becomes readily important investments to the apparent for owners as businesses business, and as such they should start up. Beekeepers are compelled to be thoroughly researched increase efficiency in the bee yard to beforehand. ensure all tasks are accomplished promptly and to make time to devote Delegating tasks to other people to other aspects of owning a business. and professionals as the business The changes that accompany grows allows the beekeeper to becoming a business owner call for focus on what they do best. beekeepers to reflect on their own personal strengths and to plan for how to address their challenges. Developing skills in time management and expense management reveals the areas of the business where the beekeeper's skill set is most valuable; this is the area in which they can have the greatest influence. If there are skills that the beekeeper is lacking, they can delegate tasks to other people as the business grows. Professional services, such as lawyers, accountants, insurers, marketing specialists, designers, and manufacturers, may be better equipped to offset a beekeeper's workload in these areas. Hiring help to manage office needs or processing and packaging products can be intimidating to a new business owner, but remember that a beekeeper's skill set best serves the business in the bee yard with the bees — increasing bee health, product yield, and population growth. In this video, Mary Kate Wheeler, NYS Beekeeper Tech Team Agricultural Economist, discusses the shifts in perspectives that beekeepers must undergo in order to run a successful business. Video Transcript Let's talk about some of the differences between being a producer and being a business manager. So, a lot of you folks, if you're taking this course, have considerable experience beekeeping and managing your hives and your apiaries. And what I want to highlight is that there's a really big change in perspective when you go from keeping bees as a hobby or as a fun activity, to actually managing a beekeeping business. So as a producer, your focus is really on the logistics of managing your apiaries and your colonies, producing really incredible products. So a lot of hobbyists are really product focused. Managing the health of your bees and keeping them healthy and happy. And a lot of the focus is kind of on the the present moment, really the experience of working with bees. So in contrast, when you move over to managing a business, you want to have more of an

entrepreneurial perspective. One of the key differences is that instead of focusing so much on the products, you really have to focus on the customer, understanding the needs and the desires of your target customer, and how you can deliver your product to them in a really unique and innovative way. Because that's what's going to create a unique value and a unique position for your business. So on top of that, an entrepreneur is thinking not only about production, but also about a bunch of other complex systems that the business requires to be sustained and successful. And the entrepreneur is also thinking ahead, thinking out into the future, and not getting lost in day to day operations, but pursuing kind of a strategy and a vision to be successful in the business. So, I want to tell you guys about this great book. It's like a business advice book for small businesses that was first put out back in the 80s, it's called the E Myth. So in this case, E stands for entrepreneurial, the entrepreneurial myth. And this book is based on the premise that there's a myth in our culture in America, that sort of says anybody can start a business. If you have a technical skill or area of expertise, that is easy for you to just go into business for yourself. Forget having a boss, you can run the show, you can be in charge and be successful. But the statistics don't actually back that up because a lot of small businesses don't succeed. And a lot really struggle, and many actually fail within the first few years. So this book, The E Myth, sort of explores some of the reasons for why that happens. One of the most interesting ideas in the book is this concept of the three business personalities. And so, the three are, there's a technician, the manager, and the entrepreneur. You can think about the technician as being a doer, a tinkerer, somebody who loves to get their hands dirty. That's solving problems, getting things done, and just loves to be immersed in their work itself. So the strength of the technician is really execution, getting things done. In terms of the manager, this is a person who's really organized, very pragmatic, practical, loves planning, thinking about systems, and creating structure and organization. So, if you're somebody that loves spreadsheets and record keeping, you might be a manager at heart. And the strength of the manager is bringing stability to the business. Helping to be sure that the business activities can be sustained over time. So the third personality, the entrepreneur, is the visionary. The big picture thinker. This person loves change, thrives in super chaotic environments, craves control, and can articulate a vision or concept that other people can rally around. And so the strength in a business setting of the entrepreneur is that this person can really drive the business forward and identify opportunities for growth. So the tricky thing, is that any business needs all three of these perspectives. And as a small business, you may just be one person in charge. Right? And so it's really important to have an awareness of your own personality and what your strengths and weaknesses are. And this framework is really helpful. Right? Because you can think about, are you more of a technician, a manager, or an entrepreneur? If you're one of these, or if you are most comfortable in one of these areas, can you build skills in another area and bring that into your business? And if not, who else can you bring in? Can you in an employee, maybe a partner or an adviser, that has a complementary perspective that can help you get all three of those perspectives in your business. So, now I want to talk a little bit about some of the business functions, core business functions, that, again, are different, or one of the major differences between beekeeping as a hobby and a fun activity, versus beekeeping as a business. So, we already talked a little bit about production, and that fits into what we call operations. So, operations is the first function of a business. And that really has to do with the logistics of how you're producing things, or products and services for your clients. And as beekeepers, you already have a lot of experience with this. So it's probably the area you're most familiar with in terms of

thinking about how to plan for your operations. So, the next thing, after you're producing something in a business, you have to think about how am I going to sell this? Who are my target customers? How am I going to get the product to them? What am I going to charge? How is it going to be packaged? And that all fits into the second core business function, which is marketing. So, marketing has a lot to do with connecting your product with your target consumer, and telling your story and communicating the value of what your product provides. Okay, so you've got your production and you get your sales, but the next question is how are you going to pay for all of this? So, how are you going to finance your production? How are you going to finance that marketing piece? And that brings us to the third function of a business, which is finance. And so that has to do with managing income and expenses, managing your cash flow so that you're never short. It has to do with projecting, budgeting, projecting your profits, and keeping track of all your financial records. So those are three core functions. On top of that, I also usually add management, sometimes as administration or even human resources. And the management function has to do with making sure all of the other functions are coordinated with each other, and it also has to do with additional business considerations. Thinking about things like legal issues, your business structure, regulatory issues, taxes, insurance, managing risk, managing people and labor in your operation. So those are all sort of comprise that fourth function, which is management. And then the final function that's critical for any business, long term, is leadership. And this function sort of encompasses all of the rest of them. So leadership requires one to step out of the day-to-day activities of the business and think about what is it that you really doing? What are your goals for the business? What's your vision? Where are you headed? And how well is your business currently meeting those goals? Is the business actually going in the right direction at the moment, or do you need to change course? And so, that leadership function is often easy to lose track of in the hustle and bustle of your day to day operations, but it's critical in terms of ensuring that the business takes you where you want to go in the end. So to wrap up, those are some of the key differences between beekeeping as a hobby and beekeeping as a business. Having the right perspective where you are thinking like an entrepreneur and focusing our customer needs. And also just keeping track of all the various business functions not just your production, but also your marketing, finance, management piece, and then trying to keep one eye on the future and on the big picture direction of your business. The necessity for record keeping and timely inspections increases exponentially as a beekeeper starts a business and their colony numbers increase. The days of being able to go into the yard only on the clearest, warmest days to accomplish tasks will be over. The business of beekeeping requires working the bees in less-than-ideal conditions on cloudy, windy, or even cool days. Business owners need to be even more selective and conscientious beekeepers. As a hobbyist, it makes sense to spend time trying to nurse a struggling colony along for a season. It is an informative experience, and therefore time well spent. From a business standpoint, it is important to realize the threshold where this is no longer the case. It often makes more sense to cull or merge struggling colonies than it does to keep supporting them through the whole season. Culling or merging allows the beekeeper's time to be spent on colonies that demonstrate a better chance of survival or of producing a big honey yield. Eliminating weak colonies sounds harsh, but the reality is that it reduces the chance of those sick or failing colonies affecting the other colonies in the yard. As a business owner, time is likely best spent helping the healthiest colonies maintain that health, as it translates to increased honey, wax, and bee yields. While the business is starting, the beekeeper is likely to experience sticker shock from purchases of large

amounts of equipment or large pieces of equipment. A business plan is invaluable at this point, as it helps them to visualize how these large costs spread out over time. What seems like a large purchase — such as 400 hive bodies, 400 supers, and 4,000 of their respective frames during a seasonal sale from an equipment manufacturer or supplier — is spread out over the course of 10 to 30 years, depending on how well the equipment is maintained. Compound this example with the fact that manufacturers tend to increase the price of goods 1–10% each year in response to inflation and market factors. One-time large purchase orders of durable goods become a prudent choice — as long as it does not exceed the growth plan for the business. There are incentives for business owners to make purchases in this manner. Business expenses are often tax deductible, and beekeepers may choose to have equipment depreciate over time, making the initial expense easier to manage in future years. Seek counsel from a certified accountant on business expenditures and financial planning. An accountant's services can seem expensive, but they should save time (and eventually money) and provide the business with better results than if a beekeeper were to attempt to manage these details by themselves. Beekeepers are also encouraged to talk to other beekeepers and visit their operations to hear their opinions on specific equipment, as they may offer invaluable advice. Overall, be prudent with purchases, and buy equipment that can grow with the operation. Equipment that increases throughput often pays for itself in the time and labor it saves compared to undersized, cheaper-priced equipment.

READ

Deciding on Business Structure

■ One of the more technical yet Key Points ambiguous aspects of starting a There are a variety of ways business — beekeeping or otherwise — businesses can be registered, and is how the owner structures the beekeepers should explore each in business. The structure of the business detail to determine which method and how it operates is largely best suits their goals. dependent on the regulations and laws in place wherever the beekeeper is doing business. It is important to be mindful of not only where a beekeeper physically sets up shop, but also where the products are distributed. In all cases, the business interests are always best served by being informed of local laws, customs, practices, and limitations. In the U.S. — and New York specifically — there are four main categories to register and define a business with the state and federal government: sole proprietorship, partnership, limited liability company (LLC), and corporation (Inc.). Each entity has pros and cons associated with it. The main factors to consider with each setup are the tax structure, the benefits afforded to the business, and the amount of financial and legal liability each structure/entity allows to pass through to the owner(s). The other factor to consider is the cost associated with registering the business as any one of the structures or switching between types. This information is presented purely as a helpful resource to beekeepers who are curious about growing and starting a beekeeping business; the final exam series will not ask questions about different business structure details. Sole Proprietorships Sole proprietorships are the most basic form of business structure. There is no legal distinction between the owner and the business — the owner is responsible for any taxes, litigation, and liability due to operating the business. This is often the easiest form of business to set up, with minimal paperwork, minimal fees, and minimal time to do so. However, this structure carries a large amount of risk to the business owner. If a charge comes against the business, such as a personal injury case or tax collection, assets of the owner can be seized in order to satisfy the charges against the business.

Partnerships are similar to sole proprietorships in the sense that the risks, taxes, and liabilities are all directly linked to the owners. The amount of risk can depend on how this structure is set up in the beginning. Not all partnerships are equal in the distribution of risk. Partnerships involve multiple parties (partners can be people, investors, and other business entities), and so they can require legal assistance to start. This assistance is mainly for defining the ■ Partnerships ■ roles and responsibilities of each partner, as well as the liability that each partner assumes as business is carried out. Limited Liability Companies Limited liability companies are a hybrid entity that affords similar tax and liability benefits to the owner as a corporation, while requiring slightly less paperwork and legal fees to maintain over time than a corporate structure. LLCs are able to file taxes with the federal government as a corporation. LLCs, as their name implies, offer liability protection to the owners in the event of bankruptcy or litigation against the company. If the business collapses, the owner is not responsible to pay for any business accounts or taxes that have not been personally guaranteed. The downsides of an LLC include the highest initial cost to set up, and there are additional paperwork and steps in New York to announce the formation of the business. The business is required to file a semiannual report with the state. However, maintenance costs are minimal and, over time, cheaper than incorporation. Corporation is the last of the business structures. There are three main types of corporations: S corporation, C corporation, and B corporation. The most pertinent to beekeeping are S and B corporations. S corporations are small corporate entities defined by having fewer than 100 shareholders. S corps allow all corporate income and losses to be passed through to the shareholders, who report these directly on their personal income tax returns at the income tax rate. This allows the owners to avoid corporate tax, which many see as a "double taxation." C corporations are taxed separately from their shareholders to whom they are responsible, and B corporations are similar in structure to C corps in this respect. B corps differ from C corps in the sense that they are intended to provide a public or societal good or mission, and most treat employees as shareholders. All corporate structures offer liability protection for shareholders in the case of action taken against the company, be it financial, legal, or otherwise. ■ Corporation ■ Corporations require rigid structure, dedication to turning a profit to benefit shareholders and are subject to filing corporate documents on a regular basis to interested parties. Corporations that fail to follow these practices are subject to action by their shareholders, such as upper management being replaced (i.e., the original owners/founders), the business being shuttered for not turning a profit or adhering to the corporate goals, or the business being dissolved to reduce shareholder burden for bankruptcy. The startup cost for an incorporation (forming a corporate entity) is less than that of an LLC, with similar protections. However, the maintenance costs increase as the corporation grows. Registering a Business Registering a business can be a daunting task, but the benefits of doing so far outweigh the risks of operating an unregistered business. Registered businesses are eligible for many tax benefits and qualify to receive resources made available by the federal and state governments that are designed to help business owners start, grow, and expand. The USDA offers many loan and grant opportunities to small business owners, as does New York State. Creditors and financial institutions are more likely to lend on favorable terms to registered businesses, and investors are more comfortable investing in a registered business. Lastly, consumers often view a registered business as being more reliable and trustworthy, paying a higher premium for products and brands that leverage this image accordingly. ■

READ

Programs and Resources to Help Beekeepers Overview

Today, more resources and programs exist to help beekeepers than ever before. Some programs help beekeepers maintain healthy colonies, while others help them mitigate risks and manage their businesses. Depending on your operation size and beekeeping goals, some of these programs might be helpful for you (even if you don't own a beekeeping business). Others may not suit you but are nonetheless good to be aware of in the event a fellow beekeeper turns to you for support. Many of these programs are not well advertised and may not be known in bee clubs. In this course, we will present a brief overview of U.S. programs as well as programs specific to New York State to help beekeepers. Beekeepers from other regions are encouraged to check in with their state or provincial apiculturist and local extension associates to find similar resources that can help them. We will not ask questions about these programs in the Master Beekeeper Program Exam Series. We realize this information is not helpful for everyone, but it can be helpful to some. NYS Beekeeper Tech Team

WATCH

Programs to Help With Colony Health

New York State Beekeeper Tech Team The NYS Beekeeper Tech Team is fully funded by the state Pollinator Protection Plan to provide three years of colony health sampling to 25 beekeepers across the state. Colonies are sampled twice per year (early summer and late summer) for Varroa mites, Nosema, and pesticides. In addition, beekeepers receive counseling on best management practices and develop an annual plan to maximize colony health and productivity. Enrollment is on a first-come/first-served basis to operations based in New York State that bring in annual revenues of at least \$1,000. Priority is given to commercial beekeepers as well as beekeepers who have consistently experienced unsustainable colony losses. After three years, beekeepers graduate from the program and new beekeepers are enrolled. Website:

<https://pollinator.cals.cornell.edu/resources/nys-beekeeper-tech-team>

WATCH

Programs to Help With Your Beekeeping Business

USDA Farm Service Agency Loan Programs National Program The USDA offers a variety of loan and grant programs for which beekeepers can apply to help them establish or operate their business. Some loans, such as the "Beginning Farmer and Rancher Development Loan Program" are appropriate for beekeepers that have been in business for fewer than 10 years. Loans for "Socially Disadvantaged Farmers and Ranchers" allow farmers to buy and operate family-sized farms; eligible groups include "American Indians or Alaskan Natives, Asians, Blacks or African-Americans, Native Hawaiians or other Pacific Islanders, Hispanics, and women." Depending on the loan, it can be used to help cover costs of purchasing buildings, land, equipment, livestock, chemicals, feed, and insurance. "Emergency Loans" can help beekeepers recover from unexpected production losses as well as livestock and equipment losses from natural disasters or disease quarantine. There are a variety of these grant programs available, with more information and options found at the website below. Website:

<https://www.fsa.usda.gov/programs-and-services/farm-loan-programs/index> NY FarmNet New York State Program NY FarmNet is a government-funded program that provides financial analysis, business

planning, and psychological counseling to beekeepers struggling with their businesses. High colony losses can be devastating emotionally and financially, so FarmNet representatives offer help free of charge. In addition, NY FarmNet can help with business transitioning for beekeepers nearing retirement. NY FarmNet also offers a webinar course through eCornell on business and succession planning. Website: <https://www.nyfarmnet.org> In this video, Dan Welch, Extension Support Specialist for NY FarmNet, discusses how the program can help beekeepers. Video Transcript FarmNet is a program based out of Cornell that's supported by New York State to assist farm families throughout the state with business transitions, maybe from one generation to another, looking at profitability, and assisting sometimes with some of the thornier issues that come up when you're working close together with family and friends that can sometimes lead to conflict. And if those situations aren't dealt with, that can have negative impacts on business. So we try to be a resource for those farms and help them to continue to make good business decisions so their farms are viable into the future. New York FarmNet is a 100% free and confidential service that's available to beekeepers and any other farm in New York State. We are supported through the state budget by the Department of Ag and Markets and New York State Office of Mental Health. And that enables us to go out to any farm, and we come right to the operation, so you don't have to come down to Ithaca to meet with us. One of the other services that we offer is helping individuals and families work together in developing techniques to manage some of the stress that comes along with keeping bees. I know here in New York, a lot of beekeepers have been losing a lot of their colonies, which causes both financial and mental stress, and can also stress relationships. Our family consultants are able to work with beekeepers to give them good techniques for managing that stress so that you can focus on your business and continue to grow. If you're just looking for more information, you can go to our website, which is www.nyfarmnet.org. Or you can call 1-800 number, and that number is 1-800-547-3276, and that's a number where there's always a person. If it's outside the office hours, it will go to an answering service where there's a real, live human being. And then shortly after we get that message, one of our consultants will come back to schedule a meeting with you. New York State Beekeeper Tech Team: Financial Analysis and Business Benchmarking Program New York State Program In addition to investigating a beekeeper's colony health and management practices, the New York State Beekeeper Tech Team provides financial analysis and business benchmarking for beekeepers. Beekeepers enrolled in the program meet with an agricultural economist to develop financial statements that allow beekeepers to understand the financial position of their business, which in turn allows them to make informed business decisions. The program also offers record-keeping assistance and provides beekeepers with a report on how their businesses are doing compared to similarly sized beekeepers that are also participating in the program. Website: <https://pollinator.cals.cornell.edu/nys-beekeeper-tech-team/>

MODULE WRAP-UP

Turning Bees Into Business

In this module, you explored the shift in mindset that the transition from hobbyist to business owner requires. You examined the factors beekeepers can consider when growing their operation and beginning a beekeeping business.

Being a Leader in Your Community

Community involvement is a natural part of keeping bees; after all, our animals of interest are themselves organized as complex social organisms. Indeed, we need to be involved with other beekeepers, as honey bees fly well beyond our property lines. Foragers travel two to three miles in each direction from their colony to collect nectar and pollen, and virgin queens can fly about eight miles in search of drones. This behavior, combined with the density of our human population, means that the territories of our apiaries often overlap. Our problems become our neighbors' problems, just as our solutions may also work for our neighbors. For the sake of healthy colonies and maintaining genetic diversity, we should offer and accept ideas from our neighboring beekeepers. Our success with honey bees also directly benefits from relationships with scientists, gardeners, farmers, the public, honey lovers, and everyone with an interest in bees. Nearly all Master Beekeeper Programs are developed with public education and community service as a core value, and Cornell's is no exception. We encourage you to provide volunteer service in your community, in whatever way is most fulfilling to you. While our program does not require you to log volunteer hours, we do hope you step up to be an ambassador for honey bee and pollinator health. In this module, we will discuss leadership skills and explore opportunities for serving your communities.

READ

Mentoring

■ Being a mentor allows beekeepers to reflect on their early lessons in mentoring new beekeepers and fosters their own valuable way to share knowledge and curiosity. It forces them to slow their pace and think about what they are mentoring requires skills in doing when they open a hive or beekeeping, teaching, and purchase new equipment. Mentors communicate, who articulate what is seen, what is done, and why should remain open to beekeepers should review how comments and questions. They will much time they have available to discover occasional holes in their mentor before committing to any reasoning or perhaps a fact that needs mentees to be checked, and they may be grateful for the insight. There is a lot for a newcomer to the apiary to absorb. Finding the right level of exposure and appropriate tasks for an apprentice is one of the first challenges met by a mentor. You might start with a lazy conversation in lawn chairs beside some hives on a fair day for an introduction to a mentor-mentee relationship. This way, you can share personal insights and ambitions with apiculture, as well as a preliminary lesson in observing entrance behavior. Becoming familiar with bees in the air and in the hive, as they buzz about and land on people, helps newcomers understand the various sounds of bees, one of many skills required to work a colony with gentleness and composure. During one or two years of mentorship, new beekeepers gradually gain perspective into seasonality and other details of timing. It is often necessary to experience a few seasons to absorb such knowledge, just as it was for you! It may take several more seasons to be able to apply creative decision making to colony management. As the apprentice develops a commitment to beekeeping, they will become a contributing member to the beekeeping community. Apprentice beekeepers may provide healthy pollinators for crops and natural vegetation, contribute to the local economy through their businesses, and may even educate some new beekeepers. Investing in the next generation of beekeeping through mentoring can be a very rewarding way for a beekeeper to

spend their time. Being a beekeeping mentor requires a number of skills and qualities, including (1) a knowledge of the art and science of beekeeping; (2) experience in beekeeping; (3) being an effective communicator; and (4) being a patient and compassionate teacher. Master Beekeepers, by definition, fulfill the first two criteria. For some, mentoring might come naturally. Many beekeepers are happy and excited to share their knowledge, expertise, and guidance with new beekeepers. If mentoring is not for you, that is OK! Master Beekeepers can contribute to the community and beekeeping industry in a variety of ways beyond mentoring. Mentoring is a form of informal teaching through modeling methods and providing opportunities for hands-on learning. Learning through continuous dialogue is natural and effective in the apiary and honey house. The beekeeper can describe the rationale of management choices and manipulations, while the mentee can ask questions about any gaps in understanding. It is common to hear experts describe how much beginners teach them simply by asking questions that require articulate, thoughtful answers or a little extra research. Group mentorships can also be effective in beekeeping clubs. Beginners and experts alike build their knowledge and refine their skills with a synergistic collective intelligence, seeking advice from peers in their groups. Every local beekeeping association should have a mentoring program. If yours does not, one of the first impacts you can make as a Master Beekeeper is to help start one. To find mentees, advertise your contact information on your club's website and social media channels. Preparation Start small. If you have never mentored before, begin with one beekeeper. Decide whether you want the mentee to learn and practice on your colonies, whether you want to travel and teach with their colony, or a combination. Some mentors prefer to work with their mentees' colonies and get an understanding of the needs of those colonies to best answer any questions they have, while others appreciate an extra hand with their own apiary chores! Mentoring goes beyond arriving at a new beekeeper's apiary and giving them an hour to ask you questions. It is guiding them through the process of learning at an appropriate pace. If the mentee has specific things they would like to learn, have them tell you up front so that you can work those topics into your sessions. Availability New beekeepers are excited and passionate about beekeeping. They generally have many questions and are grateful to have someone to consult for answers. From the start, you may want to set contact boundaries. Decide upon your schedule of meeting times and determine what times it is OK for your mentee to call you outside of the sessions. When you are meeting with your mentee, be approachable and patient. Foster an environment where they feel comfortable asking questions. It can take considerable time and energy to be a one-stop shop for all questions and conversations about beekeeping. Encourage your mentee to seek out other sources of information. Suggest that they join a beekeeping club if they have not already, turn to the local honey bee extension associate, sign up for introductory classes (either online or in person), subscribe to beekeeping journals, and read books. We recommend the following introductory books: "Honey Bee Biology and Beekeeping" by Dewey Caron and Lawrence John Connor "The Beekeeper's Handbook" by Diana Sammataro and Alphonse Avitabile "Beekeeping for Dummies" by Howland Blackiston (a good introductory guide despite the title) Feedback is important — and do not forget the positive kind! Recognize your mentee when they are handling frames gently and identifying colony items correctly, and provide useful, constructive criticism if they slip up. Do this in an honest, encouraging manner. As leaders and educators, mentors should tell their mentees what they need to hear, not what they want to hear. Be compassionate when communicating advice and suggestions.

TOOL

A Sample Mentoring Schedule

To help you get started, we have **■** Download the Tool prepared a downloadable guide for the You can download the Mentoring first year of mentoring entitled Schedule for a First-Year Mentoring Schedule for a First-Year Mentee here. Mentee. This guide discusses the activities to work on and topics to discuss during a beekeeper's first season from April to November. In this scenario, the mentor usually visits the mentee's apiary for one- to two- hour sessions, and the two meet every second week. Of course, your mentoring schedule may be very different from this. You may wish to meet more or less frequently or cover different topics. Do what feels most natural to you.

WATCH

Community Service

■ Master Beekeepers are valued for Key Points community services like speaking to Master Beekeepers can be leaders organizations, mentoring new in their communities in many beekeepers, providing public ways, including working with education with lectures for schools and other beekeepers, students, civic organizations, conducting hands- researchers, the government, and on workshops, and discussing current the public. issues with the media — all activities to benefit the public good and the environment. Beekeepers can learn how their customers benefit from having honey at their table, who plants pollinator gardens, and which farmers are finding ways to reduce their use of pesticides; in turn, they can foster ties with all who support pollinator protection efforts. The benefits extend to the beekeepers themselves as they enjoy fulfillment from service and teaching. The ever-changing world of apiculture is a rich environment for lifelong learning. There are several ways that Master Beekeepers can volunteer in their communities. Depending on their skills, comfort level, and time availability, some avenues may be more appropriate than others. We encourage beekeepers to explore the suggestions below. Speaking to Media Honey bee issues are a hot topic in media in recent years. Journalists frequently seek beekeepers to interview for TV, news, and radio segments, but unfortunately they sometimes perpetuate inaccurate information. One common example is colony collapse disorder. Although it no longer has a major effect on honey bee health in the U.S., many stories are still centered on colony collapse disorder. In reality, Varroa mites, viruses and other diseases, management practices, pesticides, and economic pressures are bigger issues facing bees today. Most of what the public learns about bees is from the media, so an informed beekeeper's voice can make an impact on the community's understanding of the current status of bees. If beekeepers are interested in speaking to the media, they can submit their name at their club and local university as a contact person for when those institutions receive inquiries. **■** Contributing to Policy Discussions **■** With the increasing awareness of pollinator health declines, new policies and regulations are being developed to help mitigate bee health issues. Pollinator protection plans are being created (and continuously updated) at the state level, and now several counties and cities are developing their own strategies to help bees. During the formulation of new policies and programs, beekeepers, scientists, other pollinator stakeholders, and government legislators discuss the issues and possible solutions for better bee health. Beekeepers are often invited to speak on behalf of the beekeeping industry and voice the issues affecting beekeepers in their region. Master Beekeepers can contribute effectively to these discussions. Delivering Presentations to Schools Educating children about bees is, above all, fun! The students may grow up to be beekeepers, environmental scientists, farmers, landscape architects, or

politicians; no matter what they do, the more they recognize the importance of agriculture, pollinators, and their habitats, the better. Some teachers advertise opportunities to have beekeepers give presentations and demonstrations to their students. Depending on the age of the children a beekeeper visits, they can create a simple presentation about bees and beekeeping, then combine it with a fun activity, story time, or show-and-tell with equipment and an observation hive. Beekeepers can also invite classes to take a field trip to their honey house to see the ins and outs of extracting honey. Students learn where honey comes from, how the bees make it, and how people harvest it. ■ Before working in schools, beekeepers should speak to the teacher about protocols regarding food (especially sweets!), bee allergies, and stings. Taking on a Leadership Role in a Bee Club Bee clubs are the heart of the hobbyist beekeeping community, with members of all experiences, ages, and lifestyles meeting together with a common goal. They are a great source of information for beekeepers, as well as friendships and partnerships that can last a long time. In this video, Shelley Stuart, former president of the Finger Lakes Beekeepers Club, discusses the dynamic community of beekeeping clubs. Video Transcript As a new beekeeper, and even as a beekeeper with a few years under your belt, I find that bee clubs can be really helpful in building a bee community. And it's not just about getting information, it's about sharing your own expertise. I was president of the Finger Lakes Bee Club for about five years, I think it was. When I started, I had two years of beekeeping under my belt, so I wasn't particularly expert in beekeeping, but I could organize a meeting. And ■ what I found with the beekeeping community is that everybody shares their information and experience, their microclimates, their tips, their tricks, and it really encourages a vibrant beekeeping community. When folks that are in your local area say, "This worked for me, but it may not work for you," or "I'm also up on a windy hill, and here is what I've done to successfully overwinter my bees," or "I'm seeing the goldenrod flow start now and my hive is putting on weight," or, "For the spring, I'm seeing drone cells in my hives, so it's time to make splits." And I think that information flow and sharing of information not only makes for better beekeepers in your area, but eventually, ultimately, will fill your area with better bees. And you'll see that in your own hives with the genetics because your neighbor's drones are going to be mating with your queens, and vice versa, and the stronger your greater area of bees are, then the stronger your own hives will be. Every club is different, but most meet once a month to discuss honey bee news, management practices appropriate for the current season and climate, and upcoming events in the community. Most offer mentorship opportunities to partner experienced beekeepers with new beekeepers. Here is a list of additional activities and resources provided by some clubs: Host speakers to talk about current research or specialized topics in beekeeping. Hold workshops for club members on topics such as introductory beekeeping, monitoring for Varroa mites, and wintering colonies. Share equipment and resources, including honey extractors, microscopes, Varroa mite vaporizers, and beekeeping books. Order nucs/packages or mite treatments in bulk to sell or distribute to members. Develop breeding programs to improve the genetics in the local beekeeping community. Set up booths at festivals and fairs to promote honey bees and sell honey from various club members. Hold honey judging competitions. Hold conferences (typically state- or province-wide bee clubs). Host social events like potlucks and honey tastings. Put together floats for community parades. Get involved in apicultural research. Support pollinator protection organizations. If beekeepers have the time and interest, joining the club as an officer or board member can be an effective way to influence the quality of the club and the benefits it offers its members. Clubs often need many individuals to help them run efficiently. The

leadership roles are diverse, including organizing meetings, managing membership and finances, sending out and responding to emails, and managing social media. Giving Presentations With their knowledge and expertise, Master Beekeepers are in a position to help educate others through public speaking. Presenting at clubs, workshops, and bee conferences (such as the annual Eastern Apicultural Society or American Beekeeping Federation conferences) can allow them to share valuable information with other beekeepers from different regions or experience levels. Beekeepers who wish to give talks should choose topics that are important to them, on which they feel confident speaking, and that are interesting or valuable to others. Later in this module, we will discuss how to prepare and deliver an effective presentation. Presentations are not restricted to beekeeping organizations but could also be beneficial to gardening clubs, agricultural organizations, horticultural societies, ornithology associations, golf clubs, and nature groups. The written word can reach a much larger audience than a presentation with a limited audience. Consider submitting something to American Bee Journal or Bee Culture, newsletters for regional or state beekeeping clubs, or the local paper. If beekeepers have ideas for articles, they should reach out to magazine or newspaper editors to pitch them. ■ Writing Articles ■ Volunteering at Universities ■ Universities are the hubs for honey bee research and extension. These institutions often have lots of projects underway and are grateful for all the volunteer help they can get. The Dyce Lab, for instance, could not effectively function without help from volunteers! There are many volunteer opportunities maintaining research and extension colonies, supporting scientific experiments, teaching workshops, responding to beekeeper inquiries through email or phone, creating handouts and educational materials, and managing social media. If beekeepers want to be on the cutting edge of research or extension, they can contact their nearest academic institution to see what opportunities are available and where they will best fit in.

READ

Participating in Research

■ Pollinator health research has grown in Key Points recent years, and one of the most Citizen science projects harness impactful ways that Master the efforts of many people to Beekeepers can give back to the collect data on the health of honey community is through participating in bees and other pollinators. the research. While scientists are aware Many universities and institutions of the factors impacting honey bee regularly seek out beekeepers to health, we have yet to understand their participate in pollinator health relative importance and develop projects. solutions. To investigate the real-world situations of honey bees, institutions Responding to Bee Informed are turning to beekeepers to help Partnership’s annual colony loss provide data. Beekeepers do not have survey provides invaluable data for to be trained in scientific research to understanding honey bee health participate. problems. There are two major ways that beekeepers can take part in research: participating in citizen science and responding to beekeeper surveys. While beekeepers must devote some time to collecting samples or filling out surveys, the benefits are numerous, as the data can provide insight into a variety of questions: How many colonies were lost in the area last winter? What were the main causes of mortality? What are the disease levels and pesticide risks in the area? When are nearby colonies swarming? Answers to such questions can allow beekeepers to make more informed management decisions in their own bee yards. Citizen Science Citizen science is scientific research conducted in part by non-scientists. The process usually involves citizens collecting and submitting scientific data

and, as such, can foster collaborative efforts between beekeepers and scientists. Citizen science allows for the study of bees in a variety of environments where they are normally found or managed. Unlike strictly controlled scientific experiments, this encompasses the real-world variations encountered by bees. Citizen science is a good approach for understanding trends that are widely distributed across a state or country because many volunteers can gather much more information in a short period of time than a few scientists could. There are several ongoing citizen science projects on honey bees and other pollinators in the U.S.

New York Bee Wellness The nonprofit organization New York Bee Wellness collects data on swarm dates, mite counts, and virus levels to post on virtual interactive maps. Beekeepers simply enter their swarm and mite information online but must send in samples of bees at four time points across the year to determine virus levels.

■ **Bumble Bee Watch** Bumble Bee Watch is a citizen science project tracking North America's bumble bee populations. Citizens photograph bumble bees, identify the species, and upload the photos to a website where experts verify their identifications. In this way, bumble bee abundance and diversity can be tracked continuously across the U.S. and Canada, particularly for rare or endangered species such as the rusty-patched bumble bee.

National USDA Animal and Plant Health Inspection Service (APHIS) Survey The National APHIS Survey is a government program that surveys beekeepers' colonies for parasites, pathogens, and pesticides. It also actively monitors the presence or absence of invasive threats to honey bee health (such as the *Tropilaelaps* parasite and slow bee paralysis virus). This program is conducted through the state apiculturist and state inspectors, who select different beekeepers each year to participate.

Cornell University Projects Each year, the Dyce Lab recruits beekeepers to participate in various research projects. In 2015 and 2019, beekeepers and growers participated in an apple pollination pesticide study. Every year since 2016, the lab samples colonies of approximately 25 beekeepers participating in the NYS Beekeeper Tech Team for Varroa, Nosema, and pesticides to understand these threats as well as beekeepers' management practices. The projects for each summer are determined the previous winter and spring and are advertised on the Pollinator Network website and the Dyce Lab Facebook page. Beekeepers from New York that are interested in participating in Cornell University research are encouraged to check these avenues regularly.

Surveys At this point, all information about colony losses across the country are from voluntary beekeeper survey data. Beekeeper observations about colony losses and management practices are critical, as they are often cited by scientists and media to provide an overview of the status of honey bees. These data can even be helpful in persuading grant reviewers and legislators to provide ■ research funding for projects about bees. The more data that is gathered, the more complete the picture is for honey bee health and the beekeeping industry.

New York Bee Wellness One survey that is specific to New York is released each year by New York Bee Wellness; it asks participants about their colony losses, winter management practices, and suspected causes of colony deaths.

READ

Delivering Effective Presentations

■ If beekeepers plan to give talks to bee Key Points clubs, conferences, schools, or other A good presentation includes venues, it is important to work toward simple slides, compelling delivering effective presentations. information, and effective Being an effective presenter is not a communication. quality with which people are born but It is helpful to develop an outline a skill that can be developed

and prior to creating a presentation. continuously improved. The final exam series of the Cornell University Master Beekeeping Program requires students to give a 15-minute presentation on a topic in beekeeping. Students will be graded on this presentation. The considerations that we present here will be important for beekeepers to use when they are preparing their presentations for their final exam, as well as when they are giving presentations to the community as Master Beekeepers. An effective speaker can accomplish the following: Present compelling information in a logical order. Use effective visuals. Speak clearly and convey enthusiasm. Let's explore each in more detail.

Present Compelling Information in a Logical Order An effective presentation, like a good story, needs to be easy for the audience to follow. The audience should be able to identify the key points and understand how it relates back to the original question or purpose. Once the audience gets lost, it is hard for them to maintain their focus on the rest of the talk. An audience can quickly fall behind if the information is presented at a level that is above their comprehension. To avoid this, learn who your audience is ahead of time. If an audience consists of beginner beekeepers, assume they do not know anything about bees or beekeeping when you create your presentation; define and explain everything. If the audience has a few years of beekeeping experience, do a quick recap of the prerequisite knowledge they should have before you dive into your topic. For instance, if you are going to give a presentation on the impacts of neonicotinoids on honey bee health, start by simply introducing what neonicotinoids are. While most people in the audience may already have a general idea of this pesticide, there might be some who have heard of neonicotinoids but do not actually know what they are. Bring everyone to the same level of understanding early in the presentation, then build up to new information together. It's important to remind beekeepers that once they have a few years of experience, they have gained specialized knowledge that others do not yet have. Start simple. Before you create your accompanying presentation slides, develop an outline of your presentation. This keeps the presentation logical and to the point. A presentation outline includes the following information:

- Description of the audience.**
- Presentation title.**
- Introduction section:** Introduce the specific question or purpose to be addressed in the presentation. Identify any information the audience needs to know before getting to the body of the talk; this ensures everyone has the same level of understanding from the beginning. Some presenters like to also include a brief overview of the presentation in the introduction so the audience has a preview of what is to come. The length of the introduction depends on the length of the talk. For a 15-minute talk, aim to give a four-minute introduction. For a one-hour talk, aim to give a 10-minute introduction.
- The body of the talk:** This is the main focus of the talk. In this section, list the important points to cover and identify the sources of information that are included to support the key points. Information should be presented with a logical flow, using smooth transitions to get from one main point to the next. Keep information interesting and supportive of your main question or purpose. Try not to stray into tangents or focus on irrelevant material. Think about personal experiences to share during the main body of the talk. By nature, people like stories and can connect easily with a speaker who shares an anecdote related to the topic. For a 15-minute talk, aim to spend about nine minutes on the body. For a one-hour talk, aim to spend about 45 minutes on the body.
- Conclusion:** Repeat the main findings and overall take-home message of the talk. Clearly identify how this information applies to the audience and what they can do (if anything) after hearing the presentation. For a 15-minute talk, a two-minute conclusion should suffice. For a one-hour talk, a five-minute conclusion is ideal. Creating this outline will help speakers stay focused and organized. It works to ensure that the information

presented is relevant to the main topic and presented thoughtfully. Use Effective Visuals Effective visuals are a great accompaniment to a presentation; after all, a picture is worth a thousand words. Include many photos and visuals in the presentation, and reduce text to only include a few keywords. Any words that the audience may not be familiar with should also be shown as text to make them more comprehensible and memorable; e.g., "superorganism," ■ "neonicotinoids," "queen pheromone," and "waggle dance." Use large font in colors that are complementary to the rest of the presentation's color scheme. Having little to no text on slides directs the audience's attention to what the speaker is saying, rather than tempting them to concentrate on the slides. It also prevents the speaker from relying on looking at the slides too much. Instead, the speaker can look at the audience, make eye contact with individuals, and engage them. Make sure to always include photo credits for photos that are not your own. Not sure how many slides to create for your presentation? Anticipate spending 30 seconds to one minute on each slide. Slides with more elaborate illustrations or diagrams take longer. Speak Clearly and Convey Enthusiasm An effective presentation engages the audience from the beginning. Get their attention with some sort of hook; maybe a thoughtful question posed to the audience, a relevant humorous anecdote, or a powerful photo. The speaker's voice should be clear and projected so that the person in the last row can hear them. Slow down when speaking. Nervous speakers tend to ■ speak very quickly; while this is normal, it unfortunately makes it more difficult for the audience to understand them and keep up with the information. The speaker can slow down by taking advantage of natural pauses. Blank air can feel a bit strange to the speaker, but it is helpful to have a few seconds of silence in between main points to allow the audience a moment of reflection. This is also an ideal time for the speaker to take a sip of water or gather their thoughts for how they are going to transition to the next slide. Conveying enthusiasm is the easiest way to interest an audience in a topic. Excitement is contagious. Enthusiastic speakers are not monotone; instead, their voices naturally vary in tempo along with rises and falls in pitch. Some speakers find ways to further engage their audience and keep them focused on the presentation. Posing a question or two to the audience during the presentation can make them feel more like they are participating in a discussion rather than simply watching someone talk. Lastly, always rehearse presentations before they are delivered. Practicing helps speakers stay within their time limit, transition seamlessly between points, and remain on track. It can also reduce your anxiety if you are nervous about public speaking. Practice in front of friends or family that can give constructive feedback, or by yourself, to become familiar with the content and be confident in your delivery. Steve Jobs, the late cofounder of Apple and world-renowned public speaker, rehearsed every one of his presentations for hours before giving them to the public. An Example of an Effective Presentation An example of an effective presentation is this 15-minute TED talk by Dr. Marla Spivak. Note how she uses many of the suggestions we discussed on this page. She engages the audience from the start by showing a captivating photo that illustrates the importance of bees. She uses effective visuals without relying on text on a slide. All the text is limited to a few keywords. Marla speaks clearly and slowly, and she makes use of natural pauses. She tells a few stories and personal anecdotes to which the audience can relate. Her four main points on why bees are declining are clearly stated: diseases and parasites, pesticides, monocultures, and flowerless landscapes. In her conclusion, she recaps the issues harming bees, explains to the audience why this issue matters, and gives examples of what they can do to help. This is a very effective presentation.

Being a Leader in Your Community

In this module, you discussed leadership skills and explored opportunities for serving your communities.

Keeping Up With Science

By studying to be a Master Beekeeper, you are already motivated to learn best beekeeping practices and to understand the biology of the honey bee. After this course is over, you will want to stay up to date as new information becomes available on topics such as disease, nutrition, behavior, and the condition of the local environment where your bees live. Being familiar with current science findings can benefit your management practices and will better inform your discussions with others in the beekeeping community. Furthermore, any educational work that you do as a Master Beekeeper is facilitated if you are prepared to answer questions about and discuss new findings. You can learn about recent science advances without a science degree. In this module, we will look at some ways to find, comprehend, and evaluate science.

READ

Where to Find New Information

There are two main reasons to look for bee science information. Beekeepers either browse to learn what is new or hunt for information on a specific topic. In both cases, there are three avenues that beekeepers can explore when seeking new information about bees and beekeeping. Popular Science Articles Popular science articles appear in newspapers, news sites, and journals. These pieces highlight research without delving into too much detail. The authors are science writers who generally quote an expert or experts to support the information they present. Primary Literature Primary literature, in the form of academic journal articles, is where scientific discoveries first appear in print. Authors of the primary literature are the researchers who did the work. The results and methodology necessary to repeat the study appear in a primary research paper. Primary literature can be challenging to read because it is written for other scientists, not laypeople. Even scientists from different disciplines may have difficulty following a primary literature paper outside of their own field of study. However, beekeepers can certainly gain confidence in reading these articles with practice. Reviews Reviews are summaries of multiple primary research papers dealing with the topic of interest — they are not the original research. Reviews are also published in academic journals and are reputable sources of information. Scientific reviews are typically written for the “generalist scientist”; anyone who has taken general science courses in college is likely able to understand a review, even if they are not experts on the topic being discussed. A good review is an efficient way to find much of the available information on a topic.

READ

Popular Science Articles

■ “Did you see that article claiming 75% Key Points of all flying insects in Europe are Critically reading popular science dead?!” articles takes practice. This is the type of exchange Media articles often

sensationalize beekeepers may engage in with their scientific findings. friends. In order to be an informed Some popular science journals and participant in a conversation about a magazines are more reputable trending topic in honey bee science, than others. beekeepers have to find the popular sources and read them, then dig deeper: Locate the original research cited within those articles, get copies of the original papers, and evaluate them. Browsing the internet for bee science news requires more scrutiny for authenticity than running a specialized academic search on a specific topic. It is important to look past the headlines and think critically while reading the article. Often, popular science articles are deliberately sensationalized in order to increase readership; for example, the headline in Newsweek on this topic is "Flying Insect Numbers Fall By 75%, Threatening 'Ecological Armageddon.'" Most would agree that this title is sensationalized! The actual work is frequently fairly pedestrian and far less earth shaking than the headlines make it out to be. We learn this by going to the primary literature sources and evaluating them directly. The primary avenue of most beekeepers for learning about new findings in bee science is through popular science, as their easy-to-read style and incorporation of photos and videos makes them fun and engaging to consume. They are also frequently distributed through email and posted on social media sites. Popular reviews are the easiest to understand because they were written for a lay audience; however, not all science writers do justice to the research papers about which they write. Readers need to look at the original work as well for a deeper understanding. Let's examine how to perform a popular science news search. Searching for Reputable Popular Science Articles To receive new articles without having to regularly search for them, beekeepers can customize their interests in Google News. After they enter a few key words under "Your Interests," the search engine pulls up related articles each time they sign in to check Google News. This feature lists every type of article available on the subjects they chose, so it is a necessity to be able to evaluate the sources. Searching within Google News is more effective at pulling up news stories than if beekeepers were to broadly search in Google. Before wasting time reading a poor source, scan the journals and newspapers that published various versions of the story and select those whose coverage is likely to be the best. Some of the most trustworthy news sites that use evidenced-based reporting are Nature News, National Geographic, Smithsonian, New Scientist, The Atlantic, Live Science, The Guardian, BBC, The Scientist, Science News, and Discover. The reasons why they are good sources include: These publications focus on science, not politics. The articles are written for the intelligent layperson. Data are presented, not merely opinions. Methodology is explained so readers can broadly understand how the study was conducted. The news articles do not make grand, sweeping conclusions based on the one primary literature study they are reporting. Information about the authors of the study, their institutions, and the journal in which the study was published is included in the news article. ■ It is a good idea to browse at least two of the articles on a topic, as this helps readers find and ignore biased reporting. It is also useful to see how various stakeholders present the same basic facts. But what if readers do not know which news source is best? They can spend a few minutes each time they do a search investigating one or two of the unknown journals or newspapers that come up in their results, allowing them to move faster through the list of options they have the next time they do a search. Beekeepers may find that certain news sites are not real organizations or that they have a specific agenda. Searching for news organizations on Wikipedia or the business- and organization-reviewing website Sitejabber can help beekeepers navigate which news outlets are trustworthy. Absorbing the Main Points When reading news articles, extract

information about the researchers, the institution where the work was performed, the method of how they conducted the study, and the major findings of the study. Let's use the news article we introduced at the top of this page that covers the massive insect die-off in Europe. We will start by reading the article in *The Scientist*, which was published on October 18, 2017; it is titled "Germany sees drastic decrease in insects." Follow the link and read the article. Try jotting down some key points, and then compare them to our list below. Key Points From the Article in *The Scientist*:

1. The research is published in PLoS ONE. This is an open-access journal, meaning the public is able to read the primary paper since this journal does not require a paid subscription to access it.
2. The authors are experts. They include plant ecologists from the Netherlands and some colleagues from the Entomological Society in Krefeld, Germany, who have been collecting data on flying insect biomass for two decades.
3. Researchers sampled 63 protected nature areas throughout Germany over 27 years and found a 76% decrease of flying insect mass.
4. The methodology is described. Here, the technique used to collect insects consisted of deploying "malaise traps," specialized mesh tents that direct flying insects toward a chamber with collection bottles filled with alcohol. ■ They are transparent about the data collection: 37 of the locations were only sampled once, and 26 locations at least twice, with many years in between. All of the sampling sites are lowlands regions amid agricultural lands and human-populated areas.
5. The major finding was that after 20 years, the scientists began to see a drastic decrease in insects of about 80%, so they re-sampled certain locations to ensure that their information was accurate. The trend continued, and by midsummer the decline had reached 82%.
6. The reasons behind the decline are not fully understood. Weather and temperature patterns, plant inventories, and the type of preserve were all examined for associations, but no factor alone could explain the disappearance of the flying insects. Climate change could not explain the results either.
7. At the end of this paper, the original citation in PLoS ONE is given, so readers can pull up the primary research article. If readers are not scientists, it might be best to continue to evaluate the work by reading more of the popular science articles rather than heading straight to the research paper in PLoS ONE. Each one should have a different synopsis to help readers absorb the key findings. If, for example, we compare the article in *The Scientist* with another one reporting the same study in *Science Daily*, we see that some points are omitted, while new ones appear. The article in *Science Daily* was published on October 19, 2017; it is titled "Three-quarters of the total insect population lost in protected nature reserves." Follow the link and read the article. Additional Key Points From the Article in *Science Daily*:

1. The article provides information on previous work in this field: Previous research has shown an overall pattern of decline in insect diversity and abundance but has focused on single species or taxonomic groups, whereas this study looked at overall insect biomass over an extended period of 27 years.
2. There is further extrapolation into the reasons behind the insect decline: The authors believe large-scale factors are involved in this loss. After reading several articles from reputable news sites, beekeepers may wish to read the original scientific article. One reason to go to PLoS ONE to read the article would be to look at the actual data as well as how the authors presented and interpreted their findings. Readers can also find all the other articles that have been referenced in the study to provide additional sources of information on the topic.

READ

Obtaining Reputable Scientific Articles

■ The insect die-off paper is easily Key Points obtained online through open access While most scientific journals to the journal PLoS ONE. Open-access require subscriptions, beekeepers journals like this one make a profit by can access many for free. charging the authors a fee to publish Authentic, high-quality papers are within it. Research that was publicly found in peer-reviewed journals. funded is also available for free on PubMed when the term “open access[filter]” is included in the search bar. Beekeepers can also search within the search engine Google Scholar to find articles of interest. By searching these three avenues — within open-access journals, the PubMed database, or Google Scholar — beekeepers are able to locate and download many scientific articles at no charge. Review articles published in peer-reviewed scientific journals can also be searched in this exact same manner. However, most research papers are in journals that require a subscription to access, or readers must physically be in a university library that owns a subscription in order to read the paper. If beekeepers want to read a paper that is behind a journal paywall, their local public library may be able to assist them through interlibrary loan. Alternatively, they can try to contact the authors, who are usually willing to send them a PDF copy of their paper. Determining Authenticity Once beekeepers have obtained the scientific paper, they are ready to begin reading! When combing through the pages, critique the quality and rigor of the work, as well as the content of the paper. An important feature of reputable journals is that each research article has been read by at least two experts in the field and comments have been returned to the authors for improvement of the paper before it is published. Through this method, known as peer review, inferior or incomplete work is culled, and readers who are not experts can trust that the published research represents the highest standards and best experimental approach available. Journals can be verified as peer reviewed by visiting their website and reading about the publication. For instance, we can establish that PLoS ONE is peer reviewed by visiting the “About” section on their website, where there is detailed information about their peer- review process, giving us confidence that we have an authentic source. Be aware that there are hundreds of “desktop publishers” of science journals with impressive-sounding names that publish insufficiently peer-reviewed and inferior work. When in doubt, search the web for information about the source. When reading articles, be on guard against sources that are attempting to persuade readers, promote products or institutions, or sell products. The funding source for the research should be outlined in the paper. The researchers should declare any conflicts of interest for their work (or if there are none). Scientific articles should be unbiased and present information based on evidence. Lastly, do not trust a source that does not identify the authors or provenance of the work.

READ

Developing Skills in Reading Scientific Articles —

Understanding Layout ■ Scientific papers are organized in a Key Points very stylized way. These papers are Articles are organized in a precise written for science colleagues, not for way that helps readers navigate the general public, and therefore may content. be difficult for a layperson to follow. Readers can avoid getting bogged down in technical details by focusing on the way science papers are organized into sections. The sections of a research paper include, in this precise order1: 1. Abstract: A brief summary of the topic, the approach, and key findings. The abstract is always freely available, even to those who do not own a subscription to the journal. By reading the abstract, a beekeeper can quickly identify the take-home message, and they can determine if they want to continue reading the

rest of the article. 2. Introduction: Relevant background information on the topic is summarized with citations (references) to key studies that led up to the current work. The introduction usually concludes with a statement of purpose, or hypothesis, for the paper. 3. Materials and Methods: Techniques used in the experiment are described in detail in the Methods section. All special materials are referenced, and information on where they were obtained is given. Any detailed method that has been described elsewhere may be referenced with a citation to the original description. Any reader should be able to repeat the entire experiment based on the methods stated here. It is important that all studies and experiments be repeatable by other scientists. 4. Results: Major findings are presented in the form of tables, graphs, mathematical analyses (if appropriate), and photographs. Each graphic includes a descriptive legend, and all findings of the study are detailed without interpretation. Results that are statistically significant are mentioned here. In scientific papers, the word “significant” has a very precise definition. Every time this word is mentioned, it means that the results are very unlikely to be found by chance (usually less than a 5% chance). Authors most often present significant findings by reporting a “p-value.” A p-value less than 0.05 is considered significant — there is less than a 5% chance the same outcome would be observed by random chance. In experimental studies, a statistically significant result gives scientists confidence that the outcomes they measure are a response to the treatment they provided. 5. Discussion: The major findings presented in the results are interpreted. The authors contextualize their work with reference to previously published studies. In this section, authors speculate on the significance of their findings. If there are any limitations to the study, they are often outlined here. This section usually closes with a brief set of conclusions drawn from the study and a suggestion for future investigation of the topic. 6. Citations (also known as References or Bibliography): Any paper or papers that were cited in the article or detail techniques used by the authors is listed in the Citations. This makes it possible for readers to locate these articles in the event that they wish to read them. 1 Some journals put the Materials and Methods section at the end of the paper. Interpreting Papers Interpreting papers does not come naturally to anyone; it is truly a skill that requires practice. For beekeepers just being introduced to reading primary science articles, it is helpful to pause and reflect on what they have just read as they go through an article. This can help them think critically throughout the article and gauge whether or not they are able to logically follow the content. With particularly dense articles, it is normal to require reading the paper several times to fully digest all the information. Be patient with yourself. Scientific papers often use a lot of jargon. Write down any words that you do not recognize so you can look them up afterward. It is not necessary to fully understand the statistical analyses of a paper in order to learn the major findings and the take-home message. Here are some examples of reflection questions beekeepers can ask as they read a paper. Practicing reading scientific articles in this manner helps to strengthen skills in critical thinking: Interpreting Scientific Papers After reading this section... ...Try to answer these questions. What are the specific questions being tested? Why are the authors interested in testing Introduction these questions? What is the extent of background knowledge pertaining to this topic? How are the authors specifically testing this question? Methods What outcomes are they measuring (e.g., colony survival, honey production, virus levels)? Do the results help answer the specific questions they tested? Are any of the results significant? Results What are the figures and tables in a paper describing? How might I interpret these results? How did the authors interpret these findings? Did the paper reference any other articles that support or dispute their findings? Discussion Do I agree with the author's interpretation? Are there any

limitations to this study? What are the follow-up questions to research in this field? In rereading the abstract after I have finished the paper, does it make clear sense to me? Abstract Would I be able to provide a similar summary after reading the paper?

READ

Developing Skills in Reading Scientific Articles —

Interpreting Papers ■ Interpreting papers does not come Key Points naturally to anyone; it is truly a skill that Readers should think critically requires practice. For beekeepers just when interpreting scientific being introduced to reading primary articles. science articles, it is helpful to pause Reading articles with proficiency is and reflect on what they have just read a skill that requires time and as they go through an article. This can practice. help them think critically throughout the article and gauge whether or not they are able to logically follow the content. With particularly dense articles, it is normal to require reading the paper several times to fully digest all the information. Be patient with yourself. Scientific papers often use a lot of jargon. Write down any words that you do not recognize so you can look them up afterward. It is not necessary to fully understand the statistical analyses of a paper in order to learn the major findings and the take-home message. Here are some examples of reflection questions beekeepers can ask as they read a paper. Practicing reading scientific articles in this manner helps to strengthen skills in critical thinking: Interpreting Scientific Papers After reading this section... ..Try to answer these questions. What are the specific questions being tested? Why are the authors interested in testing Introduction these questions? What is the extent of background knowledge pertaining to this topic? How are the authors specifically testing this question? Methods What outcomes are they measuring (e.g., colony survival, honey production, virus levels)? Do the results help answer the specific questions they tested? Are any of the results significant? Results What are the figures and tables in a paper describing? How might I interpret these results? How did the authors interpret these findings? Did the paper reference any other articles that support or dispute their findings? Discussion Do I agree with the author's interpretation? Are there any limitations to this study? What are the follow-up questions to research in this field? In rereading the abstract after I have finished the paper, does it make clear sense to me? Abstract Would I be able to provide a similar summary after reading the paper?

WATCH

Developing Skills in Reading Scientific Articles

In the following video, Courtesy Assistant Professor and fourth-generation beekeeper Dr. Christina Wahl takes us through reading and interpreting a scientific article. The article she will use as an example is the PLoS One article on insect declines that we have been following throughout this module. After viewing the video, download the paper and read through it. Reflect on whether or not you are able to understand the main points in the article and interpret the tables and figures. Video Transcript We're going to be looking at the science paper that you've already been reading about in this module from PLoS ONE, and that paper is titled "More than 75 percent decline over 27 years in total flying insect biomass in protected areas." This paper was written by quite a number of people all working in Europe on the phenomenon of how many insects are flying near the ground over periods of time in Germany. How do we approach a science paper? This is the research paper, this is the primary

literature that we have on the topic here. You want to read it differently than a book because it's written for professionals, it's not written for the average person on the street. So it's going to be difficult to comprehend, especially if you try to start at the beginning and plod through it like you would a newspaper article. So I'm going to suggest how to go about reading this. The first thing to do is to look at the abstract. In a very short amount of text, sometimes 150 to 250 words, the entire study is summarized. And from the abstract that you read on this paper, you're going to see that it's an observational study, it's not a controlled experiment, it doesn't test a hypothesis, it's making notes about the natural environment in a controlled and disciplined way. The bottom line in the abstract here is that as much as 82 percent of flying insect biomass was lost over a period of 27 years in Europe. The average loss was 76 percent, but a midsummer decline of up to 82 percent was seen. So, that's what our abstract's telling us. Now, the introduction is the next section of the paper, and normally that gives you the background of the study, why it was worth doing, what other people have found in the past, and sort of concludes with a little here's where we're going with this. We're going to skip over that for now. And then we get to materials and methods, which is the next section of a literature paper, primary literature paper. And here I take a little time and take a quick look. So there's the data section, and it describes how the biomass data was acquired. Now, biomass is weight of biology, so what they did was collect insects, dry them according to a specific protocol, and then weigh that biomass to see how much there was, again, over time in different locations. Then they described what their locations were, they talk about how they collected the insects, they show photos of the type of traps they used, and they also have some table data on how the experiments were set up. So we're going to look at one table and a couple of figures out of this paper in some detail and try and get a sense for you of how to evaluate these things as best you can. And don't feel bad if you don't understand them because even the experts in other fields that try to read a paper in this category may have trouble. So in table one, it's an overview of malaise trap sample sizes. Now, the malaise trap is the kind of device they use to collect the insects. and I notice, reading the columns from left to right, that they have a column of years, the next column is the number of locations they collected from in those years, the next column over is the number of locations that have been set sample previously in those years. A lot of zeros there. They didn't realize until the last ten years or so that this was going to be important data; they were collecting it for some other reason. They said at some point, "Oh my gosh, there are fewer and fewer insects here," so then they turned it into a real study. The number of samples is in the next column, and that varies from a low of 4 to a high of 348 in 2014 here. Then comes the mean exposure time, which is simply a list of numbers. 146.6, 228.5, what does that mean? To find out, you have to go to the table legend, where you learn that it's exposure time in days. And then finally, you have the standard deviation of exposure time. If you don't understand descriptive statistics or basic statistics, there are very good short explanations online, Wikipedia and so forth have these. But for this purpose the standard deviation is the range plus and minus around the mean that gives you the range within which half your data is going to be found. So if the value is, say, 13 then that means subtract 13 and add 13 to the mean and you get the range for all the data. The next couple of tables deal with some fairly sophisticated statistics that would take us quite a bit of time and effort to wade into, and for now, let's just don't. And we'll go to figure two. Figure two is the temporal distribution of insect biomass. So when I look at a graph, I hope when you look at graphs the first thing you do is you look at the x-axis, that horizontal axis which gives you the independent variable. This is the information that's going to change

no matter what else happens. Something like time, a for instance, which is on the x-axis and both halves of this figure. The upper half A is in years and the lower half B is in months. The y-axis is the dependent variable, that's the variable that's changing according to whatever is happening on the x-axis. And in both of the cases A and B here, we see that the y-axis is biomass, which is measured in grams per day. Okay, so real quickly then, if we look at a part A on this figure we see that between 1990 and 2010 there is a trend line that's going downhill for biomass. So biomass is being lost on a steady basis over that period of time. In part B we see that, in terms of an annual change in biomass, from March is the first month here, December is the last month on this graph. We see a kind of increase and then a decrease, which is what you expect in a temperate climate like Germany where we don't see too many things flying around in January, but in June and July things are very busy. So, lots of bugs flying around in the middle of the summer, not so many in November. And that is consistent throughout the whole study, all 27 years. But what we see in the color coded part of this graph is that the earlier years are in the darker blue and the later years, like 2016, is in orange. And so, from that you can see that the curve is dampening, meaning that the height of the season when there should be the most biomass, June, July and August, is much diminished compared to years past. So, while the overall cycle of biomass change is still there, the amount of change has been reduced quite a bit. There's much less biomass in figure two. Now, figures three and four present versions of that data in different ways, trying to pull a little more information out of the same data set that was collected. And you should look at those and see if you can understand what they're trying to show you and how it differs from the other figures, but I'm going to go right ahead here to figure five, which is titled "Marginal effects of temporal changes and considered co-variants on insect biomass." Now, you'll remember that we went over a lot of complicated statistics earlier. Those statistics were designed to evaluate the possible correlative factors that might have influenced the decline in biomass. So it could have been has it been climate change, temperature changing over time, has it been a decrease in tree cover in the areas that were examined, could it be the amount of herbs growing on the ground? Whatever it was, they had all these different variables to look at and they came up with different ways of testing those. And the result is, one way of looking at it, is presented here in figure five, where the x-axis is the zero line in the middle and the y-axis shows you the contribution to insect biomass development, which is either positive above the zero or negative below the zero. And if you look across the graph you'll see that temperature has a positive effect on biomass, which doesn't surprise me because sure, in June and July temperatures are warm, lots of insects are flying. So temperature has a positive correlation with insect biomass. Another one that's positive is the amount of nitrogen, probably increasing the amount of plant matter, and that makes insects happy. And arable land. If there's more arable land, there's more bugs. That's fine, too. But all of them hover around zero, more closely than the unknown bar over there on the left, which you'll see is extremely negative. How so, why is that extreme? Look at the double hash bar in the middle of that column and you'll see that the scale is so much larger that they had a kind of collapse to fit it here. What that's telling us is that unknown factors are the most strongly correlated with the change in insect biomass. They don't know why the insect biomass was declining, and their statistics didn't fish anything out, either. So we don't need to go back and understand all those statistics. Let's go on to the discussion now. And in reading the discussion, what I generally do is I read the first sentences of each paragraph, which in a well written paper should pretty much tell you what the content of that paragraph is, and if I find that the paper is very readable and I

don't have to struggle too hard with it, I'll actually read the whole discussion more carefully. When you do that with this paper you will find that, indeed, what we just concluded from those two figures we looked at is what they concluded from the whole project. That they've seen this extreme decline in biomass, but that they weren't able to correlate it with any of the variables expected that might account for it. And what they propose is that some of the variables that they didn't look at could influence this. And they didn't look at pesticide use, the increasing amount of monoculture in the area. They didn't look at herbicides, or fertilizers, or any other kind of human activity in the area. Because they were working with nature preserves and forested land, they didn't think there would be a problem with that in those regions, but now they're considering maybe there is some drift over of effect and they should go back and do that analysis. They also mentioned in the discussion that there has only been one other study that looks at the decline of insect biomass, and it wasn't anywhere near as extensive as this one, but there was also another study done with vertebrates, animals with backbones, that shows there's been a large decline in biomass among vertebrates. So I would at this point say, "Wow, I want to read more." I could go to the bibliography and find those other studies, the other one on insects and the one on vertebrates, that have also been shown to decline over time and go to those papers and read them. And that's what the bibliography is for. So, in summary then, what you want to do with a primary research paper is read the abstract, skim over the methods to get a gist of some sense of what was going on there. Then look at the figures in more detail. Read a little bit of the discussion. If it seems like an accessible paper, read the whole thing. And by all means, go back, as we reviewed earlier in module, and find a good popular review that was written by an expert or as a result of an interview with somebody who could help interpret the paper for the lay person. And as you probably can tell, what we found in popular reviews was very good as far as getting just about all the information we needed out of this paper in PLoS ONE.

READ

Distinguishing Science From Pseudoscience

■ The scientific method is the process by which scientists conduct experiments. Fallacious arguments are mistakes. This method relies on a series of steps in reasoning. 1. Scientists make observations of the natural world around them. The risk of fallacious argument is low in a peer-reviewed journal. 2. They formulate hypotheses to explain these observations. The term "pseudoscience" refers to 3. They test these hypotheses by work based on fallacious gathering data and/or conducting experiments. 4. They refine, alter, expand, or reject A scientific theory is a well-founded explanation based on continuous data collection. 5. They develop a scientific theory — a repeatedly tested explanation of the natural world based on extensive scientific evidence. Note that this definition is substantially different from the common vernacular use of the word "theory," which simply means a speculative guess. One of the chief reasons that scientists around the world rely on the use of the scientific method is that a properly designed experiment avoids fallacious arguments, or mistakes in reasoning, and its findings help move our collective understanding toward the truth. Good science is published in peer-reviewed journals and stands the test of time; it is referenced and repeatable by others. There is much fake science online, and some is perpetuated in the beekeeping industry through the promotion of certain apitherapy products. Be critical (in the sense of critical thinking; not mean or disparaging) about

conclusions drawn by peers conversing at bee clubs, in online blogs, and in popular science articles that appear at first glance to be quite scientific. Pseudoscience involves beliefs or statements that claim to be scientific and factual but in reality contain fallacies in their arguments. Always examine the evidence behind claims to distinguish between scientific fact and fiction. It is important to be able to recognize a fallacious argument. Perhaps beekeepers will be better able to recognize a fallacy by considering a few examples. Example 1: “My spring honey is dandelion honey because dandelions are blooming when the bees are foraging.” In this example, there is insufficient evidence to infer a causal link between dandelion bloom and the type of honey that bees are producing. It is possible that there are other plants in bloom that are contributing to the honey. Example 2: “I lost 200 hives as a result of the local farmer’s increased use of pesticides.” In this example, there is no evidence provided to show a link between the pesticides and those hive deaths. They do not provide any evidence that the farmer’s pesticide usage has increased. The beekeeper speculates that this is the cause, but there are other causes that should also be investigated. Example 3: “Neonicotinoids must be killing my bees because I have Varroa mites under control.” In this example, these arguments are presented as though there are only two competing explanations, when, in fact, there are more than two. The bees could be dying for other reasons, or a combination of reasons, such as disease or poor nutrition.

WATCH

Exploring Your Curiosity: How to Design Your Own Study

■ Beekeepers are a wide range of people with diverse backgrounds, opinions, lifestyles, and personalities. However, if encouraged! to test questions in their own bee yards, there is one trait that beekeepers share, it is curiosity. Beekeepers by nature tend to be curious about bees and test questions by comparing and the inner workings of the colony, treatment groups to control the enticing products they produce, groups, their role in the surrounding ecosystem, and what factors and management practices can make a colony either flourish or perish. In these experiments, all factors remain constant. Beekeepers may find it quite empowering to learn what works best for their bees through simple experiments, and they do not have to be scientists to begin testing their questions in their own bee yards. One component of the Master Beekeeper final exam series is to deliver a 15-minute presentation on a research topic. Beekeepers have two options: Option 1: They can select a topic from a list provided by the facilitator to thoroughly research online, through books, journals, or other avenues. Option 2: They can conduct their own research experiment or study on which to present. The research projects we introduce in this course are controlled experimental projects. The projects are considered “experimental” because the beekeeper is manipulating one group (i.e., applying a “treatment” to one group) of colonies and recording the resulting effects. The projects are considered “controlled” because all other aspects remain the same for colonies, except for what the beekeeper wishes to test. In addition, there is a “control group”; this group does not receive a treatment and serves as a benchmark. Controlled experimental studies are different from observational studies, where researchers do not control any variables but instead simply observe and collect data on how colonies behave in the natural world. Even if students do not choose to conduct their own research project for their final exam, understanding the basic steps of these projects can allow them to test questions they may have in the future. Beekeepers need a minimum of two colonies to perform controlled experiments, but the power

in the accuracy of results is much improved if more colonies are tested as there is a lot of natural variation in honey bee colonies; no two are identical! The Questions That Can Be Tested Are Limitless There are infinite opportunities when it comes to conducting research projects. Beekeepers can choose to investigate questions that are relevant to their own operation, goals, or business. Some examples of questions that beekeepers may be interested in testing with a controlled experiment include: Which bee stock performs better in my region with regard to Varroa mite levels? Does using a honey excluder reduce honey production? Does applying pollen patties to my colonies in February boost the colony population in time for apple pollination in May? Does the combination of breaking the brood cycle plus applying a mite treatment reduce Varroa levels more so than applying a mite treatment alone? Does wrapping colonies for winter improve overwintering success compared to colonies that are not wrapped? Do my colonies experience more disease and lower brood production after they come out of blueberry pollination? Each of these questions can be tested quite simply in the bee yard, yet the results can help beekeepers make meaningful decisions that can affect the amount of time or money they spend on a specific practice. For instance, beekeepers may find that they do not need to spend money on winter wraps every year, or that the time investment of breaking the brood cycle is worthwhile to reduce mite levels. Let's go through a hypothetical example to see how a beekeeper could test a question in their bee yard, then we will summarize the main considerations to take into account for any experiment. The question we will investigate is the first example in this list: "Which bee stock performs better in my region with regard to Varroa mite levels?" Video Transcript Let's go through an example of a beekeeper designing and conducting his own experiment so that you can see how to set out and test your own beekeeping questions. A beekeeper, let's call him Matt, has been spending lots of time and money treating his bees for Varroa mites in recent years. But despite his efforts, he continues to lose colonies every year from high infestations. He's been reading a lot about different bee stocks for mite resistance, but he doesn't know which ones will work best in his operation. And he's heard some mixed reviews from other beekeepers who've tried them. Matt is most interested in two stocks, Varroa sensitive hygienic bees and Russian bees, and he has 12 colonies that he can use in his experiment. The specific questions he wants to set out to test are as follows. First, which bee stock requires the fewest mite treatments? And second, which bee stock experiences the greatest overwintering success? The outcomes that he is going to measure to test these questions are the number of mite treatments needed in the season and survival the following spring. The first thing Matt needs to do is divide up his colonies into three groups. One group will test the Varroa sensitive hygienic bees, the next group will test the Russian bees, and the third group will be his current bee stock. The current bee stock is considered his control group. This group is the normal, or usual, state for his operation and it doesn't receive a treatment. In this case, a Varroa-resistant stock. Having this control group will allow him to see whether the Varroa-resistant stocks are better than what he's currently using. His 12 colonies can be divided up equally so that there's four colonies in the Varroa sensitive hygienic group, 4 in the Russian group, and 4 in the control group. Matt gets on the phone and he calls up some reputable queen breeders and orders 4 Varroa sensitive hygienic queens, 4 Russian queens, and 4 queens from his current supplier. He wants to requeen all 12 of his colonies to treat them identically throughout the experiment. The act of requeening, regardless of the stock, may influence Varroa mite levels, so to be sure that the response that we're seeing is from the actual genetic stock and not from the act of requeening in general, Matt will requeen all of his colonies. Each colony is

numbered so that he can easily keep track of them in his notebook. Before the experiment begins, he considers all other factors that might skew his results and he figures out ways to address them ahead of time. For instance, Matt knows that Varroa populations can also be influenced by bee drift, and he doesn't want this to confound his results. So he separates the groups of colonies from each other by placing them in three separate nearby apiaries. And he adds unique labels on the front of each hive to minimize drift. He makes sure that the apiaries are close enough, though, so that colonies are still able to experience the same climate and the same access to food sources. The population strength of bees can also influence Varroa. If any of his colonies are particularly weak to begin with, he'll first equalize them by swapping frames of bees and brood so that they all start out the experiment with a roughly equal population. He also monitors his colonies for Varroa mites before the experiment begins to ensure that they are all starting with roughly equal mite numbers. If needed, he treats his colonies to bring them all below the treatment threshold for the start of the experiment. In late May his queens arrive. He marks them all with different colors for each group to keep track of them, and he requeens all of his colonies. The experiment has officially begun. Throughout the season, Matt takes monthly Varroa mite samples as he normally does in his beekeeping operation. If colonies are above the treatment threshold, he treats on an apiary basis, again, as he normally does in his operation. Each group is managed identically for Varroa. He keeps records on how many treatments each group receives. Occasionally, he spots the queens and ensures that they're the same ones that he introduced. In all other respects besides the genetic stock and Varroa treatments, Matt manages all his colonies identically. For instance, if they need feeding, he feeds them all. When it's time to winter, he winters them all in the same manner. After he winters his colonies, he observes which ones survived over winter, and he records survivorship for each hive. Matt has done a fantastic job controlling all factors in his experiment. Now it's spring and Matt has collected data on which colonies survived, and it's time to take a look at his results. Here are his data in his record book. Mite levels are recorded in number of mites per 100 bees. The way that he's organized his notebooks show us his mite levels and whether or not he needed to treat throughout the months of July and October. The first column are his 12 colonies. The first four are the Varroa sensitive hygienic colonies, followed by the next four, which are his Russian colonies, followed by the last four, which are his control colonies. In July, he monitors them all, but it turns out that both the Varroa sensitive hygienic colonies and Russian colonies are all below the treatment threshold. He does not need to apply a treatment to that group. However, in his control group, there are a few colonies that are above the treatment threshold, so he needs to apply a treatment to these. In August, we see the same trend. He doesn't need to treat his Varroa sensitive hygienic group or his Russian group, but he does need to treat his control group. In September, it's the first time he needs to treat his Varroa sensitive hygienic group. He has at least one colony that's above the treatment threshold. He does not need to treat the last two groups. In October, he doesn't need to treat his Varroa sensitive hygienic group at his low mite levels, but he does need to treat his Russian group for the first time. There's at least one colony that's above the treatment threshold. And he also needs to treat his control group for a third time because, again, there's high mites in at least one colony. The final column is denoting survival over winter. It says Yes if that colony is alive in the spring and No if that colony isn't. At this point scientists perform statistical analyses to determine whether there are significant differences among groups. Statistics helps you determine the probability of finding the same results by pure chance. If there's a very low probability, usually less than 5 percent, of finding

the same results by chance, but it's highly suggestive that the findings were a result of the treatment. In this example, the statistics find that the likelihood of winter survival or number of mite treatments needed is actually really low by pure chance, then it's likely to be related to the genetic stock that Matt tested. It's important to note that master beekeepers do not need to be statisticians, and performing statistics goes well beyond the requirements for this course and the final exams. If you're familiar with stats and like to formally test your data, please do so. But in this example we're just going to base it on his notebook observations. So, by looking at the data, we can see that the Varroa sensitive hygienic stock had the best overall winter survival. All four out of four colonies survived. The Russian stock group had the next best survival. Three and four colonies made it until spring. The control group did the poorest, with only one of the four colonies surviving. As for the number of mite treatments that were needed, both the Russian stock and Varroa sensitive hygienic stock were tied. Each required only one treatment throughout the beekeeping season. On the other hand, the control group required three treatments. This information is incredibly useful for Matt. He now has info about which stocks work best in his operation conditions and under his management practices. Although it's only one beekeeping season's worth of data and it's not looking at very many colonies, it still gives them the confidence to order Varroa sensitive hygienic queens for his operation next year to see how they continue to perform. This experiment suggests that he might save time and money on mite treatments, and he might experience greater overwinter survival. After watching this example, you can see the steps and considerations that are required for performing a controlled experiment. The steps are similar regardless of the question that is asked. Step 1: Determine the question to ask and the outcomes to test (e.g., honey production, overwintering survival, frames of brood) Step 2: Organize colonies into groups. For a controlled experiment, there must be a control group that does not receive a "treatment." Step 3: Control for all factors other than the treatment. Ensure colonies begin with the same population and access to resources, and manage colonies in an identical manner throughout the experiment. Step 4: Record measurements. Step 5: Interpret results. Scientists use statistics to help find meaningful differences in the data; however, it is not necessary to do this for the Master Beekeeper Program. Take some time to think about how you could set up and test the other five examples provided near the top of the page. What types of questions could you imagine wanting to find answers to in your bee yard? Begin to gain confidence in thinking scientifically about testing your questions.

MODULE WRAP-UP

Keeping Up With Science

In this module, you looked at some ways to find, comprehend, and evaluate science. You discovered that you can explore recent science advances without a science degree. Downloadable Resources You may wish to download and save these files for future reference. Marketing Local Honey In New York State Hive Stocking Rates for Pollination in the U.S. Pollination Map of the U.S. Hive Rental Rates in the U.S. Sample Pollination Services Contract Doses of Neonicotinoids That Cause Sublethal Impacts on Honey Bees NYS Beekeeper Tech Team Inventory & Financial Analysis Handbook New York Risk Management Programs for Beekeepers Sample Mentoring Schedule for a First-Year Mentee Recommended Readings There are more books written about honey bees than any other animal — besides humans — on the planet. With so many to choose from, we've compiled a list to help guide you in your readings. There are no required readings for this course; all examination questions will be

based solely on the online material. However, if you are interested in delving deeper into topics we've discussed, here are some great resources: Honey "A Book of Honey" (1980) by Eva Crane Pollination Services "Crop Pollination by Bees" (2000) by K.S. Delaplane and D.F. Mayer "Insect Pollination of Cultivated Crop Plants: USDA Agricultural Handbook" (1976) by S.E. McGregor "Pollination in the Agricultural Landscape: Best Management Practices for Crop Pollination" (2012) by Thomas S. Woodcock Confidence in Communicating About Pesticides "How Neonicotinoids Can Kill Bees: The Science Behind the Role These Insecticides Play in Harming Bees, 2nd Edition" (2016) by Jennifer Hopwood, Aimee Code, Mace Vaughan, David Biddinger, Matthew Shepherd, Scott Hoffman Black, Eric Lee-Mäder, and Celeste Mazzacano "Worldwide Integrated Assessment of the Impacts of Systemic Pesticides on Biodiversity and Ecosystems" (2015) by The Task Force on Systemic Pesticides "Pollinator Protection: A Bee and Pesticide Handbook" (1990) by Carl A. Johansen and Daniel F. Mayer Turning Bees Into Business "In Business with Bees" (2018) by Kim Flottum "The E Myth" (1986) by Michael E. Gerber "Fearless Farm Finances" (2012) by Midwest Organic and Sustainable Education Service Being a Leader in Your Community "The Art of Public Speaking" (2014) by Stephen E. Lucas Keeping Up With Science "The Scientific Endeavor: A Primer on Scientific Principles and Practice" (2000) by Jeffrey A. Lee "A Beginner's Guide to the Scientific Method, 3rd Edition" (2004) by Stephen S. Carey

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NY FarmNet

Photo of New York FarmNet Logo: New York FarmNet

WATCH

Developing Skills in Reading Scientific Articles

All photos and illustrations: Hallman et al. 2017

WATCH

Participating in Research (Karen's management survey)

Screenshots of Bee Informed Partnership website: Bee Informed Partnership Credits for Illustrations Marketing Honey in New York State This information booklet was created by Cornell Cooperative Extension Hive Stocking Rates for Pollination in the US Data from this illustration are from Cost of Pollination (2016), The USDA National Agricultural Statistics Service Pollination Map of the US Data from this illustration are from Cost of Pollination (2016), The USDA National Agricultural Statistics Service Hive Rental Rates in the US Data from this illustration are from Cost of Pollination (2016), The USDA National Agricultural Statistics Service Doses of Neonicotinoids that Cause Sublethal Impacts on Honey Bees Data from this illustration are from Hopwood et al. (2016). How neonicotinoids can kill bees: the science behind the role these insects play in harming bees. 2nd ed. Beekeeping Business Record Booklet This information booklet was created by Mary Kate Wheeler New York Risk Management Programs for Beekeepers This information booklet was created by Jenny Ifft